Symbolic gate atlas

Exact identities. Explicit phase policies. Native entangler counts.

327 targets × 32 basis panels

Each target has one page with 32 basis panels in alphabetical order. Blue labels denote literal matrix equality. Brown labels denote a projective circuit with its exact scalar factor. Both appear when the projective construction is strictly cheaper. Physical GP operations are forbidden.
E counts native two-qubit primitives, L counts one-qubit operations. Scores are displayed rounded; the archive stores exact expressions. Objective version 2 clamps negative denominator penalties; an exact compatibility audit preserves the earlier retained scores. Only an explicit matching certificate establishes a minimum entangler count. A verified upper bound does not establish minimum local cost. A minimum uniform E budget concerns a circuit for the whole parameterized family: a certified hard member forces that budget, while special parameter values can require fewer pulses. Published-cap notes identify constructions above known universal two-qubit bounds; these remain optimization gaps. Proof conditions, source hashes and attempt histories remain in the archive.
Open cases are shown as open. Proved obstructions preserve the supplied basis and may be conditional for parameterized families. Register constructors use the recorded integer-width or matrix fixtures. These pages do not claim a solution for every register size.
Circuits are drawn by Qiskit's LaTeX drawer. Long circuits continue on successive rows in time order. An explicit $[G]^{\times n}$ box denotes n consecutive identical primitive operations; their full physical count remains in E.
Long exact expressions use numbered η_j labels defined in the expression glossary. No parameter is rounded or dropped.
The separate Clifford volume lists every exact phase in each distinct one-qubit alphabet, with 24 circuits per page. Identical alphabets shared by multiple presets are printed once.

Archive SHA256: e4d7c8086e4361b97120819ab7bf42bb5e8fd4c95f5b4c87386063a2dfa1a8e5

Alphabetical target index

Links lead to the 32-basis page for each target.

ACX 4
B 5
Barenco 6
C3SX 7
C3X 8
C4X 9
Canonical 10
CCP 11
CCX 12
CCXPow 13
CCY 14
CCZ 15
CCZPow 16
CCZS 17
CH 18
CNOT 19
Controlled 20
CP 21
CR 22
CROT 23
CRX 24
CRY 25
CRZ 26
CS 27
CSdg 28
CSWAP 29
CSX 30
CSXdg 31
CT 32
CTdg 33
CU 34
CU1 35
CU3 36
CX 37
CXPow 38
CY 39
CZ 40
CZPow 41
CZPowGate 42
CZS 43
DB 44
DCX 45
Deutsch 46
Diagonal 47
ECR 48
fFredkin 49
fSim 50
FSWAP 51
FSWAPPow 52
Givens 53
GP 54
GPi 55
GPi2 56
H 57
H_q1 58
H_q2 59
H_q3 60
H_q4 61
H_q5 62
H_q6 63
H_q7 64
HPow 65
HS 66
HS_q1 67
HS_q2 68
HS_q3 69
HS_q4 70
HS_q5 71
HS_q6 72
HS_q7 73
HSdg 74
HSdg_q1 75
HSdg_q2 76
HSdg_q3 77
HSdg_q4 78
HSdg_q5 79
HSdg_q6 80
HSdg_q7 81
HSX 82
HSX_q1 83
HSX_q2 84
HSX_q3 85
HSX_q4 86
HSX_q5 87
HSX_q6 88
HSX_q7 89
HSY 90
HSY_q1 91
HSY_q2 92
HSY_q3 93
HSY_q4 94
HSY_q5 95
HSY_q6 96
HSY_q7 97
HX 98
HX_q1 99
HX_q2 100
HX_q3 101
HX_q4 102
HX_q5 103
HX_q6 104
HX_q7 105
HY 106
HY_q1 107
HY_q2 108
HY_q3 109
HY_q4 110
HY_q5 111
HY_q6 112
HY_q7 113
HZ 114
HZ_q1 115
HZ_q2 116
HZ_q3 117
HZ_q4 118
HZ_q5 119
HZ_q6 120
HZ_q7 121
I 122
I_q1 123
I_q2 124
I_q3 125
I_q4 126
I_q5 127
I_q6 128
I_q7 129
Identity 130
IonQMS 131
IonQMSMax 132
IonQZZ 133
iSWAP 134
iSWAP_pow 135
ISWAPPow 136
LinearFunction 137
Magic 138
Margolus 139
MCMT 140
MCPhase 141
MCX 142
MS 143
P 144
Pauli 145
PauliRotation 146
Peres 147
Permutation 148
PhasedFSim 149
PhasedISWAPPow 150
PhasedX 151
PhasedXPow 152
PhasedXZ 153
QFT 154
QFTdg 155
R 156
RC3X 157
RCCX 158
RV 159
Rx 160
Rxmy 161
Rxx 162
Rxy 163
Rxz 164
Ry 165
Ryy 166
Rz 167
Rzx 168
Rzz 169
S 170
S_q1 171
S_q2 172
S_q3 173
S_q4 174
S_q5 175
S_q6 176
S_q7 177
Sdg 178
Sdg_q1 179
Sdg_q2 180
Sdg_q3 181
Sdg_q4 182
Sdg_q5 183
Sdg_q6 184
Sdg_q7 185
SH 186
SH_q1 187
SH_q2 188
SH_q3 189
SH_q4 190
SH_q5 191
SH_q6 192
SH_q7 193
SHX 194
SHX_q1 195
SHX_q2 196
SHX_q3 197
SHX_q4 198
SHX_q5 199
SHX_q6 200
SHX_q7 201
SHY 202
SHY_q1 203
SHY_q2 204
SHY_q3 205
SHY_q4 206
SHY_q5 207
SHY_q6 208
SHY_q7 209
SHZ 210
SHZ_q1 211
SHZ_q2 212
SHZ_q3 213
SHZ_q4 214
SHZ_q5 215
SHZ_q6 216
SHZ_q7 217
SiSWAP 218
SiSWAPdg 219
SqrtH 220
SqrtHdg 221
SSWAP 222
SSWAPdg 223
SWAP 224
SWAPPow 225
SX 226
Sx 227
SX_q1 228
Sx_q1 229
SX_q2 230
Sx_q2 231
SX_q3 232
Sx_q3 233
SX_q4 234
Sx_q4 235
SX_q5 236
Sx_q5 237
SX_q6 238
Sx_q6 239
SX_q7 240
Sx_q7 241
Sxdg 242
Sxdg_q1 243
Sxdg_q2 244
Sxdg_q3 245
Sxdg_q4 246
Sxdg_q5 247
Sxdg_q6 248
Sxdg_q7 249
SxY 250
SxY_q1 251
SxY_q2 252
SxY_q3 253
SxY_q4 254
SxY_q5 255
SxY_q6 256
SxY_q7 257
SxZ 258
SxZ_q1 259
SxZ_q2 260
SxZ_q3 261
SxZ_q4 262
SxZ_q5 263
SxZ_q6 264
SxZ_q7 265
SY 266
Sy 267
SY_q1 268
SY_q2 269
SY_q3 270
SY_q4 271
SY_q5 272
SY_q6 273
SY_q7 274
SYC 275
Sydg 276
T 277
Tdg 278
TK2 279
Toffoli 280
U 281
U1 282
U2 283
U3 284
UC 285
UCRX 286
UCRY 287
UCRZ 288
Unitary 289
WalshHadamard 290
X 291
X_q1 292
X_q2 293
X_q3 294
X_q4 295
X_q5 296
X_q6 297
X_q7 298
XPow 299
XX 300
XX90 301
xx_minus_yy 302
xx_plus_yy 303
XXPow 304
Y 305
Y_q1 306
Y_q2 307
Y_q3 308
Y_q4 309
Y_q5 310
Y_q6 311
Y_q7 312
YPow 313
YY 314
YY90 315
YYPow 316
Z 317
Z_q1 318
Z_q2 319
Z_q3 320
Z_q4 321
Z_q5 322
Z_q6 323
Z_q7 324
ZPow 325
ZZ 326
ZZ90 327
ZZMax 328
ZZPhase 329
ZZPow 330

Native basis key

Every panel preserves its supplied alphabet. Native units and macro counts matter.

Preset	Entangler	Local alphabet	Scope / units
aqt_rxx	Rxx	R, Rz	exact continuous
berkeley_b	B	U3	exact continuous
clifford_only	CX	H, S	Clifford analysis/control; not universal.
clifford_t_cx	CX	H, S, T, Tdg	Dense universal; exact only for representable targets, not arbitrary symbolic angles.
clifford_t_cz	CZ	H, S, T, Tdg	approximate discrete
cx_h_phase	CX	H, P	exact continuous
cx_rz_ry	CX	Rz, Ry	exact continuous
cx_u	CX	U3	exact continuous
cz_rz_rx	CZ	Rz, Rx	exact continuous
cz_u	CZ	U3	exact continuous
google_cz	CZ	PhasedXZ	exact continuous
google_fsim	fSim	PhasedXZ	exact continuous
google_sqrt_iswap	SiSWAP	PhasedXZ	exact continuous
google_syc	SYC	PhasedXZ	exact continuous
ibm_cx	CX	Rz, Sx, X	exact continuous
ibm_cz	CZ	Rz, Sx, X	exact continuous
ibm_ecr	ECR	Rz, Sx, X	exact continuous
ionq_aria	IonQMS	GPi, GPi2	Native turns; interaction theta restricted to [0,1/4] on hardware.
ionq_aria_full	IonQMSMax	GPi, GPi2	exact continuous
ionq_forte	IonQZZ	GPi, GPi2	exact continuous
iqm_cz	CZ	R	R(theta,phi) represents PRX in radians; IQM API units require conversion.
iswap_u	iSWAP	U3	exact continuous
quantinuum_zzmax	ZZMax	PhasedX, Rz	exact continuous
quantinuum_zzphase	ZZPhase	PhasedX, Rz	exact continuous
rigetti_cz	CZ	Rz, Sx, X	Sx differs from native Rx(pi/2) by a tracked global phase.
sqrt_iswap_u	SiSWAP	U3	exact continuous
sqrt_swap_u	SSWAP	U3	exact continuous
tket_tk2	TK2	PhasedX, Rz	Compiler macro count; expand TK2 before comparing physical pulse counts.
xx90_u	XX90	U3	exact continuous
xx_rz_ry	Rxx	Rz, Ry	exact continuous
zx_u	Rzx	U3	exact continuous
zz_u	Rzz	U3	exact continuous

ACX

2 qubits / 32 preserved native bases / alphabetical panel order

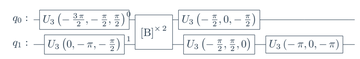
aqt_rxx

PROJECTIVE E=1 / L=7 / score ≈ 19.226 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

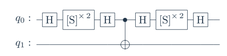
berkeley_b

EXACT PHASE E=2 / L=5 / score ≈ 19.630 — proved E minimum



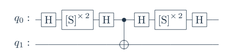
clifford_only

EXACT PHASE E=1 / L=8 / score ≈ 20.000 — proved E minimum



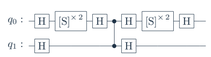
clifford_t_cx

EXACT PHASE E=1 / L=8 / score ≈ 20.000 — proved E minimum



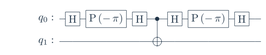
clifford_t_cz

EXACT PHASE E=1 / L=10 / score ≈ 24.000 — proved E minimum



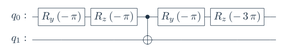
cx_h_phase

EXACT PHASE E=1 / L=6 / score ≈ 16.004 — proved E minimum



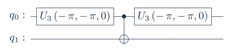
cx_rz_ry

EXACT PHASE E=1 / L=4 / score ≈ 12.008 — proved E minimum



cx_u

EXACT PHASE E=1 / L=2 / score ≈ 8.012 — proved E minimum



cz_rz_rx

EXACT PHASE E=1 / L=8 / score ≈ 21.216 — proved E minimum

PROJECTIVE E=1 / L=4 / score ≈ 12.808 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_u

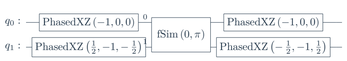
EXACT PHASE E=1 / L=2 / score ≈ 8.412 — proved E minimum

google_cz

EXACT PHASE E=1 / L=4 / score ≈ 12.824 — proved E minimum

google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 12.828 — proved E minimum



google_sqrt_iswap

EXACT PHASE E=2 / L=5 / score ≈ 19.830 — proved E minimum

PROJECTIVE E=2 / L=5 / score ≈ 19.230 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=6 / L=6 / score ≈ 36.836 — E upper bound

ibm_cx

EXACT PHASE E=1 / L=2 / score ≈ 8.000 — proved E minimum

ibm_cz

EXACT PHASE E=1 / L=4 / score ≈ 12.404 — proved E minimum

ibm_ecr

EXACT PHASE E=1 / L=6 / score ≈ 16.202 — proved E minimum

ionq_aria

PROJECTIVE E=1 / L=7 / score ≈ 19.420 — proved E minimum

$U = \frac{\sqrt{2}(1+i)}{2} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=1 / L=7 / score ≈ 19.018 — proved E minimum

$U = \frac{\sqrt{2}(1+i)}{2} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=1 / L=8 / score ≈ 21.818 — proved E minimum

$U = \frac{\sqrt{2}(-1-i)}{2} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

EXACT PHASE E=1 / L=6 / score ≈ 16.824 — proved E minimum

iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 17.024 — proved E minimum

PROJECTIVE E=2 / L=3 / score ≈ 15.018 — proved E minimum

$U = \eta_1\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=1 / L=7 / score ≈ 19.022 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=1 / L=7 / score ≈ 19.224 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=1 / L=8 / score ≈ 20.404 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=2 / L=5 / score ≈ 19.230 — proved E minimum

sqrt_swap_u

EXACT PHASE E=2 / L=6 / score ≈ 22.036 — proved E minimum

PROJECTIVE E=2 / L=6 / score ≈ 21.236 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=7 / score ≈ 19.230 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=1 / L=5 / score ≈ 15.230 — proved E minimum

xx_rz_ry

PROJECTIVE E=1 / L=8 / score ≈ 21.418 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=1 / L=4 / score ≈ 13.026 — proved E minimum

PROJECTIVE E=1 / L=3 / score ≈ 11.020 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=1 / L=5 / score ≈ 15.832 — proved E minimum

PROJECTIVE E=1 / L=3 / score ≈ 11.020 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

B

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=2 / L=34 / score ≈ 83.904 — proved E minimum

PROJECTIVE E=2 / L=16 / score ≈ 43.858 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

berkeley_b

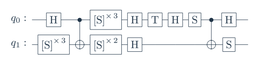
EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

clifford_only

Exact: Proved obstruction: not a clifford
Projective: Proved obstruction: not a clifford

clifford_t_cx

PROJECTIVE E=2 / L=16 / score ≈ 40.000 — proved E minimum



$U = e^{-\frac{i\pi}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: outside clifford t number field

clifford_t_cz

PROJECTIVE E=2 / L=18 / score ≈ 44.000 — proved E minimum

$U = e^{-\frac{i\pi}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: outside clifford t number field

cx_h_phase

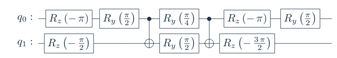
EXACT PHASE E=2 / L=19 / score ≈ 47.220 — proved E minimum

PROJECTIVE E=2 / L=11 / score ≈ 31.212 — proved E minimum

$U = e^{-\frac{i\pi}{8}} C$. Scalar metadata; no physical GP.

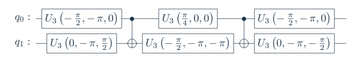
cx_rz_ry

EXACT PHASE E=2 / L=8 / score ≈ 25.416 — proved E minimum



cx_u

EXACT PHASE E=2 / L=6 / score ≈ 21.436 — proved E minimum

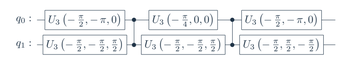


cz_rz_rx

EXACT PHASE E=2 / L=9 / score ≈ 27.818 — proved E minimum

cz_u

EXACT PHASE E=2 / L=6 / score ≈ 22.636 — proved E minimum

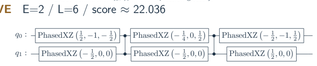


google_cz

EXACT PHASE E=2 / L=8 / score ≈ 26.709 — proved E minimum

Exact definitions: η_0, η_4 in the expression glossary.

PROJECTIVE E=2 / L=6 / score ≈ 22.036 — proved E minimum



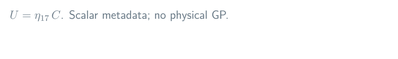
$U = e^{\frac{3i\pi}{8}} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=2 / score ≈ 13.220 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=2 / L=4 / score ≈ 18.275 — proved E minimum



Exact definitions: $\eta_5, \eta_6, \eta_7, \eta_8, \eta_9$ in the expression glossary.

PROJECTIVE E=2 / L=3 / score ≈ 15.009 — proved E minimum

Exact definitions: η_5, η_9 in the expression glossary.

$U = \eta_{10} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=16 / score ≈ 83.757 — E upper bound

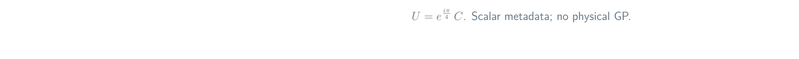
Exact definitions: η_3, η_4 in the expression glossary.

PROJECTIVE E=12 / L=14 / score ≈ 79.084 — E upper bound

$U = (-1)^{\frac{3}{8}} C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=13 / score ≈ 35.008 — proved E minimum

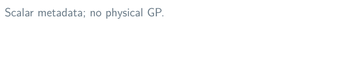


PROJECTIVE E=2 / L=9 / score ≈ 26.402 — proved E minimum

$U = \sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=15 / score ≈ 39.010 — proved E minimum



PROJECTIVE E=2 / L=15 / score ≈ 38.808 — proved E minimum

$U = (-1)^{\frac{3}{4}} C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=14 / score ≈ 38.220 — proved E minimum

PROJECTIVE E=2 / L=14 / score ≈ 37.816 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=17 / score ≈ 46.646 — proved E minimum

PROJECTIVE E=2 / L=17 / score ≈ 46.246 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=17 / score ≈ 45.842 — proved E minimum

PROJECTIVE E=2 / L=17 / score ≈ 45.442 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=19 / score ≈ 51.442 — proved E minimum

PROJECTIVE E=2 / L=19 / score ≈ 51.042 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=8 / score ≈ 26.032 — proved E minimum

iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 22.636 — proved E minimum

quantinuum_zzmax

EXACT PHASE E=5 / L=27 / score ≈ 80.286 — E upper bound

PROJECTIVE E=2 / L=8 / score ≈ 26.028 — proved E minimum

$U = \frac{\sqrt[8]{-1}\sqrt{2}(1+i)}{2} C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=2 / L=18 / score ≈ 48.656 — proved E minimum

PROJECTIVE E=2 / L=15 / score ≈ 41.852 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=2 / L=15 / score ≈ 39.010 — proved E minimum

PROJECTIVE E=2 / L=15 / score ≈ 38.808 — proved E minimum

$U = (-1)^{\frac{3}{4}} C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=2 / L=3 / score ≈ 15.218 — proved E minimum

Exact definitions: η_{11}, η_{12} in the expression glossary.

sqrt_swap_u

EXACT PHASE E=4 / L=16 / score ≈ 53.896 — E upper bound

Exact definitions: η_{11}, η_{12} in the expression glossary.

PROJECTIVE E=4 / L=14 / score ≈ 49.884 — E upper bound

$U = -(-1)^{\frac{5}{8}} C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.606 — proved E minimum

xx90_u

EXACT PHASE E=2 / L=7 / score ≈ 23.042 — proved E minimum

Exact definitions: η_{15}, η_{16} in the expression glossary.

PROJECTIVE E=2 / L=5 / score ≈ 19.030 — proved E minimum

$U = \eta_{17} C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=2 / L=14 / score ≈ 39.032 — proved E minimum

zx_u

EXACT PHASE E=2 / L=8 / score ≈ 28.252 — proved E minimum

PROJECTIVE E=2 / L=6 / score ≈ 23.440 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=2 / L=8 / score ≈ 27.452 — proved E minimum

PROJECTIVE E=2 / L=6 / score ≈ 23.040 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

Barenco (α, ϕ, θ)

2 qubits / 32 preserved native bases / alphabetical panel order

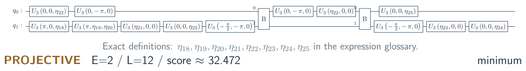
aqt_rxx

PROJECTIVE E=2 / L=15 / score ≈ 41.250 — E upper bound

$U = -ie^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

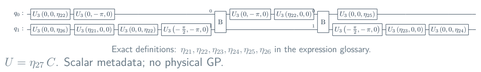
berkeley_b

EXACT PHASE E=2 / L=13 / score ≈ 34.878 — minimum uniform E budget



Exact definitions: $\eta_{18}, \eta_{19}, \eta_{20}, \eta_{21}, \eta_{22}, \eta_{23}, \eta_{24}, \eta_{25}$ in the expression glossary.

PROJECTIVE E=2 / L=12 / score ≈ 32.472 — minimum uniform E budget



Exact definitions: $\eta_{21}, \eta_{22}, \eta_{23}, \eta_{24}, \eta_{25}, \eta_{26}$ in the expression glossary.

$U = \eta_{27}\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=12 / score ≈ 32.816 — minimum uniform E budget

cx_rz_ry

PROJECTIVE E=2 / L=5 / score ≈ 18.410 — minimum uniform E budget

$U = e^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=2 / L=5 / score ≈ 18.430 — minimum uniform E budget

cz_rz_rx

PROJECTIVE E=2 / L=14 / score ≈ 37.628 — minimum uniform E budget

$U = e^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=9 / score ≈ 27.254 — minimum uniform E budget

google_cz

EXACT PHASE E=2 / L=9 / score ≈ 30.105 — minimum uniform E budget

google_fsim

EXACT PHASE E=2 / L=9 / score ≈ 30.113 — E upper bound

google_sqrt_iswap

EXACT PHASE E=2 / L=11 / score ≈ 31.727 — minimum uniform E budget

Exact definitions: $\eta_{28}, \eta_{29}, \eta_{30}, \eta_{31}, \eta_{32}, \eta_{33}, \eta_{34}, \eta_{35}, \eta_{36}, \eta_{37}, \eta_{38}, \eta_{39}, \eta_{40}, \eta_{41}, \eta_{42}$ in the expression glossary.

PROJECTIVE E=2 / L=10 / score ≈ 29.721 — minimum uniform E budget

Exact definitions: $\eta_{32}, \eta_{33}, \eta_{34}, \eta_{35}, \eta_{36}, \eta_{37}, \eta_{38}, \eta_{39}, \eta_{40}, \eta_{41}, \eta_{42}, \eta_{43}, \eta_{44}$ in the expression glossary.

$U = \eta_{45}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=13 / score ≈ 78.129 — E upper bound

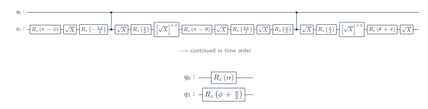
ibm_cx

PROJECTIVE E=2 / L=10 / score ≈ 28.412 — minimum uniform E budget

$U = -e^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

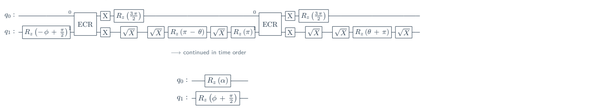
PROJECTIVE E=2 / L=20 / score ≈ 49.220 — minimum uniform E budget



$U = ie^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=2 / L=18 / score ≈ 44.816 — minimum uniform E budget



$U = -e^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=2 / L=24 / score ≈ 63.311 — E upper bound

$U = ie^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

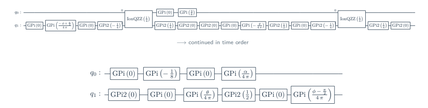
ionq_aria_full

PROJECTIVE E=2 / L=24 / score ≈ 62.507 — minimum uniform E budget

$U = ie^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

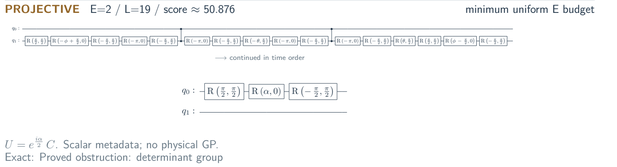
PROJECTIVE E=2 / L=26 / score ≈ 68.107 — E upper bound



$U = -ie^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=2 / L=19 / score ≈ 50.876 — minimum uniform E budget



$U = e^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=4 / L=13 / score ≈ 45.278 — E upper bound

quantinuum_zzmax

PROJECTIVE E=2 / L=14 / score ≈ 39.501 — minimum uniform E budget

$U = -ie^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

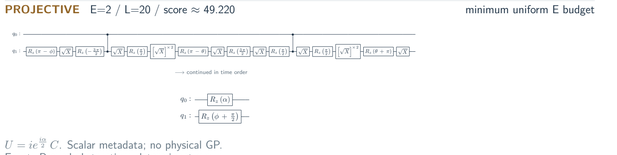
quantinuum_zzphase

PROJECTIVE E=2 / L=14 / score ≈ 39.905 — E upper bound

$U = -ie^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=2 / L=20 / score ≈ 49.220 — minimum uniform E budget



$U = ie^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=2 / L=11 / score ≈ 32.066 — minimum uniform E budget

Exact definitions: $\eta_{18}, \eta_{21}, \eta_{23}, \eta_{24}, \eta_{25}, \eta_{46}, \eta_{47}, \eta_{48}, \eta_{49}, \eta_{50}$ in the expression glossary.

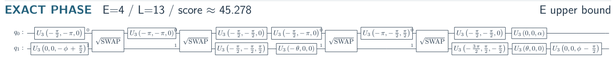
PROJECTIVE E=2 / L=10 / score ≈ 30.060 — minimum uniform E budget

Exact definitions: $\eta_{21}, \eta_{23}, \eta_{24}, \eta_{25}, \eta_{26}, \eta_{48}, \eta_{49}, \eta_{50}$ in the expression glossary.

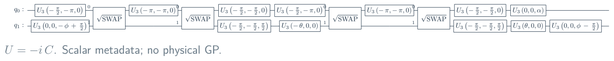
$U = \eta_{51}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=13 / score ≈ 45.278 — E upper bound



PROJECTIVE E=4 / L=13 / score ≈ 44.878 — E upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

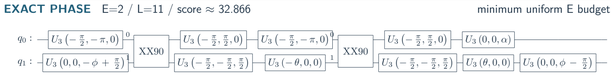
tket_tk2

PROJECTIVE E=2 / L=15 / score ≈ 41.919 — E upper bound

$U = -ie^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

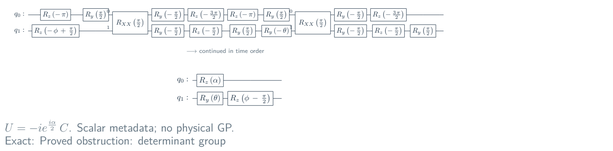
xx90_u

EXACT PHASE E=2 / L=11 / score ≈ 32.866 — minimum uniform E budget



xx_rz_ry

PROJECTIVE E=2 / L=19 / score ≈ 49.242 — E upper bound



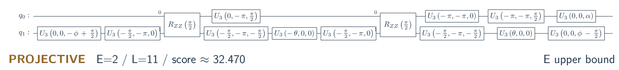
$U = -ie^{\frac{i\alpha}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

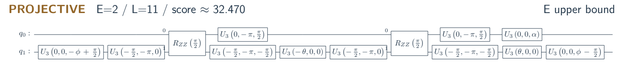
EXACT PHASE E=2 / L=9 / score ≈ 28.458 — E upper bound

zz_u

EXACT PHASE E=2 / L=12 / score ≈ 34.476 — E upper bound



PROJECTIVE E=2 / L=11 / score ≈ 32.470 — E upper bound



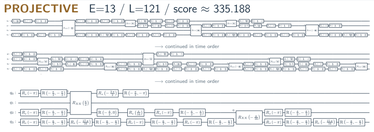
$U = i\,C$. Scalar metadata; no physical GP.

C3SX

4 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

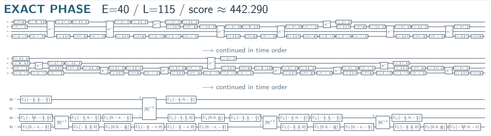
PROJECTIVE E=13 / L=121 / score ≈ 335.188 — E upper bound



$U = -(-1)^{\frac{17}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=40 / L=115 / score ≈ 442.290 — E upper bound



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

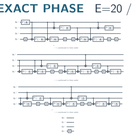
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

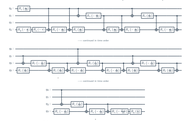
cx_h_phase

EXACT PHASE E=20 / L=35 / score ≈ 166.842 — E upper bound



cx_rz_ry

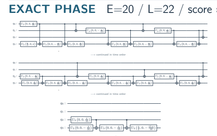
PROJECTIVE E=20 / L=24 / score ≈ 145.248 — E upper bound



$U = e^{-\frac{31i\pi}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

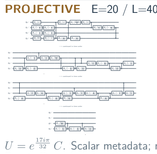
cx_u

EXACT PHASE E=20 / L=22 / score ≈ 141.332 — E upper bound



cz_rz_rx

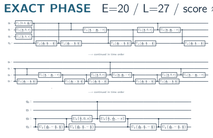
PROJECTIVE E=20 / L=40 / score ≈ 179.680 — E upper bound



$U = e^{\frac{17i\pi}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=20 / L=27 / score ≈ 158.962 — E upper bound

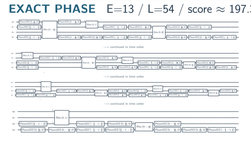


google_cz

EXACT PHASE E=20 / L=75 / score ≈ 268.850 — E upper bound

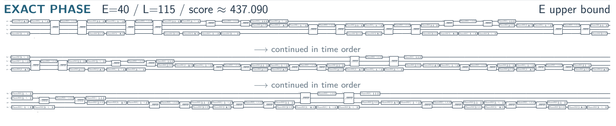
google_fsim

EXACT PHASE E=13 / L=54 / score ≈ 197.376 — E upper bound



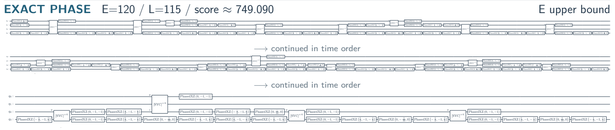
google_sqrt_iswap

EXACT PHASE E=40 / L=115 / score ≈ 437.090 — E upper bound

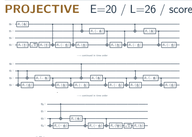


google_syc

EXACT PHASE E=120 / L=115 / score ≈ 749.090 — E upper bound

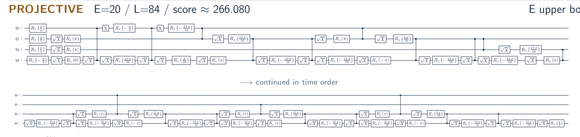


ibm_cx

PROJECTIVE E=20 / L=26 / score ≈ 149.448 — E upper bound

$U = e^{\frac{17i\pi}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

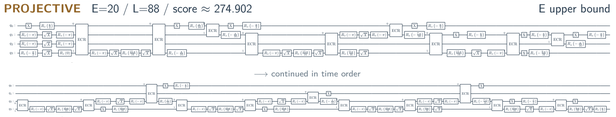
PROJECTIVE E=20 / L=84 / score ≈ 266.080 — E upper bound



$U = e^{-\frac{31i\pi}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

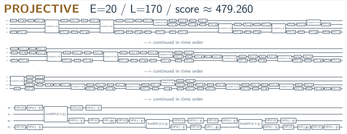
PROJECTIVE E=20 / L=88 / score ≈ 274.902 — E upper bound



$U = e^{-\frac{15i\pi}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

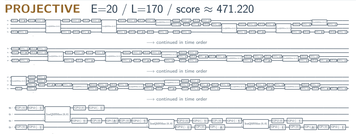
PROJECTIVE E=20 / L=170 / score ≈ 479.260 — E upper bound



$U = -\sqrt[32]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

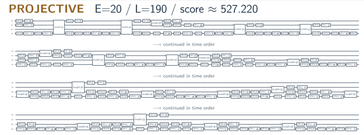
PROJECTIVE E=20 / L=170 / score ≈ 471.220 — E upper bound



$U = -\sqrt[32]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

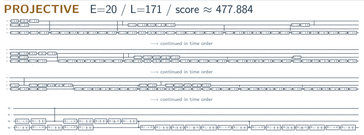
PROJECTIVE E=20 / L=190 / score ≈ 527.220 — E upper bound



$U = -\sqrt[32]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

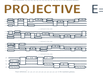
PROJECTIVE E=20 / L=171 / score ≈ 477.884 — E upper bound



$U = -\sqrt[32]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

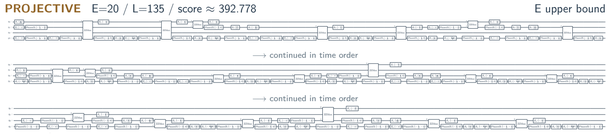
iswap_u

EXACT PHASE E=26 / L=41 / score ≈ 200.446 — E upper bound

PROJECTIVE E=26 / L=40 / score ≈ 196.440 — E upper bound

$U = \eta_{66}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

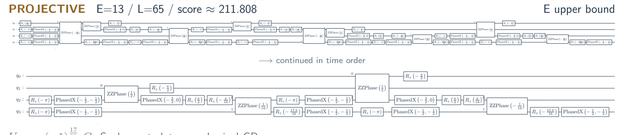
PROJECTIVE E=20 / L=135 / score ≈ 392.778 — E upper bound



$U = \sqrt[32]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

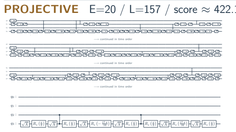
PROJECTIVE E=13 / L=65 / score ≈ 211.808 — E upper bound



$U = -(-1)^{\frac{17}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

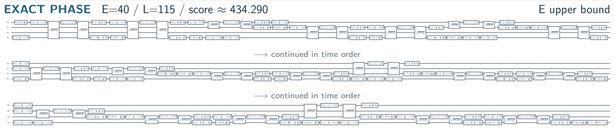
PROJECTIVE E=20 / L=157 / score ≈ 422.154 — E upper bound



$U = e^{\frac{1i\pi}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

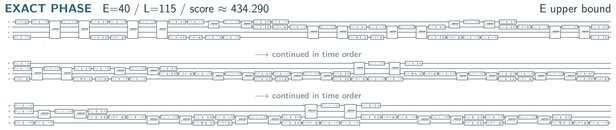
sqrt_iswap_u

EXACT PHASE E=40 / L=115 / score ≈ 434.290 — E upper bound



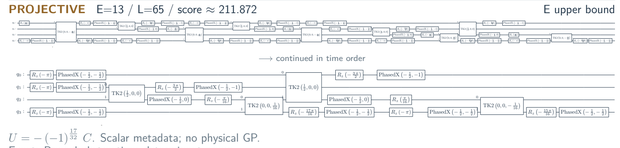
sqrt_swap_u

EXACT PHASE E=40 / L=115 / score ≈ 434.290 — E upper bound



tket_tk2

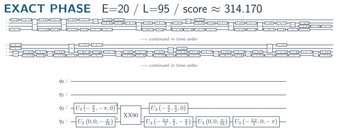
PROJECTIVE E=13 / L=65 / score ≈ 211.872 — E upper bound



$U = -(-1)^{\frac{17}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

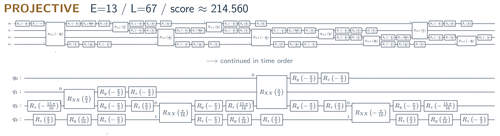
xx90_u

EXACT PHASE E=20 / L=95 / score ≈ 314.170 — E upper bound



xx_rz_ry

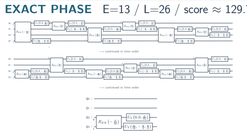
PROJECTIVE E=13 / L=67 / score ≈ 214.560 — E upper bound



$U = e^{\frac{17i\pi}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

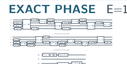
zx_u

EXACT PHASE E=13 / L=26 / score ≈ 129.782 — E upper bound

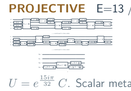


zz_u

EXACT PHASE E=13 / L=29 / score ≈ 135.000 — E upper bound



PROJECTIVE E=13 / L=27 / score ≈ 128.988 — E upper bound



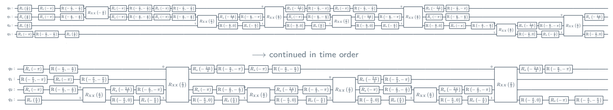
$U = e^{\frac{15i\pi}{32}}\, C$. Scalar metadata; no physical GP.

C3X

4 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

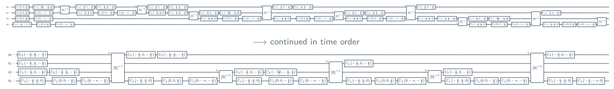
PROJECTIVE E=13 / L=86 / score ≈ 250.082 — E upper bound



$U = \sqrt[16]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=28 / L=73 / score ≈ 290.238 — E upper bound



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

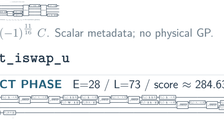
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

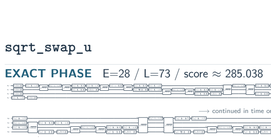
cx_h_phase

EXACT PHASE E=14 / L=17 / score ≈ 99.030 — E upper bound



cx_rz_ry

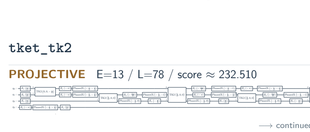
PROJECTIVE E=14 / L=19 / score ≈ 103.438 — E upper bound



$U = -\sqrt[16]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

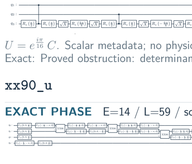
cx_u

EXACT PHASE E=14 / L=17 / score ≈ 99.502 — E upper bound



cz_rz_rx

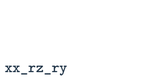
PROJECTIVE E=14 / L=27 / score ≈ 120.854 — E upper bound



$U = e^{\frac{9i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=14 / L=19 / score ≈ 96.714 — E upper bound

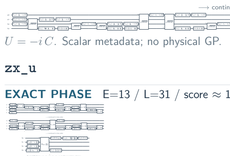


PROJECTIVE E=14 / L=17 / score ≈ 91.502 — E upper bound

$U = \eta_{73}\,C$. Scalar metadata; no physical GP.

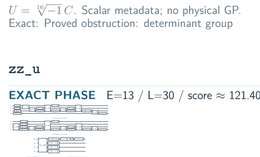
google_cz

EXACT PHASE E=14 / L=45 / score ≈ 167.270 — E upper bound



google_fsim

EXACT PHASE E=13 / L=42 / score ≈ 156.704 — E upper bound



google_sqrt_iswap

EXACT PHASE E=28 / L=73 / score ≈ 285.038 — E upper bound

google_syc

EXACT PHASE E=84 / L=73 / score ≈ 503.438 — E upper bound

ibm_cx

PROJECTIVE E=14 / L=21 / score ≈ 107.434 — E upper bound

$U = e^{\frac{7i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=14 / L=51 / score ≈ 168.042 — E upper bound

$U = e^{-\frac{7i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=14 / L=70 / score ≈ 208.084 — E upper bound

$U = e^{-\frac{15i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=14 / L=104 / score ≈ 299.692 — E upper bound

$U = -(-1)^{\frac{9}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=14 / L=104 / score ≈ 294.064 — E upper bound

$U = -(-1)^{\frac{9}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=14 / L=118 / score ≈ 333.264 — E upper bound

$U = (-1)^{\frac{9}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=14 / L=105 / score ≈ 299.420 — E upper bound

$U = -\sqrt[16]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=26 / L=44 / score ≈ 208.264 — E upper bound

PROJECTIVE E=26 / L=42 / score ≈ 202.052 — E upper bound

$U = (-1)^{\frac{11}{16}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=14 / L=78 / score ≈ 236.016 — E upper bound

$U = -(-1)^{\frac{9}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=13 / L=68 / score ≈ 212.414 — E upper bound

$U = \sqrt[16]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=14 / L=83 / score ≈ 237.294 — E upper bound

$U = e^{\frac{7i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=28 / L=73 / score ≈ 284.638 — E upper bound

sqrt_swap_u

EXACT PHASE E=28 / L=73 / score ≈ 285.038 — E upper bound

PROJECTIVE E=28 / L=73 / score ≈ 284.638 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=13 / L=78 / score ≈ 232.510 — E upper bound

$U = \sqrt[16]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=14 / L=59 / score ≈ 200.554 — E upper bound

xx_rz_ry

PROJECTIVE E=13 / L=80 / score ≈ 236.786 — E upper bound

$U = e^{-\frac{15i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=13 / L=31 / score ≈ 128.612 — E upper bound

PROJECTIVE E=13 / L=29 / score ≈ 122.400 — E upper bound

$U = -(-1)^{\frac{3}{16}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=13 / L=30 / score ≈ 121.406 — E upper bound

PROJECTIVE E=13 / L=28 / score ≈ 115.594 — E upper bound

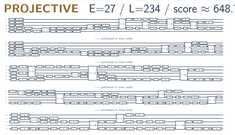
$U = \eta_{83}\,C$. Scalar metadata; no physical GP.

C4X

5 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

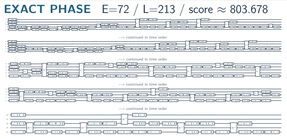
PROJECTIVE E=27 / L=234 / score ≈ 648.754 — E upper bound



$U = (-1)^{\frac{17}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=72 / L=213 / score ≈ 803.678 — E upper bound



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

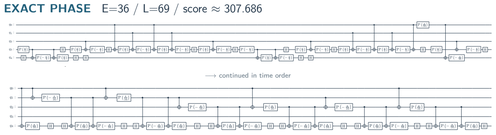
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

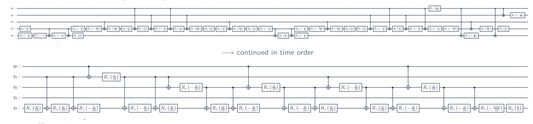
cx_h_phase

EXACT PHASE E=36 / L=69 / score ≈ 307.686 — E upper bound



cx_rz_ry

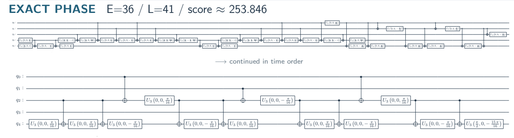
PROJECTIVE E=36 / L=54 / score ≈ 279.708 — E upper bound



$U = e^{\frac{i\pi}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

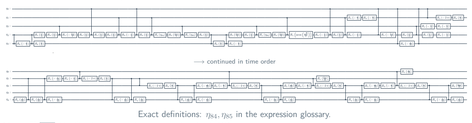
cx_u

EXACT PHASE E=36 / L=41 / score ≈ 253.846 — E upper bound



cz_rz_rx

PROJECTIVE E=36 / L=69 / score ≈ 310.372 — E upper bound

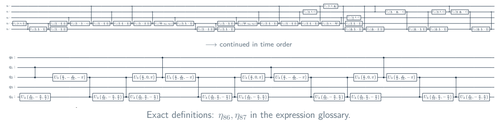


Exact definitions: η_{b1}, η_{b5} in the expression glossary.

$U = \sqrt[32]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

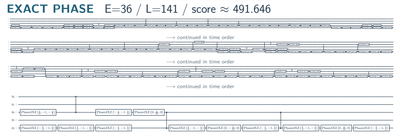
EXACT PHASE E=36 / L=46 / score ≈ 275.510 — E upper bound



Exact definitions: η_{b6}, η_{b7} in the expression glossary.

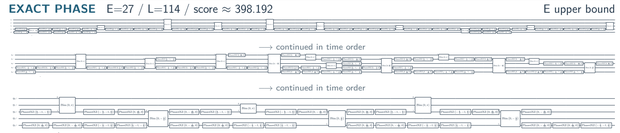
google_cz

EXACT PHASE E=36 / L=141 / score ≈ 491.646 — E upper bound



google_fsim

EXACT PHASE E=27 / L=114 / score ≈ 398.192 — E upper bound

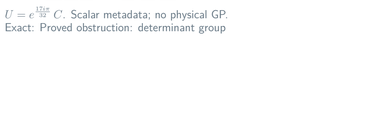


google_sqrt_iswap

EXACT PHASE E=72 / L=213 / score ≈ 794.478 — E upper bound

google_syc

EXACT PHASE E=216 / L=213 / score ≈ 1356.078 — E upper bound



ibm_cx

PROJECTIVE E=36 / L=59 / score ≈ 288.898 — E upper bound

$U = e^{\frac{17i\pi}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=36 / L=130 / score ≈ 429.746 — E upper bound

$U = e^{\frac{17i\pi}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=36 / L=169 / score ≈ 513.392 — E upper bound

$U = e^{-\frac{7i\pi}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=36 / L=318 / score ≈ 884.452 — E upper bound

$U = -\sqrt[32]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=36 / L=318 / score ≈ 869.980 — E upper bound

$U = -\sqrt[32]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=36 / L=354 / score ≈ 970.780 — E upper bound

$U = -\sqrt[32]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=36 / L=325 / score ≈ 894.500 — E upper bound

$U = -\sqrt[32]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=54 / L=79 / score ≈ 404.674 — E upper bound

Exact definitions: $\eta_{b1}, \eta_{b2}, \eta_{b3}, \eta_{b4}, \eta_{b5}, \eta_{b6}, \eta_{b7}, \eta_{b8}, \eta_{b9}, \eta_{b10}, \eta_{b11}, \eta_{b12}, \eta_{b13}, \eta_{b14}, \eta_{b15}, \eta_{b16}$ in the expression glossary.

PROJECTIVE E=54 / L=78 / score ≈ 400.668 — E upper bound

$U = \eta_{b7}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=36 / L=241 / score ≈ 698.678 — E upper bound

$U = \sqrt[32]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=27 / L=151 / score ≈ 468.080 — E upper bound

$U = (-1)^{\frac{1}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=36 / L=130 / score ≈ 429.746 — E upper bound

$U = e^{\frac{17i\pi}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=72 / L=213 / score ≈ 789.278 — E upper bound

sqrt_swap_u

EXACT PHASE E=72 / L=213 / score ≈ 789.278 — E upper bound

tket_tk2

PROJECTIVE E=27 / L=162 / score ≈ 490.246 — E upper bound

$U = (-1)^{\frac{1}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=36 / L=177 / score ≈ 573.062 — E upper bound

xx_rz_ry

PROJECTIVE E=27 / L=147 / score ≈ 453.948 — E upper bound

$U = e^{\frac{17i\pi}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=27 / L=56 / score ≈ 265.190 — E upper bound

PROJECTIVE E=27 / L=54 / score ≈ 259.978 — E upper bound

$U = e^{\frac{7i\pi}{8}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=27 / L=55 / score ≈ 243.618 — E upper bound

Exact definitions: $\eta_{b1}, \eta_{b2}, \eta_{b3}, \eta_{b4}, \eta_{b5}, \eta_{b6}$ in the expression glossary.

PROJECTIVE E=27 / L=53 / score ≈ 237.606 — E upper bound

Exact definitions: $\eta_{b1}, \eta_{b2}, \eta_{b3}, \eta_{b4}, \eta_{b5}, \eta_{b6}$ in the expression glossary.

$U = (-1)^{\frac{1}{32}}\, C$. Scalar metadata; no physical GP.

Canonical (a, b, c)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

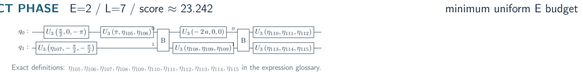
EXACT PHASE E=3 / L=41 / score ≈ 102.928 — minimum uniform E budget

PROJECTIVE E=3 / L=20 / score ≈ 56.872 — minimum uniform E budget

$U = \eta_{104}\, C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=2 / L=7 / score ≈ 23.242 — minimum uniform E budget



Exact definitions: $\eta_{105}, \eta_{106}, \eta_{107}, \eta_{108}, \eta_{109}, \eta_{110}, \eta_{111}, \eta_{112}, \eta_{113}, \eta_{114}, \eta_{115}$ in the expression glossary.

PROJECTIVE E=2 / L=6 / score ≈ 21.236 — minimum uniform E budget

Exact definitions: $\eta_{107}, \eta_{108}, \eta_{109}, \eta_{110}, \eta_{111}, \eta_{112}, \eta_{113}, \eta_{114}, \eta_{115}, \eta_{116}$ in the expression glossary.

$U = \eta_{117}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

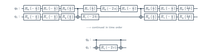
EXACT PHASE E=3 / L=20 / score ≈ 52.620 — minimum uniform E budget

PROJECTIVE E=3 / L=12 / score ≈ 36.612 — minimum uniform E budget

$U = e^{i(a-b+c)}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=4 / L=17 / score ≈ 52.834 — minimum uniform E budget

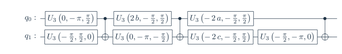


PROJECTIVE E=3 / L=11 / score ≈ 36.222 — minimum uniform E budget

$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

cx_u

EXACT PHASE E=3 / L=7 / score ≈ 28.242 — minimum uniform E budget



PROJECTIVE E=3 / L=5 / score ≈ 24.230 — minimum uniform E budget

$U = e^{ic}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=4 / L=28 / score ≈ 77.256 — minimum uniform E budget

PROJECTIVE E=3 / L=11 / score ≈ 35.822 — minimum uniform E budget

$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=3 / L=10 / score ≈ 34.660 — minimum uniform E budget

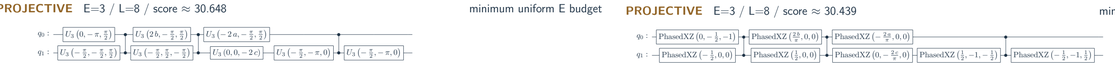
PROJECTIVE E=3 / L=8 / score ≈ 30.648 — minimum uniform E budget

$U = e^{ic}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=3 / L=10 / score ≈ 35.111 — minimum uniform E budget

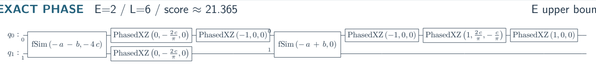
PROJECTIVE E=3 / L=8 / score ≈ 30.439 — minimum uniform E budget



$U = e^{i(a-b+c)}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=6 / score ≈ 21.365 — E upper bound



PROJECTIVE E=2 / L=4 / score ≈ 16.693 — E upper bound

$U = e^{ic}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=6 / L=19 / score ≈ 69.165 — E upper bound

PROJECTIVE E=6 / L=17 / score ≈ 63.893 — E upper bound

$U = \eta_{118}\, C$. Scalar metadata; no physical GP.
Published uniform cap: $E \leq 3$. Current construction exceeds it; optimization remains.

google_syc

EXACT PHASE E=18 / L=19 / score ≈ 115.365 — E upper bound

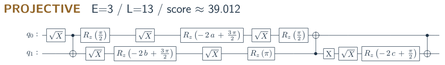
PROJECTIVE E=18 / L=17 / score ≈ 110.693 — E upper bound

$U = \eta_{119}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=3 / L=16 / score ≈ 45.216 — minimum uniform E budget

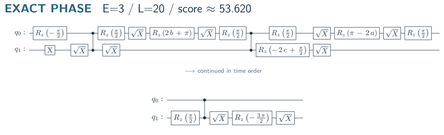
PROJECTIVE E=3 / L=13 / score ≈ 39.012 — minimum uniform E budget



$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=3 / L=20 / score ≈ 53.620 — minimum uniform E budget



ibm_ecr

EXACT PHASE E=3 / L=24 / score ≈ 61.018 — minimum uniform E budget

PROJECTIVE E=3 / L=22 / score ≈ 57.214 — minimum uniform E budget

$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

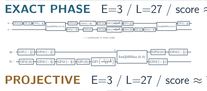
EXACT PHASE E=3 / L=27 / score ≈ 74.463 — minimum uniform E budget

PROJECTIVE E=3 / L=27 / score ≈ 74.263 — minimum uniform E budget

$U = \frac{\sqrt[3]{-1}\sqrt{2}(-1+i)}{2}\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=3 / L=27 / score ≈ 73.257 — minimum uniform E budget

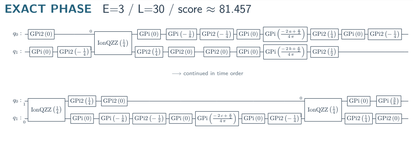


PROJECTIVE E=3 / L=27 / score ≈ 73.057 — minimum uniform E budget

$U = \frac{\sqrt[3]{-1}\sqrt{2}(-1+i)}{2}\, C$. Scalar metadata; no physical GP.

ionq_forte

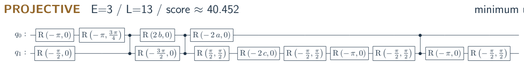
EXACT PHASE E=3 / L=30 / score ≈ 81.457 — minimum uniform E budget



iqm_cz

EXACT PHASE E=4 / L=27 / score ≈ 75.708 — minimum uniform E budget

PROJECTIVE E=3 / L=13 / score ≈ 40.452 — minimum uniform E budget



$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

iswap_u

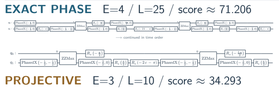
EXACT PHASE E=3 / L=13 / score ≈ 41.278 — minimum uniform E budget

PROJECTIVE E=3 / L=10 / score ≈ 35.260 — minimum uniform E budget

$U = \frac{\sqrt{2}(1+i)e^{ib}}{2}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=4 / L=25 / score ≈ 71.206 — E upper bound

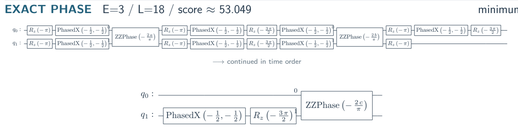


PROJECTIVE E=3 / L=10 / score ≈ 34.293 — minimum uniform E budget

$U = -i\, C$. Scalar metadata; no physical GP.

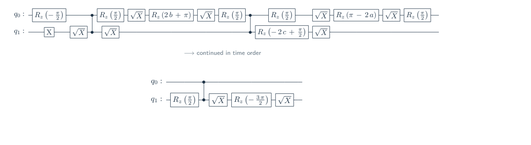
quantinuum_zzphase

EXACT PHASE E=3 / L=18 / score ≈ 53.049 — minimum uniform E budget



rigetti_cz

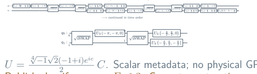
EXACT PHASE E=3 / L=20 / score ≈ 53.620 — minimum uniform E budget



sqrt_iswap_u

EXACT PHASE E=6 / L=19 / score ≈ 67.914 — E upper bound

PROJECTIVE E=6 / L=17 / score ≈ 63.902 — E upper bound



$U = \frac{\sqrt[3]{-1}\sqrt{2}(-1+i)e^{ic}}{2}\, C$. Scalar metadata; no physical GP.
Published uniform cap: $E \leq 3$. Current construction exceeds it; optimization remains.

sqrt_swap_u

EXACT PHASE E=6 / L=19 / score ≈ 68.714 — E upper bound

PROJECTIVE E=6 / L=17 / score ≈ 63.902 — E upper bound

$U = -\sqrt[4]{-1}e^{ic}\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.997 — minimum uniform E budget

xx90_u

EXACT PHASE E=3 / L=10 / score ≈ 33.860 — minimum uniform E budget

PROJECTIVE E=3 / L=8 / score ≈ 29.448 — minimum uniform E budget

$U = ie^{-ib}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

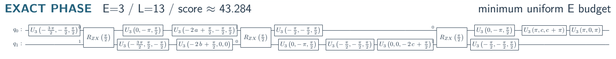
EXACT PHASE E=3 / L=41 / score ≈ 98.888 — minimum uniform E budget

PROJECTIVE E=3 / L=31 / score ≈ 80.068 — minimum uniform E budget

$U = \eta_{104}\, C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=3 / L=13 / score ≈ 43.284 — minimum uniform E budget

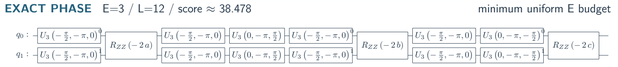


PROJECTIVE E=3 / L=11 / score ≈ 39.272 — minimum uniform E budget

$U = \frac{\sqrt[3]{-1}\sqrt{2}(1-i)e^{ic}}{2}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=3 / L=12 / score ≈ 38.478 — minimum uniform E budget

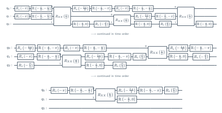


CCP (θ)

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

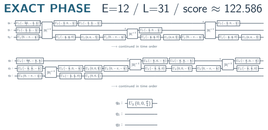
PROJECTIVE E=6 / L=37 / score ≈ 108.122 — E upper bound



$U = ie^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=12 / L=31 / score ≈ 122.586 — E upper bound



$U = -ie^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

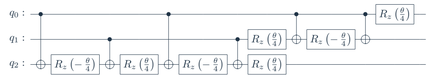
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=6 / L=7 / score ≈ 40.814 — E upper bound

cx_rz_ry

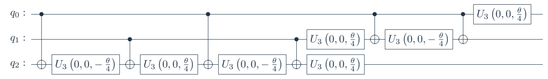
PROJECTIVE E=6 / L=7 / score ≈ 40.814 — E upper bound



$U = e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=6 / L=7 / score ≈ 40.842 — E upper bound



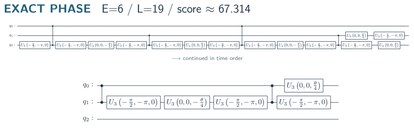
cz_rz_rx

PROJECTIVE E=6 / L=33 / score ≈ 98.066 — E upper bound

$U = e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

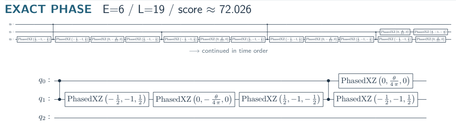
cz_u

EXACT PHASE E=6 / L=19 / score ≈ 67.314 — E upper bound



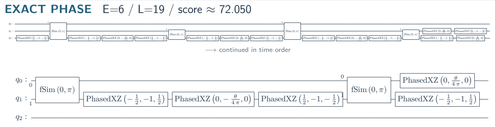
google_cz

EXACT PHASE E=6 / L=19 / score ≈ 72.026 — E upper bound



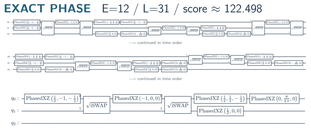
google_fsim

EXACT PHASE E=6 / L=19 / score ≈ 72.050 — E upper bound



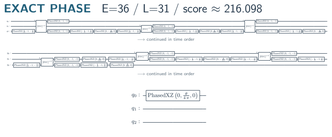
google_sqrt_iswap

EXACT PHASE E=12 / L=31 / score ≈ 122.498 — E upper bound



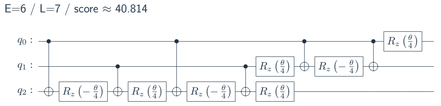
google_syc

EXACT PHASE E=36 / L=31 / score ≈ 216.098 — E upper bound



ibm_cx

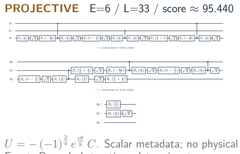
PROJECTIVE E=6 / L=7 / score ≈ 40.814 — E upper bound



$U = e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

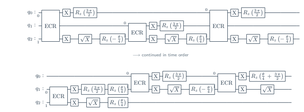
PROJECTIVE E=6 / L=33 / score ≈ 95.440 — E upper bound



$U = -(-1)^{\frac{3}{4}} e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

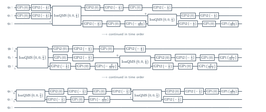
PROJECTIVE E=6 / L=30 / score ≈ 87.824 — E upper bound



$U = e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=6 / L=44 / score ≈ 128.436 — E upper bound



$U = -ie^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

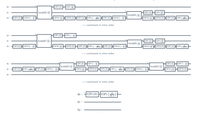
ionq_aria_full

PROJECTIVE E=6 / L=44 / score ≈ 126.024 — E upper bound

$U = -ie^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=6 / L=50 / score ≈ 142.824 — E upper bound



$U = ie^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

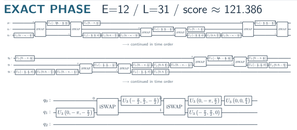
iqm_cz

PROJECTIVE E=6 / L=45 / score ≈ 127.380 — E upper bound

$U = e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=12 / L=31 / score ≈ 121.386 — E upper bound



quantinuum_zzmax

PROJECTIVE E=6 / L=31 / score ≈ 94.686 — E upper bound

$U = ie^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

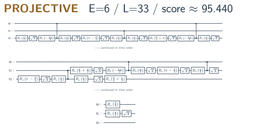
quantinuum_zzphase

PROJECTIVE E=6 / L=31 / score ≈ 95.898 — E upper bound

$U = ie^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

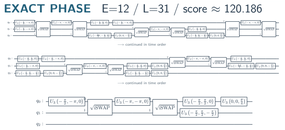
PROJECTIVE E=6 / L=33 / score ≈ 95.440 — E upper bound



$U = -(-1)^{\frac{3}{4}} e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

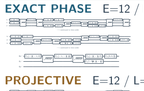
sqrt_iswap_u

EXACT PHASE E=12 / L=31 / score ≈ 120.186 — E upper bound



sqrt_swap_u

EXACT PHASE E=12 / L=31 / score ≈ 120.586 — E upper bound

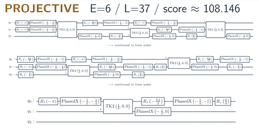


PROJECTIVE E=12 / L=31 / score ≈ 120.186 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

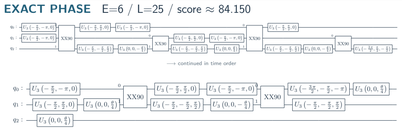
PROJECTIVE E=6 / L=37 / score ≈ 108.146 — E upper bound



$U = ie^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

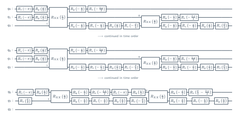
xx90_u

EXACT PHASE E=6 / L=25 / score ≈ 84.150 — E upper bound



xx_rz_ry

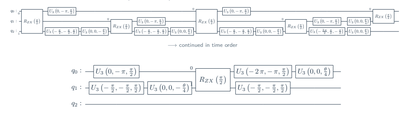
PROJECTIVE E=6 / L=48 / score ≈ 131.108 — E upper bound



$U = ie^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=6 / L=19 / score ≈ 70.926 — E upper bound



zz_u

EXACT PHASE E=6 / L=26 / score ≈ 84.968 — E upper bound

PROJECTIVE E=6 / L=25 / score ≈ 82.962 — E upper bound

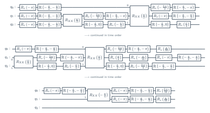
$U = -iC$. Scalar metadata; no physical GP.

CCX

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

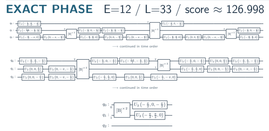
PROJECTIVE E=5 / L=40 / score ≈ 112.926 — E upper bound



$U = \sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=12 / L=33 / score ≈ 126.998 — E upper bound



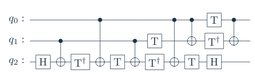
$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

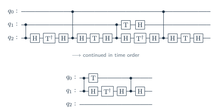
clifford_t_cx

EXACT PHASE E=6 / L=9 / score ≈ 42.000 — proved E minimum



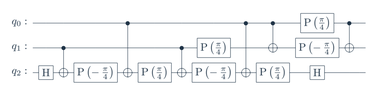
clifford_t_cz

EXACT PHASE E=6 / L=19 / score ≈ 62.000 — proved E minimum



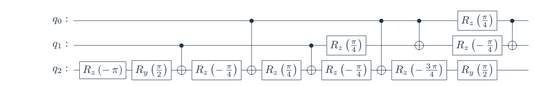
cx_h_phase

EXACT PHASE E=6 / L=9 / score ≈ 44.814 — proved E minimum



cx_rz_ry

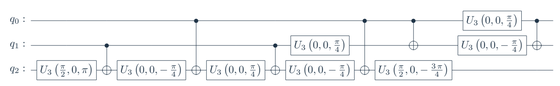
PROJECTIVE E=6 / L=10 / score ≈ 47.220 — proved E minimum



$U = -\sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

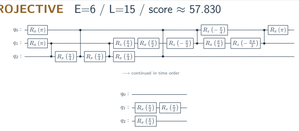
cx_u

EXACT PHASE E=6 / L=8 / score ≈ 43.248 — proved E minimum



cz_rz_rx

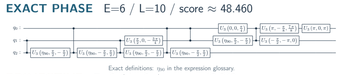
PROJECTIVE E=6 / L=15 / score ≈ 57.830 — proved E minimum



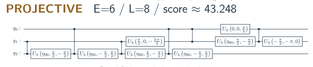
$U = e^{\frac{7i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=6 / L=10 / score ≈ 48.460 — proved E minimum



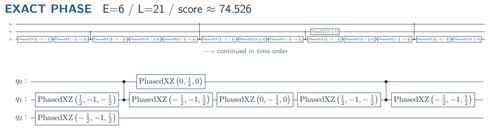
PROJECTIVE E=6 / L=8 / score ≈ 43.248 — proved E minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

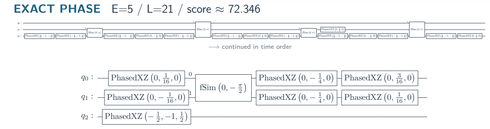
google_cz

EXACT PHASE E=6 / L=21 / score ≈ 74.526 — proved E minimum



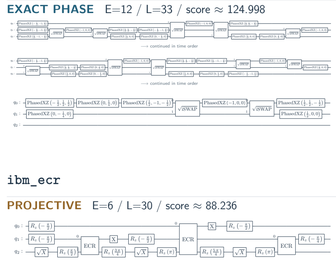
google_fsim

EXACT PHASE E=5 / L=21 / score ≈ 72.346 — E upper bound



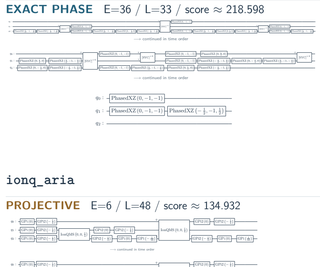
google_sqrt_iswap

EXACT PHASE E=12 / L=33 / score ≈ 124.998 — E upper bound



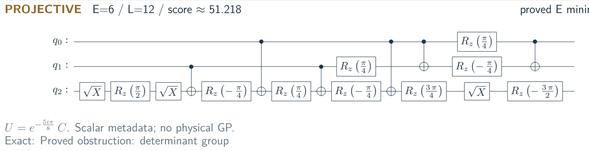
google_syc

EXACT PHASE E=36 / L=33 / score ≈ 218.598 — E upper bound



ibm_cx

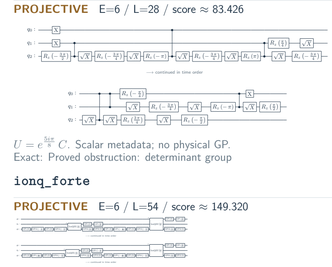
PROJECTIVE E=6 / L=12 / score ≈ 51.218 — proved E minimum



$U = e^{\frac{5i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

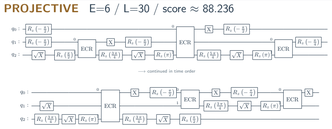
PROJECTIVE E=6 / L=28 / score ≈ 83.426 — proved E minimum



$U = e^{\frac{5i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

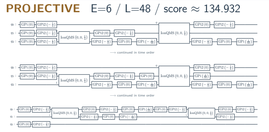
PROJECTIVE E=6 / L=30 / score ≈ 88.236 — proved E minimum



$U = e^{\frac{3i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

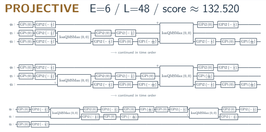
PROJECTIVE E=6 / L=48 / score ≈ 134.932 — E upper bound



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

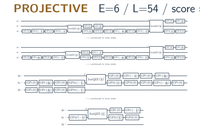
PROJECTIVE E=6 / L=48 / score ≈ 132.520 — E upper bound



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

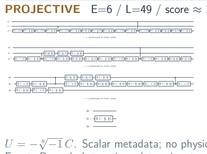
PROJECTIVE E=6 / L=54 / score ≈ 149.320 — E upper bound



$U = (-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=6 / L=49 / score ≈ 136.196 — proved E minimum



$U = -\sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

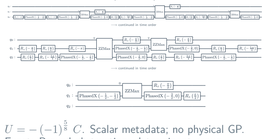
EXACT PHASE E=10 / L=18 / score ≈ 83.108 — E upper bound

PROJECTIVE E=10 / L=16 / score ≈ 79.096 — E upper bound

$U = (-1)^{\frac{3}{8}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

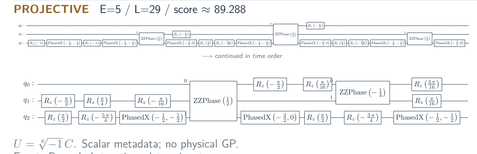
PROJECTIVE E=6 / L=35 / score ≈ 103.498 — proved E minimum



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

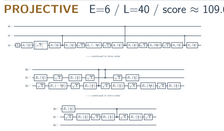
PROJECTIVE E=5 / L=29 / score ≈ 89.288 — E upper bound



$U = \sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

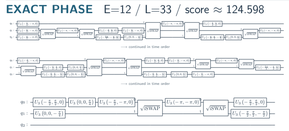
PROJECTIVE E=6 / L=40 / score ≈ 109.642 — proved E minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

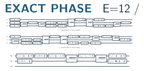
sqrt_iswap_u

EXACT PHASE E=12 / L=33 / score ≈ 124.598 — E upper bound

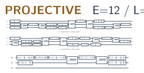


sqrt_swap_u

EXACT PHASE E=12 / L=33 / score ≈ 124.998 — E upper bound



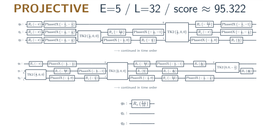
PROJECTIVE E=12 / L=33 / score ≈ 124.598 — E upper bound



$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

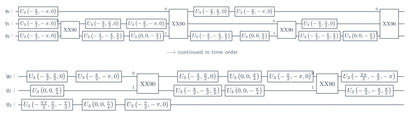
PROJECTIVE E=5 / L=32 / score ≈ 95.322 — E upper bound



$U = \sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

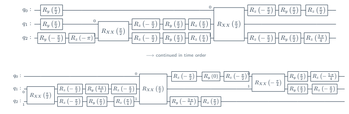
xx90_u

EXACT PHASE E=6 / L=27 / score ≈ 88.562 — proved E minimum



xx_rz_ry

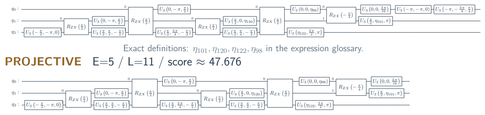
PROJECTIVE E=5 / L=31 / score ≈ 90.272 — E upper bound



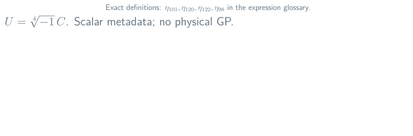
$U = e^{-\frac{7i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=5 / L=13 / score ≈ 52.488 — E upper bound



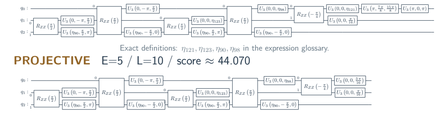
PROJECTIVE E=5 / L=11 / score ≈ 47.676 — E upper bound



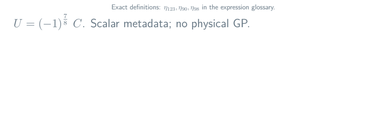
$U = \sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=5 / L=12 / score ≈ 48.482 — E upper bound



PROJECTIVE E=5 / L=10 / score ≈ 44.070 — E upper bound



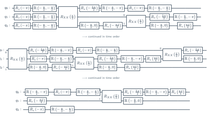
$U = (-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.

CCXPow $(t, global_{shift})$

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

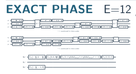
PROJECTIVE E=6 / L=41 / score ≈ 116.934 — E upper bound



$U = \eta_{124} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=12 / L=35 / score ≈ 131.010 — E upper bound



PROJECTIVE E=12 / L=33 / score ≈ 126.998 — E upper bound

$U = e^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

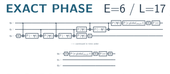
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

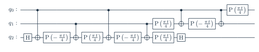
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=6 / L=17 / score ≈ 60.822 — E upper bound



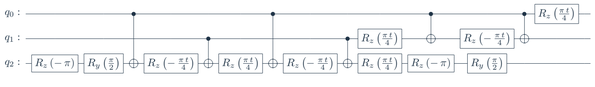
PROJECTIVE E=6 / L=9 / score ≈ 44.814 — E upper bound



$U = e^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=6 / L=11 / score ≈ 49.222 — E upper bound

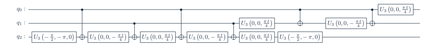


$U = -e^{i\pi t \left(global_{shift} + \frac{1}{8}\right)}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=6 / L=11 / score ≈ 49.266 — E upper bound

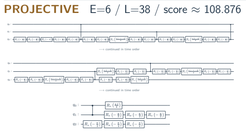
PROJECTIVE E=6 / L=9 / score ≈ 45.254 — E upper bound



$U = e^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=6 / L=38 / score ≈ 108.876 — E upper bound



$U = -e^{i\pi t \left(global_{shift} + \frac{1}{8}\right)}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=6 / L=23 / score ≈ 75.738 — E upper bound

PROJECTIVE E=6 / L=21 / score ≈ 71.726 — E upper bound

$U = e^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=6 / L=23 / score ≈ 78.538 — E upper bound

PROJECTIVE E=6 / L=21 / score ≈ 74.526 — E upper bound

$U = e^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

google_fsim

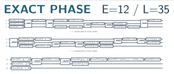
EXACT PHASE E=6 / L=23 / score ≈ 78.562 — E upper bound

PROJECTIVE E=6 / L=21 / score ≈ 74.550 — E upper bound

$U = e^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=12 / L=35 / score ≈ 129.010 — E upper bound

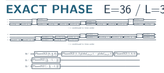


PROJECTIVE E=12 / L=33 / score ≈ 124.998 — E upper bound

$U = ie^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=36 / L=35 / score ≈ 222.610 — E upper bound

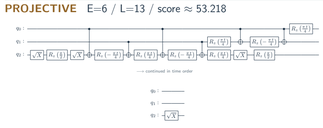


PROJECTIVE E=36 / L=33 / score ≈ 218.598 — E upper bound

$U = e^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

ibm_cx

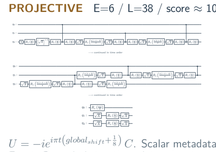
PROJECTIVE E=6 / L=13 / score ≈ 53.218 — E upper bound



$U = e^{\frac{i\pi t \left(8 g l o b a l_{shift} + 1\right)}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

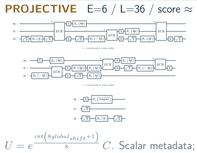
PROJECTIVE E=6 / L=38 / score ≈ 105.642 — E upper bound



$U = -ie^{i\pi t \left(global_{shift} + \frac{1}{8}\right)}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

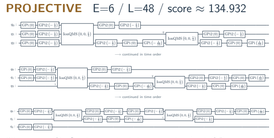
PROJECTIVE E=6 / L=36 / score ≈ 100.228 — E upper bound



$U = e^{\frac{i\pi t \left(8 g l o b a l_{shift} + 1\right)}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

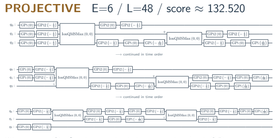
PROJECTIVE E=6 / L=48 / score ≈ 134.932 — E upper bound



$U = \eta_{124} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

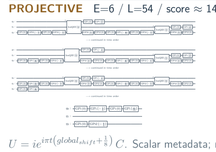
PROJECTIVE E=6 / L=48 / score ≈ 132.520 — E upper bound



$U = \eta_{124} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=6 / L=54 / score ≈ 149.320 — E upper bound



$U = ie^{i\pi t \left(global_{shift} + \frac{1}{8}\right)}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

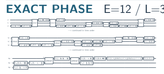
iqm_cz

PROJECTIVE E=6 / L=49 / score ≈ 136.196 — E upper bound

$U = -e^{i\pi t \left(global_{shift} + \frac{1}{8}\right)}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=12 / L=35 / score ≈ 129.810 — E upper bound

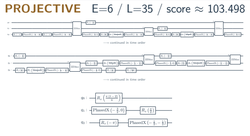


PROJECTIVE E=12 / L=33 / score ≈ 125.798 — E upper bound

$U = e^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

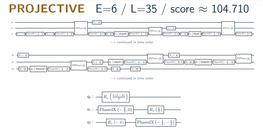
PROJECTIVE E=6 / L=35 / score ≈ 103.498 — E upper bound



$U = \eta_{124} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

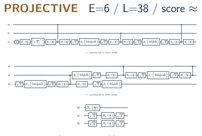
PROJECTIVE E=6 / L=35 / score ≈ 104.710 — E upper bound



$U = \eta_{124} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

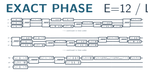
PROJECTIVE E=6 / L=38 / score ≈ 105.642 — E upper bound



$U = -ie^{i\pi t \left(global_{shift} + \frac{1}{8}\right)}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=12 / L=35 / score ≈ 128.610 — E upper bound

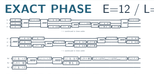


PROJECTIVE E=12 / L=33 / score ≈ 124.598 — E upper bound

$U = e^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=12 / L=35 / score ≈ 129.010 — E upper bound

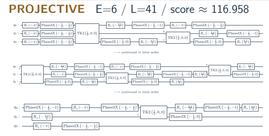


PROJECTIVE E=12 / L=33 / score ≈ 124.598 — E upper bound

$U = -ie^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=6 / L=41 / score ≈ 116.958 — E upper bound



$U = \eta_{124} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

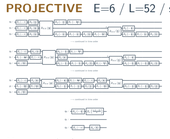
EXACT PHASE E=6 / L=29 / score ≈ 92.574 — E upper bound

PROJECTIVE E=6 / L=27 / score ≈ 88.562 — E upper bound

$U = -e^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=6 / L=52 / score ≈ 139.516 — E upper bound



$U = \eta_{124} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=6 / L=23 / score ≈ 79.350 — E upper bound

PROJECTIVE E=6 / L=21 / score ≈ 75.338 — E upper bound

$U = -e^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=6 / L=30 / score ≈ 93.392 — E upper bound

PROJECTIVE E=6 / L=27 / score ≈ 87.374 — E upper bound

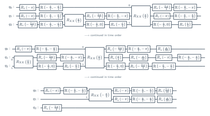
$U = -ie^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

CCY

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

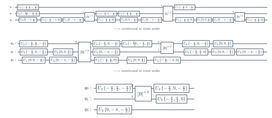
PROJECTIVE E=5 / L=41 / score ≈ 115.328 — E upper bound



$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=12 / L=35 / score ≈ 131.410 — E upper bound

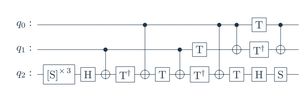


clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

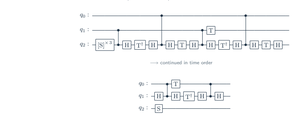
clifford_t_cx

EXACT PHASE E=6 / L=13 / score ≈ 50.000 — proved E minimum



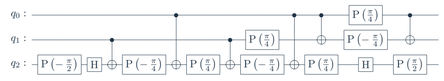
clifford_t_cz

EXACT PHASE E=6 / L=23 / score ≈ 70.000 — proved E minimum



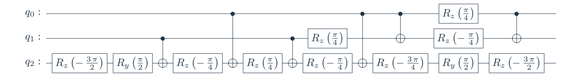
cx_h_phase

EXACT PHASE E=6 / L=11 / score ≈ 49.218 — proved E minimum



cx_rz_ry

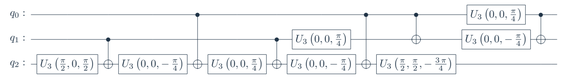
PROJECTIVE E=6 / L=11 / score ≈ 49.622 — proved E minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

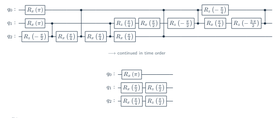
cx_u

EXACT PHASE E=6 / L=8 / score ≈ 43.648 — proved E minimum



cz_rz_rx

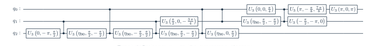
PROJECTIVE E=6 / L=17 / score ≈ 62.234 — proved E minimum



$U = e^{-\frac{7i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

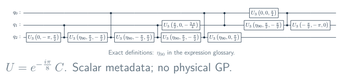
cz_u

EXACT PHASE E=6 / L=11 / score ≈ 50.466 — proved E minimum



Exact definitions: η_{66} in the expression glossary.

PROJECTIVE E=6 / L=9 / score ≈ 45.254 — proved E minimum

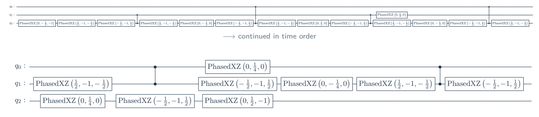


Exact definitions: η_{66} in the expression glossary.

$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

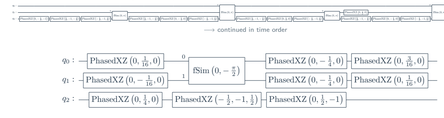
google_cz

EXACT PHASE E=6 / L=23 / score ≈ 78.938 — proved E minimum



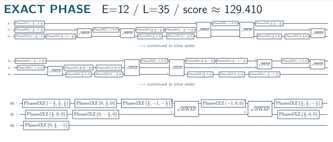
google_fsim

EXACT PHASE E=5 / L=23 / score ≈ 76.758 — E upper bound



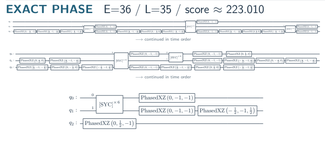
google_sqrt_iswap

EXACT PHASE E=12 / L=35 / score ≈ 129.410 — E upper bound



google_syc

EXACT PHASE E=36 / L=35 / score ≈ 223.010 — E upper bound



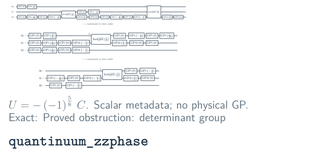
ibm_cx

PROJECTIVE E=6 / L=11 / score ≈ 49.018 — proved E minimum

$U = e^{-\frac{3i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

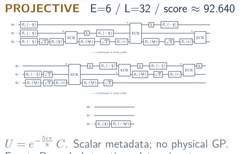
PROJECTIVE E=6 / L=26 / score ≈ 79.022 — proved E minimum



$U = e^{-\frac{3i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

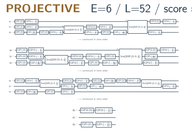
PROJECTIVE E=6 / L=32 / score ≈ 92.640 — proved E minimum



$U = e^{-\frac{9i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=6 / L=52 / score ≈ 144.140 — E upper bound



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

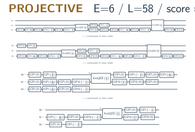
ionq_aria_full

PROJECTIVE E=6 / L=52 / score ≈ 141.728 — E upper bound

$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

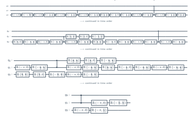
PROJECTIVE E=6 / L=58 / score ≈ 158.528 — E upper bound



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=6 / L=53 / score ≈ 145.012 — proved E minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

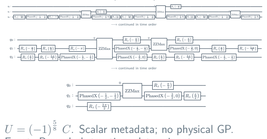
EXACT PHASE E=10 / L=18 / score ≈ 83.108 — E upper bound

PROJECTIVE E=10 / L=16 / score ≈ 79.096 — E upper bound

$U = -(-1)^{\frac{3}{8}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

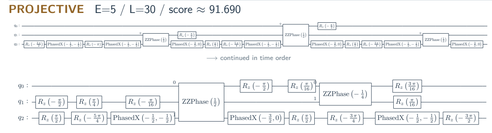
PROJECTIVE E=6 / L=36 / score ≈ 105.900 — proved E minimum



$U = (-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

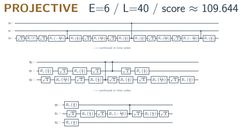
PROJECTIVE E=5 / L=30 / score ≈ 91.690 — E upper bound



$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

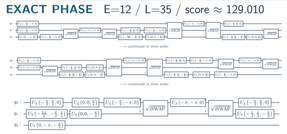
PROJECTIVE E=6 / L=40 / score ≈ 109.644 — proved E minimum



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

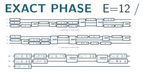
sqrt_iswap_u

EXACT PHASE E=12 / L=35 / score ≈ 129.010 — E upper bound



sqrt_swap_u

EXACT PHASE E=12 / L=35 / score ≈ 129.410 — E upper bound

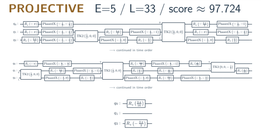


PROJECTIVE E=12 / L=35 / score ≈ 129.010 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

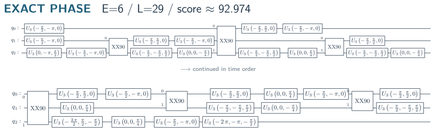
PROJECTIVE E=5 / L=33 / score ≈ 97.724 — E upper bound



$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

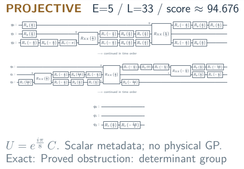
xx90_u

EXACT PHASE E=6 / L=29 / score ≈ 92.974 — proved E minimum



xx_rz_ry

PROJECTIVE E=5 / L=33 / score ≈ 94.676 — E upper bound



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=5 / L=13 / score ≈ 51.688 — E upper bound

Exact definitions: $\eta_{101}, \eta_{120}, \eta_{121}, \eta_{122}, \eta_{98}$ in the expression glossary.

PROJECTIVE E=5 / L=11 / score ≈ 47.676 — E upper bound

Exact definitions: $\eta_{101}, \eta_{120}, \eta_{122}, \eta_{98}$ in the expression glossary.

$U = \sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=5 / L=13 / score ≈ 50.488 — E upper bound

Exact definitions: $\eta_{121}, \eta_{123}, \eta_{80}, \eta_{98}$ in the expression glossary.

PROJECTIVE E=5 / L=11 / score ≈ 46.076 — E upper bound

Exact definitions: $\eta_{123}, \eta_{81}, \eta_{98}$ in the expression glossary.

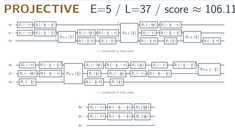
$U = (-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.

CCZ

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

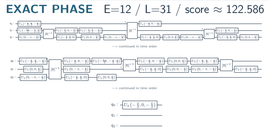
PROJECTIVE E=5 / L=37 / score ≈ 106.116 — E upper bound



$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=12 / L=31 / score ≈ 122.586 — E upper bound



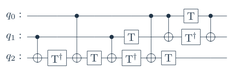
$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

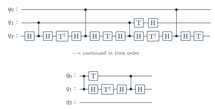
clifford_t_cx

EXACT PHASE E=6 / L=7 / score ≈ 38.000 — proved E minimum



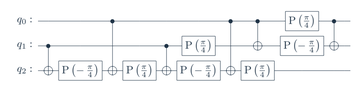
clifford_t_cz

EXACT PHASE E=6 / L=19 / score ≈ 62.000 — proved E minimum



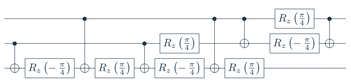
cx_h_phase

EXACT PHASE E=6 / L=7 / score ≈ 40.814 — proved E minimum



cx_rz_ry

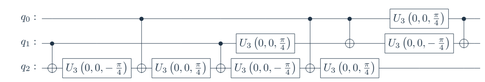
PROJECTIVE E=6 / L=7 / score ≈ 40.814 — proved E minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

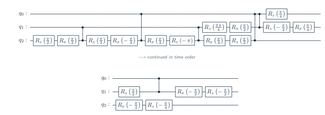
cx_u

EXACT PHASE E=6 / L=7 / score ≈ 40.842 — proved E minimum



cz_rz_rx

PROJECTIVE E=6 / L=18 / score ≈ 64.836 — proved E minimum

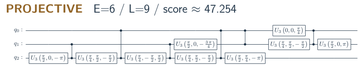


$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=6 / L=11 / score ≈ 52.066 — proved E minimum

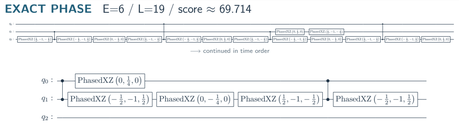
PROJECTIVE E=6 / L=9 / score ≈ 47.254 — proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

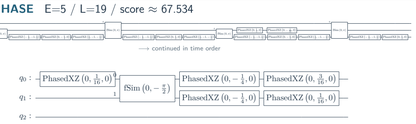
google_cz

EXACT PHASE E=6 / L=19 / score ≈ 69.714 — proved E minimum



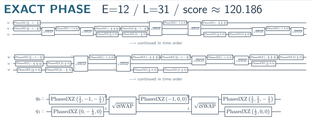
google_fsim

EXACT PHASE E=5 / L=19 / score ≈ 67.534 — E upper bound



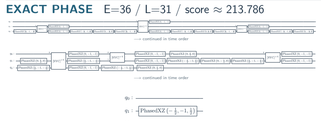
google_sqrt_iswap

EXACT PHASE E=12 / L=31 / score ≈ 120.186 — E upper bound



google_syc

EXACT PHASE E=36 / L=31 / score ≈ 213.786 — E upper bound



ibm_cx

PROJECTIVE E=6 / L=7 / score ≈ 40.814 — proved E minimum

$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

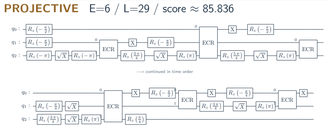
ibm_cz

PROJECTIVE E=6 / L=29 / score ≈ 85.228 — proved E minimum

$U = e^{-\frac{3i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

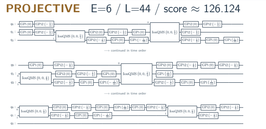
PROJECTIVE E=6 / L=29 / score ≈ 85.836 — proved E minimum



$U = e^{\frac{5i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

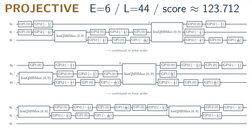
PROJECTIVE E=6 / L=44 / score ≈ 126.124 — E upper bound



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=6 / L=44 / score ≈ 123.712 — E upper bound



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=6 / L=50 / score ≈ 140.512 — E upper bound

$U = (-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=6 / L=45 / score ≈ 127.380 — proved E minimum

$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

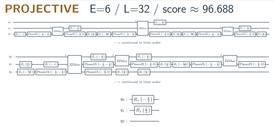
EXACT PHASE E=10 / L=18 / score ≈ 83.508 — E upper bound

PROJECTIVE E=10 / L=16 / score ≈ 79.096 — E upper bound

$U = (-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

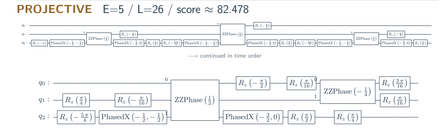
PROJECTIVE E=6 / L=32 / score ≈ 96.688 — proved E minimum



$U = (-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

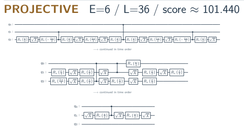
PROJECTIVE E=5 / L=26 / score ≈ 82.478 — E upper bound



$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

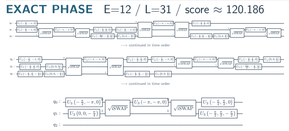
PROJECTIVE E=6 / L=36 / score ≈ 101.440 — proved E minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

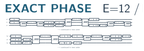
sqrt_iswap_u

EXACT PHASE E=12 / L=31 / score ≈ 120.186 — E upper bound



sqrt_swap_u

EXACT PHASE E=12 / L=31 / score ≈ 120.586 — E upper bound

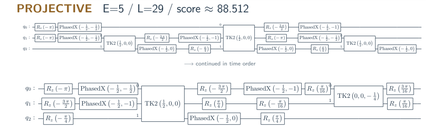


PROJECTIVE E=12 / L=31 / score ≈ 120.186 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

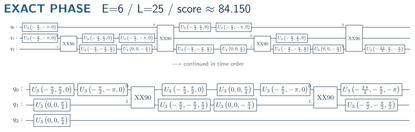
PROJECTIVE E=5 / L=29 / score ≈ 88.512 — E upper bound



$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

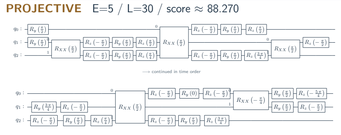
xx90_u

EXACT PHASE E=6 / L=25 / score ≈ 84.150 — proved E minimum



xx_rz_ry

PROJECTIVE E=5 / L=30 / score ≈ 88.270 — E upper bound



$U = e^{-\frac{7i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

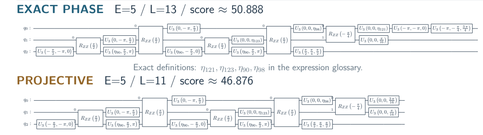
EXACT PHASE E=5 / L=14 / score ≈ 55.894 — E upper bound

PROJECTIVE E=5 / L=12 / score ≈ 50.682 — E upper bound

$U = e^{\frac{5i\pi}{8}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=5 / L=13 / score ≈ 50.888 — E upper bound



Exact definitions: $\eta_{123}, \eta_{60}, \eta_{36}$ in the expression glossary.

PROJECTIVE E=5 / L=11 / score ≈ 46.876 — E upper bound

Exact definitions: $\eta_{123}, \eta_{60}, \eta_{84}$ in the expression glossary.

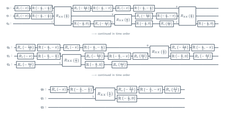
$U = (-1)^{\frac{3}{8}}\,C$. Scalar metadata; no physical GP.

CCZPow $(t, global_{shift})$

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

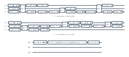
PROJECTIVE E=6 / L=37 / score ≈ 108.122 — E upper bound



$U = ie^{i\pi t\left(global_{shift}+\frac{1}{8}\right)}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=12 / L=33 / score ≈ 126.598 — E upper bound



PROJECTIVE E=12 / L=31 / score ≈ 122.586 — E upper bound

$U = e^{i\pi g lobal_{shift}t}C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

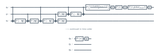
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

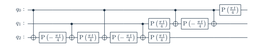
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=6 / L=14 / score ≈ 54.820 — E upper bound



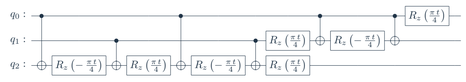
PROJECTIVE E=6 / L=7 / score ≈ 40.814 — E upper bound



$U = e^{i\pi g lobal_{shift}t}C$. Scalar metadata; no physical GP.

cx_rz_ry

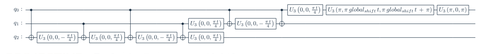
PROJECTIVE E=6 / L=7 / score ≈ 40.814 — E upper bound



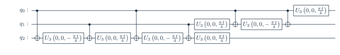
$U = e^{\frac{i\pi t\left(8global_{shift}+1\right)}{8}}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=6 / L=9 / score ≈ 44.854 — E upper bound



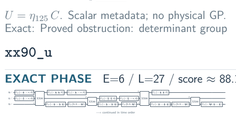
PROJECTIVE E=6 / L=7 / score ≈ 40.842 — E upper bound



$U = e^{i\pi g lobal_{shift}t}C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=6 / L=33 / score ≈ 98.066 — E upper bound

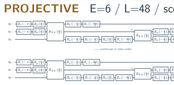


$U = e^{\frac{i\pi t\left(8global_{shift}+1\right)}{8}}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=6 / L=21 / score ≈ 71.326 — E upper bound

PROJECTIVE E=6 / L=19 / score ≈ 67.314 — E upper bound



$U = e^{i\pi g lobal_{shift}t}C$. Scalar metadata; no physical GP.

google_cz

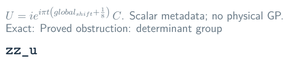
EXACT PHASE E=6 / L=21 / score ≈ 73.726 — E upper bound

PROJECTIVE E=6 / L=19 / score ≈ 69.714 — E upper bound

$U = e^{i\pi g lobal_{shift}t}C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=6 / L=21 / score ≈ 73.750 — E upper bound



PROJECTIVE E=6 / L=19 / score ≈ 69.738 — E upper bound

$U = e^{i\pi g lobal_{shift}t}C$. Scalar metadata; no physical GP.

google_sqrt_iswap

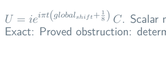
EXACT PHASE E=12 / L=33 / score ≈ 124.198 — E upper bound

PROJECTIVE E=12 / L=31 / score ≈ 120.186 — E upper bound

$U = ie^{i\pi g lobal_{shift}t}C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=36 / L=33 / score ≈ 217.798 — E upper bound



PROJECTIVE E=36 / L=31 / score ≈ 213.786 — E upper bound

$U = e^{i\pi g lobal_{shift}t}C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=6 / L=7 / score ≈ 40.814 — E upper bound

$U = e^{\frac{i\pi t\left(8global_{shift}+1\right)}{8}}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=6 / L=33 / score ≈ 95.440 — E upper bound

$U = \eta_{125}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=6 / L=30 / score ≈ 87.824 — E upper bound

$U = e^{\frac{i\pi t\left(8global_{shift}+1\right)}{8}}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=6 / L=44 / score ≈ 126.124 — E upper bound

$U = \eta_{124}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=6 / L=44 / score ≈ 123.712 — E upper bound

$U = \eta_{124}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=6 / L=50 / score ≈ 140.512 — E upper bound

$U = ie^{i\pi t\left(global_{shift}+\frac{1}{8}\right)}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=6 / L=45 / score ≈ 127.380 — E upper bound

$U = e^{\frac{i\pi t\left(8global_{shift}+1\right)}{8}}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=12 / L=33 / score ≈ 125.398 — E upper bound

PROJECTIVE E=12 / L=31 / score ≈ 121.386 — E upper bound

$U = e^{i\pi g lobal_{shift}t}C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=6 / L=31 / score ≈ 94.686 — E upper bound

$U = ie^{i\pi t\left(global_{shift}+\frac{1}{8}\right)}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=6 / L=31 / score ≈ 95.898 — E upper bound

$U = ie^{i\pi t\left(global_{shift}+\frac{1}{8}\right)}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=6 / L=33 / score ≈ 95.440 — E upper bound

$U = \eta_{125}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=12 / L=33 / score ≈ 124.198 — E upper bound

PROJECTIVE E=12 / L=31 / score ≈ 120.186 — E upper bound

$U = e^{i\pi g lobal_{shift}t}C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=12 / L=33 / score ≈ 124.598 — E upper bound

PROJECTIVE E=12 / L=31 / score ≈ 120.186 — E upper bound

$U = -ie^{i\pi g lobal_{shift}t}C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=6 / L=37 / score ≈ 108.146 — E upper bound

$U = ie^{i\pi t\left(global_{shift}+\frac{1}{8}\right)}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=6 / L=27 / score ≈ 88.162 — E upper bound

PROJECTIVE E=6 / L=25 / score ≈ 84.150 — E upper bound

$U = -e^{i\pi g lobal_{shift}t}C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=6 / L=48 / score ≈ 131.108 — E upper bound

$U = ie^{i\pi t\left(global_{shift}+\frac{1}{8}\right)}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=6 / L=21 / score ≈ 74.938 — E upper bound

PROJECTIVE E=6 / L=19 / score ≈ 70.926 — E upper bound

$U = -e^{i\pi g lobal_{shift}t}C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=6 / L=28 / score ≈ 88.980 — E upper bound

PROJECTIVE E=6 / L=25 / score ≈ 82.962 — E upper bound

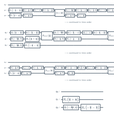
$U = -ie^{i\pi g lobal_{shift}t}C$. Scalar metadata; no physical GP.

CCZS (θ, ϕ, γ)

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

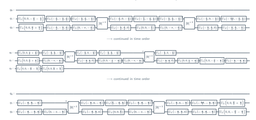
PROJECTIVE E=6 / L=51 / score ≈ 139.370 — E upper bound



$U = ie^{\frac{i(\gamma+\pi)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=12 / L=47 / score ≈ 162.082 — E upper bound



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

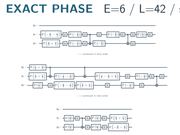
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

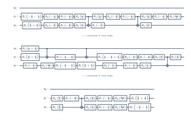
cx_h_phase

EXACT PHASE E=6 / L=42 / score ≈ 114.044 — E upper bound



cx_rz_ry

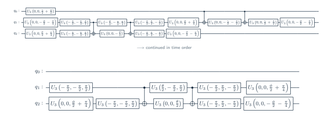
PROJECTIVE E=6 / L=42 / score ≈ 118.084 — E upper bound



$U = e^{\frac{i(\gamma+\pi)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

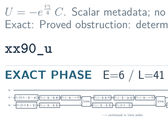
cx_u

EXACT PHASE E=6 / L=23 / score ≈ 80.338 — E upper bound



cz_rz_rx

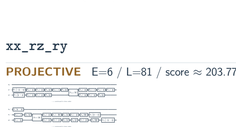
PROJECTIVE E=6 / L=53 / score ≈ 142.306 — E upper bound



$U = e^{\frac{i(\gamma+\pi)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

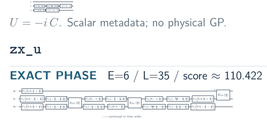
cz_u

EXACT PHASE E=6 / L=35 / score ≈ 106.810 — E upper bound



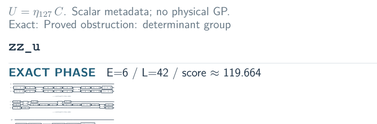
google_cz

EXACT PHASE E=6 / L=35 / score ≈ 110.164 — E upper bound



google_fsim

EXACT PHASE E=4 / L=15 / score ≈ 55.539 — E upper bound

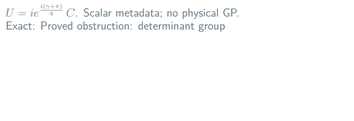


google_sqrt_iswap

EXACT PHASE E=12 / L=47 / score ≈ 160.636 — E upper bound

google_syc

EXACT PHASE E=36 / L=47 / score ≈ 254.236 — E upper bound



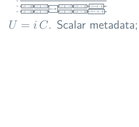
ibm_cx

PROJECTIVE E=6 / L=34 / score ≈ 97.236 — E upper bound

$U = -e^{\frac{i(\gamma+\pi)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=6 / L=67 / score ≈ 165.660 — E upper bound



$U = -e^{\frac{\pi i}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=6 / L=54 / score ≈ 137.640 — E upper bound

$U = -e^{\frac{i(\gamma+\pi)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=6 / L=72 / score ≈ 194.534 — E upper bound

$U = \eta_{126} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=6 / L=72 / score ≈ 192.122 — E upper bound

$U = \eta_{126} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=6 / L=78 / score ≈ 208.922 — E upper bound

$U = ie^{\frac{i(\gamma+\pi)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=6 / L=73 / score ≈ 191.692 — E upper bound

$U = e^{\frac{i(\gamma+\pi)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=8 / L=31 / score ≈ 105.586 — E upper bound

quantinuum_zzmax

PROJECTIVE E=6 / L=49 / score ≈ 134.803 — E upper bound

$U = ie^{\frac{i(\gamma+\pi)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=6 / L=49 / score ≈ 136.015 — E upper bound

$U = ie^{\frac{i(\gamma+\pi)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=6 / L=67 / score ≈ 165.660 — E upper bound

$U = -e^{\frac{\pi i}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=12 / L=47 / score ≈ 159.682 — E upper bound

sqrt_swap_u

EXACT PHASE E=12 / L=47 / score ≈ 160.082 — E upper bound

PROJECTIVE E=12 / L=47 / score ≈ 159.682 — E upper bound

$U = -i C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=4 / L=20 / score ≈ 64.397 — E upper bound

$U = \eta_{127} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=6 / L=41 / score ≈ 123.646 — E upper bound

xx_rz_ry

PROJECTIVE E=6 / L=81 / score ≈ 203.774 — E upper bound

$U = ie^{\frac{i(\gamma+\pi)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=6 / L=35 / score ≈ 110.422 — E upper bound

zz_u

EXACT PHASE E=6 / L=42 / score ≈ 119.664 — E upper bound

PROJECTIVE E=6 / L=41 / score ≈ 117.658 — E upper bound

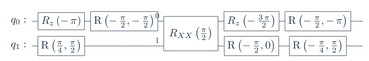
$U = i C$. Scalar metadata; no physical GP.

CH

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

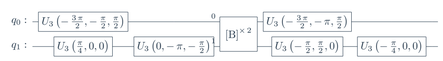
PROJECTIVE E=1 / L=7 / score ≈ 20.426 — proved E minimum



$U = \frac{\sqrt{2}i(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=6 / score ≈ 22.436 — proved E minimum



clifford_only

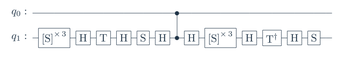
Exact: Proved obstruction: not a clifford
Projective: Proved obstruction: not a clifford

clifford_t_cx

EXACT PHASE E=1 / L=14 / score ≈ 32.000 — proved E minimum

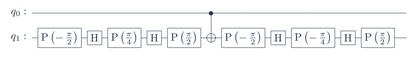
clifford_t_cz

EXACT PHASE E=1 / L=16 / score ≈ 36.000 — proved E minimum



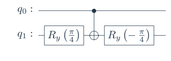
cx_h_phase

EXACT PHASE E=1 / L=10 / score ≈ 25.612 — proved E minimum



cx_rz_ry

EXACT PHASE E=1 / L=2 / score ≈ 8.804 — proved E minimum

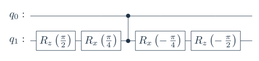


cx_u

EXACT PHASE E=1 / L=2 / score ≈ 8.812 — proved E minimum

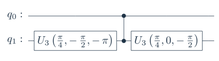
cz_rz_rx

EXACT PHASE E=1 / L=4 / score ≈ 13.208 — proved E minimum



cz_u

EXACT PHASE E=1 / L=2 / score ≈ 9.212 — proved E minimum



google_cz

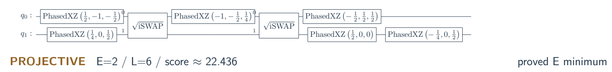
EXACT PHASE E=1 / L=4 / score ≈ 14.024 — proved E minimum

google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 14.028 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=2 / L=6 / score ≈ 23.036 — proved E minimum

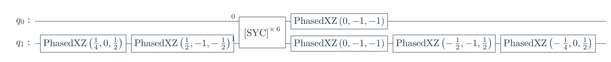


PROJECTIVE E=2 / L=6 / score ≈ 22.436 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=6 / L=6 / score ≈ 38.036 — E upper bound



ibm_cx

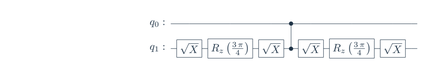
EXACT PHASE E=1 / L=9 / score ≈ 22.810 — proved E minimum

PROJECTIVE E=1 / L=5 / score ≈ 14.806 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

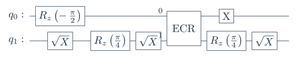
ibm_cz

EXACT PHASE E=1 / L=6 / score ≈ 16.804 — proved E minimum



ibm_ecr

EXACT PHASE E=1 / L=7 / score ≈ 19.006 — proved E minimum



ionq_aria

PROJECTIVE E=1 / L=13 / score ≈ 33.432 — proved E minimum

$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=1 / L=13 / score ≈ 33.030 — proved E minimum

$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

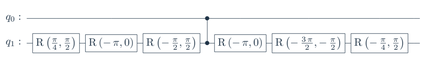
ionq_forte

PROJECTIVE E=1 / L=14 / score ≈ 35.830 — proved E minimum

$U = \frac{\sqrt{2}(-1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

EXACT PHASE E=1 / L=6 / score ≈ 18.024 — proved E minimum

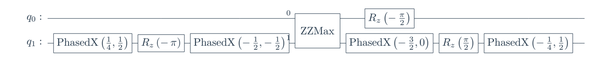


iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 22.236 — proved E minimum

quantinuum_zzmax

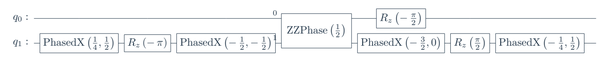
PROJECTIVE E=1 / L=7 / score ≈ 20.222 — proved E minimum



$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=1 / L=7 / score ≈ 20.424 — proved E minimum

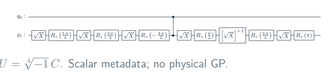


$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=1 / L=14 / score ≈ 33.212 — proved E minimum

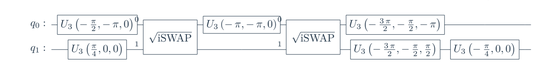
PROJECTIVE E=1 / L=13 / score ≈ 31.412 — proved E minimum



$U = \sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

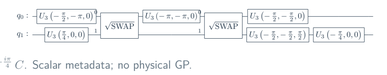
EXACT PHASE E=2 / L=6 / score ≈ 22.036 — proved E minimum



sqrt_swap_u

EXACT PHASE E=2 / L=6 / score ≈ 22.836 — proved E minimum

PROJECTIVE E=2 / L=6 / score ≈ 22.036 — proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

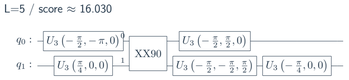
tket_tk2

PROJECTIVE E=1 / L=7 / score ≈ 20.430 — proved E minimum

$U = \frac{\sqrt{2}i(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

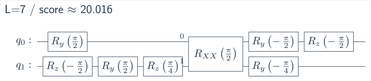
xx90_u

EXACT PHASE E=1 / L=5 / score ≈ 16.030 — proved E minimum



xx_rz_ry

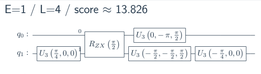
PROJECTIVE E=1 / L=7 / score ≈ 20.016 — proved E minimum



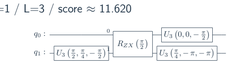
$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=1 / L=4 / score ≈ 13.826 — proved E minimum



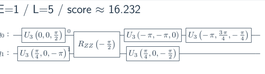
PROJECTIVE E=1 / L=3 / score ≈ 11.620 — proved E minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=1 / L=5 / score ≈ 16.232 — proved E minimum



PROJECTIVE E=1 / L=3 / score ≈ 11.420 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

CNOT

2 qubits / 32 preserved native bases / alphabetical panel order

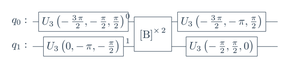
aqt_rxx

PROJECTIVE E=1 / L=5 / score ≈ 15.218 — proved E minimum

$U = \frac{\sqrt{2}i(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=4 / score ≈ 17.624 — proved E minimum



clifford_only

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

clifford_t_cx

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

clifford_t_cz

EXACT PHASE E=1 / L=2 / score ≈ 8.000 — proved E minimum

cx_h_phase

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

cx_rz_ry

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

cx_u

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

cz_rz_rx

EXACT PHASE E=1 / L=4 / score ≈ 12.808 — proved E minimum

cz_u

EXACT PHASE E=1 / L=2 / score ≈ 8.412 — proved E minimum

google_cz

EXACT PHASE E=1 / L=2 / score ≈ 8.812 — proved E minimum

google_fsim

EXACT PHASE E=1 / L=2 / score ≈ 8.816 — proved E minimum

google_sqrt_iswap

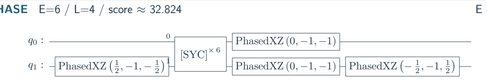
EXACT PHASE E=2 / L=4 / score ≈ 17.824 — proved E minimum

PROJECTIVE E=2 / L=4 / score ≈ 17.224 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=6 / L=4 / score ≈ 32.824 — E upper bound



ibm_cx

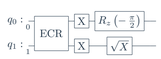
EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

ibm_cz

EXACT PHASE E=1 / L=6 / score ≈ 16.404 — proved E minimum

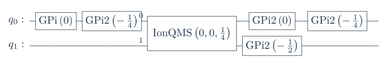
ibm_ecr

EXACT PHASE E=1 / L=4 / score ≈ 12.202 — proved E minimum



ionq_aria

PROJECTIVE E=1 / L=5 / score ≈ 15.416 — proved E minimum



$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=1 / L=5 / score ≈ 15.014 — proved E minimum

$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=1 / L=6 / score ≈ 17.814 — proved E minimum

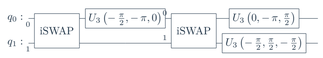
$U = \frac{\sqrt{2}(-1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

EXACT PHASE E=1 / L=4 / score ≈ 12.816 — proved E minimum

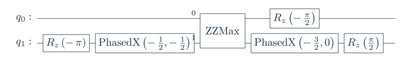
iswap_u

EXACT PHASE E=2 / L=3 / score ≈ 15.018 — proved E minimum



quantinuum_zzmax

PROJECTIVE E=1 / L=5 / score ≈ 15.014 — proved E minimum



$U = \frac{\sqrt{2}i(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

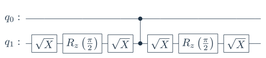
quantinuum_zzphase

PROJECTIVE E=1 / L=5 / score ≈ 15.216 — proved E minimum

$U = \frac{\sqrt{2}i(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

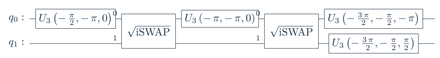
rigetti_cz

EXACT PHASE E=1 / L=6 / score ≈ 16.404 — proved E minimum



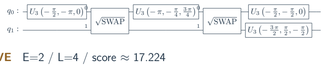
sqrt_iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 17.224 — proved E minimum



sqrt_swap_u

EXACT PHASE E=2 / L=4 / score ≈ 18.024 — proved E minimum



PROJECTIVE E=2 / L=4 / score ≈ 17.224 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

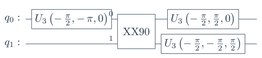
tket_tk2

PROJECTIVE E=1 / L=5 / score ≈ 15.222 — proved E minimum

$U = \frac{\sqrt{2}i(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

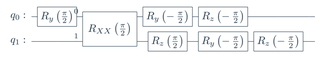
xx90_u

EXACT PHASE E=1 / L=3 / score ≈ 11.218 — proved E minimum



xx_rz_ry

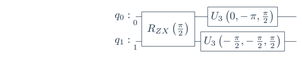
PROJECTIVE E=1 / L=6 / score ≈ 17.414 — proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

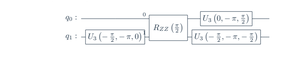
EXACT PHASE E=1 / L=2 / score ≈ 9.014 — proved E minimum



zz_u

EXACT PHASE E=1 / L=4 / score ≈ 13.426 — proved E minimum

PROJECTIVE E=1 / L=3 / score ≈ 11.020 — proved E minimum



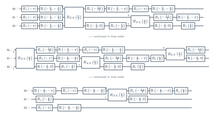
$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

Controlled

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

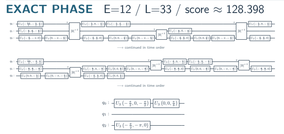
PROJECTIVE E=6 / L=41 / score ≈ 118.334 — E upper bound



$U = -(-1)^{\frac{9}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=12 / L=33 / score ≈ 128.398 — E upper bound



$U = -(-1)^{\frac{9}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

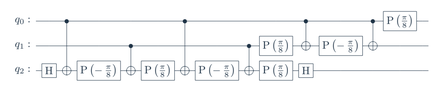
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

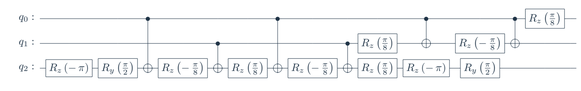
cx_h_phase

EXACT PHASE E=6 / L=9 / score ≈ 46.214 — E upper bound



cx_rz_ry

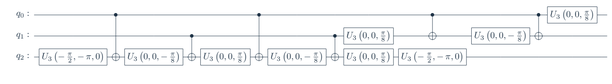
PROJECTIVE E=6 / L=11 / score ≈ 50.622 — E upper bound



$U = -\sqrt[16]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

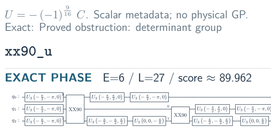
cx_u

EXACT PHASE E=6 / L=9 / score ≈ 46.654 — E upper bound



cz_rz_rx

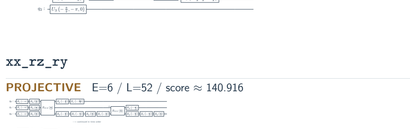
PROJECTIVE E=6 / L=38 / score ≈ 110.276 — E upper bound



$U = -\sqrt[16]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=6 / L=21 / score ≈ 73.126 — E upper bound

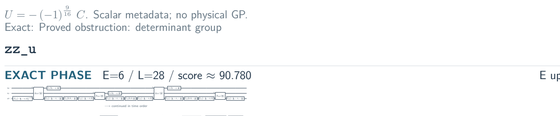


google_cz

EXACT PHASE E=6 / L=21 / score ≈ 75.926 — E upper bound

google_fsim

EXACT PHASE E=6 / L=21 / score ≈ 75.950 — E upper bound



google_sqrt_iswap

EXACT PHASE E=12 / L=33 / score ≈ 126.398 — E upper bound

google_syc

EXACT PHASE E=36 / L=33 / score ≈ 219.998 — E upper bound

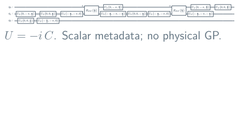
ibm_cx

PROJECTIVE E=6 / L=13 / score ≈ 54.618 — E upper bound

$U = e^{\frac{15}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=6 / L=38 / score ≈ 107.042 — E upper bound



$U = -(-1)^{\frac{9}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=6 / L=36 / score ≈ 101.628 — E upper bound

$U = e^{\frac{17}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=6 / L=48 / score ≈ 136.332 — E upper bound

$U = -(-1)^{\frac{9}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=6 / L=48 / score ≈ 133.920 — E upper bound

$U = -(-1)^{\frac{9}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=6 / L=54 / score ≈ 150.720 — E upper bound

$U = (-1)^{\frac{9}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=6 / L=49 / score ≈ 137.596 — E upper bound

$U = -\sqrt[16]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=12 / L=33 / score ≈ 127.198 — E upper bound

quantinuum_zzmax

PROJECTIVE E=6 / L=35 / score ≈ 104.898 — E upper bound

$U = -(-1)^{\frac{9}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=6 / L=35 / score ≈ 106.110 — E upper bound

$U = -(-1)^{\frac{9}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=6 / L=38 / score ≈ 107.042 — E upper bound

$U = -(-1)^{\frac{9}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=12 / L=33 / score ≈ 125.998 — E upper bound

sqrt_swap_u

EXACT PHASE E=12 / L=33 / score ≈ 126.398 — E upper bound

PROJECTIVE E=12 / L=33 / score ≈ 125.998 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=6 / L=52 / score ≈ 140.916 — E upper bound

$U = -(-1)^{\frac{9}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=6 / L=21 / score ≈ 76.738 — E upper bound

zz_u

EXACT PHASE E=6 / L=28 / score ≈ 90.780 — E upper bound

PROJECTIVE E=6 / L=27 / score ≈ 88.774 — E upper bound

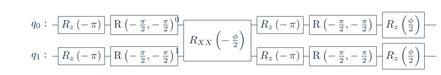
$U = -iC$. Scalar metadata; no physical GP.

CP (ϕ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

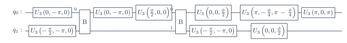
PROJECTIVE E=1 / L=10 / score ≈ 26.230 — minimum uniform E budget



$U = e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=9 / score ≈ 27.854 — minimum uniform E budget



PROJECTIVE E=2 / L=7 / score ≈ 23.042 — minimum uniform E budget

$U = e^{-\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

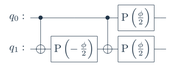
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

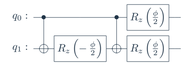
cx_h_phase

EXACT PHASE E=2 / L=3 / score ≈ 14.606 — minimum uniform E budget



cx_rz_ry

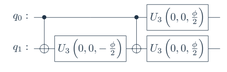
PROJECTIVE E=2 / L=3 / score ≈ 14.606 — minimum uniform E budget



$U = e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=2 / L=3 / score ≈ 14.618 — minimum uniform E budget



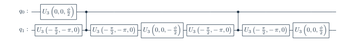
cz_rz_rx

PROJECTIVE E=2 / L=8 / score ≈ 25.616 — minimum uniform E budget

$U = -e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=7 / score ≈ 23.442 — minimum uniform E budget



PROJECTIVE E=2 / L=5 / score ≈ 19.430 — minimum uniform E budget

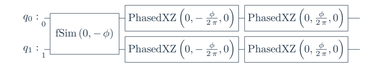
$U = e^{-\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=2 / L=5 / score ≈ 20.421 — minimum uniform E budget

google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 14.149 — minimum uniform E budget



google_sqrt_iswap

EXACT PHASE E=2 / L=6 / score ≈ 24.809 — minimum uniform E budget

Exact definitions: $\eta_{128}, \eta_{129}, \eta_{130}, \eta_{131}, \eta_{132}, \eta_{133}, \eta_{134}, \eta_{135}, \eta_{136}$ in the expression glossary.

PROJECTIVE E=2 / L=5 / score ≈ 21.542 — minimum uniform E budget

Exact definitions: $\eta_{131}, \eta_{132}, \eta_{135}, \eta_{136}, \eta_{137}$ in the expression glossary.

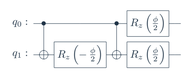
$U = \eta_{138}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=11 / score ≈ 73.257 — E upper bound

ibm_cx

PROJECTIVE E=2 / L=3 / score ≈ 14.606 — minimum uniform E budget



$U = e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=2 / L=12 / score ≈ 33.414 — minimum uniform E budget

$U = -(-1)^{\frac{3}{4}} e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=2 / L=6 / score ≈ 20.608 — minimum uniform E budget

$U = ie^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=2 / L=16 / score ≈ 45.635 — E upper bound

$U = ie^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=2 / L=16 / score ≈ 44.831 — minimum uniform E budget

$U = ie^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

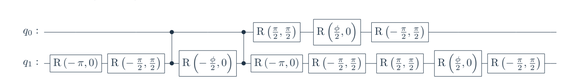
ionq_forte

PROJECTIVE E=1 / L=4 / score ≈ 14.601 — minimum uniform E budget

$U = e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

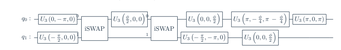
PROJECTIVE E=2 / L=11 / score ≈ 33.044 — minimum uniform E budget



$U = -e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=2 / L=8 / score ≈ 25.848 — minimum uniform E budget

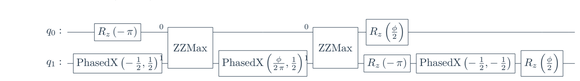


PROJECTIVE E=2 / L=6 / score ≈ 21.036 — minimum uniform E budget

$U = e^{-\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

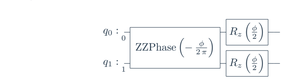
PROJECTIVE E=2 / L=7 / score ≈ 23.950 — minimum uniform E budget



$U = -ie^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

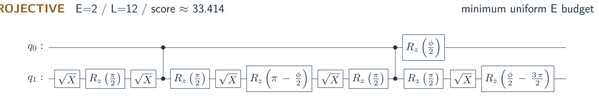
PROJECTIVE E=1 / L=2 / score ≈ 8.936 — minimum uniform E budget



$U = e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=2 / L=12 / score ≈ 33.414 — minimum uniform E budget



$U = -(-1)^{\frac{3}{4}} e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 23.036 — minimum uniform E budget

Exact definitions: $\eta_{139}, \eta_{140}, \eta_{141}, \eta_{142}, \eta_{143}, \eta_{144}, \eta_{145}$ in the expression glossary.

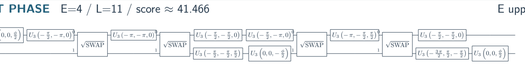
PROJECTIVE E=2 / L=5 / score ≈ 20.430 — minimum uniform E budget

Exact definitions: $\eta_{143}, \eta_{144}, \eta_{145}$ in the expression glossary.

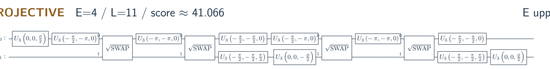
$U = \eta_{146}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=11 / score ≈ 41.466 — E upper bound



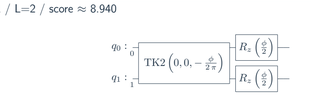
PROJECTIVE E=4 / L=11 / score ≈ 41.066 — E upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=2 / score ≈ 8.940 — minimum uniform E budget



$U = e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

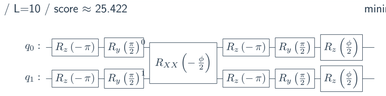
EXACT PHASE E=2 / L=8 / score ≈ 26.248 — minimum uniform E budget

PROJECTIVE E=2 / L=6 / score ≈ 21.036 — minimum uniform E budget

$U = ie^{-\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=1 / L=10 / score ≈ 25.422 — minimum uniform E budget



$U = e^{\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=1 / L=6 / score ≈ 17.838 — minimum uniform E budget

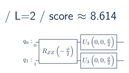
PROJECTIVE E=1 / L=4 / score ≈ 13.026 — minimum uniform E budget

$U = e^{-\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=1 / L=4 / score ≈ 13.426 — minimum uniform E budget

PROJECTIVE E=1 / L=2 / score ≈ 8.614 — minimum uniform E budget



$U = e^{-\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.

CR (θ, ϕ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

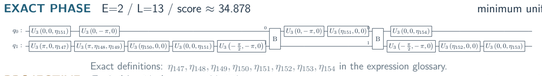
EXACT PHASE E=3 / L=16 / score ≈ 47.658 — E upper bound

PROJECTIVE E=2 / L=14 / score ≈ 39.648 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

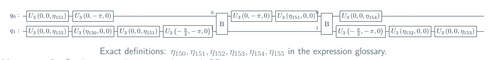
berkeley_b

EXACT PHASE E=2 / L=13 / score ≈ 34.878 — minimum uniform E budget



Exact definitions: $\eta_{147}, \eta_{148}, \eta_{149}, \eta_{150}, \eta_{151}, \eta_{152}, \eta_{153}, \eta_{154}$ in the expression glossary.

PROJECTIVE E=2 / L=12 / score ≈ 32.472 — minimum uniform E budget



Exact definitions: $\eta_{150}, \eta_{151}, \eta_{152}, \eta_{153}, \eta_{154}, \eta_{155}$ in the expression glossary.

$U = \eta_{156}\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=11 / score ≈ 31.214 — minimum uniform E budget

cx_rz_ry

EXACT PHASE E=2 / L=4 / score ≈ 16.808 — minimum uniform E budget

cx_u

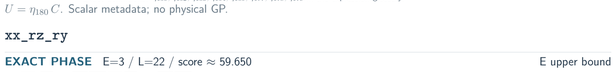
EXACT PHASE E=2 / L=4 / score ≈ 16.824 — minimum uniform E budget

cz_rz_rx

EXACT PHASE E=2 / L=13 / score ≈ 35.826 — minimum uniform E budget

cz_u

EXACT PHASE E=2 / L=8 / score ≈ 25.648 — minimum uniform E budget



google_cz

EXACT PHASE E=2 / L=8 / score ≈ 28.169 — minimum uniform E budget

google_fsim

EXACT PHASE E=2 / L=8 / score ≈ 28.177 — E upper bound

google_sqrt_iswap

EXACT PHASE E=2 / L=11 / score ≈ 31.727 — minimum uniform E budget

Exact definitions: $\eta_{157}, \eta_{158}, \eta_{159}, \eta_{160}, \eta_{161}, \eta_{162}, \eta_{163}, \eta_{164}, \eta_{165}, \eta_{166}, \eta_{167}, \eta_{168}, \eta_{169}, \eta_{170}, \eta_{171}$ in the expression glossary.

PROJECTIVE E=2 / L=10 / score ≈ 29.721 — minimum uniform E budget

Exact definitions: $\eta_{161}, \eta_{162}, \eta_{163}, \eta_{164}, \eta_{165}, \eta_{166}, \eta_{167}, \eta_{168}, \eta_{169}, \eta_{170}, \eta_{171}, \eta_{172}, \eta_{173}$ in the expression glossary.

$U = \eta_{174}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=12 / score ≈ 76.193 — E upper bound

ibm_cx

EXACT PHASE E=2 / L=10 / score ≈ 28.812 — minimum uniform E budget

PROJECTIVE E=2 / L=9 / score ≈ 26.810 — minimum uniform E budget

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=19 / score ≈ 48.022 — minimum uniform E budget

PROJECTIVE E=2 / L=19 / score ≈ 47.618 — minimum uniform E budget

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=17 / score ≈ 43.214 — minimum uniform E budget

ionq_aria

EXACT PHASE E=2 / L=22 / score ≈ 59.377 — E upper bound

PROJECTIVE E=2 / L=22 / score ≈ 58.977 — E upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=22 / score ≈ 58.573 — minimum uniform E budget

PROJECTIVE E=2 / L=22 / score ≈ 58.173 — minimum uniform E budget

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=24 / score ≈ 64.173 — E upper bound

PROJECTIVE E=2 / L=24 / score ≈ 63.773 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=16 / score ≈ 44.464 — minimum uniform E budget

iswap_u

EXACT PHASE E=4 / L=12 / score ≈ 43.672 — E upper bound

quantinuum_zzmax

EXACT PHASE E=5 / L=32 / score ≈ 92.157 — E upper bound

PROJECTIVE E=2 / L=13 / score ≈ 37.899 — minimum uniform E budget

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=3 / L=15 / score ≈ 46.309 — E upper bound

PROJECTIVE E=2 / L=13 / score ≈ 38.303 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=2 / L=19 / score ≈ 48.022 — minimum uniform E budget

PROJECTIVE E=2 / L=19 / score ≈ 47.618 — minimum uniform E budget

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=2 / L=11 / score ≈ 32.066 — minimum uniform E budget

Exact definitions: $\eta_{147}, \eta_{150}, \eta_{152}, \eta_{153}, \eta_{154}, \eta_{175}, \eta_{176}, \eta_{177}, \eta_{178}, \eta_{179}$ in the expression glossary.

PROJECTIVE E=2 / L=10 / score ≈ 30.060 — minimum uniform E budget

Exact definitions: $\eta_{150}, \eta_{152}, \eta_{153}, \eta_{154}, \eta_{155}, \eta_{177}, \eta_{178}, \eta_{179}$ in the expression glossary.

$U = \eta_{180}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=12 / score ≈ 43.672 — E upper bound

PROJECTIVE E=4 / L=12 / score ≈ 43.272 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=5 / L=34 / score ≈ 97.201 — E upper bound

PROJECTIVE E=2 / L=14 / score ≈ 40.317 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

xx90_u

EXACT PHASE E=2 / L=10 / score ≈ 31.260 — minimum uniform E budget

xx_rz_ry

EXACT PHASE E=3 / L=22 / score ≈ 59.650 — E upper bound

PROJECTIVE E=2 / L=18 / score ≈ 47.640 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=2 / L=8 / score ≈ 26.852 — E upper bound

zz_u

EXACT PHASE E=2 / L=11 / score ≈ 32.870 — E upper bound

PROJECTIVE E=2 / L=10 / score ≈ 30.864 — E upper bound

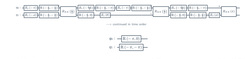
$U = i\,C$. Scalar metadata; no physical GP.

CROT (θ, ϕ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=3 / L=17 / score ≈ 49.260 — E upper bound



PROJECTIVE E=2 / L=15 / score ≈ 41.250 — E upper bound

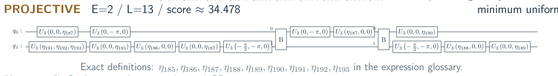
$U = -iC$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=2 / L=14 / score ≈ 36.484 — minimum uniform E budget

Exact definitions: $\eta_{181}, \eta_{182}, \eta_{183}, \eta_{184}, \eta_{185}, \eta_{186}, \eta_{187}, \eta_{188}, \eta_{189}, \eta_{190}$ in the expression glossary.

PROJECTIVE E=2 / L=13 / score ≈ 34.478 — minimum uniform E budget



Exact definitions: $\eta_{185}, \eta_{186}, \eta_{187}, \eta_{188}, \eta_{189}, \eta_{190}, \eta_{191}, \eta_{192}, \eta_{193}$ in the expression glossary.

$U = \eta_{194} C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

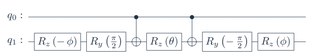
EXACT PHASE E=2 / L=13 / score ≈ 34.414 — minimum uniform E budget

PROJECTIVE E=2 / L=5 / score ≈ 18.006 — minimum uniform E budget

$U = e^{-\frac{i\theta}{2}} C$. Scalar metadata; no physical GP.

cx_rz_ry

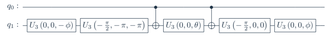
EXACT PHASE E=2 / L=5 / score ≈ 18.410 — minimum uniform E budget



cx_u

EXACT PHASE E=2 / L=7 / score ≈ 22.842 — minimum uniform E budget

PROJECTIVE E=2 / L=5 / score ≈ 18.430 — minimum uniform E budget



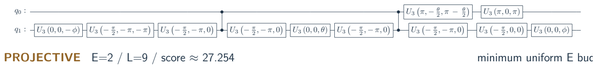
$U = e^{-\frac{i\theta}{2}} C$. Scalar metadata; no physical GP.

cz_rz_rx

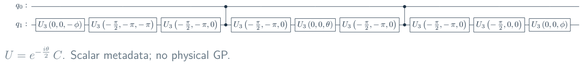
EXACT PHASE E=2 / L=11 / score ≈ 31.622 — minimum uniform E budget

cz_u

EXACT PHASE E=2 / L=11 / score ≈ 31.666 — minimum uniform E budget



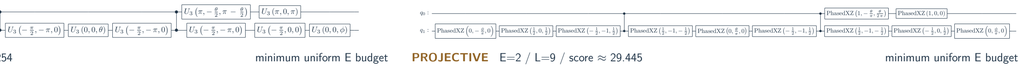
PROJECTIVE E=2 / L=9 / score ≈ 27.254 — minimum uniform E budget



$U = e^{-\frac{i\theta}{2}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=2 / L=11 / score ≈ 34.317 — minimum uniform E budget



PROJECTIVE E=2 / L=9 / score ≈ 29.445 — minimum uniform E budget

$U = e^{-\frac{i\theta}{2}} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=11 / score ≈ 34.325 — E upper bound

PROJECTIVE E=2 / L=9 / score ≈ 29.453 — E upper bound

$U = e^{-\frac{i\theta}{2}} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=2 / L=12 / score ≈ 34.063 — minimum uniform E budget

Exact definitions: $\eta_{195}, \eta_{196}, \eta_{197}, \eta_{198}, \eta_{199}, \eta_{200}, \eta_{201}, \eta_{202}, \eta_{203}, \eta_{204}, \eta_{205}, \eta_{206}, \eta_{207}, \eta_{208}, \eta_{209}, \eta_{210}, \eta_{211}, \eta_{212}$ in the expression glossary.

PROJECTIVE E=2 / L=11 / score ≈ 32.057 — minimum uniform E budget

Exact definitions: $\eta_{200}, \eta_{201}, \eta_{202}, \eta_{203}, \eta_{204}, \eta_{205}, \eta_{206}, \eta_{207}, \eta_{208}, \eta_{209}, \eta_{210}, \eta_{211}, \eta_{212}, \eta_{213}, \eta_{214}, \eta_{215}$ in the expression glossary.

$U = \eta_{216} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=15 / score ≈ 82.341 — E upper bound

PROJECTIVE E=12 / L=13 / score ≈ 77.469 — E upper bound

$U = e^{-\frac{i\theta}{2}} C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=8 / score ≈ 24.810 — minimum uniform E budget

PROJECTIVE E=2 / L=7 / score ≈ 22.810 — minimum uniform E budget

$U = -iC$. Scalar metadata; no physical GP.

ibm_cz

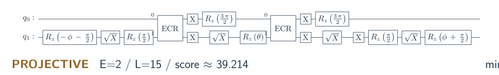
EXACT PHASE E=2 / L=17 / score ≈ 43.218 — minimum uniform E budget

PROJECTIVE E=2 / L=15 / score ≈ 39.218 — minimum uniform E budget

$U = -iC$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=16 / score ≈ 41.214 — minimum uniform E budget



PROJECTIVE E=2 / L=15 / score ≈ 39.214 — minimum uniform E budget

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=18 / score ≈ 50.239 — E upper bound

PROJECTIVE E=2 / L=18 / score ≈ 49.839 — E upper bound

$U = iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=18 / score ≈ 49.435 — minimum uniform E budget

PROJECTIVE E=2 / L=18 / score ≈ 49.035 — minimum uniform E budget

$U = iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=20 / score ≈ 55.035 — E upper bound

PROJECTIVE E=2 / L=20 / score ≈ 54.635 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

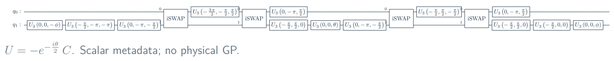
iqm_cz

EXACT PHASE E=2 / L=19 / score ≈ 50.876 — minimum uniform E budget

iswap_u

EXACT PHASE E=4 / L=15 / score ≈ 49.690 — E upper bound

PROJECTIVE E=4 / L=13 / score ≈ 45.278 — E upper bound



$U = -e^{-\frac{i\theta}{2}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=5 / L=33 / score ≈ 93.098 — E upper bound

PROJECTIVE E=2 / L=14 / score ≈ 38.840 — minimum uniform E budget

$U = -iC$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=3 / L=16 / score ≈ 47.250 — E upper bound

PROJECTIVE E=2 / L=14 / score ≈ 39.244 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=2 / L=17 / score ≈ 43.218 — minimum uniform E budget

PROJECTIVE E=2 / L=15 / score ≈ 39.218 — minimum uniform E budget

$U = -iC$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=2 / L=12 / score ≈ 34.072 — minimum uniform E budget

Exact definitions: $\eta_{181}, \eta_{182}, \eta_{185}, \eta_{186}, \eta_{188}, \eta_{189}, \eta_{190}, \eta_{217}, \eta_{218}, \eta_{219}, \eta_{220}, \eta_{221}$ in the expression glossary.

PROJECTIVE E=2 / L=11 / score ≈ 32.066 — minimum uniform E budget

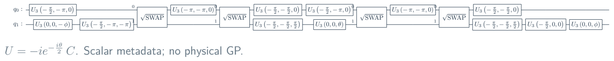
Exact definitions: $\eta_{185}, \eta_{186}, \eta_{188}, \eta_{189}, \eta_{190}, \eta_{191}, \eta_{192}, \eta_{193}, \eta_{219}, \eta_{220}, \eta_{221}$ in the expression glossary.

$U = \eta_{222} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=15 / score ≈ 49.690 — E upper bound

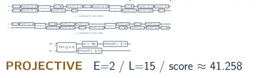
PROJECTIVE E=4 / L=13 / score ≈ 44.878 — E upper bound



$U = -ie^{-\frac{i\theta}{2}} C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=5 / L=35 / score ≈ 98.142 — E upper bound

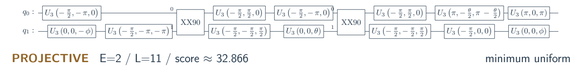


PROJECTIVE E=2 / L=15 / score ≈ 41.258 — E upper bound

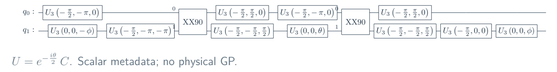
$U = -iC$. Scalar metadata; no physical GP.

xx90_u

EXACT PHASE E=2 / L=13 / score ≈ 37.278 — minimum uniform E budget



PROJECTIVE E=2 / L=11 / score ≈ 32.866 — minimum uniform E budget

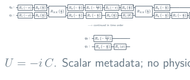


$U = e^{-\frac{i\theta}{2}} C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=3 / L=23 / score ≈ 61.252 — E upper bound

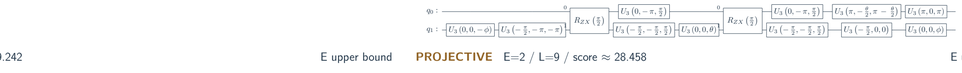
PROJECTIVE E=2 / L=19 / score ≈ 49.242 — E upper bound



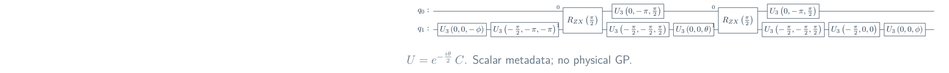
$U = -iC$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=2 / L=11 / score ≈ 32.870 — E upper bound



PROJECTIVE E=2 / L=9 / score ≈ 28.458 — E upper bound



$U = e^{-\frac{i\theta}{2}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=2 / L=14 / score ≈ 38.888 — E upper bound

PROJECTIVE E=2 / L=11 / score ≈ 32.470 — E upper bound

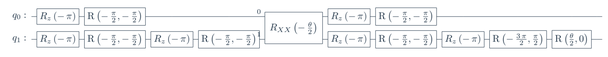
$U = ie^{-\frac{i\theta}{2}} C$. Scalar metadata; no physical GP.

CRX (θ)

2 qubits / 32 preserved native bases / alphabetical panel order

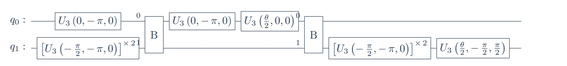
aqt_rxx

EXACT PHASE E=1 / L=13 / score ≈ 32.842 — minimum uniform E budget



berkeley_b

EXACT PHASE E=2 / L=8 / score ≈ 25.648 — minimum uniform E budget



clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

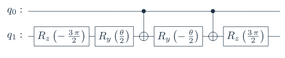
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=10 / score ≈ 29.212 — minimum uniform E budget

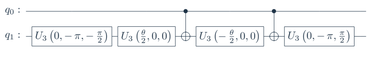
cx_rz_ry

EXACT PHASE E=2 / L=4 / score ≈ 16.808 — minimum uniform E budget



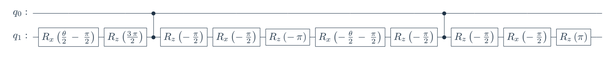
cx_u

EXACT PHASE E=2 / L=4 / score ≈ 16.824 — minimum uniform E budget



cz_rz_rx

EXACT PHASE E=2 / L=10 / score ≈ 29.620 — minimum uniform E budget



cz_u

EXACT PHASE E=2 / L=8 / score ≈ 25.648 — minimum uniform E budget

google_cz

EXACT PHASE E=2 / L=8 / score ≈ 27.509 — minimum uniform E budget

google_fsim

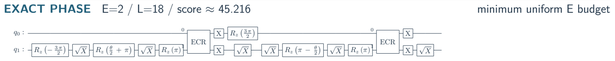
EXACT PHASE E=2 / L=8 / score ≈ 27.517 — E upper bound

google_sqrt_iswap

EXACT PHASE E=2 / L=11 / score ≈ 31.727 — minimum uniform E budget

Exact definitions: η_{162}, η_{166}, η_{223}, η_{224}, η_{225}, η_{226}, η_{227}, η_{228}, η_{229}, η_{230}, η_{231}, η_{232}, η_{233}, η_{234}, η_{235} in the expression glossary.

PROJECTIVE E=2 / L=10 / score ≈ 29.721 — minimum uniform E budget



Exact definitions: η_{162}, η_{166}, η_{227}, η_{228}, η_{229}, η_{230}, η_{231}, η_{232}, η_{233}, η_{234}, η_{235}, η_{236}, η_{237} in the expression glossary.

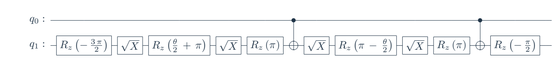
$U = \eta_{238}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=12 / score ≈ 75.533 — E upper bound

ibm_cx

EXACT PHASE E=2 / L=10 / score ≈ 28.812 — minimum uniform E budget



ibm_cz

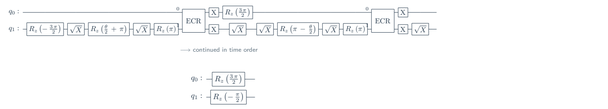
EXACT PHASE E=2 / L=19 / score ≈ 48.022 — minimum uniform E budget

PROJECTIVE E=2 / L=18 / score ≈ 46.020 — minimum uniform E budget

$U = -1\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=18 / score ≈ 45.216 — minimum uniform E budget



ionq_aria

EXACT PHASE E=2 / L=22 / score ≈ 58.717 — E upper bound

PROJECTIVE E=2 / L=22 / score ≈ 58.317 — E upper bound

$U = i\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=22 / score ≈ 57.913 — minimum uniform E budget

PROJECTIVE E=2 / L=22 / score ≈ 57.513 — minimum uniform E budget

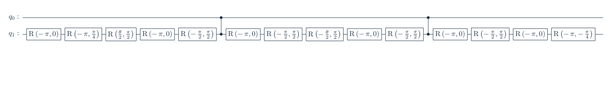
$U = i\, C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=1 / L=8 / score ≈ 23.279 — minimum uniform E budget

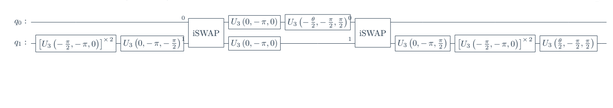
iqm_cz

EXACT PHASE E=2 / L=14 / score ≈ 39.256 — minimum uniform E budget



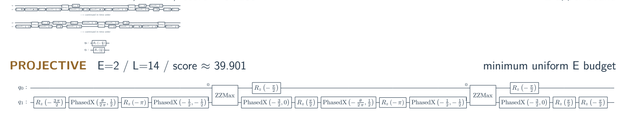
iswap_u

EXACT PHASE E=2 / L=10 / score ≈ 30.460 — minimum uniform E budget



quantinuum_zzmax

EXACT PHASE E=5 / L=33 / score ≈ 94.159 — E upper bound

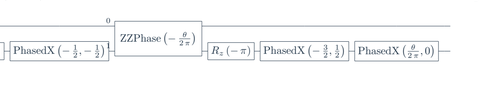


PROJECTIVE E=2 / L=14 / score ≈ 39.901 — minimum uniform E budget

$U = i\, C$. Scalar metadata; no physical GP.

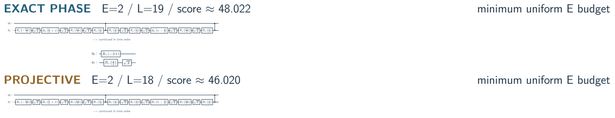
quantinuum_zzphase

EXACT PHASE E=1 / L=5 / score ≈ 15.879 — minimum uniform E budget



rigetti_cz

EXACT PHASE E=2 / L=19 / score ≈ 48.022 — minimum uniform E budget

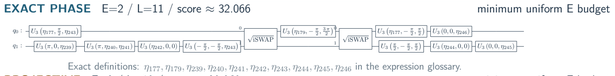


PROJECTIVE E=2 / L=18 / score ≈ 46.020 — minimum uniform E budget

$U = -1\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=2 / L=11 / score ≈ 32.066 — minimum uniform E budget



Exact definitions: η_{177}, η_{179}, η_{239}, η_{240}, η_{241}, η_{242}, η_{243}, η_{244}, η_{245}, η_{246} in the expression glossary.

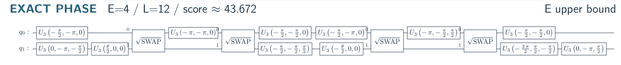
PROJECTIVE E=2 / L=10 / score ≈ 30.060 — minimum uniform E budget

Exact definitions: η_{177}, η_{179}, η_{242}, η_{243}, η_{244}, η_{245}, η_{246}, η_{247} in the expression glossary.

$U = \eta_{248}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=12 / score ≈ 43.672 — E upper bound



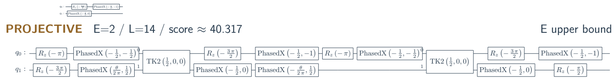
PROJECTIVE E=4 / L=12 / score ≈ 43.272 — E upper bound

$U = -i\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=5 / L=34 / score ≈ 97.201 — E upper bound

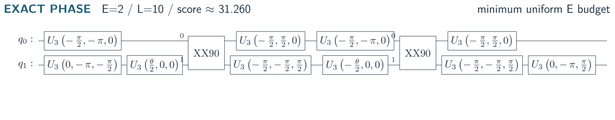
PROJECTIVE E=2 / L=14 / score ≈ 40.317 — E upper bound



$U = i\, C$. Scalar metadata; no physical GP.

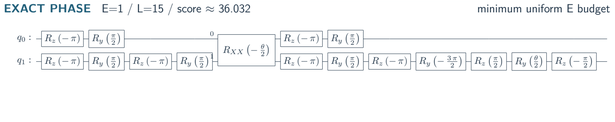
xx90_u

EXACT PHASE E=2 / L=10 / score ≈ 31.260 — minimum uniform E budget



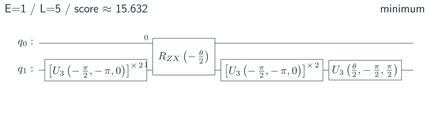
xx_rz_ry

EXACT PHASE E=1 / L=15 / score ≈ 36.032 — minimum uniform E budget



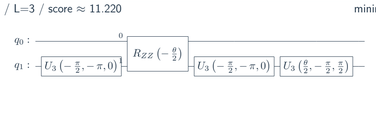
zx_u

EXACT PHASE E=1 / L=5 / score ≈ 15.632 — minimum uniform E budget



zz_u

EXACT PHASE E=1 / L=3 / score ≈ 11.220 — minimum uniform E budget



CRY (θ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=1 / L=14 / score ≈ 35.444 — minimum uniform E budget

berkeley_b

EXACT PHASE E=2 / L=10 / score ≈ 29.660 — minimum uniform E budget

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

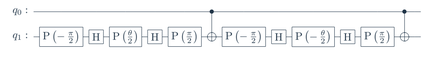
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

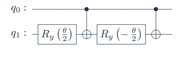
cx_h_phase

EXACT PHASE E=2 / L=10 / score ≈ 29.212 — minimum uniform E budget



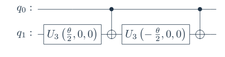
cx_rz_ry

EXACT PHASE E=2 / L=2 / score ≈ 12.404 — minimum uniform E budget



cx_u

EXACT PHASE E=2 / L=2 / score ≈ 12.412 — minimum uniform E budget

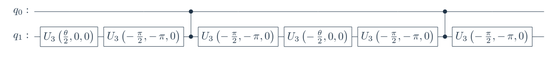


cz_rz_rx

EXACT PHASE E=2 / L=11 / score ≈ 32.022 — minimum uniform E budget

cz_u

EXACT PHASE E=2 / L=6 / score ≈ 21.236 — minimum uniform E budget

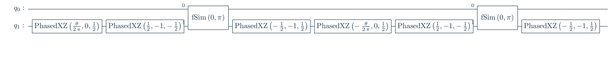


google_cz

EXACT PHASE E=2 / L=6 / score ≈ 23.097 — minimum uniform E budget

google_fsim

EXACT PHASE E=2 / L=6 / score ≈ 23.105 — E upper bound



google_sqrt_iswap

EXACT PHASE E=2 / L=11 / score ≈ 31.727 — minimum uniform E budget

Exact definitions: η_{162}, η_{166}, η_{225}, η_{226}, η_{227}, η_{228}, η_{229}, η_{230}, η_{231}, η_{234}, η_{235}, η_{249}, η_{250}, η_{251}, η_{252} in the expression glossary.

PROJECTIVE E=2 / L=10 / score ≈ 29.721 — minimum uniform E budget

Exact definitions: η_{162}, η_{166}, η_{227}, η_{228}, η_{229}, η_{230}, η_{231}, η_{234}, η_{235}, η_{251}, η_{252}, η_{253}, η_{254} in the expression glossary.

$U = \eta_{238}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=10 / score ≈ 71.121 — E upper bound

ibm_cx

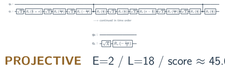
EXACT PHASE E=2 / L=9 / score ≈ 26.410 — minimum uniform E budget

PROJECTIVE E=2 / L=8 / score ≈ 24.408 — minimum uniform E budget

$U = -1\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=18 / score ≈ 46.020 — minimum uniform E budget

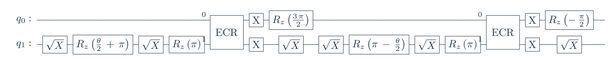


PROJECTIVE E=2 / L=18 / score ≈ 45.616 — minimum uniform E budget

$U = i\, C$. Scalar metadata; no physical GP.

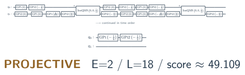
ibm_ecr

EXACT PHASE E=2 / L=16 / score ≈ 40.812 — minimum uniform E budget



ionq_aria

EXACT PHASE E=2 / L=18 / score ≈ 49.509 — E upper bound



PROJECTIVE E=2 / L=18 / score ≈ 49.109 — E upper bound

$U = i\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=18 / score ≈ 48.705 — minimum uniform E budget

PROJECTIVE E=2 / L=18 / score ≈ 48.305 — minimum uniform E budget

$U = i\, C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=1 / L=12 / score ≈ 31.887 — minimum uniform E budget

iqm_cz

EXACT PHASE E=2 / L=10 / score ≈ 30.440 — minimum uniform E budget

iswap_u

EXACT PHASE E=2 / L=12 / score ≈ 34.472 — minimum uniform E budget

quantinuum_zzmax

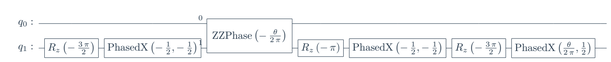
EXACT PHASE E=5 / L=31 / score ≈ 89.755 — E upper bound

PROJECTIVE E=2 / L=12 / score ≈ 35.497 — minimum uniform E budget

$U = -i\, C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=1 / L=6 / score ≈ 18.481 — minimum uniform E budget



rigetti_cz

EXACT PHASE E=2 / L=18 / score ≈ 46.020 — minimum uniform E budget

PROJECTIVE E=2 / L=18 / score ≈ 45.616 — minimum uniform E budget

$U = i\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=2 / L=11 / score ≈ 32.066 — minimum uniform E budget

Exact definitions: η_{177}, η_{179}, η_{230}, η_{241}, η_{242}, η_{243}, η_{244}, η_{246}, η_{255}, η_{256} in the expression glossary.

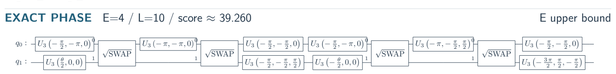
PROJECTIVE E=2 / L=10 / score ≈ 30.060 — minimum uniform E budget

Exact definitions: η_{177}, η_{179}, η_{242}, η_{243}, η_{244}, η_{246}, η_{256}, η_{257} in the expression glossary.

$U = \eta_{248}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=10 / score ≈ 39.260 — E upper bound



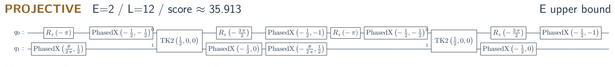
PROJECTIVE E=4 / L=10 / score ≈ 38.860 — E upper bound

$U = -i\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=5 / L=32 / score ≈ 92.797 — E upper bound

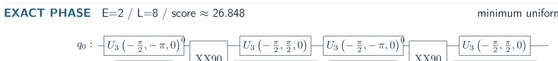
PROJECTIVE E=2 / L=12 / score ≈ 35.913 — E upper bound



$U = -i\, C$. Scalar metadata; no physical GP.

xx90_u

EXACT PHASE E=2 / L=8 / score ≈ 26.848 — minimum uniform E budget

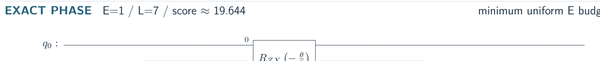


xx_rz_ry

EXACT PHASE E=1 / L=14 / score ≈ 34.030 — minimum uniform E budget

zx_u

EXACT PHASE E=1 / L=7 / score ≈ 19.644 — minimum uniform E budget



zz_u

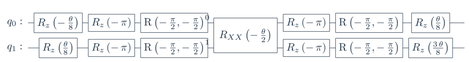
EXACT PHASE E=1 / L=5 / score ≈ 15.232 — minimum uniform E budget

CRZ (θ)

2 qubits / 32 preserved native bases / alphabetical panel order

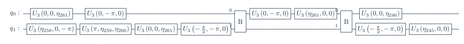
aqt_rxx

EXACT PHASE E=1 / L=12 / score ≈ 32.234 — minimum uniform E budget



berkeley_b

EXACT PHASE E=2 / L=11 / score ≈ 30.866 — minimum uniform E budget



Exact definitions: $\eta_{215}, \eta_{246}, \eta_{258}, \eta_{259}, \eta_{260}, \eta_{261}$ in the expression glossary.

PROJECTIVE E=2 / L=10 / score ≈ 28.460 — minimum uniform E budget

Exact definitions: $\eta_{245}, \eta_{246}, \eta_{217}, \eta_{261}$ in the expression glossary.

$U = \eta_{262}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=2 / score ≈ 12.404 — minimum uniform E budget

cx_rz_ry

EXACT PHASE E=2 / L=2 / score ≈ 12.404 — minimum uniform E budget

cx_u

EXACT PHASE E=2 / L=2 / score ≈ 12.412 — minimum uniform E budget

cz_rz_rx

EXACT PHASE E=2 / L=10 / score ≈ 31.620 — minimum uniform E budget

cz_u

EXACT PHASE E=2 / L=6 / score ≈ 21.236 — minimum uniform E budget

google_cz

EXACT PHASE E=2 / L=6 / score ≈ 22.697 — minimum uniform E budget

google_fsim

EXACT PHASE E=1 / L=6 / score ≈ 20.822 — minimum uniform E budget

google_sqrt_iswap

EXACT PHASE E=2 / L=9 / score ≈ 27.715 — minimum uniform E budget

Exact definitions: $\eta_{162}, \eta_{166}, \eta_{228}, \eta_{229}, \eta_{230}, \eta_{232}, \eta_{234}, \eta_{235}, \eta_{263}, \eta_{264}, \eta_{265}$ in the expression glossary.

PROJECTIVE E=2 / L=8 / score ≈ 25.709 — minimum uniform E budget

Exact definitions: $\eta_{162}, \eta_{166}, \eta_{228}, \eta_{229}, \eta_{230}, \eta_{232}, \eta_{234}, \eta_{235}, \eta_{236}$ in the expression glossary.

$U = \eta_{238}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=10 / score ≈ 70.721 — E upper bound

ibm_cx

EXACT PHASE E=2 / L=2 / score ≈ 12.404 — minimum uniform E budget

ibm_cz

EXACT PHASE E=2 / L=11 / score ≈ 31.414 — minimum uniform E budget

PROJECTIVE E=2 / L=11 / score ≈ 31.212 — minimum uniform E budget

$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=10 / score ≈ 28.808 — minimum uniform E budget

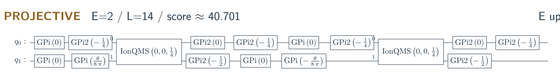
PROJECTIVE E=2 / L=8 / score ≈ 26.612 — minimum uniform E budget

$U = i\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=14 / score ≈ 41.101 — E upper bound

PROJECTIVE E=2 / L=14 / score ≈ 40.701 — E upper bound



$U = i\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=14 / score ≈ 40.297 — minimum uniform E budget

PROJECTIVE E=2 / L=14 / score ≈ 39.897 — minimum uniform E budget

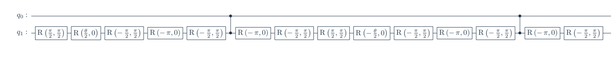
$U = i\, C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=1 / L=2 / score ≈ 9.667 — minimum uniform E budget

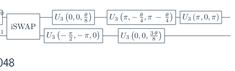
iqm_cz

EXACT PHASE E=2 / L=14 / score ≈ 39.656 — minimum uniform E budget



iswap_u

EXACT PHASE E=2 / L=10 / score ≈ 31.860 — minimum uniform E budget



PROJECTIVE E=2 / L=8 / score ≈ 27.048 — minimum uniform E budget

$U = e^{-\frac{i\theta}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=5 / L=27 / score ≈ 82.210 — E upper bound

PROJECTIVE E=2 / L=8 / score ≈ 27.952 — minimum uniform E budget

$U = -i\, C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=1 / L=4 / score ≈ 14.940 — minimum uniform E budget

rigetti_cz

EXACT PHASE E=2 / L=11 / score ≈ 31.414 — minimum uniform E budget

PROJECTIVE E=2 / L=11 / score ≈ 31.212 — minimum uniform E budget

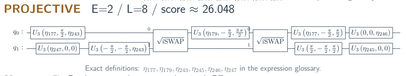
$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=2 / L=9 / score ≈ 28.054 — minimum uniform E budget

Exact definitions: $\eta_{177}, \eta_{178}, \eta_{240}, \eta_{241}, \eta_{243}, \eta_{244}, \eta_{246}$ in the expression glossary.

PROJECTIVE E=2 / L=8 / score ≈ 26.048 — minimum uniform E budget



Exact definitions: $\eta_{177}, \eta_{178}, \eta_{243}, \eta_{244}, \eta_{247}$ in the expression glossary.

$U = \eta_{248}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=10 / score ≈ 39.260 — E upper bound

PROJECTIVE E=4 / L=10 / score ≈ 38.860 — E upper bound

$U = -i\, C$. Scalar metadata; no physical GP.

tket_tk2

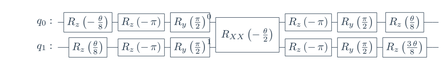
EXACT PHASE E=1 / L=4 / score ≈ 14.944 — minimum uniform E budget

xx90_u

EXACT PHASE E=2 / L=8 / score ≈ 26.848 — minimum uniform E budget

xx_rz_ry

EXACT PHASE E=1 / L=12 / score ≈ 31.426 — minimum uniform E budget



zx_u

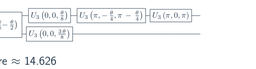
EXACT PHASE E=1 / L=8 / score ≈ 23.850 — minimum uniform E budget

PROJECTIVE E=1 / L=6 / score ≈ 19.038 — minimum uniform E budget

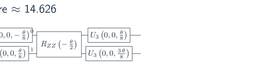
$U = e^{-\frac{i\theta}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=1 / L=6 / score ≈ 19.438 — minimum uniform E budget



PROJECTIVE E=1 / L=4 / score ≈ 14.626 — minimum uniform E budget



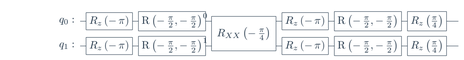
$U = e^{-\frac{i\theta}{4}}\, C$. Scalar metadata; no physical GP.

CS

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=1 / L=10 / score ≈ 26.830 proved E minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=6 / score ≈ 21.436 proved E minimum

Exact definitions: $\eta_{260}, \eta_{267}, \eta_2, \eta_{26}$ in the expression glossary.

PROJECTIVE E=2 / L=4 / score ≈ 17.224 proved E minimum

Exact definitions: η_{90} in the expression glossary.

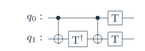
$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

EXACT PHASE E=2 / L=3 / score ≈ 14.000 proved E minimum

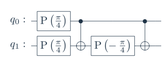


clifford_t_cz

EXACT PHASE E=2 / L=7 / score ≈ 22.000 proved E minimum

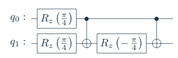
cx_h_phase

EXACT PHASE E=2 / L=3 / score ≈ 15.206 proved E minimum



cx_rz_ry

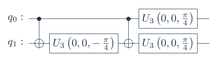
PROJECTIVE E=2 / L=3 / score ≈ 15.206 proved E minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

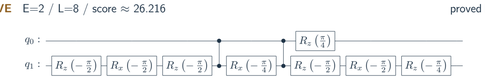
cx_u

EXACT PHASE E=2 / L=3 / score ≈ 15.218 proved E minimum



cz_rz_rx

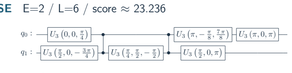
PROJECTIVE E=2 / L=8 / score ≈ 26.216 proved E minimum



$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=6 / score ≈ 23.236 proved E minimum



PROJECTIVE E=2 / L=4 / score ≈ 18.024 proved E minimum

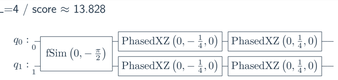
$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=2 / L=5 / score ≈ 20.030 proved E minimum

google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 13.828 proved E minimum

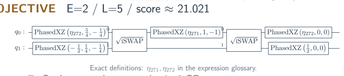


google_sqrt_iswap

EXACT PHASE E=2 / L=6 / score ≈ 24.687 proved E minimum

Exact definitions: $\eta_{268}, \eta_{269}, \eta_{270}, \eta_{271}, \eta_{272}$ in the expression glossary.

PROJECTIVE E=2 / L=5 / score ≈ 21.021 proved E minimum



Exact definitions: η_{271}, η_{272} in the expression glossary.

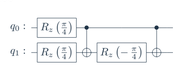
$U = \eta_{273}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=11 / score ≈ 72.866 E upper bound

ibm_cx

PROJECTIVE E=2 / L=3 / score ≈ 15.206 proved E minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=2 / L=8 / score ≈ 25.408 proved E minimum

$U = e^{-\frac{7i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=2 / L=6 / score ≈ 21.208 proved E minimum

$U = (-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=2 / L=16 / score ≈ 45.244 E upper bound

$U = (-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=2 / L=16 / score ≈ 44.440 proved E minimum

$U = (-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

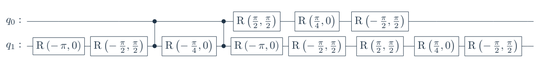
ionq_forte

PROJECTIVE E=1 / L=4 / score ≈ 14.210 proved E minimum

$U = \eta_{274}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

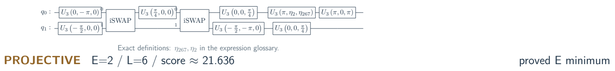
PROJECTIVE E=2 / L=11 / score ≈ 33.644 proved E minimum



$U = -\sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=2 / L=8 / score ≈ 25.648 proved E minimum



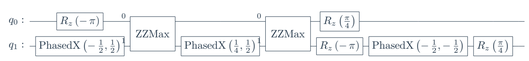
Exact definitions: η_{267}, η_2 in the expression glossary.

PROJECTIVE E=2 / L=6 / score ≈ 21.636 proved E minimum

$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=2 / L=7 / score ≈ 24.220 proved E minimum



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

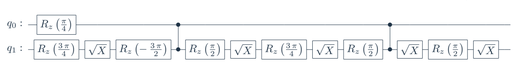
quantinuum_zzphase

PROJECTIVE E=1 / L=2 / score ≈ 9.206 proved E minimum

$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=2 / L=12 / score ≈ 34.014 proved E minimum



$U = -(-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 23.836 proved E minimum

Exact definitions: $\eta_{275}, \eta_{276}, \eta_{277}$ in the expression glossary.

PROJECTIVE E=2 / L=5 / score ≈ 20.830 proved E minimum

Exact definitions: η_{276}, η_{277} in the expression glossary.

$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=11 / score ≈ 42.066 E upper bound

PROJECTIVE E=4 / L=11 / score ≈ 41.666 E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

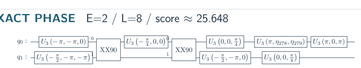
tket_tk2

PROJECTIVE E=1 / L=2 / score ≈ 9.210 proved E minimum

$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

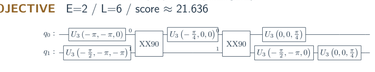
xx90_u

EXACT PHASE E=2 / L=8 / score ≈ 25.648 proved E minimum



Exact definitions: η_{274}, η_{275} in the expression glossary.

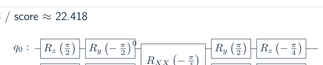
PROJECTIVE E=2 / L=6 / score ≈ 21.636 proved E minimum



$U = (-1)^{\frac{3}{8}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=1 / L=8 / score ≈ 22.418 proved E minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

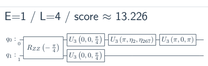
EXACT PHASE E=1 / L=5 / score ≈ 16.832 proved E minimum

PROJECTIVE E=1 / L=3 / score ≈ 11.620 proved E minimum

$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

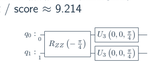
zz_u

EXACT PHASE E=1 / L=4 / score ≈ 13.226 proved E minimum



Exact definitions: η_{267}, η_2 in the expression glossary.

PROJECTIVE E=1 / L=2 / score ≈ 9.214 proved E minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

CSdg

2 qubits / 32 preserved native bases / alphabetical panel order

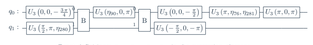
aqt_rxx

PROJECTIVE E=1 / L=10 / score ≈ 26.830 proved E minimum

$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=7 / score ≈ 23.042 proved E minimum



Exact definitions: $\eta_{260}, \eta_{283}, \eta_{75}, \eta_{90}$ in the expression glossary.

PROJECTIVE E=2 / L=5 / score ≈ 19.430 proved E minimum

Exact definitions: η_{90} in the expression glossary.

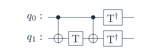
$U = \sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

EXACT PHASE E=2 / L=3 / score ≈ 14.000 proved E minimum

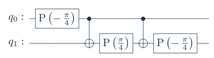


clifford_t_cz

EXACT PHASE E=2 / L=7 / score ≈ 22.000 proved E minimum

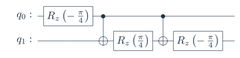
cx_h_phase

EXACT PHASE E=2 / L=3 / score ≈ 15.206 proved E minimum



cx_rz_ry

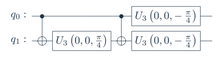
PROJECTIVE E=2 / L=3 / score ≈ 15.206 proved E minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=2 / L=3 / score ≈ 15.218 proved E minimum



cz_rz_rx

PROJECTIVE E=2 / L=8 / score ≈ 26.216 proved E minimum

$U = (-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=6 / score ≈ 23.236 proved E minimum

PROJECTIVE E=2 / L=4 / score ≈ 18.024 proved E minimum

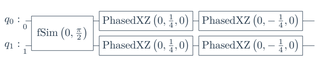
$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=2 / L=5 / score ≈ 20.030 proved E minimum

google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 13.828 proved E minimum



google_sqrt_iswap

EXACT PHASE E=2 / L=6 / score ≈ 24.687 proved E minimum

Exact definitions: $\eta_{282}, \eta_{283}, \eta_{284}, \eta_{285}, \eta_{286}$ in the expression glossary.

PROJECTIVE E=2 / L=5 / score ≈ 21.021 proved E minimum

Exact definitions: η_{285}, η_{286} in the expression glossary.

$U = \eta_{287}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=11 / score ≈ 72.866 E upper bound

ibm_cx

PROJECTIVE E=2 / L=3 / score ≈ 15.206 proved E minimum

$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=2 / L=10 / score ≈ 29.412 proved E minimum

$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=2 / L=6 / score ≈ 21.208 proved E minimum

$U = (-1)^{\frac{3}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=1 / L=12 / score ≈ 31.830 proved E minimum

$U = \eta_{288}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=2 / L=16 / score ≈ 44.440 proved E minimum

$U = (-1)^{\frac{3}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

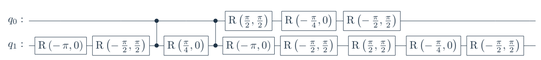
ionq_forte

PROJECTIVE E=1 / L=4 / score ≈ 14.210 proved E minimum

$U = \eta_{288}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=2 / L=11 / score ≈ 33.644 proved E minimum



$U = (-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=2 / L=8 / score ≈ 25.648 proved E minimum

Exact definitions: η_{260}, η_{266} in the expression glossary.

PROJECTIVE E=2 / L=6 / score ≈ 21.636 proved E minimum

$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=2 / L=7 / score ≈ 24.220 proved E minimum

$U = -(-1)^{\frac{3}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

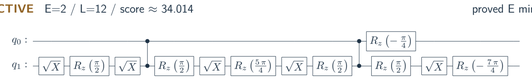
quantinuum_zzphase

PROJECTIVE E=1 / L=2 / score ≈ 9.206 proved E minimum

$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=2 / L=12 / score ≈ 34.014 proved E minimum



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 23.836 proved E minimum

Exact definitions: $\eta_{291}, \eta_{292}, \eta_{293}$ in the expression glossary.

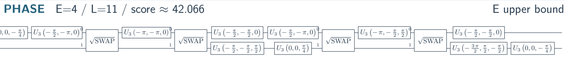
PROJECTIVE E=2 / L=5 / score ≈ 20.830 proved E minimum

Exact definitions: η_{292}, η_{293} in the expression glossary.

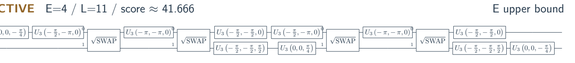
$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=11 / score ≈ 42.066 E upper bound



PROJECTIVE E=4 / L=11 / score ≈ 41.666 E upper bound



$U = -iC$. Scalar metadata; no physical GP.

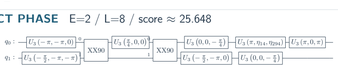
tket_tk2

PROJECTIVE E=1 / L=2 / score ≈ 9.210 proved E minimum

$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=2 / L=8 / score ≈ 25.648 proved E minimum



Exact definitions: η_{14}, η_{264} in the expression glossary.

PROJECTIVE E=2 / L=6 / score ≈ 21.636 proved E minimum

$U = (-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.

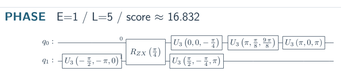
xx_rz_ry

PROJECTIVE E=1 / L=8 / score ≈ 22.418 proved E minimum

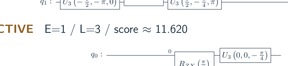
$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=1 / L=5 / score ≈ 16.832 proved E minimum



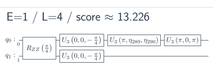
PROJECTIVE E=1 / L=3 / score ≈ 11.620 proved E minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=1 / L=4 / score ≈ 13.226 proved E minimum



Exact definitions: η_{289}, η_{290} in the expression glossary.

PROJECTIVE E=1 / L=2 / score ≈ 9.214 proved E minimum

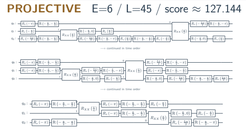
$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

CSWAP

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

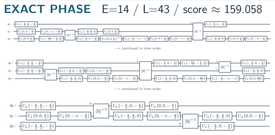
PROJECTIVE E=6 / L=45 / score ≈ 127.144 — E upper bound



$U = \frac{\sqrt[3]{-1}\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=14 / L=43 / score ≈ 159.058 — E upper bound



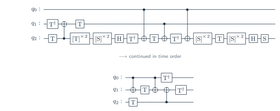
$U = \frac{\sqrt[3]{-1}\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

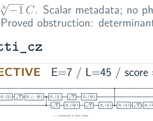
clifford_t_cx

EXACT PHASE E=7 / L=22 / score ≈ 72.000 — E upper bound



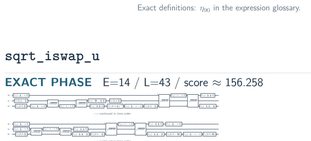
clifford_t_cz

EXACT PHASE E=7 / L=34 / score ≈ 96.000 — E upper bound



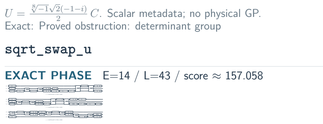
cx_h_phase

EXACT PHASE E=7 / L=14 / score ≈ 60.624 — E upper bound



cx_rz_ry

PROJECTIVE E=7 / L=14 / score ≈ 61.028 — E upper bound



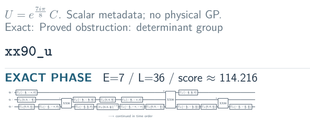
$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=7 / L=10 / score ≈ 53.060 — E upper bound

cz_rz_rx

PROJECTIVE E=7 / L=17 / score ≈ 66.434 — E upper bound



$U = e^{\frac{7i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

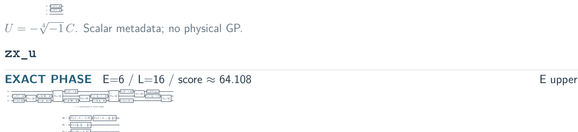
EXACT PHASE E=7 / L=12 / score ≈ 59.472 — E upper bound

PROJECTIVE E=7 / L=10 / score ≈ 54.260 — E upper bound

$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

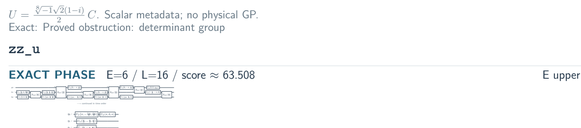
google_cz

EXACT PHASE E=7 / L=29 / score ≈ 97.774 — E upper bound



google_fsim

EXACT PHASE E=6 / L=26 / score ≈ 86.780 — E upper bound



google_sqrt_iswap

EXACT PHASE E=14 / L=43 / score ≈ 157.258 — E upper bound

PROJECTIVE E=14 / L=43 / score ≈ 156.658 — E upper bound

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=42 / L=43 / score ≈ 265.858 — E upper bound

$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cx

PROJECTIVE E=7 / L=13 / score ≈ 58.422 — E upper bound

$U = e^{-\frac{3i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=7 / L=28 / score ≈ 87.228 — E upper bound

$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=7 / L=36 / score ≈ 104.242 — E upper bound

$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=7 / L=63 / score ≈ 175.168 — E upper bound

$U = \frac{\sqrt[3]{-1}\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=7 / L=63 / score ≈ 172.354 — E upper bound

$U = \frac{\sqrt[3]{-1}\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=7 / L=70 / score ≈ 191.954 — E upper bound

$U = \frac{\sqrt[3]{-1}\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=7 / L=69 / score ≈ 188.276 — E upper bound

$U = -\sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=12 / L=20 / score ≈ 97.720 — E upper bound

Exact definitions: η_{90} in the expression glossary.

quantinuum_zzmax

PROJECTIVE E=7 / L=45 / score ≈ 130.522 — E upper bound

$U = \frac{\sqrt[3]{-1}\sqrt{2}(-1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=6 / L=35 / score ≈ 105.506 — E upper bound

$U = \frac{\sqrt[3]{-1}\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=7 / L=45 / score ≈ 125.656 — E upper bound

$U = e^{\frac{7i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=14 / L=43 / score ≈ 156.258 — E upper bound

sqrt_swap_u

EXACT PHASE E=14 / L=43 / score ≈ 157.058 — E upper bound

PROJECTIVE E=14 / L=43 / score ≈ 156.258 — E upper bound

$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=6 / L=37 / score ≈ 109.544 — E upper bound

$U = \frac{\sqrt[3]{-1}\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=7 / L=36 / score ≈ 114.216 — E upper bound

xx_rz_ry

PROJECTIVE E=6 / L=35 / score ≈ 103.482 — E upper bound

$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=6 / L=16 / score ≈ 64.108 — E upper bound

PROJECTIVE E=6 / L=14 / score ≈ 60.096 — E upper bound

$U = \eta_{296}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=6 / L=16 / score ≈ 63.508 — E upper bound

PROJECTIVE E=6 / L=14 / score ≈ 58.696 — E upper bound

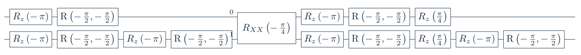
$U = \frac{\sqrt[3]{-1}\sqrt{2}(1+i)^5}{8}\,C$. Scalar metadata; no physical GP.

CSX

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

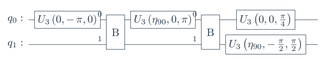
PROJECTIVE E=1 / L=14 / score ≈ 35.642 — proved E minimum



$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=4 / score ≈ 16.824 — proved E minimum



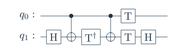
Exact definitions: η_{90} in the expression glossary.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

EXACT PHASE E=2 / L=5 / score ≈ 18.000 — proved E minimum

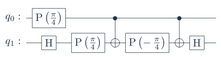


clifford_t_cz

EXACT PHASE E=2 / L=9 / score ≈ 26.000 — proved E minimum

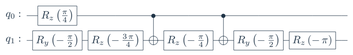
cx_h_phase

EXACT PHASE E=2 / L=5 / score ≈ 19.206 — proved E minimum



cx_rz_ry

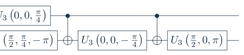
PROJECTIVE E=2 / L=6 / score ≈ 21.612 — proved E minimum



$U = e^{-\frac{7i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

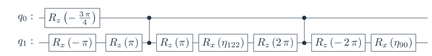
cx_u

EXACT PHASE E=2 / L=4 / score ≈ 17.624 — proved E minimum



cz_rz_rx

PROJECTIVE E=2 / L=8 / score ≈ 24.416 — proved E minimum

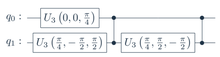


Exact definitions: η_{122}, η_{90} in the expression glossary.

$U = (-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=3 / score ≈ 16.018 — proved E minimum



google_cz

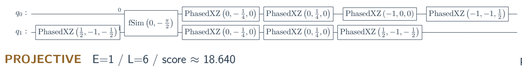
EXACT PHASE E=2 / L=9 / score ≈ 29.054 — proved E minimum

PROJECTIVE E=2 / L=7 / score ≈ 24.842 — proved E minimum

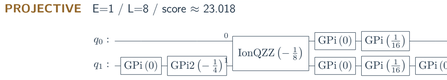
$U = -i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=8 / score ≈ 22.852 — proved E minimum



PROJECTIVE E=1 / L=6 / score ≈ 18.640 — proved E minimum



$U = -i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=2 / L=8 / score ≈ 29.499 — proved E minimum

Exact definitions: $\eta_{271}, \eta_{272}, \eta_{298}, \eta_{299}$ in the expression glossary.

PROJECTIVE E=2 / L=7 / score ≈ 25.833 — proved E minimum

Exact definitions: η_{271}, η_{272} in the expression glossary.

$U = \eta_{273}\,C$. Scalar metadata; no physical GP.

google_syc

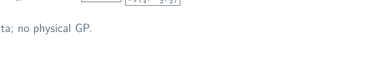
EXACT PHASE E=12 / L=15 / score ≈ 81.890 — E upper bound

PROJECTIVE E=12 / L=13 / score ≈ 77.678 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=2 / L=8 / score ≈ 25.812 — proved E minimum



$U = e^{\frac{5i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=2 / L=13 / score ≈ 35.812 — proved E minimum

$U = e^{-\frac{7i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=2 / L=12 / score ≈ 33.612 — proved E minimum

$U = (-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=2 / L=20 / score ≈ 54.052 — E upper bound

$U = (-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=2 / L=20 / score ≈ 53.248 — proved E minimum

$U = (-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=1 / L=8 / score ≈ 23.018 — proved E minimum

$U = \eta_{274}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=2 / L=15 / score ≈ 42.460 — proved E minimum

$U = \sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=2 / L=5 / score ≈ 19.230 — proved E minimum

Exact definitions: η_{90} in the expression glossary.

quantinuum_zzmax

PROJECTIVE E=2 / L=11 / score ≈ 33.032 — proved E minimum

$U = (-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=1 / L=6 / score ≈ 18.018 — proved E minimum

$U = -\sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=2 / L=18 / score ≈ 46.418 — proved E minimum

$U = -(-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=2 / L=8 / score ≈ 28.448 — proved E minimum

Exact definitions: η_{276}, η_{277} in the expression glossary.

PROJECTIVE E=2 / L=7 / score ≈ 25.242 — proved E minimum

Exact definitions: η_{276}, η_{277} in the expression glossary.

$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=15 / score ≈ 50.490 — E upper bound

PROJECTIVE E=4 / L=13 / score ≈ 46.078 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=6 / score ≈ 18.022 — proved E minimum

$U = -\sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=2 / L=10 / score ≈ 31.260 — proved E minimum

PROJECTIVE E=2 / L=8 / score ≈ 26.048 — proved E minimum

$U = (-1)^{\frac{3}{8}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=1 / L=6 / score ≈ 18.014 — proved E minimum

$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=1 / L=5 / score ≈ 16.432 — proved E minimum

PROJECTIVE E=1 / L=3 / score ≈ 11.620 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=1 / L=4 / score ≈ 14.026 — proved E minimum

PROJECTIVE E=1 / L=4 / score ≈ 13.626 — proved E minimum

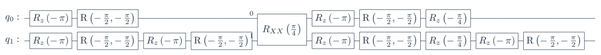
$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

CSXdg

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

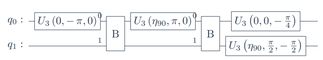
PROJECTIVE E=1 / L=14 / score ≈ 35.642 — proved E minimum



$U = (-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=4 / score ≈ 16.824 — proved E minimum



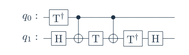
Exact definitions: η_{90} in the expression glossary.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

EXACT PHASE E=2 / L=5 / score ≈ 18.000 — proved E minimum

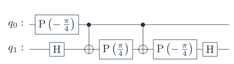


clifford_t_cz

EXACT PHASE E=2 / L=9 / score ≈ 26.000 — proved E minimum

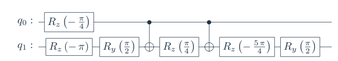
cx_h_phase

EXACT PHASE E=2 / L=5 / score ≈ 19.206 — proved E minimum



cx_rz_ry

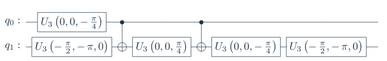
PROJECTIVE E=2 / L=6 / score ≈ 21.612 — proved E minimum



$U = (-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=2 / L=5 / score ≈ 19.630 — proved E minimum



cz_rz_rx

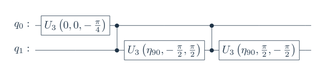
PROJECTIVE E=2 / L=6 / score ≈ 20.412 — proved E minimum

Exact definitions: η_{90} in the expression glossary.

$U = -(-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=3 / score ≈ 15.218 — proved E minimum

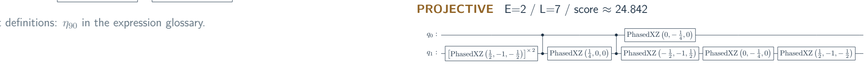


Exact definitions: η_{90} in the expression glossary.

google_cz

EXACT PHASE E=2 / L=9 / score ≈ 29.054 — proved E minimum

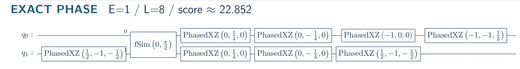
PROJECTIVE E=2 / L=7 / score ≈ 24.842 — proved E minimum



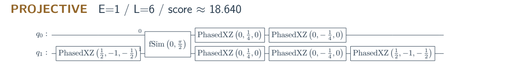
$U = -i\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=8 / score ≈ 22.852 — proved E minimum



PROJECTIVE E=1 / L=6 / score ≈ 18.640 — proved E minimum



$U = -i\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=2 / L=11 / score ≈ 36.117 — proved E minimum

Exact definitions: $\eta_{271}, \eta_{272}, \eta_{300}, \eta_{301}$ in the expression glossary.

PROJECTIVE E=2 / L=5 / score ≈ 20.951 — proved E minimum

Exact definitions: $\eta_{265}, \eta_{266}, \eta_{302}$ in the expression glossary.

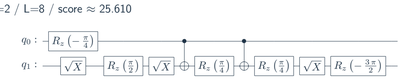
$U = \eta_{303}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=13 / score ≈ 77.678 — E upper bound

ibm_cx

PROJECTIVE E=2 / L=8 / score ≈ 25.610 — proved E minimum



$U = e^{-\frac{7i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=2 / L=11 / score ≈ 31.214 — proved E minimum

Exact definitions: η_{304} in the expression glossary.

$U = e^{-\frac{i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=2 / L=12 / score ≈ 33.612 — proved E minimum

$U = (-1)^{\frac{3}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

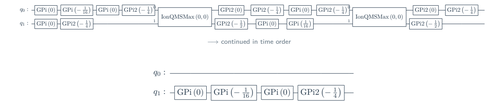
ionq_aria

PROJECTIVE E=1 / L=16 / score ≈ 40.638 — proved E minimum

$U = \eta_{288}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=2 / L=20 / score ≈ 53.248 — proved E minimum



$U = (-1)^{\frac{3}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

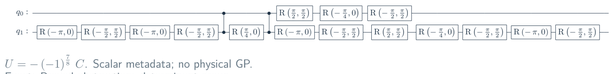
ionq_forte

PROJECTIVE E=1 / L=8 / score ≈ 23.018 — proved E minimum

$U = \eta_{288}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=2 / L=15 / score ≈ 42.460 — proved E minimum



$U = -(-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

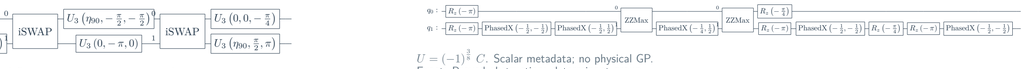
iswap_u

EXACT PHASE E=2 / L=5 / score ≈ 19.230 — proved E minimum

Exact definitions: η_{90} in the expression glossary.

quantinuum_zzmax

PROJECTIVE E=2 / L=11 / score ≈ 33.032 — proved E minimum



$U = (-1)^{\frac{3}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

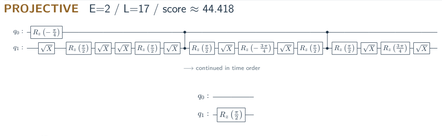
quantinuum_zzphase

PROJECTIVE E=1 / L=6 / score ≈ 18.018 — proved E minimum

$U = (-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=2 / L=17 / score ≈ 44.418 — proved E minimum



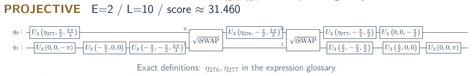
$U = e^{-\frac{i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=2 / L=11 / score ≈ 34.666 — proved E minimum

Exact definitions: η_{276}, η_{277} in the expression glossary.

PROJECTIVE E=2 / L=10 / score ≈ 31.460 — proved E minimum

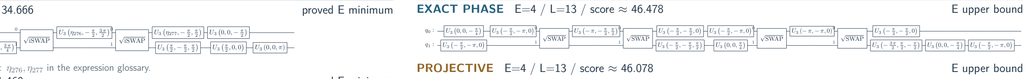


Exact definitions: η_{276}, η_{277} in the expression glossary.

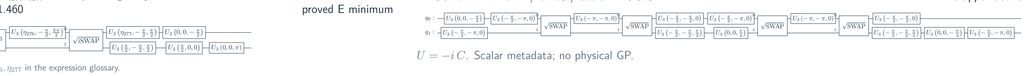
$U = e^{-\frac{i\pi}{8}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=13 / score ≈ 46.478 — E upper bound



PROJECTIVE E=4 / L=13 / score ≈ 46.078 — E upper bound



$U = -i\, C$. Scalar metadata; no physical GP.

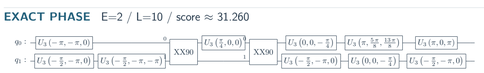
tket_tk2

PROJECTIVE E=1 / L=6 / score ≈ 18.022 — proved E minimum

$U = (-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=2 / L=10 / score ≈ 31.260 — proved E minimum

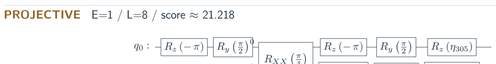


PROJECTIVE E=2 / L=8 / score ≈ 26.048 — proved E minimum

$U = (-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=1 / L=8 / score ≈ 21.218 — proved E minimum



Exact definitions: η_{305}, η_{90} in the expression glossary.

$U = \eta_{306}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

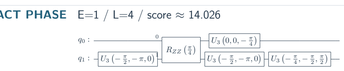
zx_u

EXACT PHASE E=1 / L=2 / score ≈ 9.214 — proved E minimum

Exact definitions: η_{90} in the expression glossary.

zz_u

EXACT PHASE E=1 / L=4 / score ≈ 14.026 — proved E minimum



PROJECTIVE E=1 / L=3 / score ≈ 11.220 — proved E minimum

Exact definitions: η_{90} in the expression glossary.

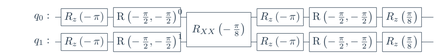
$U = \eta_{308}\, C$. Scalar metadata; no physical GP.

CT

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=1 / L=10 / score ≈ 27.430 — proved E minimum



$U = e^{\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=8 / score ≈ 25.448 — proved E minimum

Exact definitions: $\eta_{09}, \eta_{10}, \eta_{11}, \eta_{12}, \eta_{11}$ in the expression glossary.

PROJECTIVE E=2 / L=6 / score ≈ 21.636 — proved E minimum

Exact definitions: η_{04} in the expression glossary.

$U = e^{-\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

cx_h_phase

EXACT PHASE E=2 / L=3 / score ≈ 15.806 — proved E minimum

cx_rz_ry

PROJECTIVE E=2 / L=3 / score ≈ 15.806 — proved E minimum

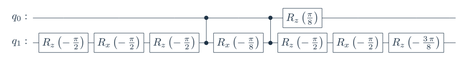
$U = e^{\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=2 / L=3 / score ≈ 15.818 — proved E minimum

cz_rz_rx

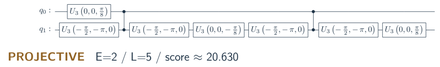
PROJECTIVE E=2 / L=8 / score ≈ 26.816 — proved E minimum



$U = -\sqrt[16]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=7 / score ≈ 24.642 — proved E minimum

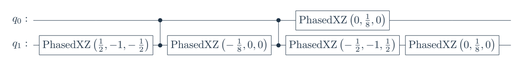


PROJECTIVE E=2 / L=5 / score ≈ 20.630 — proved E minimum

$U = e^{-\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.

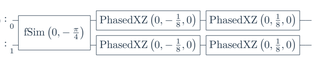
google_cz

EXACT PHASE E=2 / L=5 / score ≈ 20.630 — proved E minimum



google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 14.828 — proved E minimum

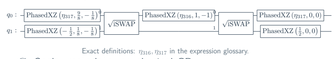


google_sqrt_iswap

EXACT PHASE E=2 / L=6 / score ≈ 25.887 — proved E minimum

Exact definitions: $\eta_{013}, \eta_{014}, \eta_{015}, \eta_{016}, \eta_{017}$ in the expression glossary.

PROJECTIVE E=2 / L=5 / score ≈ 21.821 — proved E minimum



Exact definitions: η_{016}, η_{017} in the expression glossary.

$U = \eta_{318}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=11 / score ≈ 73.466 — E upper bound

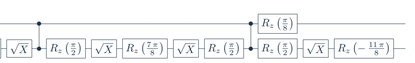
ibm_cx

PROJECTIVE E=2 / L=3 / score ≈ 15.806 — proved E minimum

$U = e^{\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=2 / L=12 / score ≈ 34.614 — proved E minimum



$U = -(-1)^{\frac{13}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=2 / L=6 / score ≈ 21.808 — proved E minimum

$U = (-1)^{\frac{9}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=2 / L=16 / score ≈ 45.844 — E upper bound

$U = (-1)^{\frac{9}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=2 / L=16 / score ≈ 45.040 — proved E minimum

$U = (-1)^{\frac{9}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=1 / L=4 / score ≈ 14.810 — proved E minimum

$U = e^{\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=2 / L=11 / score ≈ 34.244 — proved E minimum

$U = -\sqrt[16]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

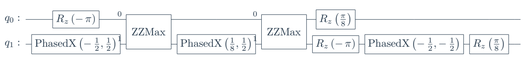
EXACT PHASE E=2 / L=8 / score ≈ 27.848 — proved E minimum

PROJECTIVE E=2 / L=6 / score ≈ 22.236 — proved E minimum

$U = e^{-\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=2 / L=7 / score ≈ 24.820 — proved E minimum



$U = -(-1)^{\frac{9}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=1 / L=2 / score ≈ 9.806 — proved E minimum

$U = e^{\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=2 / L=12 / score ≈ 34.614 — proved E minimum

$U = -(-1)^{\frac{13}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 24.636 — proved E minimum

Exact definitions: $\eta_{019}, \eta_{020}, \eta_{021}$ in the expression glossary.

PROJECTIVE E=2 / L=5 / score ≈ 21.230 — proved E minimum

Exact definitions: η_{020}, η_{021} in the expression glossary.

$U = e^{-\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=11 / score ≈ 42.666 — E upper bound

PROJECTIVE E=4 / L=11 / score ≈ 42.266 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=2 / score ≈ 9.810 — proved E minimum

$U = e^{\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=2 / L=8 / score ≈ 28.248 — proved E minimum

PROJECTIVE E=2 / L=6 / score ≈ 22.636 — proved E minimum

$U = -(-1)^{\frac{15}{16}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=1 / L=8 / score ≈ 23.018 — proved E minimum

$U = e^{\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

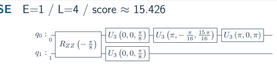
EXACT PHASE E=1 / L=6 / score ≈ 19.838 — proved E minimum

PROJECTIVE E=1 / L=4 / score ≈ 14.226 — proved E minimum

$U = e^{-\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=1 / L=4 / score ≈ 15.426 — proved E minimum



PROJECTIVE E=1 / L=2 / score ≈ 9.814 — proved E minimum

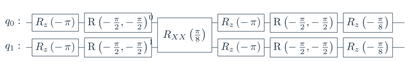
$U = e^{-\frac{i\pi}{16}}\,C$. Scalar metadata; no physical GP.

CTdg

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

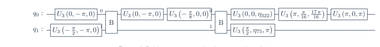
PROJECTIVE E=1 / L=10 / score ≈ 27.430 proved E minimum



$U = e^{-\frac{i\pi}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

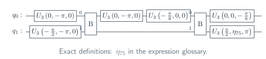
berkeley_b

EXACT PHASE E=2 / L=8 / score ≈ 27.448 proved E minimum



Exact definitions: η_{322}, η_{75} in the expression glossary.

PROJECTIVE E=2 / L=6 / score ≈ 21.636 proved E minimum



Exact definitions: η_{75} in the expression glossary.

$U = e^{\frac{i\pi}{16}}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

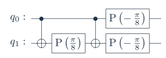
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

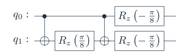
cx_h_phase

EXACT PHASE E=2 / L=3 / score ≈ 15.806 proved E minimum



cx_rz_ry

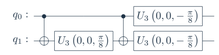
PROJECTIVE E=2 / L=3 / score ≈ 15.806 proved E minimum



$U = e^{-\frac{i\pi}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

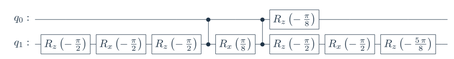
cx_u

EXACT PHASE E=2 / L=3 / score ≈ 15.818 proved E minimum



cz_rz_rx

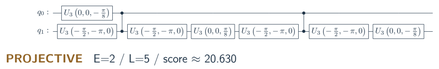
PROJECTIVE E=2 / L=8 / score ≈ 26.816 proved E minimum



$U = (-1)^{\frac{15}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=7 / score ≈ 24.642 proved E minimum



PROJECTIVE E=2 / L=5 / score ≈ 20.630 proved E minimum

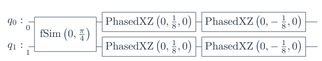
$U = e^{\frac{i\pi}{16}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=2 / L=5 / score ≈ 20.630 proved E minimum

google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 14.828 proved E minimum



google_sqrt_iswap

EXACT PHASE E=2 / L=6 / score ≈ 25.887 proved E minimum

Exact definitions: $\eta_{323}, \eta_{324}, \eta_{325}, \eta_{236}, \eta_{327}$ in the expression glossary.

PROJECTIVE E=2 / L=5 / score ≈ 21.821 proved E minimum

Exact definitions: η_{236}, η_{327} in the expression glossary.

$U = \eta_{328}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=11 / score ≈ 73.466 E upper bound

ibm_cx

PROJECTIVE E=2 / L=3 / score ≈ 15.806 proved E minimum

$U = e^{-\frac{i\pi}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=2 / L=12 / score ≈ 34.614 proved E minimum

$U = -(-1)^{\frac{11}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

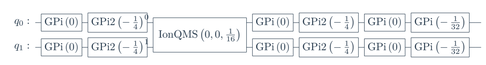
ibm_ecr

PROJECTIVE E=2 / L=10 / score ≈ 30.212 proved E minimum

$U = e^{-\frac{9i\pi}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=1 / L=12 / score ≈ 32.430 proved E minimum



$U = e^{-\frac{i\pi}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=2 / L=16 / score ≈ 45.440 proved E minimum

$U = (-1)^{\frac{15}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=1 / L=4 / score ≈ 14.810 proved E minimum

$U = e^{-\frac{i\pi}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

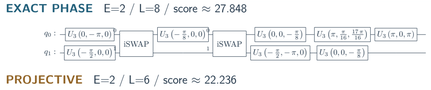
iqm_cz

PROJECTIVE E=2 / L=11 / score ≈ 34.244 proved E minimum

$U = (-1)^{\frac{15}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=2 / L=8 / score ≈ 27.848 proved E minimum



PROJECTIVE E=2 / L=6 / score ≈ 22.236 proved E minimum

$U = e^{\frac{i\pi}{16}}\, C$. Scalar metadata; no physical GP.

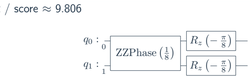
quantinuum_zzmax

PROJECTIVE E=2 / L=7 / score ≈ 24.820 proved E minimum

$U = -(-1)^{\frac{7}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=1 / L=2 / score ≈ 9.806 proved E minimum



$U = e^{-\frac{i\pi}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=2 / L=12 / score ≈ 34.614 proved E minimum

$U = -(-1)^{\frac{11}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 24.636 proved E minimum

Exact definitions: $\eta_{320}, \eta_{230}, \eta_{321}$ in the expression glossary.

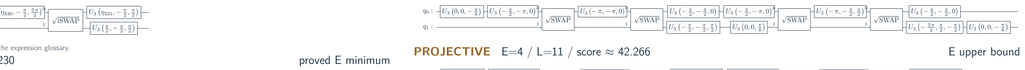
PROJECTIVE E=2 / L=5 / score ≈ 21.230 proved E minimum

Exact definitions: η_{230}, η_{321} in the expression glossary.

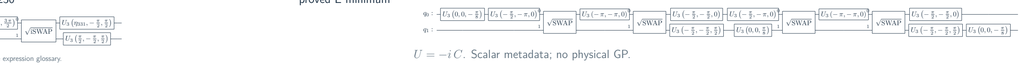
$U = e^{\frac{i\pi}{16}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=11 / score ≈ 42.666 E upper bound



PROJECTIVE E=4 / L=11 / score ≈ 42.266 E upper bound



$U = -iC$. Scalar metadata; no physical GP.

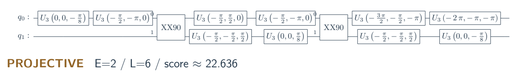
tket_tk2

PROJECTIVE E=1 / L=2 / score ≈ 9.810 proved E minimum

$U = e^{-\frac{i\pi}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=2 / L=10 / score ≈ 32.260 proved E minimum



PROJECTIVE E=2 / L=6 / score ≈ 22.636 proved E minimum

$U = \sqrt[16]{-1}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=1 / L=8 / score ≈ 23.018 proved E minimum

$U = e^{-\frac{i\pi}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

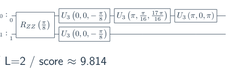
EXACT PHASE E=1 / L=6 / score ≈ 19.838 proved E minimum

PROJECTIVE E=1 / L=4 / score ≈ 14.226 proved E minimum

$U = e^{\frac{i\pi}{16}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=1 / L=4 / score ≈ 15.426 proved E minimum



PROJECTIVE E=1 / L=2 / score ≈ 9.814 proved E minimum

$U = e^{\frac{i\pi}{16}}\, C$. Scalar metadata; no physical GP.

CU $(\theta, \phi, lbda, \gamma)$

2 qubits / 32 preserved native bases / alphabetical panel order

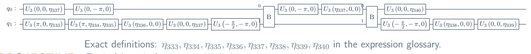
aqt_rxx

PROJECTIVE E=2 / L=16 / score ≈ 43.852 — E upper bound

$U = \eta_{332}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=13 / score ≈ 35.078 — minimum uniform E budget



Exact definitions: $\eta_{333}, \eta_{334}, \eta_{335}, \eta_{336}, \eta_{337}, \eta_{338}, \eta_{339}, \eta_{340}$ in the expression glossary.

PROJECTIVE E=2 / L=12 / score ≈ 32.672 — minimum uniform E budget

Exact definitions: $\eta_{336}, \eta_{337}, \eta_{338}, \eta_{339}, \eta_{340}, \eta_{341}$ in the expression glossary.

$U = \eta_{342}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=12 / score ≈ 33.616 — minimum uniform E budget

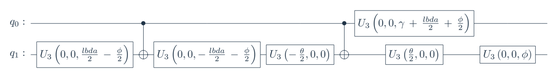
cx_rz_ry

PROJECTIVE E=2 / L=6 / score ≈ 21.012 — minimum uniform E budget

$U = e^{\frac{i(2\gamma+lbda+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=2 / L=6 / score ≈ 21.036 — minimum uniform E budget



cz_rz_rx

PROJECTIVE E=2 / L=14 / score ≈ 38.628 — minimum uniform E budget

$U = e^{\frac{i(2\gamma+lbda+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=10 / score ≈ 29.860 — minimum uniform E budget

google_cz

EXACT PHASE E=2 / L=10 / score ≈ 33.042 — minimum uniform E budget

google_fsim

EXACT PHASE E=2 / L=10 / score ≈ 33.050 — E upper bound

google_sqrt_iswap

EXACT PHASE E=2 / L=11 / score ≈ 31.927 — minimum uniform E budget

Exact definitions: $\eta_{343}, \eta_{344}, \eta_{345}, \eta_{346}, \eta_{347}, \eta_{348}, \eta_{349}, \eta_{350}, \eta_{351}, \eta_{352}, \eta_{353}, \eta_{354}, \eta_{355}, \eta_{356}, \eta_{357}$ in the expression glossary.

PROJECTIVE E=2 / L=10 / score ≈ 29.921 — minimum uniform E budget

Exact definitions: $\eta_{347}, \eta_{348}, \eta_{349}, \eta_{350}, \eta_{351}, \eta_{352}, \eta_{353}, \eta_{354}, \eta_{355}, \eta_{356}, \eta_{357}, \eta_{358}, \eta_{359}$ in the expression glossary.

$U = \eta_{360}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=14 / score ≈ 81.066 — E upper bound

ibm_cx

PROJECTIVE E=2 / L=11 / score ≈ 31.014 — minimum uniform E budget

$U = -e^{\frac{i(2\gamma+lbda+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

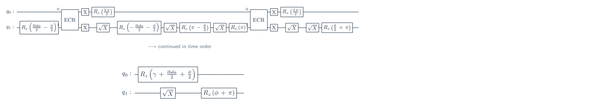
ibm_cz

PROJECTIVE E=2 / L=20 / score ≈ 50.022 — minimum uniform E budget

$U = \eta_{361}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

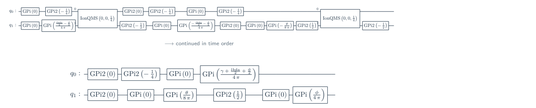
PROJECTIVE E=2 / L=19 / score ≈ 47.418 — minimum uniform E budget



$U = -e^{\frac{i(2\gamma+lbda+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

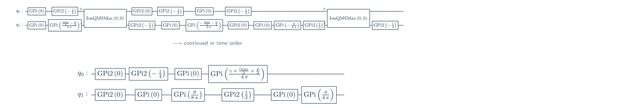
PROJECTIVE E=2 / L=26 / score ≈ 68.646 — E upper bound



$U = \eta_{362}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

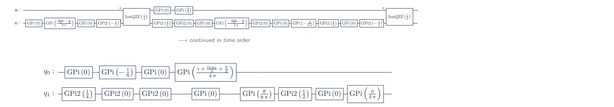
PROJECTIVE E=2 / L=26 / score ≈ 67.842 — minimum uniform E budget



$U = \eta_{362}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

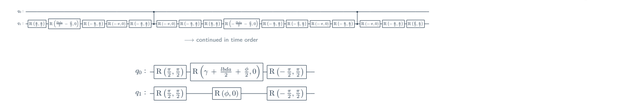
PROJECTIVE E=2 / L=28 / score ≈ 73.442 — E upper bound



$U = \eta_{332}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=2 / L=22 / score ≈ 58.288 — minimum uniform E budget



$U = e^{\frac{i(2\gamma+lbda+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=4 / L=14 / score ≈ 47.884 — E upper bound

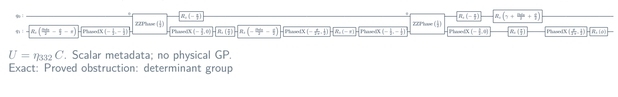
quantinuum_zzmax

PROJECTIVE E=2 / L=15 / score ≈ 42.103 — minimum uniform E budget

$U = \eta_{332}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

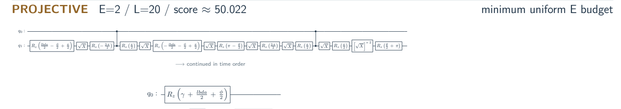
PROJECTIVE E=2 / L=15 / score ≈ 42.507 — E upper bound



$U = \eta_{332}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

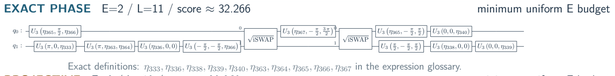
PROJECTIVE E=2 / L=20 / score ≈ 50.022 — minimum uniform E budget



$U = \eta_{361}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=2 / L=11 / score ≈ 32.266 — minimum uniform E budget



Exact definitions: $\eta_{333}, \eta_{336}, \eta_{338}, \eta_{339}, \eta_{340}, \eta_{363}, \eta_{364}, \eta_{365}, \eta_{366}, \eta_{367}$ in the expression glossary.

PROJECTIVE E=2 / L=10 / score ≈ 30.260 — minimum uniform E budget

Exact definitions: $\eta_{336}, \eta_{338}, \eta_{339}, \eta_{340}, \eta_{341}, \eta_{365}, \eta_{366}, \eta_{367}$ in the expression glossary.

$U = \eta_{368}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=14 / score ≈ 47.884 — E upper bound

PROJECTIVE E=4 / L=14 / score ≈ 47.484 — E upper bound

$U = -i\, C$. Scalar metadata; no physical GP.

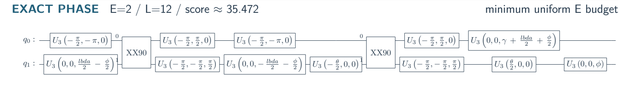
tket_tk2

PROJECTIVE E=2 / L=16 / score ≈ 44.521 — E upper bound

$U = \eta_{332}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

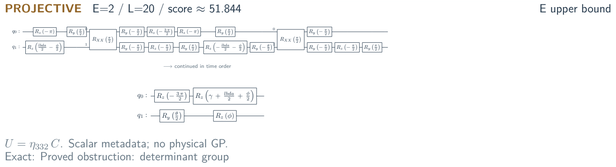
xx90_u

EXACT PHASE E=2 / L=12 / score ≈ 35.472 — minimum uniform E budget



xx_rz_ry

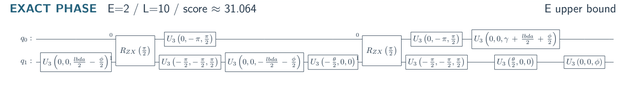
PROJECTIVE E=2 / L=20 / score ≈ 51.844 — E upper bound



$U = \eta_{332}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

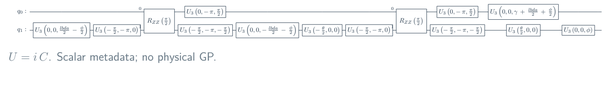
EXACT PHASE E=2 / L=10 / score ≈ 31.064 — E upper bound



zz_u

EXACT PHASE E=2 / L=13 / score ≈ 37.082 — E upper bound

PROJECTIVE E=2 / L=12 / score ≈ 35.076 — E upper bound



$U = i\, C$. Scalar metadata; no physical GP.

CU1 (θ)

2 qubits / 32 preserved native bases / alphabetical panel order

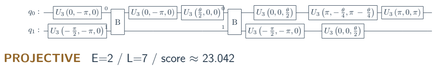
aqt_rxx

PROJECTIVE E=1 / L=10 / score ≈ 26.230 — minimum uniform E budget

$U = e^{i\frac{i\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=9 / score ≈ 27.854 — minimum uniform E budget



PROJECTIVE E=2 / L=7 / score ≈ 23.042 — minimum uniform E budget

$U = e^{-i\frac{i\theta}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

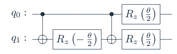
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=3 / score ≈ 14.606 — minimum uniform E budget

cx_rz_ry

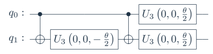
PROJECTIVE E=2 / L=3 / score ≈ 14.606 — minimum uniform E budget



$U = e^{i\frac{i\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

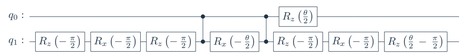
cx_u

EXACT PHASE E=2 / L=3 / score ≈ 14.618 — minimum uniform E budget



cz_rz_rx

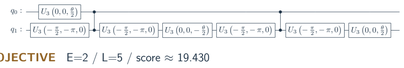
PROJECTIVE E=2 / L=8 / score ≈ 25.616 — minimum uniform E budget



$U = -e^{\frac{i\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=7 / score ≈ 23.442 — minimum uniform E budget

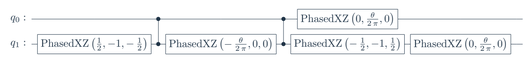


PROJECTIVE E=2 / L=5 / score ≈ 19.430 — minimum uniform E budget

$U = e^{-i\frac{i\theta}{4}}\,C$. Scalar metadata; no physical GP.

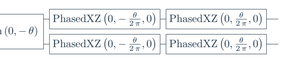
google_cz

EXACT PHASE E=2 / L=5 / score ≈ 20.421 — minimum uniform E budget



google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 14.149 — minimum uniform E budget



google_sqrt_iswap

EXACT PHASE E=2 / L=6 / score ≈ 24.809 — minimum uniform E budget

Exact definitions: $\eta_{369}, \eta_{370}, \eta_{371}, \eta_{372}, \eta_{373}, \eta_{374}, \eta_{375}, \eta_{376}, \eta_{377}$ in the expression glossary.

PROJECTIVE E=2 / L=5 / score ≈ 21.542 — minimum uniform E budget

Exact definitions: $\eta_{374}, \eta_{375}, \eta_{376}, \eta_{377}, \eta_{378}$ in the expression glossary.

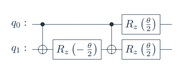
$U = \eta_{379}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=11 / score ≈ 73.257 — E upper bound

ibm_cx

PROJECTIVE E=2 / L=3 / score ≈ 14.606 — minimum uniform E budget



$U = e^{\frac{i\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=2 / L=12 / score ≈ 33.414 — minimum uniform E budget

$U = -(-1)^{\frac{1}{4}}\,e^{\frac{i\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=2 / L=6 / score ≈ 20.608 — minimum uniform E budget

$U = ie^{\frac{i\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=2 / L=16 / score ≈ 45.635 — E upper bound

$U = ie^{\frac{i\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=2 / L=16 / score ≈ 44.831 — minimum uniform E budget

$U = ie^{\frac{i\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=1 / L=4 / score ≈ 14.601 — minimum uniform E budget

$U = e^{\frac{i\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=2 / L=11 / score ≈ 33.044 — minimum uniform E budget

$U = -e^{\frac{i\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=2 / L=8 / score ≈ 25.848 — minimum uniform E budget

PROJECTIVE E=2 / L=6 / score ≈ 21.036 — minimum uniform E budget

$U = e^{-i\frac{i\theta}{4}}\,C$. Scalar metadata; no physical GP.

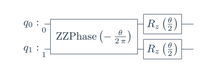
quantinuum_zzmax

PROJECTIVE E=2 / L=7 / score ≈ 23.950 — minimum uniform E budget

$U = -ie^{\frac{i\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=1 / L=2 / score ≈ 8.936 — minimum uniform E budget



$U = e^{\frac{i\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=2 / L=12 / score ≈ 33.414 — minimum uniform E budget

$U = -(-1)^{\frac{1}{4}}\,e^{\frac{i\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 23.036 — minimum uniform E budget

Exact definitions: $\eta_{380}, \eta_{381}, \eta_{382}, \eta_{383}, \eta_{384}, \eta_{385}, \eta_{386}$ in the expression glossary.

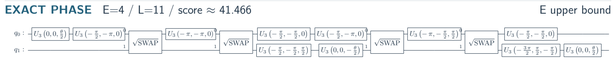
PROJECTIVE E=2 / L=5 / score ≈ 20.430 — minimum uniform E budget

Exact definitions: $\eta_{384}, \eta_{385}, \eta_{386}$ in the expression glossary.

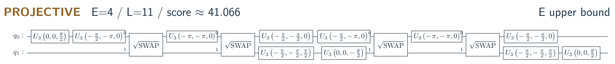
$U = \eta_{387}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=11 / score ≈ 41.466 — E upper bound



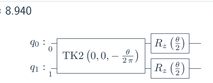
PROJECTIVE E=4 / L=11 / score ≈ 41.066 — E upper bound



$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=2 / score ≈ 8.940 — minimum uniform E budget

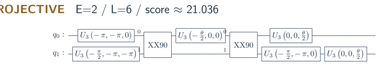


$U = e^{\frac{i\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=2 / L=8 / score ≈ 26.248 — minimum uniform E budget

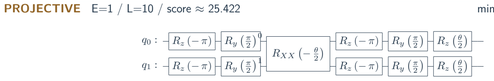
PROJECTIVE E=2 / L=6 / score ≈ 21.036 — minimum uniform E budget



$U = ie^{-i\frac{i\theta}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=1 / L=10 / score ≈ 25.422 — minimum uniform E budget



$U = e^{i\frac{i\theta}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

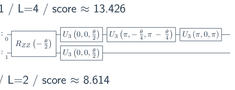
EXACT PHASE E=1 / L=6 / score ≈ 17.838 — minimum uniform E budget

PROJECTIVE E=1 / L=4 / score ≈ 13.026 — minimum uniform E budget

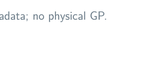
$U = e^{-i\frac{i\theta}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=1 / L=4 / score ≈ 13.426 — minimum uniform E budget



PROJECTIVE E=1 / L=2 / score ≈ 8.614 — minimum uniform E budget



$U = e^{-i\frac{i\theta}{4}}\,C$. Scalar metadata; no physical GP.

CU3 $(\theta, \phi, lbda)$

2 qubits / 32 preserved native bases / alphabetical panel order

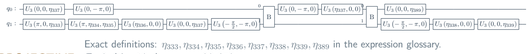
aqt_rxx

PROJECTIVE E=2 / L=16 / score ≈ 43.852 — E upper bound

$U = \eta_{388}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=13 / score ≈ 35.078 — minimum uniform E budget



Exact definitions: $\eta_{333}, \eta_{334}, \eta_{335}, \eta_{336}, \eta_{337}, \eta_{338}, \eta_{339}, \eta_{389}$ in the expression glossary.

PROJECTIVE E=2 / L=12 / score ≈ 32.672 — minimum uniform E budget

Exact definitions: $\eta_{336}, \eta_{337}, \eta_{338}, \eta_{339}, \eta_{341}, \eta_{389}$ in the expression glossary.

$U = \eta_{342}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

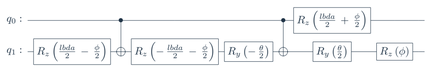
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=12 / score ≈ 33.616 — minimum uniform E budget

cx_rz_ry

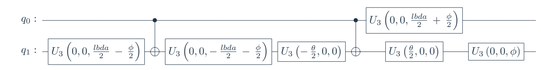
PROJECTIVE E=2 / L=6 / score ≈ 21.012 — minimum uniform E budget



$U = e^{\frac{i(lbda+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=2 / L=6 / score ≈ 21.036 — minimum uniform E budget



cz_rz_rx

PROJECTIVE E=2 / L=14 / score ≈ 38.628 — minimum uniform E budget

$U = e^{\frac{i(lbda+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=10 / score ≈ 29.860 — minimum uniform E budget

google_cz

EXACT PHASE E=2 / L=10 / score ≈ 33.042 — minimum uniform E budget

google_fsim

EXACT PHASE E=2 / L=10 / score ≈ 33.050 — E upper bound

google_sqrt_iswap

EXACT PHASE E=2 / L=11 / score ≈ 31.927 — minimum uniform E budget

Exact definitions: $\eta_{343}, \eta_{344}, \eta_{345}, \eta_{346}, \eta_{347}, \eta_{348}, \eta_{349}, \eta_{350}, \eta_{351}, \eta_{352}, \eta_{353}, \eta_{354}, \eta_{355}, \eta_{390}, \eta_{391}$ in the expression glossary.

PROJECTIVE E=2 / L=10 / score ≈ 29.921 — minimum uniform E budget

Exact definitions: $\eta_{347}, \eta_{348}, \eta_{349}, \eta_{350}, \eta_{351}, \eta_{352}, \eta_{353}, \eta_{354}, \eta_{355}, \eta_{358}, \eta_{359}, \eta_{390}, \eta_{391}$ in the expression glossary.

$U = \eta_{360}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=14 / score ≈ 81.066 — E upper bound

ibm_cx

PROJECTIVE E=2 / L=11 / score ≈ 31.014 — minimum uniform E budget

$U = -e^{\frac{i(lbda+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=2 / L=20 / score ≈ 50.022 — minimum uniform E budget

$U = \eta_{392}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=2 / L=19 / score ≈ 47.418 — minimum uniform E budget

$U = -e^{\frac{i(lbda+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=2 / L=26 / score ≈ 68.646 — E upper bound

$U = \eta_{393}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=2 / L=26 / score ≈ 67.842 — minimum uniform E budget

$U = \eta_{393}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=2 / L=28 / score ≈ 73.442 — E upper bound

$U = \eta_{388}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=2 / L=22 / score ≈ 58.288 — minimum uniform E budget

$U = e^{\frac{i(lbda+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=4 / L=14 / score ≈ 47.884 — E upper bound

quantinuum_zzmax

PROJECTIVE E=2 / L=15 / score ≈ 42.103 — minimum uniform E budget

$U = \eta_{388}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=2 / L=15 / score ≈ 42.507 — E upper bound

$U = \eta_{388}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=2 / L=20 / score ≈ 50.022 — minimum uniform E budget

$U = \eta_{392}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=2 / L=11 / score ≈ 32.266 — minimum uniform E budget

Exact definitions: $\eta_{333}, \eta_{336}, \eta_{338}, \eta_{339}, \eta_{363}, \eta_{364}, \eta_{365}, \eta_{366}, \eta_{367}, \eta_{389}$ in the expression glossary.

PROJECTIVE E=2 / L=10 / score ≈ 30.260 — minimum uniform E budget

Exact definitions: $\eta_{336}, \eta_{338}, \eta_{339}, \eta_{341}, \eta_{365}, \eta_{366}, \eta_{367}, \eta_{389}$ in the expression glossary.

$U = \eta_{368}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=14 / score ≈ 47.884 — E upper bound

PROJECTIVE E=4 / L=14 / score ≈ 47.484 — E upper bound

$U = -i\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=2 / L=16 / score ≈ 44.521 — E upper bound

$U = \eta_{388}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=2 / L=12 / score ≈ 35.472 — minimum uniform E budget

xx_rz_ry

PROJECTIVE E=2 / L=20 / score ≈ 51.844 — E upper bound

$U = \eta_{388}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=2 / L=10 / score ≈ 31.064 — E upper bound

zz_u

EXACT PHASE E=2 / L=13 / score ≈ 37.082 — E upper bound

PROJECTIVE E=2 / L=12 / score ≈ 35.076 — E upper bound

$U = i\, C$. Scalar metadata; no physical GP.

CX

2 qubits / 32 preserved native bases / alphabetical panel order

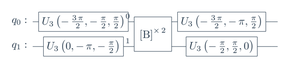
aqt_rxx

PROJECTIVE E=1 / L=5 / score ≈ 15.218 — proved E minimum

$U = \frac{\sqrt{2}i(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=4 / score ≈ 17.624 — proved E minimum



clifford_only

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

clifford_t_cx

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

clifford_t_cz

EXACT PHASE E=1 / L=2 / score ≈ 8.000 — proved E minimum

cx_h_phase

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

cx_rz_ry

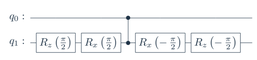
EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

cx_u

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

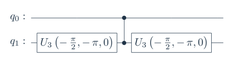
cz_rz_rx

EXACT PHASE E=1 / L=4 / score ≈ 12.808 — proved E minimum



cz_u

EXACT PHASE E=1 / L=2 / score ≈ 8.412 — proved E minimum



google_cz

EXACT PHASE E=1 / L=2 / score ≈ 8.812 — proved E minimum

google_fsim

EXACT PHASE E=1 / L=2 / score ≈ 8.816 — proved E minimum

google_sqrt_iswap

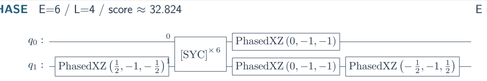
EXACT PHASE E=2 / L=4 / score ≈ 17.824 — proved E minimum

PROJECTIVE E=2 / L=4 / score ≈ 17.224 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=6 / L=4 / score ≈ 32.824 — E upper bound



ibm_cx

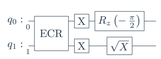
EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

ibm_cz

EXACT PHASE E=1 / L=6 / score ≈ 16.404 — proved E minimum

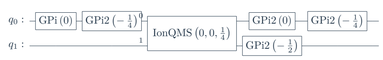
ibm_ecr

EXACT PHASE E=1 / L=4 / score ≈ 12.202 — proved E minimum



ionq_aria

PROJECTIVE E=1 / L=5 / score ≈ 15.416 — proved E minimum



$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=1 / L=5 / score ≈ 15.014 — proved E minimum

$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=1 / L=6 / score ≈ 17.814 — proved E minimum

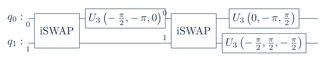
$U = \frac{\sqrt{2}(-1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

EXACT PHASE E=1 / L=4 / score ≈ 12.816 — proved E minimum

iswap_u

EXACT PHASE E=2 / L=3 / score ≈ 15.018 — proved E minimum



quantinuum_zzmax

PROJECTIVE E=1 / L=5 / score ≈ 15.014 — proved E minimum

$U = \frac{\sqrt{2}i(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

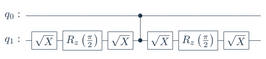
quantinuum_zzphase

PROJECTIVE E=1 / L=5 / score ≈ 15.216 — proved E minimum

$U = \frac{\sqrt{2}i(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

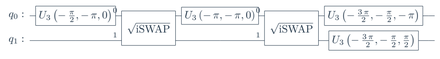
rigetti_cz

EXACT PHASE E=1 / L=6 / score ≈ 16.404 — proved E minimum



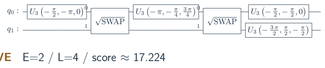
sqrt_iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 17.224 — proved E minimum

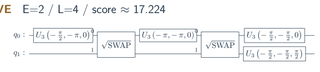


sqrt_swap_u

EXACT PHASE E=2 / L=4 / score ≈ 18.024 — proved E minimum



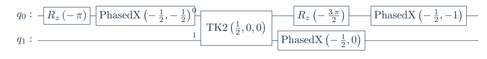
PROJECTIVE E=2 / L=4 / score ≈ 17.224 — proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

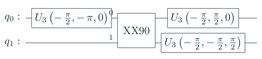
PROJECTIVE E=1 / L=5 / score ≈ 15.222 — proved E minimum



$U = \frac{\sqrt{2}i(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

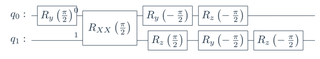
xx90_u

EXACT PHASE E=1 / L=3 / score ≈ 11.218 — proved E minimum



xx_rz_ry

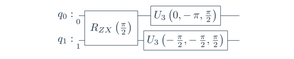
PROJECTIVE E=1 / L=6 / score ≈ 17.414 — proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

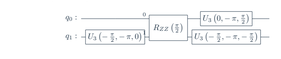
EXACT PHASE E=1 / L=2 / score ≈ 9.014 — proved E minimum



zz_u

EXACT PHASE E=1 / L=4 / score ≈ 13.426 — proved E minimum

PROJECTIVE E=1 / L=3 / score ≈ 11.020 — proved E minimum



$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

CXPow $(t, global_{shift})$

2 qubits / 32 preserved native bases / alphabetical panel order

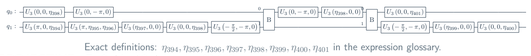
aqt_rxx

PROJECTIVE E=2 / L=15 / score ≈ 41.850 — E upper bound

$U = ie^{i\pi t\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=13 / score ≈ 35.078 — minimum uniform E budget



Exact definitions: $\eta_{394}, \eta_{395}, \eta_{396}, \eta_{397}, \eta_{398}, \eta_{399}, \eta_{400}, \eta_{401}$ in the expression glossary.

PROJECTIVE E=2 / L=12 / score ≈ 32.672 — minimum uniform E budget

Exact definitions: $\eta_{397}, \eta_{398}, \eta_{399}, \eta_{400}, \eta_{401}, \eta_{402}$ in the expression glossary.

$U = \eta_{403}\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=18 / score ≈ 45.420 — minimum uniform E budget

PROJECTIVE E=2 / L=11 / score ≈ 31.414 — minimum uniform E budget

$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

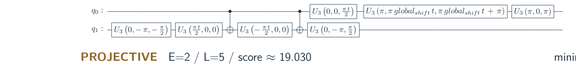
cx_rz_ry

PROJECTIVE E=2 / L=5 / score ≈ 19.010 — minimum uniform E budget

$U = -e^{i\pi t\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=2 / L=7 / score ≈ 23.042 — minimum uniform E budget

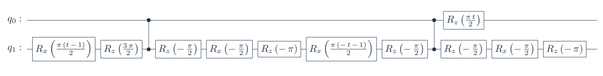


PROJECTIVE E=2 / L=5 / score ≈ 19.030 — minimum uniform E budget

$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

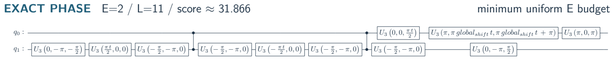
PROJECTIVE E=2 / L=11 / score ≈ 31.822 — minimum uniform E budget



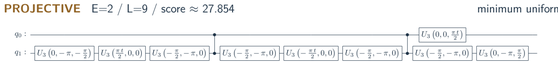
$U = -e^{i\pi t\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=11 / score ≈ 31.866 — minimum uniform E budget



PROJECTIVE E=2 / L=9 / score ≈ 27.854 — minimum uniform E budget



$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

google_cz

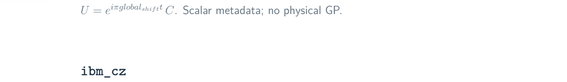
EXACT PHASE E=2 / L=11 / score ≈ 33.066 — minimum uniform E budget

PROJECTIVE E=2 / L=9 / score ≈ 29.054 — minimum uniform E budget

$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=11 / score ≈ 33.074 — E upper bound



PROJECTIVE E=2 / L=9 / score ≈ 29.062 — E upper bound

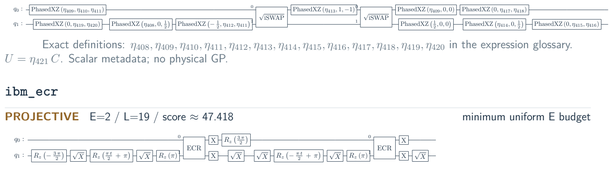
$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=2 / L=11 / score ≈ 31.927 — minimum uniform E budget

Exact definitions: $\eta_{404}, \eta_{405}, \eta_{406}, \eta_{407}, \eta_{408}, \eta_{409}, \eta_{410}, \eta_{411}, \eta_{412}, \eta_{413}, \eta_{414}, \eta_{415}, \eta_{416}, \eta_{417}, \eta_{418}$ in the expression glossary.

PROJECTIVE E=2 / L=10 / score ≈ 29.921 — minimum uniform E budget



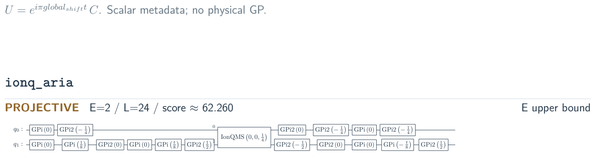
Exact definitions: $\eta_{408}, \eta_{409}, \eta_{410}, \eta_{411}, \eta_{412}, \eta_{413}, \eta_{414}, \eta_{415}, \eta_{416}, \eta_{417}, \eta_{418}, \eta_{419}, \eta_{420}$ in the expression glossary.

$U = \eta_{421}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=15 / score ≈ 81.090 — E upper bound

PROJECTIVE E=12 / L=13 / score ≈ 77.078 — E upper bound



$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

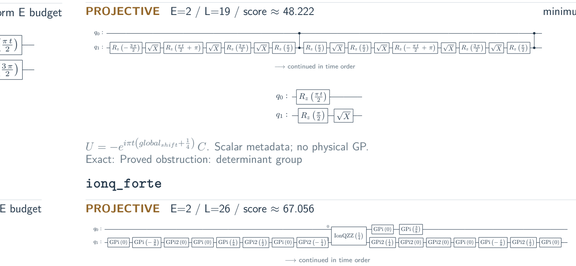
ibm_cx

PROJECTIVE E=2 / L=11 / score ≈ 31.014 — minimum uniform E budget

$U = -e^{i\pi t\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

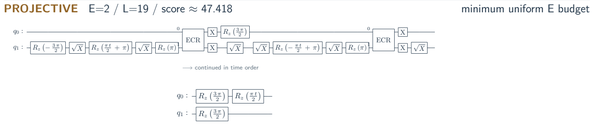
PROJECTIVE E=2 / L=19 / score ≈ 48.222 — minimum uniform E budget



$U = -e^{i\pi t\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

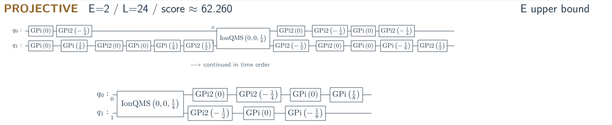
PROJECTIVE E=2 / L=19 / score ≈ 47.418 — minimum uniform E budget



$U = -e^{i\pi t\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=2 / L=24 / score ≈ 62.260 — E upper bound



$U = ie^{i\pi t\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

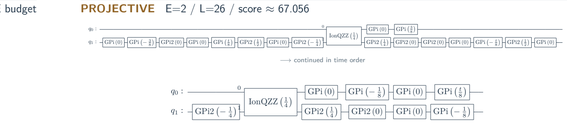
ionq_aria_full

PROJECTIVE E=2 / L=24 / score ≈ 61.456 — minimum uniform E budget

$U = ie^{i\pi t\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

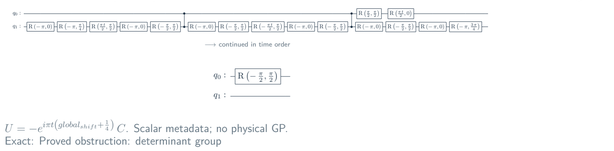
PROJECTIVE E=2 / L=26 / score ≈ 67.056 — E upper bound



$U = ie^{i\pi t\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=2 / L=17 / score ≈ 46.268 — minimum uniform E budget

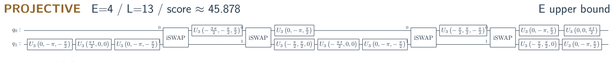


$U = -e^{i\pi t\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=4 / L=15 / score ≈ 49.890 — E upper bound

PROJECTIVE E=4 / L=13 / score ≈ 45.878 — E upper bound



$U = -e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

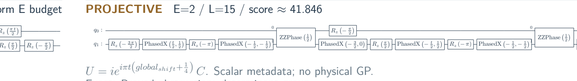
quantinuum_zzmax

PROJECTIVE E=2 / L=15 / score ≈ 41.442 — minimum uniform E budget

$U = ie^{i\pi t\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

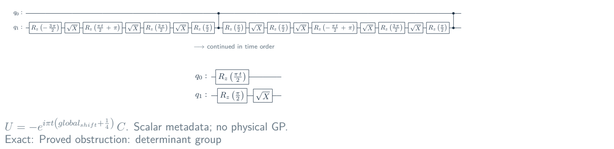
PROJECTIVE E=2 / L=15 / score ≈ 41.846 — E upper bound



$U = ie^{i\pi t\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

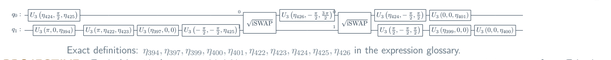
PROJECTIVE E=2 / L=19 / score ≈ 48.222 — minimum uniform E budget



$U = -e^{i\pi t\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=2 / L=11 / score ≈ 32.266 — minimum uniform E budget



Exact definitions: $\eta_{394}, \eta_{397}, \eta_{399}, \eta_{400}, \eta_{401}, \eta_{422}, \eta_{423}, \eta_{424}, \eta_{425}, \eta_{426}$ in the expression glossary.

PROJECTIVE E=2 / L=10 / score ≈ 30.260 — minimum uniform E budget

Exact definitions: $\eta_{397}, \eta_{399}, \eta_{400}, \eta_{401}, \eta_{402}, \eta_{424}, \eta_{425}, \eta_{426}$ in the expression glossary.

$U = \eta_{427}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

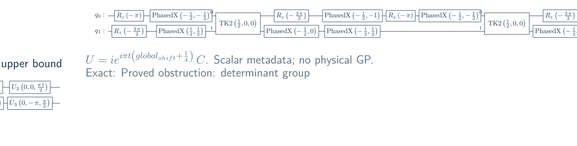
EXACT PHASE E=4 / L=15 / score ≈ 49.890 — E upper bound

PROJECTIVE E=4 / L=13 / score ≈ 45.478 — E upper bound

$U = -ie^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=2 / L=15 / score ≈ 41.858 — E upper bound



$U = ie^{i\pi t\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

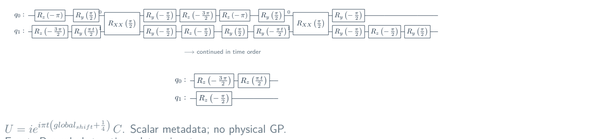
EXACT PHASE E=2 / L=13 / score ≈ 37.478 — minimum uniform E budget

PROJECTIVE E=2 / L=11 / score ≈ 33.466 — minimum uniform E budget

$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=2 / L=19 / score ≈ 49.842 — E upper bound



$U = ie^{i\pi t\left(global_{shift}+\frac{1}{2}\right)}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=2 / L=11 / score ≈ 33.070 — E upper bound

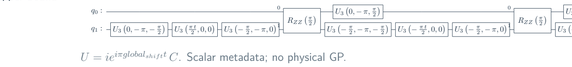
PROJECTIVE E=2 / L=9 / score ≈ 29.058 — E upper bound

$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=2 / L=14 / score ≈ 39.088 — E upper bound

PROJECTIVE E=2 / L=11 / score ≈ 33.070 — E upper bound



$U = ie^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

CY

2 qubits / 32 preserved native bases / alphabetical panel order

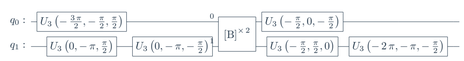
aqt_rxx

PROJECTIVE E=1 / L=7 / score ≈ 19.622 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

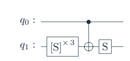
berkeley_b

EXACT PHASE E=2 / L=6 / score ≈ 22.036 — proved E minimum



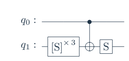
clifford_only

EXACT PHASE E=1 / L=4 / score ≈ 12.000 — proved E minimum



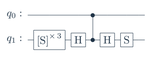
clifford_t_cx

EXACT PHASE E=1 / L=4 / score ≈ 12.000 — proved E minimum



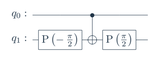
clifford_t_cz

EXACT PHASE E=1 / L=6 / score ≈ 16.000 — proved E minimum



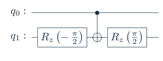
cx_h_phase

EXACT PHASE E=1 / L=2 / score ≈ 8.404 — proved E minimum



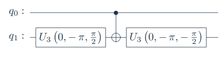
cx_rz_ry

EXACT PHASE E=1 / L=2 / score ≈ 8.404 — proved E minimum



cx_u

EXACT PHASE E=1 / L=2 / score ≈ 8.412 — proved E minimum



cz_rz_rx

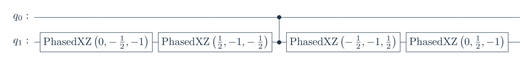
EXACT PHASE E=1 / L=2 / score ≈ 8.404 — proved E minimum

cz_u

EXACT PHASE E=1 / L=2 / score ≈ 8.812 — proved E minimum

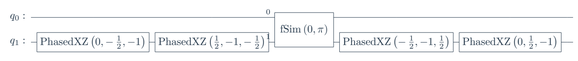
google_cz

EXACT PHASE E=1 / L=4 / score ≈ 13.224 — proved E minimum



google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 13.228 — proved E minimum



google_sqrt_iswap

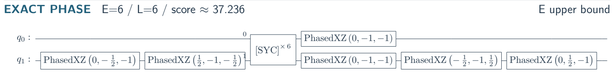
EXACT PHASE E=2 / L=6 / score ≈ 22.236 — proved E minimum

PROJECTIVE E=2 / L=6 / score ≈ 21.636 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

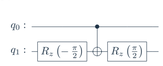
google_syc

EXACT PHASE E=6 / L=6 / score ≈ 37.236 — E upper bound



ibm_cx

EXACT PHASE E=1 / L=2 / score ≈ 8.404 — proved E minimum



ibm_cz

EXACT PHASE E=1 / L=5 / score ≈ 14.404 — proved E minimum

ibm_ecr

EXACT PHASE E=1 / L=6 / score ≈ 16.606 — proved E minimum

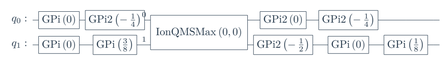
ionq_aria

PROJECTIVE E=1 / L=9 / score ≈ 24.624 — proved E minimum

$U = \frac{\sqrt{2}(-1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

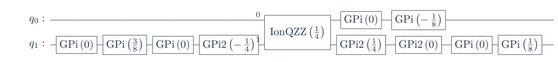
PROJECTIVE E=1 / L=9 / score ≈ 24.222 — proved E minimum



$U = \frac{\sqrt{2}(-1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

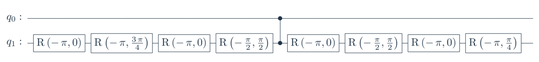
PROJECTIVE E=1 / L=10 / score ≈ 27.022 — proved E minimum



$U = \frac{\sqrt{2}(-1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

EXACT PHASE E=1 / L=8 / score ≈ 21.632 — proved E minimum



iswap_u

EXACT PHASE E=2 / L=3 / score ≈ 15.018 — proved E minimum

quantinuum_zzmax

PROJECTIVE E=1 / L=6 / score ≈ 17.416 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

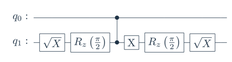
quantinuum_zzphase

PROJECTIVE E=1 / L=6 / score ≈ 17.618 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=1 / L=5 / score ≈ 14.404 — proved E minimum



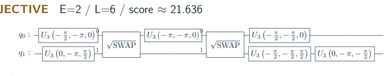
sqrt_iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 21.636 — proved E minimum

sqrt_swap_u

EXACT PHASE E=2 / L=6 / score ≈ 22.436 — proved E minimum

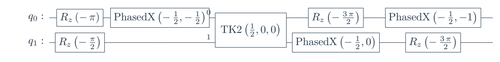
PROJECTIVE E=2 / L=6 / score ≈ 21.636 — proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

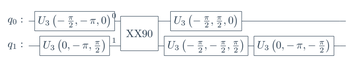
PROJECTIVE E=1 / L=7 / score ≈ 19.626 — proved E minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=1 / L=5 / score ≈ 15.630 — proved E minimum



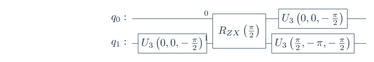
xx_rz_ry

PROJECTIVE E=1 / L=6 / score ≈ 17.414 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

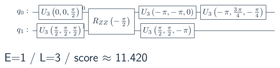
zx_u

EXACT PHASE E=1 / L=3 / score ≈ 11.020 — proved E minimum



zz_u

EXACT PHASE E=1 / L=5 / score ≈ 16.232 — proved E minimum



PROJECTIVE E=1 / L=3 / score ≈ 11.420 — proved E minimum

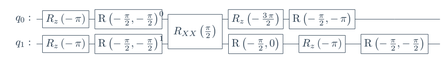
$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

CZ

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=1 / L=9 / score ≈ 24.030 — proved E minimum



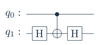
$U = \frac{\sqrt{2}(1-i)}{2} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=6 / score ≈ 22.036 — proved E minimum

clifford_only

EXACT PHASE E=1 / L=2 / score ≈ 8.000 — proved E minimum



clifford_t_cx

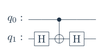
EXACT PHASE E=1 / L=2 / score ≈ 8.000 — proved E minimum

clifford_t_cz

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

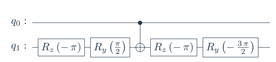
cx_h_phase

EXACT PHASE E=1 / L=2 / score ≈ 8.000 — proved E minimum



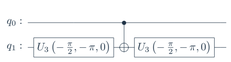
cx_rz_ry

EXACT PHASE E=1 / L=4 / score ≈ 12.408 — proved E minimum



cx_u

EXACT PHASE E=1 / L=2 / score ≈ 8.412 — proved E minimum



cz_rz_rx

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

cz_u

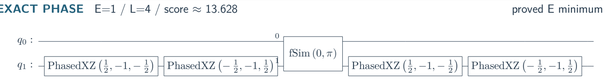
EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

google_cz

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 13.628 — proved E minimum



google_sqrt_iswap

EXACT PHASE E=2 / L=6 / score ≈ 22.636 — proved E minimum

PROJECTIVE E=2 / L=6 / score ≈ 22.036 — proved E minimum

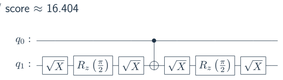
$U = \frac{\sqrt{2}(1-i)}{2} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=6 / L=6 / score ≈ 37.636 — E upper bound

ibm_cx

EXACT PHASE E=1 / L=6 / score ≈ 16.404 — proved E minimum



ibm_cz

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

ibm_ecr

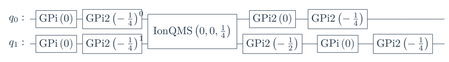
EXACT PHASE E=1 / L=10 / score ≈ 24.606 — proved E minimum

PROJECTIVE E=1 / L=7 / score ≈ 18.808 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=1 / L=9 / score ≈ 24.224 — proved E minimum



$U = \frac{\sqrt{2}(1+i)}{2} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

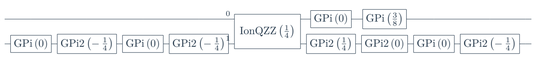
ionq_aria_full

PROJECTIVE E=1 / L=9 / score ≈ 23.822 — proved E minimum

$U = \frac{\sqrt{2}(1+i)}{2} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=1 / L=10 / score ≈ 26.622 — proved E minimum



$U = \frac{\sqrt{2}(-1-i)}{2} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

iswap_u

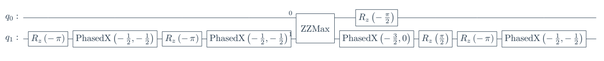
EXACT PHASE E=2 / L=5 / score ≈ 19.430 — proved E minimum

PROJECTIVE E=2 / L=4 / score ≈ 17.024 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

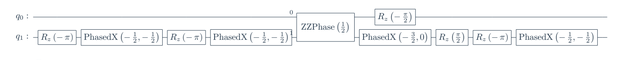
PROJECTIVE E=1 / L=9 / score ≈ 23.826 — proved E minimum



$U = \frac{\sqrt{2}(1-i)}{2} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=1 / L=9 / score ≈ 24.028 — proved E minimum



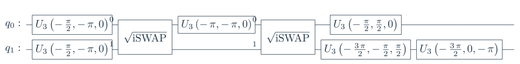
$U = \frac{\sqrt{2}(1-i)}{2} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 21.636 — proved E minimum



sqrt_swap_u

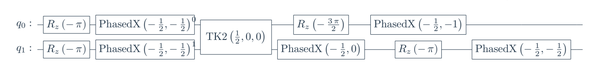
EXACT PHASE E=2 / L=6 / score ≈ 22.436 — proved E minimum

PROJECTIVE E=2 / L=6 / score ≈ 21.636 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

tket_tk2

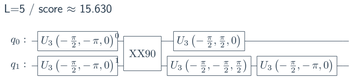
PROJECTIVE E=1 / L=9 / score ≈ 24.034 — proved E minimum



$U = \frac{\sqrt{2}(1-i)}{2} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

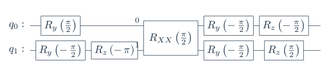
xx90_u

EXACT PHASE E=1 / L=5 / score ≈ 15.630 — proved E minimum



xx_rz_ry

PROJECTIVE E=1 / L=7 / score ≈ 19.416 — proved E minimum

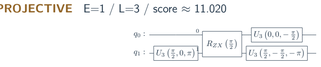


$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=1 / L=4 / score ≈ 13.426 — proved E minimum

PROJECTIVE E=1 / L=3 / score ≈ 11.020 — proved E minimum

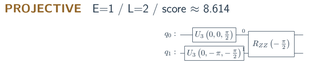


$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=1 / L=4 / score ≈ 13.426 — proved E minimum

PROJECTIVE E=1 / L=2 / score ≈ 8.614 — proved E minimum



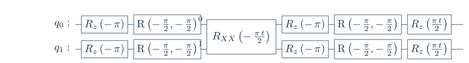
$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

CZPow $(t, global_{shift})$

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

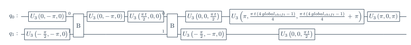
PROJECTIVE E=1 / L=10 / score ≈ 26.230 — minimum uniform E budget



$U = e^{\frac{i\pi\left(4global_{shift}+1\right)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=9 / score ≈ 27.854 — minimum uniform E budget



PROJECTIVE E=2 / L=7 / score ≈ 23.042 — minimum uniform E budget

$U = e^{\frac{i\pi\left(4global_{shift}-1\right)}{4}} C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=10 / score ≈ 28.612 — minimum uniform E budget

PROJECTIVE E=2 / L=3 / score ≈ 14.606 — minimum uniform E budget

$U = e^{i\pi global_{shift}t} C$. Scalar metadata; no physical GP.

cx_rz_ry

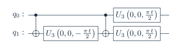
PROJECTIVE E=2 / L=3 / score ≈ 14.606 — minimum uniform E budget

$U = e^{\frac{i\pi\left(4global_{shift}+1\right)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=2 / L=5 / score ≈ 18.630 — minimum uniform E budget

PROJECTIVE E=2 / L=3 / score ≈ 14.618 — minimum uniform E budget



$U = e^{i\pi global_{shift}t} C$. Scalar metadata; no physical GP.

cz_rz_rx

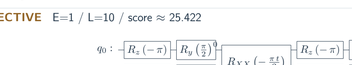
PROJECTIVE E=2 / L=8 / score ≈ 25.616 — minimum uniform E budget

$U = -e^{\frac{i\pi\left(4global_{shift}+1\right)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=7 / score ≈ 24.242 — minimum uniform E budget

PROJECTIVE E=2 / L=5 / score ≈ 19.430 — minimum uniform E budget



$U = e^{\frac{i\pi\left(4global_{shift}-1\right)}{4}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=2 / L=7 / score ≈ 23.442 — minimum uniform E budget

PROJECTIVE E=2 / L=5 / score ≈ 19.430 — minimum uniform E budget

$U = e^{i\pi global_{shift}t} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=6 / score ≈ 16.840 — minimum uniform E budget

PROJECTIVE E=1 / L=4 / score ≈ 12.828 — minimum uniform E budget

$U = e^{i\pi global_{shift}t} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=2 / L=9 / score ≈ 27.915 — minimum uniform E budget

Exact definitions: $\eta_{409}, \eta_{410}, \eta_{411}, \eta_{412}, \eta_{413}, \eta_{415}, \eta_{417}, \eta_{418}, \eta_{428}, \eta_{429}, \eta_{430}$ in the expression glossary.

PROJECTIVE E=2 / L=8 / score ≈ 25.909 — minimum uniform E budget

Exact definitions: $\eta_{409}, \eta_{410}, \eta_{411}, \eta_{412}, \eta_{413}, \eta_{415}, \eta_{417}, \eta_{418}, \eta_{419}$ in the expression glossary.
$U = \eta_{421} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=13 / score ≈ 76.278 — E upper bound

PROJECTIVE E=12 / L=11 / score ≈ 72.266 — E upper bound

$U = e^{i\pi global_{shift}t} C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=2 / L=3 / score ≈ 14.606 — minimum uniform E budget

$U = e^{\frac{i\pi\left(4global_{shift}+1\right)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=2 / L=12 / score ≈ 33.414 — minimum uniform E budget

$U = \eta_{431} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=2 / L=6 / score ≈ 20.608 — minimum uniform E budget

$U = ie^{\frac{i\pi\left(4global_{shift}+1\right)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=2 / L=16 / score ≈ 44.644 — E upper bound

$U = ie^{i\pi\left(global_{shift}+\frac{1}{4}\right)} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=2 / L=16 / score ≈ 43.840 — minimum uniform E budget

$U = ie^{i\pi\left(global_{shift}+\frac{1}{4}\right)} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=2 / L=18 / score ≈ 49.440 — E upper bound

$U = -ie^{i\pi\left(global_{shift}+\frac{1}{4}\right)} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=2 / L=11 / score ≈ 33.044 — minimum uniform E budget

$U = -e^{\frac{i\pi\left(4global_{shift}+1\right)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=2 / L=8 / score ≈ 25.848 — minimum uniform E budget

PROJECTIVE E=2 / L=6 / score ≈ 21.036 — minimum uniform E budget

$U = e^{\frac{i\pi\left(4global_{shift}-1\right)}{4}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=2 / L=7 / score ≈ 23.620 — minimum uniform E budget

$U = -ie^{\frac{i\pi\left(4global_{shift}+1\right)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=1 / L=2 / score ≈ 8.606 — minimum uniform E budget

$U = e^{\frac{i\pi\left(4global_{shift}+1\right)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=2 / L=12 / score ≈ 33.414 — minimum uniform E budget

$U = \eta_{431} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=2 / L=9 / score ≈ 28.254 — minimum uniform E budget

Exact definitions: $\eta_{400}, \eta_{401}, \eta_{422}, \eta_{423}, \eta_{424}, \eta_{425}, \eta_{426}, \eta_{432}$ in the expression glossary.

PROJECTIVE E=2 / L=8 / score ≈ 26.248 — minimum uniform E budget

Exact definitions: $\eta_{400}, \eta_{401}, \eta_{402}, \eta_{424}, \eta_{425}, \eta_{426}$ in the expression glossary.
$U = \eta_{427} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=13 / score ≈ 45.478 — E upper bound

PROJECTIVE E=4 / L=11 / score ≈ 41.066 — E upper bound

$U = -ie^{i\pi global_{shift}t} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=2 / score ≈ 8.610 — minimum uniform E budget

$U = e^{\frac{i\pi\left(4global_{shift}+1\right)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=2 / L=8 / score ≈ 26.248 — minimum uniform E budget

PROJECTIVE E=2 / L=6 / score ≈ 21.036 — minimum uniform E budget

$U = ie^{\frac{i\pi\left(4global_{shift}-1\right)}{4}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=1 / L=10 / score ≈ 25.422 — minimum uniform E budget

$U = e^{\frac{i\pi\left(4global_{shift}+1\right)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=1 / L=6 / score ≈ 17.838 — minimum uniform E budget

PROJECTIVE E=1 / L=4 / score ≈ 13.026 — minimum uniform E budget

$U = e^{\frac{i\pi\left(4global_{shift}-1\right)}{4}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=1 / L=4 / score ≈ 13.426 — minimum uniform E budget

PROJECTIVE E=1 / L=2 / score ≈ 8.614 — minimum uniform E budget

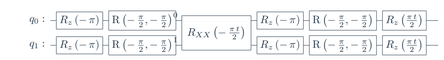
$U = e^{\frac{i\pi\left(4global_{shift}-1\right)}{4}} C$. Scalar metadata; no physical GP.

CZPowGate (t)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

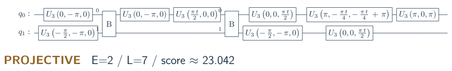
PROJECTIVE E=1 / L=10 / score ≈ 26.230 — minimum uniform E budget



$U = e^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=9 / score ≈ 27.854 — minimum uniform E budget



PROJECTIVE E=2 / L=7 / score ≈ 23.042 — minimum uniform E budget

$U = e^{-\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

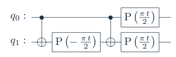
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

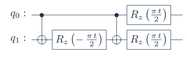
cx_h_phase

EXACT PHASE E=2 / L=3 / score ≈ 14.606 — minimum uniform E budget



cx_rz_ry

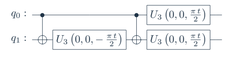
PROJECTIVE E=2 / L=3 / score ≈ 14.606 — minimum uniform E budget



$U = e^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=2 / L=3 / score ≈ 14.618 — minimum uniform E budget



cz_rz_rx

PROJECTIVE E=2 / L=8 / score ≈ 25.616 — minimum uniform E budget

$U = -e^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=7 / score ≈ 23.442 — minimum uniform E budget

PROJECTIVE E=2 / L=5 / score ≈ 19.430 — minimum uniform E budget

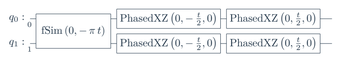
$U = e^{-\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=2 / L=5 / score ≈ 19.430 — minimum uniform E budget

google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 12.828 — minimum uniform E budget



google_sqrt_iswap

EXACT PHASE E=2 / L=6 / score ≈ 24.809 — minimum uniform E budget

Exact definitions: $\eta_{433}, \eta_{434}, \eta_{435}, \eta_{436}, \eta_{437}, \eta_{438}, \eta_{439}, \eta_{440}, \eta_{441}$ in the expression glossary.

PROJECTIVE E=2 / L=5 / score ≈ 21.542 — minimum uniform E budget

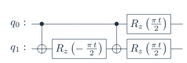
Exact definitions: $\eta_{428}, \eta_{429}, \eta_{430}, \eta_{431}, \eta_{432}$ in the expression glossary.
$U = \eta_{443}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=11 / score ≈ 72.266 — E upper bound

ibm_cx

PROJECTIVE E=2 / L=3 / score ≈ 14.606 — minimum uniform E budget



$U = e^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=2 / L=12 / score ≈ 33.414 — minimum uniform E budget

$U = -(-1)^{\frac{3}{4}}\,e^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=2 / L=6 / score ≈ 20.608 — minimum uniform E budget

$U = ie^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=2 / L=16 / score ≈ 44.644 — E upper bound

$U = ie^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=2 / L=16 / score ≈ 43.840 — minimum uniform E budget

$U = ie^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=1 / L=4 / score ≈ 13.610 — minimum uniform E budget

$U = e^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=2 / L=11 / score ≈ 33.044 — minimum uniform E budget

$U = -e^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=2 / L=8 / score ≈ 25.848 — minimum uniform E budget

PROJECTIVE E=2 / L=6 / score ≈ 21.036 — minimum uniform E budget

$U = e^{-\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.

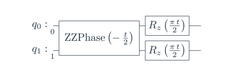
quantinuum_zzmax

PROJECTIVE E=2 / L=7 / score ≈ 23.620 — minimum uniform E budget

$U = -ie^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=1 / L=2 / score ≈ 8.606 — minimum uniform E budget



$U = e^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=2 / L=12 / score ≈ 33.414 — minimum uniform E budget

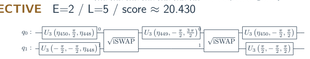
$U = -(-1)^{\frac{3}{4}}\,e^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 23.036 — minimum uniform E budget

Exact definitions: $\eta_{444}, \eta_{445}, \eta_{446}, \eta_{447}, \eta_{448}, \eta_{449}, \eta_{450}$ in the expression glossary.

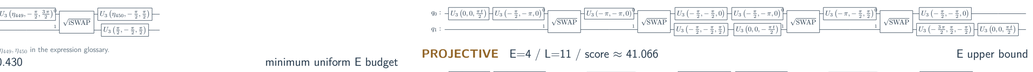
PROJECTIVE E=2 / L=5 / score ≈ 20.430 — minimum uniform E budget



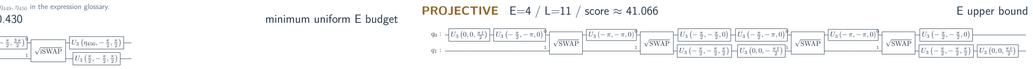
Exact definitions: $\eta_{438}, \eta_{439}, \eta_{440}$ in the expression glossary.
$U = \eta_{451}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=11 / score ≈ 41.466 — E upper bound



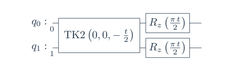
PROJECTIVE E=4 / L=11 / score ≈ 41.066 — E upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=2 / score ≈ 8.610 — minimum uniform E budget

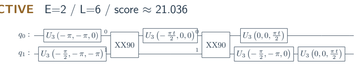


$U = e^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=2 / L=8 / score ≈ 26.248 — minimum uniform E budget

PROJECTIVE E=2 / L=6 / score ≈ 21.036 — minimum uniform E budget



$U = ie^{-\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.

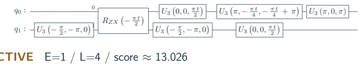
xx_rz_ry

PROJECTIVE E=1 / L=10 / score ≈ 25.422 — minimum uniform E budget

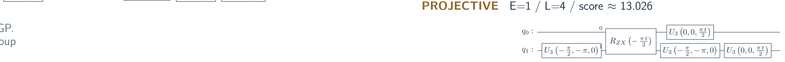
$U = e^{\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=1 / L=6 / score ≈ 17.838 — minimum uniform E budget



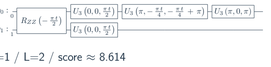
PROJECTIVE E=1 / L=4 / score ≈ 13.026 — minimum uniform E budget



$U = e^{-\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=1 / L=4 / score ≈ 13.426 — minimum uniform E budget



PROJECTIVE E=1 / L=2 / score ≈ 8.614 — minimum uniform E budget

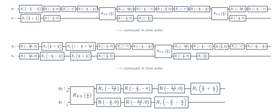
$U = e^{-\frac{i\pi t}{4}}\,C$. Scalar metadata; no physical GP.

CZS (θ, ϕ, γ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

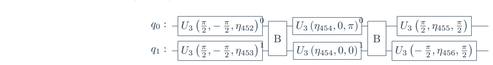
PROJECTIVE E=4 / L=40 / score ≈ 106.532 — E upper bound



$U = -ie^{\frac{3\pi}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=6 / score ≈ 21.636 — minimum uniform E budget



Exact definitions: $\eta_{452}, \eta_{453}, \eta_{454}, \eta_{455}, \eta_{456}$ in the expression glossary.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

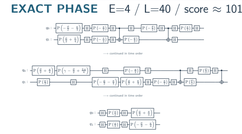
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

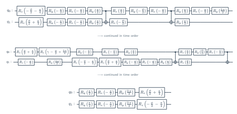
cx_h_phase

EXACT PHASE E=4 / L=40 / score ≈ 101.640 — E upper bound



cx_rz_ry

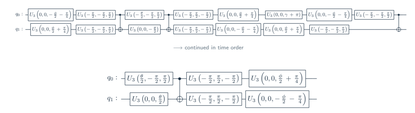
PROJECTIVE E=4 / L=40 / score ≈ 105.680 — E upper bound



$U = e^{\frac{i(\gamma+\pi)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

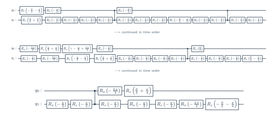
cx_u

EXACT PHASE E=4 / L=21 / score ≈ 67.726 — E upper bound



cz_rz_rx

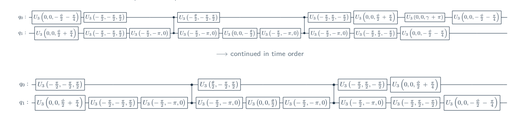
PROJECTIVE E=4 / L=42 / score ≈ 110.084 — E upper bound



$U = e^{\frac{i(\gamma+\pi)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

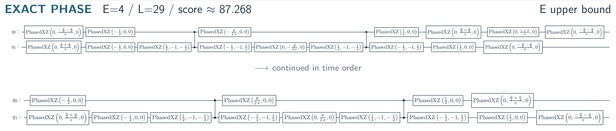
cz_u

EXACT PHASE E=4 / L=29 / score ≈ 85.374 — E upper bound



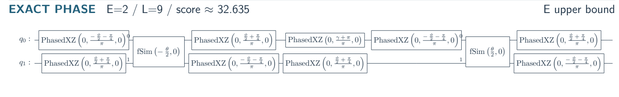
google_cz

EXACT PHASE E=4 / L=29 / score ≈ 87.268 — E upper bound



google_fsim

EXACT PHASE E=2 / L=9 / score ≈ 32.635 — E upper bound



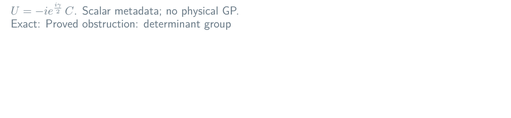
google_sqrt_iswap

EXACT PHASE E=2 / L=6 / score ≈ 21.757 — minimum uniform E budget

Exact definitions: $\eta_{457}, \eta_{458}, \eta_{459}, \eta_{460}, \eta_{461}, \eta_{462}, \eta_{463}, \eta_{464}, \eta_{465}, \eta_{466}, \eta_{467}, \eta_{468}$ in the expression glossary.

google_syc

EXACT PHASE E=24 / L=37 / score ≈ 183.316 — E upper bound



ibm_cx

PROJECTIVE E=4 / L=32 / score ≈ 84.832 — E upper bound

$U = -e^{\frac{i(\gamma+\pi)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=4 / L=54 / score ≈ 130.448 — E upper bound

$U = -e^{\frac{3\pi}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=4 / L=44 / score ≈ 108.832 — E upper bound

$U = -e^{\frac{i(\gamma+\pi)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=4 / L=58 / score ≈ 153.634 — E upper bound

$U = -ie^{\frac{3\pi}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=4 / L=58 / score ≈ 152.026 — E upper bound

$U = -ie^{\frac{3\pi}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=4 / L=62 / score ≈ 163.226 — E upper bound

$U = -ie^{\frac{3\pi}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=4 / L=59 / score ≈ 151.836 — E upper bound

$U = e^{\frac{i(\gamma+\pi)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=4 / L=21 / score ≈ 66.126 — E upper bound

quantinuum_zzmax

PROJECTIVE E=4 / L=38 / score ≈ 102.373 — E upper bound

$U = -ie^{\frac{3\pi}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=4 / L=38 / score ≈ 103.181 — E upper bound

$U = -ie^{\frac{3\pi}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=4 / L=54 / score ≈ 130.448 — E upper bound

$U = -e^{\frac{3\pi}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 20.036 — minimum uniform E budget

Exact definitions: $\eta_{469}, \eta_{470}, \eta_{471}, \eta_{472}, \eta_{473}, \eta_{474}$ in the expression glossary.

sqrt_swap_u

EXACT PHASE E=8 / L=37 / score ≈ 120.622 — E upper bound

tket_tk2

PROJECTIVE E=2 / L=8 / score ≈ 29.349 — E upper bound

$U = e^{\frac{i(\gamma+\pi)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=4 / L=33 / score ≈ 96.598 — E upper bound

xx_rz_ry

PROJECTIVE E=4 / L=66 / score ≈ 162.540 — E upper bound

$U = -ie^{\frac{3\pi}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=4 / L=29 / score ≈ 87.782 — E upper bound

zz_u

EXACT PHASE E=4 / L=33 / score ≈ 91.006 — E upper bound

DB

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

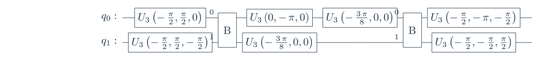
EXACT PHASE E=2 / L=34 / score ≈ 84.504 — proved E minimum

PROJECTIVE E=2 / L=16 / score ≈ 44.458 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=2 / L=7 / score ≈ 25.242 — proved E minimum



clifford_only

Exact: Proved obstruction: not a clifford
Projective: Proved obstruction: not a clifford

clifford_t_cx

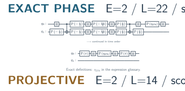
Exact: Proved obstruction: outside clifford t number field
Projective: Proved obstruction: outside clifford t number field

clifford_t_cz

Exact: Proved obstruction: outside clifford t number field
Projective: Proved obstruction: outside clifford t number field

cx_h_phase

EXACT PHASE E=2 / L=22 / score ≈ 54.424 — proved E minimum

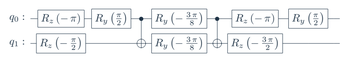


PROJECTIVE E=2 / L=14 / score ≈ 38.416 — proved E minimum

$U = e^{\frac{3i\pi}{8}}\,C$. Scalar metadata; no physical GP.

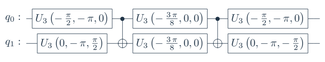
cx_rz_ry

EXACT PHASE E=2 / L=8 / score ≈ 26.016 — proved E minimum



cx_u

EXACT PHASE E=2 / L=6 / score ≈ 22.036 — proved E minimum



cz_rz_rx

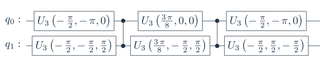
EXACT PHASE E=2 / L=12 / score ≈ 35.224 — proved E minimum

PROJECTIVE E=2 / L=11 / score ≈ 33.622 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

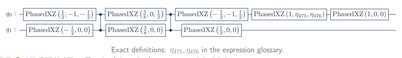
cz_u

EXACT PHASE E=2 / L=6 / score ≈ 23.236 — proved E minimum



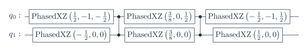
google_cz

EXACT PHASE E=2 / L=8 / score ≈ 27.309 — proved E minimum



Exact definitions: η_{475}, η_{476} in the expression glossary.

PROJECTIVE E=2 / L=6 / score ≈ 22.636 — proved E minimum



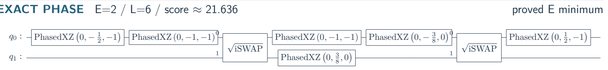
$U = e^{-\frac{3i\pi}{8}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=0 / score ≈ 4.604 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=2 / L=6 / score ≈ 21.636 — proved E minimum



google_syc

EXACT PHASE E=12 / L=16 / score ≈ 84.357 — E upper bound

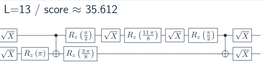
Exact definitions: η_{475}, η_{476} in the expression glossary.

PROJECTIVE E=12 / L=14 / score ≈ 79.684 — E upper bound

$U = e^{-\frac{3i\pi}{8}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=13 / score ≈ 35.612 — proved E minimum

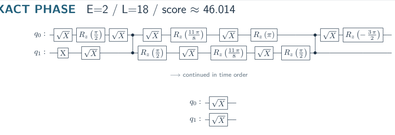


PROJECTIVE E=2 / L=12 / score ≈ 33.608 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

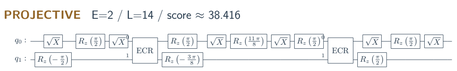
EXACT PHASE E=2 / L=18 / score ≈ 46.014 — proved E minimum



ibm_ecr

EXACT PHASE E=2 / L=14 / score ≈ 38.820 — proved E minimum

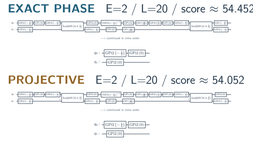
PROJECTIVE E=2 / L=14 / score ≈ 38.416 — proved E minimum



$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=20 / score ≈ 54.452 — proved E minimum

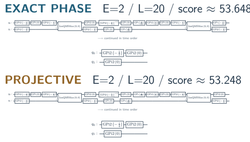


PROJECTIVE E=2 / L=20 / score ≈ 54.052 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=20 / score ≈ 53.648 — proved E minimum

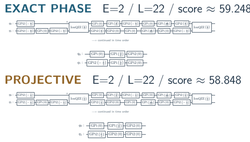


PROJECTIVE E=2 / L=20 / score ≈ 53.248 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=22 / score ≈ 59.248 — proved E minimum



PROJECTIVE E=2 / L=22 / score ≈ 58.848 — proved E minimum

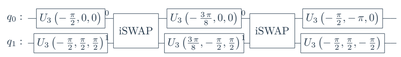
$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=8 / score ≈ 26.632 — proved E minimum

iswap_u

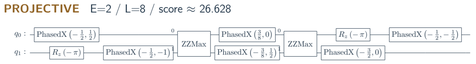
EXACT PHASE E=2 / L=6 / score ≈ 23.236 — proved E minimum



quantinuum_zzmax

EXACT PHASE E=5 / L=27 / score ≈ 80.886 — E upper bound

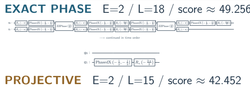
PROJECTIVE E=2 / L=8 / score ≈ 26.628 — proved E minimum



$U = \frac{\sqrt[8]{-1}\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=2 / L=18 / score ≈ 49.256 — proved E minimum

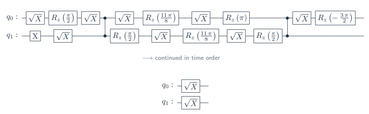


PROJECTIVE E=2 / L=15 / score ≈ 42.452 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

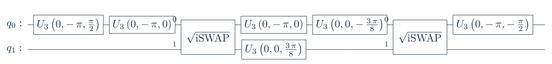
rigetti_cz

EXACT PHASE E=2 / L=18 / score ≈ 46.014 — proved E minimum



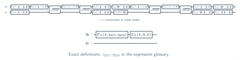
sqrt_iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 21.636 — proved E minimum



sqrt_swap_u

EXACT PHASE E=4 / L=16 / score ≈ 54.496 — E upper bound



Exact definitions: η_{475}, η_{476} in the expression glossary.

PROJECTIVE E=4 / L=14 / score ≈ 50.484 — E upper bound

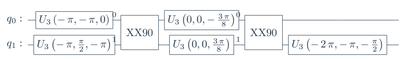
$U = -(-1)^{\frac{5}{16}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 5.206 — proved E minimum

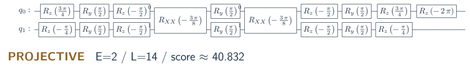
xx90_u

EXACT PHASE E=2 / L=5 / score ≈ 19.630 — proved E minimum



xx_rz_ry

EXACT PHASE E=2 / L=15 / score ≈ 42.834 — proved E minimum

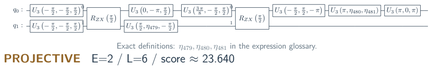


PROJECTIVE E=2 / L=14 / score ≈ 40.832 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=2 / L=8 / score ≈ 27.652 — proved E minimum



Exact definitions: $\eta_{479}, \eta_{480}, \eta_{481}$ in the expression glossary.

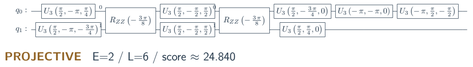
PROJECTIVE E=2 / L=6 / score ≈ 23.640 — proved E minimum

Exact definitions: η_{479} in the expression glossary.

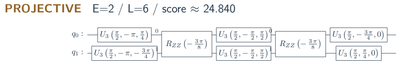
$U = -(-1)^{\frac{13}{16}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=2 / L=8 / score ≈ 29.252 — proved E minimum



PROJECTIVE E=2 / L=6 / score ≈ 24.840 — proved E minimum



$U = -i\,C$. Scalar metadata; no physical GP.

DCX

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

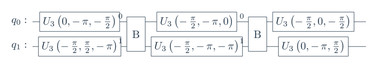
EXACT PHASE E=3 / L=12 / score ≈ 38.446 — E upper bound

PROJECTIVE E=2 / L=10 / score ≈ 30.436 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=2 / L=6 / score ≈ 21.636 — proved E minimum



clifford_only

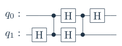
EXACT PHASE E=2 / L=0 / score ≈ 8.000 — proved E minimum

clifford_t_cx

EXACT PHASE E=2 / L=0 / score ≈ 8.000 — proved E minimum

clifford_t_cz

EXACT PHASE E=2 / L=4 / score ≈ 16.000 — proved E minimum



cx_h_phase

EXACT PHASE E=2 / L=0 / score ≈ 8.000 — proved E minimum

cx_rz_ry

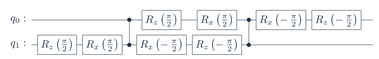
EXACT PHASE E=2 / L=0 / score ≈ 8.000 — proved E minimum

cx_u

EXACT PHASE E=2 / L=0 / score ≈ 8.000 — proved E minimum

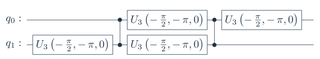
cz_rz_rx

EXACT PHASE E=2 / L=8 / score ≈ 25.616 — proved E minimum



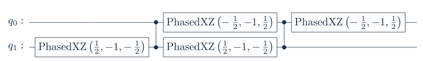
cz_u

EXACT PHASE E=2 / L=4 / score ≈ 16.824 — proved E minimum



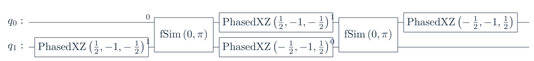
google_cz

EXACT PHASE E=2 / L=4 / score ≈ 17.624 — proved E minimum



google_fsim

EXACT PHASE E=2 / L=4 / score ≈ 17.632 — E upper bound



google_sqrt_iswap

EXACT PHASE E=2 / L=4 / score ≈ 17.024 — proved E minimum

google_syc

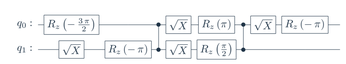
EXACT PHASE E=12 / L=8 / score ≈ 65.648 — E upper bound

ibm_cx

EXACT PHASE E=2 / L=0 / score ≈ 8.000 — proved E minimum

ibm_cz

EXACT PHASE E=2 / L=9 / score ≈ 26.410 — proved E minimum



ibm_ecr

EXACT PHASE E=2 / L=8 / score ≈ 24.404 — proved E minimum

PROJECTIVE E=2 / L=7 / score ≈ 22.408 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=10 / score ≈ 31.232 — proved E minimum

PROJECTIVE E=2 / L=10 / score ≈ 30.832 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=10 / score ≈ 30.428 — proved E minimum

PROJECTIVE E=2 / L=10 / score ≈ 30.028 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

ionq_forte

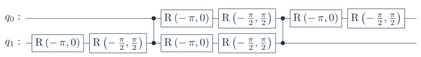
EXACT PHASE E=2 / L=12 / score ≈ 36.028 — proved E minimum

PROJECTIVE E=2 / L=12 / score ≈ 35.628 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=8 / score ≈ 25.632 — proved E minimum



iswap_u

EXACT PHASE E=1 / L=3 / score ≈ 10.818 — proved E minimum

quantinuum_zzmax

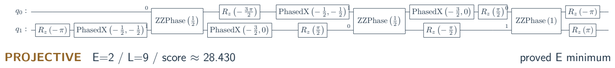
EXACT PHASE E=5 / L=25 / score ≈ 75.478 — E upper bound

PROJECTIVE E=2 / L=6 / score ≈ 21.220 — proved E minimum

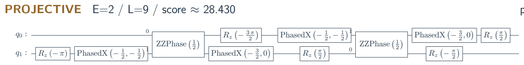
$U = -iC$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=3 / L=11 / score ≈ 36.436 — E upper bound



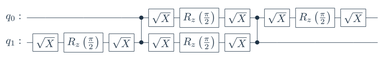
PROJECTIVE E=2 / L=9 / score ≈ 28.430 — proved E minimum



$U = -iC$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=2 / L=12 / score ≈ 32.808 — proved E minimum



sqrt_iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 16.824 — proved E minimum

sqrt_swap_u

EXACT PHASE E=4 / L=8 / score ≈ 34.848 — E upper bound

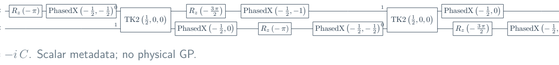
PROJECTIVE E=4 / L=8 / score ≈ 34.448 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=5 / L=30 / score ≈ 87.328 — E upper bound

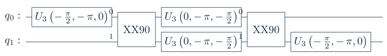
PROJECTIVE E=2 / L=10 / score ≈ 30.444 — E upper bound



$U = -iC$. Scalar metadata; no physical GP.

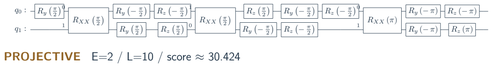
xx90_u

EXACT PHASE E=2 / L=4 / score ≈ 16.824 — proved E minimum

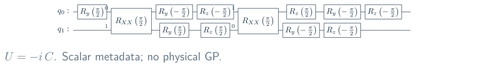


xx_rz_ry

EXACT PHASE E=3 / L=14 / score ≈ 42.434 — E upper bound



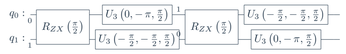
PROJECTIVE E=2 / L=10 / score ≈ 30.424 — proved E minimum



$U = -iC$. Scalar metadata; no physical GP.

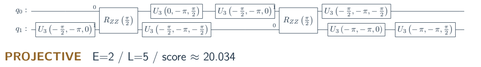
zx_u

EXACT PHASE E=2 / L=4 / score ≈ 18.028 — proved E minimum

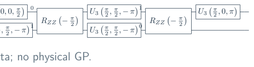


zz_u

EXACT PHASE E=2 / L=7 / score ≈ 24.046 — proved E minimum



PROJECTIVE E=2 / L=5 / score ≈ 20.034 — proved E minimum



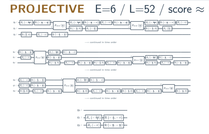
$U = -iC$. Scalar metadata; no physical GP.

Deutsch (θ)

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

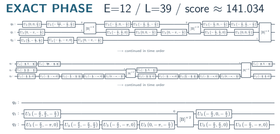
PROJECTIVE E=6 / L=52 / score ≈ 140.576 — E upper bound



$U = (-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=12 / L=39 / score ≈ 141.034 — E upper bound



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

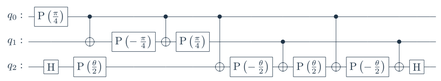
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

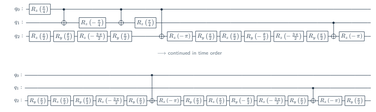
cx_h_phase

EXACT PHASE E=6 / L=9 / score ≈ 44.014 — E upper bound



cx_rz_ry

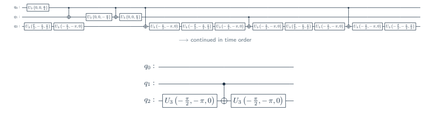
PROJECTIVE E=6 / L=27 / score ≈ 83.254 — E upper bound



$U = e^{\frac{i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

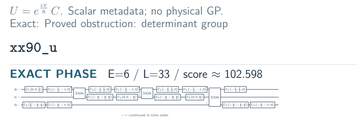
cx_u

EXACT PHASE E=6 / L=15 / score ≈ 59.290 — E upper bound



cz_rz_rx

PROJECTIVE E=6 / L=16 / score ≈ 59.832 — E upper bound



$U = e^{\frac{i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

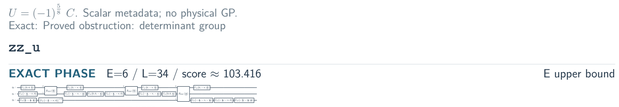
EXACT PHASE E=6 / L=11 / score ≈ 50.466 — E upper bound

google_cz

EXACT PHASE E=6 / L=11 / score ≈ 50.987 — E upper bound

google_fsim

EXACT PHASE E=6 / L=27 / score ≈ 89.507 — E upper bound

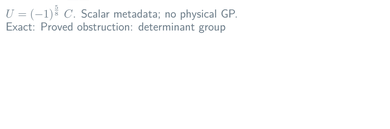


google_sqrt_iswap

EXACT PHASE E=12 / L=39 / score ≈ 139.955 — E upper bound

google_syc

EXACT PHASE E=36 / L=39 / score ≈ 233.555 — E upper bound



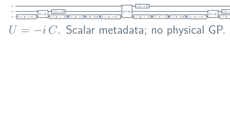
ibm_cx

PROJECTIVE E=6 / L=40 / score ≈ 107.846 — E upper bound

$U = -(-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=6 / L=32 / score ≈ 92.640 — E upper bound



$U = e^{\frac{i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=6 / L=64 / score ≈ 157.058 — E upper bound

$U = -(-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=6 / L=68 / score ≈ 181.093 — E upper bound

$U = -(-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=6 / L=68 / score ≈ 178.681 — E upper bound

$U = -(-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=6 / L=74 / score ≈ 195.481 — E upper bound

$U = (-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=6 / L=21 / score ≈ 72.084 — E upper bound

$U = e^{\frac{i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=12 / L=39 / score ≈ 139.834 — E upper bound

quantinuum_zzmax

PROJECTIVE E=6 / L=52 / score ≈ 140.673 — E upper bound

$U = (-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=6 / L=52 / score ≈ 141.885 — E upper bound

$U = (-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=6 / L=32 / score ≈ 92.640 — E upper bound

$U = e^{\frac{i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=12 / L=39 / score ≈ 138.634 — E upper bound

sqrt_swap_u

EXACT PHASE E=12 / L=39 / score ≈ 139.034 — E upper bound

PROJECTIVE E=12 / L=39 / score ≈ 138.634 — E upper bound

$U = -i\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=6 / L=52 / score ≈ 141.921 — E upper bound

$U = (-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=6 / L=33 / score ≈ 102.598 — E upper bound

xx_rz_ry

PROJECTIVE E=6 / L=68 / score ≈ 173.748 — E upper bound

$U = (-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=6 / L=27 / score ≈ 89.374 — E upper bound

zz_u

EXACT PHASE E=6 / L=34 / score ≈ 103.416 — E upper bound

PROJECTIVE E=6 / L=33 / score ≈ 101.410 — E upper bound

$U = -i\, C$. Scalar metadata; no physical GP.

Diagonal

2 qubits / 32 preserved native bases / alphabetical panel order

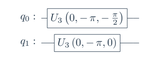
aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = (-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

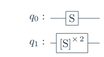


clifford_only

EXACT PHASE E=0 / L=3 / score ≈ 6.000 — proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.000 — proved E minimum

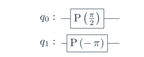


clifford_t_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.000 — proved E minimum

cx_h_phase

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum



cx_rz_ry

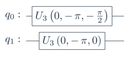
EXACT PHASE E=3 / L=12 / score ≈ 38.424 — E upper bound

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = (-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum



cz_rz_rx

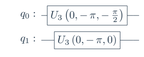
EXACT PHASE E=3 / L=24 / score ≈ 64.448 — E upper bound

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = (-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.

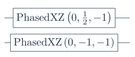
cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum



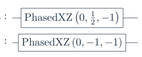
google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum



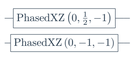
google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum



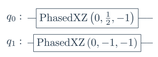
google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum



google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum



ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.406 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = -(-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.406 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = -(-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.

ibm_ecr

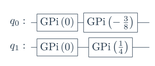
EXACT PHASE E=0 / L=6 / score ≈ 12.406 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = -(-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=4 / score ≈ 9.008 — proved E minimum



$U = -(-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

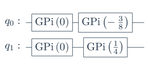
ionq_aria_full

PROJECTIVE E=0 / L=4 / score ≈ 9.008 — proved E minimum

$U = -(-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=4 / score ≈ 9.008 — proved E minimum

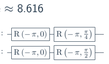


$U = -(-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

EXACT PHASE E=3 / L=22 / score ≈ 60.488 — E upper bound

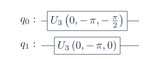
PROJECTIVE E=0 / L=4 / score ≈ 8.616 — proved E minimum



$U = (-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.

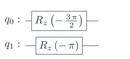
iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum



quantinuum_zzmax

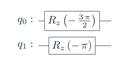
PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum



$U = (-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum



$U = (-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

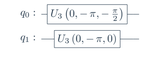
EXACT PHASE E=0 / L=6 / score ≈ 12.406 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = -(-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.

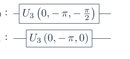
sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum



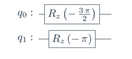
sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum



tket_tk2

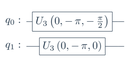
PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum



$U = (-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

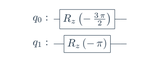
xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum



xx_rz_ry

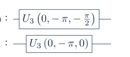
PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum



$U = (-1)^{\frac{3}{4}}\ C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

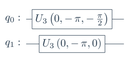
zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum



zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum



ECR

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

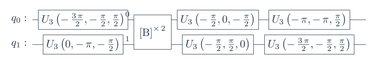
EXACT PHASE E=2 / L=9 / score ≈ 27.836 — E upper bound

PROJECTIVE E=1 / L=7 / score ≈ 19.826 — proved E minimum

$U = \frac{\sqrt[4]{-1}\sqrt{2}(1+i)}{2}\, C$. Scalar metadata; no physical GP.

berkeley_b

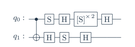
EXACT PHASE E=2 / L=6 / score ≈ 22.436 — proved E minimum



clifford_only

EXACT PHASE E=1 / L=10 / score ≈ 24.000 — proved E minimum

PROJECTIVE E=1 / L=8 / score ≈ 20.000 — proved E minimum

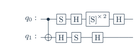


$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=1 / L=9 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=1 / L=8 / score ≈ 20.000 — proved E minimum



$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=1 / L=11 / score ≈ 26.000 — proved E minimum

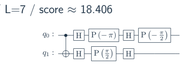
PROJECTIVE E=1 / L=10 / score ≈ 24.000 — proved E minimum

$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=1 / L=7 / score ≈ 18.608 — proved E minimum

PROJECTIVE E=1 / L=7 / score ≈ 18.406 — proved E minimum



$U = \sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

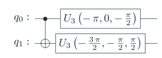
EXACT PHASE E=4 / L=16 / score ≈ 51.032 — E upper bound

PROJECTIVE E=1 / L=5 / score ≈ 14.810 — proved E minimum

$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

cx_u

EXACT PHASE E=1 / L=2 / score ≈ 8.812 — proved E minimum



cz_rz_rx

EXACT PHASE E=4 / L=29 / score ≈ 79.458 — E upper bound

PROJECTIVE E=1 / L=7 / score ≈ 19.214 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=1 / L=4 / score ≈ 13.224 — proved E minimum

PROJECTIVE E=1 / L=3 / score ≈ 10.818 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=1 / L=4 / score ≈ 13.424 — proved E minimum

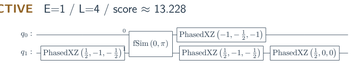
PROJECTIVE E=1 / L=4 / score ≈ 13.224 — proved E minimum

$U = (-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 13.428 — proved E minimum

PROJECTIVE E=1 / L=4 / score ≈ 13.228 — proved E minimum



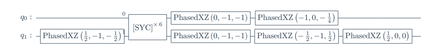
$U = (-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

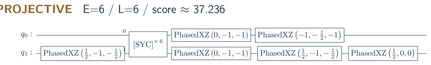
EXACT PHASE E=2 / L=6 / score ≈ 21.636 — proved E minimum

google_syc

EXACT PHASE E=6 / L=6 / score ≈ 37.436 — E upper bound



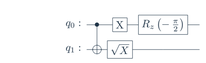
PROJECTIVE E=6 / L=6 / score ≈ 37.236 — E upper bound



$U = (-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

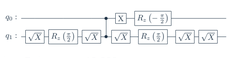
ibm_cx

EXACT PHASE E=1 / L=3 / score ≈ 10.202 — proved E minimum



ibm_cz

EXACT PHASE E=1 / L=9 / score ≈ 22.606 — proved E minimum



PROJECTIVE E=1 / L=7 / score ≈ 18.808 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

ionq_aria

EXACT PHASE E=1 / L=7 / score ≈ 20.420 — proved E minimum

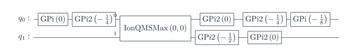
PROJECTIVE E=1 / L=7 / score ≈ 20.020 — proved E minimum

$U = \frac{\sqrt[4]{-1}\sqrt{2}(1+i)}{2}\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=1 / L=7 / score ≈ 20.018 — proved E minimum

PROJECTIVE E=1 / L=7 / score ≈ 19.618 — proved E minimum



$U = \frac{\sqrt[4]{-1}\sqrt{2}(1+i)}{2}\, C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=1 / L=8 / score ≈ 22.818 — proved E minimum

PROJECTIVE E=1 / L=8 / score ≈ 22.418 — proved E minimum

$U = \frac{\sqrt[4]{-1}\sqrt{2}(1+i)}{2}\, C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=4 / L=24 / score ≈ 69.296 — E upper bound

PROJECTIVE E=1 / L=6 / score ≈ 17.424 — proved E minimum

$U = (-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

iswap_u

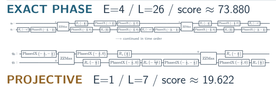
EXACT PHASE E=2 / L=3 / score ≈ 14.618 — proved E minimum

PROJECTIVE E=2 / L=3 / score ≈ 14.218 — proved E minimum

$U = -i\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=4 / L=26 / score ≈ 73.880 — E upper bound



PROJECTIVE E=1 / L=7 / score ≈ 19.622 — proved E minimum

$U = \frac{\sqrt[4]{-1}\sqrt{2}(1+i)}{2}\, C$. Scalar metadata; no physical GP.

quantinuum_zzphase

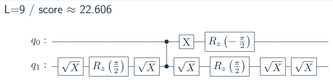
EXACT PHASE E=2 / L=9 / score ≈ 27.830 — E upper bound

PROJECTIVE E=1 / L=7 / score ≈ 19.824 — proved E minimum

$U = \frac{\sqrt[4]{-1}\sqrt{2}(1+i)}{2}\, C$. Scalar metadata; no physical GP.

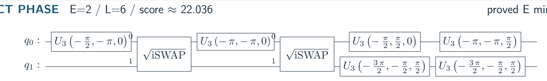
rigetti_cz

EXACT PHASE E=1 / L=9 / score ≈ 22.606 — proved E minimum



sqrt_iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 22.036 — proved E minimum



sqrt_swap_u

EXACT PHASE E=2 / L=6 / score ≈ 22.636 — proved E minimum

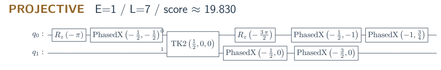
PROJECTIVE E=2 / L=6 / score ≈ 22.036 — proved E minimum

$U = (-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=4 / L=27 / score ≈ 76.714 — E upper bound

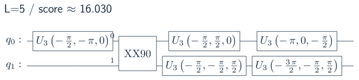
PROJECTIVE E=1 / L=7 / score ≈ 19.830 — proved E minimum



$U = \frac{\sqrt[4]{-1}\sqrt{2}(1+i)}{2}\, C$. Scalar metadata; no physical GP.

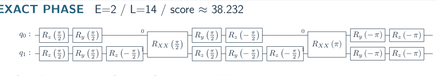
xx90_u

EXACT PHASE E=1 / L=5 / score ≈ 16.030 — proved E minimum



xx_rz_ry

EXACT PHASE E=2 / L=14 / score ≈ 38.232 — E upper bound

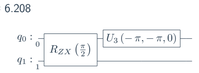


PROJECTIVE E=1 / L=10 / score ≈ 26.222 — proved E minimum

$U = i\, C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=1 / L=1 / score ≈ 6.208 — proved E minimum



zz_u

EXACT PHASE E=1 / L=3 / score ≈ 10.620 — proved E minimum

fFredkin

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=17 / L=162 / score ≈ 434.912 — E upper bound

PROJECTIVE E=5 / L=49 / score ≈ 131.362 — E upper bound

$U = -\frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=10 / L=42 / score ≈ 143.052 — E upper bound

clifford_only

Exact: Proved obstruction: not a clifford
Projective: Proved obstruction: not a clifford

clifford_t_cx

EXACT PHASE E=5 / L=50 / score ≈ 120.000 — E upper bound

PROJECTIVE E=5 / L=42 / score ≈ 104.000 — E upper bound

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

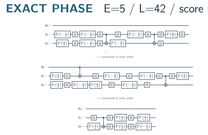
EXACT PHASE E=5 / L=52 / score ≈ 124.000 — E upper bound

PROJECTIVE E=5 / L=48 / score ≈ 116.000 — E upper bound

$U = -1\,C$. Scalar metadata; no physical GP.

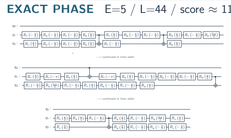
cx_h_phase

EXACT PHASE E=5 / L=42 / score ≈ 110.440 — E upper bound



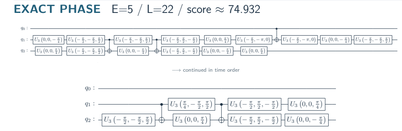
cx_rz_ry

EXACT PHASE E=5 / L=44 / score ≈ 118.888 — E upper bound



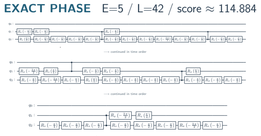
cx_u

EXACT PHASE E=5 / L=22 / score ≈ 74.932 — E upper bound



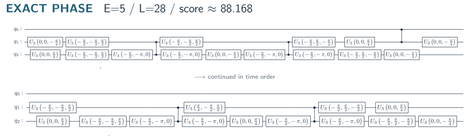
cz_rz_rx

EXACT PHASE E=5 / L=42 / score ≈ 114.884 — E upper bound



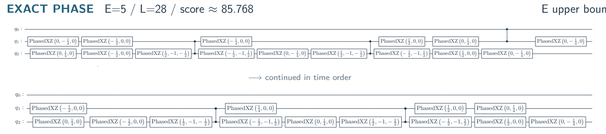
cz_u

EXACT PHASE E=5 / L=28 / score ≈ 88.168 — E upper bound



google_cz

EXACT PHASE E=5 / L=28 / score ≈ 85.768 — E upper bound



google_fsim

EXACT PHASE E=3 / L=12 / score ≈ 41.684 — E upper bound

google_sqrt_iswap

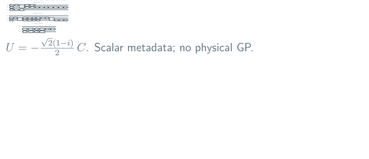
EXACT PHASE E=4 / L=16 / score ≈ 53.896 — E upper bound

PROJECTIVE E=4 / L=16 / score ≈ 53.296 — E upper bound

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=30 / L=42 / score ≈ 215.452 — E upper bound



ibm_cx

EXACT PHASE E=5 / L=38 / score ≈ 102.036 — E upper bound

PROJECTIVE E=5 / L=37 / score ≈ 100.036 — E upper bound

$U = \sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=5 / L=54 / score ≈ 135.652 — E upper bound

PROJECTIVE E=5 / L=54 / score ≈ 135.248 — E upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=5 / L=54 / score ≈ 134.238 — E upper bound

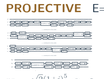
PROJECTIVE E=5 / L=53 / score ≈ 132.238 — E upper bound

$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=17 / L=161 / score ≈ 440.224 — E upper bound

PROJECTIVE E=5 / L=65 / score ≈ 169.960 — E upper bound

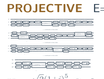


$U = \frac{\sqrt{2}(1+i)^5}{8}\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=17 / L=161 / score ≈ 433.390 — E upper bound

PROJECTIVE E=5 / L=65 / score ≈ 167.950 — E upper bound



$U = \frac{\sqrt{2}(1+i)^5}{8}\,C$. Scalar metadata; no physical GP.

ionq_forte

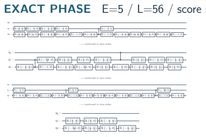
EXACT PHASE E=17 / L=178 / score ≈ 480.990 — E upper bound

PROJECTIVE E=5 / L=70 / score ≈ 181.950 — E upper bound

$U = \frac{\sqrt{2}i(1-i)}{2}\,C$. Scalar metadata; no physical GP.

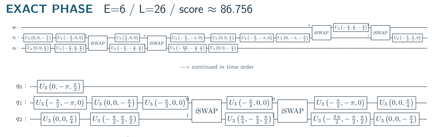
iqm_cz

EXACT PHASE E=5 / L=56 / score ≈ 149.824 — E upper bound



iswap_u

EXACT PHASE E=6 / L=26 / score ≈ 86.756 — E upper bound



quantinuum_zzmax

EXACT PHASE E=17 / L=117 / score ≈ 333.334 — E upper bound

PROJECTIVE E=5 / L=47 / score ≈ 126.338 — E upper bound

$U = -\frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=17 / L=120 / score ≈ 343.970 — E upper bound

PROJECTIVE E=5 / L=47 / score ≈ 127.348 — E upper bound

$U = -\frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

rigetti_cz

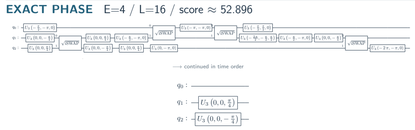
EXACT PHASE E=5 / L=54 / score ≈ 135.652 — E upper bound

PROJECTIVE E=5 / L=54 / score ≈ 135.248 — E upper bound

$U = i\,C$. Scalar metadata; no physical GP.

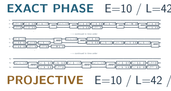
sqrt_iswap_u

EXACT PHASE E=4 / L=16 / score ≈ 52.896 — E upper bound

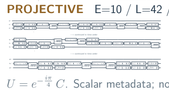


sqrt_swap_u

EXACT PHASE E=10 / L=42 / score ≈ 141.852 — E upper bound



PROJECTIVE E=10 / L=42 / score ≈ 141.052 — E upper bound



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

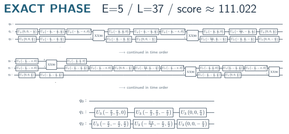
EXACT PHASE E=15 / L=96 / score ≈ 280.772 — E upper bound

PROJECTIVE E=3 / L=16 / score ≈ 50.860 — E upper bound

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

xx90_u

EXACT PHASE E=5 / L=37 / score ≈ 111.022 — E upper bound



xx_rz_ry

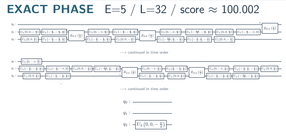
EXACT PHASE E=17 / L=186 / score ≈ 477.406 — E upper bound

PROJECTIVE E=5 / L=77 / score ≈ 191.164 — E upper bound

$U = -\frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

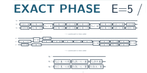
zx_u

EXACT PHASE E=5 / L=32 / score ≈ 100.002 — E upper bound



zz_u

EXACT PHASE E=5 / L=38 / score ≈ 107.638 — E upper bound



PROJECTIVE E=5 / L=37 / score ≈ 105.232 — E upper bound

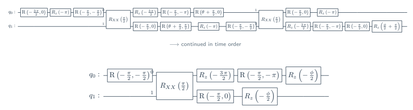
$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

fSim (θ, ϕ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

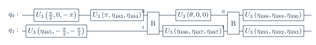
PROJECTIVE E=3 / L=22 / score ≈ 61.276 — minimum uniform E budget



$U = \eta_{482}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=7 / score ≈ 23.442 — minimum uniform E budget



Exact definitions: $\eta_{483}, \eta_{484}, \eta_{485}, \eta_{486}, \eta_{487}, \eta_{488}, \eta_{489}, \eta_{490}, \eta_{491}, \eta_{492}, \eta_{493}$ in the expression glossary.

PROJECTIVE E=2 / L=6 / score ≈ 21.436 — minimum uniform E budget

Exact definitions: $\eta_{485}, \eta_{486}, \eta_{487}, \eta_{488}, \eta_{489}, \eta_{490}, \eta_{491}, \eta_{492}, \eta_{493}, \eta_{494}$ in the expression glossary.

$U = \eta_{495}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

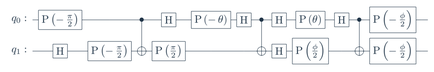
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

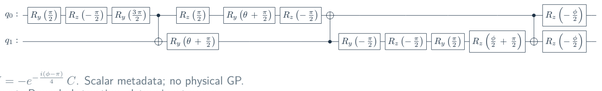
cx_h_phase

EXACT PHASE E=3 / L=14 / score ≈ 41.216 — minimum uniform E budget



cx_rz_ry

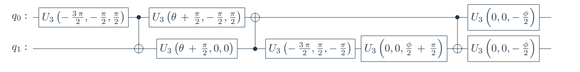
PROJECTIVE E=3 / L=13 / score ≈ 40.626 — minimum uniform E budget



$U = -e^{-\frac{i(\phi-\pi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

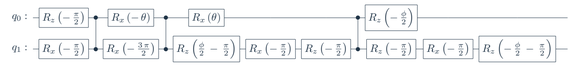
cx_u

EXACT PHASE E=3 / L=7 / score ≈ 28.642 — minimum uniform E budget



cz_rz_rx

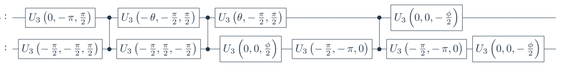
PROJECTIVE E=3 / L=12 / score ≈ 38.024 — minimum uniform E budget



$U = -(-1)^{\frac{3}{4}}\, e^{-\frac{i\phi}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

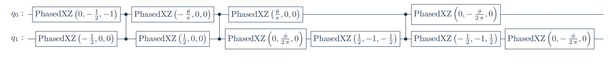
cz_u

EXACT PHASE E=3 / L=10 / score ≈ 35.260 — minimum uniform E budget



google_cz

EXACT PHASE E=3 / L=10 / score ≈ 35.711 — minimum uniform E budget



google_fsim

EXACT PHASE E=1 / L=0 / score ≈ 4.004 — minimum uniform E budget

google_sqrt_iswap

EXACT PHASE E=6 / L=21 / score ≈ 74.238 — E upper bound

PROJECTIVE E=6 / L=19 / score ≈ 68.965 — E upper bound

$U = \frac{\sqrt{2}(1-i)e^{-i\theta}}{2}\, C$. Scalar metadata; no physical GP.
Published uniform cap: $E \leq 3$. Current construction exceeds it; optimization remains.

google_syc

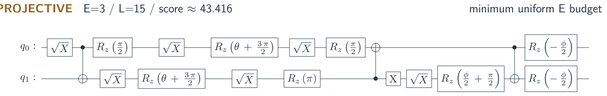
EXACT PHASE E=18 / L=21 / score ≈ 120.438 — E upper bound

PROJECTIVE E=18 / L=19 / score ≈ 115.765 — E upper bound

$U = -ie^{-i\theta}\, C$. Scalar metadata; no physical GP.

ibm_cx

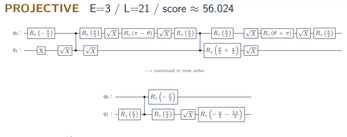
PROJECTIVE E=3 / L=15 / score ≈ 43.416 — minimum uniform E budget



$U = -e^{-\frac{i(\phi-\pi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

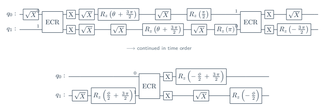
PROJECTIVE E=3 / L=21 / score ≈ 56.024 — minimum uniform E budget



$U = \sqrt[4]{-1}e^{-\frac{i\phi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

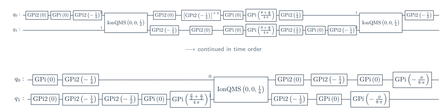
PROJECTIVE E=3 / L=23 / score ≈ 59.416 — minimum uniform E budget



$U = -ie^{-\frac{i(\phi-\pi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

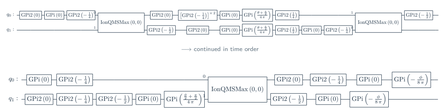
PROJECTIVE E=3 / L=31 / score ≈ 84.131 — minimum uniform E budget



$U = \eta_{496}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

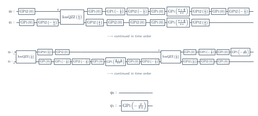
PROJECTIVE E=3 / L=31 / score ≈ 82.925 — minimum uniform E budget



$U = \eta_{496}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=3 / L=34 / score ≈ 91.325 — minimum uniform E budget



$U = \eta_{496}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

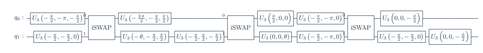
PROJECTIVE E=3 / L=19 / score ≈ 54.676 — minimum uniform E budget

$U = -(-1)^{\frac{3}{4}}\, e^{-\frac{i\phi}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=3 / L=15 / score ≈ 46.690 — minimum uniform E budget

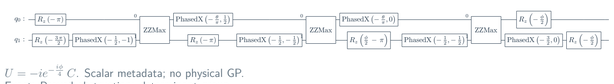
PROJECTIVE E=3 / L=12 / score ≈ 39.872 — minimum uniform E budget



$U = \eta_{497}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=3 / L=12 / score ≈ 38.897 — minimum uniform E budget



$U = -ie^{-\frac{i\phi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

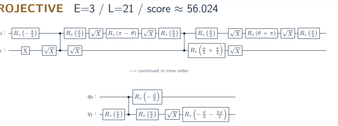
quantinuum_zzphase

PROJECTIVE E=3 / L=20 / score ≈ 57.653 — minimum uniform E budget

$U = e^{-\frac{i\phi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

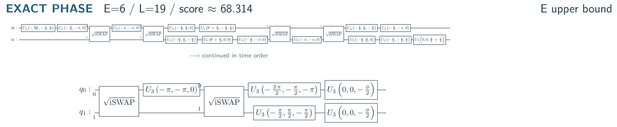
PROJECTIVE E=3 / L=21 / score ≈ 56.024 — minimum uniform E budget



$U = \sqrt[4]{-1}e^{-\frac{i\phi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=6 / L=19 / score ≈ 68.314 — E upper bound



Published uniform cap: $E \leq 3$. Current construction exceeds it; optimization remains.

sqrt_swap_u

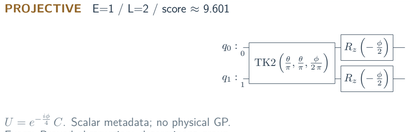
EXACT PHASE E=6 / L=19 / score ≈ 69.114 — E upper bound

PROJECTIVE E=6 / L=19 / score ≈ 68.314 — E upper bound

$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

tket_tk2

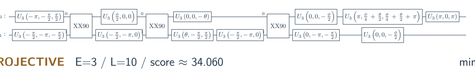
PROJECTIVE E=1 / L=2 / score ≈ 9.601 — minimum uniform E budget



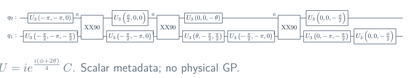
$U = e^{-\frac{i\phi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=3 / L=12 / score ≈ 39.272 — minimum uniform E budget



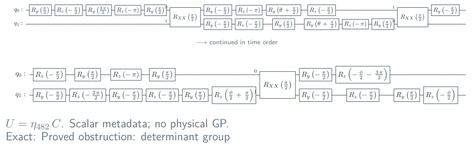
PROJECTIVE E=3 / L=10 / score ≈ 34.060 — minimum uniform E budget



$U = ie^{\frac{i(\phi+2\theta)}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

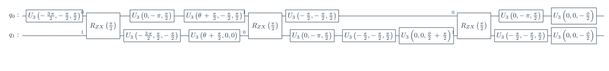
PROJECTIVE E=3 / L=32 / score ≈ 82.270 — minimum uniform E budget



$U = \eta_{482}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=3 / L=13 / score ≈ 43.684 — minimum uniform E budget



zz_u

EXACT PHASE E=3 / L=16 / score ≈ 47.902 — minimum uniform E budget

PROJECTIVE E=3 / L=14 / score ≈ 43.090 — minimum uniform E budget

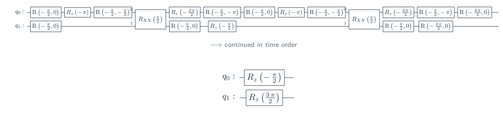
$U = e^{\frac{i\phi}{4}}\, C$. Scalar metadata; no physical GP.

FSWAP

2 qubits / 32 preserved native bases / alphabetical panel order

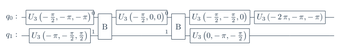
aqt_rxx

EXACT PHASE E=2 / L=18 / score ≈ 48.062 — proved E minimum



berkeley_b

EXACT PHASE E=2 / L=6 / score ≈ 21.436 — proved E minimum



PROJECTIVE E=2 / L=5 / score ≈ 19.030 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

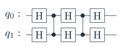
EXACT PHASE E=2 / L=4 / score ≈ 16.000 — proved E minimum

clifford_t_cx

EXACT PHASE E=2 / L=4 / score ≈ 16.000 — proved E minimum

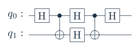
clifford_t_cz

EXACT PHASE E=2 / L=6 / score ≈ 20.000 — proved E minimum



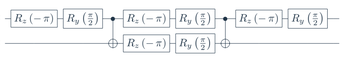
cx_h_phase

EXACT PHASE E=2 / L=4 / score ≈ 16.000 — proved E minimum



cx_rz_ry

EXACT PHASE E=2 / L=8 / score ≈ 24.816 — proved E minimum

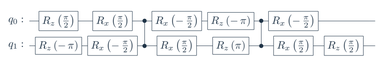


cx_u

EXACT PHASE E=2 / L=4 / score ≈ 16.824 — proved E minimum

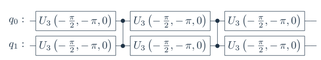
cz_rz_rx

EXACT PHASE E=2 / L=11 / score ≈ 31.622 — proved E minimum



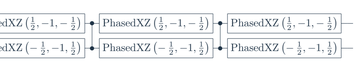
cz_u

EXACT PHASE E=2 / L=6 / score ≈ 21.236 — proved E minimum



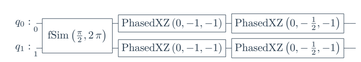
google_cz

EXACT PHASE E=2 / L=6 / score ≈ 22.436 — proved E minimum



google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 12.628 — proved E minimum

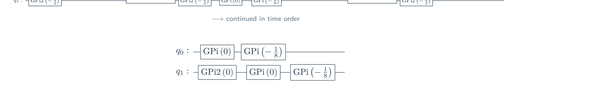


google_sqrt_iswap

EXACT PHASE E=2 / L=2 / score ≈ 12.412 — proved E minimum

google_syc

EXACT PHASE E=12 / L=16 / score ≈ 83.296 — E upper bound



ibm_cx

EXACT PHASE E=2 / L=12 / score ≈ 32.808 — proved E minimum

PROJECTIVE E=2 / L=11 / score ≈ 30.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=17 / score ≈ 42.418 — proved E minimum

PROJECTIVE E=2 / L=13 / score ≈ 34.414 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=10 / score ≈ 28.808 — proved E minimum

PROJECTIVE E=2 / L=9 / score ≈ 26.808 — proved E minimum

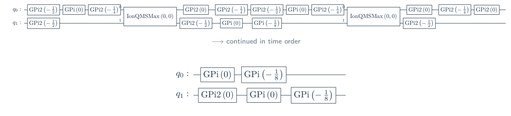
$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=21 / score ≈ 55.254 — proved E minimum

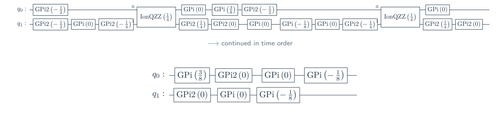
ionq_aria_full

EXACT PHASE E=2 / L=21 / score ≈ 54.450 — proved E minimum



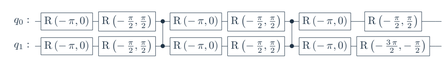
ionq_forte

EXACT PHASE E=2 / L=23 / score ≈ 60.050 — proved E minimum



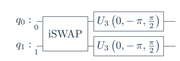
iqm_cz

EXACT PHASE E=2 / L=12 / score ≈ 34.448 — proved E minimum



iswap_u

EXACT PHASE E=1 / L=2 / score ≈ 8.412 — proved E minimum



quantinuum_zzmax

EXACT PHASE E=2 / L=17 / score ≈ 45.652 — proved E minimum

PROJECTIVE E=2 / L=10 / score ≈ 30.032 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

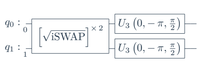
EXACT PHASE E=2 / L=17 / score ≈ 46.056 — proved E minimum

rigetti_cz

EXACT PHASE E=2 / L=18 / score ≈ 45.212 — proved E minimum

sqrt_iswap_u

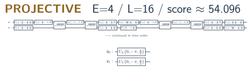
EXACT PHASE E=2 / L=2 / score ≈ 12.412 — proved E minimum



sqrt_swap_u

EXACT PHASE E=4 / L=16 / score ≈ 54.896 — E upper bound

PROJECTIVE E=4 / L=16 / score ≈ 54.096 — E upper bound



$U = -(-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=2 / score ≈ 8.810 — proved E minimum

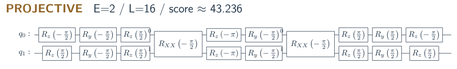
xx90_u

EXACT PHASE E=2 / L=2 / score ≈ 12.412 — proved E minimum

xx_rz_ry

EXACT PHASE E=2 / L=32 / score ≈ 78.468 — proved E minimum

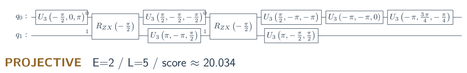
PROJECTIVE E=2 / L=16 / score ≈ 43.236 — proved E minimum



$U = -i\,C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=2 / L=7 / score ≈ 24.846 — proved E minimum

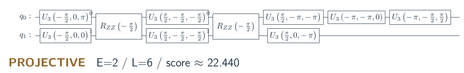


PROJECTIVE E=2 / L=5 / score ≈ 20.034 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=2 / L=8 / score ≈ 26.852 — proved E minimum



PROJECTIVE E=2 / L=6 / score ≈ 22.440 — proved E minimum

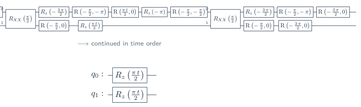
$U = i\,C$. Scalar metadata; no physical GP.

FSWAPPow $(t, global_{shift})$

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=2 / L=18 / score ≈ 48.062 — minimum uniform E budget



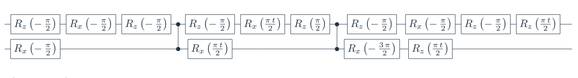
$U = -ie^{i\pi t\left(global_{shift}t+\frac{1}{2}\right)}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cz_rz_rx

PROJECTIVE E=2 / L=14 / score ≈ 38.828 — minimum uniform E budget



$U = e^{\frac{i\pi t\left(2global_{shift}t+1\right)}{2}}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

google_sqrt_iswap

EXACT PHASE E=2 / L=10 / score ≈ 29.260 — E upper bound

PROJECTIVE E=2 / L=8 / score ≈ 25.248 — E upper bound

$U = e^{i\pi global_{shift}t}C$. Scalar metadata; no physical GP.

ibm_ecr

PROJECTIVE E=2 / L=15 / score ≈ 39.818 — minimum uniform E budget

$U = ie^{\frac{i\pi t\left(2global_{shift}t+1\right)}{2}}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=2 / L=14 / score ≈ 39.856 — minimum uniform E budget

$U = e^{\frac{i\pi t\left(2global_{shift}t+1\right)}{2}}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=2 / L=20 / score ≈ 49.618 — minimum uniform E budget

$U = -e^{\frac{i\pi t\left(2global_{shift}t+1\right)}{2}}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

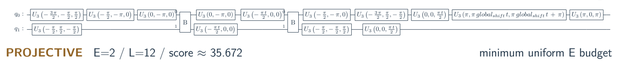
EXACT PHASE E=2 / L=9 / score ≈ 27.254 — minimum uniform E budget

PROJECTIVE E=2 / L=7 / score ≈ 23.242 — minimum uniform E budget

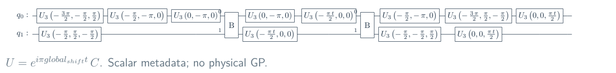
$U = \eta_{500}C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=2 / L=14 / score ≈ 39.684 — minimum uniform E budget



PROJECTIVE E=2 / L=12 / score ≈ 35.672 — minimum uniform E budget

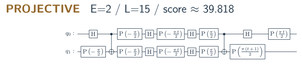


$U = e^{i\pi global_{shift}t}C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=2 / L=22 / score ≈ 53.824 — minimum uniform E budget

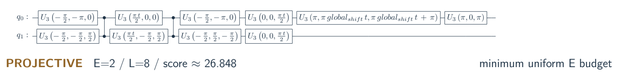
PROJECTIVE E=2 / L=15 / score ≈ 39.818 — minimum uniform E budget



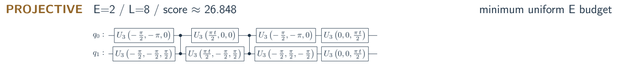
$U = e^{\frac{i\pi t\left(2global_{shift}t+1\right)}{2}}C$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=2 / L=10 / score ≈ 30.860 — minimum uniform E budget



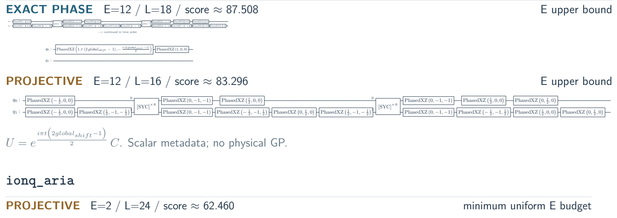
PROJECTIVE E=2 / L=8 / score ≈ 26.848 — minimum uniform E budget



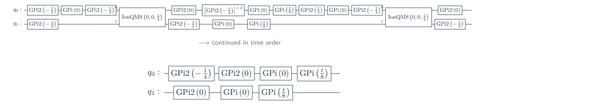
$U = e^{i\pi global_{shift}t}C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=18 / score ≈ 87.508 — E upper bound



PROJECTIVE E=12 / L=16 / score ≈ 83.296 — E upper bound



$U = e^{\frac{i\pi t\left(2global_{shift}t-1\right)}{2}}C$. Scalar metadata; no physical GP.

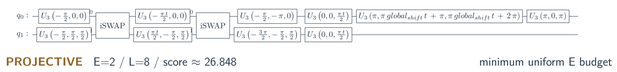
ionq_aria

PROJECTIVE E=2 / L=24 / score ≈ 62.460 — minimum uniform E budget

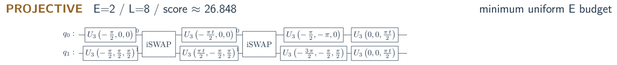
$U = ie^{i\pi t\left(global_{shift}t+\frac{1}{2}\right)}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=2 / L=10 / score ≈ 30.860 — minimum uniform E budget



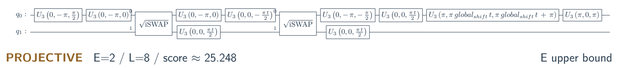
PROJECTIVE E=2 / L=8 / score ≈ 26.848 — minimum uniform E budget



$U = \eta_{498}C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=2 / L=10 / score ≈ 29.260 — E upper bound



PROJECTIVE E=2 / L=8 / score ≈ 25.248 — E upper bound

$U = e^{i\pi global_{shift}t}C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=2 / L=31 / score ≈ 76.066 — minimum uniform E budget

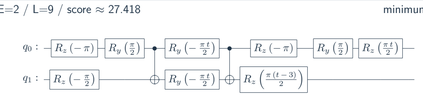
$U = -ie^{i\pi t\left(global_{shift}t+\frac{1}{2}\right)}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_rz_ry

PROJECTIVE E=2 / L=9 / score ≈ 27.418 — minimum uniform E budget



$U = e^{\frac{i\pi t\left(2global_{shift}t+1\right)}{2}}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

google_cz

EXACT PHASE E=2 / L=10 / score ≈ 30.460 — minimum uniform E budget

PROJECTIVE E=2 / L=8 / score ≈ 26.248 — minimum uniform E budget

$U = e^{\frac{i\pi t\left(2global_{shift}t-1\right)}{2}}C$. Scalar metadata; no physical GP.

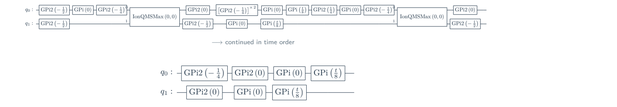
ibm_cx

PROJECTIVE E=2 / L=14 / score ≈ 37.212 — minimum uniform E budget

$U = ie^{i\pi t\left(global_{shift}t+\frac{1}{2}\right)}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

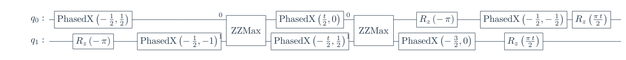
PROJECTIVE E=2 / L=24 / score ≈ 61.656 — minimum uniform E budget



$U = ie^{i\pi t\left(global_{shift}t+\frac{1}{2}\right)}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzmax

PROJECTIVE E=2 / L=10 / score ≈ 30.232 — minimum uniform E budget



$U = \eta_{499}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_swap_u

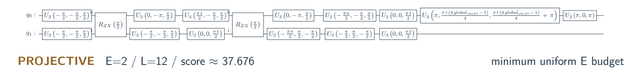
EXACT PHASE E=4 / L=18 / score ≈ 59.308 — E upper bound

PROJECTIVE E=4 / L=16 / score ≈ 54.096 — E upper bound

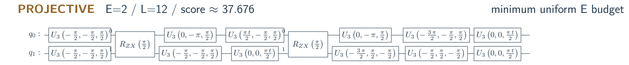
$U = -ie^{i\pi t\left(global_{shift}t-\frac{1}{4}\right)}C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=2 / L=14 / score ≈ 42.488 — minimum uniform E budget



PROJECTIVE E=2 / L=12 / score ≈ 37.676 — minimum uniform E budget



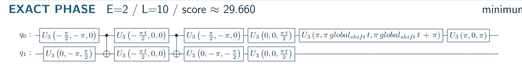
$U = e^{i\pi t\left(global_{shift}t-\frac{1}{4}\right)}C$. Scalar metadata; no physical GP.

clifford_t_cx

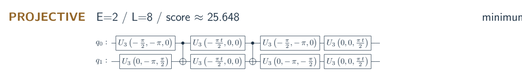
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_u

EXACT PHASE E=2 / L=10 / score ≈ 29.660 — minimum uniform E budget



PROJECTIVE E=2 / L=8 / score ≈ 25.648 — minimum uniform E budget

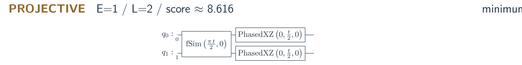


$U = e^{i\pi global_{shift}t}C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 12.628 — minimum uniform E budget

PROJECTIVE E=1 / L=2 / score ≈ 8.616 — minimum uniform E budget



$U = e^{i\pi global_{shift}t}C$. Scalar metadata; no physical GP.

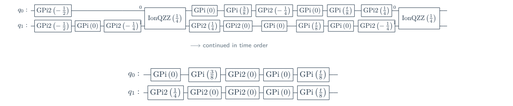
ibm_cz

PROJECTIVE E=2 / L=20 / score ≈ 49.618 — minimum uniform E budget

$U = -e^{\frac{i\pi t\left(2global_{shift}t+1\right)}{2}}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

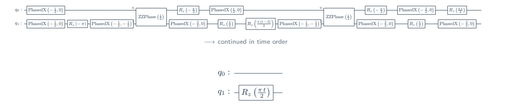
PROJECTIVE E=2 / L=26 / score ≈ 67.256 — minimum uniform E budget



$U = ie^{i\pi t\left(global_{shift}t+\frac{1}{2}\right)}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=2 / L=17 / score ≈ 46.056 — minimum uniform E budget



$U = -ie^{i\pi t\left(global_{shift}t+\frac{1}{2}\right)}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

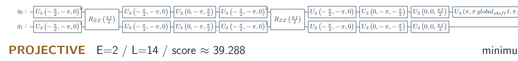
tket_tk2

PROJECTIVE E=1 / L=2 / score ≈ 8.810 — minimum uniform E budget

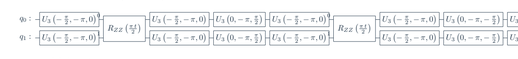
$U = e^{\frac{i\pi t\left(2global_{shift}t+1\right)}{2}}C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zz_u

EXACT PHASE E=2 / L=16 / score ≈ 43.300 — minimum uniform E budget



PROJECTIVE E=2 / L=14 / score ≈ 39.288 — minimum uniform E budget



$U = e^{i\pi global_{shift}t}C$. Scalar metadata; no physical GP.

Givens (θ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=2 / L=37 / score ≈ 90.710 — minimum uniform E budget

PROJECTIVE E=2 / L=21 / score ≈ 54.866 — minimum uniform E budget

$U = i\,C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=2 / L=14 / score ≈ 40.484 — minimum uniform E budget

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

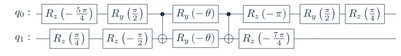
EXACT PHASE E=2 / L=24 / score ≈ 58.628 — minimum uniform E budget

PROJECTIVE E=2 / L=17 / score ≈ 44.622 — minimum uniform E budget

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

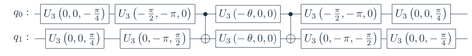
cx_rz_ry

EXACT PHASE E=2 / L=10 / score ≈ 30.220 — minimum uniform E budget



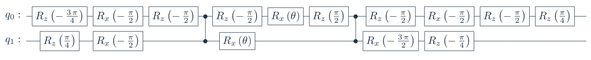
cx_u

EXACT PHASE E=2 / L=10 / score ≈ 30.460 — minimum uniform E budget



cz_rz_rx

EXACT PHASE E=2 / L=15 / score ≈ 41.430 — minimum uniform E budget



cz_u

EXACT PHASE E=2 / L=10 / score ≈ 31.660 — minimum uniform E budget

google_cz

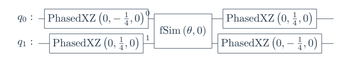
EXACT PHASE E=2 / L=12 / score ≈ 36.393 — minimum uniform E budget

PROJECTIVE E=2 / L=10 / score ≈ 31.721 — minimum uniform E budget

$U = e^{-i\theta}\,C$. Scalar metadata; no physical GP.

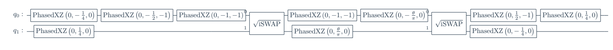
google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 13.628 — minimum uniform E budget



google_sqrt_iswap

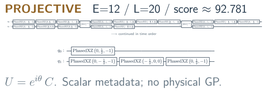
EXACT PHASE E=2 / L=10 / score ≈ 30.721 — minimum uniform E budget



google_syc

EXACT PHASE E=12 / L=22 / score ≈ 97.453 — E upper bound

PROJECTIVE E=12 / L=20 / score ≈ 92.781 — E upper bound



$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=19 / score ≈ 47.624 — minimum uniform E budget

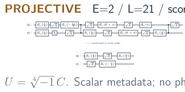
PROJECTIVE E=2 / L=18 / score ≈ 45.624 — minimum uniform E budget

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=22 / score ≈ 54.422 — minimum uniform E budget

PROJECTIVE E=2 / L=21 / score ≈ 52.422 — minimum uniform E budget



$U = \sqrt[4]{-\mathbb{I}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=16 / score ≈ 42.824 — minimum uniform E budget

PROJECTIVE E=2 / L=16 / score ≈ 42.622 — minimum uniform E budget

$U = -(-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=36 / score ≈ 90.145 — minimum uniform E budget

PROJECTIVE E=2 / L=36 / score ≈ 89.745 — minimum uniform E budget

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=36 / score ≈ 89.341 — minimum uniform E budget

PROJECTIVE E=2 / L=36 / score ≈ 88.941 — minimum uniform E budget

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

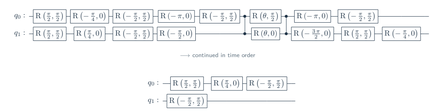
EXACT PHASE E=2 / L=38 / score ≈ 94.941 — minimum uniform E budget

PROJECTIVE E=2 / L=38 / score ≈ 94.541 — minimum uniform E budget

$U = i\,C$. Scalar metadata; no physical GP.

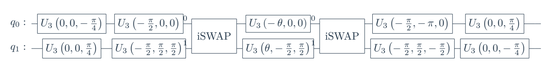
iqm_cz

EXACT PHASE E=2 / L=20 / score ≈ 54.280 — minimum uniform E budget



iswap_u

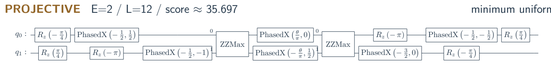
EXACT PHASE E=2 / L=10 / score ≈ 31.660 — minimum uniform E budget



quantinuum_zzmax

EXACT PHASE E=5 / L=31 / score ≈ 89.955 — E upper bound

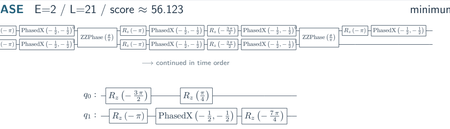
PROJECTIVE E=2 / L=12 / score ≈ 35.697 — minimum uniform E budget



$U = \frac{\sqrt[8]{-1}\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=2 / L=21 / score ≈ 56.123 — minimum uniform E budget



rigetti_cz

EXACT PHASE E=2 / L=22 / score ≈ 54.422 — minimum uniform E budget

PROJECTIVE E=2 / L=21 / score ≈ 52.422 — minimum uniform E budget

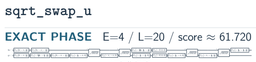
$U = \sqrt[4]{-\mathbb{I}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

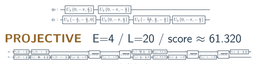
EXACT PHASE E=2 / L=10 / score ≈ 30.060 — minimum uniform E budget

sqrt_swap_u

EXACT PHASE E=4 / L=20 / score ≈ 61.720 — E upper bound



PROJECTIVE E=4 / L=20 / score ≈ 61.320 — E upper bound



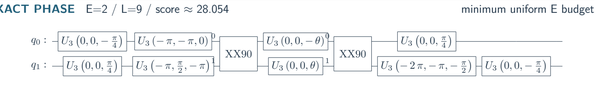
$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=4 / score ≈ 14.275 — minimum uniform E budget

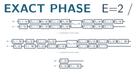
xx90_u

EXACT PHASE E=2 / L=9 / score ≈ 28.054 — minimum uniform E budget



xx_rz_ry

EXACT PHASE E=2 / L=37 / score ≈ 87.478 — minimum uniform E budget



PROJECTIVE E=2 / L=29 / score ≈ 71.662 — minimum uniform E budget

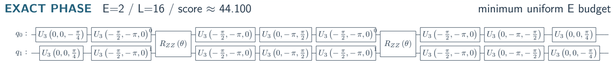
$U = i\,C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=2 / L=16 / score ≈ 44.900 — minimum uniform E budget

zz_u

EXACT PHASE E=2 / L=16 / score ≈ 44.100 — minimum uniform E budget



GP

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

clifford_only

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_t_cx

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_t_cz

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.008 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.673 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.673 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.673 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.673 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = e^{i\theta}\,C$. Scalar metadata; no physical GP.

GPi (ϕ)

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=3 / score ≈ 6.008 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.006 proved E minimum

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.006 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.006 proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

google_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

google_fsim

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

google_syc

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.004 proved E minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.004 proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.004 proved E minimum

ionq_aria

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved E minimum

ionq_forte

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved E minimum

iqm_cz

PROJECTIVE E=0 / L=7 / score ≈ 15.628 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=3 / score ≈ 6.008 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=3 / score ≈ 6.008 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.004 proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

tket_tk2

PROJECTIVE E=0 / L=3 / score ≈ 6.008 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.006 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

zz_u

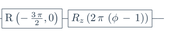
EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

GPi2 (ϕ)

1 qubits / 32 preserved native bases / alphabetical panel order

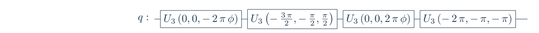
aqt_rxx

EXACT PHASE E=0 / L=3 / score ≈ 6.208 proved E minimum



berkeley_b

EXACT PHASE E=0 / L=4 / score ≈ 8.624 proved E minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.618 proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=11 / score ≈ 22.812 proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.206 proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=5 / score ≈ 10.610 proved E minimum

cx_u

EXACT PHASE E=0 / L=4 / score ≈ 8.624 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.618 proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

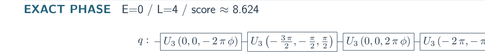
cz_rz_rx

EXACT PHASE E=0 / L=3 / score ≈ 6.206 proved E minimum

cz_u

EXACT PHASE E=0 / L=4 / score ≈ 8.624 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.618 proved E minimum



$U = -1\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum

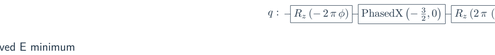
PROJECTIVE E=0 / L=3 / score ≈ 6.218 proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.218 proved E minimum



$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.218 proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.218 proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.610 proved E minimum

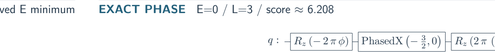
PROJECTIVE E=0 / L=3 / score ≈ 6.004 proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.610 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.004 proved E minimum



$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=9 / score ≈ 18.610 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.004 proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved E minimum

ionq_forte

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved E minimum

iqm_cz

EXACT PHASE E=0 / L=8 / score ≈ 17.832 proved E minimum

PROJECTIVE E=0 / L=7 / score ≈ 15.828 proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

iswap_u

EXACT PHASE E=0 / L=4 / score ≈ 8.624 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.618 proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=3 / score ≈ 6.208 proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=3 / score ≈ 6.208 proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.610 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.004 proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=4 / score ≈ 8.624 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.618 proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=4 / score ≈ 8.624 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.618 proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=3 / score ≈ 6.208 proved E minimum

xx90_u

EXACT PHASE E=0 / L=4 / score ≈ 8.624 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.618 proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=5 / score ≈ 10.610 proved E minimum

zx_u

EXACT PHASE E=0 / L=4 / score ≈ 8.624 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.618 proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=4 / score ≈ 8.624 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.618 proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

H

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

clifford_only

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved E minimum

clifford_t_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved E minimum

cx_h_phase

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved E minimum

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

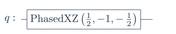
PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

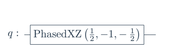


$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum



$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

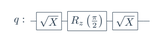
EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved E minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved E minimum



ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved E minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved E minimum

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

H_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=5 / score ≈ 10.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

ibm_cx

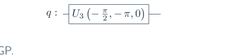
EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

H_q2

1 qubits / 32 preserved native bases / alphabetical panel order

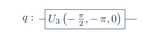
aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum



$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

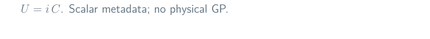
EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.408 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

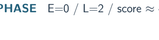
cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum



$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.806 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.806 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.806 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.408 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

H_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=13 / score ≈ 26.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.808 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

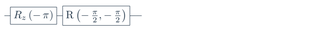
$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

H_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

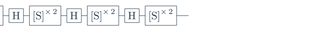
EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.008 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

H_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=15 / score ≈ 30.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=13 / score ≈ 26.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=13 / score ≈ 26.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.808 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

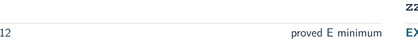
PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum



$U = -1\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

H_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=13 / score ≈ 26.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.408 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.406 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.406 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.406 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.408 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

H_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

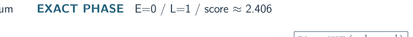
EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum



google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

ibm_cx

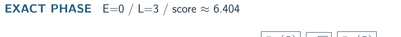
EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HPow $(t, global_{shift})$

1 qubits / 32 preserved native bases / alphabetical panel order

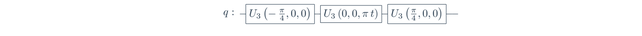
aqt_rxx

PROJECTIVE E=0 / L=3 / score ≈ 7.210 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.818 proved E minimum

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

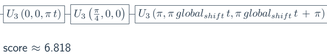
clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=16 / score ≈ 33.216 proved E minimum

PROJECTIVE E=0 / L=9 / score ≈ 19.210 proved E minimum



$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.806 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.818 proved E minimum

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=5 / score ≈ 11.210 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.818 proved E minimum

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=5 / score ≈ 11.230 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 7.218 proved E minimum

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=5 / score ≈ 11.230 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 7.218 proved E minimum

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=5 / score ≈ 11.230 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 7.218 proved E minimum

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=5 / score ≈ 11.230 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 7.218 proved E minimum

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=0 / L=8 / score ≈ 16.808 proved E minimum

$U = -e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=0 / L=8 / score ≈ 16.808 proved E minimum

$U = -e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=0 / L=8 / score ≈ 16.808 proved E minimum

$U = -e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=0 / L=10 / score ≈ 22.420 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=10 / score ≈ 22.420 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=10 / score ≈ 22.420 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

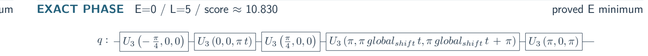
iqm_cz

PROJECTIVE E=0 / L=5 / score ≈ 12.020 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.818 proved E minimum

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=3 / score ≈ 7.210 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=3 / score ≈ 7.210 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=0 / L=8 / score ≈ 16.808 proved E minimum

$U = -e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.818 proved E minimum

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.818 proved E minimum

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=3 / score ≈ 7.210 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.818 proved E minimum

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.806 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.818 proved E minimum

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.818 proved E minimum

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

HS

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

clifford_only

EXACT PHASE E=0 / L=2 / score ≈ 4.000 proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.000 proved E minimum

clifford_t_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.000 proved E minimum

cx_h_phase

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved E minimum

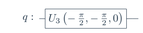
cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

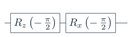
cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum



cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

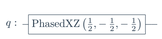
cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

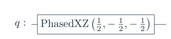


$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

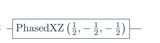


$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

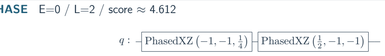
PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved E minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved E minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

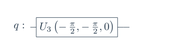
EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum



tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

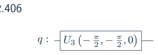
$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum



HS_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.406 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

HS_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.208 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

HS_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.808 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HS_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

clifford_only

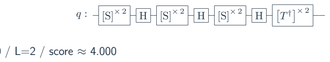
EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.208 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

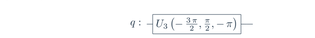
cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum



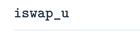
cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

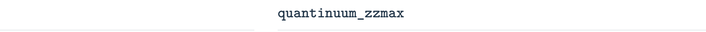
cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum



google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

HS_q5

1 qubits / 32 preserved native bases / alphabetical panel order

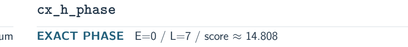
aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

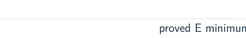
EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.808 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.806 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.806 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.806 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HS_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.206 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

HS_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

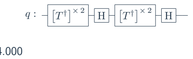
EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

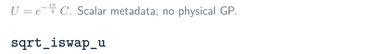
EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HSdg

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

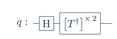
EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

clifford_only

EXACT PHASE E=0 / L=4 / score ≈ 8.000 proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.000 proved E minimum

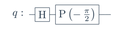


clifford_t_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.000 proved E minimum

cx_h_phase

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved E minimum



cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

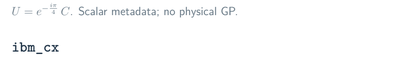
cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

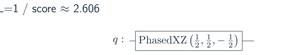
PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

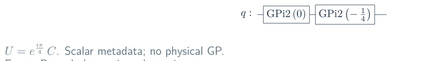
EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved E minimum

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved E minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

HSdg_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=4 / score ≈ 8.000 — proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

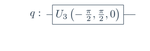
EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HSdg_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

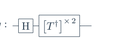
PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum



$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.206 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

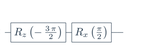
EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

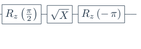
PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum



$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

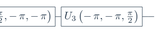
PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum



$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

HSdg_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved E minimum

berkeley_b

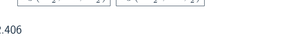
EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved E minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.808 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved E minimum

cx_u

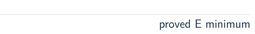
EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved E minimum



cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 proved E minimum

iswap_u

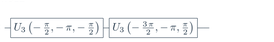
EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved E minimum



quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved E minimum

rigetti_cz

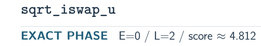
EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

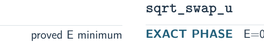
EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved E minimum



xx90_u

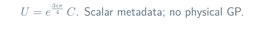
EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved E minimum



zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HSdg_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.208 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

HSdg_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=14 / score ≈ 28.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.808 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HSdg_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.208 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

HSdg_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

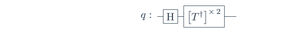
PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum



$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.406 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

HSX

1 qubits / 32 preserved native bases / alphabetical panel order

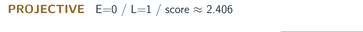
aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

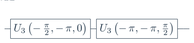
PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved E minimum



$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

HSX_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C.$ Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C.$ Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C.$ Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C.$ Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.806 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C.$ Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

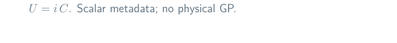
EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C.$ Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum



$U = i\,C.$ Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

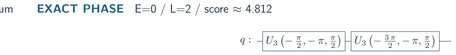
EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

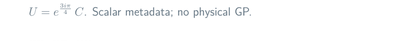
$U = e^{\frac{3i\pi}{4}}\,C.$ Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum



$U = i\,C.$ Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C.$ Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C.$ Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C.$ Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C.$ Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C.$ Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C.$ Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C.$ Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C.$ Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C.$ Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C.$ Scalar metadata; no physical GP.

HSX_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.206 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

HSX_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved E minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved E minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.806 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

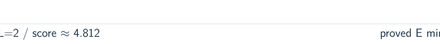
cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved E minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved E minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 proved E minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved E minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved E minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HSX_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.206 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

HSX_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

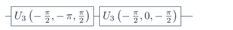
EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum



PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.406 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HSX_q6

1 qubits / 32 preserved native bases / alphabetical panel order

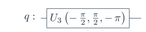
aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum



clifford_only

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.000 — proved E minimum

clifford_t_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.000 — proved E minimum

cx_h_phase

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

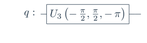
cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum



cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_cz

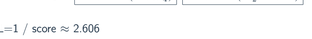
EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

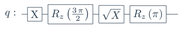
EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

HSX_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.000 — proved E minimum

clifford_t_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.000 — proved E minimum

cx_h_phase

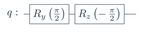
EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

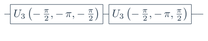
cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

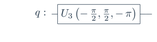


cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

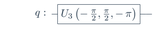
PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HSY

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

clifford_only

EXACT PHASE E=0 / L=4 / score ≈ 8.000 proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.000 proved E minimum

clifford_t_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.000 proved E minimum

cx_h_phase

EXACT PHASE E=0 / L=3 / score ≈ 6.204 proved E minimum

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.204 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.204 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

HSY_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

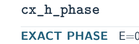
PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum



$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

clifford_t_cz

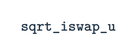
EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.406 — proved E minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

HSY_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.206 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

HSY_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

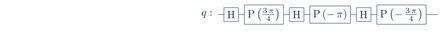
PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

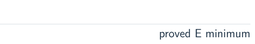
EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

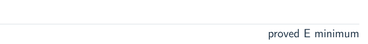
EXACT PHASE E=0 / L=6 / score ≈ 12.806 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

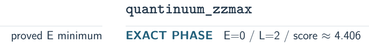
cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum



cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.806 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.806 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.806 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HSY_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.206 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

HSY_q5

1 qubits / 32 preserved native bases / alphabetical panel order

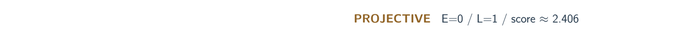
aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.806 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

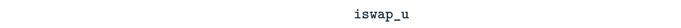
PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum



$U = -1\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HSY_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

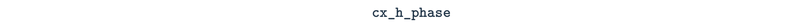
PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum



$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

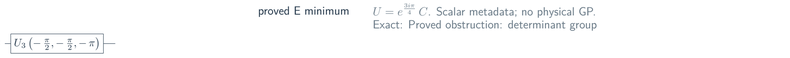
PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum



$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

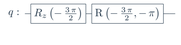
$U = -i\,C$. Scalar metadata; no physical GP.

HSY_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum



berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

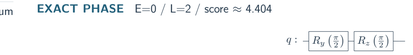
cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HX

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

berkeley_b

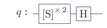
EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

clifford_only

EXACT PHASE E=0 / L=3 / score ≈ 6.000 — proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.000 — proved E minimum



clifford_t_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.000 — proved E minimum

cx_h_phase

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved E minimum

cx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

cz_rz_rx

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

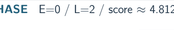
EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.202 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.202 — proved E minimum

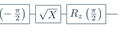
PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.202 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum



$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=1 / score ≈ 2.402 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=1 / score ≈ 2.402 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=1 / score ≈ 2.402 — proved E minimum

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.202 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

HX_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

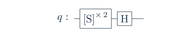
PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum



$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cz

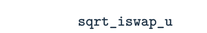
EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

HX_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.406 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

HX_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.806 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HX_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.006 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

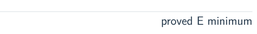
EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

cz_rz_rx

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum



cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.202 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.202 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.202 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.202 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

HX_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=13 / score ≈ 26.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.806 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

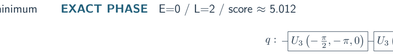
cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

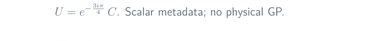
$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

HX_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.406 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

HX_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

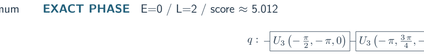
PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

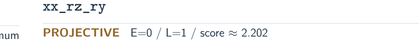
PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HY

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

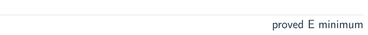
PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

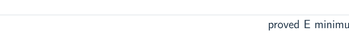


$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.406 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.004 — proved E minimum



$U = -i\,C$. Scalar metadata; no physical GP.

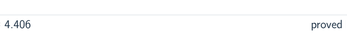
cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum



$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.806 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.806 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.806 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.408 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

HY_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

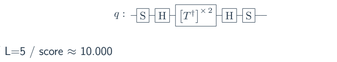
EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

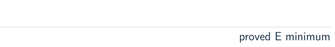


PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

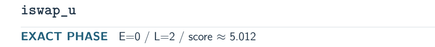


PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

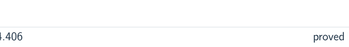


PROJECTIVE E=0 / L=3 / score ≈ 6.004 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

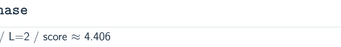
PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HY_q2

1 qubits / 32 preserved native bases / alphabetical panel order

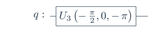
aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum



clifford_only

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved E minimum

clifford_t_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved E minimum

cx_h_phase

EXACT PHASE E=0 / L=3 / score ≈ 6.004 proved E minimum

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

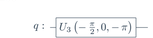
cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum



google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved E minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved E minimum

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

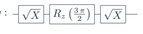
quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

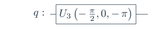
rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved E minimum



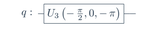
sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum



sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum



tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved E minimum

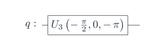
$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum



HY_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.004 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_rz_ry

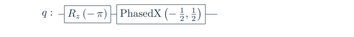
PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum



$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

HY_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

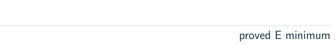
PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

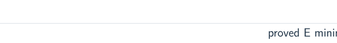


$U = iC$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.406 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.004 — proved E minimum



$U = iC$. Scalar metadata; no physical GP.

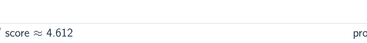
cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum



$U = iC$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

cz_u

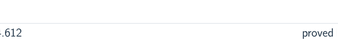
EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.406 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.406 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.406 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.408 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

HY_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

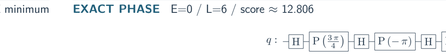
PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.806 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.004 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

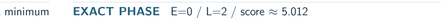
PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HY_q6

1 qubits / 32 preserved native bases / alphabetical panel order

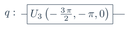
aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum



clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.004 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.004 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_cz

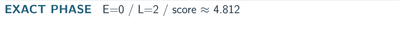
EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

HY_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.806 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.004 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HZ

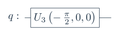
1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum



clifford_only

EXACT PHASE E=0 / L=3 / score ≈ 6.000 — proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.000 — proved E minimum

clifford_t_cz

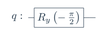
EXACT PHASE E=0 / L=3 / score ≈ 6.000 — proved E minimum

cx_h_phase

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved E minimum

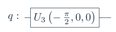
cx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum



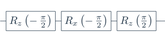
cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum



cz_rz_rx

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum



cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.202 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.202 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.202 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=1 / score ≈ 2.402 — proved E minimum

ionq_aria_full

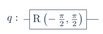
EXACT PHASE E=0 / L=1 / score ≈ 2.402 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=1 / score ≈ 2.402 — proved E minimum

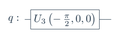
iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum



iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum



quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.202 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

HZ_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HZ_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

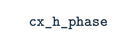
PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum



$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

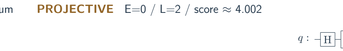
EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.406 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

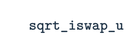
PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum



$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

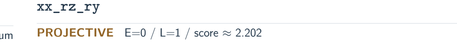
PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

HZ_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.806 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

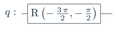
$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

HZ_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum



berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

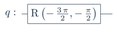
PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum



$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.006 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

cz_rz_rx

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.202 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.202 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.202 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.202 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

HZ_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=13 / score ≈ 26.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.806 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

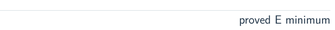
rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

HZ_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

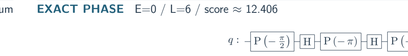
PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum



$U = -iC$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.406 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

cz_rz_rx

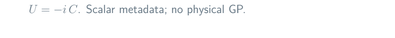
PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum



$U = -iC$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

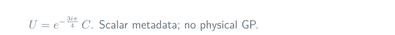
PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum



$U = e^{-\frac{3i\pi}{4}}C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

HZ_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

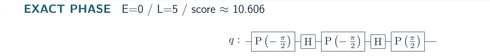
PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

berkeley_b

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

clifford_only

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

clifford_t_cx

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

clifford_t_cz

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

cx_h_phase

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

cx_rz_ry

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

cx_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

cz_rz_rx

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

cz_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

google_cz

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

google_fsim

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

google_sqrt_iswap

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

google_syc

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

ibm_cx

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

ibm_cz

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

ibm_ecr

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

ionq_aria

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

ionq_aria_full

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

ionq_forte

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

iqm_cz

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

iswap_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

quantinuum_zzmax

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

quantinuum_zzphase

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

rigetti_cz

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

sqrt_iswap_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

sqrt_swap_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

tket_tk2

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

xx90_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

xx_rz_ry

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

zx_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

zz_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

l_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

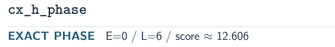
PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum



PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

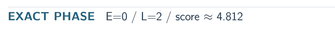
PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum



PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

l_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.408 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

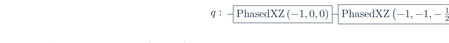
cz_rz_rx

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum



PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.004 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.004 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.004 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.206 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.206 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.206 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.004 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = i\,C.$ Scalar metadata; no physical GP.

I_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=14 / score ≈ 28.000 proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.808 proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

I_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.008 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.004 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

l_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

$q:$ —[$U_3(-\pi,-\pi,0)$]—[$U_3(-\pi,\frac{\pi}{4},-\frac{3\pi}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=16 / score ≈ 32.000 proved E minimum

$q:$ —[S]$^{\times 3}$—[H]—[S]$^{\times 2}$—[H]—[S]$^{\times 2}$—[H]—[S]$^{\times 3}$—[H]—[S]—[H]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=14 / score ≈ 28.000 proved E minimum

$q:$ —[S]$^{\times 2}$—[H]—[S]$^{\times 2}$—[H]—[S]$^{\times 2}$—[T]—[H]—[S]$^{\times 2}$—[H]—[T]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=14 / score ≈ 28.000 proved E minimum

$q:$ —[S]$^{\times 2}$—[H]—[S]$^{\times 2}$—[H]—[S]$^{\times 2}$—[T]—[H]—[S]$^{\times 2}$—[H]—[T]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.808 proved E minimum

$q:$ —[H]—[P$(-\pi)$]—[H]—[P$(-\frac{3\pi}{4})$]—[H]—[P$(-\pi)$]—[H]—[P$(-\frac{3\pi}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

$q:$ —[$U_3(-\pi,-\pi,0)$]—[$U_3(-\pi,\frac{\pi}{4},-\frac{3\pi}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

$q:$ —[$U_3(-\pi,-\pi,0)$]—[$U_3(-\pi,\frac{\pi}{4},-\frac{3\pi}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

$q:$ —[PhasedXZ$(-1,0,0)$]—[PhasedXZ$(-1,\frac{1}{2},\frac{3}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

$q:$ —[PhasedXZ$(-1,0,0)$]—[PhasedXZ$(-1,\frac{1}{2},\frac{3}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

$q:$ —[PhasedXZ$(-1,0,0)$]—[PhasedXZ$(-1,\frac{1}{2},\frac{3}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

$q:$ —[PhasedXZ$(-1,0,0)$]—[PhasedXZ$(-1,\frac{1}{2},\frac{3}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.606 proved E minimum

$q:$ —[$\sqrt{X}$]—[$R_z(\frac{\pi}{2})$]—[$\sqrt{X}$]—[$R_z(-\frac{\pi}{2})$]—[$\sqrt{X}$]—[X]—[$R_z(\frac{3\pi}{2})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 proved E minimum

$q:$ —[$\sqrt{X}$]—[$R_z(\frac{\pi}{2})$]—[$\sqrt{X}$]—[$R_z(-\frac{\pi}{2})$]—[$\sqrt{X}$]—[X]—[$R_z(\frac{3\pi}{2})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=7 / score ≈ 14.606 proved E minimum

$q:$ —[$\sqrt{X}$]—[$R_z(\frac{\pi}{2})$]—[$\sqrt{X}$]—[$R_z(-\frac{\pi}{2})$]—[$\sqrt{X}$]—[X]—[$R_z(\frac{3\pi}{2})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

$q:$ —[$U_3(-\pi,-\pi,0)$]—[$U_3(-\pi,\frac{\pi}{4},-\frac{3\pi}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 proved E minimum

$q:$ —[$\sqrt{X}$]—[$R_z(\frac{\pi}{2})$]—[$\sqrt{X}$]—[$R_z(-\frac{\pi}{2})$]—[$\sqrt{X}$]—[X]—[$R_z(\frac{3\pi}{2})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

$q:$ —[$U_3(-\pi,-\pi,0)$]—[$U_3(-\pi,\frac{\pi}{4},-\frac{3\pi}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

$q:$ —[$U_3(-\pi,-\pi,0)$]—[$U_3(-\pi,\frac{\pi}{4},-\frac{3\pi}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

$q:$ —[$U_3(-\pi,-\pi,0)$]—[$U_3(-\pi,\frac{\pi}{4},-\frac{3\pi}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

$q:$ —[$U_3(-\pi,-\pi,0)$]—[$U_3(-\pi,\frac{\pi}{4},-\frac{3\pi}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

$q:$ —[$U_3(-\pi,-\pi,0)$]—[$U_3(-\pi,\frac{\pi}{4},-\frac{3\pi}{4})$]—

PROJECTIVE E=0 / L=0 / score ≈ 0.000 proved E minimum

$q:$ —

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

l_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=14 / score ≈ 28.000 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.408 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.004 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.004 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.004 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.006 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.006 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.006 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.004 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

l_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

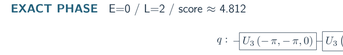
PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum



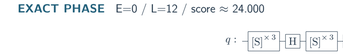
PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved E minimum



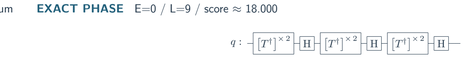
PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum



PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

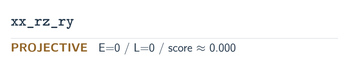
PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum



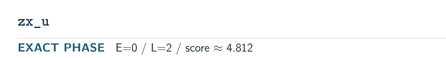
PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum



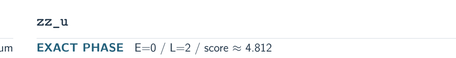
PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum



PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=0 / score ≈ 0.000 — proved E minimum

$q:$ —

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Identity

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

berkeley_b

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

clifford_only

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

clifford_t_cz

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

cx_h_phase

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

cx_rz_ry

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

cx_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

cz_rz_rx

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

cz_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

google_cz

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

google_fsim

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

google_syc

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

ibm_cx

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

ibm_cz

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

ionq_aria

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

ionq_forte

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

iqm_cz

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

iswap_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

quantinuum_zzmax

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

tket_tk2

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

xx90_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

xx_rz_ry

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

zx_u

EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

zz_u

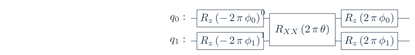
EXACT PHASE E=0 / L=0 / score ≈ 0.000 proved E minimum

IonQMS (ϕ_0, ϕ_1, θ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=1 / L=4 / score ≈ 12.010 — minimum uniform E budget



berkeley_b

EXACT PHASE E=2 / L=13 / score ≈ 35.278 — minimum uniform E budget

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

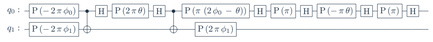
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=14 / score ≈ 36.016 — minimum uniform E budget

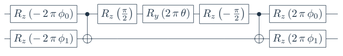


PROJECTIVE E=2 / L=7 / score ≈ 22.010 — minimum uniform E budget

$U = e^{-i\pi\theta}\, C$. Scalar metadata; no physical GP.

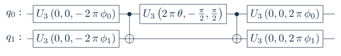
cx_rz_ry

EXACT PHASE E=2 / L=7 / score ≈ 22.414 — minimum uniform E budget



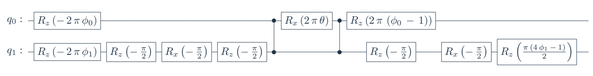
cx_u

EXACT PHASE E=2 / L=5 / score ≈ 18.430 — minimum uniform E budget



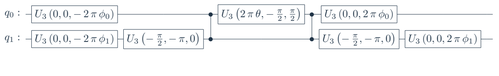
cz_rz_rx

EXACT PHASE E=2 / L=10 / score ≈ 29.220 — minimum uniform E budget



cz_u

EXACT PHASE E=2 / L=7 / score ≈ 22.842 — minimum uniform E budget



google_cz

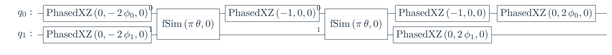
EXACT PHASE E=2 / L=9 / score ≈ 26.854 — minimum uniform E budget

PROJECTIVE E=2 / L=7 / score ≈ 22.842 — minimum uniform E budget

$U = e^{-i\pi\theta}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=6 / score ≈ 20.044 — E upper bound



google_sqrt_iswap

EXACT PHASE E=2 / L=16 / score ≈ 43.287 — minimum uniform E budget

Exact definitions: $\eta_{501}, \eta_{502}, \eta_{503}, \eta_{504}, \eta_{505}, \eta_{506}, \eta_{507}, \eta_{508}, \eta_{509}, \eta_{510}, \eta_{511}, \eta_{512}, \eta_{513}, \eta_{514}, \eta_{515}, \eta_{516}, \eta_{517}, \eta_{518}, \eta_{519}, \eta_{520}$ in the expression glossary.

PROJECTIVE E=2 / L=15 / score ≈ 41.281 — minimum uniform E budget

Exact definitions: $\eta_{505}, \eta_{506}, \eta_{507}, \eta_{508}, \eta_{509}, \eta_{510}, \eta_{511}, \eta_{512}, \eta_{513}, \eta_{514}, \eta_{515}, \eta_{516}, \eta_{517}, \eta_{518}, \eta_{519}, \eta_{520}$ in the expression glossary.
$U = \eta_{521}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=19 / score ≈ 89.314 — E upper bound

PROJECTIVE E=12 / L=17 / score ≈ 85.302 — E upper bound

$U = e^{-i\pi\theta}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=13 / score ≈ 34.418 — minimum uniform E budget

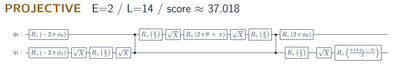
PROJECTIVE E=2 / L=9 / score ≈ 26.414 — minimum uniform E budget

$U = i\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=14 / score ≈ 37.220 — minimum uniform E budget

PROJECTIVE E=2 / L=14 / score ≈ 37.018 — minimum uniform E budget



$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=12 / score ≈ 32.016 — minimum uniform E budget

PROJECTIVE E=2 / L=8 / score ≈ 24.012 — minimum uniform E budget

$U = i\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=1 / L=0 / score ≈ 4.006 — minimum uniform E budget

ionq_aria_full

EXACT PHASE E=2 / L=28 / score ≈ 69.064 — minimum uniform E budget

PROJECTIVE E=2 / L=28 / score ≈ 68.664 — minimum uniform E budget

$U = i\, C$. Scalar metadata; no physical GP.

ionq_forte

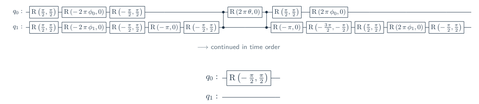
EXACT PHASE E=2 / L=30 / score ≈ 74.664 — E upper bound

PROJECTIVE E=2 / L=30 / score ≈ 74.264 — E upper bound

$U = -i\, C$. Scalar metadata; no physical GP.

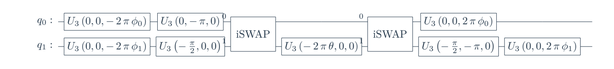
iqm_cz

EXACT PHASE E=2 / L=17 / score ≈ 46.068 — minimum uniform E budget



iswap_u

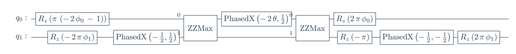
EXACT PHASE E=2 / L=8 / score ≈ 24.448 — minimum uniform E budget



quantinuum_zzmax

EXACT PHASE E=5 / L=27 / score ≈ 79.280 — E upper bound

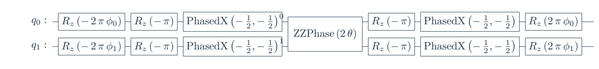
PROJECTIVE E=2 / L=8 / score ≈ 25.022 — minimum uniform E budget



$U = -i\, C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=1 / L=12 / score ≈ 29.634 — minimum uniform E budget



rigetti_cz

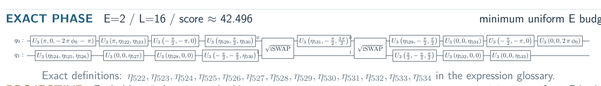
EXACT PHASE E=2 / L=14 / score ≈ 37.220 — minimum uniform E budget

PROJECTIVE E=2 / L=14 / score ≈ 37.018 — minimum uniform E budget

$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

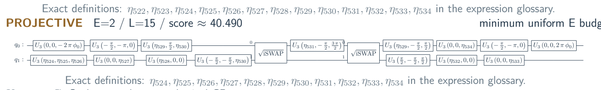
sqrt_iswap_u

EXACT PHASE E=2 / L=16 / score ≈ 42.496 — minimum uniform E budget



Exact definitions: $\eta_{522}, \eta_{523}, \eta_{524}, \eta_{525}, \eta_{526}, \eta_{527}, \eta_{528}, \eta_{529}, \eta_{530}, \eta_{531}, \eta_{532}, \eta_{533}, \eta_{534}$ in the expression glossary.

PROJECTIVE E=2 / L=15 / score ≈ 40.490 — minimum uniform E budget



Exact definitions: $\eta_{524}, \eta_{525}, \eta_{526}, \eta_{527}, \eta_{528}, \eta_{529}, \eta_{530}, \eta_{531}, \eta_{532}, \eta_{533}, \eta_{534}$ in the expression glossary.
$U = \eta_{535}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

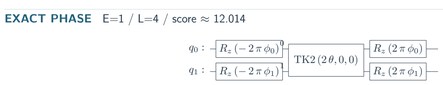
EXACT PHASE E=4 / L=19 / score ≈ 57.714 — E upper bound

PROJECTIVE E=4 / L=17 / score ≈ 53.302 — E upper bound

$U = -ie^{-i\pi\theta}\, C$. Scalar metadata; no physical GP.

tket_tk2

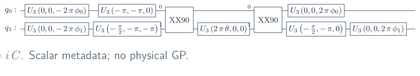
EXACT PHASE E=1 / L=4 / score ≈ 12.014 — minimum uniform E budget



xx90_u

EXACT PHASE E=2 / L=8 / score ≈ 24.848 — minimum uniform E budget

PROJECTIVE E=2 / L=8 / score ≈ 24.448 — minimum uniform E budget



$U = i\, C$. Scalar metadata; no physical GP.

xx_rz_ry

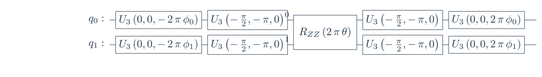
EXACT PHASE E=1 / L=4 / score ≈ 12.010 — minimum uniform E budget

zx_u

EXACT PHASE E=1 / L=10 / score ≈ 25.262 — minimum uniform E budget

zz_u

EXACT PHASE E=1 / L=8 / score ≈ 20.850 — minimum uniform E budget



IonQMSMax (ϕ_0, ϕ_1)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=1 / L=4 / score ≈ 12.210 — proved E minimum

berkeley_b

EXACT PHASE E=2 / L=12 / score ≈ 33.672 — proved E minimum

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

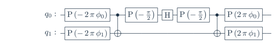
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=7 / score ≈ 22.412 — E upper bound



PROJECTIVE E=2 / L=7 / score ≈ 22.210 — E upper bound

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

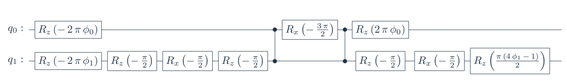
EXACT PHASE E=2 / L=7 / score ≈ 22.614 — E upper bound

cx_u

EXACT PHASE E=2 / L=5 / score ≈ 18.630 — E upper bound

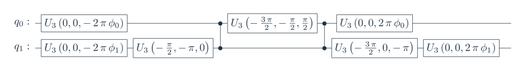
cz_rz_rx

EXACT PHASE E=2 / L=10 / score ≈ 29.420 — E upper bound



cz_u

EXACT PHASE E=2 / L=7 / score ≈ 23.042 — E upper bound



google_cz

EXACT PHASE E=2 / L=8 / score ≈ 25.448 — E upper bound

PROJECTIVE E=2 / L=7 / score ≈ 23.042 — E upper bound

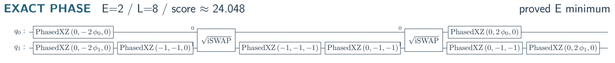
$U = -(-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=6 / score ≈ 20.844 — E upper bound

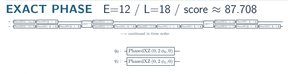
google_sqrt_iswap

EXACT PHASE E=2 / L=8 / score ≈ 24.048 — proved E minimum



google_syc

EXACT PHASE E=12 / L=18 / score ≈ 87.708 — E upper bound



PROJECTIVE E=12 / L=17 / score ≈ 85.502 — E upper bound

$U = -(-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=10 / score ≈ 28.614 — E upper bound

PROJECTIVE E=2 / L=5 / score ≈ 18.008 — E upper bound

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=10 / score ≈ 28.612 — E upper bound

ibm_ecr

EXACT PHASE E=2 / L=8 / score ≈ 24.210 — E upper bound

PROJECTIVE E=2 / L=7 / score ≈ 22.412 — E upper bound

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=28 / score ≈ 70.268 — E upper bound

PROJECTIVE E=2 / L=28 / score ≈ 69.868 — E upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=1 / L=0 / score ≈ 4.004 — minimum uniform E budget

ionq_forte

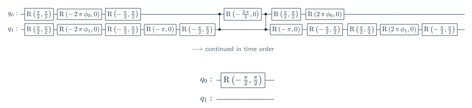
EXACT PHASE E=2 / L=30 / score ≈ 75.064 — E upper bound

PROJECTIVE E=2 / L=30 / score ≈ 74.664 — E upper bound

$U = i\,C$. Scalar metadata; no physical GP.

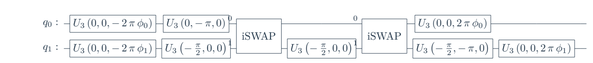
iqm_cz

EXACT PHASE E=2 / L=17 / score ≈ 46.268 — E upper bound



iswap_u

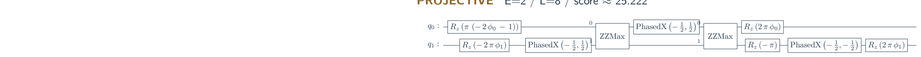
EXACT PHASE E=2 / L=8 / score ≈ 24.648 — proved E minimum



quantinuum_zzmax

EXACT PHASE E=5 / L=27 / score ≈ 79.480 — E upper bound

PROJECTIVE E=2 / L=8 / score ≈ 25.222 — E upper bound



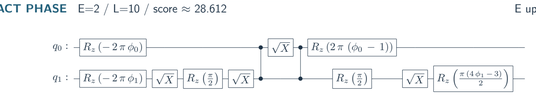
$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=1 / L=12 / score ≈ 29.834 — proved E minimum

rigetti_cz

EXACT PHASE E=2 / L=10 / score ≈ 28.612 — E upper bound



sqrt_iswap_u

EXACT PHASE E=2 / L=8 / score ≈ 24.048 — proved E minimum

sqrt_swap_u

EXACT PHASE E=4 / L=17 / score ≈ 54.302 — E upper bound

PROJECTIVE E=4 / L=17 / score ≈ 53.502 — E upper bound

$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=4 / score ≈ 12.214 — proved E minimum

xx90_u

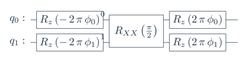
EXACT PHASE E=2 / L=8 / score ≈ 25.048 — E upper bound

PROJECTIVE E=2 / L=8 / score ≈ 24.648 — E upper bound

$U = i\,C$. Scalar metadata; no physical GP.

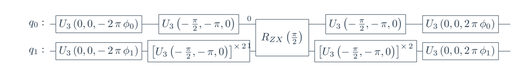
xx_rz_ry

EXACT PHASE E=1 / L=4 / score ≈ 12.210 — proved E minimum



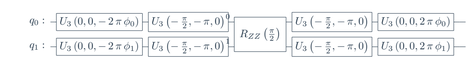
zx_u

EXACT PHASE E=1 / L=10 / score ≈ 25.462 — proved E minimum



zz_u

EXACT PHASE E=1 / L=8 / score ≈ 21.050 — proved E minimum



IonQZZ (θ)

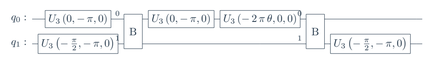
2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=1 / L=8 / score ≈ 21.626 — minimum uniform E budget

berkeley_b

EXACT PHASE E=2 / L=5 / score ≈ 18.430 — minimum uniform E budget



clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

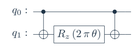
EXACT PHASE E=2 / L=9 / score ≈ 26.010 — minimum uniform E budget

PROJECTIVE E=2 / L=1 / score ≈ 10.002 — minimum uniform E budget

$U = e^{-i\pi\theta}\,C$. Scalar metadata; no physical GP.

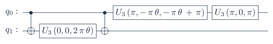
cx_rz_ry

EXACT PHASE E=2 / L=1 / score ≈ 10.002 — minimum uniform E budget



cx_u

EXACT PHASE E=2 / L=3 / score ≈ 14.018 — minimum uniform E budget

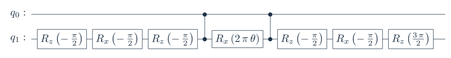


PROJECTIVE E=2 / L=1 / score ≈ 10.006 — minimum uniform E budget

$U = e^{-i\pi\theta}\,C$. Scalar metadata; no physical GP.

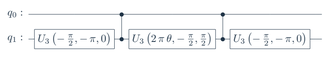
cz_rz_rx

EXACT PHASE E=2 / L=7 / score ≈ 23.214 — minimum uniform E budget



cz_u

EXACT PHASE E=2 / L=3 / score ≈ 14.818 — minimum uniform E budget



google_cz

EXACT PHASE E=2 / L=5 / score ≈ 18.830 — minimum uniform E budget

PROJECTIVE E=2 / L=3 / score ≈ 14.818 — minimum uniform E budget

$U = e^{-i\pi\theta}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 12.028 — minimum uniform E budget

PROJECTIVE E=1 / L=2 / score ≈ 8.016 — minimum uniform E budget

$U = e^{-i\pi\theta}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=2 / L=10 / score ≈ 30.581 — minimum uniform E budget

Exact definitions: $\eta_{536}, \eta_{537}, \eta_{538}, \eta_{539}, \eta_{540}, \eta_{541}, \eta_{542}, \eta_{543}, \eta_{544}, \eta_{545}, \eta_{546}, \eta_{547}, \eta_{548}, \eta_{549}$ in the expression glossary.

PROJECTIVE E=2 / L=9 / score ≈ 28.575 — minimum uniform E budget

Exact definitions: $\eta_{541}, \eta_{542}, \eta_{543}, \eta_{544}, \eta_{545}, \eta_{546}, \eta_{547}, \eta_{548}, \eta_{549}, \eta_{550}, \eta_{551}, \eta_{552}$ in the expression glossary.

$U = \eta_{553}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=11 / score ≈ 71.666 — E upper bound

PROJECTIVE E=12 / L=9 / score ≈ 67.654 — E upper bound

$U = e^{-i\pi\theta}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=1 / score ≈ 10.002 — minimum uniform E budget

ibm_cz

EXACT PHASE E=2 / L=11 / score ≈ 31.214 — minimum uniform E budget

PROJECTIVE E=2 / L=11 / score ≈ 30.810 — minimum uniform E budget

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=8 / score ≈ 24.008 — minimum uniform E budget

PROJECTIVE E=2 / L=4 / score ≈ 16.004 — minimum uniform E budget

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

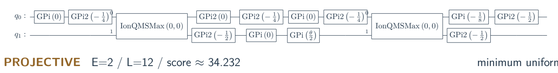
EXACT PHASE E=2 / L=12 / score ≈ 35.436 — E upper bound

PROJECTIVE E=2 / L=12 / score ≈ 35.036 — E upper bound

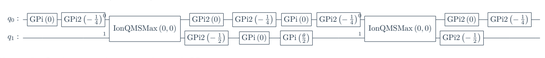
$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=12 / score ≈ 34.632 — minimum uniform E budget



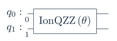
PROJECTIVE E=2 / L=12 / score ≈ 34.232 — minimum uniform E budget



$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=1 / L=0 / score ≈ 4.002 — minimum uniform E budget

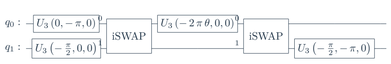


iqm_cz

EXACT PHASE E=2 / L=5 / score ≈ 18.820 — minimum uniform E budget

iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 16.424 — minimum uniform E budget



quantinuum_zzmax

EXACT PHASE E=5 / L=24 / score ≈ 73.274 — E upper bound

PROJECTIVE E=2 / L=5 / score ≈ 19.016 — minimum uniform E budget

$U = -i\,C$. Scalar metadata; no physical GP.

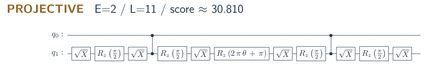
quantinuum_zzphase

EXACT PHASE E=1 / L=0 / score ≈ 4.002 — minimum uniform E budget

rigetti_cz

EXACT PHASE E=2 / L=11 / score ≈ 31.214 — minimum uniform E budget

PROJECTIVE E=2 / L=11 / score ≈ 30.810 — minimum uniform E budget



$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=2 / L=10 / score ≈ 30.060 — minimum uniform E budget

Exact definitions: $\eta_{554}, \eta_{555}, \eta_{556}, \eta_{557}, \eta_{558}, \eta_{559}, \eta_{560}, \eta_{561}, \eta_{562}, \eta_{563}$ in the expression glossary.

PROJECTIVE E=2 / L=9 / score ≈ 28.054 — minimum uniform E budget

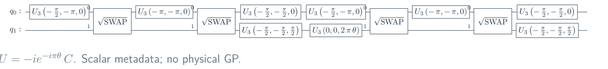
Exact definitions: $\eta_{558}, \eta_{559}, \eta_{560}, \eta_{561}, \eta_{562}, \eta_{563}, \eta_{564}, \eta_{565}$ in the expression glossary.

$U = \eta_{566}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=11 / score ≈ 40.866 — E upper bound

PROJECTIVE E=4 / L=9 / score ≈ 36.454 — E upper bound



$U = -ie^{-i\pi\theta}\,C$. Scalar metadata; no physical GP.

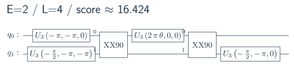
tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.006 — minimum uniform E budget

xx90_u

EXACT PHASE E=2 / L=4 / score ≈ 16.824 — minimum uniform E budget

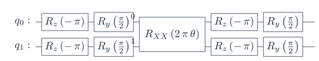
PROJECTIVE E=2 / L=4 / score ≈ 16.424 — minimum uniform E budget



$U = i\,C$. Scalar metadata; no physical GP.

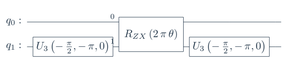
xx_rz_ry

EXACT PHASE E=1 / L=8 / score ≈ 20.818 — minimum uniform E budget



zx_u

EXACT PHASE E=1 / L=2 / score ≈ 8.414 — minimum uniform E budget



zz_u

EXACT PHASE E=1 / L=0 / score ≈ 4.002 — minimum uniform E budget

iSWAP

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

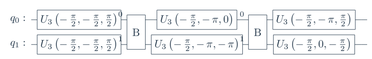
EXACT PHASE E=2 / L=34 / score ≈ 83.704 — proved E minimum

PROJECTIVE E=2 / L=16 / score ≈ 43.658 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=2 / L=6 / score ≈ 22.436 — proved E minimum



clifford_only

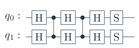
EXACT PHASE E=2 / L=6 / score ≈ 20.000 — proved E minimum

clifford_t_cx

EXACT PHASE E=2 / L=6 / score ≈ 20.000 — proved E minimum

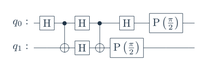
clifford_t_cz

EXACT PHASE E=2 / L=8 / score ≈ 24.000 — proved E minimum



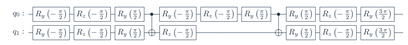
cx_h_phase

EXACT PHASE E=2 / L=6 / score ≈ 20.404 — proved E minimum



cx_rz_ry

EXACT PHASE E=2 / L=16 / score ≈ 43.232 — proved E minimum

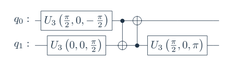


PROJECTIVE E=2 / L=5 / score ≈ 18.810 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_u

EXACT PHASE E=2 / L=3 / score ≈ 14.818 — proved E minimum

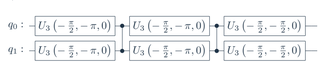


cz_rz_rx

EXACT PHASE E=2 / L=10 / score ≈ 29.620 — proved E minimum

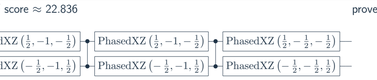
cz_u

EXACT PHASE E=2 / L=6 / score ≈ 21.636 — proved E minimum



google_cz

EXACT PHASE E=2 / L=6 / score ≈ 22.836 — proved E minimum



google_fsim

EXACT PHASE E=1 / L=0 / score ≈ 4.204 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=2 / L=0 / score ≈ 8.000 — proved E minimum

google_syc

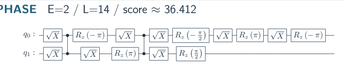
EXACT PHASE E=12 / L=14 / score ≈ 78.884 — E upper bound

ibm_cx

EXACT PHASE E=2 / L=7 / score ≈ 22.810 — proved E minimum

ibm_cz

EXACT PHASE E=2 / L=14 / score ≈ 36.412 — proved E minimum



PROJECTIVE E=2 / L=10 / score ≈ 28.408 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

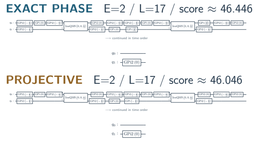
EXACT PHASE E=2 / L=10 / score ≈ 28.810 — proved E minimum

PROJECTIVE E=2 / L=10 / score ≈ 28.808 — proved E minimum

$U = \eta_{296}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=17 / score ≈ 46.446 — proved E minimum

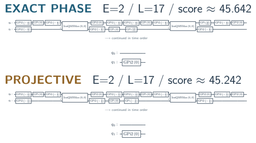


PROJECTIVE E=2 / L=17 / score ≈ 46.046 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=17 / score ≈ 45.642 — proved E minimum

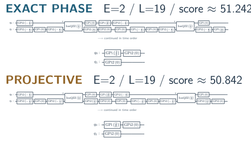


PROJECTIVE E=2 / L=17 / score ≈ 45.242 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=19 / score ≈ 51.242 — proved E minimum

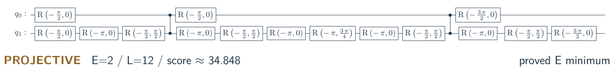


PROJECTIVE E=2 / L=19 / score ≈ 50.842 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=15 / score ≈ 41.060 — proved E minimum



PROJECTIVE E=2 / L=12 / score ≈ 34.848 — proved E minimum

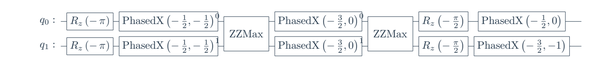
$U = -i\,C$. Scalar metadata; no physical GP.

iswap_u

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

quantinuum_zzmax

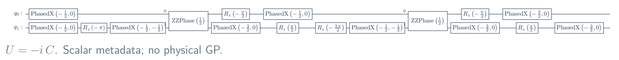
EXACT PHASE E=2 / L=10 / score ≈ 30.032 — proved E minimum



quantinuum_zzphase

EXACT PHASE E=2 / L=18 / score ≈ 48.456 — proved E minimum

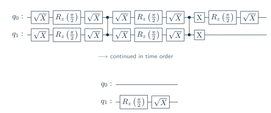
PROJECTIVE E=2 / L=15 / score ≈ 41.652 — proved E minimum



$U = -i\,C$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=2 / L=18 / score ≈ 45.212 — proved E minimum



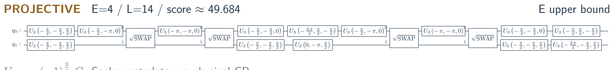
sqrt_iswap_u

EXACT PHASE E=2 / L=0 / score ≈ 8.000 — proved E minimum

sqrt_swap_u

EXACT PHASE E=4 / L=14 / score ≈ 50.484 — E upper bound

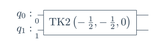
PROJECTIVE E=4 / L=14 / score ≈ 49.684 — E upper bound



$U = -(-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.406 — proved E minimum



xx90_u

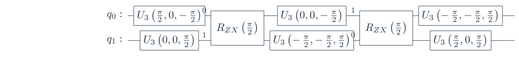
EXACT PHASE E=2 / L=4 / score ≈ 16.824 — proved E minimum

xx_rz_ry

EXACT PHASE E=2 / L=13 / score ≈ 36.830 — proved E minimum

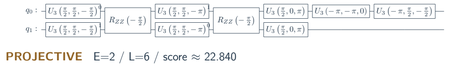
zx_u

EXACT PHASE E=2 / L=6 / score ≈ 22.840 — proved E minimum

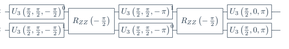


zz_u

EXACT PHASE E=2 / L=8 / score ≈ 27.252 — proved E minimum



PROJECTIVE E=2 / L=6 / score ≈ 22.840 — proved E minimum



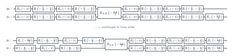
$U = -i\,C$. Scalar metadata; no physical GP.

iSWAP_pow (α)

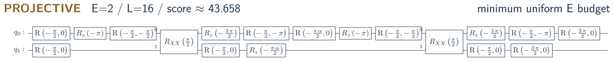
2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=2 / L=34 / score ≈ 83.704 — minimum uniform E budget



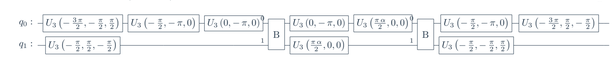
PROJECTIVE E=2 / L=16 / score ≈ 43.658 — minimum uniform E budget



$U = -iC$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=2 / L=10 / score ≈ 31.260 — minimum uniform E budget



clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

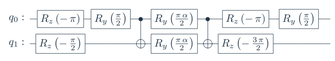
EXACT PHASE E=2 / L=22 / score ≈ 54.024 — minimum uniform E budget

PROJECTIVE E=2 / L=14 / score ≈ 37.616 — minimum uniform E budget

$U = e^{-\frac{i\pi\alpha}{2}} C$. Scalar metadata; no physical GP.

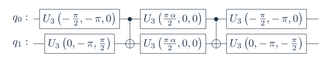
cx_rz_ry

EXACT PHASE E=2 / L=8 / score ≈ 25.216 — minimum uniform E budget



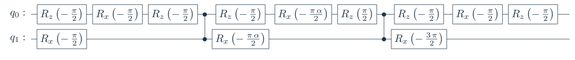
cx_u

EXACT PHASE E=2 / L=6 / score ≈ 21.236 — minimum uniform E budget



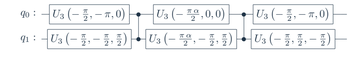
cz_rz_rx

EXACT PHASE E=2 / L=12 / score ≈ 34.424 — minimum uniform E budget



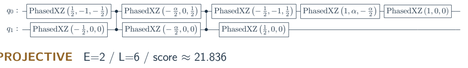
cz_u

EXACT PHASE E=2 / L=6 / score ≈ 22.436 — minimum uniform E budget

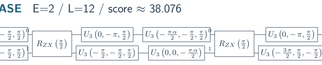


google_cz

EXACT PHASE E=2 / L=8 / score ≈ 26.048 — minimum uniform E budget



PROJECTIVE E=2 / L=6 / score ≈ 21.836 — minimum uniform E budget



$U = e^{\frac{i\pi\alpha}{2}} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=0 / score ≈ 4.204 — minimum uniform E budget

google_sqrt_iswap

EXACT PHASE E=2 / L=6 / score ≈ 20.836 — minimum uniform E budget

google_syc

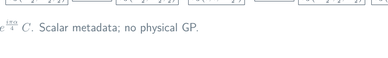
EXACT PHASE E=12 / L=16 / score ≈ 83.096 — E upper bound

PROJECTIVE E=12 / L=14 / score ≈ 78.884 — E upper bound

$U = e^{\frac{i\pi\alpha}{2}} C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=13 / score ≈ 34.812 — minimum uniform E budget



PROJECTIVE E=2 / L=12 / score ≈ 32.808 — minimum uniform E budget

$U = iC$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=18 / score ≈ 45.214 — minimum uniform E budget

ibm_ecr

EXACT PHASE E=2 / L=14 / score ≈ 38.020 — minimum uniform E budget

PROJECTIVE E=2 / L=14 / score ≈ 37.616 — minimum uniform E budget

$U = iC$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=20 / score ≈ 53.652 — minimum uniform E budget

PROJECTIVE E=2 / L=20 / score ≈ 53.252 — minimum uniform E budget

$U = iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=20 / score ≈ 52.848 — minimum uniform E budget

PROJECTIVE E=2 / L=20 / score ≈ 52.448 — minimum uniform E budget

$U = iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=22 / score ≈ 58.448 — minimum uniform E budget

PROJECTIVE E=2 / L=22 / score ≈ 58.048 — minimum uniform E budget

$U = iC$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=8 / score ≈ 25.832 — minimum uniform E budget

iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 22.436 — minimum uniform E budget

quantinuum_zzmax

EXACT PHASE E=5 / L=27 / score ≈ 80.086 — E upper bound

PROJECTIVE E=2 / L=8 / score ≈ 25.828 — minimum uniform E budget

$U = \frac{\sqrt[4]{-1}\sqrt{2}(1+i)}{2} C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=2 / L=18 / score ≈ 48.456 — minimum uniform E budget

PROJECTIVE E=2 / L=15 / score ≈ 41.652 — minimum uniform E budget

$U = -iC$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=2 / L=18 / score ≈ 45.214 — minimum uniform E budget

sqrt_iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 20.836 — minimum uniform E budget

sqrt_swap_u

EXACT PHASE E=4 / L=16 / score ≈ 54.896 — E upper bound

PROJECTIVE E=4 / L=14 / score ≈ 49.684 — E upper bound

$U = -ie^{\frac{i\pi\alpha}{4}} C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.406 — minimum uniform E budget

xx90_u

EXACT PHASE E=2 / L=5 / score ≈ 18.830 — minimum uniform E budget

xx_rz_ry

EXACT PHASE E=2 / L=34 / score ≈ 80.472 — minimum uniform E budget

PROJECTIVE E=2 / L=29 / score ≈ 71.662 — minimum uniform E budget

$U = -iC$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=2 / L=12 / score ≈ 38.076 — minimum uniform E budget

PROJECTIVE E=2 / L=10 / score ≈ 33.264 — minimum uniform E budget

$U = e^{\frac{i\pi\alpha}{4}} C$. Scalar metadata; no physical GP.

zz_u

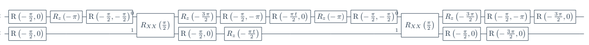
EXACT PHASE E=2 / L=12 / score ≈ 34.876 — minimum uniform E budget

ISWAPPow $(t, global_{shift})$

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=2 / L=16 / score ≈ 43.658 — minimum uniform E budget



$U = -ie^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=12 / score ≈ 35.272 — minimum uniform E budget

PROJECTIVE E=2 / L=10 / score ≈ 31.260 — minimum uniform E budget

$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

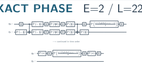
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=22 / score ≈ 54.024 — minimum uniform E budget

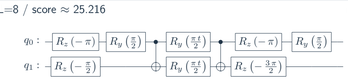


PROJECTIVE E=2 / L=14 / score ≈ 37.616 — minimum uniform E budget

$U = e^{\frac{i\pi\left(2global_{shift}t-1\right)}{2}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=2 / L=8 / score ≈ 25.216 — minimum uniform E budget

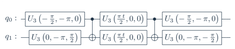


$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=2 / L=8 / score ≈ 25.248 — minimum uniform E budget

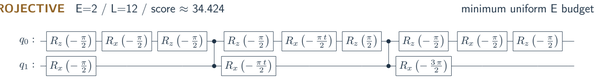
PROJECTIVE E=2 / L=6 / score ≈ 21.236 — minimum uniform E budget



$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

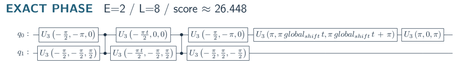
PROJECTIVE E=2 / L=12 / score ≈ 34.424 — minimum uniform E budget



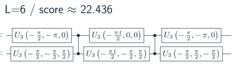
$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=8 / score ≈ 26.448 — minimum uniform E budget



PROJECTIVE E=2 / L=6 / score ≈ 22.436 — minimum uniform E budget



$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=2 / L=8 / score ≈ 26.048 — minimum uniform E budget

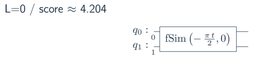
PROJECTIVE E=2 / L=6 / score ≈ 21.836 — minimum uniform E budget

$U = e^{\frac{i\pi\left(2global_{shift}t+1\right)}{2}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=2 / score ≈ 8.216 — minimum uniform E budget

PROJECTIVE E=1 / L=0 / score ≈ 4.204 — minimum uniform E budget



$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=2 / L=8 / score ≈ 24.848 — minimum uniform E budget

PROJECTIVE E=2 / L=6 / score ≈ 20.836 — minimum uniform E budget

$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=16 / score ≈ 83.096 — E upper bound

PROJECTIVE E=12 / L=14 / score ≈ 78.884 — E upper bound

$U = e^{\frac{i\pi\left(2global_{shift}t+1\right)}{2}}\,C$. Scalar metadata; no physical GP.

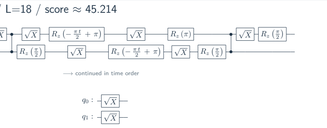
ibm_cx

PROJECTIVE E=2 / L=12 / score ≈ 32.808 — minimum uniform E budget

$U = ie^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

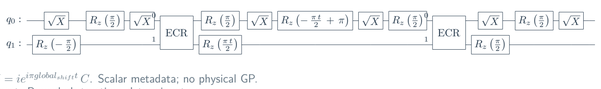
PROJECTIVE E=2 / L=18 / score ≈ 45.214 — minimum uniform E budget



$U = -e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

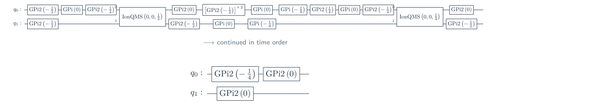
PROJECTIVE E=2 / L=14 / score ≈ 37.616 — minimum uniform E budget



$U = ie^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=2 / L=20 / score ≈ 53.252 — minimum uniform E budget



$U = -ie^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

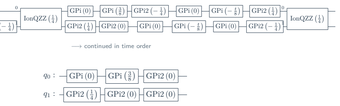
ionq_aria_full

PROJECTIVE E=2 / L=20 / score ≈ 52.448 — minimum uniform E budget

$U = ie^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

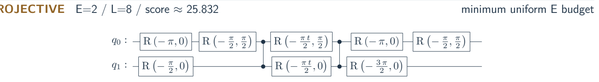
PROJECTIVE E=2 / L=22 / score ≈ 58.048 — minimum uniform E budget



$U = ie^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

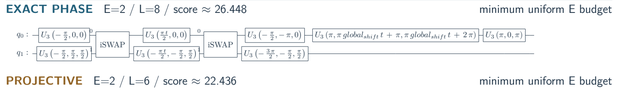
PROJECTIVE E=2 / L=8 / score ≈ 25.832 — minimum uniform E budget



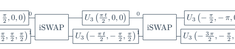
$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=2 / L=8 / score ≈ 26.448 — minimum uniform E budget



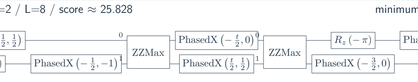
PROJECTIVE E=2 / L=6 / score ≈ 22.436 — minimum uniform E budget



$U = \eta_{498}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

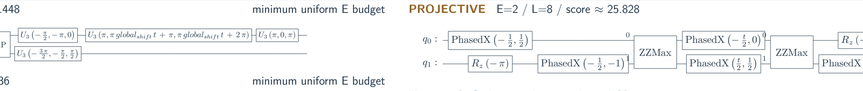
PROJECTIVE E=2 / L=8 / score ≈ 25.828 — minimum uniform E budget



$U = \eta_{567}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=2 / L=15 / score ≈ 41.652 — minimum uniform E budget



$U = -ie^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=2 / L=18 / score ≈ 45.214 — minimum uniform E budget

$U = -e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=2 / L=8 / score ≈ 24.848 — minimum uniform E budget

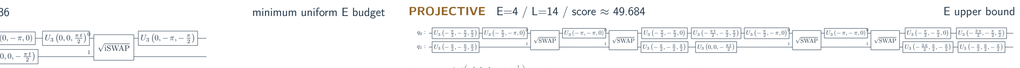
PROJECTIVE E=2 / L=6 / score ≈ 20.836 — minimum uniform E budget

$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=16 / score ≈ 54.896 — E upper bound

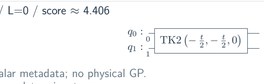
PROJECTIVE E=4 / L=14 / score ≈ 49.684 — E upper bound



$U = -ie^{i\pi\left(global_{shift}t+\frac{1}{4}\right)}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=0 / score ≈ 4.406 — minimum uniform E budget



$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

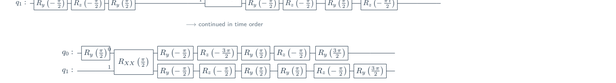
EXACT PHASE E=2 / L=7 / score ≈ 22.842 — minimum uniform E budget

PROJECTIVE E=2 / L=5 / score ≈ 18.830 — minimum uniform E budget

$U = \eta_{500}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=2 / L=29 / score ≈ 71.662 — minimum uniform E budget

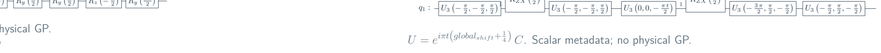


$U = -ie^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=2 / L=12 / score ≈ 38.076 — minimum uniform E budget

PROJECTIVE E=2 / L=10 / score ≈ 33.264 — minimum uniform E budget



$U = e^{i\pi\left(global_{shift}t+\frac{1}{4}\right)}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=2 / L=14 / score ≈ 38.888 — minimum uniform E budget

PROJECTIVE E=2 / L=12 / score ≈ 34.876 — minimum uniform E budget

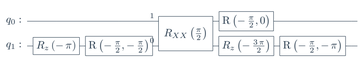
$U = e^{i\pi global_{shift}t}\,C$. Scalar metadata; no physical GP.

LinearFunction

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

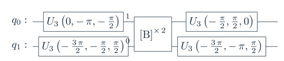
PROJECTIVE E=1 / L=5 / score ≈ 15.218 — proved E minimum



$U = \frac{\sqrt{2}i(1+i)}{2}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=4 / score ≈ 17.624 — proved E minimum



clifford_only

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

clifford_t_cx

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

clifford_t_cz

EXACT PHASE E=1 / L=2 / score ≈ 8.000 — proved E minimum

cx_h_phase

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

cx_rz_ry

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

cx_u

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

cz_rz_rx

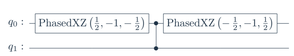
EXACT PHASE E=1 / L=6 / score ≈ 17.212 — proved E minimum

cz_u

EXACT PHASE E=1 / L=2 / score ≈ 8.412 — proved E minimum

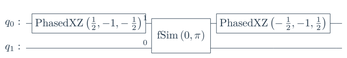
google_cz

EXACT PHASE E=1 / L=2 / score ≈ 8.812 — proved E minimum



google_fsim

EXACT PHASE E=1 / L=2 / score ≈ 8.816 — proved E minimum



google_sqrt_iswap

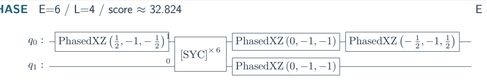
EXACT PHASE E=2 / L=4 / score ≈ 17.824 — proved E minimum

PROJECTIVE E=2 / L=4 / score ≈ 17.224 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=6 / L=4 / score ≈ 32.824 — E upper bound

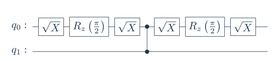


ibm_cx

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

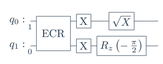
ibm_cz

EXACT PHASE E=1 / L=6 / score ≈ 16.404 — proved E minimum



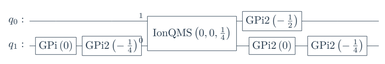
ibm_ecr

EXACT PHASE E=1 / L=4 / score ≈ 12.202 — proved E minimum



ionq_aria

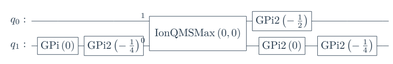
PROJECTIVE E=1 / L=5 / score ≈ 15.416 — proved E minimum



$U = \frac{\sqrt{2}(1+i)}{2}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=1 / L=5 / score ≈ 15.014 — proved E minimum



$U = \frac{\sqrt{2}(1+i)}{2}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

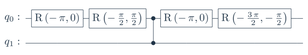
ionq_forte

PROJECTIVE E=1 / L=6 / score ≈ 17.814 — proved E minimum

$U = \frac{\sqrt{2}(-1-i)}{2}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

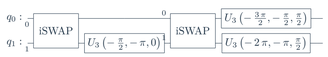
iqm_cz

EXACT PHASE E=1 / L=4 / score ≈ 12.816 — proved E minimum



iswap_u

EXACT PHASE E=2 / L=3 / score ≈ 15.018 — proved E minimum



quantinuum_zzmax

PROJECTIVE E=1 / L=5 / score ≈ 15.014 — proved E minimum

$U = \frac{\sqrt{2}i(1+i)}{2}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=1 / L=5 / score ≈ 15.216 — proved E minimum

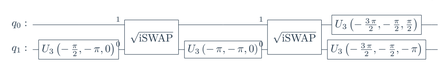
$U = \frac{\sqrt{2}i(1+i)}{2}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=1 / L=6 / score ≈ 16.404 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 17.224 — proved E minimum



sqrt_swap_u

EXACT PHASE E=2 / L=4 / score ≈ 18.024 — proved E minimum

PROJECTIVE E=2 / L=4 / score ≈ 17.224 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=5 / score ≈ 15.222 — proved E minimum

$U = \frac{\sqrt{2}i(1+i)}{2}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=1 / L=3 / score ≈ 11.218 — proved E minimum

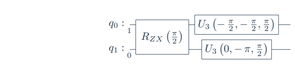
xx_rz_ry

PROJECTIVE E=1 / L=7 / score ≈ 19.416 — proved E minimum

$U = \frac{\sqrt{2}i(1+i)}{2}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

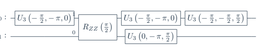
zx_u

EXACT PHASE E=1 / L=2 / score ≈ 9.014 — proved E minimum



zz_u

EXACT PHASE E=1 / L=4 / score ≈ 13.426 — proved E minimum



PROJECTIVE E=1 / L=3 / score ≈ 11.020 — proved E minimum

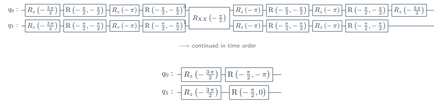
$U = \frac{\sqrt{2}(1+i)}{2}\, C$. Scalar metadata; no physical GP.

Magic

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=1 / L=21 / score ≈ 50.864 — proved E minimum



$U = \frac{\sqrt{2}(-1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_t_cz

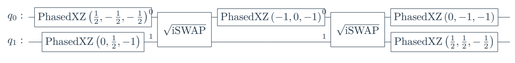
EXACT PHASE E=1 / L=17 / score ≈ 38.000 — proved E minimum

cz_rz_rx

EXACT PHASE E=1 / L=6 / score ≈ 17.012 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=2 / L=5 / score ≈ 19.430 — proved E minimum



ibm_ecr

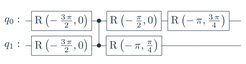
EXACT PHASE E=1 / L=8 / score ≈ 20.810 — proved E minimum

PROJECTIVE E=1 / L=8 / score ≈ 20.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

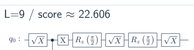
iqm_cz

EXACT PHASE E=1 / L=5 / score ≈ 15.420 — proved E minimum



rigetti_cz

EXACT PHASE E=1 / L=9 / score ≈ 22.606 — proved E minimum

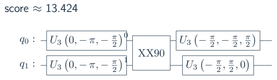


PROJECTIVE E=1 / L=6 / score ≈ 16.404 — proved E minimum

$U = \frac{\sqrt{2}(-1+i)}{2}\,C$. Scalar metadata; no physical GP.

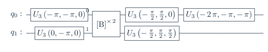
xx90_u

EXACT PHASE E=1 / L=4 / score ≈ 13.424 — proved E minimum

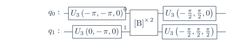


berkeley_b

EXACT PHASE E=2 / L=5 / score ≈ 19.030 — proved E minimum



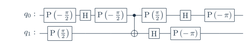
PROJECTIVE E=2 / L=4 / score ≈ 17.024 — proved E minimum



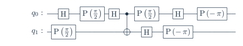
$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=1 / L=9 / score ≈ 22.812 — proved E minimum



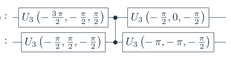
PROJECTIVE E=1 / L=9 / score ≈ 22.610 — proved E minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=1 / L=4 / score ≈ 13.824 — proved E minimum



google_cz

EXACT PHASE E=1 / L=4 / score ≈ 13.224 — proved E minimum

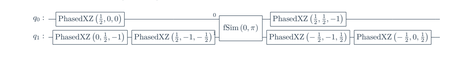
PROJECTIVE E=1 / L=4 / score ≈ 13.024 — proved E minimum

$U = \frac{\sqrt{2}(-1+i)}{2}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=7 / score ≈ 20.246 — proved E minimum

PROJECTIVE E=1 / L=6 / score ≈ 18.040 — proved E minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

google_syc

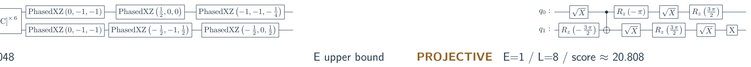
EXACT PHASE E=6 / L=9 / score ≈ 44.254 — E upper bound

PROJECTIVE E=6 / L=8 / score ≈ 42.048 — E upper bound

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=1 / L=9 / score ≈ 22.608 — proved E minimum



PROJECTIVE E=1 / L=8 / score ≈ 20.808 — proved E minimum

$U = \frac{\sqrt{2}(-1+i)}{2}\,C$. Scalar metadata; no physical GP.

ibm_cz

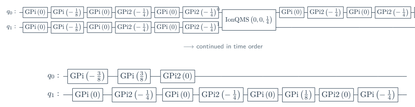
EXACT PHASE E=1 / L=9 / score ≈ 22.606 — proved E minimum

PROJECTIVE E=1 / L=6 / score ≈ 16.404 — proved E minimum

$U = \frac{\sqrt{2}(-1+i)}{2}\,C$. Scalar metadata; no physical GP.

ionq_aria

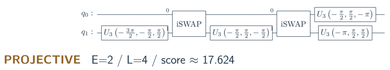
PROJECTIVE E=1 / L=28 / score ≈ 67.062 — proved E minimum



$U = (-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 18.024 — proved E minimum

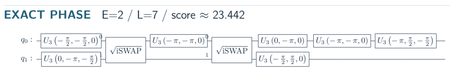


PROJECTIVE E=2 / L=4 / score ≈ 17.624 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=2 / L=7 / score ≈ 23.442 — proved E minimum

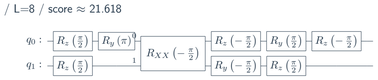


PROJECTIVE E=2 / L=5 / score ≈ 19.030 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=1 / L=8 / score ≈ 21.618 — proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_only

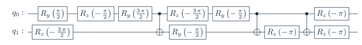
EXACT PHASE E=1 / L=13 / score ≈ 30.000 — proved E minimum

PROJECTIVE E=1 / L=11 / score ≈ 26.000 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=3 / L=10 / score ≈ 33.420 — E upper bound



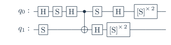
PROJECTIVE E=1 / L=7 / score ≈ 19.414 — proved E minimum

$U = \eta_1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=1 / L=13 / score ≈ 30.000 — proved E minimum

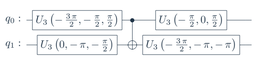
PROJECTIVE E=1 / L=11 / score ≈ 26.000 — proved E minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

cx_u

EXACT PHASE E=1 / L=4 / score ≈ 13.424 — proved E minimum



google_fsim

PROJECTIVE E=1 / L=6 / score ≈ 18.040 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=1 / L=9 / score ≈ 22.606 — proved E minimum

PROJECTIVE E=1 / L=6 / score ≈ 16.404 — proved E minimum

$U = \frac{\sqrt{2}(-1+i)}{2}\,C$. Scalar metadata; no physical GP.

ionq_aria_full

PROJECTIVE E=1 / L=11 / score ≈ 28.626 — proved E minimum

$U = \sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

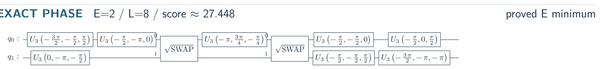
quantinuum_zzmax

PROJECTIVE E=1 / L=5 / score ≈ 15.018 — proved E minimum

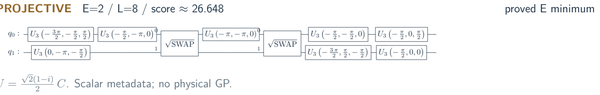
$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_swap_u

EXACT PHASE E=2 / L=8 / score ≈ 27.448 — proved E minimum



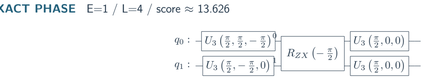
PROJECTIVE E=2 / L=8 / score ≈ 26.648 — proved E minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

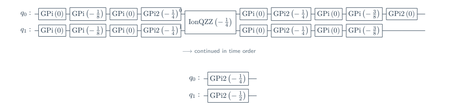
zx_u

EXACT PHASE E=1 / L=4 / score ≈ 13.626 — proved E minimum



ionq_forte

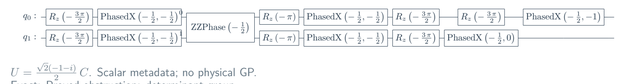
PROJECTIVE E=1 / L=19 / score ≈ 47.040 — proved E minimum



$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

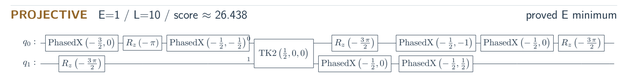
PROJECTIVE E=1 / L=13 / score ≈ 33.240 — proved E minimum



$U = \frac{\sqrt{2}(-1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

tket_tk2

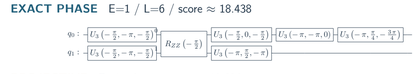
PROJECTIVE E=1 / L=10 / score ≈ 26.438 — proved E minimum



$U = \sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zz_u

EXACT PHASE E=1 / L=6 / score ≈ 18.438 — proved E minimum



PROJECTIVE E=1 / L=4 / score ≈ 13.626 — proved E minimum

$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

Margolus

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

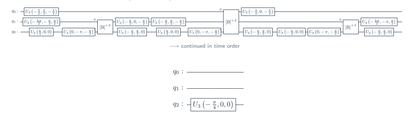
EXACT PHASE E=15 / L=99 / score ≈ 285.934 — E upper bound

PROJECTIVE E=3 / L=19 / score ≈ 56.070 — E upper bound

$U = \frac{\sqrt{2}i(1-i)}{4}\,C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=6 / L=16 / score ≈ 62.496 — E upper bound

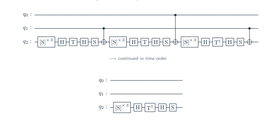


clifford_only

Exact: Proved obstruction: not a clifford
Projective: Proved obstruction: not a clifford

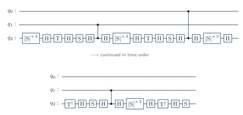
clifford_t_cx

EXACT PHASE E=3 / L=28 / score ≈ 68.000 — proved E minimum



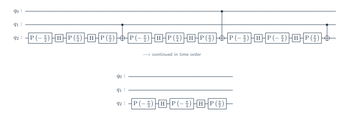
clifford_t_cz

EXACT PHASE E=3 / L=34 / score ≈ 80.000 — proved E minimum



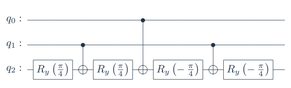
cx_h_phase

EXACT PHASE E=3 / L=20 / score ≈ 55.224 — proved E minimum



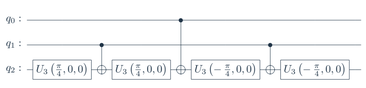
cx_rz_ry

EXACT PHASE E=3 / L=4 / score ≈ 21.608 — proved E minimum



cx_u

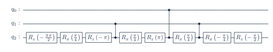
EXACT PHASE E=3 / L=4 / score ≈ 21.624 — proved E minimum



cz_rz_rx

EXACT PHASE E=3 / L=9 / score ≈ 32.018 — proved E minimum

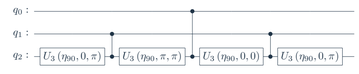
PROJECTIVE E=3 / L=8 / score ≈ 30.016 — proved E minimum



$U = -1\,C$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=3 / L=4 / score ≈ 20.024 — proved E minimum



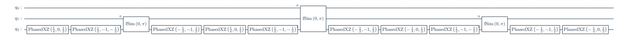
Exact definitions: η_{90} in the expression glossary.

google_cz

EXACT PHASE E=3 / L=10 / score ≈ 36.860 — proved E minimum

google_fsim

EXACT PHASE E=3 / L=10 / score ≈ 36.872 — E upper bound



google_sqrt_iswap

EXACT PHASE E=6 / L=16 / score ≈ 62.696 — E upper bound

PROJECTIVE E=6 / L=16 / score ≈ 62.096 — E upper bound

$U = \frac{\sqrt{2}(1-i)^2}{4}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=18 / L=16 / score ≈ 108.896 — E upper bound

ibm_cx

EXACT PHASE E=3 / L=14 / score ≈ 42.420 — proved E minimum

PROJECTIVE E=3 / L=10 / score ≈ 34.416 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=3 / L=18 / score ≈ 49.612 — proved E minimum

ibm_ecr

EXACT PHASE E=3 / L=28 / score ≈ 70.222 — proved E minimum

PROJECTIVE E=3 / L=19 / score ≈ 51.822 — proved E minimum

$U = \eta_{571}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=15 / L=127 / score ≈ 352.144 — E upper bound

PROJECTIVE E=3 / L=31 / score ≈ 82.280 — E upper bound

$U = \frac{\sqrt{2}(1+i)^3}{4}\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=15 / L=127 / score ≈ 346.114 — E upper bound

PROJECTIVE E=3 / L=31 / score ≈ 81.074 — E upper bound

$U = \frac{\sqrt{2}(1+i)^3}{4}\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=15 / L=142 / score ≈ 388.114 — E upper bound

PROJECTIVE E=3 / L=34 / score ≈ 89.474 — E upper bound

$U = \frac{\sqrt{2}(1+i)^3}{4}\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=3 / L=16 / score ≈ 48.864 — proved E minimum

iswap_u

EXACT PHASE E=6 / L=11 / score ≈ 49.866 — E upper bound

Exact definitions: $\eta_{307}, \eta_{572}, \eta_{90}$ in the expression glossary.

quantinuum_zzmax

EXACT PHASE E=15 / L=113 / score ≈ 310.550 — E upper bound

PROJECTIVE E=3 / L=19 / score ≈ 55.458 — proved E minimum

$U = \frac{\sqrt{2}i(1-i)}{2}\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=15 / L=89 / score ≈ 265.484 — E upper bound

PROJECTIVE E=3 / L=19 / score ≈ 56.064 — E upper bound

$U = \frac{\sqrt{2}i(1-i)}{2}\,C$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=3 / L=31 / score ≈ 77.630 — proved E minimum

PROJECTIVE E=3 / L=31 / score ≈ 77.428 — proved E minimum

$U = -(-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=6 / L=16 / score ≈ 61.296 — E upper bound

sqrt_swap_u

EXACT PHASE E=6 / L=16 / score ≈ 62.096 — E upper bound

PROJECTIVE E=6 / L=16 / score ≈ 61.296 — E upper bound

$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=15 / L=99 / score ≈ 285.994 — E upper bound

PROJECTIVE E=3 / L=19 / score ≈ 56.082 — E upper bound

$U = \frac{\sqrt{2}i(1-i)}{2}\,C$. Scalar metadata; no physical GP.

xx90_u

EXACT PHASE E=3 / L=13 / score ≈ 43.278 — proved E minimum

xx_rz_ry

EXACT PHASE E=15 / L=127 / score ≈ 342.884 — E upper bound

PROJECTIVE E=3 / L=21 / score ≈ 57.448 — E upper bound

$U = \frac{\sqrt{2}i(-1+i)}{2}\,C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=3 / L=9 / score ≈ 33.460 — E upper bound

Exact definitions: $\eta_{570}, \eta_{574}, \eta_{575}, \eta_{576}$ in the expression glossary.

PROJECTIVE E=3 / L=7 / score ≈ 28.248 — E upper bound

Exact definitions: $\eta_{574}, \eta_{575}, \eta_{576}$ in the expression glossary.

$U = \eta_{308}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=3 / L=9 / score ≈ 32.460 — E upper bound

Exact definitions: $\eta_{577}, \eta_{578}, \eta_{90}$ in the expression glossary.

PROJECTIVE E=3 / L=7 / score ≈ 27.848 — E upper bound

Exact definitions: η_{90} in the expression glossary.

$U = \frac{\sqrt{2}(1+i)^3}{4}\,C$. Scalar metadata; no physical GP.

MCMT

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

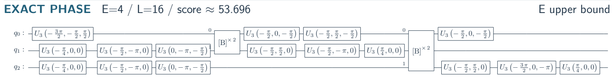
EXACT PHASE E=3 / L=24 / score ≈ 66.486 — E upper bound

PROJECTIVE E=2 / L=22 / score ≈ 58.476 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=4 / L=16 / score ≈ 53.696 — E upper bound

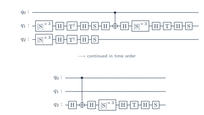


clifford_only

Exact: Proved obstruction: not a clifford
Projective: Proved obstruction: not a clifford

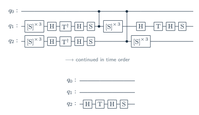
clifford_t_cx

EXACT PHASE E=2 / L=32 / score ≈ 72.000 — proved E minimum



clifford_t_cz

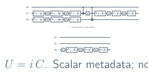
EXACT PHASE E=2 / L=28 / score ≈ 64.000 — proved E minimum



cx_h_phase

EXACT PHASE E=2 / L=24 / score ≈ 59.224 — proved E minimum

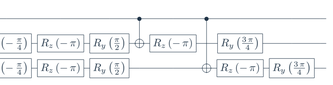
PROJECTIVE E=2 / L=22 / score ≈ 55.224 — proved E minimum



$U = i\,C$. Scalar metadata; no physical GP.

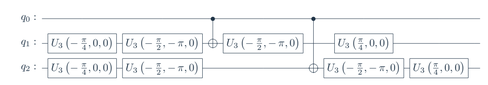
cx_rz_ry

EXACT PHASE E=2 / L=10 / score ≈ 30.020 — proved E minimum



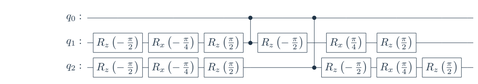
cx_u

EXACT PHASE E=2 / L=8 / score ≈ 26.448 — proved E minimum



cz_rz_rx

EXACT PHASE E=2 / L=12 / score ≈ 35.224 — proved E minimum

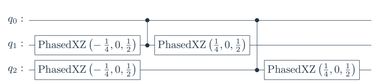


cz_u

EXACT PHASE E=2 / L=4 / score ≈ 17.624 — proved E minimum

google_cz

EXACT PHASE E=2 / L=4 / score ≈ 18.424 — proved E minimum

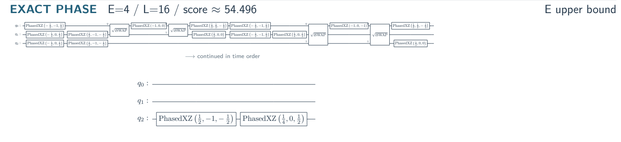


google_fsim

EXACT PHASE E=2 / L=12 / score ≈ 37.680 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=4 / L=16 / score ≈ 54.496 — E upper bound

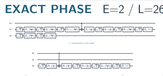


google_syc

EXACT PHASE E=12 / L=16 / score ≈ 85.696 — E upper bound

ibm_cx

EXACT PHASE E=2 / L=26 / score ≈ 63.228 — proved E minimum

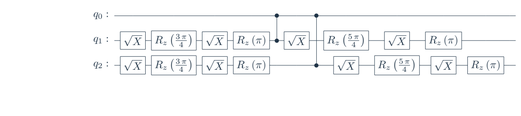


PROJECTIVE E=2 / L=26 / score ≈ 62.824 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

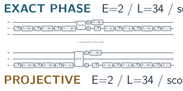
ibm_cz

EXACT PHASE E=2 / L=16 / score ≈ 41.616 — proved E minimum

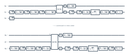


ibmq_ecr

EXACT PHASE E=2 / L=34 / score ≈ 79.632 — proved E minimum



PROJECTIVE E=2 / L=34 / score ≈ 79.228 — proved E minimum



$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=34 / score ≈ 84.880 — proved E minimum

PROJECTIVE E=2 / L=34 / score ≈ 84.480 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

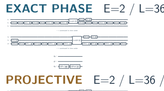
EXACT PHASE E=2 / L=34 / score ≈ 84.076 — proved E minimum

PROJECTIVE E=2 / L=34 / score ≈ 83.676 — proved E minimum

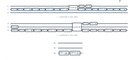
$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=36 / score ≈ 89.676 — proved E minimum



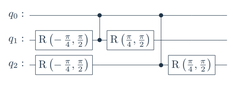
PROJECTIVE E=2 / L=36 / score ≈ 89.276 — proved E minimum



$U = -i\,C$. Scalar metadata; no physical GP.

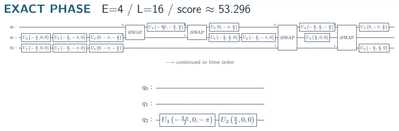
iqm_cz

EXACT PHASE E=2 / L=4 / score ≈ 18.416 — proved E minimum



iswap_u

EXACT PHASE E=4 / L=16 / score ≈ 53.296 — E upper bound



quantinuum_zzmax

EXACT PHASE E=5 / L=41 / score ≈ 112.326 — E upper bound

PROJECTIVE E=2 / L=22 / score ≈ 58.068 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

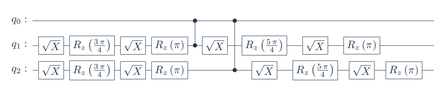
EXACT PHASE E=3 / L=24 / score ≈ 66.478 — E upper bound

PROJECTIVE E=2 / L=22 / score ≈ 58.472 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=2 / L=16 / score ≈ 41.616 — proved E minimum



sqrt_iswap_u

EXACT PHASE E=4 / L=16 / score ≈ 52.896 — E upper bound

sqrt_swap_u

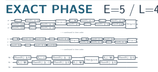
EXACT PHASE E=4 / L=16 / score ≈ 53.296 — E upper bound

PROJECTIVE E=4 / L=16 / score ≈ 52.896 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=5 / L=42 / score ≈ 115.368 — E upper bound

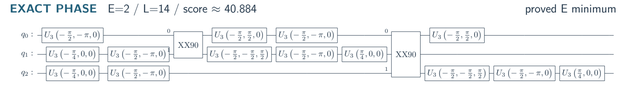


PROJECTIVE E=2 / L=22 / score ≈ 58.484 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

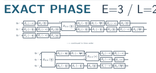
xx90_u

EXACT PHASE E=2 / L=14 / score ≈ 40.884 — proved E minimum



xx_rz_ry

EXACT PHASE E=3 / L=28 / score ≈ 72.862 — E upper bound

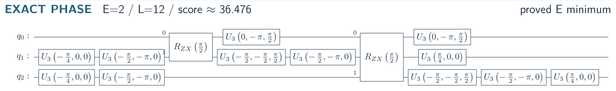


PROJECTIVE E=2 / L=24 / score ≈ 60.852 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

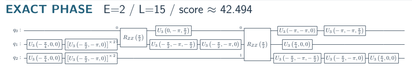
zx_u

EXACT PHASE E=2 / L=12 / score ≈ 36.476 — proved E minimum

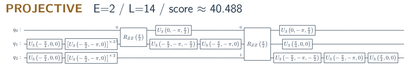


zz_u

EXACT PHASE E=2 / L=15 / score ≈ 42.494 — proved E minimum



PROJECTIVE E=2 / L=14 / score ≈ 40.488 — proved E minimum



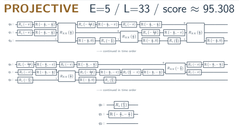
$U = i\,C$. Scalar metadata; no physical GP.

MCPhase

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=5 / L=33 / score ≈ 95.308 — E upper bound



$U = -e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=12 / L=31 / score ≈ 122.586 — E upper bound

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

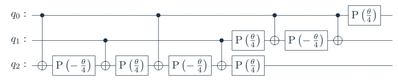
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

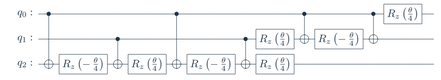
cx_h_phase

EXACT PHASE E=6 / L=7 / score ≈ 40.814 — E upper bound



cx_rz_ry

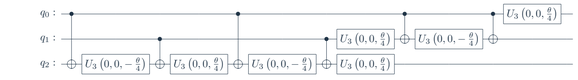
PROJECTIVE E=6 / L=7 / score ≈ 40.814 — E upper bound



$U = e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

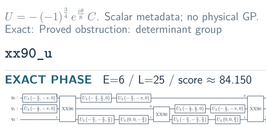
cx_u

EXACT PHASE E=6 / L=7 / score ≈ 40.842 — E upper bound



cz_rz_rx

PROJECTIVE E=6 / L=33 / score ≈ 98.066 — E upper bound



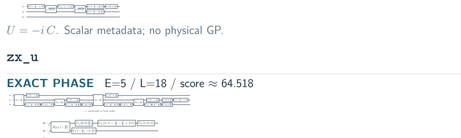
$U = e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=6 / L=19 / score ≈ 67.314 — E upper bound

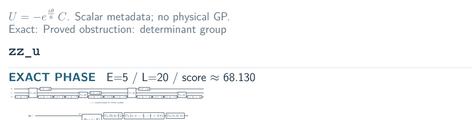
google_cz

EXACT PHASE E=6 / L=19 / score ≈ 72.026 — E upper bound



google_fsim

EXACT PHASE E=5 / L=16 / score ≈ 61.358 — E upper bound

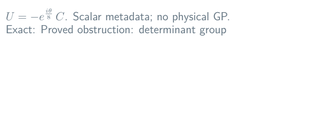


google_sqrt_iswap

EXACT PHASE E=12 / L=31 / score ≈ 122.498 — E upper bound

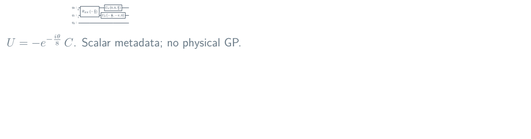
google_syc

EXACT PHASE E=36 / L=31 / score ≈ 216.098 — E upper bound



ibm_cx

PROJECTIVE E=6 / L=7 / score ≈ 40.814 — E upper bound



$U = e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

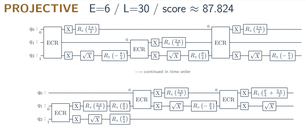
ibm_cz

PROJECTIVE E=6 / L=33 / score ≈ 95.440 — E upper bound

$U = -(-1)^{\frac{3}{4}} e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

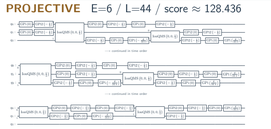
PROJECTIVE E=6 / L=30 / score ≈ 87.824 — E upper bound



$U = e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

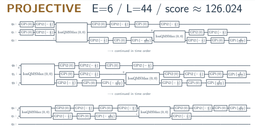
PROJECTIVE E=6 / L=44 / score ≈ 128.436 — E upper bound



$U = -ie^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

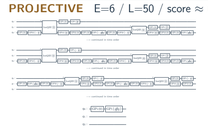
PROJECTIVE E=6 / L=44 / score ≈ 126.024 — E upper bound



$U = -ie^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=6 / L=50 / score ≈ 142.824 — E upper bound



$U = ie^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

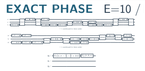
iqm_cz

PROJECTIVE E=6 / L=45 / score ≈ 127.380 — E upper bound

$U = e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=10 / L=28 / score ≈ 106.168 — E upper bound



PROJECTIVE E=10 / L=26 / score ≈ 100.956 — E upper bound

$U = -e^{-\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.

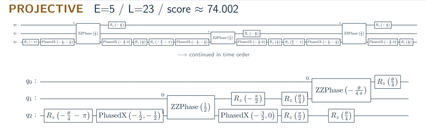
quantinuum_zzmax

PROJECTIVE E=6 / L=31 / score ≈ 94.686 — E upper bound

$U = ie^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=5 / L=23 / score ≈ 74.002 — E upper bound



$U = -e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

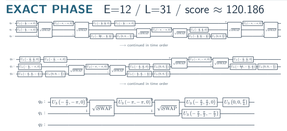
rigetti_cz

PROJECTIVE E=6 / L=33 / score ≈ 95.440 — E upper bound

$U = -(-1)^{\frac{3}{4}} e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

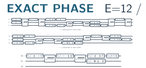
sqrt_iswap_u

EXACT PHASE E=12 / L=31 / score ≈ 120.186 — E upper bound



sqrt_swap_u

EXACT PHASE E=12 / L=31 / score ≈ 120.586 — E upper bound

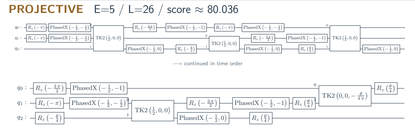


PROJECTIVE E=12 / L=31 / score ≈ 120.186 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

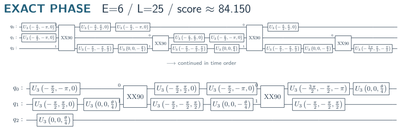
PROJECTIVE E=5 / L=26 / score ≈ 80.036 — E upper bound



$U = -e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

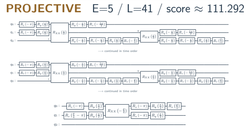
xx90_u

EXACT PHASE E=6 / L=25 / score ≈ 84.150 — E upper bound



xx_rz_ry

PROJECTIVE E=5 / L=41 / score ≈ 111.292 — E upper bound



$U = -e^{\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=5 / L=18 / score ≈ 64.518 — E upper bound

PROJECTIVE E=5 / L=16 / score ≈ 59.306 — E upper bound

$U = -e^{-\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=5 / L=20 / score ≈ 68.130 — E upper bound

PROJECTIVE E=5 / L=18 / score ≈ 62.918 — E upper bound

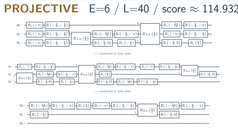
$U = -e^{-\frac{i\theta}{8}} C$. Scalar metadata; no physical GP.

MCX

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

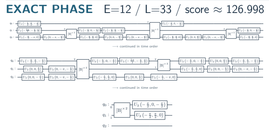
PROJECTIVE E=6 / L=40 / score ≈ 114.932 — E upper bound



$U = -(-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=12 / L=33 / score ≈ 126.998 — E upper bound

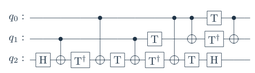


clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

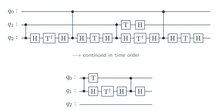
clifford_t_cx

EXACT PHASE E=6 / L=9 / score ≈ 42.000 — E upper bound



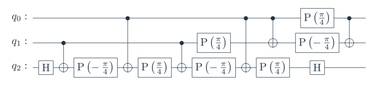
clifford_t_cz

EXACT PHASE E=6 / L=19 / score ≈ 62.000 — E upper bound



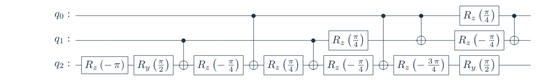
cx_h_phase

EXACT PHASE E=6 / L=9 / score ≈ 44.814 — E upper bound



cx_rz_ry

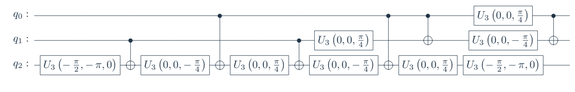
PROJECTIVE E=6 / L=10 / score ≈ 47.220 — E upper bound



$U = -\sqrt[8]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

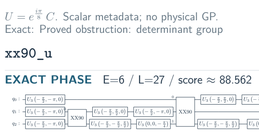
cx_u

EXACT PHASE E=6 / L=9 / score ≈ 45.254 — E upper bound



cz_rz_rx

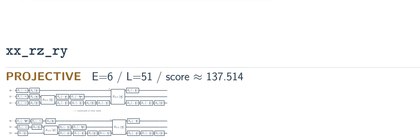
PROJECTIVE E=6 / L=40 / score ≈ 113.280 — E upper bound



$U = -\sqrt[8]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=6 / L=21 / score ≈ 71.726 — E upper bound



google_cz

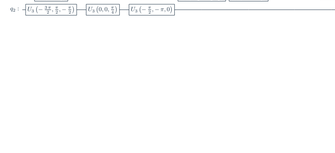
EXACT PHASE E=6 / L=21 / score ≈ 74.526 — E upper bound

google_fsim

EXACT PHASE E=6 / L=21 / score ≈ 74.550 — E upper bound

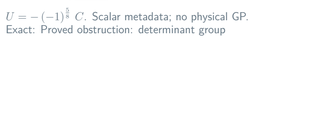
google_sqrt_iswap

EXACT PHASE E=12 / L=33 / score ≈ 124.998 — E upper bound



google_syc

EXACT PHASE E=36 / L=33 / score ≈ 218.598 — E upper bound



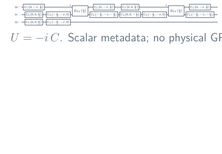
ibm_cx

PROJECTIVE E=6 / L=12 / score ≈ 51.218 — E upper bound

$U = e^{-\frac{3\pi i}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=6 / L=40 / score ≈ 109.642 — E upper bound



$U = e^{\frac{1\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=6 / L=35 / score ≈ 98.228 — E upper bound

$U = e^{-\frac{3\pi i}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=6 / L=48 / score ≈ 134.932 — E upper bound

$U = -(-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=6 / L=48 / score ≈ 132.520 — E upper bound

$U = -(-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=6 / L=54 / score ≈ 149.320 — E upper bound

$U = (-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=6 / L=49 / score ≈ 136.196 — E upper bound

$U = -\sqrt[8]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=12 / L=33 / score ≈ 125.798 — E upper bound

quantinuum_zzmax

PROJECTIVE E=6 / L=35 / score ≈ 103.498 — E upper bound

$U = -(-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=6 / L=35 / score ≈ 104.710 — E upper bound

$U = -(-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=6 / L=40 / score ≈ 109.642 — E upper bound

$U = e^{\frac{1\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=12 / L=33 / score ≈ 124.598 — E upper bound

sqrt_swap_u

EXACT PHASE E=12 / L=33 / score ≈ 124.998 — E upper bound

PROJECTIVE E=12 / L=33 / score ≈ 124.598 — E upper bound

$U = -i\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=6 / L=40 / score ≈ 114.956 — E upper bound

$U = -(-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=6 / L=27 / score ≈ 88.562 — E upper bound

xx_rz_ry

PROJECTIVE E=6 / L=51 / score ≈ 137.514 — E upper bound

$U = -(-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=6 / L=21 / score ≈ 75.338 — E upper bound

zz_u

EXACT PHASE E=6 / L=28 / score ≈ 89.380 — E upper bound

PROJECTIVE E=6 / L=27 / score ≈ 87.374 — E upper bound

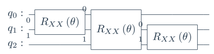
$U = -i\, C$. Scalar metadata; no physical GP.

MS

3 qubits / 32 preserved native bases / alphabetical panel order

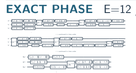
aqt_rxx

EXACT PHASE E=3 / L=0 / score ≈ 12.006 — E upper bound



berkeley_b

EXACT PHASE E=12 / L=41 / score ≈ 142.646 — E upper bound



PROJECTIVE E=12 / L=39 / score ≈ 138.234 — E upper bound

$U = e^{-\frac{3i\theta}{2}}\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

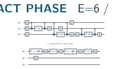
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=6 / L=21 / score ≈ 66.414 — E upper bound

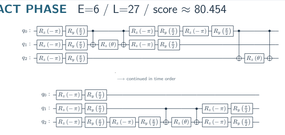


PROJECTIVE E=6 / L=13 / score ≈ 50.006 — E upper bound

$U = e^{-\frac{3i\theta}{2}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=6 / L=27 / score ≈ 80.454 — E upper bound



cx_u

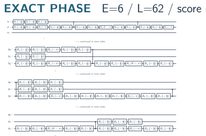
EXACT PHASE E=6 / L=17 / score ≈ 60.902 — E upper bound

PROJECTIVE E=6 / L=15 / score ≈ 56.490 — E upper bound

$U = e^{-\frac{3i\theta}{2}}\,C$. Scalar metadata; no physical GP.

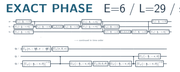
cz_rz_rx

EXACT PHASE E=6 / L=62 / score ≈ 158.524 — E upper bound



cz_u

EXACT PHASE E=6 / L=29 / score ≈ 87.374 — E upper bound



PROJECTIVE E=6 / L=27 / score ≈ 82.962 — E upper bound

$U = e^{-\frac{3i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=6 / L=29 / score ≈ 93.625 — E upper bound

PROJECTIVE E=6 / L=27 / score ≈ 88.753 — E upper bound

$U = e^{-\frac{3i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=6 / L=6 / score ≈ 37.260 — E upper bound

google_sqrt_iswap

EXACT PHASE E=12 / L=41 / score ≈ 144.097 — E upper bound

PROJECTIVE E=12 / L=39 / score ≈ 139.225 — E upper bound

$U = ie^{-\frac{3i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_syc

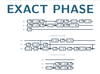
EXACT PHASE E=36 / L=41 / score ≈ 237.697 — E upper bound

PROJECTIVE E=36 / L=39 / score ≈ 232.825 — E upper bound

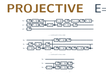
$U = e^{-\frac{3i\theta}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=6 / L=38 / score ≈ 102.836 — E upper bound



PROJECTIVE E=6 / L=38 / score ≈ 102.432 — E upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=6 / L=63 / score ≈ 154.864 — E upper bound

PROJECTIVE E=6 / L=62 / score ≈ 152.866 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=6 / L=62 / score ≈ 152.048 — E upper bound

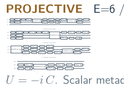
PROJECTIVE E=6 / L=62 / score ≈ 151.644 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=6 / L=60 / score ≈ 159.947 — E upper bound

PROJECTIVE E=6 / L=60 / score ≈ 159.547 — E upper bound

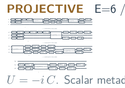


$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=6 / L=60 / score ≈ 157.535 — E upper bound

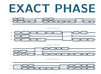
PROJECTIVE E=6 / L=60 / score ≈ 157.135 — E upper bound



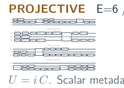
$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=6 / L=66 / score ≈ 174.335 — E upper bound



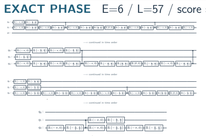
PROJECTIVE E=6 / L=66 / score ≈ 173.935 — E upper bound



$U = i\,C$. Scalar metadata; no physical GP.

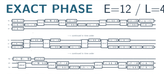
iqm_cz

EXACT PHASE E=6 / L=57 / score ≈ 150.228 — E upper bound



iswap_u

EXACT PHASE E=12 / L=41 / score ≈ 141.446 — E upper bound



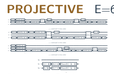
PROJECTIVE E=12 / L=39 / score ≈ 137.034 — E upper bound

$U = e^{-\frac{3i\theta}{2}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=9 / L=73 / score ≈ 197.214 — E upper bound

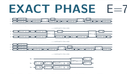
PROJECTIVE E=6 / L=54 / score ≈ 142.956 — E upper bound



$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=7 / L=56 / score ≈ 152.174 — E upper bound



PROJECTIVE E=6 / L=54 / score ≈ 144.168 — E upper bound

$U = i\,C$. Scalar metadata; no physical GP.

rigetti_cz

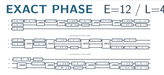
EXACT PHASE E=6 / L=63 / score ≈ 154.864 — E upper bound

PROJECTIVE E=6 / L=62 / score ≈ 152.866 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=12 / L=41 / score ≈ 140.246 — E upper bound

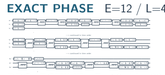


PROJECTIVE E=12 / L=39 / score ≈ 135.834 — E upper bound

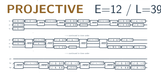
$U = e^{-\frac{3i\theta}{2}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=12 / L=41 / score ≈ 140.646 — E upper bound



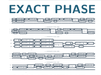
PROJECTIVE E=12 / L=39 / score ≈ 135.834 — E upper bound



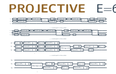
$U = -ie^{-\frac{3i\theta}{2}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=9 / L=77 / score ≈ 207.094 — E upper bound



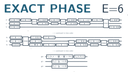
PROJECTIVE E=6 / L=57 / score ≈ 150.210 — E upper bound



$U = i\,C$. Scalar metadata; no physical GP.

xx90_u

EXACT PHASE E=6 / L=35 / score ≈ 104.210 — E upper bound

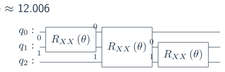


PROJECTIVE E=6 / L=33 / score ≈ 99.798 — E upper bound

$U = -e^{-\frac{3i\theta}{2}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=3 / L=0 / score ≈ 12.006 — E upper bound



zx_u

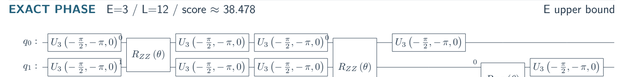
EXACT PHASE E=6 / L=29 / score ≈ 90.986 — E upper bound

PROJECTIVE E=6 / L=27 / score ≈ 86.574 — E upper bound

$U = -e^{-\frac{3i\theta}{2}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=3 / L=12 / score ≈ 38.478 — E upper bound



P (θ)

1 qubits / 32 preserved native bases / alphabetical panel order

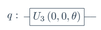
aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved E minimum

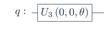
cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum



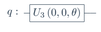
cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum



google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.336 proved E minimum

google_fsim

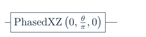
EXACT PHASE E=0 / L=1 / score ≈ 2.336 proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.336 proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.336 proved E minimum



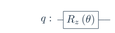
ibm_cx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum



$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.734 proved E minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.734 proved E minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.734 proved E minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

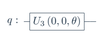
iqm_cz

PROJECTIVE E=0 / L=3 / score ≈ 6.812 proved E minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum



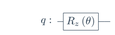
quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum



$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

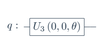
rigetti_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum



sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

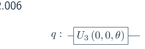
$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum



Pauli

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=1 / L=5 / score ≈ 14.220 — E upper bound

PROJECTIVE E=0 / L=3 / score ≈ 6.210 — E upper bound

$U = iC$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=0 / L=3 / score ≈ 6.418 — E upper bound

PROJECTIVE E=0 / L=3 / score ≈ 6.018 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=14 / score ≈ 28.000 — E upper bound

PROJECTIVE E=0 / L=12 / score ≈ 24.000 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=14 / score ≈ 28.000 — E upper bound

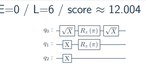
PROJECTIVE E=0 / L=12 / score ≈ 24.000 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=14 / score ≈ 28.000 — E upper bound

PROJECTIVE E=0 / L=12 / score ≈ 24.000 — E upper bound

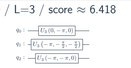


$U = -iC$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=9 / score ≈ 18.410 — E upper bound

PROJECTIVE E=0 / L=8 / score ≈ 16.008 — E upper bound



$U = -iC$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=2 / L=7 / score ≈ 22.014 — E upper bound

PROJECTIVE E=0 / L=4 / score ≈ 8.008 — E upper bound

$U = iC$. Scalar metadata; no physical GP.

cx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 — E upper bound

PROJECTIVE E=0 / L=3 / score ≈ 6.018 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=2 / L=14 / score ≈ 37.628 — E upper bound

PROJECTIVE E=0 / L=4 / score ≈ 8.008 — E upper bound

$U = iC$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 — E upper bound

PROJECTIVE E=0 / L=3 / score ≈ 6.018 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

google_cz

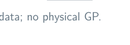
EXACT PHASE E=0 / L=3 / score ≈ 6.218 — E upper bound

PROJECTIVE E=0 / L=3 / score ≈ 6.018 — E upper bound

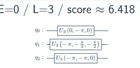
$U = -iC$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=3 / score ≈ 6.218 — E upper bound



PROJECTIVE E=0 / L=3 / score ≈ 6.018 — E upper bound



$U = -iC$. Scalar metadata; no physical GP.

google_sqrt_iswap

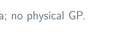
EXACT PHASE E=0 / L=3 / score ≈ 6.218 — E upper bound

PROJECTIVE E=0 / L=3 / score ≈ 6.018 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=3 / score ≈ 6.218 — E upper bound



PROJECTIVE E=0 / L=3 / score ≈ 6.018 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.004 — E upper bound

PROJECTIVE E=0 / L=4 / score ≈ 8.004 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.004 — E upper bound

PROJECTIVE E=0 / L=4 / score ≈ 8.004 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.004 — E upper bound

PROJECTIVE E=0 / L=4 / score ≈ 8.004 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=5 / score ≈ 10.810 — E upper bound

PROJECTIVE E=0 / L=4 / score ≈ 8.808 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=5 / score ≈ 10.810 — E upper bound

PROJECTIVE E=0 / L=4 / score ≈ 8.808 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=5 / score ≈ 10.810 — E upper bound

PROJECTIVE E=0 / L=4 / score ≈ 8.808 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=18 / score ≈ 46.672 — E upper bound

PROJECTIVE E=0 / L=4 / score ≈ 8.416 — E upper bound

$U = iC$. Scalar metadata; no physical GP.

iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 — E upper bound

PROJECTIVE E=0 / L=3 / score ≈ 6.018 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=3 / L=22 / score ≈ 60.468 — E upper bound

PROJECTIVE E=0 / L=3 / score ≈ 6.210 — E upper bound

$U = iC$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=1 / L=5 / score ≈ 14.216 — E upper bound

PROJECTIVE E=0 / L=3 / score ≈ 6.210 — E upper bound

$U = iC$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.004 — E upper bound

PROJECTIVE E=0 / L=4 / score ≈ 8.004 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 — E upper bound

PROJECTIVE E=0 / L=3 / score ≈ 6.018 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 — E upper bound

PROJECTIVE E=0 / L=3 / score ≈ 6.018 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=3 / L=23 / score ≈ 63.094 — E upper bound

PROJECTIVE E=0 / L=3 / score ≈ 6.210 — E upper bound

$U = iC$. Scalar metadata; no physical GP.

xx90_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 — E upper bound

PROJECTIVE E=0 / L=3 / score ≈ 6.018 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=1 / L=8 / score ≈ 20.018 — E upper bound

PROJECTIVE E=0 / L=4 / score ≈ 8.808 — E upper bound

$U = iC$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 — E upper bound

PROJECTIVE E=0 / L=3 / score ≈ 6.018 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 — E upper bound

PROJECTIVE E=0 / L=3 / score ≈ 6.018 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

PauliRotation

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=4 / L=29 / score ≈ 80.098 — E upper bound

PROJECTIVE E=3 / L=27 / score ≈ 72.088 — E upper bound

$U = iC$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=8 / L=25 / score ≈ 90.150 — E upper bound

PROJECTIVE E=8 / L=23 / score ≈ 85.738 — E upper bound

$U = -e^{-\frac{i\theta}{2}}C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

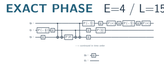
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=4 / L=15 / score ≈ 46.814 — E upper bound

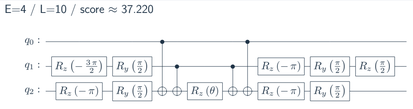


PROJECTIVE E=4 / L=7 / score ≈ 30.406 — E upper bound

$U = e^{-\frac{i\theta}{2}}C$. Scalar metadata; no physical GP.

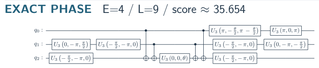
cx_rz_ry

EXACT PHASE E=4 / L=10 / score ≈ 37.220 — E upper bound

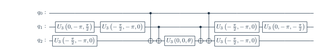


cx_u

EXACT PHASE E=4 / L=9 / score ≈ 35.654 — E upper bound



PROJECTIVE E=4 / L=7 / score ≈ 31.242 — E upper bound



$U = e^{-\frac{i\theta}{2}}C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=4 / L=30 / score ≈ 80.860 — E upper bound

cz_u

EXACT PHASE E=4 / L=17 / score ≈ 53.302 — E upper bound

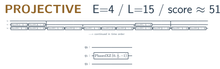
PROJECTIVE E=4 / L=15 / score ≈ 48.890 — E upper bound

$U = e^{-\frac{i\theta}{2}}C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=4 / L=17 / score ≈ 56.493 — E upper bound

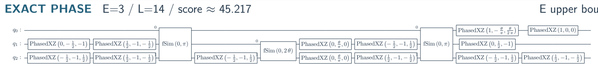
PROJECTIVE E=4 / L=15 / score ≈ 51.620 — E upper bound



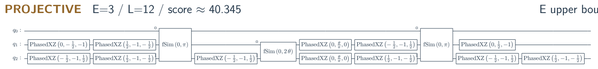
$U = e^{-\frac{i\theta}{2}}C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=3 / L=14 / score ≈ 45.217 — E upper bound



PROJECTIVE E=3 / L=12 / score ≈ 40.345 — E upper bound



$U = e^{-\frac{i\theta}{2}}C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=8 / L=25 / score ≈ 90.141 — E upper bound

PROJECTIVE E=8 / L=23 / score ≈ 85.268 — E upper bound

$U = -e^{-\frac{i\theta}{2}}C$. Scalar metadata; no physical GP.

google_syc

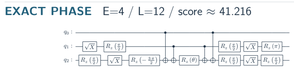
EXACT PHASE E=24 / L=25 / score ≈ 152.541 — E upper bound

PROJECTIVE E=24 / L=23 / score ≈ 147.668 — E upper bound

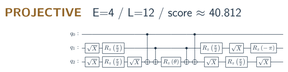
$U = e^{-\frac{i\theta}{2}}C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=4 / L=12 / score ≈ 41.216 — E upper bound



PROJECTIVE E=4 / L=12 / score ≈ 40.812 — E upper bound



$U = -iC$. Scalar metadata; no physical GP.

ibm_cz

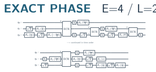
EXACT PHASE E=4 / L=30 / score ≈ 78.436 — E upper bound

PROJECTIVE E=4 / L=30 / score ≈ 78.434 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=4 / L=28 / score ≈ 74.024 — E upper bound

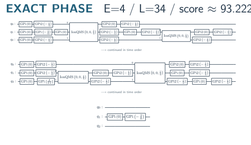


PROJECTIVE E=4 / L=28 / score ≈ 73.620 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

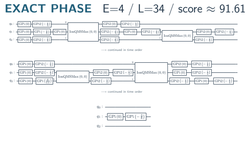
ionq_aria

EXACT PHASE E=4 / L=34 / score ≈ 93.222 — E upper bound



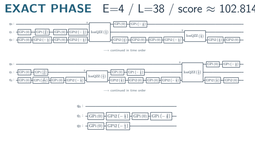
ionq_aria_full

EXACT PHASE E=4 / L=34 / score ≈ 91.614 — E upper bound



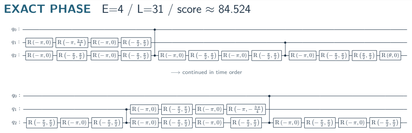
ionq_forte

EXACT PHASE E=4 / L=38 / score ≈ 102.814 — E upper bound



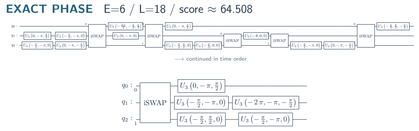
iqm_cz

EXACT PHASE E=4 / L=31 / score ≈ 84.524 — E upper bound



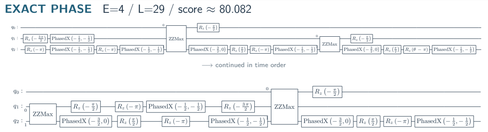
iswap_u

EXACT PHASE E=6 / L=18 / score ≈ 64.508 — E upper bound



quantinuum_zzmax

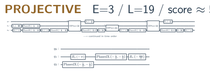
EXACT PHASE E=4 / L=29 / score ≈ 80.082 — E upper bound



quantinuum_zzphase

EXACT PHASE E=4 / L=21 / score ≈ 62.796 — E upper bound

PROJECTIVE E=3 / L=19 / score ≈ 54.790 — E upper bound



$U = iC$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=4 / L=30 / score ≈ 78.436 — E upper bound

PROJECTIVE E=4 / L=30 / score ≈ 78.434 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=8 / L=25 / score ≈ 88.550 — E upper bound

PROJECTIVE E=8 / L=23 / score ≈ 84.138 — E upper bound

$U = -e^{-\frac{i\theta}{2}}C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=8 / L=25 / score ≈ 88.550 — E upper bound

PROJECTIVE E=8 / L=23 / score ≈ 84.138 — E upper bound

$U = e^{-\frac{i\theta}{2}}C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=4 / L=30 / score ≈ 82.916 — E upper bound

PROJECTIVE E=3 / L=19 / score ≈ 54.806 — E upper bound

$U = iC$. Scalar metadata; no physical GP.

xx90_u

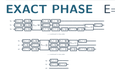
EXACT PHASE E=4 / L=21 / score ≈ 64.526 — E upper bound

PROJECTIVE E=4 / L=19 / score ≈ 60.114 — E upper bound

$U = -e^{-\frac{i\theta}{2}}C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=4 / L=35 / score ≈ 90.878 — E upper bound

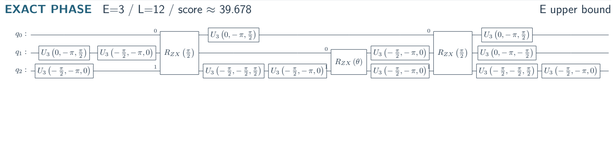


PROJECTIVE E=3 / L=31 / score ≈ 78.868 — E upper bound

$U = iC$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=3 / L=12 / score ≈ 39.678 — E upper bound



zz_u

EXACT PHASE E=3 / L=13 / score ≈ 41.284 — E upper bound

PROJECTIVE E=3 / L=12 / score ≈ 39.278 — E upper bound

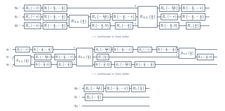
$U = iC$. Scalar metadata; no physical GP.

Peres

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

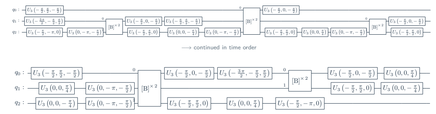
PROJECTIVE E=5 / L=35 / score ≈ 99.714 — E upper bound



$U = \frac{\sqrt[8]{-1}\sqrt{2}(-1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=10 / L=29 / score ≈ 109.374 — E upper bound

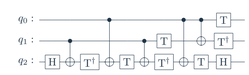


clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

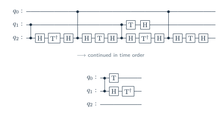
clifford_t_cx

EXACT PHASE E=5 / L=9 / score ≈ 38.000 — proved E minimum



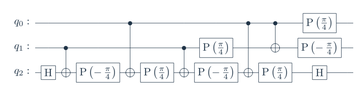
clifford_t_cz

EXACT PHASE E=5 / L=17 / score ≈ 54.000 — proved E minimum



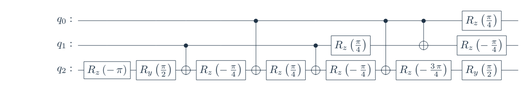
cx_h_phase

EXACT PHASE E=5 / L=9 / score ≈ 40.814 — proved E minimum



cx_rz_ry

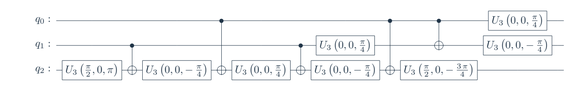
PROJECTIVE E=5 / L=10 / score ≈ 43.220 — proved E minimum



$U = -\sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

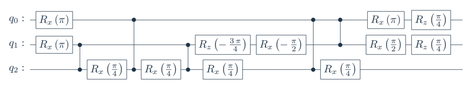
cx_u

EXACT PHASE E=5 / L=8 / score ≈ 39.248 — proved E minimum



cz_rz_rx

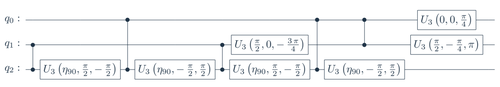
PROJECTIVE E=5 / L=12 / score ≈ 47.224 — proved E minimum



$U = e^{-\frac{3i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

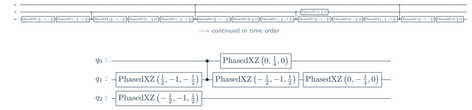
EXACT PHASE E=5 / L=7 / score ≈ 37.242 — proved E minimum



Exact definitions: η_{90} in the expression glossary.

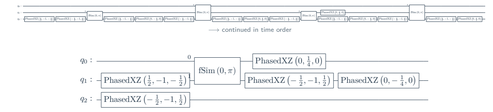
google_cz

EXACT PHASE E=5 / L=19 / score ≈ 65.714 — proved E minimum



google_fsim

EXACT PHASE E=5 / L=19 / score ≈ 65.734 — E upper bound



google_sqrt_iswap

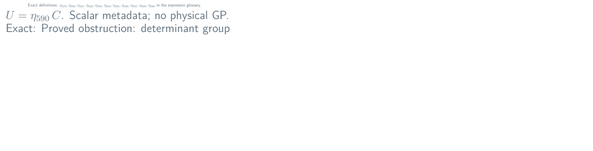
EXACT PHASE E=10 / L=29 / score ≈ 108.374 — E upper bound

PROJECTIVE E=10 / L=29 / score ≈ 107.774 — E upper bound

$U = \frac{\sqrt{2}(1-i)^5}{8}\,C$. Scalar metadata; no physical GP.

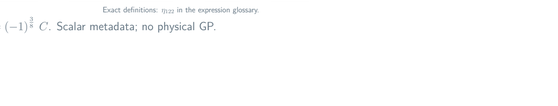
google_syc

EXACT PHASE E=30 / L=29 / score ≈ 185.774 — E upper bound



ibm_cx

PROJECTIVE E=5 / L=12 / score ≈ 47.218 — proved E minimum



$U = e^{-\frac{3i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=5 / L=26 / score ≈ 75.226 — proved E minimum

$U = e^{-\frac{3i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=5 / L=31 / score ≈ 86.026 — proved E minimum

$U = (-1)^{\frac{3}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=5 / L=43 / score ≈ 119.516 — E upper bound

$U = \frac{\sqrt[8]{-1}\sqrt{2}(1+i)^5}{8}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=5 / L=43 / score ≈ 117.506 — E upper bound

$U = \frac{\sqrt[8]{-1}\sqrt{2}(1+i)^5}{8}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=5 / L=48 / score ≈ 131.506 — E upper bound

$U = \frac{\sqrt[8]{-1}\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=5 / L=45 / score ≈ 123.380 — proved E minimum

$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=10 / L=18 / score ≈ 84.908 — E upper bound

Exact definitions: η_{90} in the expression glossary.

PROJECTIVE E=10 / L=16 / score ≈ 79.696 — E upper bound

$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=5 / L=31 / score ≈ 90.486 — proved E minimum

$U = \frac{\sqrt[8]{-1}\sqrt{2}(-1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=5 / L=31 / score ≈ 91.496 — E upper bound

$U = \frac{\sqrt[8]{-1}\sqrt{2}(-1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=5 / L=35 / score ≈ 95.238 — proved E minimum

$U = -(-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=10 / L=29 / score ≈ 107.374 — E upper bound

sqrt_swap_u

EXACT PHASE E=10 / L=29 / score ≈ 108.174 — E upper bound

PROJECTIVE E=10 / L=29 / score ≈ 107.374 — E upper bound

$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=5 / L=35 / score ≈ 99.734 — E upper bound

$U = \frac{\sqrt[8]{-1}\sqrt{2}(-1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=5 / L=24 / score ≈ 77.344 — proved E minimum

xx_rz_ry

PROJECTIVE E=5 / L=35 / score ≈ 94.080 — E upper bound

$U = \eta_{2090}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=5 / L=13 / score ≈ 53.288 — E upper bound

Exact definitions: η_{222} in the expression glossary.

PROJECTIVE E=5 / L=11 / score ≈ 48.076 — E upper bound

Exact definitions: η_{222} in the expression glossary.

$U = (-1)^{\frac{3}{8}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=5 / L=12 / score ≈ 49.082 — E upper bound

Exact definitions: η_{90} in the expression glossary.

PROJECTIVE E=5 / L=10 / score ≈ 44.270 — E upper bound

Exact definitions: η_{90} in the expression glossary.

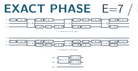
$U = \frac{\sqrt{2}(1+i)^5}{8}\,C$. Scalar metadata; no physical GP.

Permutation

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=7 / L=32 / score ≈ 99.318 — E upper bound



PROJECTIVE E=6 / L=30 / score ≈ 91.308 — E upper bound

$U = i\,C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=12 / L=24 / score ≈ 105.744 — E upper bound

clifford_only

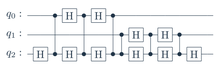
EXACT PHASE E=6 / L=0 / score ≈ 24.000 — E upper bound

clifford_t_cx

EXACT PHASE E=6 / L=0 / score ≈ 24.000 — E upper bound

clifford_t_cz

EXACT PHASE E=6 / L=10 / score ≈ 44.000 — E upper bound



cx_h_phase

EXACT PHASE E=6 / L=0 / score ≈ 24.000 — E upper bound

cx_rz_ry

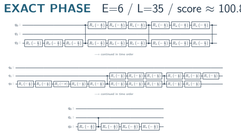
EXACT PHASE E=6 / L=0 / score ≈ 24.000 — E upper bound

cx_u

EXACT PHASE E=6 / L=0 / score ≈ 24.000 — E upper bound

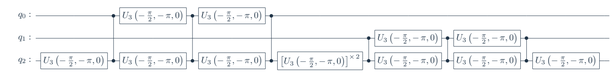
cz_rz_rx

EXACT PHASE E=6 / L=35 / score ≈ 100.870 — E upper bound



cz_u

EXACT PHASE E=6 / L=12 / score ≈ 50.472 — E upper bound



google_cz

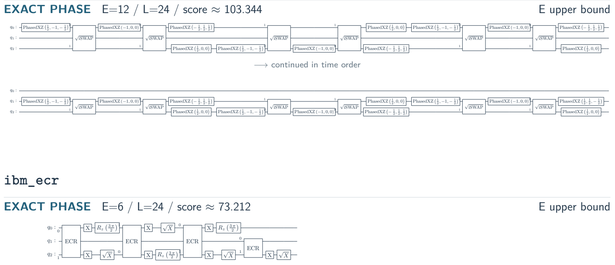
EXACT PHASE E=6 / L=12 / score ≈ 52.872 — E upper bound

google_fsim

EXACT PHASE E=6 / L=12 / score ≈ 52.896 — E upper bound

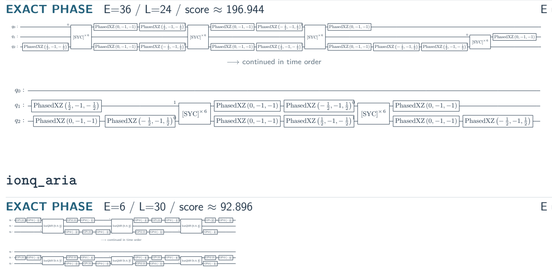
google_sqrt_iswap

EXACT PHASE E=12 / L=24 / score ≈ 103.344 — E upper bound



google_syc

EXACT PHASE E=36 / L=24 / score ≈ 196.944 — E upper bound

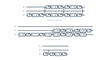


ibm_cx

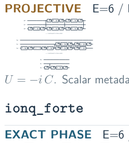
EXACT PHASE E=6 / L=0 / score ≈ 24.000 — E upper bound

ibm_cz

EXACT PHASE E=6 / L=35 / score ≈ 96.830 — E upper bound



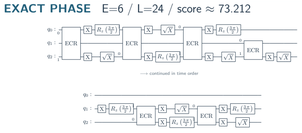
PROJECTIVE E=6 / L=35 / score ≈ 96.426 — E upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=6 / L=24 / score ≈ 73.212 — E upper bound



ionq_aria

EXACT PHASE E=6 / L=30 / score ≈ 92.896 — E upper bound

PROJECTIVE E=6 / L=30 / score ≈ 92.496 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

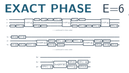
EXACT PHASE E=6 / L=30 / score ≈ 90.484 — E upper bound

PROJECTIVE E=6 / L=30 / score ≈ 90.084 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=6 / L=36 / score ≈ 107.284 — E upper bound

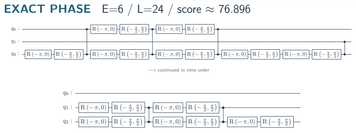


PROJECTIVE E=6 / L=36 / score ≈ 106.884 — E upper bound

$U = i\,C$. Scalar metadata; no physical GP.

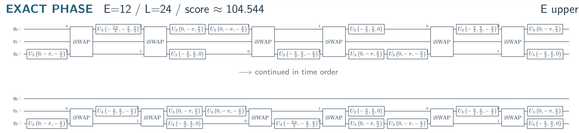
iqm_cz

EXACT PHASE E=6 / L=24 / score ≈ 76.896 — E upper bound



iswap_u

EXACT PHASE E=12 / L=24 / score ≈ 104.544 — E upper bound



quantinuum_zzmax

EXACT PHASE E=9 / L=45 / score ≈ 136.334 — E upper bound

PROJECTIVE E=6 / L=26 / score ≈ 82.076 — E upper bound

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=7 / L=28 / score ≈ 91.294 — E upper bound

PROJECTIVE E=6 / L=26 / score ≈ 83.288 — E upper bound

$U = i\,C$. Scalar metadata; no physical GP.

rigetti_cz

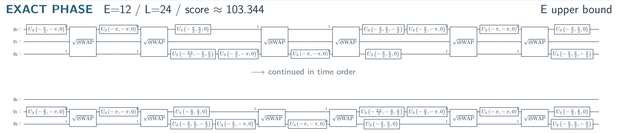
EXACT PHASE E=6 / L=35 / score ≈ 96.830 — E upper bound

PROJECTIVE E=6 / L=35 / score ≈ 96.426 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

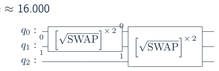
sqrt_iswap_u

EXACT PHASE E=12 / L=24 / score ≈ 103.344 — E upper bound



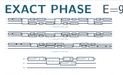
sqrt_swap_u

EXACT PHASE E=4 / L=0 / score ≈ 16.000 — E upper bound



tket_tk2

EXACT PHASE E=9 / L=50 / score ≈ 148.216 — E upper bound

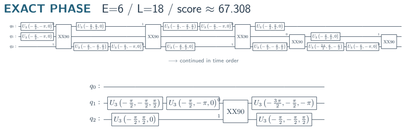


PROJECTIVE E=6 / L=30 / score ≈ 91.332 — E upper bound

$U = i\,C$. Scalar metadata; no physical GP.

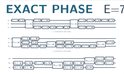
xx90_u

EXACT PHASE E=6 / L=18 / score ≈ 67.308 — E upper bound

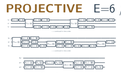


xx_rz_ry

EXACT PHASE E=7 / L=46 / score ≈ 128.506 — E upper bound



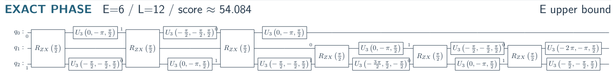
PROJECTIVE E=6 / L=42 / score ≈ 116.496 — E upper bound



$U = i\,C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=6 / L=12 / score ≈ 54.084 — E upper bound



zz_u

EXACT PHASE E=6 / L=19 / score ≈ 68.126 — E upper bound

PROJECTIVE E=6 / L=18 / score ≈ 66.120 — E upper bound

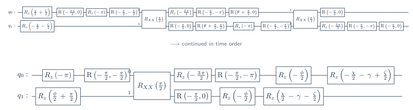
$U = -i\,C$. Scalar metadata; no physical GP.

PhasedFSim $(\theta, \zeta, \chi, \gamma, \phi)$

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

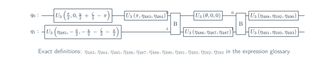
PROJECTIVE E=3 / L=26 / score ≈ 70.084 — minimum uniform E budget



$U = \eta_{591}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

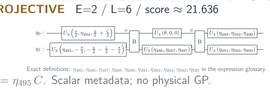
berkeley_b

EXACT PHASE E=2 / L=7 / score ≈ 23.642 — minimum uniform E budget



Exact definitions: $\eta_{611}, \eta_{614}, \eta_{615}, \eta_{596}, \eta_{617}, \eta_{598}, \eta_{599}, \eta_{601}, \eta_{603}, \eta_{620}, \eta_{605}, \eta_{606}$ in the expression glossary.

PROJECTIVE E=2 / L=6 / score ≈ 21.636 — minimum uniform E budget



Exact definitions: $\eta_{611}, \eta_{596}, \eta_{617}, \eta_{598}, \eta_{599}, \eta_{601}, \eta_{603}, \eta_{620}, \eta_{605}, \eta_{606}$ in the expression glossary.

$U = \eta_{495}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

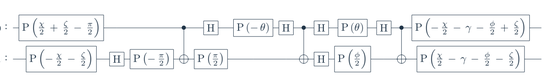
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

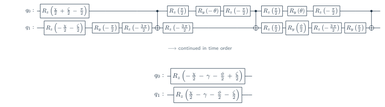
cx_h_phase

EXACT PHASE E=3 / L=15 / score ≈ 43.418 — minimum uniform E budget



cx_rz_ry

PROJECTIVE E=3 / L=17 / score ≈ 49.034 — minimum uniform E budget



$U = \eta_{594}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=3 / L=11 / score ≈ 37.466 — minimum uniform E budget

PROJECTIVE E=3 / L=11 / score ≈ 37.266 — minimum uniform E budget

$U = e^{\frac{i\phi}{2}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

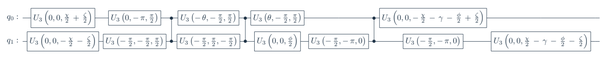
PROJECTIVE E=3 / L=13 / score ≈ 40.226 — minimum uniform E budget

Exact definitions: η_{595} in the expression glossary.

$U = \eta_{594}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

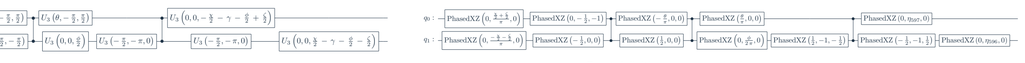
cz_u

EXACT PHASE E=3 / L=12 / score ≈ 39.672 — minimum uniform E budget



google_cz

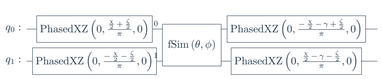
EXACT PHASE E=3 / L=12 / score ≈ 40.784 — minimum uniform E budget



Exact definitions: η_{596}, η_{597} in the expression glossary.

google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 14.149 — minimum uniform E budget



google_sqrt_iswap

EXACT PHASE E=6 / L=25 / score ≈ 84.383 — E upper bound

PROJECTIVE E=6 / L=23 / score ≈ 79.111 — E upper bound

$U = \frac{\sqrt{2}(1-i)e^{-i\theta}}{2}\, C$. Scalar metadata; no physical GP.
Published uniform cap: $E \leq 3$. Current construction exceeds it; optimization remains.

google_syc

EXACT PHASE E=18 / L=25 / score ≈ 130.583 — E upper bound

PROJECTIVE E=18 / L=23 / score ≈ 125.911 — E upper bound

$U = -ie^{-i\theta}\, C$. Scalar metadata; no physical GP.

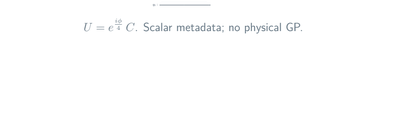
ibm_cx

PROJECTIVE E=3 / L=19 / score ≈ 52.224 — minimum uniform E budget

$U = -e^{-\frac{i(\phi_1+\phi_2-\pi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=3 / L=22 / score ≈ 58.226 — minimum uniform E budget



$U = \sqrt[4]{-1}e^{-\frac{i(\phi_1+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=3 / L=27 / score ≈ 68.224 — minimum uniform E budget

$U = -ie^{-\frac{i(\phi_1+\phi_2-\pi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=3 / L=39 / score ≈ 103.869 — minimum uniform E budget

$U = \eta_{599}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=3 / L=39 / score ≈ 102.663 — minimum uniform E budget

$U = \eta_{599}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=3 / L=42 / score ≈ 111.063 — minimum uniform E budget

$U = \eta_{599}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=3 / L=25 / score ≈ 68.700 — minimum uniform E budget

$U = \eta_{594}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=3 / L=17 / score ≈ 51.102 — minimum uniform E budget

PROJECTIVE E=3 / L=14 / score ≈ 44.284 — minimum uniform E budget

$U = \eta_{497}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=3 / L=13 / score ≈ 41.299 — minimum uniform E budget

$U = -ie^{-\frac{i(\lambda_1+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=3 / L=22 / score ≈ 62.057 — minimum uniform E budget

$U = e^{-\frac{i(\lambda_1+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=3 / L=22 / score ≈ 58.226 — minimum uniform E budget

Exact definitions: η_{598} in the expression glossary.

$U = \sqrt[4]{-1}e^{-\frac{i(\lambda_1+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=6 / L=23 / score ≈ 77.138 — E upper bound

Published uniform cap: $E \leq 3$. Current construction exceeds it; optimization remains.

sqrt_swap_u

EXACT PHASE E=6 / L=23 / score ≈ 77.938 — E upper bound

PROJECTIVE E=6 / L=23 / score ≈ 77.138 — E upper bound

$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=4 / score ≈ 14.005 — minimum uniform E budget

$U = e^{-\frac{i(\lambda_1+\phi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=3 / L=14 / score ≈ 43.684 — minimum uniform E budget

PROJECTIVE E=3 / L=12 / score ≈ 38.472 — minimum uniform E budget

$U = ie^{\frac{i(\phi+2\theta)}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=3 / L=36 / score ≈ 91.078 — minimum uniform E budget

$U = \eta_{591}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=3 / L=17 / score ≈ 52.508 — minimum uniform E budget

zz_u

EXACT PHASE E=3 / L=18 / score ≈ 52.314 — minimum uniform E budget

PROJECTIVE E=3 / L=16 / score ≈ 47.502 — minimum uniform E budget

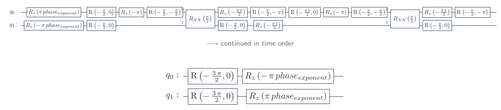
$U = e^{\frac{i\phi}{4}}\, C$. Scalar metadata; no physical GP.

PhasedISWAPPow $(phase_{exponent}, t, global_{shift})$

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=2 / L=20 / score ≈ 51.666 — minimum uniform E budget



$U = -ie^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cz_rz_rx

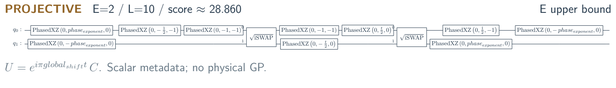
PROJECTIVE E=2 / L=15 / score ≈ 40.430 — minimum uniform E budget

$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

google_sqrt_iswap

EXACT PHASE E=2 / L=12 / score ≈ 32.872 — E upper bound

PROJECTIVE E=2 / L=10 / score ≈ 28.860 — E upper bound



$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

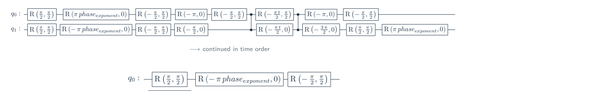
ibm_ecr

PROJECTIVE E=2 / L=16 / score ≈ 41.822 — minimum uniform E budget

$U = -\left(-1\right)^{\frac{3}{4}} e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

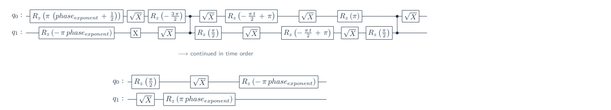
PROJECTIVE E=2 / L=20 / score ≈ 53.080 — minimum uniform E budget



$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=2 / L=21 / score ≈ 51.422 — minimum uniform E budget

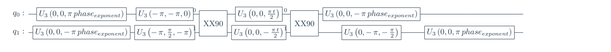


$U = \sqrt[4]{-1} e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=2 / L=11 / score ≈ 30.866 — minimum uniform E budget

PROJECTIVE E=2 / L=9 / score ≈ 26.854 — minimum uniform E budget



$U = \eta_{500}\, C$. Scalar metadata; no physical GP.

berkeley_b

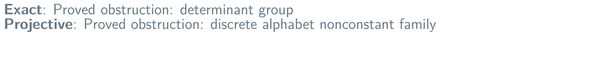
EXACT PHASE E=2 / L=16 / score ≈ 43.296 — minimum uniform E budget

PROJECTIVE E=2 / L=14 / score ≈ 39.284 — minimum uniform E budget

$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=2 / L=24 / score ≈ 58.028 — minimum uniform E budget



PROJECTIVE E=2 / L=17 / score ≈ 43.622 — minimum uniform E budget

$U = e^{\frac{i\pi\left(2 global_{shift} - 1\right)}{2}}\, C$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=2 / L=12 / score ≈ 34.472 — minimum uniform E budget

PROJECTIVE E=2 / L=10 / score ≈ 30.460 — minimum uniform E budget

$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=20 / score ≈ 91.120 — E upper bound

PROJECTIVE E=12 / L=18 / score ≈ 86.908 — E upper bound

$U = e^{\frac{i\pi\left(2 global_{shift} + 1\right)}{2}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=2 / L=33 / score ≈ 79.670 — minimum uniform E budget

$U = -ie^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_rz_ry

PROJECTIVE E=2 / L=10 / score ≈ 29.220 — minimum uniform E budget

$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

google_cz

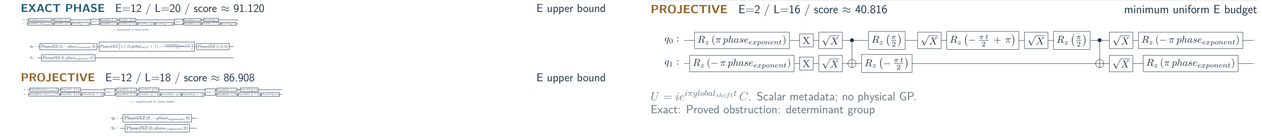
EXACT PHASE E=2 / L=12 / score ≈ 34.072 — minimum uniform E budget

PROJECTIVE E=2 / L=10 / score ≈ 29.860 — minimum uniform E budget

$U = e^{\frac{i\pi\left(2 global_{shift} + 1\right)}{2}}\, C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=2 / L=16 / score ≈ 40.816 — minimum uniform E budget



$U = ie^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=2 / L=21 / score ≈ 51.422 — minimum uniform E budget

$U = \sqrt[4]{-1} e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

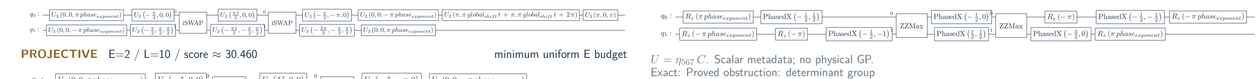
ionq_aria

PROJECTIVE E=2 / L=28 / score ≈ 70.868 — minimum uniform E budget

$U = ie^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=2 / L=12 / score ≈ 34.472 — minimum uniform E budget

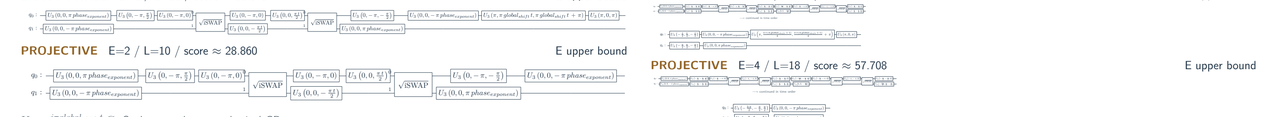


PROJECTIVE E=2 / L=10 / score ≈ 30.460 — minimum uniform E budget

$U = \eta_{498}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=2 / L=12 / score ≈ 32.872 — E upper bound

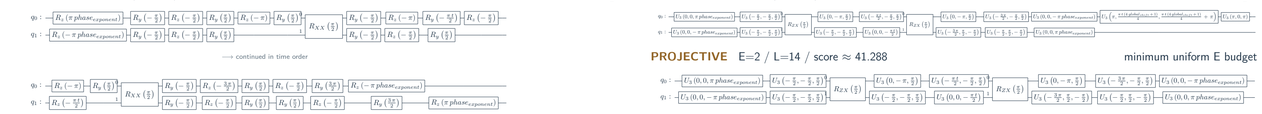


PROJECTIVE E=2 / L=10 / score ≈ 28.860 — E upper bound

$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=2 / L=33 / score ≈ 79.670 — minimum uniform E budget



$U = -ie^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=2 / L=28 / score ≈ 70.064 — minimum uniform E budget

$U = ie^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzmax

PROJECTIVE E=2 / L=12 / score ≈ 33.836 — minimum uniform E budget

$U = \eta_{567}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_swap_u

EXACT PHASE E=4 / L=20 / score ≈ 62.920 — E upper bound

PROJECTIVE E=4 / L=18 / score ≈ 57.708 — E upper bound

$U = -ie^{i\pi\left(global_{shift} t + \frac{1}{4}\right)}\, C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=2 / L=16 / score ≈ 46.100 — minimum uniform E budget

PROJECTIVE E=2 / L=14 / score ≈ 41.288 — minimum uniform E budget

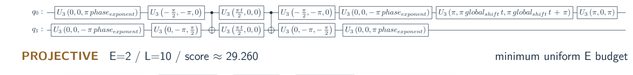
$U = e^{i\pi\left(global_{shift} t + \frac{1}{4}\right)}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

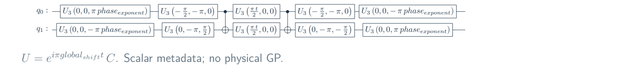
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_u

EXACT PHASE E=2 / L=12 / score ≈ 33.272 — minimum uniform E budget



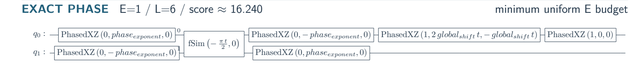
PROJECTIVE E=2 / L=10 / score ≈ 29.260 — minimum uniform E budget



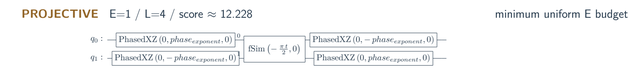
$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=6 / score ≈ 16.240 — minimum uniform E budget



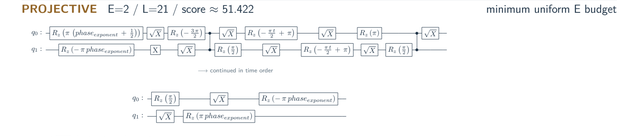
PROJECTIVE E=1 / L=4 / score ≈ 12.228 — minimum uniform E budget



$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

ibm_cz

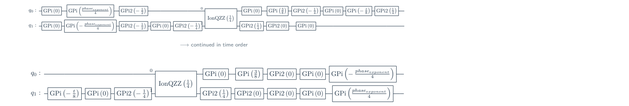
PROJECTIVE E=2 / L=21 / score ≈ 51.422 — minimum uniform E budget



$U = \sqrt[4]{-1} e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

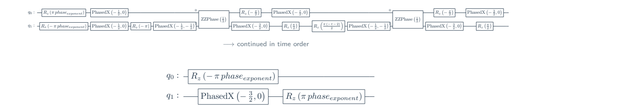
PROJECTIVE E=2 / L=30 / score ≈ 75.664 — minimum uniform E budget



$U = ie^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=2 / L=19 / score ≈ 49.660 — minimum uniform E budget



$U = -ie^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

tket_tk2

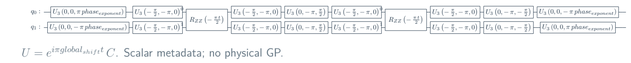
PROJECTIVE E=1 / L=4 / score ≈ 12.414 — minimum uniform E budget

$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zz_u

EXACT PHASE E=2 / L=18 / score ≈ 46.912 — minimum uniform E budget

PROJECTIVE E=2 / L=16 / score ≈ 42.900 — minimum uniform E budget



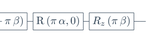
$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

PhasedX (α, β)

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=3 / score ≈ 6.008 — proved E minimum



berkeley_b

EXACT PHASE E=0 / L=3 / score ≈ 6.418 — proved E minimum

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

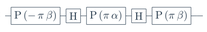
clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=12 / score ≈ 24.412 — proved E minimum

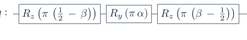
PROJECTIVE E=0 / L=5 / score ≈ 10.006 — proved E minimum



$U = e^{-i\frac{\pi\alpha}{2}}\,C$. Scalar metadata; no physical GP.

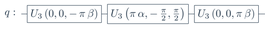
cx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.406 — proved E minimum



cx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 — proved E minimum



cz_rz_rx

EXACT PHASE E=0 / L=3 / score ≈ 6.006 — proved E minimum

cz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 — proved E minimum

google_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.230 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.018 — proved E minimum

$U = e^{-i\frac{\pi\alpha}{2}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=5 / score ≈ 10.230 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.018 — proved E minimum

$U = e^{-i\frac{\pi\alpha}{2}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=5 / score ≈ 10.230 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.018 — proved E minimum

$U = e^{-i\frac{\pi\alpha}{2}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=5 / score ≈ 10.230 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.018 — proved E minimum

$U = e^{-i\frac{\pi\alpha}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.410 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.410 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=9 / score ≈ 18.410 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=8 / score ≈ 18.016 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=8 / score ≈ 18.016 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=8 / score ≈ 18.016 — proved E minimum

iqm_cz

EXACT PHASE E=0 / L=7 / score ≈ 15.628 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 — proved E minimum

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.004 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.004 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.410 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 — proved E minimum

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.004 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 — proved E minimum

xx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.406 — proved E minimum

zx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 — proved E minimum

zz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 — proved E minimum

PhasedXPow $(t, phase_{exponent}, global_{shift})$

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=3 / score ≈ 6.008 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

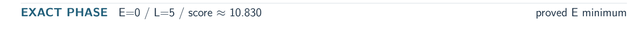
clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=12 / score ≈ 24.012 proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.006 proved E minimum



$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

cx_rz_ry

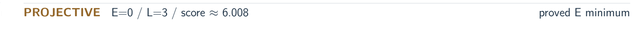
PROJECTIVE E=0 / L=3 / score ≈ 6.406 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved E minimum



$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

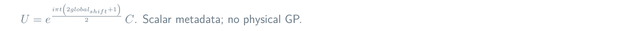
cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.006 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.030 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.018 proved E minimum

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=5 / score ≈ 10.030 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.018 proved E minimum

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=5 / score ≈ 10.030 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.018 proved E minimum

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=5 / score ≈ 10.030 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.018 proved E minimum

$U = e^{i\pi global_{shift} t} C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved E minimum

$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved E minimum

$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved E minimum

$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=0 / L=8 / score ≈ 18.016 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=8 / score ≈ 18.016 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=8 / score ≈ 18.016 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=7 / score ≈ 15.628 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=3 / score ≈ 6.008 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=3 / score ≈ 6.008 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved E minimum

$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=3 / score ≈ 6.008 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.406 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

PhasedXZ $(x_{exponent}, z_{exponent}, axis_{phaseexponent})$

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=3 / score ≈ 6.008 proved E minimum

$U = e^{\frac{i\pi\left(x_{exponent}+z_{exponent}\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved E minimum

$U = e^{\frac{i\pi x_{exponent}}{2}}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.006 proved E minimum

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.406 proved E minimum

Exact definitions: η_{601} in the expression glossary.
$U = e^{\frac{i\pi\left(x_{exponent}+z_{exponent}\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

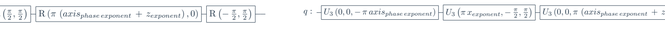
EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved E minimum

$U = e^{\frac{i\pi x_{exponent}}{2}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.006 proved E minimum



$U = e^{\frac{i\pi\left(x_{exponent}+z_{exponent}\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved E minimum

$U = e^{\frac{i\pi x_{exponent}}{2}}\, C$. Scalar metadata; no physical GP.

google_cz

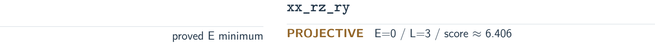
EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum



google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

ibm_cx

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved E minimum

Exact definitions: η_{602} in the expression glossary.
$U = \eta_{603}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved E minimum

Exact definitions: η_{602} in the expression glossary.
$U = \eta_{603}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved E minimum

Exact definitions: η_{602} in the expression glossary.
$U = \eta_{603}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=0 / L=8 / score ≈ 18.016 proved E minimum

$U = e^{\frac{i\pi\left(x_{exponent}+z_{exponent}\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=8 / score ≈ 18.016 proved E minimum

$U = e^{\frac{i\pi\left(x_{exponent}+z_{exponent}\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=8 / score ≈ 18.016 proved E minimum

$U = e^{\frac{i\pi\left(x_{exponent}+z_{exponent}\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=7 / score ≈ 15.628 proved E minimum

$U = e^{\frac{i\pi\left(x_{exponent}+z_{exponent}\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved E minimum

$U = e^{\frac{i\pi x_{exponent}}{2}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=3 / score ≈ 6.008 proved E minimum

$U = e^{\frac{i\pi\left(x_{exponent}+z_{exponent}\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=3 / score ≈ 6.008 proved E minimum

$U = e^{\frac{i\pi\left(x_{exponent}+z_{exponent}\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved E minimum

Exact definitions: η_{602} in the expression glossary.
$U = \eta_{603}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved E minimum

$U = e^{\frac{i\pi x_{exponent}}{2}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved E minimum

$U = e^{\frac{i\pi x_{exponent}}{2}}\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=3 / score ≈ 6.008 proved E minimum

$U = e^{\frac{i\pi\left(x_{exponent}+z_{exponent}\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved E minimum

$U = e^{\frac{i\pi x_{exponent}}{2}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.406 proved E minimum

Exact definitions: η_{601} in the expression glossary.
$U = e^{\frac{i\pi\left(x_{exponent}+z_{exponent}\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved E minimum

$U = e^{\frac{i\pi x_{exponent}}{2}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=5 / score ≈ 10.830 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.418 proved E minimum

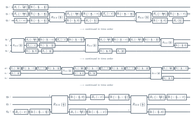
$U = e^{\frac{i\pi x_{exponent}}{2}}\, C$. Scalar metadata; no physical GP.

QFT

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

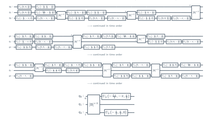
PROJECTIVE E=9 / L=57 / score ≈ 166.392 — E upper bound



$U = \eta_{604}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=18 / L=48 / score ≈ 187.488 — E upper bound



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

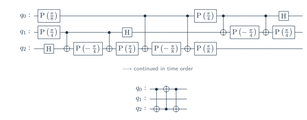
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

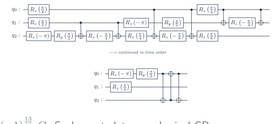
cx_h_phase

EXACT PHASE E=9 / L=12 / score ≈ 64.218 — E upper bound



cx_rz_ry

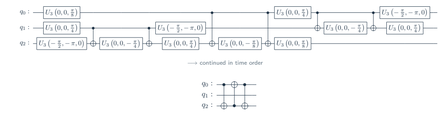
PROJECTIVE E=9 / L=15 / score ≈ 70.830 — E upper bound



$U = (-1)^{\frac{13}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

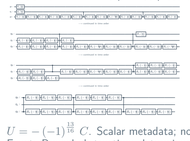
cx_u

EXACT PHASE E=9 / L=12 / score ≈ 64.872 — E upper bound



cz_rz_rx

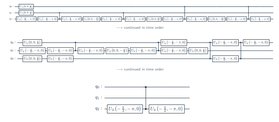
PROJECTIVE E=9 / L=63 / score ≈ 177.126 — E upper bound



$U = -(-1)^{\frac{13}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

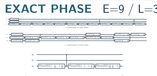
cz_u

EXACT PHASE E=9 / L=30 / score ≈ 104.580 — E upper bound

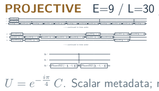


google_cz

EXACT PHASE E=9 / L=31 / score ≈ 111.186 — E upper bound



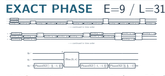
PROJECTIVE E=9 / L=30 / score ≈ 108.780 — E upper bound



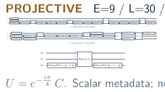
$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=9 / L=31 / score ≈ 111.222 — E upper bound



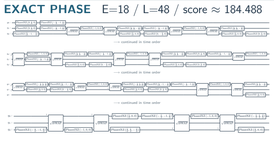
PROJECTIVE E=9 / L=30 / score ≈ 108.816 — E upper bound



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

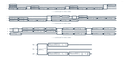
google_sqrt_iswap

EXACT PHASE E=18 / L=48 / score ≈ 184.488 — E upper bound



google_syc

EXACT PHASE E=54 / L=49 / score ≈ 327.294 — E upper bound

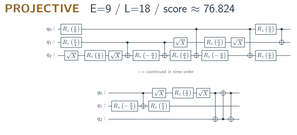


PROJECTIVE E=54 / L=48 / score ≈ 324.888 — E upper bound

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cx

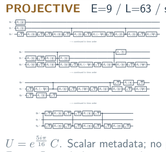
PROJECTIVE E=9 / L=18 / score ≈ 76.824 — E upper bound



$U = e^{\frac{3i\pi}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

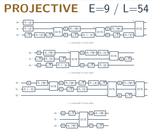
PROJECTIVE E=9 / L=63 / score ≈ 171.066 — E upper bound



$U = e^{\frac{3i\pi}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

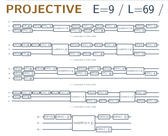
PROJECTIVE E=9 / L=54 / score ≈ 150.642 — E upper bound



$U = -(-1)^{\frac{3}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

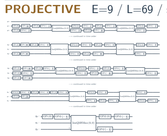
PROJECTIVE E=9 / L=69 / score ≈ 195.792 — E upper bound



$U = \eta_{605}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

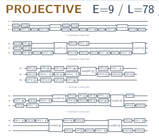
PROJECTIVE E=9 / L=69 / score ≈ 192.174 — E upper bound



$U = \eta_{605}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

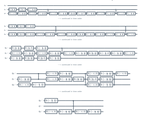
PROJECTIVE E=9 / L=78 / score ≈ 217.374 — E upper bound



$U = \eta_{604}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

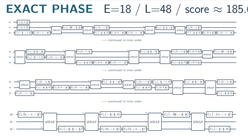
PROJECTIVE E=9 / L=69 / score ≈ 194.076 — E upper bound



$U = -(-1)^{\frac{13}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

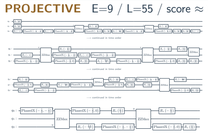
iswap_u

EXACT PHASE E=18 / L=48 / score ≈ 185.688 — E upper bound



quantinuum_zzmax

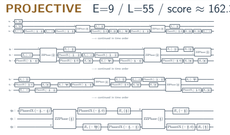
PROJECTIVE E=9 / L=55 / score ≈ 160.552 — E upper bound



$U = \eta_{604}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

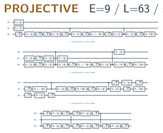
PROJECTIVE E=9 / L=55 / score ≈ 162.370 — E upper bound



$U = \eta_{604}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

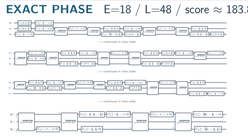
PROJECTIVE E=9 / L=63 / score ≈ 171.066 — E upper bound



$U = e^{\frac{3i\pi}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

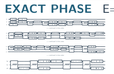
sqrt_iswap_u

EXACT PHASE E=18 / L=48 / score ≈ 183.888 — E upper bound

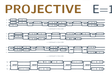


sqrt_swap_u

EXACT PHASE E=18 / L=48 / score ≈ 184.688 — E upper bound



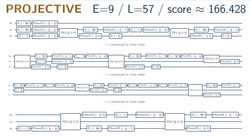
PROJECTIVE E=18 / L=48 / score ≈ 183.888 — E upper bound



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

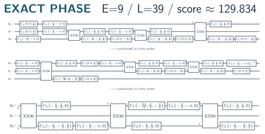
PROJECTIVE E=9 / L=57 / score ≈ 166.428 — E upper bound



$U = \eta_{604}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

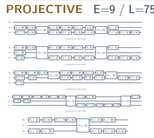
xx90_u

EXACT PHASE E=9 / L=39 / score ≈ 129.834 — E upper bound



xx_rz_ry

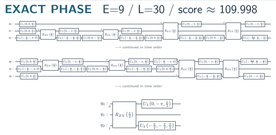
PROJECTIVE E=9 / L=75 / score ≈ 203.568 — E upper bound



$U = \eta_{604}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

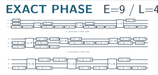
zx_u

EXACT PHASE E=9 / L=30 / score ≈ 109.998 — E upper bound

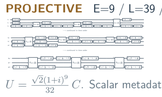


zz_u

EXACT PHASE E=9 / L=40 / score ≈ 130.458 — E upper bound



PROJECTIVE E=9 / L=39 / score ≈ 128.052 — E upper bound



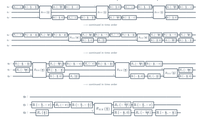
$U = \frac{\sqrt{2}(1+i)^9}{32}\, C$. Scalar metadata; no physical GP.

QFTdg

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

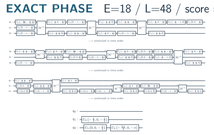
PROJECTIVE E=9 / L=55 / score ≈ 162.388 — E upper bound



$U = \eta_{6046}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=18 / L=48 / score ≈ 187.488 — E upper bound



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

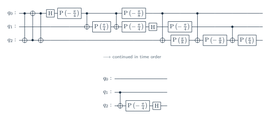
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

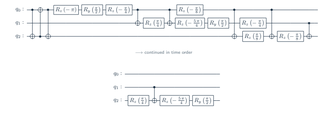
cx_h_phase

EXACT PHASE E=9 / L=12 / score ≈ 64.218 — E upper bound



cx_rz_ry

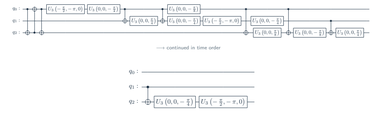
PROJECTIVE E=9 / L=13 / score ≈ 66.826 — E upper bound



$U = (-1)^{\frac{3}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

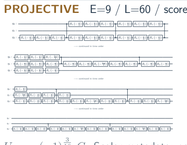
cx_u

EXACT PHASE E=9 / L=12 / score ≈ 64.872 — E upper bound



cz_rz_rx

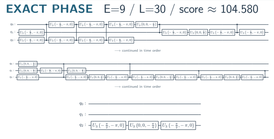
PROJECTIVE E=9 / L=60 / score ≈ 170.520 — E upper bound



$U = -(-1)^{\frac{3}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

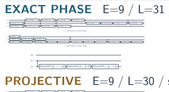
cz_u

EXACT PHASE E=9 / L=30 / score ≈ 104.580 — E upper bound



google_cz

EXACT PHASE E=9 / L=31 / score ≈ 111.186 — E upper bound

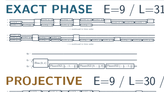


PROJECTIVE E=9 / L=30 / score ≈ 108.780 — E upper bound

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=9 / L=31 / score ≈ 111.222 — E upper bound

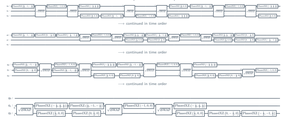


PROJECTIVE E=9 / L=30 / score ≈ 108.816 — E upper bound

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=18 / L=48 / score ≈ 184.488 — E upper bound



google_syc

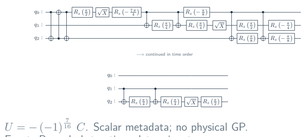
EXACT PHASE E=54 / L=49 / score ≈ 327.294 — E upper bound

PROJECTIVE E=54 / L=48 / score ≈ 324.888 — E upper bound

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cx

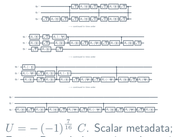
PROJECTIVE E=9 / L=15 / score ≈ 70.824 — E upper bound



$U = -(-1)^{\frac{7}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

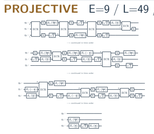
PROJECTIVE E=9 / L=60 / score ≈ 165.066 — E upper bound



$U = -(-1)^{\frac{7}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

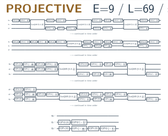
PROJECTIVE E=9 / L=49 / score ≈ 140.238 — E upper bound



$U = (-1)^{\frac{7}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

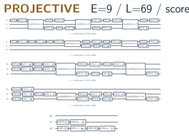
PROJECTIVE E=9 / L=69 / score ≈ 195.792 — E upper bound



$U = \eta_{6017}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

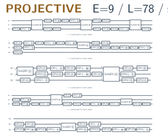
PROJECTIVE E=9 / L=69 / score ≈ 192.174 — E upper bound



$U = \eta_{6017}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

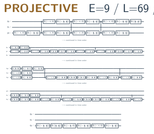
PROJECTIVE E=9 / L=78 / score ≈ 217.374 — E upper bound



$U = \eta_{6046}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

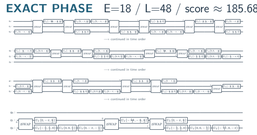
PROJECTIVE E=9 / L=69 / score ≈ 194.076 — E upper bound



$U = -(-1)^{\frac{3}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

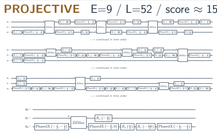
iswap_u

EXACT PHASE E=18 / L=48 / score ≈ 185.688 — E upper bound



quantinuum_zzmax

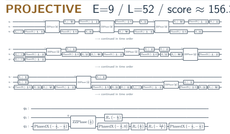
PROJECTIVE E=9 / L=52 / score ≈ 154.546 — E upper bound



$U = \eta_{6046}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

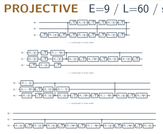
PROJECTIVE E=9 / L=52 / score ≈ 156.364 — E upper bound



$U = \eta_{6046}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

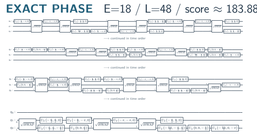
PROJECTIVE E=9 / L=60 / score ≈ 165.066 — E upper bound



$U = -(-1)^{\frac{7}{16}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

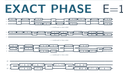
sqrt_iswap_u

EXACT PHASE E=18 / L=48 / score ≈ 183.888 — E upper bound



sqrt_swap_u

EXACT PHASE E=18 / L=48 / score ≈ 184.688 — E upper bound

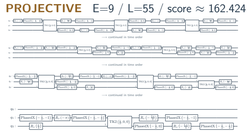


PROJECTIVE E=18 / L=48 / score ≈ 183.888 — E upper bound

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

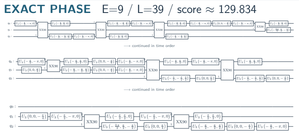
PROJECTIVE E=9 / L=55 / score ≈ 162.424 — E upper bound



$U = \eta_{6046}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

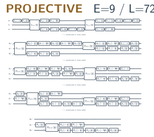
xx90_u

EXACT PHASE E=9 / L=39 / score ≈ 129.834 — E upper bound



xx_rz_ry

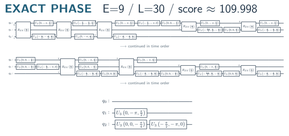
PROJECTIVE E=9 / L=72 / score ≈ 197.562 — E upper bound



$U = \eta_{6046}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

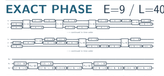
zx_u

EXACT PHASE E=9 / L=30 / score ≈ 109.998 — E upper bound



zz_u

EXACT PHASE E=9 / L=40 / score ≈ 130.458 — E upper bound



PROJECTIVE E=9 / L=39 / score ≈ 128.052 — E upper bound

$U = \frac{\sqrt{2}(1+i)^9}{32}\, C$. Scalar metadata; no physical GP.

R (θ, ϕ)

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.004 proved E minimum

berkeley_b

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved E minimum

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

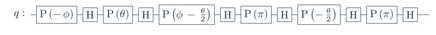
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=12 / score ≈ 24.412 proved E minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.006 proved E minimum

$U = e^{-i\frac{\theta}{2}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved E minimum

cx_u

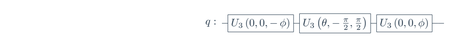
EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved E minimum

cz_rz_rx

EXACT PHASE E=0 / L=3 / score ≈ 6.006 proved E minimum

cz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved E minimum



google_cz

EXACT PHASE E=0 / L=5 / score ≈ 11.881 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 7.009 proved E minimum

$U = e^{-i\frac{\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=5 / score ≈ 11.881 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 7.009 proved E minimum

$U = e^{-i\frac{\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=5 / score ≈ 11.881 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 7.009 proved E minimum

$U = e^{-i\frac{\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=5 / score ≈ 11.881 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 7.009 proved E minimum

$U = e^{-i\frac{\theta}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.410 proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.410 proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=9 / score ≈ 18.410 proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=8 / score ≈ 19.007 proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=8 / score ≈ 19.007 proved E minimum

ionq_forte

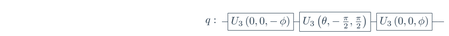
EXACT PHASE E=0 / L=8 / score ≈ 19.007 proved E minimum

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.004 proved E minimum

iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved E minimum



quantinuum_zzmax

EXACT PHASE E=0 / L=3 / score ≈ 6.338 proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=3 / score ≈ 6.338 proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.410 proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved E minimum

tket_tk2

EXACT PHASE E=0 / L=3 / score ≈ 6.338 proved E minimum

xx90_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved E minimum

xx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved E minimum

zx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved E minimum

zz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved E minimum

RC3X

4 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

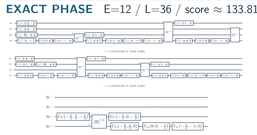
EXACT PHASE E=7 / L=45 / score ≈ 130.152 — E upper bound

PROJECTIVE E=6 / L=43 / score ≈ 122.142 — E upper bound

$U = i\,C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=12 / L=36 / score ≈ 133.816 — E upper bound

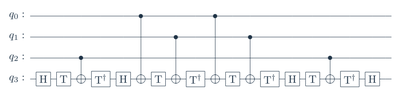


clifford_only

Exact: Proved obstruction: not a clifford
Projective: Proved obstruction: not a clifford

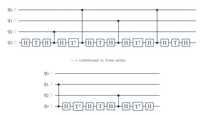
clifford_t_cx

EXACT PHASE E=6 / L=12 / score ≈ 48.000 — E upper bound



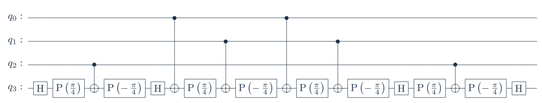
clifford_t_cz

EXACT PHASE E=6 / L=22 / score ≈ 68.000 — E upper bound



cx_h_phase

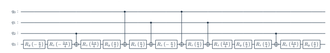
EXACT PHASE E=6 / L=12 / score ≈ 51.216 — E upper bound



cx_rz_ry

EXACT PHASE E=6 / L=13 / score ≈ 54.026 — E upper bound

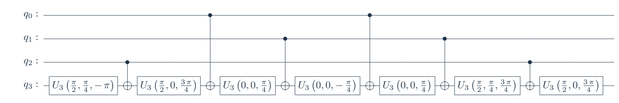
PROJECTIVE E=6 / L=12 / score ≈ 52.024 — E upper bound



$U = -1\,C$. Scalar metadata; no physical GP.

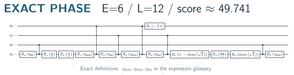
cx_u

EXACT PHASE E=6 / L=7 / score ≈ 42.042 — E upper bound

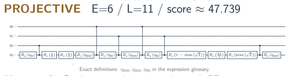


cz_rz_rx

EXACT PHASE E=6 / L=12 / score ≈ 49.741 — E upper bound



PROJECTIVE E=6 / L=11 / score ≈ 47.739 — E upper bound



$U = -1\,C$. Scalar metadata; no physical GP.

cz_u

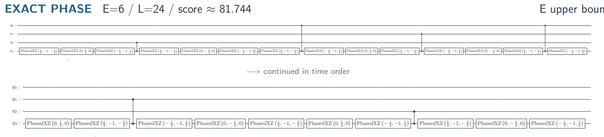
EXACT PHASE E=6 / L=9 / score ≈ 46.171 — E upper bound

PROJECTIVE E=6 / L=7 / score ≈ 40.959 — E upper bound

$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

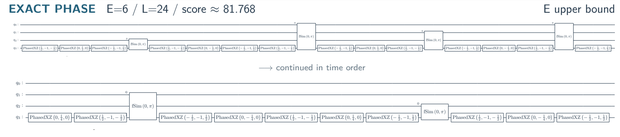
google_cz

EXACT PHASE E=6 / L=24 / score ≈ 81.744 — E upper bound



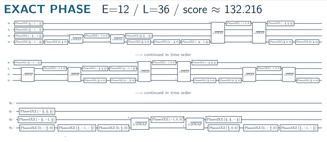
google_fsim

EXACT PHASE E=6 / L=24 / score ≈ 81.768 — E upper bound



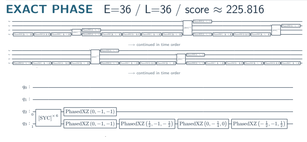
google_sqrt_iswap

EXACT PHASE E=12 / L=36 / score ≈ 132.216 — E upper bound

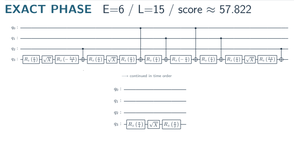


google_syc

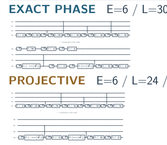
EXACT PHASE E=36 / L=36 / score ≈ 225.816 — E upper bound



ibm_cx

EXACT PHASE E=6 / L=15 / score ≈ 57.822 — E upper bound

ibm_cz

EXACT PHASE E=6 / L=30 / score ≈ 87.745 — E upper bound



PROJECTIVE E=6 / L=24 / score ≈ 75.139 — E upper bound

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=6 / L=39 / score ≈ 107.034 — E upper bound

PROJECTIVE E=6 / L=33 / score ≈ 94.636 — E upper bound

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=6 / L=54 / score ≈ 148.944 — E upper bound

PROJECTIVE E=6 / L=54 / score ≈ 148.544 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=6 / L=54 / score ≈ 146.532 — E upper bound

PROJECTIVE E=6 / L=54 / score ≈ 146.132 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

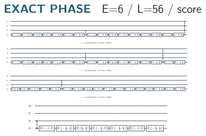
EXACT PHASE E=6 / L=60 / score ≈ 163.332 — E upper bound

PROJECTIVE E=6 / L=60 / score ≈ 162.932 — E upper bound

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=6 / L=56 / score ≈ 152.224 — E upper bound



iswap_u

EXACT PHASE E=12 / L=21 / score ≈ 99.726 — E upper bound

PROJECTIVE E=12 / L=19 / score ≈ 94.514 — E upper bound

$U = e^{\frac{7i\pi}{8}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

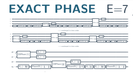
EXACT PHASE E=9 / L=57 / score ≈ 165.166 — E upper bound

PROJECTIVE E=6 / L=38 / score ≈ 110.908 — E upper bound

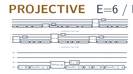
$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=7 / L=40 / score ≈ 120.126 — E upper bound



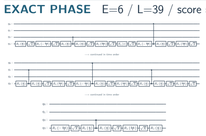
PROJECTIVE E=6 / L=38 / score ≈ 112.120 — E upper bound



$U = i\,C$. Scalar metadata; no physical GP.

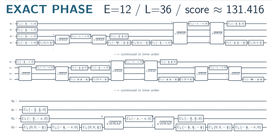
rigetti_cz

EXACT PHASE E=6 / L=39 / score ≈ 108.046 — E upper bound



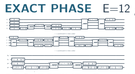
sqrt_iswap_u

EXACT PHASE E=12 / L=36 / score ≈ 131.416 — E upper bound



sqrt_swap_u

EXACT PHASE E=12 / L=36 / score ≈ 131.816 — E upper bound

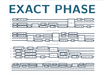


PROJECTIVE E=12 / L=36 / score ≈ 131.416 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=9 / L=63 / score ≈ 179.050 — E upper bound

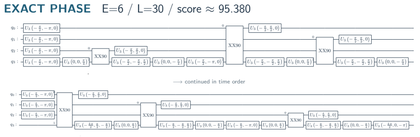


PROJECTIVE E=6 / L=43 / score ≈ 122.166 — E upper bound

$U = i\,C$. Scalar metadata; no physical GP.

xx90_u

EXACT PHASE E=6 / L=30 / score ≈ 95.380 — E upper bound



xx_rz_ry

EXACT PHASE E=7 / L=39 / score ≈ 115.892 — E upper bound

PROJECTIVE E=6 / L=35 / score ≈ 103.882 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

zx_u

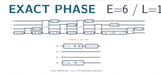
EXACT PHASE E=6 / L=15 / score ≈ 62.702 — E upper bound

PROJECTIVE E=6 / L=13 / score ≈ 57.490 — E upper bound

$U = e^{\frac{3i\pi}{8}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=6 / L=15 / score ≈ 60.019 — E upper bound



PROJECTIVE E=6 / L=13 / score ≈ 54.807 — E upper bound

$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.

RCCX

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

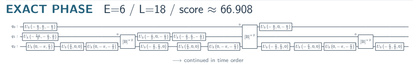
EXACT PHASE E=15 / L=101 / score ≈ 290.338 — E upper bound

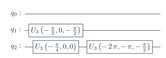
PROJECTIVE E=3 / L=21 / score ≈ 60.474 — E upper bound

$U = \frac{\sqrt{2}i(1+i)^3}{4} C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=6 / L=18 / score ≈ 66.908 — E upper bound





clifford_only

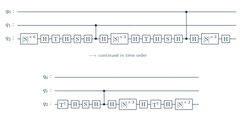
Exact: Proved obstruction: not a clifford
Projective: Proved obstruction: not a clifford

clifford_t_cx

EXACT PHASE E=3 / L=32 / score ≈ 76.000 — proved E minimum

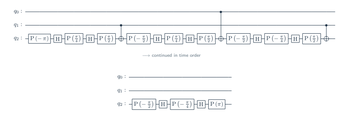
clifford_t_cz

EXACT PHASE E=3 / L=38 / score ≈ 88.000 — proved E minimum



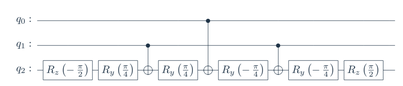
cx_h_phase

EXACT PHASE E=3 / L=20 / score ≈ 54.824 — proved E minimum



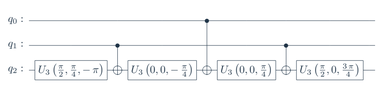
cx_rz_ry

EXACT PHASE E=3 / L=6 / score ≈ 26.012 — proved E minimum



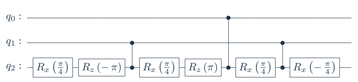
cx_u

EXACT PHASE E=3 / L=4 / score ≈ 22.024 — proved E minimum



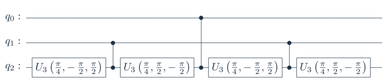
cz_rz_rx

EXACT PHASE E=3 / L=6 / score ≈ 25.612 — proved E minimum



cz_u

EXACT PHASE E=3 / L=4 / score ≈ 23.224 — proved E minimum



google_cz

EXACT PHASE E=3 / L=12 / score ≈ 41.272 — proved E minimum

google_fsim

EXACT PHASE E=3 / L=12 / score ≈ 41.284 — E upper bound

google_sqrt_iswap

EXACT PHASE E=6 / L=18 / score ≈ 67.108 — E upper bound

PROJECTIVE E=6 / L=18 / score ≈ 66.508 — E upper bound

$U = \frac{\sqrt{2}(1-i)^3}{4} C$. Scalar metadata; no physical GP.

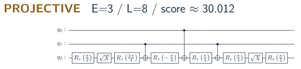
google_syc

EXACT PHASE E=18 / L=18 / score ≈ 113.308 — E upper bound

ibm_cx

EXACT PHASE E=3 / L=12 / score ≈ 38.016 — proved E minimum

PROJECTIVE E=3 / L=8 / score ≈ 30.012 — proved E minimum



$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

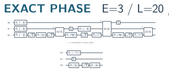
EXACT PHASE E=3 / L=17 / score ≈ 48.018 — proved E minimum

PROJECTIVE E=3 / L=16 / score ≈ 46.016 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=3 / L=20 / score ≈ 54.624 — proved E minimum



PROJECTIVE E=3 / L=19 / score ≈ 52.622 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=15 / L=131 / score ≈ 361.352 — E upper bound

PROJECTIVE E=3 / L=35 / score ≈ 91.488 — E upper bound

$U = \frac{\sqrt{2}(1+i)^3}{4} C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=15 / L=131 / score ≈ 355.322 — E upper bound

PROJECTIVE E=3 / L=35 / score ≈ 90.282 — E upper bound

$U = \frac{\sqrt{2}(1+i)^3}{4} C$. Scalar metadata; no physical GP.

ionq_forte

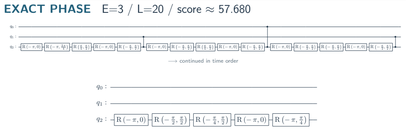
EXACT PHASE E=15 / L=146 / score ≈ 397.322 — E upper bound

PROJECTIVE E=3 / L=38 / score ≈ 98.682 — E upper bound

$U = \frac{\sqrt{2}(1+i)^3}{4} C$. Scalar metadata; no physical GP.

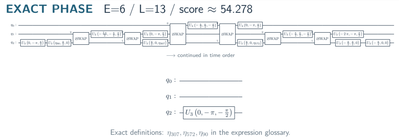
iqm_cz

EXACT PHASE E=3 / L=20 / score ≈ 57.680 — proved E minimum



iswap_u

EXACT PHASE E=6 / L=13 / score ≈ 54.278 — E upper bound



Exact definitions: $\eta_{317}, \eta_{272}, \eta_{90}$ in the expression glossary.

quantinuum_zzmax

EXACT PHASE E=15 / L=115 / score ≈ 314.954 — E upper bound

PROJECTIVE E=3 / L=21 / score ≈ 59.862 — proved E minimum

$U = \frac{\sqrt{2}i(1+i)^3}{4} C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=15 / L=91 / score ≈ 269.888 — E upper bound

PROJECTIVE E=3 / L=21 / score ≈ 60.468 — E upper bound

$U = \frac{\sqrt{2}i(1+i)^3}{4} C$. Scalar metadata; no physical GP.

rigetti_cz

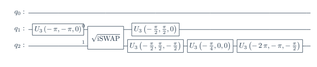
EXACT PHASE E=3 / L=32 / score ≈ 80.032 — proved E minimum

PROJECTIVE E=3 / L=32 / score ≈ 79.830 — proved E minimum

$U = -(-1)^{\frac{3}{4}} C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=6 / L=18 / score ≈ 65.708 — E upper bound



sqrt_swap_u

EXACT PHASE E=6 / L=18 / score ≈ 66.508 — E upper bound

PROJECTIVE E=6 / L=18 / score ≈ 65.708 — E upper bound

$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

tket_tk2

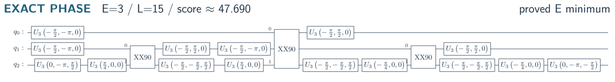
EXACT PHASE E=15 / L=101 / score ≈ 290.398 — E upper bound

PROJECTIVE E=3 / L=21 / score ≈ 60.486 — E upper bound

$U = \frac{\sqrt{2}i(1+i)^3}{4} C$. Scalar metadata; no physical GP.

xx90_u

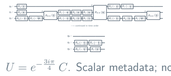
EXACT PHASE E=3 / L=15 / score ≈ 47.690 — proved E minimum



xx_rz_ry

EXACT PHASE E=15 / L=121 / score ≈ 330.272 — E upper bound

PROJECTIVE E=3 / L=19 / score ≈ 55.244 — E upper bound



$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=3 / L=9 / score ≈ 35.260 — E upper bound

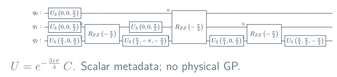
PROJECTIVE E=3 / L=7 / score ≈ 30.048 — E upper bound

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=3 / L=9 / score ≈ 34.660 — E upper bound

PROJECTIVE E=3 / L=7 / score ≈ 29.848 — E upper bound



$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

RV (vx, vy, vz)

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=3 / score ≈ 6.208 proved E minimum

Exact definitions: $\eta_{611}, \eta_{612}, \eta_{613}$ in the expression glossary.

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.012 proved E minimum

Exact definitions: $\eta_{614}, \eta_{615}, \eta_{616}, \eta_{617}$ in the expression glossary.

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

Exact definitions: $\eta_{611}, \eta_{612}, \eta_{613}$ in the expression glossary.

$U = \eta_{618}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=13 / score ≈ 26.014 proved E minimum

Exact definitions: $\eta_{612}, \eta_{619}, \eta_{620}, \eta_{621}$ in the expression glossary.

PROJECTIVE E=0 / L=5 / score ≈ 10.006 proved E minimum

Exact definitions: $\eta_{612}, \eta_{619}, \eta_{620}$ in the expression glossary.

$U = \eta_{622}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.006 proved E minimum

Exact definitions: $\eta_{611}, \eta_{612}, \eta_{613}$ in the expression glossary.

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 proved E minimum

Exact definitions: $\eta_{614}, \eta_{615}, \eta_{616}, \eta_{617}$ in the expression glossary.

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

Exact definitions: $\eta_{611}, \eta_{612}, \eta_{613}$ in the expression glossary.

$U = \eta_{618}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=3 / score ≈ 6.006 proved E minimum

Exact definitions: $\eta_{612}, \eta_{619}, \eta_{620}$ in the expression glossary.

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 proved E minimum

Exact definitions: $\eta_{614}, \eta_{615}, \eta_{616}, \eta_{617}$ in the expression glossary.

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

Exact definitions: $\eta_{611}, \eta_{612}, \eta_{613}$ in the expression glossary.

$U = \eta_{618}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.560 proved E minimum

Exact definitions: $\eta_{623}, \eta_{624}, \eta_{625}, \eta_{626}, \eta_{627}$ in the expression glossary.

PROJECTIVE E=0 / L=3 / score ≈ 6.548 proved E minimum

Exact definitions: $\eta_{623}, \eta_{624}, \eta_{625}$ in the expression glossary.

$U = \eta_{622}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=5 / score ≈ 10.560 proved E minimum

Exact definitions: $\eta_{623}, \eta_{624}, \eta_{625}, \eta_{626}, \eta_{627}$ in the expression glossary.

PROJECTIVE E=0 / L=3 / score ≈ 6.548 proved E minimum

Exact definitions: $\eta_{623}, \eta_{624}, \eta_{625}$ in the expression glossary.

$U = \eta_{622}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=5 / score ≈ 10.560 proved E minimum

Exact definitions: $\eta_{623}, \eta_{624}, \eta_{625}, \eta_{626}, \eta_{627}$ in the expression glossary.

PROJECTIVE E=0 / L=3 / score ≈ 6.548 proved E minimum

Exact definitions: $\eta_{623}, \eta_{624}, \eta_{625}$ in the expression glossary.

$U = \eta_{622}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=5 / score ≈ 10.560 proved E minimum

Exact definitions: $\eta_{623}, \eta_{624}, \eta_{625}, \eta_{626}, \eta_{627}$ in the expression glossary.

PROJECTIVE E=0 / L=3 / score ≈ 6.548 proved E minimum

Exact definitions: $\eta_{623}, \eta_{624}, \eta_{625}$ in the expression glossary.

$U = \eta_{622}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.010 proved E minimum

Exact definitions: $\eta_{611}, \eta_{628}, \eta_{629}$ in the expression glossary.

PROJECTIVE E=0 / L=5 / score ≈ 10.006 proved E minimum

Exact definitions: $\eta_{611}, \eta_{628}, \eta_{629}$ in the expression glossary.

$U = i\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.010 proved E minimum

Exact definitions: $\eta_{611}, \eta_{628}, \eta_{629}$ in the expression glossary.

PROJECTIVE E=0 / L=5 / score ≈ 10.006 proved E minimum

Exact definitions: $\eta_{611}, \eta_{628}, \eta_{629}$ in the expression glossary.

$U = i\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=9 / score ≈ 18.010 proved E minimum

Exact definitions: $\eta_{611}, \eta_{628}, \eta_{629}$ in the expression glossary.

PROJECTIVE E=0 / L=5 / score ≈ 10.006 proved E minimum

Exact definitions: $\eta_{611}, \eta_{628}, \eta_{629}$ in the expression glossary.

$U = i\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=8 / score ≈ 16.746 proved E minimum

Exact definitions: $\eta_{630}, \eta_{631}, \eta_{632}$ in the expression glossary.

ionq_aria_full

EXACT PHASE E=0 / L=8 / score ≈ 16.746 proved E minimum

Exact definitions: $\eta_{630}, \eta_{631}, \eta_{632}$ in the expression glossary.

ionq_forte

EXACT PHASE E=0 / L=8 / score ≈ 16.746 proved E minimum

Exact definitions: $\eta_{630}, \eta_{631}, \eta_{632}$ in the expression glossary.

iqm_cz

EXACT PHASE E=0 / L=7 / score ≈ 15.828 proved E minimum

Exact definitions: $\eta_{611}, \eta_{612}, \eta_{613}$ in the expression glossary.

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 proved E minimum

Exact definitions: $\eta_{614}, \eta_{615}, \eta_{616}, \eta_{617}$ in the expression glossary.

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

Exact definitions: $\eta_{611}, \eta_{612}, \eta_{613}$ in the expression glossary.

$U = \eta_{618}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=3 / score ≈ 6.538 proved E minimum

Exact definitions: $\eta_{611}, \eta_{613}, \eta_{624}$ in the expression glossary.

quantinuum_zzphase

EXACT PHASE E=0 / L=3 / score ≈ 6.538 proved E minimum

Exact definitions: $\eta_{611}, \eta_{613}, \eta_{624}$ in the expression glossary.

rigetti_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.010 proved E minimum

Exact definitions: $\eta_{611}, \eta_{628}, \eta_{629}$ in the expression glossary.

PROJECTIVE E=0 / L=5 / score ≈ 10.006 proved E minimum

Exact definitions: $\eta_{611}, \eta_{628}, \eta_{629}$ in the expression glossary.

$U = i\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 proved E minimum

Exact definitions: $\eta_{614}, \eta_{615}, \eta_{616}, \eta_{617}$ in the expression glossary.

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

Exact definitions: $\eta_{611}, \eta_{612}, \eta_{613}$ in the expression glossary.

$U = \eta_{618}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 proved E minimum

Exact definitions: $\eta_{614}, \eta_{615}, \eta_{616}, \eta_{617}$ in the expression glossary.

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

Exact definitions: $\eta_{611}, \eta_{612}, \eta_{613}$ in the expression glossary.

$U = \eta_{618}\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=3 / score ≈ 6.538 proved E minimum

Exact definitions: $\eta_{611}, \eta_{613}, \eta_{624}$ in the expression glossary.

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 proved E minimum

Exact definitions: $\eta_{614}, \eta_{615}, \eta_{616}, \eta_{617}$ in the expression glossary.

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

Exact definitions: $\eta_{611}, \eta_{612}, \eta_{613}$ in the expression glossary.

$U = \eta_{618}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.006 proved E minimum

Exact definitions: $\eta_{611}, \eta_{612}, \eta_{613}$ in the expression glossary.

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 proved E minimum

Exact definitions: $\eta_{614}, \eta_{615}, \eta_{616}, \eta_{617}$ in the expression glossary.

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

Exact definitions: $\eta_{611}, \eta_{612}, \eta_{613}$ in the expression glossary.

$U = \eta_{618}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 proved E minimum

Exact definitions: $\eta_{614}, \eta_{615}, \eta_{616}, \eta_{617}$ in the expression glossary.

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

Exact definitions: $\eta_{611}, \eta_{612}, \eta_{613}$ in the expression glossary.

$U = \eta_{618}\, C$. Scalar metadata; no physical GP.

Rx (θ)

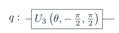
1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.004 proved E minimum

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum



clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

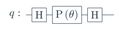
clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=11 / score ≈ 22.410 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.002 proved E minimum



$U = e^{-i\theta} C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved E minimum

cx_u

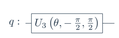
EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

cz_rz_rx

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved E minimum

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum



google_cz

EXACT PHASE E=0 / L=3 / score ≈ 7.209 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.336 proved E minimum

$U = e^{-i\theta} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=3 / score ≈ 7.209 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.336 proved E minimum

$U = e^{-i\theta} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=3 / score ≈ 7.209 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.336 proved E minimum

$U = e^{-i\theta} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=3 / score ≈ 7.209 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.336 proved E minimum

$U = e^{-i\theta} C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.410 proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.410 proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=9 / score ≈ 18.410 proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=4 / score ≈ 9.538 proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=4 / score ≈ 9.538 proved E minimum

ionq_forte

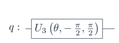
EXACT PHASE E=0 / L=4 / score ≈ 9.538 proved E minimum

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.004 proved E minimum

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum



quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.334 proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.334 proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.410 proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.334 proved E minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

xx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved E minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

zz_u

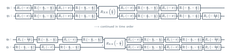
EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

Rxmy (ϕ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=2 / L=34 / score ≈ 83.704 — minimum uniform E budget

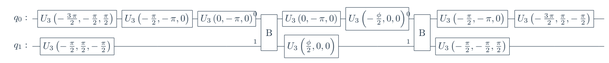


PROJECTIVE E=2 / L=19 / score ≈ 50.862 — minimum uniform E budget

$U = -iC$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=2 / L=10 / score ≈ 31.260 — minimum uniform E budget



clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

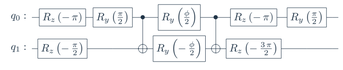
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=14 / score ≈ 37.616 — minimum uniform E budget

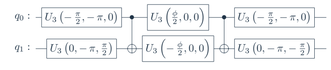
cx_rz_ry

EXACT PHASE E=2 / L=8 / score ≈ 25.216 — minimum uniform E budget



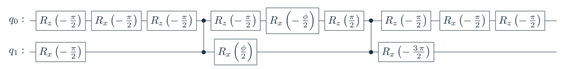
cx_u

EXACT PHASE E=2 / L=6 / score ≈ 21.236 — minimum uniform E budget



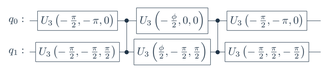
cz_rz_rx

EXACT PHASE E=2 / L=12 / score ≈ 34.424 — minimum uniform E budget



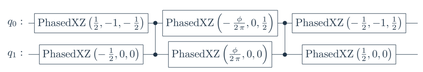
cz_u

EXACT PHASE E=2 / L=6 / score ≈ 22.436 — minimum uniform E budget



google_cz

EXACT PHASE E=2 / L=6 / score ≈ 22.497 — minimum uniform E budget



google_fsim

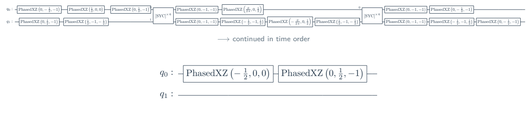
EXACT PHASE E=2 / L=2 / score ≈ 12.220 — E upper bound

google_sqrt_iswap

EXACT PHASE E=2 / L=8 / score ≈ 25.509 — minimum uniform E budget

google_syc

EXACT PHASE E=12 / L=18 / score ≈ 88.769 — E upper bound

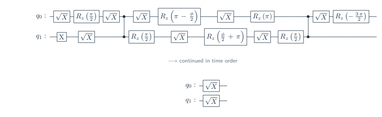


ibm_cx

EXACT PHASE E=2 / L=16 / score ≈ 41.216 — minimum uniform E budget

ibm_cz

EXACT PHASE E=2 / L=18 / score ≈ 45.214 — minimum uniform E budget



ibm_ecr

EXACT PHASE E=2 / L=14 / score ≈ 38.020 — minimum uniform E budget

PROJECTIVE E=2 / L=14 / score ≈ 37.616 — minimum uniform E budget

$U = iC$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=32 / score ≈ 81.337 — minimum uniform E budget

PROJECTIVE E=2 / L=32 / score ≈ 80.937 — minimum uniform E budget

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=32 / score ≈ 80.533 — minimum uniform E budget

PROJECTIVE E=2 / L=32 / score ≈ 80.133 — minimum uniform E budget

$U = -iC$. Scalar metadata; no physical GP.

ionq_forte

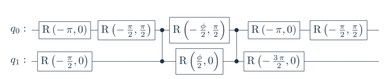
EXACT PHASE E=2 / L=34 / score ≈ 86.133 — minimum uniform E budget

PROJECTIVE E=2 / L=34 / score ≈ 85.733 — minimum uniform E budget

$U = -iC$. Scalar metadata; no physical GP.

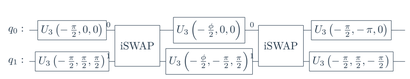
iqm_cz

EXACT PHASE E=2 / L=8 / score ≈ 25.832 — minimum uniform E budget



iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 22.436 — minimum uniform E budget



quantinuum_zzmax

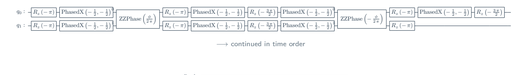
EXACT PHASE E=5 / L=27 / score ≈ 80.747 — E upper bound

PROJECTIVE E=2 / L=8 / score ≈ 26.489 — minimum uniform E budget

$U = \frac{\sqrt[3]{-1}\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

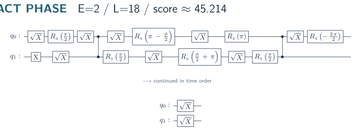
quantinuum_zzphase

EXACT PHASE E=2 / L=18 / score ≈ 49.117 — minimum uniform E budget



rigetti_cz

EXACT PHASE E=2 / L=18 / score ≈ 45.214 — minimum uniform E budget



sqrt_iswap_u

EXACT PHASE E=2 / L=8 / score ≈ 24.848 — minimum uniform E budget

sqrt_swap_u

EXACT PHASE E=4 / L=18 / score ≈ 57.708 — E upper bound

PROJECTIVE E=4 / L=18 / score ≈ 57.308 — E upper bound

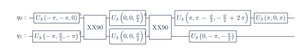
$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 5.067 — minimum uniform E budget

xx90_u

EXACT PHASE E=2 / L=7 / score ≈ 23.242 — minimum uniform E budget

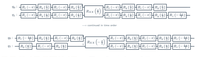


PROJECTIVE E=2 / L=5 / score ≈ 18.830 — minimum uniform E budget

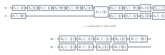
$U = \eta_{633}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=2 / L=34 / score ≈ 80.472 — minimum uniform E budget



PROJECTIVE E=2 / L=25 / score ≈ 63.254 — minimum uniform E budget



$U = -iC$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=2 / L=14 / score ≈ 40.888 — minimum uniform E budget

zz_u

EXACT PHASE E=2 / L=12 / score ≈ 34.876 — minimum uniform E budget

Rxx (θ)

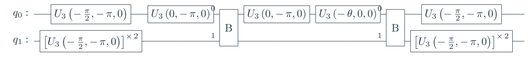
2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=1 / L=0 / score ≈ 4.002 — minimum uniform E budget

berkeley_b

EXACT PHASE E=2 / L=9 / score ≈ 27.254 — minimum uniform E budget



clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

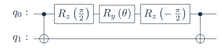
EXACT PHASE E=2 / L=11 / score ≈ 30.410 — minimum uniform E budget

PROJECTIVE E=2 / L=3 / score ≈ 14.002 — minimum uniform E budget

$U = e^{-\frac{i\theta}{2}}\, C$. Scalar metadata; no physical GP.

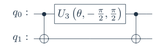
cx_rz_ry

EXACT PHASE E=2 / L=3 / score ≈ 14.406 — minimum uniform E budget



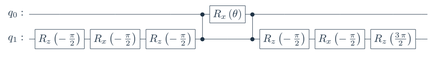
cx_u

EXACT PHASE E=2 / L=1 / score ≈ 10.406 — minimum uniform E budget



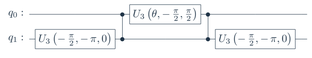
cz_rz_rx

EXACT PHASE E=2 / L=7 / score ≈ 23.214 — minimum uniform E budget



cz_u

EXACT PHASE E=2 / L=3 / score ≈ 14.818 — minimum uniform E budget



google_cz

EXACT PHASE E=2 / L=5 / score ≈ 20.021 — minimum uniform E budget

PROJECTIVE E=2 / L=3 / score ≈ 15.148 — minimum uniform E budget

$U = e^{-\frac{i\theta}{2}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=2 / score ≈ 12.420 — E upper bound

google_sqrt_iswap

EXACT PHASE E=2 / L=14 / score ≈ 38.875 — minimum uniform E budget

Exact definitions: $\eta_{214}, \eta_{634}, \eta_{635}, \eta_{636}, \eta_{637}, \eta_{638}, \eta_{639}, \eta_{640}, \eta_{641}, \eta_{642}, \eta_{643}, \eta_{644}, \eta_{645}, \eta_{646}, \eta_{647}, \eta_{648}, \eta_{649}, \eta_{650}$ in the expression glossary.

PROJECTIVE E=2 / L=13 / score ≈ 36.869 — minimum uniform E budget

Exact definitions: $\eta_{214}, \eta_{636}, \eta_{637}, \eta_{638}, \eta_{639}, \eta_{640}, \eta_{641}, \eta_{642}, \eta_{643}, \eta_{644}, \eta_{645}, \eta_{646}, \eta_{647}, \eta_{648}, \eta_{649}, \eta_{650}$ in the expression glossary.
$U = \eta_{651}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=15 / score ≈ 82.481 — E upper bound

PROJECTIVE E=12 / L=13 / score ≈ 77.608 — E upper bound

$U = e^{-\frac{i\theta}{2}}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=9 / score ≈ 26.410 — minimum uniform E budget

PROJECTIVE E=2 / L=5 / score ≈ 18.406 — minimum uniform E budget

$U = i\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=11 / score ≈ 31.214 — minimum uniform E budget

PROJECTIVE E=2 / L=11 / score ≈ 30.810 — minimum uniform E budget

$U = i\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=8 / score ≈ 24.008 — minimum uniform E budget

PROJECTIVE E=2 / L=4 / score ≈ 16.004 — minimum uniform E budget

$U = i\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=20 / score ≈ 53.582 — E upper bound

PROJECTIVE E=2 / L=20 / score ≈ 53.182 — E upper bound

$U = i\, C$. Scalar metadata; no physical GP.

ionq_aria_full

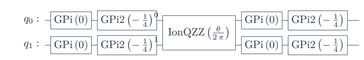
EXACT PHASE E=2 / L=20 / score ≈ 52.778 — minimum uniform E budget

PROJECTIVE E=2 / L=20 / score ≈ 52.378 — minimum uniform E budget

$U = i\, C$. Scalar metadata; no physical GP.

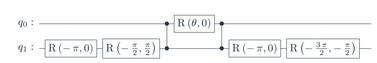
ionq_forte

EXACT PHASE E=1 / L=8 / score ≈ 22.148 — minimum uniform E budget



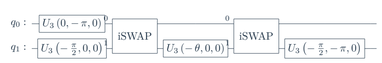
iqm_cz

EXACT PHASE E=2 / L=5 / score ≈ 18.820 — minimum uniform E budget



iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 16.424 — minimum uniform E budget



quantinuum_zzmax

EXACT PHASE E=5 / L=24 / score ≈ 73.604 — E upper bound

PROJECTIVE E=2 / L=5 / score ≈ 19.346 — minimum uniform E budget

$U = -i\, C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=1 / L=8 / score ≈ 21.956 — minimum uniform E budget

rigetti_cz

EXACT PHASE E=2 / L=11 / score ≈ 31.214 — minimum uniform E budget

PROJECTIVE E=2 / L=11 / score ≈ 30.810 — minimum uniform E budget

$U = i\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=2 / L=14 / score ≈ 38.484 — minimum uniform E budget

Exact definitions: $\eta_{652}, \eta_{653}, \eta_{654}, \eta_{655}, \eta_{656}, \eta_{657}, \eta_{658}, \eta_{659}, \eta_{660}, \eta_{661}, \eta_{662}, \eta_{663}, \eta_{664}$ in the expression glossary.

PROJECTIVE E=2 / L=13 / score ≈ 36.478 — minimum uniform E budget

Exact definitions: $\eta_{654}, \eta_{655}, \eta_{656}, \eta_{657}, \eta_{658}, \eta_{659}, \eta_{660}, \eta_{661}, \eta_{662}, \eta_{663}, \eta_{664}$ in the expression glossary.
$U = \eta_{665}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=15 / score ≈ 50.090 — E upper bound

PROJECTIVE E=4 / L=13 / score ≈ 45.278 — E upper bound

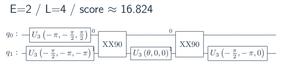
$U = -ie^{-\frac{i\theta}{2}}\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.336 — minimum uniform E budget

xx90_u

EXACT PHASE E=2 / L=4 / score ≈ 16.824 — minimum uniform E budget



PROJECTIVE E=2 / L=4 / score ≈ 16.424 — minimum uniform E budget

$U = i\, C$. Scalar metadata; no physical GP.

xx_rz_ry

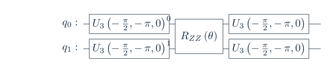
EXACT PHASE E=1 / L=0 / score ≈ 4.002 — minimum uniform E budget

zx_u

EXACT PHASE E=1 / L=6 / score ≈ 17.238 — minimum uniform E budget

zz_u

EXACT PHASE E=1 / L=4 / score ≈ 12.826 — minimum uniform E budget

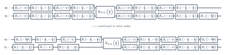


Rxy (ϕ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=2 / L=34 / score ≈ 83.704 — minimum uniform E budget

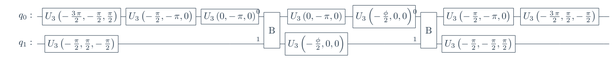


PROJECTIVE E=2 / L=19 / score ≈ 50.862 — minimum uniform E budget

$U = -iC$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=2 / L=10 / score ≈ 31.260 — minimum uniform E budget



clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

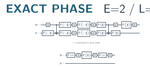
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

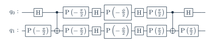
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=22 / score ≈ 54.024 — minimum uniform E budget



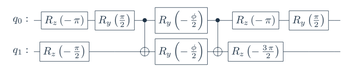
PROJECTIVE E=2 / L=14 / score ≈ 37.616 — minimum uniform E budget



$U = e^{\frac{i\phi}{2}}C$. Scalar metadata; no physical GP.

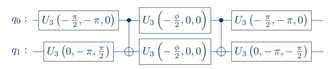
cx_rz_ry

EXACT PHASE E=2 / L=8 / score ≈ 25.216 — minimum uniform E budget



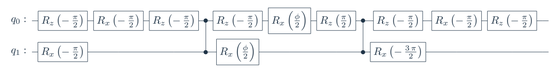
cx_u

EXACT PHASE E=2 / L=6 / score ≈ 21.236 — minimum uniform E budget



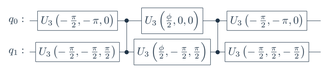
cz_rz_rx

EXACT PHASE E=2 / L=12 / score ≈ 34.424 — minimum uniform E budget



cz_u

EXACT PHASE E=2 / L=6 / score ≈ 22.436 — minimum uniform E budget



google_cz

EXACT PHASE E=2 / L=8 / score ≈ 27.369 — minimum uniform E budget

PROJECTIVE E=2 / L=6 / score ≈ 22.497 — minimum uniform E budget

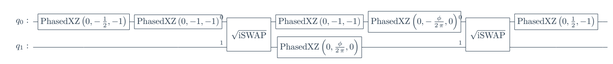
$U = e^{-\frac{i\phi}{2}}C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=0 / score ≈ 4.204 — minimum uniform E budget

google_sqrt_iswap

EXACT PHASE E=2 / L=6 / score ≈ 21.497 — minimum uniform E budget



google_syc

EXACT PHASE E=12 / L=20 / score ≈ 93.641 — E upper bound

PROJECTIVE E=12 / L=18 / score ≈ 88.769 — E upper bound

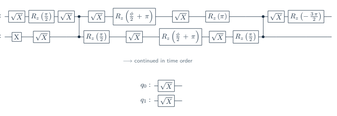
$U = e^{\frac{i\phi}{2}}C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=16 / score ≈ 41.216 — minimum uniform E budget

ibm_cz

EXACT PHASE E=2 / L=18 / score ≈ 45.214 — minimum uniform E budget



ibm_ecr

EXACT PHASE E=2 / L=14 / score ≈ 38.020 — minimum uniform E budget

PROJECTIVE E=2 / L=14 / score ≈ 37.616 — minimum uniform E budget

$U = iC$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=32 / score ≈ 81.337 — minimum uniform E budget

PROJECTIVE E=2 / L=32 / score ≈ 80.937 — minimum uniform E budget

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=32 / score ≈ 80.533 — minimum uniform E budget

PROJECTIVE E=2 / L=32 / score ≈ 80.133 — minimum uniform E budget

$U = -iC$. Scalar metadata; no physical GP.

ionq_forte

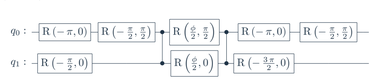
EXACT PHASE E=2 / L=34 / score ≈ 86.133 — minimum uniform E budget

PROJECTIVE E=2 / L=34 / score ≈ 85.733 — minimum uniform E budget

$U = -iC$. Scalar metadata; no physical GP.

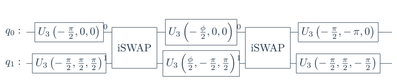
iqm_cz

EXACT PHASE E=2 / L=8 / score ≈ 25.832 — minimum uniform E budget



iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 22.436 — minimum uniform E budget



quantinuum_zzmax

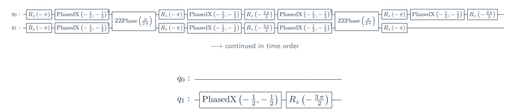
EXACT PHASE E=5 / L=27 / score ≈ 80.747 — E upper bound

PROJECTIVE E=2 / L=8 / score ≈ 26.489 — minimum uniform E budget

$U = \frac{\sqrt[4]{-1}\sqrt{2}(1+i)}{2}C$. Scalar metadata; no physical GP.

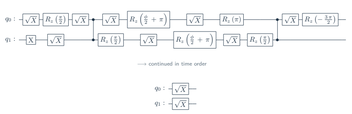
quantinuum_zzphase

EXACT PHASE E=2 / L=18 / score ≈ 49.117 — minimum uniform E budget



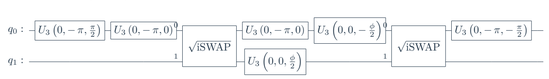
rigetti_cz

EXACT PHASE E=2 / L=18 / score ≈ 45.214 — minimum uniform E budget



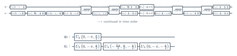
sqrt_iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 20.836 — minimum uniform E budget



sqrt_swap_u

EXACT PHASE E=4 / L=18 / score ≈ 57.708 — E upper bound



PROJECTIVE E=4 / L=18 / score ≈ 57.308 — E upper bound

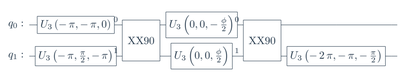
$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 5.067 — minimum uniform E budget

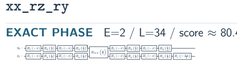
xx90_u

EXACT PHASE E=2 / L=5 / score ≈ 18.830 — minimum uniform E budget



xx_rz_ry

EXACT PHASE E=2 / L=34 / score ≈ 80.472 — minimum uniform E budget

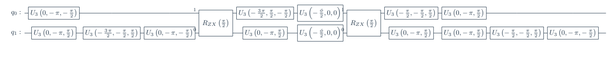


PROJECTIVE E=2 / L=27 / score ≈ 67.658 — minimum uniform E budget

$U = -iC$. Scalar metadata; no physical GP.

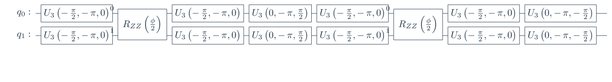
zx_u

EXACT PHASE E=2 / L=14 / score ≈ 40.888 — minimum uniform E budget



zz_u

EXACT PHASE E=2 / L=12 / score ≈ 34.876 — minimum uniform E budget



Rxz (θ)

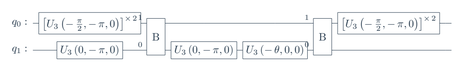
2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=1 / L=12 / score ≈ 30.438 — minimum uniform E budget

berkeley_b

EXACT PHASE E=2 / L=7 / score ≈ 22.842 — minimum uniform E budget



clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

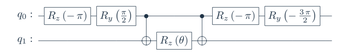
EXACT PHASE E=2 / L=11 / score ≈ 30.410 — minimum uniform E budget

PROJECTIVE E=2 / L=3 / score ≈ 14.002 — minimum uniform E budget

$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

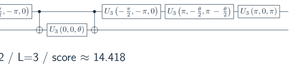
cx_rz_ry

EXACT PHASE E=2 / L=5 / score ≈ 18.410 — minimum uniform E budget



cx_u

EXACT PHASE E=2 / L=5 / score ≈ 18.830 — minimum uniform E budget

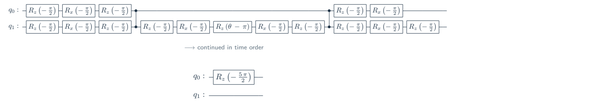


PROJECTIVE E=2 / L=3 / score ≈ 14.418 — minimum uniform E budget

$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

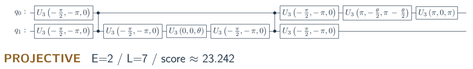
cz_rz_rx

EXACT PHASE E=2 / L=17 / score ≈ 45.234 — minimum uniform E budget



cz_u

EXACT PHASE E=2 / L=9 / score ≈ 27.654 — minimum uniform E budget



PROJECTIVE E=2 / L=7 / score ≈ 23.242 — minimum uniform E budget

$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=2 / L=9 / score ≈ 29.645 — minimum uniform E budget

PROJECTIVE E=2 / L=7 / score ≈ 24.772 — minimum uniform E budget

$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=9 / score ≈ 29.653 — E upper bound

PROJECTIVE E=2 / L=7 / score ≈ 24.780 — E upper bound

$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=2 / L=12 / score ≈ 35.393 — minimum uniform E budget

Exact definitions: $\eta_{638}, \eta_{641}, \eta_{642}, \eta_{643}, \eta_{644}, \eta_{645}, \eta_{647}, \eta_{649}, \eta_{650}, \eta_{666}, \eta_{667}, \eta_{668}, \eta_{669}, \eta_{670}$ in the expression glossary.

PROJECTIVE E=2 / L=11 / score ≈ 33.387 — minimum uniform E budget

Exact definitions: $\eta_{638}, \eta_{641}, \eta_{642}, \eta_{643}, \eta_{644}, \eta_{645}, \eta_{647}, \eta_{649}, \eta_{650}, \eta_{668}, \eta_{669}, \eta_{670}$ in the expression glossary.

$U = \eta_{671}\,C$. Scalar metadata; no physical GP.

google_syc

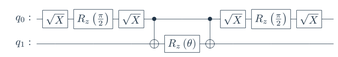
EXACT PHASE E=12 / L=13 / score ≈ 77.669 — E upper bound

PROJECTIVE E=12 / L=11 / score ≈ 72.796 — E upper bound

$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=7 / score ≈ 22.406 — minimum uniform E budget



ibm_cz

EXACT PHASE E=2 / L=17 / score ≈ 43.618 — minimum uniform E budget

PROJECTIVE E=2 / L=17 / score ≈ 43.214 — minimum uniform E budget

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=14 / score ≈ 36.810 — minimum uniform E budget

PROJECTIVE E=2 / L=13 / score ≈ 34.608 — minimum uniform E budget

$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=16 / score ≈ 44.774 — E upper bound

PROJECTIVE E=2 / L=16 / score ≈ 44.374 — E upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=16 / score ≈ 43.970 — minimum uniform E budget

PROJECTIVE E=2 / L=16 / score ≈ 43.570 — minimum uniform E budget

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

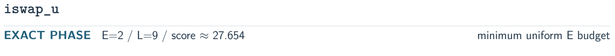
EXACT PHASE E=1 / L=4 / score ≈ 13.340 — minimum uniform E budget

iqm_cz

EXACT PHASE E=2 / L=15 / score ≈ 41.260 — minimum uniform E budget

iswap_u

EXACT PHASE E=2 / L=9 / score ≈ 27.654 — minimum uniform E budget



quantinuum_zzmax

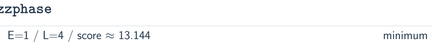
EXACT PHASE E=5 / L=32 / score ≈ 91.096 — E upper bound

PROJECTIVE E=2 / L=13 / score ≈ 36.838 — minimum uniform E budget

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=1 / L=4 / score ≈ 13.144 — minimum uniform E budget



rigetti_cz

EXACT PHASE E=2 / L=17 / score ≈ 43.618 — minimum uniform E budget

PROJECTIVE E=2 / L=17 / score ≈ 43.214 — minimum uniform E budget

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=2 / L=12 / score ≈ 34.472 — minimum uniform E budget

Exact definitions: $\eta_{657}, \eta_{659}, \eta_{660}, \eta_{663}, \eta_{664}, \eta_{672}, \eta_{673}, \eta_{674}, \eta_{675}$ in the expression glossary.

PROJECTIVE E=2 / L=11 / score ≈ 32.466 — minimum uniform E budget

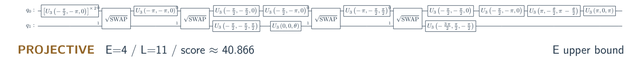
Exact definitions: $\eta_{657}, \eta_{659}, \eta_{660}, \eta_{661}, \eta_{663}, \eta_{664}, \eta_{674}, \eta_{675}$ in the expression glossary.

$U = \eta_{676}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=13 / score ≈ 45.678 — E upper bound

PROJECTIVE E=4 / L=11 / score ≈ 40.866 — E upper bound



$U = -ie^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=5 / L=35 / score ≈ 98.142 — E upper bound

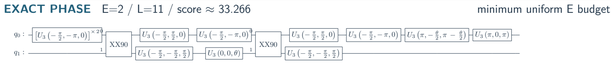
PROJECTIVE E=2 / L=15 / score ≈ 41.258 — E upper bound

$U = i\,C$. Scalar metadata; no physical GP.

xx90_u

EXACT PHASE E=2 / L=11 / score ≈ 33.266 — minimum uniform E budget

PROJECTIVE E=2 / L=9 / score ≈ 28.854 — minimum uniform E budget



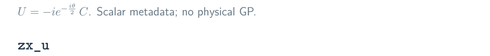
$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=1 / L=12 / score ≈ 29.226 — minimum uniform E budget

zx_u

EXACT PHASE E=1 / L=4 / score ≈ 12.826 — minimum uniform E budget



zz_u

EXACT PHASE E=1 / L=2 / score ≈ 8.414 — minimum uniform E budget

Ry (θ)

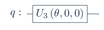
1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved E minimum

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum



clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

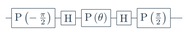
clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=12 / score ≈ 24.612 proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved E minimum



$U = e^{-i\theta} C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved E minimum

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

cz_rz_rx

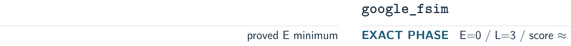
EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved E minimum

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

google_cz

EXACT PHASE E=0 / L=3 / score ≈ 7.409 proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.536 proved E minimum

$U = e^{-i\theta} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=3 / score ≈ 7.409 proved E minimum

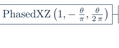
PROJECTIVE E=0 / L=1 / score ≈ 2.536 proved E minimum

$U = e^{-i\theta} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=3 / score ≈ 7.409 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.536 proved E minimum

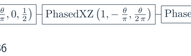


$U = e^{-i\theta} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=3 / score ≈ 7.409 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.536 proved E minimum



$U = e^{-i\theta} C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.008 proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.004 proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.008 proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.004 proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=8 / score ≈ 16.008 proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.004 proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=4 / score ≈ 8.938 proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=4 / score ≈ 8.938 proved E minimum

ionq_forte

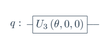
EXACT PHASE E=0 / L=4 / score ≈ 8.938 proved E minimum

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved E minimum

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum



quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.534 proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.534 proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.008 proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.004 proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.534 proved E minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved E minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

zz_u

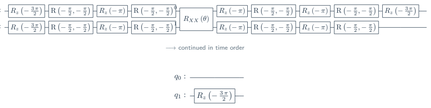
EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

Ryy (θ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=1 / L=18 / score ≈ 44.054 — minimum uniform E budget



berkeley_b

EXACT PHASE E=2 / L=13 / score ≈ 36.078 — minimum uniform E budget

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

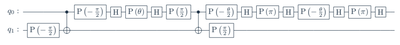
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=15 / score ≈ 39.218 — minimum uniform E budget

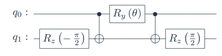


PROJECTIVE E=2 / L=7 / score ≈ 22.810 — minimum uniform E budget

$U = e^{-\frac{i\theta}{2}}\, C$. Scalar metadata; no physical GP.

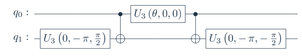
cx_rz_ry

EXACT PHASE E=2 / L=3 / score ≈ 14.406 — minimum uniform E budget



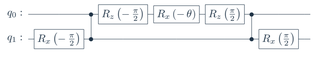
cx_u

EXACT PHASE E=2 / L=3 / score ≈ 14.418 — minimum uniform E budget



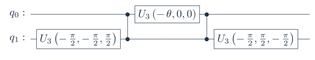
cz_rz_rx

EXACT PHASE E=2 / L=5 / score ≈ 18.810 — minimum uniform E budget



cz_u

EXACT PHASE E=2 / L=3 / score ≈ 15.218 — minimum uniform E budget



google_cz

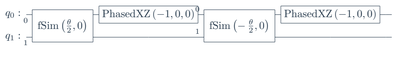
EXACT PHASE E=2 / L=5 / score ≈ 19.821 — minimum uniform E budget

PROJECTIVE E=2 / L=3 / score ≈ 14.948 — minimum uniform E budget

$U = e^{\frac{i\theta}{2}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=2 / score ≈ 12.420 — E upper bound



google_sqrt_iswap

EXACT PHASE E=2 / L=16 / score ≈ 43.287 — minimum uniform E budget

Exact definitions: $\eta_{214}, \eta_{636}, \eta_{640}, \eta_{641}, \eta_{642}, \eta_{643}, \eta_{644}, \eta_{645}, \eta_{646}, \eta_{649}, \eta_{650}, \eta_{677}, \eta_{678}, \eta_{679}, \eta_{680}, \eta_{681}, \eta_{682}, \eta_{683}$ in the expression glossary.

PROJECTIVE E=2 / L=15 / score ≈ 41.281 — minimum uniform E budget

Exact definitions: $\eta_{214}, \eta_{636}, \eta_{640}, \eta_{641}, \eta_{642}, \eta_{643}, \eta_{644}, \eta_{645}, \eta_{646}, \eta_{649}, \eta_{650}, \eta_{679}, \eta_{680}, \eta_{681}, \eta_{682}, \eta_{683}$ in the expression glossary.
$U = \eta_{651}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=19 / score ≈ 91.305 — E upper bound

PROJECTIVE E=12 / L=17 / score ≈ 86.432 — E upper bound

$U = e^{-\frac{i\theta}{2}}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=9 / score ≈ 26.410 — minimum uniform E budget

PROJECTIVE E=2 / L=6 / score ≈ 20.408 — minimum uniform E budget

$U = i\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=8 / score ≈ 24.008 — minimum uniform E budget

PROJECTIVE E=2 / L=7 / score ≈ 22.004 — minimum uniform E budget

$U = i\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=10 / score ≈ 28.812 — minimum uniform E budget

PROJECTIVE E=2 / L=7 / score ≈ 22.810 — minimum uniform E budget

$U = i\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=28 / score ≈ 71.998 — E upper bound

PROJECTIVE E=2 / L=28 / score ≈ 71.598 — E upper bound

$U = i\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=28 / score ≈ 71.194 — minimum uniform E budget

PROJECTIVE E=2 / L=28 / score ≈ 70.794 — minimum uniform E budget

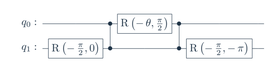
$U = i\, C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=1 / L=16 / score ≈ 40.564 — minimum uniform E budget

iqm_cz

EXACT PHASE E=2 / L=3 / score ≈ 14.612 — minimum uniform E budget



iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 17.624 — minimum uniform E budget

quantinuum_zzmax

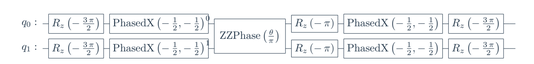
EXACT PHASE E=5 / L=24 / score ≈ 73.004 — E upper bound

PROJECTIVE E=2 / L=5 / score ≈ 18.746 — minimum uniform E budget

$U = \frac{\sqrt[4]{-1}\sqrt{2}(1+i)}{2}\, C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=1 / L=10 / score ≈ 26.760 — minimum uniform E budget



rigetti_cz

EXACT PHASE E=2 / L=8 / score ≈ 24.008 — minimum uniform E budget

PROJECTIVE E=2 / L=7 / score ≈ 22.004 — minimum uniform E budget

$U = i\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=2 / L=16 / score ≈ 42.896 — minimum uniform E budget

Exact definitions: $\eta_{652}, \eta_{653}, \eta_{654}, \eta_{658}, \eta_{659}, \eta_{660}, \eta_{661}, \eta_{662}, \eta_{664}, \eta_{684}, \eta_{685}, \eta_{686}, \eta_{687}$ in the expression glossary.

PROJECTIVE E=2 / L=15 / score ≈ 40.890 — minimum uniform E budget

Exact definitions: $\eta_{654}, \eta_{658}, \eta_{659}, \eta_{660}, \eta_{661}, \eta_{662}, \eta_{664}, \eta_{684}, \eta_{685}, \eta_{686}, \eta_{687}$ in the expression glossary.
$U = \eta_{665}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=19 / score ≈ 58.914 — E upper bound

PROJECTIVE E=4 / L=17 / score ≈ 54.102 — E upper bound

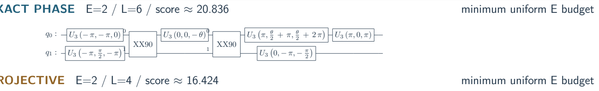
$U = -ie^{-\frac{i\theta}{2}}\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.336 — minimum uniform E budget

xx90_u

EXACT PHASE E=2 / L=6 / score ≈ 20.836 — minimum uniform E budget

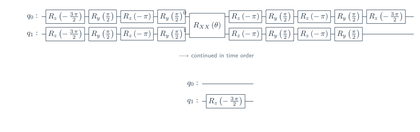


PROJECTIVE E=2 / L=4 / score ≈ 16.424 — minimum uniform E budget

$U = \eta_{688}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=1 / L=18 / score ≈ 42.438 — minimum uniform E budget

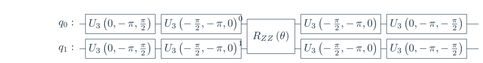


zx_u

EXACT PHASE E=1 / L=10 / score ≈ 26.062 — minimum uniform E budget

zz_u

EXACT PHASE E=1 / L=8 / score ≈ 21.650 — minimum uniform E budget



Rz (ϕ)

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved E minimum

berkeley_b

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$U = e^{-i\frac{\phi}{2}}\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.408 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = e^{-i\frac{\phi}{2}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved E minimum

cx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$U = e^{-i\frac{\phi}{2}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved E minimum

cz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$U = e^{-i\frac{\phi}{2}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=3 / score ≈ 7.209 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.336 proved E minimum

$U = e^{-i\frac{\phi}{2}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=3 / score ≈ 7.209 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.336 proved E minimum

$U = e^{-i\frac{\phi}{2}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=3 / score ≈ 7.209 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.336 proved E minimum

$U = e^{-i\frac{\phi}{2}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=3 / score ≈ 7.209 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.336 proved E minimum

$U = e^{-i\frac{\phi}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved E minimum

ibm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved E minimum

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.734 proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.734 proved E minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.734 proved E minimum

iqm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.812 proved E minimum

iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$U = e^{-i\frac{\phi}{2}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$U = e^{-i\frac{\phi}{2}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$U = e^{-i\frac{\phi}{2}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved E minimum

xx90_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$U = e^{-i\frac{\phi}{2}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved E minimum

zx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$U = e^{-i\frac{\phi}{2}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

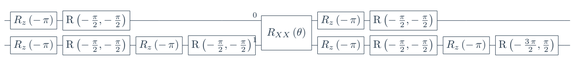
$U = e^{-i\frac{\phi}{2}}\,C$. Scalar metadata; no physical GP.

Rzx (θ)

2 qubits / 32 preserved native bases / alphabetical panel order

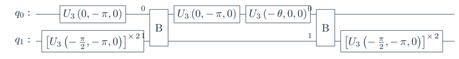
aqt_rxx

EXACT PHASE E=1 / L=12 / score ≈ 30.438 — minimum uniform E budget



berkeley_b

EXACT PHASE E=2 / L=7 / score ≈ 22.842 — minimum uniform E budget



clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

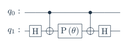
clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=11 / score ≈ 30.410 — minimum uniform E budget

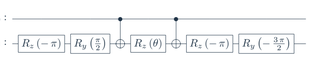
PROJECTIVE E=2 / L=3 / score ≈ 14.002 — minimum uniform E budget



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=2 / L=5 / score ≈ 18.410 — minimum uniform E budget



cx_u

EXACT PHASE E=2 / L=5 / score ≈ 18.830 — minimum uniform E budget

PROJECTIVE E=2 / L=3 / score ≈ 14.418 — minimum uniform E budget

$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=2 / L=15 / score ≈ 40.430 — minimum uniform E budget

cz_u

EXACT PHASE E=2 / L=9 / score ≈ 27.654 — minimum uniform E budget

PROJECTIVE E=2 / L=7 / score ≈ 23.242 — minimum uniform E budget

$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=2 / L=9 / score ≈ 29.645 — minimum uniform E budget

PROJECTIVE E=2 / L=7 / score ≈ 24.772 — minimum uniform E budget

$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=9 / score ≈ 29.653 — E upper bound

PROJECTIVE E=2 / L=7 / score ≈ 24.780 — E upper bound

$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=2 / L=12 / score ≈ 34.063 — minimum uniform E budget

Exact definitions: $\eta_{638}, \eta_{639}, \eta_{640}, \eta_{641}, \eta_{642}, \eta_{643}, \eta_{644}, \eta_{645}, \eta_{646}, \eta_{647}, \eta_{648}, \eta_{649}, \eta_{650}, \eta_{689}, \eta_{690}, \eta_{691}, \eta_{692}, \eta_{693}$ in the expression glossary.

PROJECTIVE E=2 / L=11 / score ≈ 32.057 — minimum uniform E budget

Exact definitions: $\eta_{214}, \eta_{636}, \eta_{637}, \eta_{638}, \eta_{639}, \eta_{640}, \eta_{641}, \eta_{642}, \eta_{643}, \eta_{644}, \eta_{645}, \eta_{646}, \eta_{647}, \eta_{648}, \eta_{649}, \eta_{650}$ in the expression glossary.
$U = \eta_{694}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=13 / score ≈ 77.669 — E upper bound

PROJECTIVE E=12 / L=11 / score ≈ 72.796 — E upper bound

$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cx

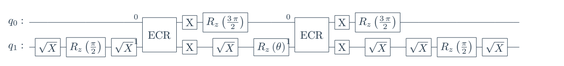
EXACT PHASE E=2 / L=7 / score ≈ 22.406 — minimum uniform E budget

ibm_cz

EXACT PHASE E=2 / L=15 / score ≈ 39.216 — minimum uniform E budget

ibm_ecr

EXACT PHASE E=2 / L=15 / score ≈ 38.810 — minimum uniform E budget



ionq_aria

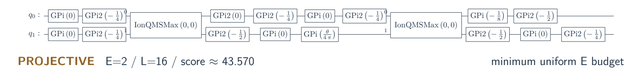
EXACT PHASE E=2 / L=16 / score ≈ 44.774 — E upper bound

PROJECTIVE E=2 / L=16 / score ≈ 44.374 — E upper bound

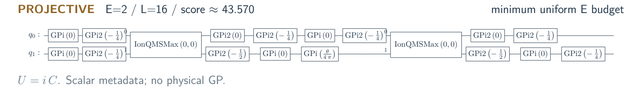
$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=16 / score ≈ 43.970 — minimum uniform E budget



PROJECTIVE E=2 / L=16 / score ≈ 43.570 — minimum uniform E budget



$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

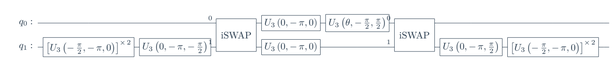
EXACT PHASE E=1 / L=4 / score ≈ 13.340 — minimum uniform E budget

iqm_cz

EXACT PHASE E=2 / L=15 / score ≈ 41.260 — minimum uniform E budget

iswap_u

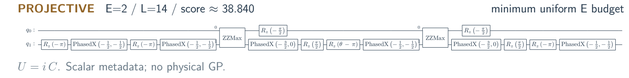
EXACT PHASE E=2 / L=9 / score ≈ 27.654 — minimum uniform E budget



quantinuum_zzmax

EXACT PHASE E=5 / L=33 / score ≈ 93.098 — E upper bound

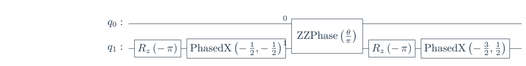
PROJECTIVE E=2 / L=14 / score ≈ 38.840 — minimum uniform E budget



$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=1 / L=4 / score ≈ 13.144 — minimum uniform E budget

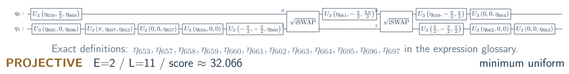


rigetti_cz

EXACT PHASE E=2 / L=15 / score ≈ 39.216 — minimum uniform E budget

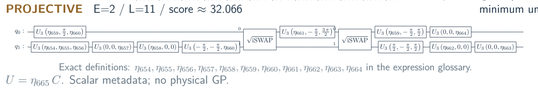
sqrt_iswap_u

EXACT PHASE E=2 / L=12 / score ≈ 34.072 — minimum uniform E budget



Exact definitions: $\eta_{653}, \eta_{657}, \eta_{658}, \eta_{659}, \eta_{660}, \eta_{661}, \eta_{662}, \eta_{663}, \eta_{664}, \eta_{695}, \eta_{696}, \eta_{697}$ in the expression glossary.

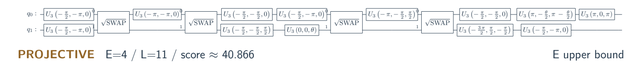
PROJECTIVE E=2 / L=11 / score ≈ 32.066 — minimum uniform E budget



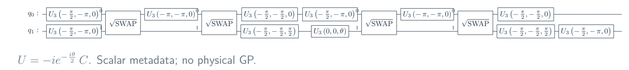
Exact definitions: $\eta_{654}, \eta_{655}, \eta_{656}, \eta_{657}, \eta_{658}, \eta_{659}, \eta_{660}, \eta_{661}, \eta_{662}, \eta_{663}, \eta_{664}$ in the expression glossary.
$U = \eta_{665}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=13 / score ≈ 45.678 — E upper bound



PROJECTIVE E=4 / L=11 / score ≈ 40.866 — E upper bound



$U = -ie^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

tket_tk2

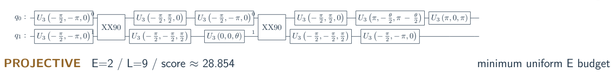
EXACT PHASE E=5 / L=35 / score ≈ 98.142 — E upper bound

PROJECTIVE E=2 / L=15 / score ≈ 41.258 — E upper bound

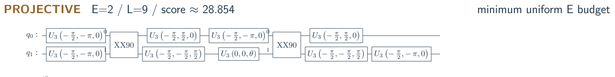
$U = i\,C$. Scalar metadata; no physical GP.

xx90_u

EXACT PHASE E=2 / L=11 / score ≈ 33.266 — minimum uniform E budget



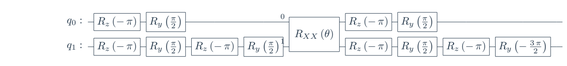
PROJECTIVE E=2 / L=9 / score ≈ 28.854 — minimum uniform E budget



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=1 / L=12 / score ≈ 29.226 — minimum uniform E budget

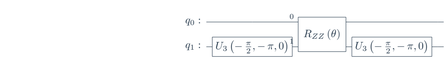


zx_u

EXACT PHASE E=1 / L=0 / score ≈ 4.002 — minimum uniform E budget

zz_u

EXACT PHASE E=1 / L=2 / score ≈ 8.414 — minimum uniform E budget

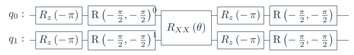


Rzz (θ)

2 qubits / 32 preserved native bases / alphabetical panel order

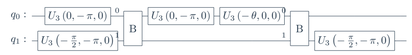
aqt_rxx

EXACT PHASE E=1 / L=8 / score ≈ 21.626 — minimum uniform E budget



berkeley_b

EXACT PHASE E=2 / L=5 / score ≈ 18.430 — minimum uniform E budget



clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

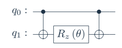
EXACT PHASE E=2 / L=9 / score ≈ 26.410 — minimum uniform E budget

PROJECTIVE E=2 / L=1 / score ≈ 10.002 — minimum uniform E budget

$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

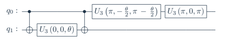
cx_rz_ry

EXACT PHASE E=2 / L=1 / score ≈ 10.002 — minimum uniform E budget



cx_u

EXACT PHASE E=2 / L=3 / score ≈ 14.418 — minimum uniform E budget

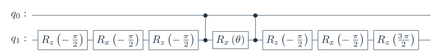


PROJECTIVE E=2 / L=1 / score ≈ 10.006 — minimum uniform E budget

$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

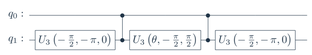
cz_rz_rx

EXACT PHASE E=2 / L=7 / score ≈ 23.214 — minimum uniform E budget



cz_u

EXACT PHASE E=2 / L=3 / score ≈ 14.818 — minimum uniform E budget



google_cz

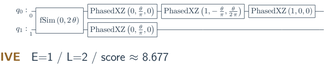
EXACT PHASE E=2 / L=5 / score ≈ 20.021 — minimum uniform E budget

PROJECTIVE E=2 / L=3 / score ≈ 15.148 — minimum uniform E budget

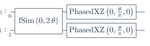
$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 13.549 — minimum uniform E budget



PROJECTIVE E=1 / L=2 / score ≈ 8.677 — minimum uniform E budget



$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=2 / L=10 / score ≈ 30.581 — minimum uniform E budget

Exact definitions: $\eta_{638}, \eta_{641}, \eta_{642}, \eta_{643}, \eta_{644}, \eta_{645}, \eta_{647}, \eta_{649}, \eta_{650}, \eta_{698}, \eta_{699}, \eta_{700}, \eta_{701}, \eta_{702}$ in the expression glossary.

PROJECTIVE E=2 / L=9 / score ≈ 28.575 — minimum uniform E budget

Exact definitions: $\eta_{638}, \eta_{641}, \eta_{642}, \eta_{643}, \eta_{644}, \eta_{645}, \eta_{647}, \eta_{649}, \eta_{650}, \eta_{668}, \eta_{669}, \eta_{670}$ in the expression glossary.

$U = \eta_{703}\,C$. Scalar metadata; no physical GP.

google_syc

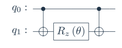
EXACT PHASE E=12 / L=11 / score ≈ 72.857 — E upper bound

PROJECTIVE E=12 / L=9 / score ≈ 67.984 — E upper bound

$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=1 / score ≈ 10.002 — minimum uniform E budget



ibm_cz

EXACT PHASE E=2 / L=11 / score ≈ 31.214 — minimum uniform E budget

PROJECTIVE E=2 / L=11 / score ≈ 30.810 — minimum uniform E budget

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=8 / score ≈ 24.008 — minimum uniform E budget

PROJECTIVE E=2 / L=4 / score ≈ 16.004 — minimum uniform E budget

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

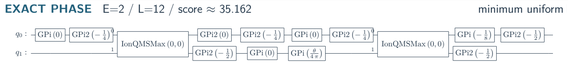
EXACT PHASE E=2 / L=12 / score ≈ 35.966 — E upper bound

PROJECTIVE E=2 / L=12 / score ≈ 35.566 — E upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=12 / score ≈ 35.162 — minimum uniform E budget

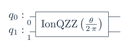


PROJECTIVE E=2 / L=12 / score ≈ 34.762 — minimum uniform E budget

$U = i\,C$. Scalar metadata; no physical GP.

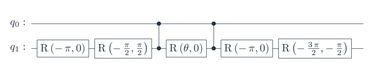
ionq_forte

EXACT PHASE E=1 / L=0 / score ≈ 4.532 — minimum uniform E budget



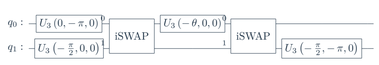
iqm_cz

EXACT PHASE E=2 / L=5 / score ≈ 18.820 — minimum uniform E budget



iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 16.424 — minimum uniform E budget



quantinuum_zzmax

EXACT PHASE E=5 / L=24 / score ≈ 73.604 — E upper bound

PROJECTIVE E=2 / L=5 / score ≈ 19.346 — minimum uniform E budget

$U = -i\,C$. Scalar metadata; no physical GP.

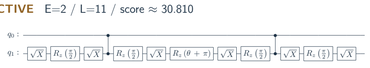
quantinuum_zzphase

EXACT PHASE E=1 / L=0 / score ≈ 4.332 — minimum uniform E budget

rigetti_cz

EXACT PHASE E=2 / L=11 / score ≈ 31.214 — minimum uniform E budget

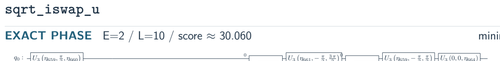
PROJECTIVE E=2 / L=11 / score ≈ 30.810 — minimum uniform E budget



$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=2 / L=10 / score ≈ 30.060 — minimum uniform E budget



Exact definitions: $\eta_{637}, \eta_{659}, \eta_{660}, \eta_{661}, \eta_{663}, \eta_{664}, \eta_{673}, \eta_{704}, \eta_{705}, \eta_{706}$ in the expression glossary.

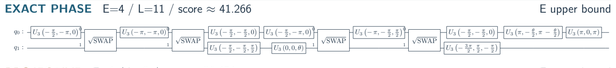
PROJECTIVE E=2 / L=9 / score ≈ 28.054 — minimum uniform E budget

Exact definitions: $\eta_{637}, \eta_{659}, \eta_{660}, \eta_{661}, \eta_{663}, \eta_{664}, \eta_{674}, \eta_{675}$ in the expression glossary.

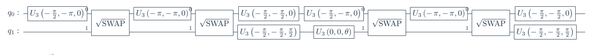
$U = \eta_{676}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=11 / score ≈ 41.266 — E upper bound



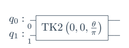
PROJECTIVE E=4 / L=9 / score ≈ 36.454 — E upper bound



$U = -ie^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

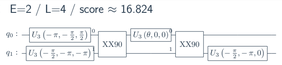
tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.336 — minimum uniform E budget

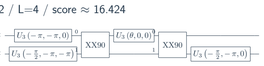


xx90_u

EXACT PHASE E=2 / L=4 / score ≈ 16.824 — minimum uniform E budget



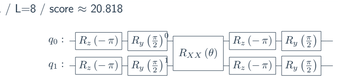
PROJECTIVE E=2 / L=4 / score ≈ 16.424 — minimum uniform E budget



$U = i\,C$. Scalar metadata; no physical GP.

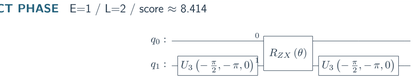
xx_rz_ry

EXACT PHASE E=1 / L=8 / score ≈ 20.818 — minimum uniform E budget



zx_u

EXACT PHASE E=1 / L=2 / score ≈ 8.414 — minimum uniform E budget



zz_u

EXACT PHASE E=1 / L=0 / score ≈ 4.002 — minimum uniform E budget

S

1 qubits / 32 preserved native bases / alphabetical panel order

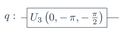
aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum



clifford_only

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved E minimum

clifford_t_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved E minimum

cx_h_phase

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved E minimum

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.404 proved E minimum

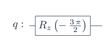
PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved E minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.404 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

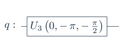
iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum



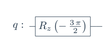
quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved E minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved E minimum

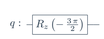
$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved E minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

S_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

S_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.208 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

S_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved E minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.808 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved E minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved E minimum

cz_u

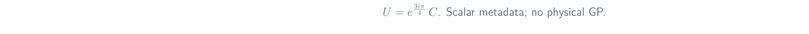
EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved E minimum

ibm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved E minimum

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved E minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 proved E minimum

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.408 proved E minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved E minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved E minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

S_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

clifford_only

EXACT PHASE E=0 / L=13 / score ≈ 26.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.208 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

S_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=13 / score ≈ 26.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.808 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

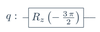
$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

S_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

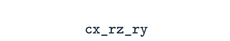
PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum



$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.206 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

S_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

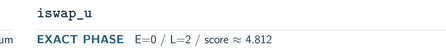
$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

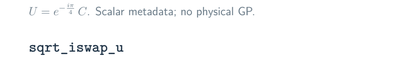
EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum



$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

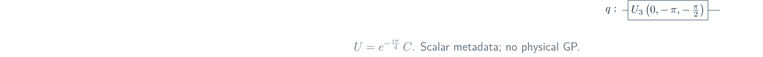
EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.408 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

Sdg

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

clifford_only

EXACT PHASE E=0 / L=3 / score ≈ 6.000 proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.000 proved E minimum

clifford_t_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.000 proved E minimum

cx_h_phase

EXACT PHASE E=0 / L=1 / score ≈ 2.202 proved E minimum

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

google_cz

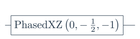
EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

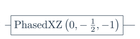
google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum



google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum



ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved E minimum

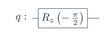
PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved E minimum

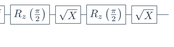
PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

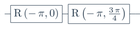
ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

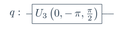
PROJECTIVE E=0 / L=2 / score ≈ 4.408 proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum



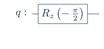
quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved E minimum

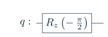
$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 proved E minimum



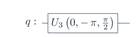
$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum



Sdg_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=5 / score ≈ 10.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

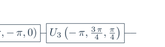
PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.408 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Sdg_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.206 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

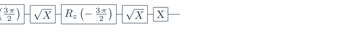
EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

Sdg_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=13 / score ≈ 26.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

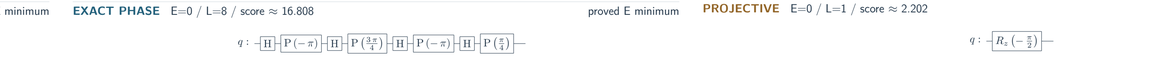
EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.808 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Sdg_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.208 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

Sdg_q5

1 qubits / 32 preserved native bases / alphabetical panel order

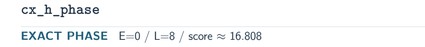
aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=15 / score ≈ 30.000 — proved E minimum

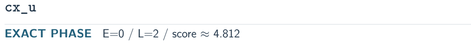
PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=13 / score ≈ 26.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=13 / score ≈ 26.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.808 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

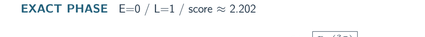
cz_rz_rx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

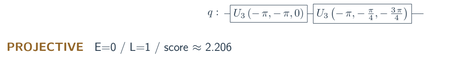
PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.408 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Sdg_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=13 / score ≈ 26.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.208 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

Sdg_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

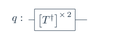
PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

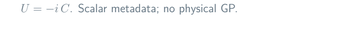
EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.406 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

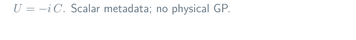
cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SH

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

clifford_only

EXACT PHASE E=0 / L=2 / score ≈ 4.000 proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.000 proved E minimum

clifford_t_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.000 proved E minimum

cx_h_phase

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved E minimum

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

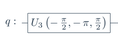
cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

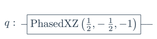
EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum



google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

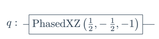


$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved E minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved E minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 proved E minimum

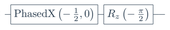
$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

quantinuum_zzmax

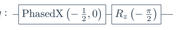
PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum



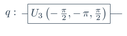
$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

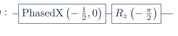


sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

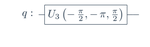
$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum



SH_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.406 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SH_q2

1 qubits / 32 preserved native bases / alphabetical panel order

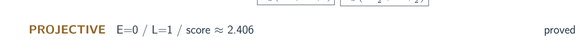
aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.208 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

SH_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.808 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

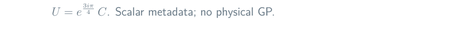
cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SH_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.208 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

SH_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved E minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.808 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved E minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved E minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.806 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.806 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.806 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 proved E minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 proved E minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 proved E minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

SH_q6

1 qubits / 32 preserved native bases / alphabetical panel order

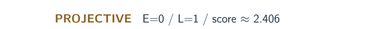
aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.206 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

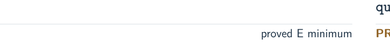
EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

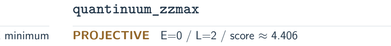


PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

SH_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

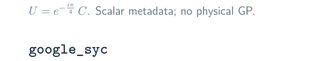
PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

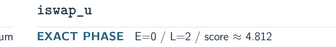
EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum



$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

SHX

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

clifford_only

EXACT PHASE E=0 / L=4 / score ≈ 8.000 proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.000 proved E minimum

clifford_t_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.000 proved E minimum

cx_h_phase

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved E minimum

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

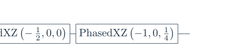
EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

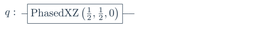
$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved E minimum

ibm_cz

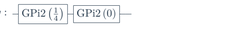
EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved E minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum



$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

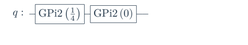
ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum



$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

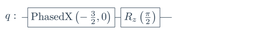
quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved E minimum

sqrt_iswap_u

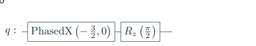
EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

tket_tk2

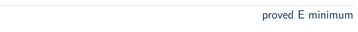
PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum



xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

SHX_q1

1 qubits / 32 preserved native bases / alphabetical panel order

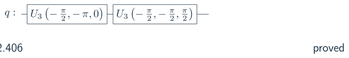
aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

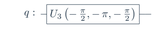
$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum



$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=4 / score ≈ 8.000 — proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

SHX_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.206 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

SHX_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.808 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

SHX_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.208 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

SHX_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

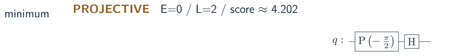
PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=14 / score ≈ 28.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved E minimum

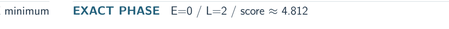
PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.808 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SHX_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.208 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

SHX_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.406 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

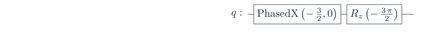
cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

SHY

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 proved E minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.206 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

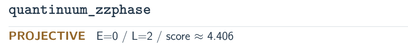
EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

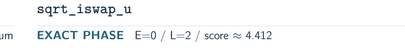


PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

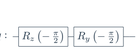
$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

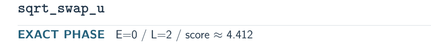
EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

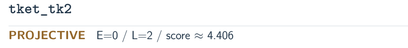
ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved E minimum



ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved E minimum



ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved E minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

SHY_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cx

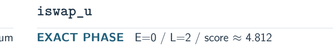
EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.406 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

SHY_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

clifford_only

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.000 — proved E minimum

clifford_t_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.000 — proved E minimum

cx_h_phase

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

SHY_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.000 — proved E minimum

clifford_t_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.000 — proved E minimum

cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SHY_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

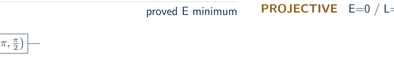
PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum



$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

SHY_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.806 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SHY_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

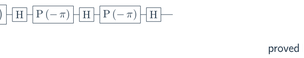
$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum



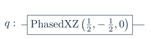
PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum



$U = -1\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.206 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

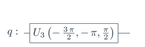
cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = i\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = i\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = i\, C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = i\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

SHY_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved E minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.806 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

SHZ

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

clifford_only

EXACT PHASE E=0 / L=4 / score ≈ 8.000 — proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.000 — proved E minimum

clifford_t_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.000 — proved E minimum

cx_h_phase

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

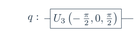
cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum



cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

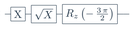
PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

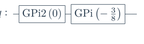
PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum



$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

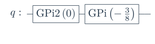
ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

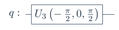
iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum



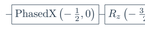
quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

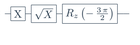


$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

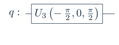
PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum



$U = -i\,C$. Scalar metadata; no physical GP.

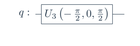
sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum



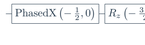
sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum



tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum



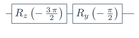
$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

xx_rz_ry

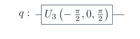
PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

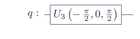
zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum



zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

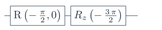


SHZ_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum



$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

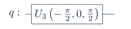
PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum



$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.406 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

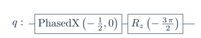
PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

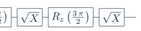


$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

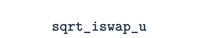
PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum



ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

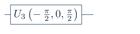
quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_swap_u

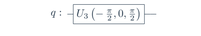
EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum



$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

SHZ_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.206 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

SHZ_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved E minimum

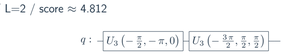
PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.806 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

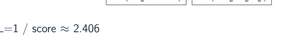
PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

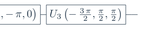
PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum



$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.806 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.806 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.806 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SHZ_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.206 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

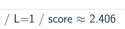
PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

SHZ_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.806 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

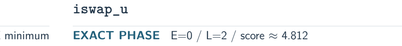
cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_syc

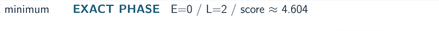
EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SHZ_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = -i\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = -i\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = -i\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved E minimum

$U = -i\, C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\, C$. Scalar metadata; no physical GP.

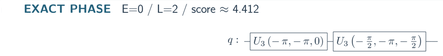
cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

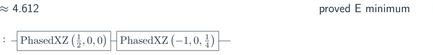


PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

ibm_cz

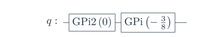
EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

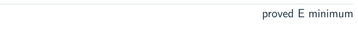
ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.604 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

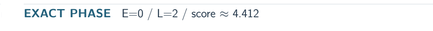
PROJECTIVE E=0 / L=2 / score ≈ 4.608 — proved E minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

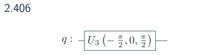
PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum



$U = -i\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.406 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = -i\, C$. Scalar metadata; no physical GP.

SHZ_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

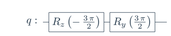
PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.000 — proved E minimum

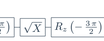
PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.000 — proved E minimum

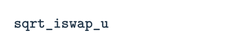
PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

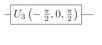
EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

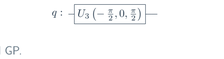


google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

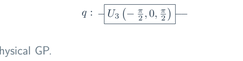


PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.604 — proved E minimum

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.608 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.406 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

SiSWAP

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

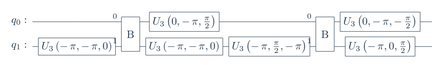
EXACT PHASE E=2 / L=34 / score ≈ 84.104 — proved E minimum

PROJECTIVE E=2 / L=16 / score ≈ 44.058 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=2 / L=6 / score ≈ 20.836 — proved E minimum



clifford_only

Exact: Proved obstruction: not a clifford
Projective: Proved obstruction: not a clifford

clifford_t_cx

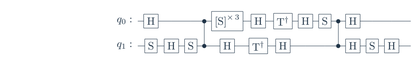
EXACT PHASE E=2 / L=18 / score ≈ 44.000 — proved E minimum

PROJECTIVE E=2 / L=16 / score ≈ 40.000 — proved E minimum

$U = -(-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=2 / L=18 / score ≈ 44.000 — proved E minimum



cx_h_phase

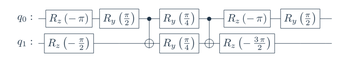
EXACT PHASE E=2 / L=16 / score ≈ 41.814 — proved E minimum

PROJECTIVE E=2 / L=14 / score ≈ 38.016 — proved E minimum

$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

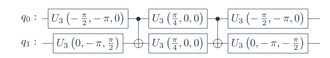
cx_rz_ry

EXACT PHASE E=2 / L=8 / score ≈ 25.616 — proved E minimum



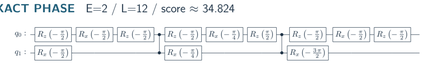
cx_u

EXACT PHASE E=2 / L=6 / score ≈ 21.636 — proved E minimum



cz_rz_rx

EXACT PHASE E=2 / L=12 / score ≈ 34.824 — proved E minimum



PROJECTIVE E=2 / L=9 / score ≈ 28.018 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

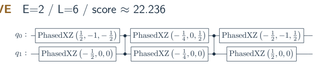
cz_u

EXACT PHASE E=2 / L=6 / score ≈ 22.836 — proved E minimum

google_cz

EXACT PHASE E=2 / L=7 / score ≈ 24.642 — proved E minimum

PROJECTIVE E=2 / L=6 / score ≈ 22.236 — proved E minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=0 / score ≈ 4.404 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

google_syc

EXACT PHASE E=12 / L=15 / score ≈ 81.690 — E upper bound

PROJECTIVE E=12 / L=14 / score ≈ 79.284 — E upper bound

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=13 / score ≈ 35.212 — proved E minimum

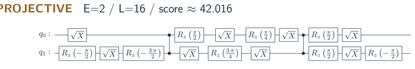
PROJECTIVE E=2 / L=12 / score ≈ 33.208 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=18 / score ≈ 45.614 — proved E minimum

PROJECTIVE E=2 / L=16 / score ≈ 42.016 — proved E minimum



$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=14 / score ≈ 38.420 — proved E minimum

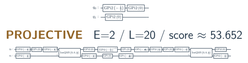
PROJECTIVE E=2 / L=14 / score ≈ 38.016 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=20 / score ≈ 54.052 — proved E minimum

PROJECTIVE E=2 / L=20 / score ≈ 53.652 — proved E minimum

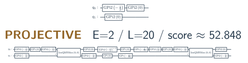


$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=20 / score ≈ 53.248 — proved E minimum

PROJECTIVE E=2 / L=20 / score ≈ 52.848 — proved E minimum

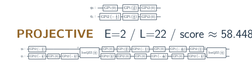


$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=22 / score ≈ 58.848 — proved E minimum

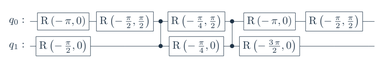
PROJECTIVE E=2 / L=22 / score ≈ 58.448 — proved E minimum



$U = i\,C$. Scalar metadata; no physical GP.

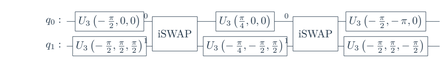
iqm_cz

EXACT PHASE E=2 / L=8 / score ≈ 26.232 — proved E minimum



iswap_u

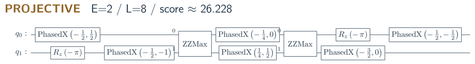
EXACT PHASE E=2 / L=6 / score ≈ 22.836 — proved E minimum



quantinuum_zzmax

EXACT PHASE E=5 / L=27 / score ≈ 80.486 — E upper bound

PROJECTIVE E=2 / L=8 / score ≈ 26.228 — proved E minimum



$U = \frac{\sqrt[8]{-1}\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

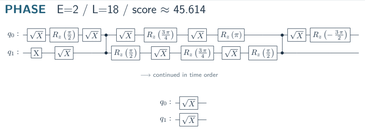
EXACT PHASE E=2 / L=18 / score ≈ 48.856 — proved E minimum

PROJECTIVE E=2 / L=15 / score ≈ 42.052 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=2 / L=18 / score ≈ 45.614 — proved E minimum



sqrt_iswap_u

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

sqrt_swap_u

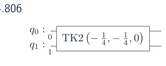
EXACT PHASE E=4 / L=16 / score ≈ 54.096 — E upper bound

PROJECTIVE E=4 / L=14 / score ≈ 50.084 — E upper bound

$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.

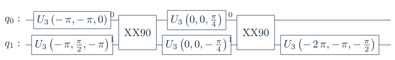
tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.806 — proved E minimum



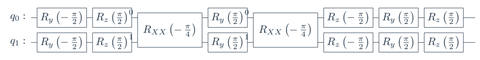
xx90_u

EXACT PHASE E=2 / L=5 / score ≈ 19.230 — proved E minimum



xx_rz_ry

EXACT PHASE E=2 / L=12 / score ≈ 35.228 — proved E minimum



zx_u

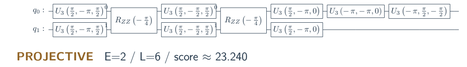
EXACT PHASE E=2 / L=8 / score ≈ 27.652 — proved E minimum

PROJECTIVE E=2 / L=6 / score ≈ 23.240 — proved E minimum

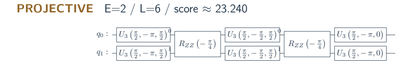
$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=2 / L=8 / score ≈ 27.652 — proved E minimum



PROJECTIVE E=2 / L=6 / score ≈ 23.240 — proved E minimum



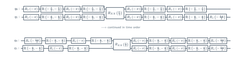
$U = -i\,C$. Scalar metadata; no physical GP.

SiSWAPdg

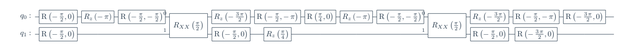
2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=2 / L=34 / score ≈ 84.104 — proved E minimum



PROJECTIVE E=2 / L=16 / score ≈ 44.058 — proved E minimum



$U = -iC$. Scalar metadata; no physical GP.

berkeley_b

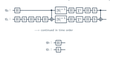
EXACT PHASE E=2 / L=6 / score ≈ 20.836 — proved E minimum

clifford_only

Exact: Proved obstruction: not a clifford
Projective: Proved obstruction: not a clifford

clifford_t_cx

EXACT PHASE E=2 / L=22 / score ≈ 52.000 — proved E minimum



PROJECTIVE E=2 / L=16 / score ≈ 40.000 — proved E minimum

$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=2 / L=24 / score ≈ 56.000 — proved E minimum

PROJECTIVE E=2 / L=18 / score ≈ 44.000 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

cx_h_phase

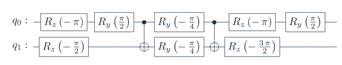
EXACT PHASE E=2 / L=16 / score ≈ 41.814 — proved E minimum

PROJECTIVE E=2 / L=14 / score ≈ 38.016 — proved E minimum

$U = e^{\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

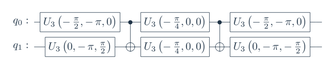
cx_rz_ry

EXACT PHASE E=2 / L=8 / score ≈ 25.616 — proved E minimum



cx_u

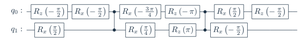
EXACT PHASE E=2 / L=6 / score ≈ 21.636 — proved E minimum



cz_rz_rx

EXACT PHASE E=2 / L=12 / score ≈ 34.824 — proved E minimum

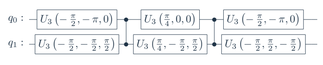
PROJECTIVE E=2 / L=10 / score ≈ 30.020 — proved E minimum



$U = iC$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=2 / L=6 / score ≈ 22.836 — proved E minimum



google_cz

EXACT PHASE E=2 / L=7 / score ≈ 24.642 — proved E minimum

PROJECTIVE E=2 / L=6 / score ≈ 22.236 — proved E minimum

$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=0 / score ≈ 4.404 — proved E minimum

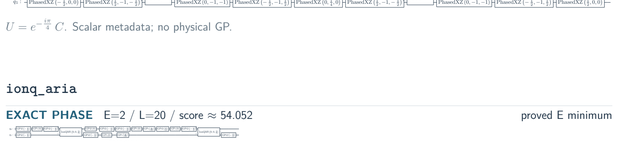
google_sqrt_iswap

EXACT PHASE E=1 / L=2 / score ≈ 8.012 — proved E minimum

google_syc

EXACT PHASE E=12 / L=15 / score ≈ 81.690 — E upper bound

PROJECTIVE E=12 / L=14 / score ≈ 79.284 — E upper bound



$U = e^{-\frac{i\pi}{4}}C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=13 / score ≈ 35.212 — proved E minimum

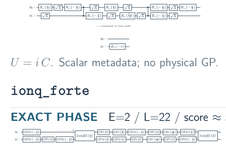
PROJECTIVE E=2 / L=12 / score ≈ 33.208 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=18 / score ≈ 45.614 — proved E minimum

PROJECTIVE E=2 / L=17 / score ≈ 44.018 — proved E minimum



$U = iC$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=14 / score ≈ 38.420 — proved E minimum

PROJECTIVE E=2 / L=14 / score ≈ 38.016 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=20 / score ≈ 54.052 — proved E minimum

PROJECTIVE E=2 / L=20 / score ≈ 53.652 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=20 / score ≈ 53.248 — proved E minimum

PROJECTIVE E=2 / L=20 / score ≈ 52.848 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=22 / score ≈ 58.848 — proved E minimum

PROJECTIVE E=2 / L=22 / score ≈ 58.448 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

iqm_cz

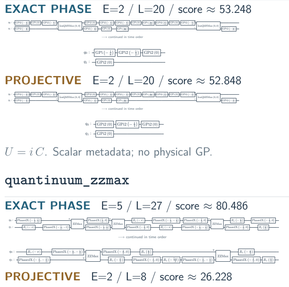
EXACT PHASE E=2 / L=8 / score ≈ 26.232 — proved E minimum

iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 22.836 — proved E minimum

quantinuum_zzmax

EXACT PHASE E=5 / L=27 / score ≈ 80.486 — E upper bound

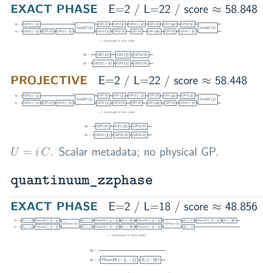


PROJECTIVE E=2 / L=8 / score ≈ 26.228 — proved E minimum

$U = \frac{\sqrt[8]{-1}\sqrt{2}(1+i)}{2}C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=2 / L=18 / score ≈ 48.856 — proved E minimum

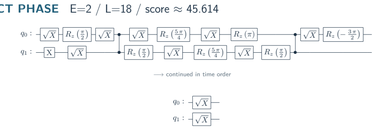


PROJECTIVE E=2 / L=16 / score ≈ 44.054 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

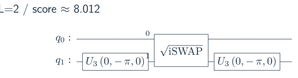
rigetti_cz

EXACT PHASE E=2 / L=18 / score ≈ 45.614 — proved E minimum



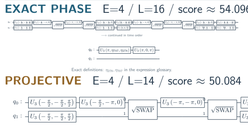
sqrt_iswap_u

EXACT PHASE E=1 / L=2 / score ≈ 8.012 — proved E minimum



sqrt_swap_u

EXACT PHASE E=4 / L=16 / score ≈ 54.096 — E upper bound

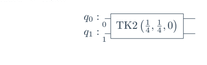


PROJECTIVE E=4 / L=14 / score ≈ 50.084 — E upper bound

$U = -(-1)^{\frac{3}{8}}C$. Scalar metadata; no physical GP.

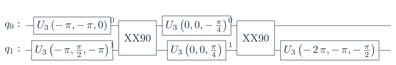
tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.806 — proved E minimum



xx90_u

EXACT PHASE E=2 / L=5 / score ≈ 19.230 — proved E minimum



xx_rz_ry

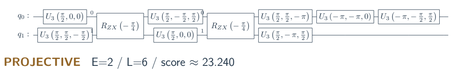
EXACT PHASE E=2 / L=13 / score ≈ 37.230 — proved E minimum

PROJECTIVE E=2 / L=12 / score ≈ 35.228 — proved E minimum

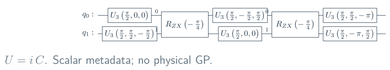
$U = -1\,C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=2 / L=8 / score ≈ 27.652 — proved E minimum



PROJECTIVE E=2 / L=6 / score ≈ 23.240 — proved E minimum

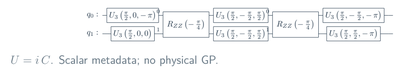


$U = iC$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=2 / L=8 / score ≈ 27.652 — proved E minimum

PROJECTIVE E=2 / L=6 / score ≈ 23.240 — proved E minimum



$U = iC$. Scalar metadata; no physical GP.

SqrtH

1 qubits / 32 preserved native bases / alphabetical panel order

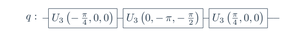
aqt_rxx

PROJECTIVE E=0 / L=3 / score ≈ 7.410 proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved E minimum



clifford_only

Exact: Proved obstruction: not a clifford
Projective: Proved obstruction: not a clifford

clifford_t_cx

EXACT PHASE E=0 / L=15 / score ≈ 30.000 proved E minimum

clifford_t_cz

EXACT PHASE E=0 / L=15 / score ≈ 30.000 proved E minimum

cx_h_phase

EXACT PHASE E=0 / L=9 / score ≈ 19.410 proved E minimum

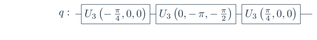
cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 7.006 proved E minimum

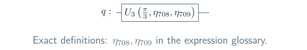
$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 3.123 proved E minimum



Exact definitions: η_{708}, η_{709} in the expression glossary.
$U = e^{i\,\mathrm{atan}\left(3-2\sqrt{2}\right)}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.323 proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 3.123 proved E minimum

Exact definitions: η_{708}, η_{709} in the expression glossary.
$U = e^{i\,\mathrm{atan}\left(3-2\sqrt{2}\right)}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=3 / score ≈ 7.418 proved E minimum

google_fsim

EXACT PHASE E=0 / L=3 / score ≈ 7.418 proved E minimum

google_sqrt_iswap

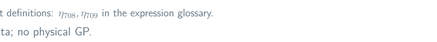
EXACT PHASE E=0 / L=3 / score ≈ 7.418 proved E minimum

google_syc

EXACT PHASE E=0 / L=3 / score ≈ 7.418 proved E minimum

ibm_cx

EXACT PHASE E=0 / L=11 / score ≈ 23.729 proved E minimum



Exact definitions: η_{709} in the expression glossary.

PROJECTIVE E=0 / L=5 / score ≈ 10.723 proved E minimum

Exact definitions: η_{710}, η_{711} in the expression glossary.
$U = -(-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=11 / score ≈ 23.729 proved E minimum

Exact definitions: η_{709} in the expression glossary.

PROJECTIVE E=0 / L=5 / score ≈ 10.723 proved E minimum

Exact definitions: η_{710}, η_{711} in the expression glossary.
$U = -(-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=11 / score ≈ 23.729 proved E minimum

Exact definitions: η_{709} in the expression glossary.

PROJECTIVE E=0 / L=5 / score ≈ 10.723 proved E minimum

Exact definitions: η_{710}, η_{711} in the expression glossary.
$U = -(-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=10 / score ≈ 22.620 proved E minimum

$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=10 / score ≈ 22.620 proved E minimum

$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=10 / score ≈ 22.620 proved E minimum

$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=4 / score ≈ 9.616 proved E minimum

$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 3.123 proved E minimum

Exact definitions: η_{708}, η_{709} in the expression glossary.
$U = e^{i\,\mathrm{atan}\left(3-2\sqrt{2}\right)}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=3 / score ≈ 7.410 proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=3 / score ≈ 7.410 proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=15 / score ≈ 31.614 proved E minimum

PROJECTIVE E=0 / L=8 / score ≈ 17.008 proved E minimum

$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved E minimum

tket_tk2

PROJECTIVE E=0 / L=3 / score ≈ 7.410 proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 7.006 proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 3.123 proved E minimum

Exact definitions: η_{708}, η_{709} in the expression glossary.
$U = e^{i\,\mathrm{atan}\left(3-2\sqrt{2}\right)}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 3.123 proved E minimum

Exact definitions: η_{708}, η_{709} in the expression glossary.
$U = e^{i\,\mathrm{atan}\left(3-2\sqrt{2}\right)}\,C$. Scalar metadata; no physical GP.

SqrtHdg

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=3 / score ≈ 7.410 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved E minimum

clifford_only

Exact: Proved obstruction: not a clifford
Projective: Proved obstruction: not a clifford

clifford_t_cx

EXACT PHASE E=0 / L=17 / score ≈ 34.000 proved E minimum

clifford_t_cz

EXACT PHASE E=0 / L=17 / score ≈ 34.000 proved E minimum

cx_h_phase

EXACT PHASE E=0 / L=9 / score ≈ 19.410 proved E minimum

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 7.006 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 3.123 proved E minimum

Exact definitions: η_{712}, η_{713} in the expression glossary.
$U = e^{-i\,\mathrm{atan}\left(3-2\sqrt{2}\right)}\,C$. Scalar metadata; no physical GP.

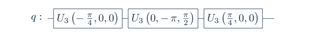
cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.323 proved E minimum

Exact definitions: η_{714} in the expression glossary.
$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved E minimum

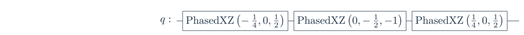


PROJECTIVE E=0 / L=1 / score ≈ 3.123 proved E minimum

Exact definitions: η_{712}, η_{713} in the expression glossary.
$U = e^{-i\,\mathrm{atan}\left(3-2\sqrt{2}\right)}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=3 / score ≈ 7.418 proved E minimum

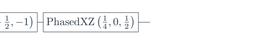


google_fsim

EXACT PHASE E=0 / L=3 / score ≈ 7.418 proved E minimum

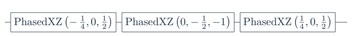
google_sqrt_iswap

EXACT PHASE E=0 / L=3 / score ≈ 7.418 proved E minimum



google_syc

EXACT PHASE E=0 / L=3 / score ≈ 7.418 proved E minimum



ibm_cx

EXACT PHASE E=0 / L=12 / score ≈ 25.729 proved E minimum

Exact definitions: η_{713}, η_{715} in the expression glossary.

PROJECTIVE E=0 / L=5 / score ≈ 10.723 proved E minimum

Exact definitions: η_{716} in the expression glossary.
$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=12 / score ≈ 25.729 proved E minimum

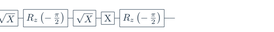
Exact definitions: η_{713}, η_{715} in the expression glossary.

PROJECTIVE E=0 / L=5 / score ≈ 10.723 proved E minimum

Exact definitions: η_{716} in the expression glossary.
$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=12 / score ≈ 25.729 proved E minimum



Exact definitions: η_{713}, η_{715} in the expression glossary.

PROJECTIVE E=0 / L=5 / score ≈ 10.723 proved E minimum

Exact definitions: η_{716} in the expression glossary.
$U = -\sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=10 / score ≈ 22.620 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=10 / score ≈ 22.620 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

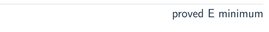
ionq_forte

PROJECTIVE E=0 / L=10 / score ≈ 22.620 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

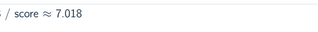
PROJECTIVE E=0 / L=4 / score ≈ 9.616 proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 3.123 proved E minimum

Exact definitions: η_{712}, η_{713} in the expression glossary.
$U = e^{-i\,\mathrm{atan}\left(3-2\sqrt{2}\right)}\,C$. Scalar metadata; no physical GP.

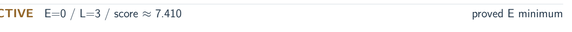
quantinuum_zzmax

PROJECTIVE E=0 / L=3 / score ≈ 7.410 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=3 / score ≈ 7.410 proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=14 / score ≈ 29.614 proved E minimum

PROJECTIVE E=0 / L=8 / score ≈ 17.008 proved E minimum

$U = (-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved E minimum

tket_tk2

PROJECTIVE E=0 / L=3 / score ≈ 7.410 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 7.006 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 3.123 proved E minimum

Exact definitions: η_{712}, η_{713} in the expression glossary.
$U = e^{-i\,\mathrm{atan}\left(3-2\sqrt{2}\right)}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=3 / score ≈ 7.018 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 3.123 proved E minimum

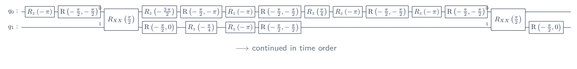
Exact definitions: η_{712}, η_{713} in the expression glossary.
$U = e^{-i\,\mathrm{atan}\left(3-2\sqrt{2}\right)}\,C$. Scalar metadata; no physical GP.

SSWAP

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=3 / L=31 / score ≈ 81.698 — proved E minimum



$U = \eta_{717}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=6 / score ≈ 22.553 — proved E minimum

PROJECTIVE E=2 / L=5 / score ≈ 20.547 — proved E minimum

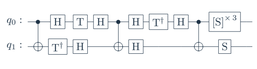
$U = \eta_{719}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

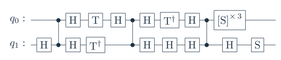
clifford_t_cx

EXACT PHASE E=3 / L=13 / score ≈ 38.000 — proved E minimum



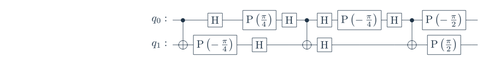
clifford_t_cz

EXACT PHASE E=3 / L=17 / score ≈ 46.000 — proved E minimum



cx_h_phase

EXACT PHASE E=3 / L=11 / score ≈ 35.610 — proved E minimum



cx_rz_ry

PROJECTIVE E=3 / L=14 / score ≈ 43.428 — proved E minimum

$U = -(-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=3 / L=9 / score ≈ 33.454 — proved E minimum

Exact definitions: $\eta_{\text{zero}}, \eta_{\text{zoo}}$ in the expression glossary.

PROJECTIVE E=3 / L=7 / score ≈ 29.442 — proved E minimum

$U = e^{\frac{i\pi}{8}}\, C$. Scalar metadata; no physical GP.

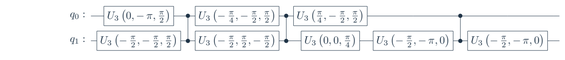
cz_rz_rx

PROJECTIVE E=3 / L=11 / score ≈ 36.822 — proved E minimum

$U = -(-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

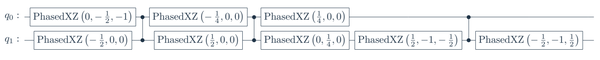
cz_u

EXACT PHASE E=3 / L=8 / score ≈ 31.848 — proved E minimum



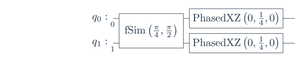
google_cz

EXACT PHASE E=3 / L=8 / score ≈ 30.648 — proved E minimum



google_fsim

EXACT PHASE E=1 / L=2 / score ≈ 9.416 — proved E minimum



google_sqrt_iswap

EXACT PHASE E=3 / L=8 / score ≈ 31.699 — proved E minimum

Exact definitions: $\eta_{720}, \eta_{721}, \eta_{722}, \eta_{9}$ in the expression glossary.

PROJECTIVE E=3 / L=7 / score ≈ 27.833 — proved E minimum

Exact definitions: η_{722}, η_{9} in the expression glossary.

$U = \eta_{723}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=18 / L=23 / score ≈ 124.538 — E upper bound

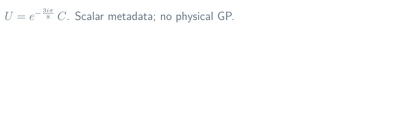
ibm_cx

PROJECTIVE E=3 / L=9 / score ≈ 31.610 — proved E minimum

$U = e^{\frac{3i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

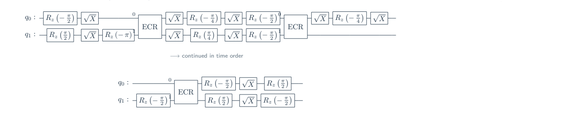
PROJECTIVE E=3 / L=17 / score ≈ 48.216 — proved E minimum



$U = e^{-\frac{3i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

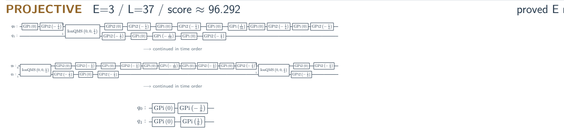
PROJECTIVE E=3 / L=23 / score ≈ 61.026 — proved E minimum



$U = e^{\frac{i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=3 / L=37 / score ≈ 96.292 — proved E minimum



$U = \eta_{724}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=3 / L=37 / score ≈ 95.086 — proved E minimum

$U = \eta_{724}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=3 / L=40 / score ≈ 103.486 — proved E minimum

$U = \eta_{724}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

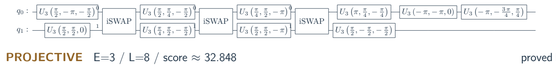
iqm_cz

PROJECTIVE E=3 / L=13 / score ≈ 41.652 — proved E minimum

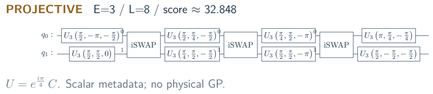
$U = -(-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=3 / L=10 / score ≈ 37.660 — proved E minimum



PROJECTIVE E=3 / L=8 / score ≈ 32.848 — proved E minimum



$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

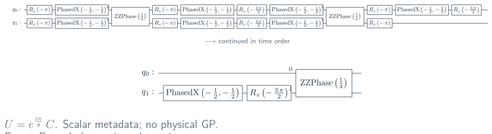
quantinuum_zzmax

PROJECTIVE E=3 / L=10 / score ≈ 34.832 — proved E minimum

$U = -(-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

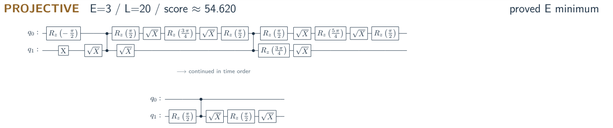
PROJECTIVE E=3 / L=18 / score ≈ 53.258 — proved E minimum



$U = e^{\frac{i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

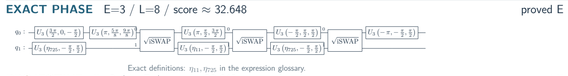
PROJECTIVE E=3 / L=20 / score ≈ 54.620 — proved E minimum



$U = -\sqrt[8]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=3 / L=8 / score ≈ 32.648 — proved E minimum



Exact definitions: η_{11}, η_{725} in the expression glossary.

PROJECTIVE E=3 / L=7 / score ≈ 29.642 — proved E minimum

Exact definitions: η_{11}, η_{725} in the expression glossary.

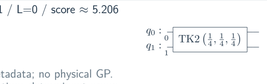
$U = e^{\frac{i\pi}{8}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

tket_tk2

PROJECTIVE E=1 / L=0 / score ≈ 5.206 — proved E minimum



$U = e^{\frac{i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=3 / L=8 / score ≈ 31.448 — proved E minimum

PROJECTIVE E=3 / L=8 / score ≈ 30.648 — proved E minimum

$U = (-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=3 / L=16 / score ≈ 48.438 — proved E minimum

$U = e^{-\frac{7i\pi}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=3 / L=8 / score ≈ 31.254 — proved E minimum

Exact definitions: η_{90} in the expression glossary.

PROJECTIVE E=3 / L=6 / score ≈ 26.042 — proved E minimum

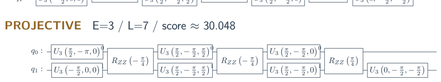
Exact definitions: η_{90} in the expression glossary.

$U = e^{\frac{3i\pi}{8}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=3 / L=9 / score ≈ 35.260 — proved E minimum

PROJECTIVE E=3 / L=7 / score ≈ 30.048 — proved E minimum



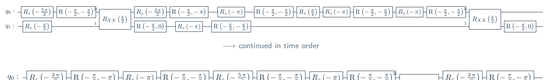
$U = e^{-\frac{3i\pi}{8}}\, C$. Scalar metadata; no physical GP.

SSWAPdg

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

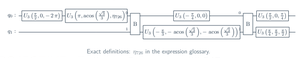
PROJECTIVE E=3 / L=30 / score ≈ 79.696 — proved E minimum



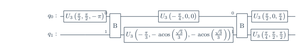
$U = \frac{\sqrt[8]{-1}\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=6 / score ≈ 22.553 — proved E minimum



PROJECTIVE E=2 / L=5 / score ≈ 20.547 — proved E minimum



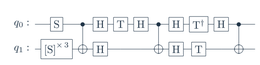
$U = \eta_{727}\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

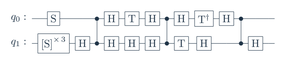
clifford_t_cx

EXACT PHASE E=3 / L=13 / score ≈ 38.000 — proved E minimum



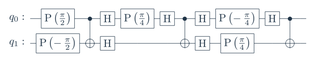
clifford_t_cz

EXACT PHASE E=3 / L=17 / score ≈ 46.000 — proved E minimum



cx_h_phase

EXACT PHASE E=3 / L=11 / score ≈ 35.610 — proved E minimum



cx_rz_ry

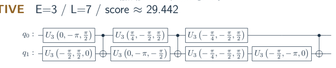
PROJECTIVE E=3 / L=14 / score ≈ 43.428 — proved E minimum

$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=3 / L=9 / score ≈ 33.454 — proved E minimum

PROJECTIVE E=3 / L=7 / score ≈ 29.442 — proved E minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

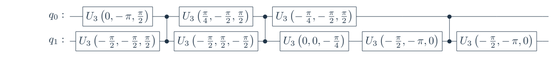
cz_rz_rx

PROJECTIVE E=3 / L=11 / score ≈ 36.822 — proved E minimum

$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

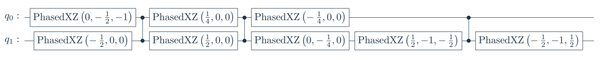
cz_u

EXACT PHASE E=3 / L=8 / score ≈ 31.848 — proved E minimum



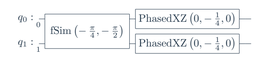
google_cz

EXACT PHASE E=3 / L=8 / score ≈ 30.648 — proved E minimum



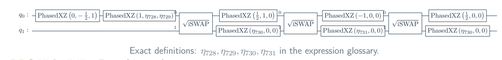
google_fsim

EXACT PHASE E=1 / L=2 / score ≈ 9.416 — proved E minimum



google_sqrt_iswap

EXACT PHASE E=3 / L=8 / score ≈ 31.699 — proved E minimum



Exact definitions: $\eta_{728}, \eta_{729}, \eta_{730}, \eta_{731}$ in the expression glossary.

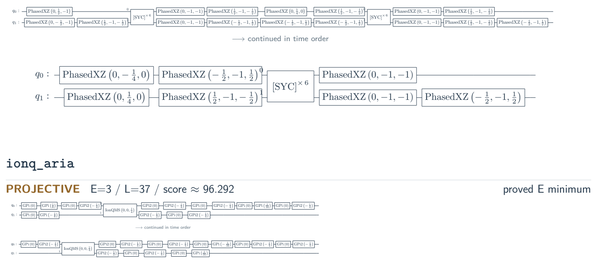
PROJECTIVE E=3 / L=7 / score ≈ 27.833 — proved E minimum

Exact definitions: η_{730}, η_{731} in the expression glossary.

$U = \eta_{732}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=18 / L=23 / score ≈ 124.538 — E upper bound



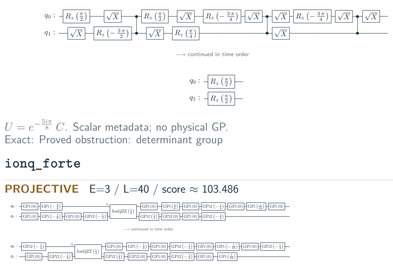
ibm_cx

PROJECTIVE E=3 / L=12 / score ≈ 37.614 — proved E minimum

$U = e^{-\frac{3i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

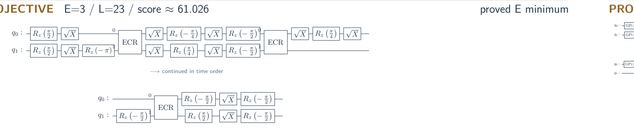
PROJECTIVE E=3 / L=17 / score ≈ 48.216 — proved E minimum



$U = e^{-\frac{5i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=3 / L=23 / score ≈ 61.026 — proved E minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

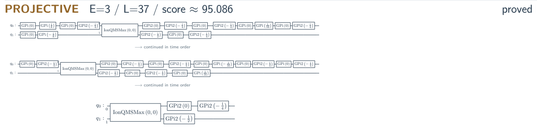
ionq_aria

PROJECTIVE E=3 / L=37 / score ≈ 96.292 — proved E minimum

$U = \frac{\sqrt[8]{-1}\sqrt{2}(-1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=3 / L=37 / score ≈ 95.086 — proved E minimum



$U = \frac{\sqrt[8]{-1}\sqrt{2}(-1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=3 / L=40 / score ≈ 103.486 — proved E minimum

$U = \frac{\sqrt[8]{-1}\sqrt{2}(-1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=3 / L=13 / score ≈ 41.652 — proved E minimum

$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=3 / L=9 / score ≈ 33.654 — proved E minimum

Exact definitions: η_{122} in the expression glossary.

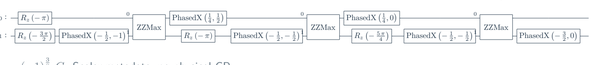
PROJECTIVE E=3 / L=7 / score ≈ 28.842 — proved E minimum

Exact definitions: η_{222} in the expression glossary.

$U = \frac{\sqrt{2}i(1+i)}{2}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

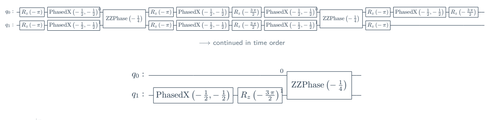
PROJECTIVE E=3 / L=10 / score ≈ 34.832 — proved E minimum



$U = -(-1)^{\frac{3}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

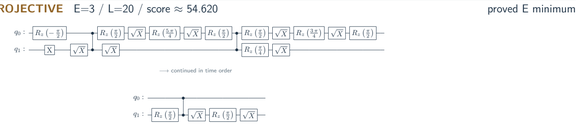
PROJECTIVE E=3 / L=18 / score ≈ 53.258 — proved E minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=3 / L=20 / score ≈ 54.620 — proved E minimum



$U = (-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=3 / L=8 / score ≈ 32.648 — proved E minimum

Exact definitions: η_{731}, η_{734} in the expression glossary.

PROJECTIVE E=3 / L=7 / score ≈ 29.642 — proved E minimum

Exact definitions: η_{733}, η_{734} in the expression glossary.

$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=3 / L=0 / score ≈ 12.000 — proved E minimum

tket_tk2

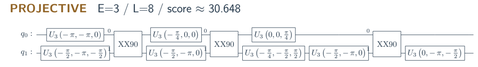
PROJECTIVE E=1 / L=0 / score ≈ 5.206 — proved E minimum

$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=3 / L=8 / score ≈ 31.448 — proved E minimum

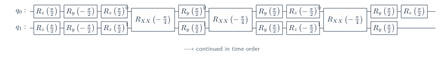
PROJECTIVE E=3 / L=8 / score ≈ 30.648 — proved E minimum



$U = \sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

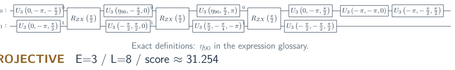
PROJECTIVE E=3 / L=16 / score ≈ 48.438 — proved E minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=3 / L=10 / score ≈ 35.666 — proved E minimum



Exact definitions: η_{93} in the expression glossary.

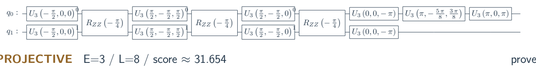
PROJECTIVE E=3 / L=8 / score ≈ 31.254 — proved E minimum

Exact definitions: η_{94} in the expression glossary.

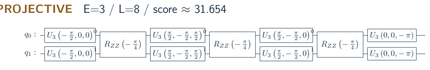
$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=3 / L=10 / score ≈ 36.866 — proved E minimum



PROJECTIVE E=3 / L=8 / score ≈ 31.654 — proved E minimum



$U = e^{-\frac{5i\pi}{8}}\,C$. Scalar metadata; no physical GP.

SWAP

2 qubits / 32 preserved native bases / alphabetical panel order

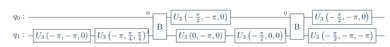
aqt_rxx

PROJECTIVE E=3 / L=15 / score ≈ 45.654 — proved E minimum

$U = \frac{\sqrt{2}i(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=7 / score ≈ 23.642 — proved E minimum



PROJECTIVE E=2 / L=6 / score ≈ 20.836 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

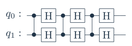
EXACT PHASE E=3 / L=0 / score ≈ 12.000 — proved E minimum

clifford_t_cx

EXACT PHASE E=3 / L=0 / score ≈ 12.000 — proved E minimum

clifford_t_cz

EXACT PHASE E=3 / L=6 / score ≈ 24.000 — proved E minimum



cx_h_phase

EXACT PHASE E=3 / L=0 / score ≈ 12.000 — proved E minimum

cx_rz_ry

EXACT PHASE E=3 / L=0 / score ≈ 12.000 — proved E minimum

cx_u

EXACT PHASE E=3 / L=0 / score ≈ 12.000 — proved E minimum

cz_rz_rx

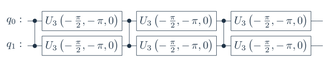
EXACT PHASE E=3 / L=7 / score ≈ 27.214 — proved E minimum

PROJECTIVE E=3 / L=6 / score ≈ 25.212 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

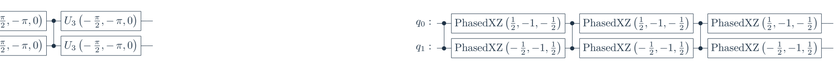
cz_u

EXACT PHASE E=3 / L=6 / score ≈ 25.236 — proved E minimum



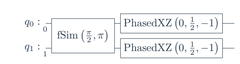
google_cz

EXACT PHASE E=3 / L=6 / score ≈ 26.436 — proved E minimum



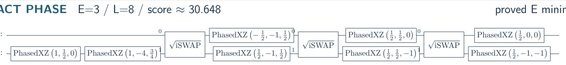
google_fsim

EXACT PHASE E=1 / L=2 / score ≈ 8.616 — proved E minimum

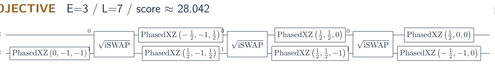


google_sqrt_iswap

EXACT PHASE E=3 / L=8 / score ≈ 30.648 — proved E minimum



PROJECTIVE E=3 / L=7 / score ≈ 28.042 — proved E minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

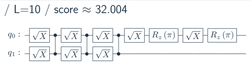
EXACT PHASE E=18 / L=12 / score ≈ 98.472 — E upper bound

ibm_cx

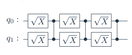
EXACT PHASE E=3 / L=0 / score ≈ 12.000 — proved E minimum

ibm_cz

EXACT PHASE E=3 / L=10 / score ≈ 32.004 — proved E minimum



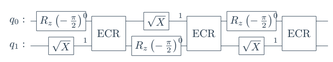
PROJECTIVE E=3 / L=6 / score ≈ 24.000 — proved E minimum



$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=3 / L=6 / score ≈ 24.606 — proved E minimum



ionq_aria

PROJECTIVE E=3 / L=15 / score ≈ 46.248 — proved E minimum

$U = \frac{\sqrt{2}(1+i)^3}{4}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

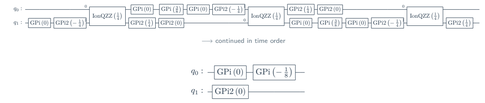
ionq_aria_full

PROJECTIVE E=3 / L=15 / score ≈ 45.042 — proved E minimum

$U = \frac{\sqrt{2}(1+i)^3}{4}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

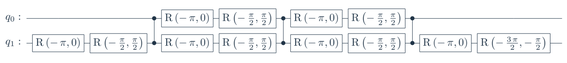
PROJECTIVE E=3 / L=18 / score ≈ 53.442 — proved E minimum



$U = \frac{\sqrt{2}(1+i)^3}{4}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

EXACT PHASE E=3 / L=12 / score ≈ 38.448 — proved E minimum



iswap_u

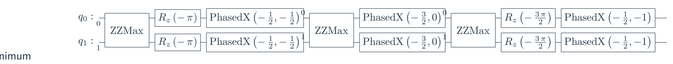
EXACT PHASE E=3 / L=5 / score ≈ 23.430 — proved E minimum

PROJECTIVE E=3 / L=4 / score ≈ 20.624 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

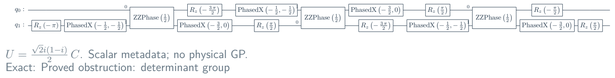
PROJECTIVE E=3 / L=10 / score ≈ 34.032 — proved E minimum



$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

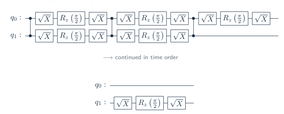
PROJECTIVE E=3 / L=13 / score ≈ 41.644 — proved E minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

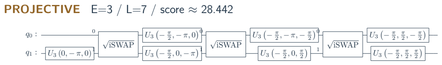
EXACT PHASE E=3 / L=18 / score ≈ 49.212 — proved E minimum



sqrt_iswap_u

EXACT PHASE E=3 / L=8 / score ≈ 31.048 — proved E minimum

PROJECTIVE E=3 / L=7 / score ≈ 28.442 — proved E minimum



$U = (-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=2 / L=0 / score ≈ 8.000 — proved E minimum

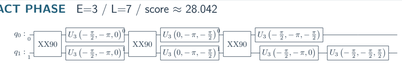
tket_tk2

PROJECTIVE E=1 / L=0 / score ≈ 4.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=3 / L=7 / score ≈ 28.042 — proved E minimum



PROJECTIVE E=3 / L=6 / score ≈ 25.636 — proved E minimum

$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=3 / L=14 / score ≈ 43.434 — proved E minimum

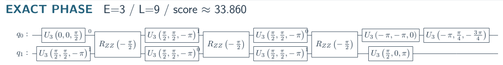
$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

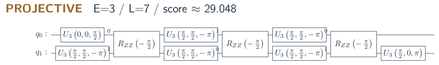
EXACT PHASE E=3 / L=6 / score ≈ 27.042 — proved E minimum

zz_u

EXACT PHASE E=3 / L=9 / score ≈ 33.860 — proved E minimum



PROJECTIVE E=3 / L=7 / score ≈ 29.048 — proved E minimum



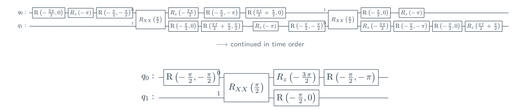
$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SWAPPow $(t, global_{shift})$

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

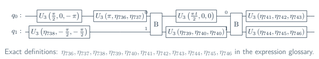
PROJECTIVE E=3 / L=20 / score ≈ 56.872 — minimum uniform E budget



$U = \eta_{735}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

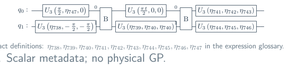
berkeley_b

EXACT PHASE E=2 / L=7 / score ≈ 23.442 — minimum uniform E budget



Exact definitions: $\eta_{736}, \eta_{737}, \eta_{738}, \eta_{739}, \eta_{740}, \eta_{741}, \eta_{742}, \eta_{743}, \eta_{744}, \eta_{745}, \eta_{746}$ in the expression glossary.

PROJECTIVE E=2 / L=6 / score ≈ 21.436 — minimum uniform E budget



Exact definitions: $\eta_{738}, \eta_{739}, \eta_{740}, \eta_{741}, \eta_{742}, \eta_{743}, \eta_{744}, \eta_{745}, \eta_{746}, \eta_{747}$ in the expression glossary.

$U = \eta_{748}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

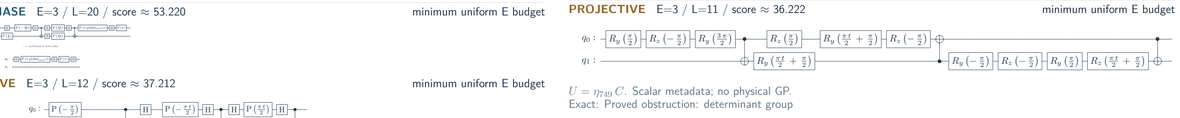
EXACT PHASE E=3 / L=20 / score ≈ 53.220 — minimum uniform E budget

PROJECTIVE E=3 / L=12 / score ≈ 37.212 — minimum uniform E budget

$U = e^{i\pi \mathrm{global}_{\mathrm{shift}} t}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

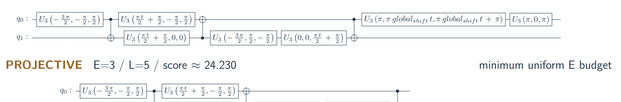
PROJECTIVE E=3 / L=11 / score ≈ 36.222 — minimum uniform E budget



$U = \eta_{749}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=3 / L=7 / score ≈ 28.242 — minimum uniform E budget

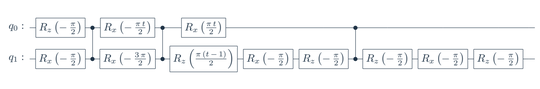


PROJECTIVE E=3 / L=5 / score ≈ 24.230 — minimum uniform E budget

$U = e^{i\pi \mathrm{global}_{\mathrm{shift}} t}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

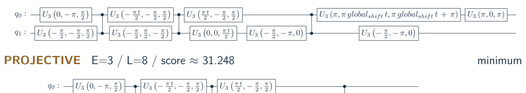
PROJECTIVE E=3 / L=11 / score ≈ 36.222 — minimum uniform E budget



$U = \eta_{431}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=3 / L=10 / score ≈ 35.260 — minimum uniform E budget



PROJECTIVE E=3 / L=8 / score ≈ 31.248 — minimum uniform E budget

$U = e^{i\pi \mathrm{global}_{\mathrm{shift}} t}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=3 / L=10 / score ≈ 34.060 — minimum uniform E budget

PROJECTIVE E=3 / L=8 / score ≈ 30.048 — minimum uniform E budget

$U = e^{i\pi \mathrm{global}_{\mathrm{shift}} t}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 12.628 — minimum uniform E budget

PROJECTIVE E=1 / L=2 / score ≈ 8.616 — minimum uniform E budget

$U = e^{i\pi \mathrm{global}_{\mathrm{shift}} t}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=6 / L=19 / score ≈ 67.514 — E upper bound

PROJECTIVE E=6 / L=17 / score ≈ 62.902 — E upper bound

$U = \eta_{751}\, C$. Scalar metadata; no physical GP.
Published uniform cap: $E \leq 3$. Current construction exceeds it; optimization remains.

google_syc

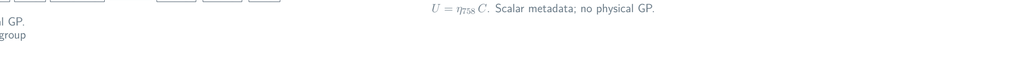
EXACT PHASE E=18 / L=19 / score ≈ 113.914 — E upper bound

PROJECTIVE E=18 / L=17 / score ≈ 109.702 — E upper bound

$U = \eta_{753}\, C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=3 / L=13 / score ≈ 39.012 — minimum uniform E budget



$U = \eta_{749}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

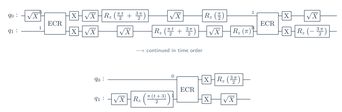
ibm_cz

PROJECTIVE E=3 / L=20 / score ≈ 54.020 — minimum uniform E budget

$U = -e^{\frac{i\pi\left(4\mathrm{global}_{\mathrm{shift}}+1\right)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

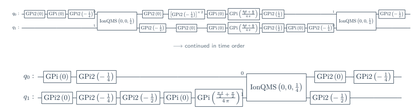
PROJECTIVE E=3 / L=22 / score ≈ 57.214 — minimum uniform E budget



$U = \eta_{754}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

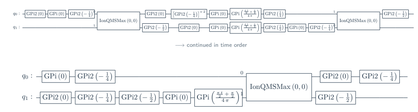
PROJECTIVE E=3 / L=27 / score ≈ 73.272 — minimum uniform E budget



$U = \eta_{755}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

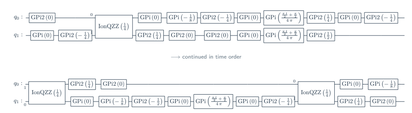
PROJECTIVE E=3 / L=27 / score ≈ 72.066 — minimum uniform E budget



$U = \eta_{755}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

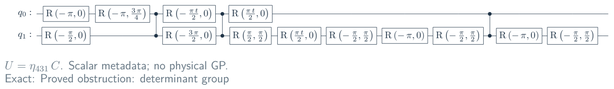
PROJECTIVE E=3 / L=30 / score ≈ 80.466 — minimum uniform E budget



$U = \eta_{755}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=3 / L=13 / score ≈ 41.052 — minimum uniform E budget

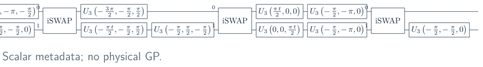


$U = \eta_{431}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=3 / L=13 / score ≈ 41.878 — minimum uniform E budget

PROJECTIVE E=3 / L=10 / score ≈ 35.860 — minimum uniform E budget



$U = \eta_{756}\, C$. Scalar metadata; no physical GP.

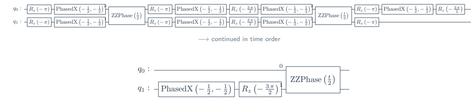
quantinuum_zzmax

PROJECTIVE E=3 / L=10 / score ≈ 34.232 — minimum uniform E budget

$U = -ie^{\frac{i\pi\left(4\mathrm{global}_{\mathrm{shift}}+1\right)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

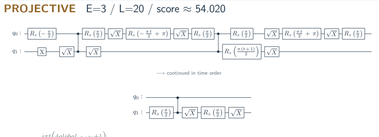
PROJECTIVE E=3 / L=18 / score ≈ 52.658 — minimum uniform E budget



$U = e^{\frac{i\pi\left(4\mathrm{global}_{\mathrm{shift}}+1\right)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=3 / L=20 / score ≈ 54.020 — minimum uniform E budget



$U = -e^{\frac{i\pi\left(4\mathrm{global}_{\mathrm{shift}}+1\right)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=6 / L=19 / score ≈ 67.914 — E upper bound

PROJECTIVE E=6 / L=17 / score ≈ 63.902 — E upper bound

$U = \eta_{757}\, C$. Scalar metadata; no physical GP.
Published uniform cap: $E \leq 3$. Current construction exceeds it; optimization remains.

sqrt_swap_u

EXACT PHASE E=6 / L=19 / score ≈ 68.714 — E upper bound

PROJECTIVE E=6 / L=17 / score ≈ 63.902 — E upper bound

$U = -\sqrt[4]{-1}e^{i\pi \mathrm{global}_{\mathrm{shift}} t}\, C$. Scalar metadata; no physical GP.

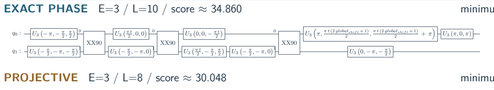
tket_tk2

PROJECTIVE E=1 / L=0 / score ≈ 4.606 — minimum uniform E budget

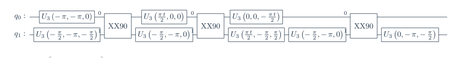
$U = e^{\frac{i\pi\left(4\mathrm{global}_{\mathrm{shift}}+1\right)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=3 / L=10 / score ≈ 34.860 — minimum uniform E budget



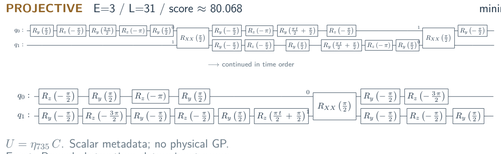
PROJECTIVE E=3 / L=8 / score ≈ 30.048 — minimum uniform E budget



$U = ie^{\frac{i\pi\left(2\mathrm{global}_{\mathrm{shift}}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

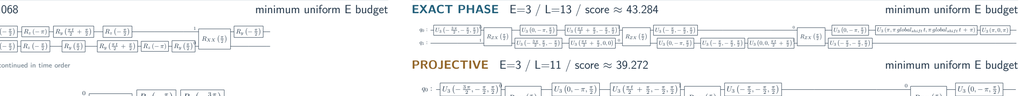
PROJECTIVE E=3 / L=31 / score ≈ 80.068 — minimum uniform E budget



$U = \eta_{735}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=3 / L=13 / score ≈ 43.284 — minimum uniform E budget

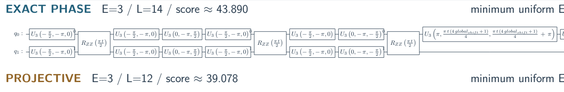


PROJECTIVE E=3 / L=11 / score ≈ 39.272 — minimum uniform E budget

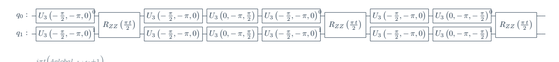
$U = \eta_{758}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=3 / L=14 / score ≈ 43.890 — minimum uniform E budget



PROJECTIVE E=3 / L=12 / score ≈ 39.078 — minimum uniform E budget



$U = e^{\frac{i\pi\left(4\mathrm{global}_{\mathrm{shift}}+1\right)}{4}}\, C$. Scalar metadata; no physical GP.

SX

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

clifford_only

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved E minimum

clifford_t_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved E minimum

cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.204 proved E minimum

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved E minimum

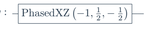
$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

cz_rz_rx

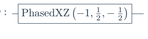
PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved E minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

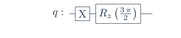
EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum



google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved E minimum

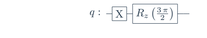


$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved E minimum



$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

Sx

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=3 / score ≈ 6.000 — proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.000 — proved E minimum

clifford_t_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.000 — proved E minimum

cx_h_phase

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_sqrt_iswap

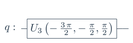
EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

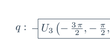
ibm_cx

EXACT PHASE E=0 / L=1 / score ≈ 2.000 — proved E minimum



ibm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.000 — proved E minimum



ibm_ecr

EXACT PHASE E=0 / L=1 / score ≈ 2.000 — proved E minimum

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{1i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{1i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{1i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.000 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SX_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

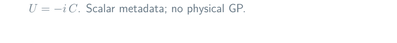
EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

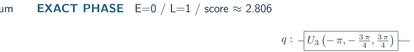
$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.806 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.204 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Sx_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.406 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

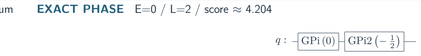
cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

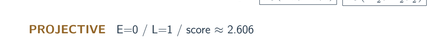
iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

SX_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

clifford_only

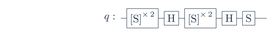
EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.206 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.204 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

Sx_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 15.008 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

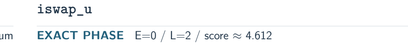
PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum



$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.004 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.004 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.004 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

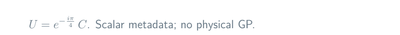
PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.004 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SX_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.806 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

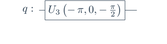
cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

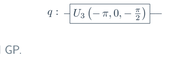


PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

ionq_aria

EXACT PHASE E=0 / L=1 / score ≈ 2.602 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=1 / score ≈ 2.602 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=1 / score ≈ 2.602 — proved E minimum

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

Sx_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.204 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

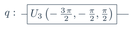
PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.408 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

cz_rz_rx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.206 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.206 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.206 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.204 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.204 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.204 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.204 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

xx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

SX_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.206 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.204 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

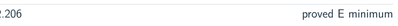
cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

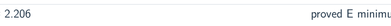


$U = -i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum



$U = -i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

Sx_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=13 / score ≈ 26.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 15.008 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

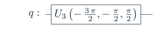
PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

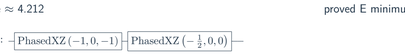
cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_fsim

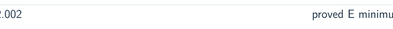
EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SX_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

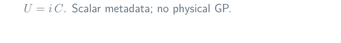
EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.806 — proved E minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = i\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = i\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = i\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = i\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = i\, C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = i\, C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = i\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

Sx_q5

1 qubits / 32 preserved native bases / alphabetical panel order

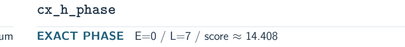
aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.408 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

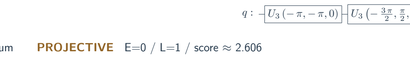
PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum



$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

SX_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

zz_u

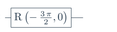
EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

Sx_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

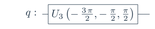


$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum



$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -i\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -i\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -i\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.206 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = -i\, C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.004 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = -i\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.004 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = -i\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.004 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = -i\, C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.004 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = -i\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

SX_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

ionq_aria

EXACT PHASE E=0 / L=1 / score ≈ 2.602 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=1 / score ≈ 2.602 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=1 / score ≈ 2.602 — proved E minimum

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Sx_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.204 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

cz_rz_rx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.204 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.204 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.204 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.204 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

xx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

Sxdg

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

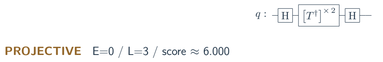
EXACT PHASE E=0 / L=5 / score ≈ 10.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.000 — proved E minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

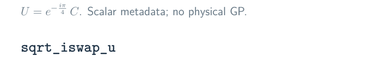
EXACT PHASE E=0 / L=4 / score ≈ 8.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=3 / score ≈ 6.202 — proved E minimum



cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.000 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.000 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.000 — proved E minimum

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.000 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Sxdg_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.204 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

clifford_only

EXACT PHASE E=0 / L=3 / score ≈ 6.000 — proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.000 — proved E minimum

clifford_t_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.000 — proved E minimum

cx_h_phase

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

cz_rz_rx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.204 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.204 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.204 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.204 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

xx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

Sxdg_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

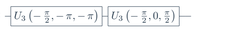
EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

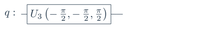
$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.206 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.004 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.004 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.004 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.004 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Sxdg_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

clifford_t_cx

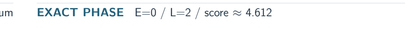
EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.408 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

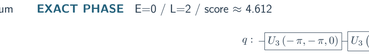
cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

Sxdg_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 15.008 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_syc

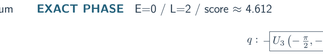
EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved E minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

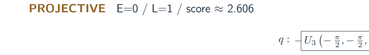
iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved E minimum

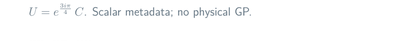
PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Sxdg_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.204 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

clifford_only

EXACT PHASE E=0 / L=13 / score ≈ 26.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

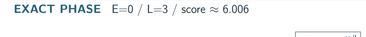
EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.408 — proved E minimum



PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

cz_rz_rx

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_cz

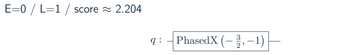
EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.006 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.006 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.006 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.204 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.204 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.204 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.204 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

xx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

Sxdg_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=13 / score ≈ 26.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 15.008 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.004 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.004 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.004 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.004 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Sxdg_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.406 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.202 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

cz_rz_rx

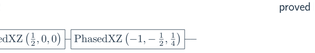
PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum



$U = -iC$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

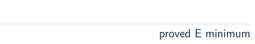
EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.004 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.004 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.004 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

SxY

1 qubits / 32 preserved native bases / alphabetical panel order

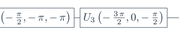
aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum



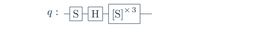
PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

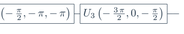
PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.206 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SxY_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.206 proved E minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.406 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved E minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved E minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

google_cz

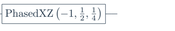
EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved E minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

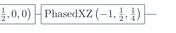


PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved E minimum

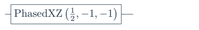
$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved E minimum



$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

google_syc

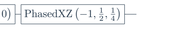
EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved E minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved E minimum

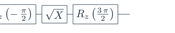


PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved E minimum

$U = e^{\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved E minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved E minimum

$U = e^{\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved E minimum

$U = e^{\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

ionq_aria

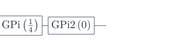
EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved E minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.408 proved E minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.206 proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.206 proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved E minimum

$U = e^{\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.206 proved E minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

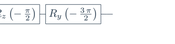
xx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved E minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum



$U = -iC$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

SxY_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

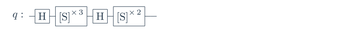
EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 proved E minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.204 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

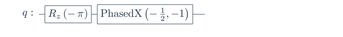
PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.002 proved E minimum

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.002 proved E minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SxY_q3

1 qubits / 32 preserved native bases / alphabetical panel order

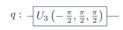
aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum



clifford_only

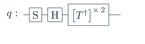
EXACT PHASE E=0 / L=5 / score ≈ 10.000 — proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.000 — proved E minimum

clifford_t_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.000 — proved E minimum



cx_h_phase

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

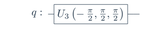
cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum



cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

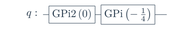
EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

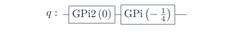


PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

SxY_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

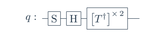
clifford_only

EXACT PHASE E=0 / L=5 / score ≈ 10.000 — proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SxY_q5

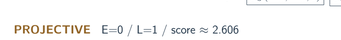
1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

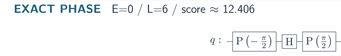
EXACT PHASE E=0 / L=2 / score ≈ 4.206 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

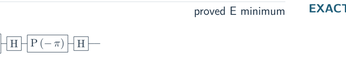


$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum



$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.406 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

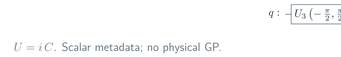
$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.406 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.406 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.406 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.408 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.206 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.206 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.206 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

SxY_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.206 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved E minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SxY_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.408 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{\frac{11i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{\frac{11i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{\frac{11i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{\frac{11i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

SxZ

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.204 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

google_sqrt_iswap

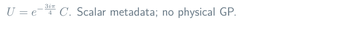
EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.002 proved E minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.002 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.002 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.002 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.206 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SxZ_q1

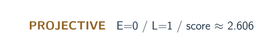
1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

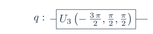
EXACT PHASE E=0 / L=2 / score ≈ 4.206 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum



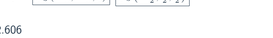
PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum



$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.406 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.406 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.406 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.406 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.408 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.206 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.206 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.206 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 — proved E minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

SxZ_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.206 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved E minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SxZ_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

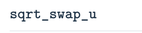
EXACT PHASE E=0 / L=7 / score ≈ 14.408 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum



$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

SxZ_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.206 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

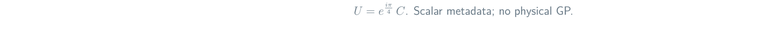
cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

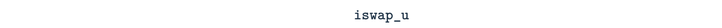
PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum



$U = -i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SxZ_q5

1 qubits / 32 preserved native bases / alphabetical panel order

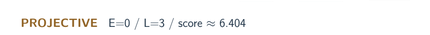
aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.206 proved E minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum



$U = iC$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_t_cx

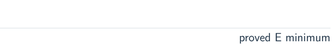
EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 proved E minimum

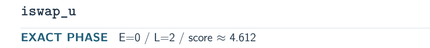


PROJECTIVE E=0 / L=4 / score ≈ 8.000 proved E minimum

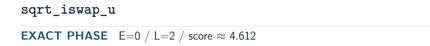
$U = -iC$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.406 proved E minimum



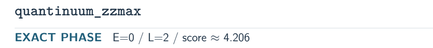
PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved E minimum



$U = -iC$. Scalar metadata; no physical GP.

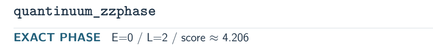
cx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved E minimum

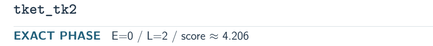


cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum



$U = iC$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.204 proved E minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = iC$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved E minimum

$U = e^{-i\frac{\pi}{4}}C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved E minimum

$U = e^{-i\frac{\pi}{4}}C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved E minimum

$U = e^{-i\frac{\pi}{4}}C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved E minimum

$U = e^{-i\frac{\pi}{4}}C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved E minimum

$U = e^{-i\frac{\pi}{4}}C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved E minimum

$U = e^{-i\frac{\pi}{4}}C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved E minimum

$U = e^{-i\frac{\pi}{4}}C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = iC$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.408 proved E minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = iC$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.206 proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.206 proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved E minimum

$U = e^{-i\frac{\pi}{4}}C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = iC$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = iC$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.206 proved E minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = iC$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=3 / score ≈ 6.606 proved E minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = iC$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 proved E minimum

$U = iC$. Scalar metadata; no physical GP.

SxZ_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved E minimum

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.606 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SxZ_q7

1 qubits / 32 preserved native bases / alphabetical panel order

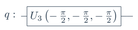
aqt_rxx

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

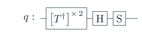


clifford_only

EXACT PHASE E=0 / L=5 / score ≈ 10.000 — proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.000 — proved E minimum



clifford_t_cz

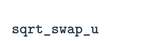
EXACT PHASE E=0 / L=4 / score ≈ 8.000 — proved E minimum

cx_h_phase

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.408 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

tket_tk2

PROJECTIVE E=0 / L=2 / score ≈ 4.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

SY

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved E minimum

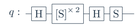
$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

clifford_only

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved E minimum

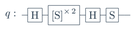


clifford_t_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved E minimum

clifford_t_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.000 proved E minimum



cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.204 proved E minimum

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved E minimum

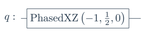
$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum



google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.602 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 proved E minimum

Sy

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

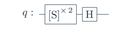
PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

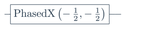
PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.404 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SY_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.806 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.204 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SY_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = i\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = i\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = i\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.206 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.204 — proved E minimum

$U = -1\, C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = -i\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

SY_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=11 / score ≈ 22.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.806 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}} C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

cz_rz_rx

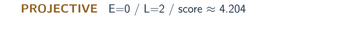
PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

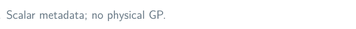
PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum



$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

ionq_aria

EXACT PHASE E=0 / L=1 / score ≈ 2.602 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=1 / score ≈ 2.602 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=1 / score ≈ 2.602 — proved E minimum

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}} C$. Scalar metadata; no physical GP.

SY_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

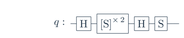
PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum



$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.206 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.204 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

SY_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.806 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

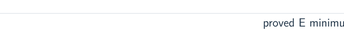
cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

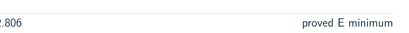


$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

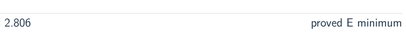


$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.204 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.404 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.204 — proved E minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SY_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.204 — proved E minimum

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.602 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.404 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.202 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

SY_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

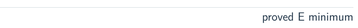
PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.000 — proved E minimum

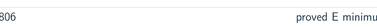


$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.404 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.204 — proved E minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

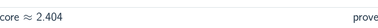
PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

ionq_aria

EXACT PHASE E=0 / L=1 / score ≈ 2.602 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=1 / score ≈ 2.602 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=1 / score ≈ 2.602 — proved E minimum

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.202 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.204 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

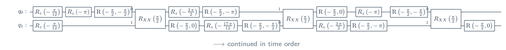
$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

SYC

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=3 / L=17 / score ≈ 51.809 — proved E minimum



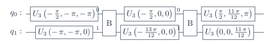
$U = \eta_{759}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=7 / score ≈ 24.076 — proved E minimum

Exact definitions: $\eta_{760}, \eta_{761}, \eta_{762}$ in the expression glossary.

PROJECTIVE E=2 / L=6 / score ≈ 22.787 — proved E minimum



$U = e^{\frac{i\pi}{24}}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

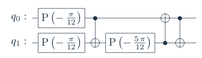
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: outside clifford t number field

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: outside clifford t number field

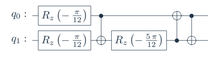
cx_h_phase

EXACT PHASE E=3 / L=3 / score ≈ 20.157 — proved E minimum



cx_rz_ry

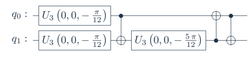
PROJECTIVE E=3 / L=3 / score ≈ 20.157 — proved E minimum



$U = e^{-\frac{7i\pi}{24}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=3 / L=3 / score ≈ 20.169 — proved E minimum



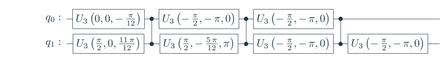
cz_rz_rx

PROJECTIVE E=3 / L=12 / score ≈ 39.975 — proved E minimum

$U = (-1)^{\frac{17}{24}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

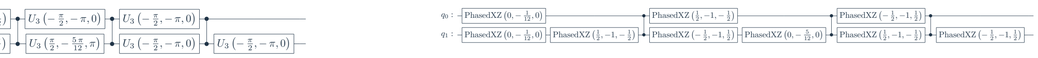
cz_u

EXACT PHASE E=3 / L=7 / score ≈ 29.393 — proved E minimum



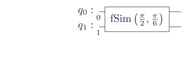
google_cz

EXACT PHASE E=3 / L=9 / score ≈ 34.605 — proved E minimum



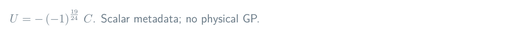
google_fsim

EXACT PHASE E=1 / L=0 / score ≈ 4.721 — proved E minimum



google_sqrt_iswap

EXACT PHASE E=3 / L=9 / score ≈ 39.025 — proved E minimum



Exact definitions: $\eta_{763}, \eta_{764}, \eta_{765}, \eta_{766}, \eta_{767}, \eta_{768}, \eta_{769}, \eta_{770}, \eta_{771}, \eta_{772}$ in the expression glossary.

PROJECTIVE E=3 / L=8 / score ≈ 34.924 — proved E minimum

Exact definitions: $\eta_{765}, \eta_{766}, \eta_{767}, \eta_{768}, \eta_{769}, \eta_{770}, \eta_{771}, \eta_{772}$ in the expression glossary.
$U = \eta_{773}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

ibm_cx

PROJECTIVE E=3 / L=3 / score ≈ 20.157 — proved E minimum

$U = e^{-\frac{7i\pi}{24}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=3 / L=14 / score ≈ 42.763 — proved E minimum

$U = e^{-\frac{19i\pi}{24}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=3 / L=15 / score ≈ 44.763 — proved E minimum

$U = (-1)^{\frac{17}{24}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=3 / L=25 / score ≈ 70.219 — proved E minimum

$U = \eta_{759}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=3 / L=21 / score ≈ 60.405 — proved E minimum

$U = \eta_{774}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=3 / L=28 / score ≈ 77.413 — proved E minimum

$U = \eta_{759}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

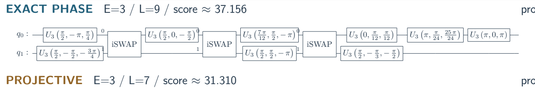
iqm_cz

PROJECTIVE E=3 / L=19 / score ≈ 56.627 — proved E minimum

$U = (-1)^{\frac{17}{24}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=3 / L=9 / score ≈ 37.156 — proved E minimum

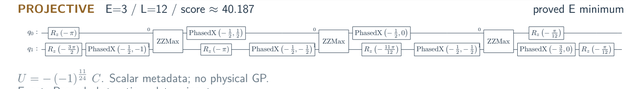


PROJECTIVE E=3 / L=7 / score ≈ 31.310 — proved E minimum

$U = e^{\frac{i\pi}{24}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=3 / L=12 / score ≈ 40.187 — proved E minimum



$U = -(-1)^{\frac{11}{24}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

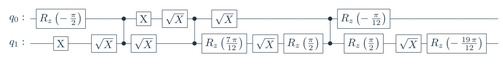
quantinuum_zzphase

PROJECTIVE E=3 / L=16 / score ≈ 49.801 — proved E minimum

$U = \eta_{759}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=3 / L=14 / score ≈ 42.763 — proved E minimum



$U = e^{-\frac{19i\pi}{24}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=3 / L=9 / score ≈ 36.322 — proved E minimum

Exact definitions: $\eta_{771}, \eta_{776}, \eta_{777}, \eta_{778}, \eta_{779}, \eta_{780}, \eta_{781}, \eta_{782}$ in the expression glossary.

PROJECTIVE E=3 / L=8 / score ≈ 32.682 — proved E minimum

Exact definitions: $\eta_{777}, \eta_{778}, \eta_{779}, \eta_{780}, \eta_{781}, \eta_{782}$ in the expression glossary.
$U = \eta_{783}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=6 / L=15 / score ≈ 60.641 — E upper bound

PROJECTIVE E=6 / L=15 / score ≈ 59.841 — E upper bound

$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

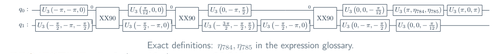
tket_tk2

PROJECTIVE E=1 / L=2 / score ≈ 10.561 — proved E minimum

$U = e^{-\frac{i\pi}{24}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

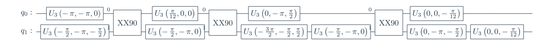
xx90_u

EXACT PHASE E=3 / L=12 / score ≈ 40.023 — proved E minimum



Exact definitions: η_{784}, η_{785} in the expression glossary.

PROJECTIVE E=3 / L=10 / score ≈ 36.011 — proved E minimum



$U = -(-1)^{\frac{19}{24}}\, C$. Scalar metadata; no physical GP.

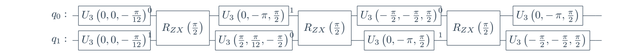
xx_rz_ry

PROJECTIVE E=3 / L=18 / score ≈ 52.993 — proved E minimum

$U = e^{\frac{11i\pi}{24}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

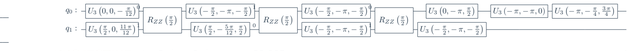
zx_u

EXACT PHASE E=3 / L=8 / score ≈ 33.005 — proved E minimum



zz_u

EXACT PHASE E=3 / L=10 / score ≈ 38.017 — proved E minimum



PROJECTIVE E=3 / L=8 / score ≈ 33.205 — proved E minimum

$U = \frac{\sqrt{2}(1+i)^3}{4}\, C$. Scalar metadata; no physical GP.

Sydg

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=9 / score ≈ 18.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

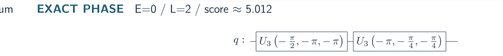
cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

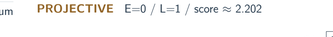
PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.404 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.404 — proved E minimum



$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.202 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 5.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.206 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

T

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved E minimum

$U = e^{\frac{i\pi}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved E minimum

clifford_t_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved E minimum

cx_h_phase

EXACT PHASE E=0 / L=1 / score ≈ 2.402 proved E minimum

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved E minimum

$U = e^{\frac{i\pi}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved E minimum

$U = e^{\frac{i\pi}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

google_fsim

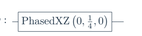
EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum



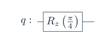
ibm_cx

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved E minimum

$U = e^{\frac{i\pi}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved E minimum



$U = e^{\frac{i\pi}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved E minimum

$U = e^{\frac{i\pi}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

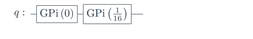
ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.804 proved E minimum

$U = e^{\frac{i\pi}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.804 proved E minimum



$U = e^{\frac{i\pi}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.804 proved E minimum

$U = e^{\frac{i\pi}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=3 / score ≈ 7.212 proved E minimum

$U = e^{\frac{i\pi}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

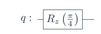
quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved E minimum

$U = e^{\frac{i\pi}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved E minimum



$U = e^{\frac{i\pi}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved E minimum

$U = e^{\frac{i\pi}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved E minimum

$U = e^{\frac{i\pi}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved E minimum

$U = e^{\frac{i\pi}{8}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

Tdg

1 qubits / 32 preserved native bases / alphabetical panel order

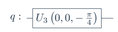
aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved E minimum

$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved E minimum

clifford_t_cz

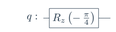
EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved E minimum

cx_h_phase

EXACT PHASE E=0 / L=1 / score ≈ 2.402 proved E minimum

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved E minimum



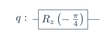
$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

cz_rz_rx

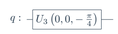
PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved E minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

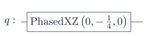
cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum



google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

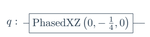


google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

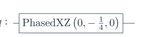
google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum



google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum



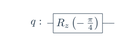
ibm_cx

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved E minimum

$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

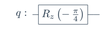
PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved E minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved E minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.804 proved E minimum

$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.804 proved E minimum

$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.804 proved E minimum

$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

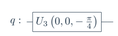
iqm_cz

PROJECTIVE E=0 / L=3 / score ≈ 7.212 proved E minimum

$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

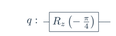
iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum



quantinuum_zzmax

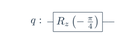
PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved E minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

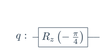
PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved E minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

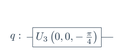
PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved E minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

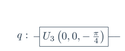
sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum



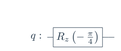
sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum



tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved E minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved E minimum

$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

zz_u

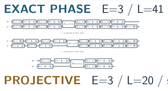
EXACT PHASE E=0 / L=1 / score ≈ 2.406 proved E minimum

TK2 (a, b, c)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=3 / L=41 / score ≈ 102.928 — minimum uniform E budget

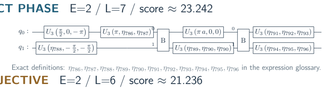


PROJECTIVE E=3 / L=20 / score ≈ 56.872 — minimum uniform E budget

$U = \eta_{104}\, C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=2 / L=7 / score ≈ 23.242 — minimum uniform E budget



Exact definitions: $\eta_{766}, \eta_{767}, \eta_{768}, \eta_{769}, \eta_{770}, \eta_{771}, \eta_{772}, \eta_{773}, \eta_{774}, \eta_{775}, \eta_{776}$ in the expression glossary.

PROJECTIVE E=2 / L=6 / score ≈ 21.236 — minimum uniform E budget

Exact definitions: $\eta_{748}, \eta_{749}, \eta_{750}, \eta_{751}, \eta_{752}, \eta_{753}, \eta_{754}, \eta_{755}, \eta_{756}, \eta_{757}$ in the expression glossary.

$U = \eta_{798}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=3 / L=20 / score ≈ 53.020 — minimum uniform E budget

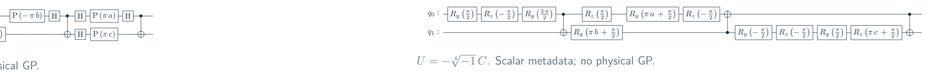
PROJECTIVE E=3 / L=12 / score ≈ 36.612 — minimum uniform E budget

$U = e^{-\frac{i\pi(a-b+c)}{2}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=4 / L=17 / score ≈ 52.834 — minimum uniform E budget

PROJECTIVE E=3 / L=11 / score ≈ 36.222 — minimum uniform E budget

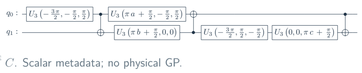


$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

cx_u

EXACT PHASE E=3 / L=7 / score ≈ 28.242 — minimum uniform E budget

PROJECTIVE E=3 / L=5 / score ≈ 24.230 — minimum uniform E budget

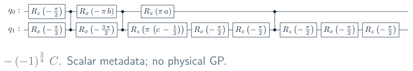


$U = e^{-\frac{i\pi c}{2}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=4 / L=28 / score ≈ 77.256 — minimum uniform E budget

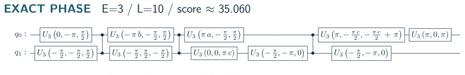
PROJECTIVE E=3 / L=11 / score ≈ 35.822 — minimum uniform E budget



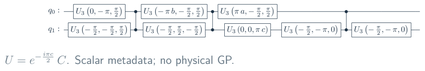
$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=3 / L=10 / score ≈ 35.060 — minimum uniform E budget



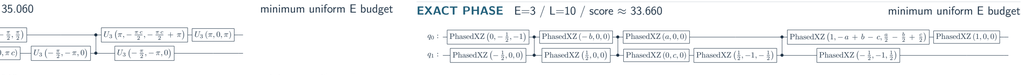
PROJECTIVE E=3 / L=8 / score ≈ 30.648 — minimum uniform E budget



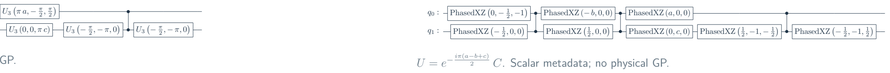
$U = e^{-\frac{i\pi c}{2}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=3 / L=10 / score ≈ 33.660 — minimum uniform E budget



PROJECTIVE E=3 / L=8 / score ≈ 29.448 — minimum uniform E budget

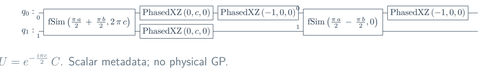


$U = e^{\frac{i\pi(a-b+c)}{2}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=6 / score ≈ 20.644 — E upper bound

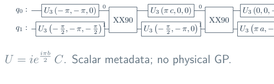
PROJECTIVE E=2 / L=4 / score ≈ 16.432 — E upper bound



$U = e^{-\frac{i\phi c}{2}}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=6 / L=19 / score ≈ 67.514 — E upper bound



PROJECTIVE E=6 / L=17 / score ≈ 62.902 — E upper bound

$U = \eta_{799}\, C$. Scalar metadata; no physical GP.
Published uniform cap: $E \leq 3$. Current construction exceeds it; optimization remains.

google_syc

EXACT PHASE E=18 / L=19 / score ≈ 113.914 — E upper bound

PROJECTIVE E=18 / L=17 / score ≈ 109.702 — E upper bound

$U = e^{\frac{i\pi(-a-b-c-1)}{2}}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=3 / L=16 / score ≈ 45.216 — minimum uniform E budget

PROJECTIVE E=3 / L=13 / score ≈ 39.012 — minimum uniform E budget

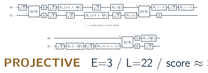
$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

ibm_cz

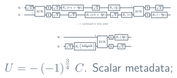
EXACT PHASE E=3 / L=20 / score ≈ 53.620 — minimum uniform E budget

ibm_ecr

EXACT PHASE E=3 / L=24 / score ≈ 61.018 — minimum uniform E budget



PROJECTIVE E=3 / L=22 / score ≈ 57.214 — minimum uniform E budget

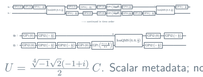


$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=3 / L=27 / score ≈ 73.472 — minimum uniform E budget

PROJECTIVE E=3 / L=27 / score ≈ 73.272 — minimum uniform E budget

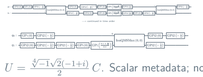


$U = \frac{\sqrt[4]{-1}\sqrt{2}(-1+i)}{2}\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=3 / L=27 / score ≈ 72.266 — minimum uniform E budget

PROJECTIVE E=3 / L=27 / score ≈ 72.066 — minimum uniform E budget



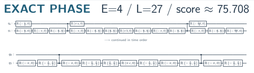
$U = \frac{\sqrt[4]{-1}\sqrt{2}(-1+i)}{2}\, C$. Scalar metadata; no physical GP.

ionq_forte

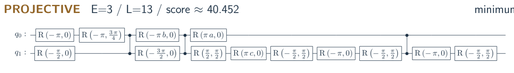
EXACT PHASE E=3 / L=30 / score ≈ 80.466 — minimum uniform E budget

iqm_cz

EXACT PHASE E=4 / L=27 / score ≈ 75.708 — minimum uniform E budget



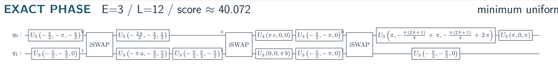
PROJECTIVE E=3 / L=13 / score ≈ 40.452 — minimum uniform E budget



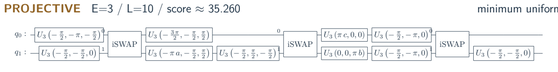
$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

iswap_u

EXACT PHASE E=3 / L=12 / score ≈ 40.072 — minimum uniform E budget



PROJECTIVE E=3 / L=10 / score ≈ 35.260 — minimum uniform E budget



$U = \eta_{800}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=4 / L=25 / score ≈ 70.876 — E upper bound

PROJECTIVE E=3 / L=10 / score ≈ 33.632 — minimum uniform E budget

$U = -i\, C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=3 / L=18 / score ≈ 52.058 — minimum uniform E budget

rigetti_cz

EXACT PHASE E=3 / L=20 / score ≈ 53.620 — minimum uniform E budget

sqrt_iswap_u

EXACT PHASE E=6 / L=19 / score ≈ 68.314 — E upper bound

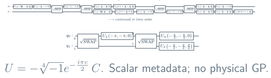
PROJECTIVE E=6 / L=17 / score ≈ 63.902 — E upper bound

$U = \eta_{801}\, C$. Scalar metadata; no physical GP.
Published uniform cap: $E \leq 3$. Current construction exceeds it; optimization remains.

sqrt_swap_u

EXACT PHASE E=6 / L=19 / score ≈ 69.114 — E upper bound

PROJECTIVE E=6 / L=17 / score ≈ 63.902 — E upper bound



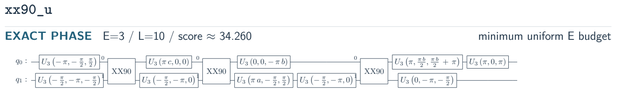
$U = -\sqrt[4]{-1}e^{-\frac{i\pi c}{2}}\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.006 — minimum uniform E budget

xx90_u

EXACT PHASE E=3 / L=10 / score ≈ 34.260 — minimum uniform E budget



PROJECTIVE E=3 / L=8 / score ≈ 29.448 — minimum uniform E budget

$U = ie^{\frac{i\pi b}{2}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

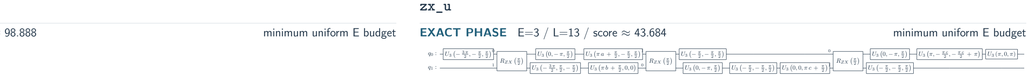
EXACT PHASE E=3 / L=41 / score ≈ 98.888 — minimum uniform E budget

PROJECTIVE E=3 / L=31 / score ≈ 80.068 — minimum uniform E budget

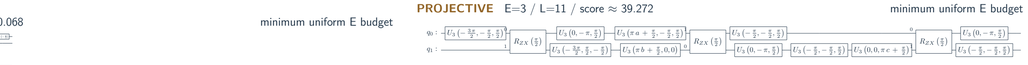
$U = \eta_{104}\, C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=3 / L=13 / score ≈ 43.684 — minimum uniform E budget



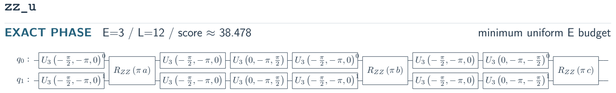
PROJECTIVE E=3 / L=11 / score ≈ 39.272 — minimum uniform E budget



$U = \eta_{802}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=3 / L=12 / score ≈ 38.478 — minimum uniform E budget

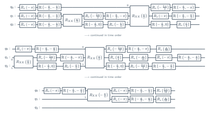


Toffoli

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

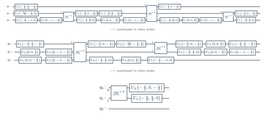
PROJECTIVE E=5 / L=40 / score ≈ 112.926 — E upper bound



$U = \sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=12 / L=33 / score ≈ 126.998 — E upper bound



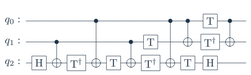
$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

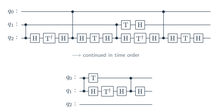
clifford_t_cx

EXACT PHASE E=6 / L=9 / score ≈ 42.000 — proved E minimum



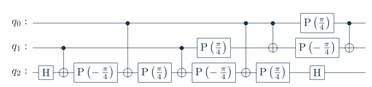
clifford_t_cz

EXACT PHASE E=6 / L=19 / score ≈ 62.000 — proved E minimum



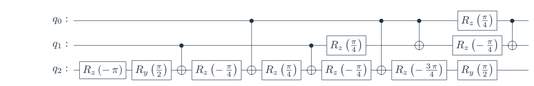
cx_h_phase

EXACT PHASE E=6 / L=9 / score ≈ 44.814 — proved E minimum



cx_rz_ry

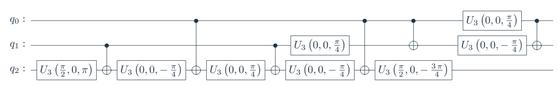
PROJECTIVE E=6 / L=10 / score ≈ 47.220 — proved E minimum



$U = -\sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

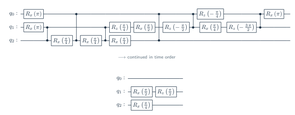
cx_u

EXACT PHASE E=6 / L=8 / score ≈ 43.248 — proved E minimum



cz_rz_rx

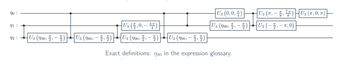
PROJECTIVE E=6 / L=15 / score ≈ 57.830 — proved E minimum



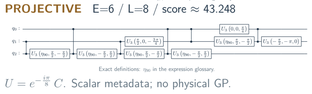
$U = e^{\frac{7i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=6 / L=10 / score ≈ 48.460 — proved E minimum



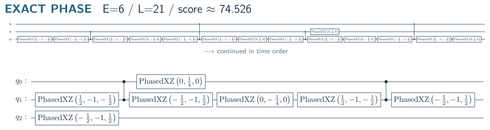
PROJECTIVE E=6 / L=8 / score ≈ 43.248 — proved E minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

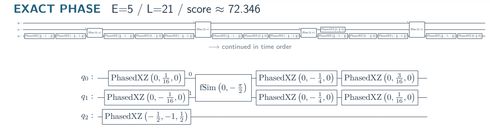
google_cz

EXACT PHASE E=6 / L=21 / score ≈ 74.526 — proved E minimum



google_fsim

EXACT PHASE E=5 / L=21 / score ≈ 72.346 — E upper bound

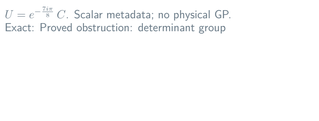


google_sqrt_iswap

EXACT PHASE E=12 / L=33 / score ≈ 124.998 — E upper bound

google_syc

EXACT PHASE E=36 / L=33 / score ≈ 218.598 — E upper bound



ibm_cx

PROJECTIVE E=6 / L=12 / score ≈ 51.218 — proved E minimum

$U = e^{-\frac{5i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

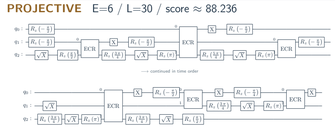
ibm_cz

PROJECTIVE E=6 / L=28 / score ≈ 83.426 — proved E minimum

$U = e^{\frac{5i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

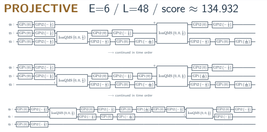
PROJECTIVE E=6 / L=30 / score ≈ 88.236 — proved E minimum



$U = e^{\frac{3i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

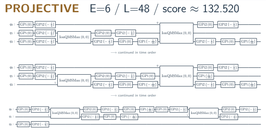
PROJECTIVE E=6 / L=48 / score ≈ 134.932 — E upper bound



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

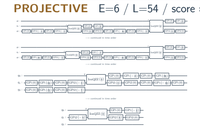
PROJECTIVE E=6 / L=48 / score ≈ 132.520 — E upper bound



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

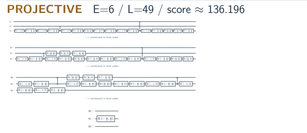
ionq_forte

PROJECTIVE E=6 / L=54 / score ≈ 149.320 — E upper bound



$U = (-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=6 / L=49 / score ≈ 136.196 — proved E minimum

$U = -\sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

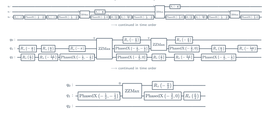
EXACT PHASE E=10 / L=18 / score ≈ 83.108 — E upper bound

PROJECTIVE E=10 / L=16 / score ≈ 79.096 — E upper bound

$U = (-1)^{\frac{3}{8}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

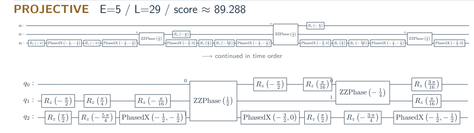
PROJECTIVE E=6 / L=35 / score ≈ 103.498 — proved E minimum



$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

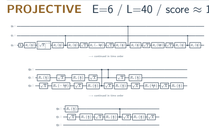
PROJECTIVE E=5 / L=29 / score ≈ 89.288 — E upper bound



$U = \sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

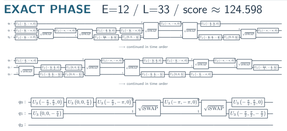
PROJECTIVE E=6 / L=40 / score ≈ 109.642 — proved E minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

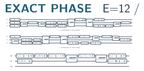
sqrt_iswap_u

EXACT PHASE E=12 / L=33 / score ≈ 124.598 — E upper bound



sqrt_swap_u

EXACT PHASE E=12 / L=33 / score ≈ 124.998 — E upper bound

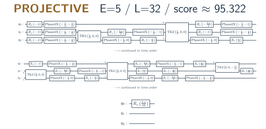


PROJECTIVE E=12 / L=33 / score ≈ 124.598 — E upper bound

$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

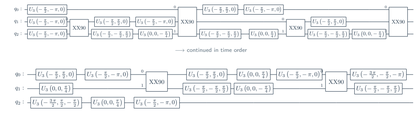
PROJECTIVE E=5 / L=32 / score ≈ 95.322 — E upper bound



$U = \sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

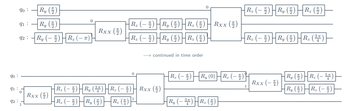
xx90_u

EXACT PHASE E=6 / L=27 / score ≈ 88.562 — proved E minimum



xx_rz_ry

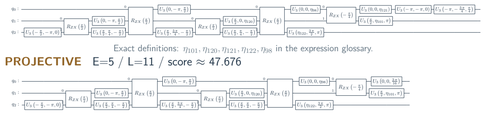
PROJECTIVE E=5 / L=31 / score ≈ 90.272 — E upper bound



$U = e^{-\frac{7i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

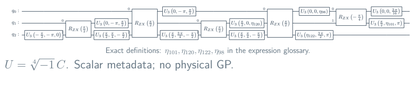
zx_u

EXACT PHASE E=5 / L=13 / score ≈ 51.688 — E upper bound



Exact definitions: $\eta_{101}, \eta_{120}, \eta_{121}, \eta_{122}, \eta_{99}$ in the expression glossary.

PROJECTIVE E=5 / L=11 / score ≈ 47.676 — E upper bound

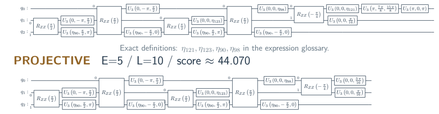


Exact definitions: $\eta_{101}, \eta_{120}, \eta_{122}, \eta_{99}$ in the expression glossary.

$U = \sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=5 / L=12 / score ≈ 48.482 — E upper bound



Exact definitions: $\eta_{121}, \eta_{123}, \eta_{80}, \eta_{99}$ in the expression glossary.

PROJECTIVE E=5 / L=10 / score ≈ 44.070 — E upper bound

Exact definitions: $\eta_{121}, \eta_{80}, \eta_{99}$ in the expression glossary.

$U = (-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.

U $(\theta, \phi, lbda)$

1 qubits / 32 preserved native bases / alphabetical panel order

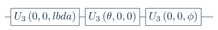
aqt_rxx

PROJECTIVE E=0 / L=3 / score ≈ 6.208 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

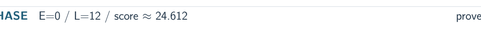
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=12 / score ≈ 24.612 proved E minimum



PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved E minimum

$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.006 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.406 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

google_cz

EXACT PHASE E=0 / L=5 / score ≈ 12.081 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 7.209 proved E minimum

$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=5 / score ≈ 12.081 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 7.209 proved E minimum

$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=5 / score ≈ 12.081 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 7.209 proved E minimum

$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=5 / score ≈ 12.081 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 7.209 proved E minimum

$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=0 / L=5 / score ≈ 10.006 proved E minimum

$U = ie^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=0 / L=5 / score ≈ 10.006 proved E minimum

$U = ie^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=0 / L=5 / score ≈ 10.006 proved E minimum

$U = ie^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=0 / L=8 / score ≈ 18.407 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=8 / score ≈ 18.407 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=8 / score ≈ 18.407 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=7 / score ≈ 15.828 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=3 / score ≈ 6.538 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=3 / score ≈ 6.538 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=0 / L=5 / score ≈ 10.006 proved E minimum

$U = ie^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

tket_tk2

PROJECTIVE E=0 / L=3 / score ≈ 6.538 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.006 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

zz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

U1 (θ)

1 qubits / 32 preserved native bases / alphabetical panel order

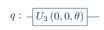
aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved E minimum

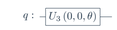
cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum



cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.336 proved E minimum

google_fsim

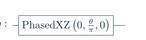
EXACT PHASE E=0 / L=1 / score ≈ 2.336 proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.336 proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.336 proved E minimum



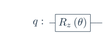
ibm_cx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum



$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.734 proved E minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.734 proved E minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.734 proved E minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

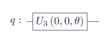
iqm_cz

PROJECTIVE E=0 / L=3 / score ≈ 6.812 proved E minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum



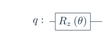
quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum



$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

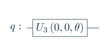
rigetti_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

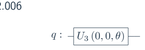
sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum



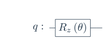
sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum



tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum



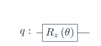
$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

xx_rz_ry

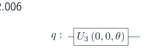
PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum



$U = e^{\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum



zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

U2 $(\phi, lbda)$

1 qubits / 32 preserved native bases / alphabetical panel order

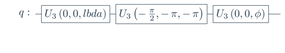
aqt_rxx

PROJECTIVE E=0 / L=3 / score ≈ 6.408 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=3 / score ≈ 6.218 proved E minimum



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

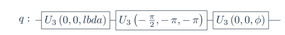
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=3 / score ≈ 6.004 proved E minimum



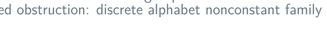
cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.206 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.218 proved E minimum



cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.606 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.218 proved E minimum

google_cz

EXACT PHASE E=0 / L=5 / score ≈ 11.691 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 7.079 proved E minimum

$U = e^{-\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=5 / score ≈ 11.691 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 7.079 proved E minimum

$U = e^{-\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=5 / score ≈ 11.691 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 7.079 proved E minimum

$U = e^{-\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=5 / score ≈ 11.691 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 7.079 proved E minimum

$U = e^{-\frac{i\phi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved E minimum

$U = e^{\frac{i(2lbda+2\phi-\pi)}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved E minimum

$U = e^{\frac{i(2lbda+2\phi-\pi)}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved E minimum

$U = e^{\frac{i(2lbda+2\phi-\pi)}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=0 / L=5 / score ≈ 11.871 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=5 / score ≈ 11.871 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=5 / score ≈ 11.871 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=7 / score ≈ 16.028 proved E minimum

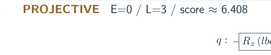
$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.218 proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=3 / score ≈ 6.408 proved E minimum



$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=3 / score ≈ 6.408 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=0 / L=3 / score ≈ 6.404 proved E minimum

$U = e^{\frac{i(2lbda+2\phi-\pi)}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.218 proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.218 proved E minimum

tket_tk2

PROJECTIVE E=0 / L=3 / score ≈ 6.408 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=3 / score ≈ 6.218 proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.206 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.218 proved E minimum

zz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.218 proved E minimum

U3 $(\theta, \phi, lbda)$

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=3 / score ≈ 6.208 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=12 / score ≈ 24.612 proved E minimum

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved E minimum

$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.006 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.406 proved E minimum

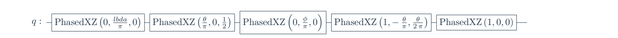
$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

google_cz

EXACT PHASE E=0 / L=5 / score ≈ 12.081 proved E minimum



PROJECTIVE E=0 / L=3 / score ≈ 7.209 proved E minimum

$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=5 / score ≈ 12.081 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 7.209 proved E minimum

$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=5 / score ≈ 12.081 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 7.209 proved E minimum

$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=5 / score ≈ 12.081 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 7.209 proved E minimum

$U = e^{-\frac{i\theta}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=0 / L=5 / score ≈ 10.006 proved E minimum

$U = ie^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=0 / L=5 / score ≈ 10.006 proved E minimum

$U = ie^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=0 / L=5 / score ≈ 10.006 proved E minimum

$U = ie^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=0 / L=8 / score ≈ 18.407 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=8 / score ≈ 18.407 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=8 / score ≈ 18.407 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=7 / score ≈ 15.828 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=3 / score ≈ 6.538 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=3 / score ≈ 6.538 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=0 / L=5 / score ≈ 10.006 proved E minimum

$U = ie^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

tket_tk2

PROJECTIVE E=0 / L=3 / score ≈ 6.538 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.006 proved E minimum

$U = e^{\frac{i(lbda+\phi)}{2}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

zz_u

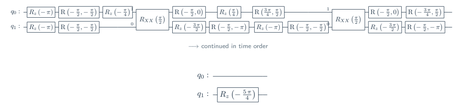
EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

UC

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

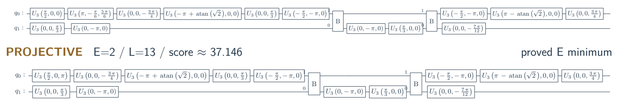
PROJECTIVE E=2 / L=17 / score ≈ 47.256 — E upper bound



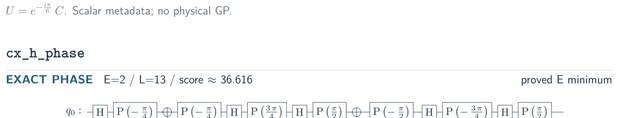
$U = (-1)^{\frac{3}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=14 / score ≈ 40.186 — proved E minimum



PROJECTIVE E=2 / L=13 / score ≈ 37.146 — proved E minimum



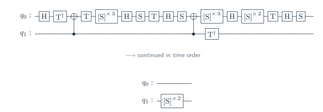
$U = e^{-\frac{i\pi}{6}}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

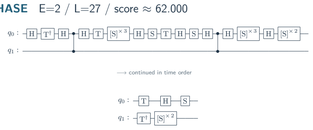
clifford_t_cx

EXACT PHASE E=2 / L=23 / score ≈ 54.000 — proved E minimum



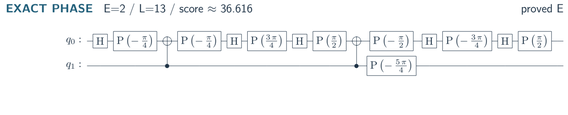
clifford_t_cz

EXACT PHASE E=2 / L=27 / score ≈ 62.000 — proved E minimum



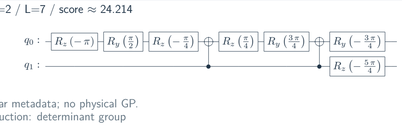
cx_h_phase

EXACT PHASE E=2 / L=13 / score ≈ 36.616 — proved E minimum



cx_rz_ry

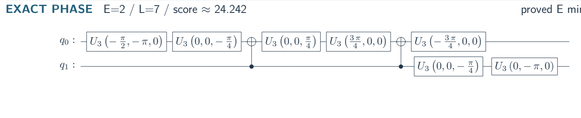
PROJECTIVE E=2 / L=7 / score ≈ 24.214 — proved E minimum



$U = (-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

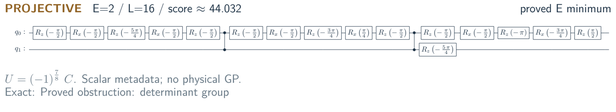
cx_u

EXACT PHASE E=2 / L=7 / score ≈ 24.242 — proved E minimum



cz_rz_rx

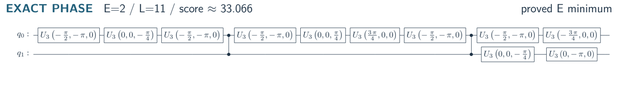
PROJECTIVE E=2 / L=16 / score ≈ 44.032 — proved E minimum



$U = (-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

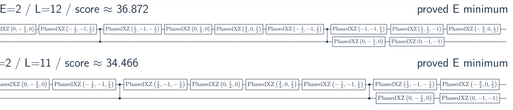
cz_u

EXACT PHASE E=2 / L=11 / score ≈ 33.066 — proved E minimum



google_cz

EXACT PHASE E=2 / L=12 / score ≈ 36.872 — proved E minimum

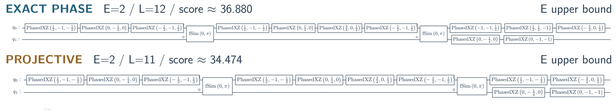


PROJECTIVE E=2 / L=11 / score ≈ 34.466 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=12 / score ≈ 36.880 — E upper bound



PROJECTIVE E=2 / L=11 / score ≈ 34.474 — E upper bound

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=2 / L=12 / score ≈ 40.590 — proved E minimum

Exact definitions: $\eta_{803}, \eta_{804}, \eta_{805}, \eta_{806}, \eta_{807}$ in the expression glossary.

PROJECTIVE E=2 / L=11 / score ≈ 37.089 — proved E minimum

Exact definitions: $\eta_{805}, \eta_{806}, \eta_{807}$ in the expression glossary.

$U = \eta_{808}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=16 / score ≈ 84.896 — E upper bound

PROJECTIVE E=12 / L=15 / score ≈ 82.490 — E upper bound

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

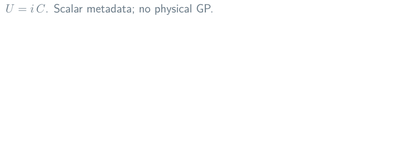
ibm_cx

PROJECTIVE E=2 / L=13 / score ≈ 36.216 — proved E minimum

$U = (-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

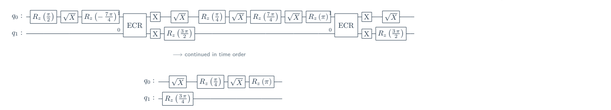
PROJECTIVE E=2 / L=22 / score ≈ 55.224 — proved E minimum



$U = (-1)^{\frac{3}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

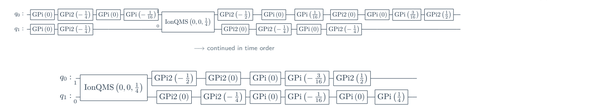
PROJECTIVE E=2 / L=21 / score ≈ 52.620 — proved E minimum



$U = (-1)^{\frac{5}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

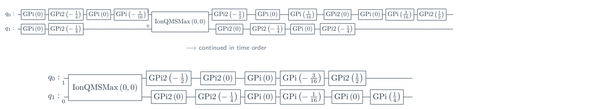
PROJECTIVE E=2 / L=28 / score ≈ 72.068 — E upper bound



$U = (-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

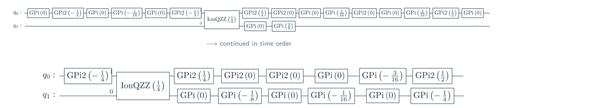
PROJECTIVE E=2 / L=28 / score ≈ 71.264 — proved E minimum



$U = (-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

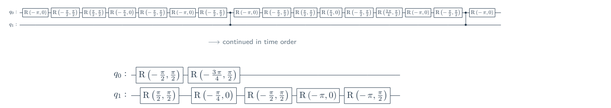
PROJECTIVE E=2 / L=30 / score ≈ 76.864 — E upper bound



$U = (-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

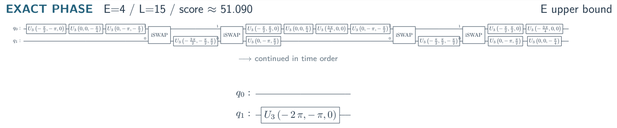
PROJECTIVE E=2 / L=23 / score ≈ 61.092 — proved E minimum



$U = (-1)^{\frac{7}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=4 / L=15 / score ≈ 51.090 — E upper bound



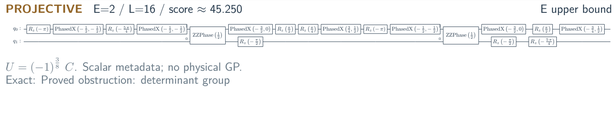
quantinuum_zzmax

PROJECTIVE E=2 / L=16 / score ≈ 44.846 — proved E minimum

$U = (-1)^{\frac{3}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=2 / L=16 / score ≈ 45.250 — E upper bound



$U = (-1)^{\frac{3}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=2 / L=22 / score ≈ 55.224 — proved E minimum

$U = (-1)^{\frac{3}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=2 / L=12 / score ≈ 38.057 — proved E minimum

Exact definitions: η_{809} in the expression glossary.

PROJECTIVE E=2 / L=11 / score ≈ 35.017 — proved E minimum

Exact definitions: η_{809} in the expression glossary.

$U = e^{-\frac{i\pi}{6}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=15 / score ≈ 51.090 — E upper bound

PROJECTIVE E=4 / L=15 / score ≈ 50.690 — E upper bound

$U = -i\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=2 / L=17 / score ≈ 47.264 — E upper bound

$U = (-1)^{\frac{3}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=2 / L=13 / score ≈ 38.678 — proved E minimum

xx_rz_ry

PROJECTIVE E=2 / L=21 / score ≈ 55.046 — E upper bound

$U = (-1)^{\frac{3}{8}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=2 / L=11 / score ≈ 34.270 — E upper bound

zz_u

EXACT PHASE E=2 / L=14 / score ≈ 40.288 — E upper bound

PROJECTIVE E=2 / L=13 / score ≈ 38.282 — E upper bound

$U = i\, C$. Scalar metadata; no physical GP.

UCRX

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=3 / L=22 / score ≈ 60.478 — E upper bound

PROJECTIVE E=2 / L=20 / score ≈ 52.468 — E upper bound

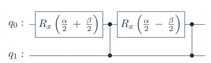
$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cz_rz_rx

EXACT PHASE E=2 / L=2 / score ≈ 12.404 — E upper bound



google_sqrt_iswap

EXACT PHASE E=2 / L=12 / score ≈ 35.593 — conditional E minimum

Exact definitions: $\eta_{819}, \eta_{820}, \eta_{821}, \eta_{822}, \eta_{823}, \eta_{824}, \eta_{825}, \eta_{826}, \eta_{827}, \eta_{828}, \eta_{829}, \eta_{830}, \eta_{831}, \eta_{832}$ in the expression glossary.

PROJECTIVE E=2 / L=11 / score ≈ 33.587 — conditional E minimum

Exact definitions: $\eta_{821}, \eta_{822}, \eta_{823}, \eta_{824}, \eta_{825}, \eta_{826}, \eta_{827}, \eta_{828}, \eta_{829}, \eta_{830}, \eta_{831}, \eta_{832}$ in the expression glossary.

$U = \eta_{833}\,C$. Scalar metadata; no physical GP.

ibm_ecr

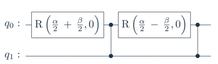
EXACT PHASE E=2 / L=27 / score ≈ 64.026 — E upper bound

PROJECTIVE E=2 / L=27 / score ≈ 63.824 — E upper bound

$U = (-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

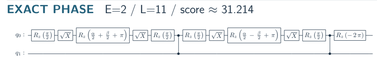
iqm_cz

EXACT PHASE E=2 / L=2 / score ≈ 12.408 — E upper bound



rigetti_cz

EXACT PHASE E=2 / L=11 / score ≈ 31.214 — E upper bound

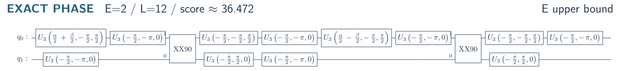


PROJECTIVE E=2 / L=10 / score ≈ 29.212 — E upper bound

$U = -1\,C$. Scalar metadata; no physical GP.

xx90_u

EXACT PHASE E=2 / L=12 / score ≈ 36.472 — E upper bound



berkeley_b

EXACT PHASE E=2 / L=14 / score ≈ 37.884 — conditional E minimum

Exact definitions: $\eta_{810}, \eta_{811}, \eta_{812}, \eta_{813}, \eta_{814}, \eta_{815}, \eta_{816}, \eta_{817}$ in the expression glossary.

PROJECTIVE E=2 / L=13 / score ≈ 35.078 — conditional E minimum

Exact definitions: $\eta_{812}, \eta_{813}, \eta_{814}, \eta_{815}, \eta_{816}, \eta_{817}$ in the expression glossary.

$U = \eta_{818}\,C$. Scalar metadata; no physical GP.

cx_h_phase

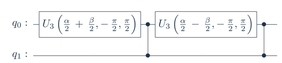
EXACT PHASE E=2 / L=12 / score ≈ 32.812 — E upper bound

PROJECTIVE E=2 / L=4 / score ≈ 16.404 — E upper bound

$U = e^{-\frac{i\phi}{2}}\,C$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=2 / L=2 / score ≈ 13.212 — E upper bound



google_syc

EXACT PHASE E=12 / L=16 / score ≈ 85.217 — E upper bound

PROJECTIVE E=12 / L=14 / score ≈ 80.345 — E upper bound

$U = e^{-\frac{i\phi}{2}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=26 / score ≈ 68.325 — E upper bound

PROJECTIVE E=2 / L=26 / score ≈ 67.925 — E upper bound

$U = i\,C$. Scalar metadata; no physical GP.

iswap_u

EXACT PHASE E=4 / L=14 / score ≈ 48.884 — E upper bound

sqrt_iswap_u

EXACT PHASE E=2 / L=12 / score ≈ 34.672 — conditional E minimum

Exact definitions: $\eta_{812}, \eta_{813}, \eta_{814}, \eta_{816}, \eta_{817}, \eta_{834}, \eta_{835}, \eta_{836}, \eta_{837}, \eta_{838}$ in the expression glossary.

PROJECTIVE E=2 / L=11 / score ≈ 32.666 — conditional E minimum

Exact definitions: $\eta_{812}, \eta_{813}, \eta_{814}, \eta_{816}, \eta_{817}, \eta_{836}, \eta_{837}, \eta_{838}$ in the expression glossary.

$U = \eta_{839}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=3 / L=30 / score ≈ 76.866 — E upper bound

PROJECTIVE E=2 / L=26 / score ≈ 64.856 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

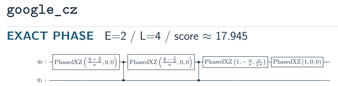
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_rz_ry

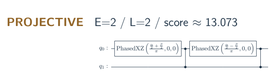
EXACT PHASE E=2 / L=12 / score ≈ 34.024 — E upper bound

google_cz

EXACT PHASE E=2 / L=4 / score ≈ 17.945 — E upper bound



PROJECTIVE E=2 / L=2 / score ≈ 13.073 — E upper bound



$U = e^{-\frac{i\phi}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=19 / score ≈ 47.622 — E upper bound

PROJECTIVE E=2 / L=19 / score ≈ 47.420 — E upper bound

$U = (-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=26 / score ≈ 67.521 — E upper bound

PROJECTIVE E=2 / L=26 / score ≈ 67.121 — E upper bound

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

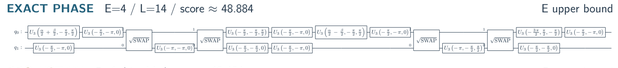
EXACT PHASE E=5 / L=39 / score ≈ 106.979 — E upper bound

PROJECTIVE E=2 / L=20 / score ≈ 52.721 — E upper bound

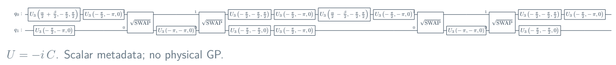
$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=14 / score ≈ 48.884 — E upper bound



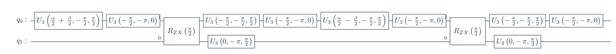
PROJECTIVE E=4 / L=14 / score ≈ 48.484 — E upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

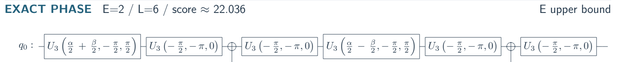
zx_u

EXACT PHASE E=2 / L=10 / score ≈ 32.064 — E upper bound



cx_u

EXACT PHASE E=2 / L=6 / score ≈ 22.036 — E upper bound



google_fsim

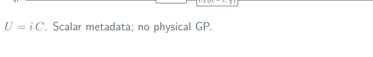
EXACT PHASE E=2 / L=12 / score ≈ 37.201 — E upper bound

PROJECTIVE E=2 / L=10 / score ≈ 32.329 — E upper bound

$U = e^{-\frac{i\phi}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=11 / score ≈ 31.214 — E upper bound



PROJECTIVE E=2 / L=10 / score ≈ 29.212 — E upper bound

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=28 / score ≈ 73.121 — E upper bound

PROJECTIVE E=2 / L=28 / score ≈ 72.721 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

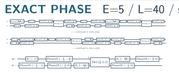
EXACT PHASE E=3 / L=22 / score ≈ 61.131 — E upper bound

PROJECTIVE E=2 / L=20 / score ≈ 53.125 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=5 / L=40 / score ≈ 110.021 — E upper bound



PROJECTIVE E=2 / L=20 / score ≈ 53.137 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=2 / L=13 / score ≈ 38.082 — E upper bound

PROJECTIVE E=2 / L=12 / score ≈ 36.076 — E upper bound

$U = i\,C$. Scalar metadata; no physical GP.

UCRY

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

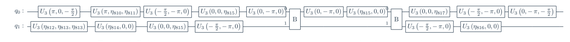
EXACT PHASE E=3 / L=14 / score ≈ 43.254 — E upper bound

PROJECTIVE E=2 / L=12 / score ≈ 35.244 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=2 / L=16 / score ≈ 42.296 — conditional E minimum



Exact definitions: $\eta_{810}, \eta_{811}, \eta_{812}, \eta_{813}, \eta_{814}, \eta_{815}, \eta_{816}, \eta_{817}$ in the expression glossary.

PROJECTIVE E=2 / L=15 / score ≈ 39.490 — conditional E minimum

Exact definitions: $\eta_{812}, \eta_{813}, \eta_{814}, \eta_{815}, \eta_{816}, \eta_{817}$ in the expression glossary.

$U = \eta_{818}\,C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

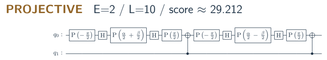
clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=18 / score ≈ 45.620 — E upper bound

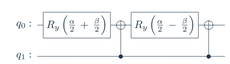
PROJECTIVE E=2 / L=10 / score ≈ 29.212 — E upper bound



$U = e^{-\frac{i\phi}{2}}\,C$. Scalar metadata; no physical GP.

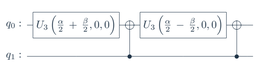
cx_rz_ry

EXACT PHASE E=2 / L=2 / score ≈ 12.404 — E upper bound



cx_u

EXACT PHASE E=2 / L=2 / score ≈ 12.412 — E upper bound

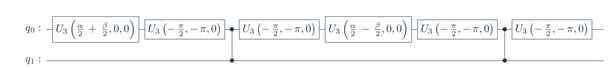


cz_rz_rx

EXACT PHASE E=2 / L=11 / score ≈ 32.022 — E upper bound

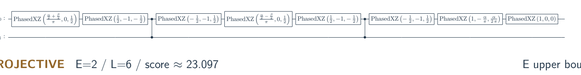
cz_u

EXACT PHASE E=2 / L=6 / score ≈ 21.236 — E upper bound



google_cz

EXACT PHASE E=2 / L=8 / score ≈ 27.969 — E upper bound



PROJECTIVE E=2 / L=6 / score ≈ 23.097 — E upper bound

$U = e^{-\frac{i\phi}{2}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=8 / score ≈ 27.977 — E upper bound

PROJECTIVE E=2 / L=6 / score ≈ 23.105 — E upper bound

$U = e^{-\frac{i\phi}{2}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=2 / L=14 / score ≈ 40.005 — conditional E minimum

Exact definitions: $\eta_{821}, \eta_{822}, \eta_{823}, \eta_{824}, \eta_{825}, \eta_{826}, \eta_{827}, \eta_{828}, \eta_{829}, \eta_{830}, \eta_{831}, \eta_{832}, \eta_{840}, \eta_{841}$ in the expression glossary.

PROJECTIVE E=2 / L=13 / score ≈ 37.999 — conditional E minimum

Exact definitions: $\eta_{821}, \eta_{822}, \eta_{823}, \eta_{824}, \eta_{825}, \eta_{826}, \eta_{827}, \eta_{828}, \eta_{829}, \eta_{830}, \eta_{831}, \eta_{832}$ in the expression glossary.

$U = \eta_{833}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=12 / score ≈ 75.993 — E upper bound

PROJECTIVE E=12 / L=10 / score ≈ 71.121 — E upper bound

$U = e^{-\frac{i\phi}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=9 / score ≈ 26.410 — E upper bound

PROJECTIVE E=2 / L=8 / score ≈ 24.408 — E upper bound

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cz

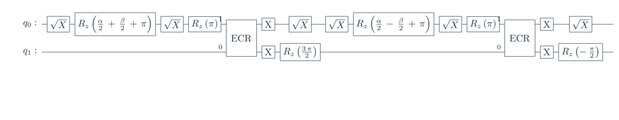
EXACT PHASE E=2 / L=18 / score ≈ 46.020 — E upper bound

PROJECTIVE E=2 / L=18 / score ≈ 45.616 — E upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=16 / score ≈ 40.812 — E upper bound



ionq_aria

EXACT PHASE E=2 / L=18 / score ≈ 49.509 — E upper bound

PROJECTIVE E=2 / L=18 / score ≈ 49.109 — E upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=18 / score ≈ 48.705 — E upper bound

PROJECTIVE E=2 / L=18 / score ≈ 48.305 — E upper bound

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=20 / score ≈ 54.305 — E upper bound

PROJECTIVE E=2 / L=20 / score ≈ 53.905 — E upper bound

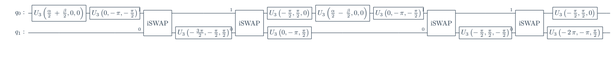
$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=10 / score ≈ 30.440 — E upper bound

iswap_u

EXACT PHASE E=4 / L=10 / score ≈ 39.260 — E upper bound



quantinuum_zzmax

EXACT PHASE E=5 / L=31 / score ≈ 89.755 — E upper bound

PROJECTIVE E=2 / L=12 / score ≈ 35.497 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=3 / L=14 / score ≈ 43.907 — E upper bound

PROJECTIVE E=2 / L=12 / score ≈ 35.901 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=2 / L=18 / score ≈ 46.020 — E upper bound

PROJECTIVE E=2 / L=18 / score ≈ 45.616 — E upper bound

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=2 / L=14 / score ≈ 39.084 — conditional E minimum

Exact definitions: $\eta_{812}, \eta_{813}, \eta_{814}, \eta_{816}, \eta_{817}, \eta_{834}, \eta_{835}, \eta_{836}, \eta_{837}, \eta_{838}$ in the expression glossary.

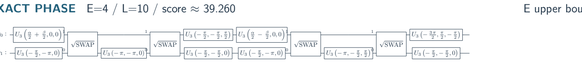
PROJECTIVE E=2 / L=13 / score ≈ 37.078 — conditional E minimum

Exact definitions: $\eta_{812}, \eta_{813}, \eta_{814}, \eta_{816}, \eta_{817}, \eta_{836}, \eta_{837}, \eta_{838}$ in the expression glossary.

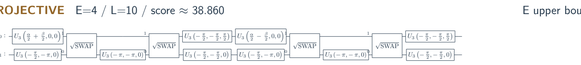
$U = \eta_{839}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=10 / score ≈ 39.260 — E upper bound



PROJECTIVE E=4 / L=10 / score ≈ 38.860 — E upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=5 / L=32 / score ≈ 92.797 — E upper bound

PROJECTIVE E=2 / L=12 / score ≈ 35.913 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

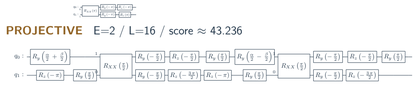
xx90_u

EXACT PHASE E=2 / L=8 / score ≈ 26.848 — E upper bound

xx_rz_ry

EXACT PHASE E=3 / L=20 / score ≈ 55.246 — E upper bound

PROJECTIVE E=2 / L=16 / score ≈ 43.236 — E upper bound



$U = -i\,C$. Scalar metadata; no physical GP.

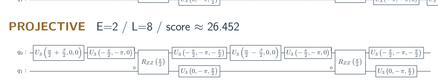
zx_u

EXACT PHASE E=2 / L=6 / score ≈ 22.440 — E upper bound

zz_u

EXACT PHASE E=2 / L=9 / score ≈ 28.458 — E upper bound

PROJECTIVE E=2 / L=8 / score ≈ 26.452 — E upper bound



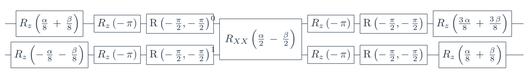
$U = i\,C$. Scalar metadata; no physical GP.

UCRZ

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=1 / L=12 / score ≈ 32.234 — minimum uniform E budget

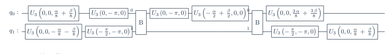


berkeley_b

EXACT PHASE E=2 / L=12 / score ≈ 33.472 — minimum uniform E budget

Exact definitions: $\eta_{811}, \eta_{814}, \eta_{815}, \eta_{816}, \eta_{817}, \eta_{842}, \eta_{843}, \eta_{844}$ in the expression glossary.

PROJECTIVE E=2 / L=9 / score ≈ 29.054 — minimum uniform E budget



$U = e^{-\frac{i(\alpha+\beta)}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

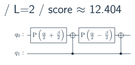
clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=10 / score ≈ 28.812 — minimum uniform E budget

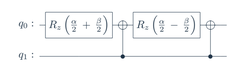
PROJECTIVE E=2 / L=2 / score ≈ 12.404 — minimum uniform E budget



$U = e^{-\frac{i\alpha}{2}}\, C$. Scalar metadata; no physical GP.

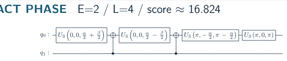
cx_rz_ry

EXACT PHASE E=2 / L=2 / score ≈ 12.404 — minimum uniform E budget



cx_u

EXACT PHASE E=2 / L=4 / score ≈ 16.824 — minimum uniform E budget

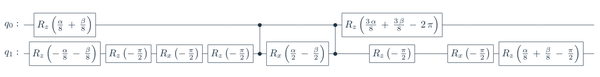


PROJECTIVE E=2 / L=2 / score ≈ 12.412 — minimum uniform E budget

$U = e^{-\frac{i\alpha}{2}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

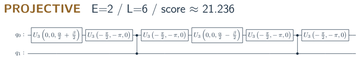
EXACT PHASE E=2 / L=10 / score ≈ 31.620 — minimum uniform E budget



cz_u

EXACT PHASE E=2 / L=8 / score ≈ 25.648 — minimum uniform E budget

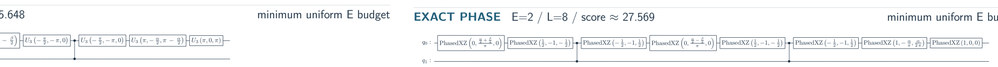
PROJECTIVE E=2 / L=6 / score ≈ 21.236 — minimum uniform E budget



$U = e^{-\frac{i\alpha}{2}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=2 / L=8 / score ≈ 27.569 — minimum uniform E budget



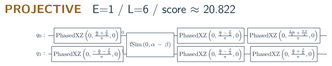
PROJECTIVE E=2 / L=6 / score ≈ 22.697 — minimum uniform E budget

$U = e^{-\frac{i\alpha}{2}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=8 / score ≈ 25.694 — minimum uniform E budget

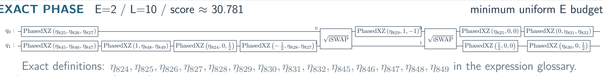
PROJECTIVE E=1 / L=6 / score ≈ 20.822 — minimum uniform E budget



$U = e^{-\frac{i\alpha}{2}}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=2 / L=10 / score ≈ 30.781 — minimum uniform E budget



Exact definitions: $\eta_{824}, \eta_{825}, \eta_{826}, \eta_{827}, \eta_{828}, \eta_{829}, \eta_{830}, \eta_{831}, \eta_{832}, \eta_{845}, \eta_{846}, \eta_{847}, \eta_{848}, \eta_{849}$ in the expression glossary.

PROJECTIVE E=2 / L=9 / score ≈ 28.775 — minimum uniform E budget

Exact definitions: $\eta_{821}, \eta_{822}, \eta_{823}, \eta_{824}, \eta_{825}, \eta_{826}, \eta_{827}, \eta_{828}, \eta_{829}, \eta_{830}, \eta_{831}, \eta_{832}$ in the expression glossary.

$U = \eta_{850}\, C$. Scalar metadata; no physical GP.

google_syc

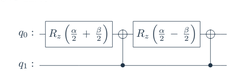
EXACT PHASE E=12 / L=12 / score ≈ 75.593 — E upper bound

PROJECTIVE E=12 / L=10 / score ≈ 70.721 — E upper bound

$U = e^{-\frac{i\alpha}{2}}\, C$. Scalar metadata; no physical GP.

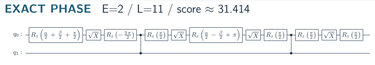
ibm_cx

EXACT PHASE E=2 / L=2 / score ≈ 12.404 — minimum uniform E budget



ibm_cz

EXACT PHASE E=2 / L=11 / score ≈ 31.414 — minimum uniform E budget

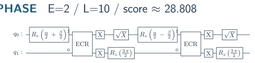


PROJECTIVE E=2 / L=11 / score ≈ 31.212 — minimum uniform E budget

$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=10 / score ≈ 28.808 — minimum uniform E budget



PROJECTIVE E=2 / L=8 / score ≈ 26.612 — minimum uniform E budget

$U = i\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=14 / score ≈ 41.101 — E upper bound

PROJECTIVE E=2 / L=14 / score ≈ 40.701 — E upper bound

$U = i\, C$. Scalar metadata; no physical GP.

ionq_aria_full

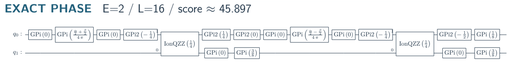
EXACT PHASE E=2 / L=14 / score ≈ 40.297 — minimum uniform E budget

PROJECTIVE E=2 / L=14 / score ≈ 39.897 — minimum uniform E budget

$U = i\, C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=16 / score ≈ 45.897 — E upper bound



PROJECTIVE E=2 / L=16 / score ≈ 45.497 — E upper bound

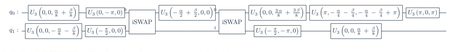
$U = -i\, C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=14 / score ≈ 39.656 — minimum uniform E budget

iswap_u

EXACT PHASE E=2 / L=10 / score ≈ 31.860 — minimum uniform E budget



PROJECTIVE E=2 / L=8 / score ≈ 27.048 — minimum uniform E budget

$U = e^{-\frac{i(\alpha+\beta)}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=5 / L=27 / score ≈ 82.210 — E upper bound

PROJECTIVE E=2 / L=8 / score ≈ 27.952 — minimum uniform E budget

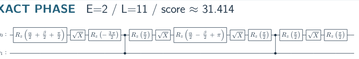
$U = -i\, C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=1 / L=4 / score ≈ 14.940 — minimum uniform E budget

rigetti_cz

EXACT PHASE E=2 / L=11 / score ≈ 31.414 — minimum uniform E budget



PROJECTIVE E=2 / L=11 / score ≈ 31.212 — minimum uniform E budget

$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=2 / L=10 / score ≈ 30.260 — minimum uniform E budget

Exact definitions: $\eta_{814}, \eta_{816}, \eta_{817}, \eta_{835}, \eta_{836}, \eta_{837}, \eta_{838}, \eta_{842}, \eta_{843}, \eta_{851}$ in the expression glossary.

PROJECTIVE E=2 / L=9 / score ≈ 28.254 — minimum uniform E budget

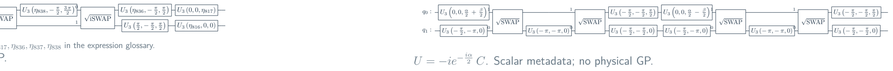
Exact definitions: $\eta_{812}, \eta_{813}, \eta_{814}, \eta_{816}, \eta_{817}, \eta_{836}, \eta_{837}, \eta_{838}$ in the expression glossary.

$U = \eta_{839}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=12 / score ≈ 43.672 — E upper bound

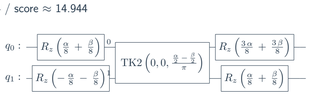
PROJECTIVE E=4 / L=10 / score ≈ 38.860 — E upper bound



$U = -ie^{-\frac{i\alpha}{2}}\, C$. Scalar metadata; no physical GP.

tket_tk2

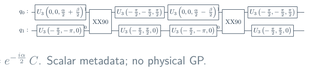
EXACT PHASE E=1 / L=4 / score ≈ 14.944 — minimum uniform E budget



xx90_u

EXACT PHASE E=2 / L=10 / score ≈ 31.260 — minimum uniform E budget

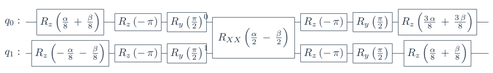
PROJECTIVE E=2 / L=8 / score ≈ 26.848 — minimum uniform E budget



$U = e^{-\frac{i\alpha}{2}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=1 / L=12 / score ≈ 31.426 — minimum uniform E budget



zx_u

EXACT PHASE E=1 / L=8 / score ≈ 23.850 — minimum uniform E budget

PROJECTIVE E=1 / L=6 / score ≈ 19.038 — minimum uniform E budget

$U = e^{-\frac{i(\alpha+\beta)}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=1 / L=6 / score ≈ 19.438 — minimum uniform E budget

PROJECTIVE E=1 / L=4 / score ≈ 14.626 — minimum uniform E budget

$U = e^{-\frac{i(\alpha+\beta)}{4}}\, C$. Scalar metadata; no physical GP.

Unitary

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=2 / L=34 / score ≈ 83.904 — proved E minimum

PROJECTIVE E=2 / L=16 / score ≈ 43.858 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

berkeley_b

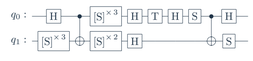
EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

clifford_only

Exact: Proved obstruction: not a clifford
Projective: Proved obstruction: not a clifford

clifford_t_cx

PROJECTIVE E=2 / L=16 / score ≈ 40.000 — proved E minimum



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: outside clifford t number field

clifford_t_cz

PROJECTIVE E=2 / L=18 / score ≈ 44.000 — proved E minimum

$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: outside clifford t number field

cx_h_phase

EXACT PHASE E=2 / L=19 / score ≈ 47.220 — proved E minimum

PROJECTIVE E=2 / L=11 / score ≈ 31.212 — proved E minimum

$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=2 / L=8 / score ≈ 25.416 — proved E minimum

cx_u

EXACT PHASE E=2 / L=6 / score ≈ 21.436 — proved E minimum

cz_rz_rx

EXACT PHASE E=2 / L=12 / score ≈ 34.624 — proved E minimum

cz_u

EXACT PHASE E=2 / L=6 / score ≈ 22.636 — proved E minimum

google_cz

EXACT PHASE E=2 / L=8 / score ≈ 26.709 — proved E minimum

PROJECTIVE E=2 / L=6 / score ≈ 22.036 — proved E minimum

$U = e^{\frac{3i\pi}{8}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=2 / score ≈ 13.220 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=2 / L=4 / score ≈ 18.275 — proved E minimum

PROJECTIVE E=2 / L=3 / score ≈ 15.009 — proved E minimum

$U = \eta_{10}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=16 / score ≈ 83.757 — E upper bound

PROJECTIVE E=12 / L=14 / score ≈ 79.084 — E upper bound

$U = (-1)^{\frac{3}{8}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=13 / score ≈ 35.008 — proved E minimum

PROJECTIVE E=2 / L=9 / score ≈ 26.402 — proved E minimum

$U = \sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=15 / score ≈ 39.010 — proved E minimum

PROJECTIVE E=2 / L=15 / score ≈ 38.808 — proved E minimum

$U = (-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=14 / score ≈ 38.220 — proved E minimum

PROJECTIVE E=2 / L=14 / score ≈ 37.816 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=17 / score ≈ 46.646 — proved E minimum

PROJECTIVE E=2 / L=17 / score ≈ 46.246 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=17 / score ≈ 45.842 — proved E minimum

PROJECTIVE E=2 / L=17 / score ≈ 45.442 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=19 / score ≈ 51.442 — proved E minimum

PROJECTIVE E=2 / L=19 / score ≈ 51.042 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=8 / score ≈ 26.032 — proved E minimum

iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 22.636 — proved E minimum

quantinuum_zzmax

EXACT PHASE E=5 / L=27 / score ≈ 80.286 — E upper bound

PROJECTIVE E=2 / L=8 / score ≈ 26.028 — proved E minimum

$U = \frac{\sqrt[8]{-1}\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=2 / L=18 / score ≈ 48.656 — proved E minimum

PROJECTIVE E=2 / L=15 / score ≈ 41.852 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

rigetti_cz

EXACT PHASE E=2 / L=15 / score ≈ 39.010 — proved E minimum

PROJECTIVE E=2 / L=15 / score ≈ 38.808 — proved E minimum

$U = (-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=2 / L=3 / score ≈ 15.218 — proved E minimum

Exact definitions: η_{11}, η_{12} in the expression glossary.

sqrt_swap_u

EXACT PHASE E=4 / L=16 / score ≈ 53.896 — E upper bound

PROJECTIVE E=4 / L=14 / score ≈ 49.884 — E upper bound

$U = -(-1)^{\frac{5}{8}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.606 — proved E minimum

xx90_u

EXACT PHASE E=2 / L=7 / score ≈ 23.042 — proved E minimum

PROJECTIVE E=2 / L=5 / score ≈ 19.030 — proved E minimum

$U = \eta_{17}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=2 / L=34 / score ≈ 80.672 — proved E minimum

PROJECTIVE E=2 / L=30 / score ≈ 74.264 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=2 / L=12 / score ≈ 37.476 — proved E minimum

PROJECTIVE E=2 / L=10 / score ≈ 33.464 — proved E minimum

$U = \sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=2 / L=12 / score ≈ 35.076 — proved E minimum

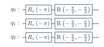
WalshHadamard

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=1 / L=8 / score ≈ 21.228 — E upper bound

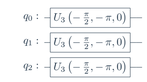
PROJECTIVE E=0 / L=6 / score ≈ 13.218 — E upper bound



$U = i\,C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=0 / L=3 / score ≈ 6.618 — E upper bound



clifford_only

EXACT PHASE E=0 / L=3 / score ≈ 6.000 — E upper bound

clifford_t_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.000 — E upper bound

clifford_t_cz

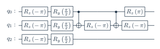
EXACT PHASE E=0 / L=3 / score ≈ 6.000 — E upper bound

cx_h_phase

EXACT PHASE E=0 / L=3 / score ≈ 6.000 — E upper bound

cx_rz_ry

EXACT PHASE E=2 / L=9 / score ≈ 26.618 — E upper bound

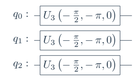


PROJECTIVE E=0 / L=6 / score ≈ 12.612 — E upper bound

$U = i\,C$. Scalar metadata; no physical GP.

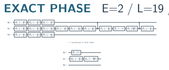
cx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.618 — E upper bound

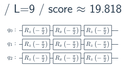


cz_rz_rx

EXACT PHASE E=2 / L=19 / score ≈ 49.438 — E upper bound



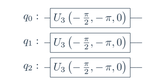
PROJECTIVE E=0 / L=9 / score ≈ 19.818 — E upper bound



$U = i\,C$. Scalar metadata; no physical GP.

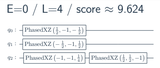
cz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.618 — E upper bound

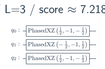


google_cz

EXACT PHASE E=0 / L=4 / score ≈ 9.624 — E upper bound



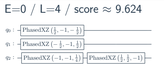
PROJECTIVE E=0 / L=3 / score ≈ 7.218 — E upper bound



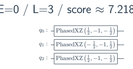
$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=4 / score ≈ 9.624 — E upper bound



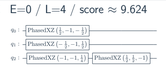
PROJECTIVE E=0 / L=3 / score ≈ 7.218 — E upper bound



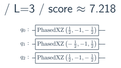
$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=4 / score ≈ 9.624 — E upper bound



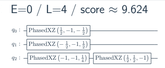
PROJECTIVE E=0 / L=3 / score ≈ 7.218 — E upper bound



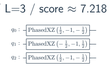
$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=4 / score ≈ 9.624 — E upper bound



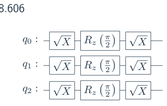
PROJECTIVE E=0 / L=3 / score ≈ 7.218 — E upper bound



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

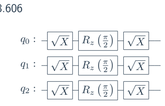
ibm_cx

EXACT PHASE E=0 / L=9 / score ≈ 18.606 — E upper bound



ibm_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.606 — E upper bound

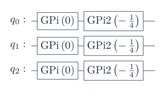


ibm_ecr

EXACT PHASE E=0 / L=9 / score ≈ 18.606 — E upper bound

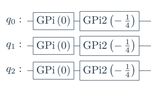
ionq_aria

EXACT PHASE E=0 / L=6 / score ≈ 13.212 — E upper bound



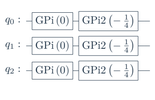
ionq_aria_full

EXACT PHASE E=0 / L=6 / score ≈ 13.212 — E upper bound



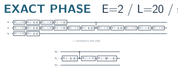
ionq_forte

EXACT PHASE E=0 / L=6 / score ≈ 13.212 — E upper bound

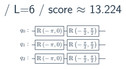


iqm_cz

EXACT PHASE E=2 / L=20 / score ≈ 51.480 — E upper bound



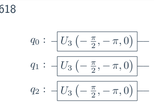
PROJECTIVE E=0 / L=6 / score ≈ 13.224 — E upper bound



$U = i\,C$. Scalar metadata; no physical GP.

iswap_u

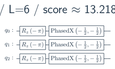
EXACT PHASE E=0 / L=3 / score ≈ 6.618 — E upper bound



quantinuum_zzmax

EXACT PHASE E=3 / L=25 / score ≈ 67.476 — E upper bound

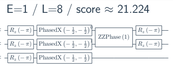
PROJECTIVE E=0 / L=6 / score ≈ 13.218 — E upper bound



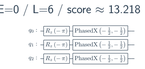
$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=1 / L=8 / score ≈ 21.224 — E upper bound



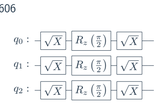
PROJECTIVE E=0 / L=6 / score ≈ 13.218 — E upper bound



$U = i\,C$. Scalar metadata; no physical GP.

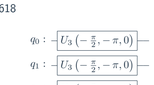
rigetti_cz

EXACT PHASE E=0 / L=9 / score ≈ 18.606 — E upper bound



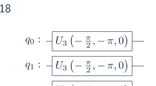
sqrt_iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.618 — E upper bound



sqrt_swap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.618 — E upper bound



tket_tk2

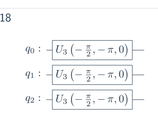
EXACT PHASE E=3 / L=26 / score ≈ 70.102 — E upper bound

PROJECTIVE E=0 / L=6 / score ≈ 13.218 — E upper bound

$U = i\,C$. Scalar metadata; no physical GP.

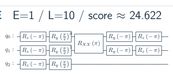
xx90_u

EXACT PHASE E=0 / L=3 / score ≈ 6.618 — E upper bound



xx_rz_ry

EXACT PHASE E=1 / L=10 / score ≈ 24.622 — E upper bound

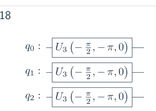


PROJECTIVE E=0 / L=6 / score ≈ 12.612 — E upper bound

$U = i\,C$. Scalar metadata; no physical GP.

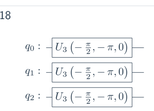
zx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.618 — E upper bound



zz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.618 — E upper bound



X

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.004 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

clifford_only

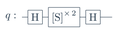
EXACT PHASE E=0 / L=4 / score ≈ 8.000 proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.000 proved E minimum

clifford_t_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.000 proved E minimum



cx_h_phase

EXACT PHASE E=0 / L=3 / score ≈ 6.002 proved E minimum

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.004 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

google_syc

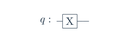
EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

ibm_cx

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved E minimum

ibm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved E minimum



ibm_ecr

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved E minimum

ionq_aria

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved E minimum

ionq_forte

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved E minimum

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.004 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.004 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.004 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.000 proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.004 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.004 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

X_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

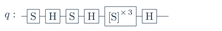
EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum



PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.806 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.002 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

X_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.004 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.406 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.002 — proved E minimum

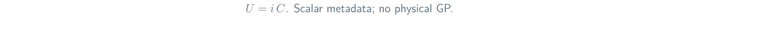
$U = iC$. Scalar metadata; no physical GP.

cx_rz_ry

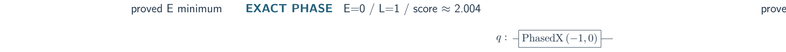
EXACT PHASE E=0 / L=2 / score ≈ 4.004 — proved E minimum

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum



$U = iC$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.004 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.004 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.004 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.004 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.004 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.004 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.004 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.004 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.004 — proved E minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = iC$. Scalar metadata; no physical GP.

X_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

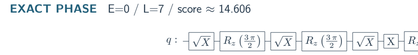
PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.806 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

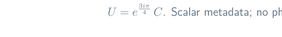
EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

X_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

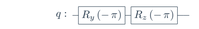
EXACT PHASE E=0 / L=5 / score ≈ 10.006 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.002 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved E minimum



$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=1 / score ≈ 2.202 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

X_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.806 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

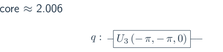
PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

X_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.004 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.406 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.002 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

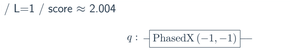
cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.004 — proved E minimum

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum



$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=5 / score ≈ 10.004 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.004 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=5 / score ≈ 10.004 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.004 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.004 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.004 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.004 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.004 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.004 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=5 / score ≈ 10.004 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.004 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.004 — proved E minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

X_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.806 — proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.004 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

XPow $(t, global_{shift})$

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.004 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=3 / score ≈ 6.818 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=11 / score ≈ 22.010 proved E minimum

PROJECTIVE E=0 / L=3 / score ≈ 6.002 proved E minimum

$U = e^{i\pi\, global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.406 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.818 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

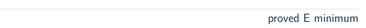
EXACT PHASE E=0 / L=3 / score ≈ 6.818 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$U = e^{i\pi\, global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$U = e^{i\pi\, global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$U = e^{i\pi\, global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

google_syc

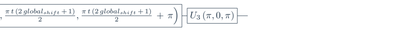
EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$U = e^{i\pi\, global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved E minimum



$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved E minimum

$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved E minimum

$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=0 / L=4 / score ≈ 9.208 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=4 / score ≈ 9.208 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=4 / score ≈ 9.208 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.004 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.818 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.004 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.004 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved E minimum

$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.818 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.818 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.004 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=3 / score ≈ 6.818 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=3 / score ≈ 6.406 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.818 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.818 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.406 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

XX

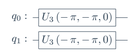
2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.008 — E upper bound

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved E minimum



clifford_only

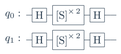
EXACT PHASE E=0 / L=8 / score ≈ 16.000 — E upper bound

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — E upper bound

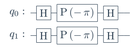
clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — E upper bound



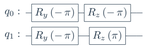
cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.004 — E upper bound



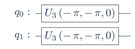
cx_rz_ry

EXACT PHASE E=0 / L=4 / score ≈ 8.008 — E upper bound



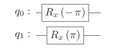
cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound



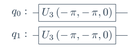
cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.004 — E upper bound



cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound



google_cz

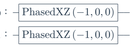
EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound

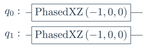
google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved E minimum



google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound



ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.000 — E upper bound

ibm_cz

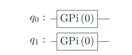
EXACT PHASE E=0 / L=2 / score ≈ 4.000 — E upper bound

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.000 — E upper bound

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.004 — E upper bound



ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.004 — E upper bound

ionq_forte

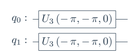
EXACT PHASE E=0 / L=2 / score ≈ 4.004 — E upper bound

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.008 — E upper bound

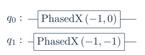
iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound



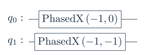
quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.008 — E upper bound



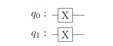
quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.008 — E upper bound



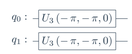
rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.000 — E upper bound



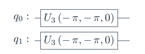
sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved E minimum



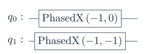
sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound



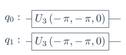
tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.008 — E upper bound



xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound

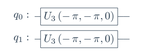


xx_rz_ry

EXACT PHASE E=0 / L=4 / score ≈ 8.008 — E upper bound

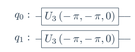
zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound



zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound



XX90

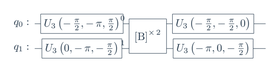
2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=1 / L=0 / score ≈ 4.202 — proved E minimum

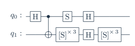
berkeley_b

EXACT PHASE E=2 / L=4 / score ≈ 17.224 — proved E minimum

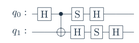


clifford_only

EXACT PHASE E=1 / L=10 / score ≈ 24.000 — proved E minimum



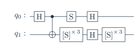
PROJECTIVE E=1 / L=6 / score ≈ 16.000 — proved E minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=1 / L=10 / score ≈ 24.000 — proved E minimum



PROJECTIVE E=1 / L=6 / score ≈ 16.000 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=1 / L=12 / score ≈ 28.000 — proved E minimum

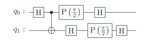
PROJECTIVE E=1 / L=6 / score ≈ 16.000 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=1 / L=6 / score ≈ 16.606 — proved E minimum

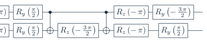
PROJECTIVE E=1 / L=6 / score ≈ 16.404 — proved E minimum



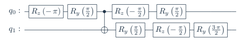
$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=2 / L=9 / score ≈ 27.018 — E upper bound



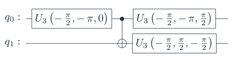
PROJECTIVE E=1 / L=7 / score ≈ 19.214 — proved E minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

cx_u

EXACT PHASE E=1 / L=3 / score ≈ 11.218 — proved E minimum



cz_rz_rx

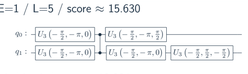
EXACT PHASE E=2 / L=21 / score ≈ 53.842 — E upper bound

PROJECTIVE E=1 / L=10 / score ≈ 26.020 — proved E minimum

$U = \frac{\sqrt{2}i(1-i)}{2}\,C$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=1 / L=5 / score ≈ 15.630 — proved E minimum

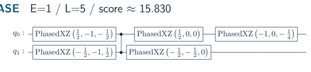


PROJECTIVE E=1 / L=4 / score ≈ 13.224 — proved E minimum

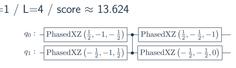
$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=1 / L=5 / score ≈ 15.830 — proved E minimum



PROJECTIVE E=1 / L=4 / score ≈ 13.624 — proved E minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=2 / score ≈ 12.820 — E upper bound

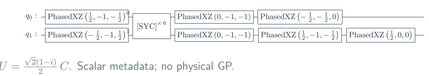
google_sqrt_iswap

EXACT PHASE E=2 / L=4 / score ≈ 16.024 — proved E minimum

google_syc

EXACT PHASE E=6 / L=8 / score ≈ 42.048 — E upper bound

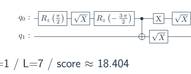
PROJECTIVE E=6 / L=7 / score ≈ 39.842 — E upper bound



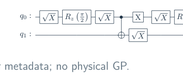
$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=1 / L=7 / score ≈ 18.606 — proved E minimum



PROJECTIVE E=1 / L=7 / score ≈ 18.404 — proved E minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=1 / L=12 / score ≈ 29.010 — proved E minimum

PROJECTIVE E=1 / L=10 / score ≈ 25.212 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=1 / L=6 / score ≈ 16.808 — proved E minimum

PROJECTIVE E=1 / L=6 / score ≈ 16.606 — proved E minimum

$U = e^{-i\frac{\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=1 / L=16 / score ≈ 39.638 — proved E minimum

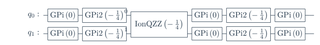
ionq_aria_full

EXACT PHASE E=1 / L=10 / score ≈ 26.024 — proved E minimum

ionq_forte

EXACT PHASE E=1 / L=11 / score ≈ 28.024 — proved E minimum

PROJECTIVE E=1 / L=10 / score ≈ 26.022 — proved E minimum



$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=18 / score ≈ 47.672 — E upper bound

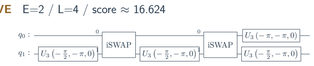
PROJECTIVE E=1 / L=8 / score ≈ 22.032 — proved E minimum

$U = \frac{\sqrt{2}i(1-i)}{2}\,C$. Scalar metadata; no physical GP.

iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 17.024 — proved E minimum

PROJECTIVE E=2 / L=4 / score ≈ 16.624 — proved E minimum



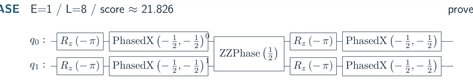
$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=1 / L=8 / score ≈ 21.624 — proved E minimum

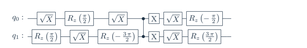
quantinuum_zzphase

EXACT PHASE E=1 / L=8 / score ≈ 21.826 — proved E minimum



rigetti_cz

EXACT PHASE E=1 / L=12 / score ≈ 29.010 — proved E minimum



PROJECTIVE E=1 / L=12 / score ≈ 28.808 — proved E minimum

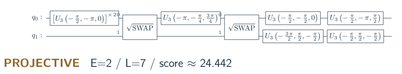
$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

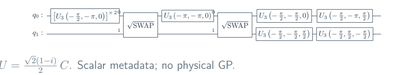
EXACT PHASE E=2 / L=4 / score ≈ 16.024 — proved E minimum

sqrt_swap_u

EXACT PHASE E=2 / L=7 / score ≈ 25.242 — proved E minimum



PROJECTIVE E=2 / L=7 / score ≈ 24.442 — proved E minimum



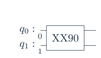
$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.206 — proved E minimum

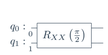
xx90_u

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum



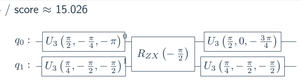
xx_rz_ry

EXACT PHASE E=1 / L=0 / score ≈ 4.202 — proved E minimum



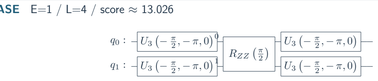
zx_u

EXACT PHASE E=1 / L=4 / score ≈ 15.026 — proved E minimum



zz_u

EXACT PHASE E=1 / L=4 / score ≈ 13.026 — proved E minimum

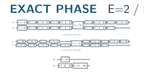


xx_minus_yy (ϕ, β)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=2 / L=37 / score ≈ 90.310 — minimum uniform E budget



PROJECTIVE E=2 / L=21 / score ≈ 54.866 — minimum uniform E budget

$U = -i\,C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=2 / L=14 / score ≈ 40.084 — minimum uniform E budget

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

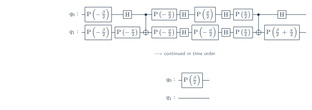
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

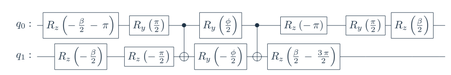
cx_h_phase

EXACT PHASE E=2 / L=17 / score ≈ 44.222 — minimum uniform E budget



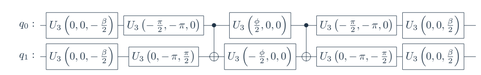
cx_rz_ry

EXACT PHASE E=2 / L=10 / score ≈ 29.820 — minimum uniform E budget



cx_u

EXACT PHASE E=2 / L=10 / score ≈ 30.060 — minimum uniform E budget

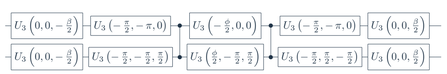


cz_rz_rx

EXACT PHASE E=2 / L=15 / score ≈ 41.030 — minimum uniform E budget

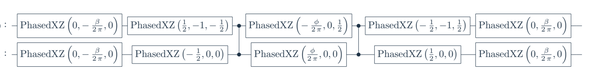
cz_u

EXACT PHASE E=2 / L=10 / score ≈ 31.260 — minimum uniform E budget



google_cz

EXACT PHASE E=2 / L=10 / score ≈ 32.642 — minimum uniform E budget



google_fsim

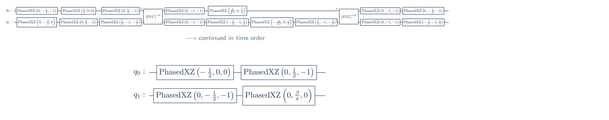
EXACT PHASE E=2 / L=6 / score ≈ 22.365 — E upper bound

google_sqrt_iswap

EXACT PHASE E=2 / L=12 / score ≈ 35.654 — minimum uniform E budget

google_syc

EXACT PHASE E=12 / L=20 / score ≈ 93.441 — E upper bound



ibm_cx

EXACT PHASE E=2 / L=19 / score ≈ 47.624 — minimum uniform E budget

PROJECTIVE E=2 / L=18 / score ≈ 45.822 — minimum uniform E budget

$U = \sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=22 / score ≈ 54.022 — minimum uniform E budget

PROJECTIVE E=2 / L=21 / score ≈ 52.022 — minimum uniform E budget

$U = \sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

ibm_ecr

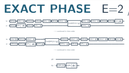
EXACT PHASE E=2 / L=16 / score ≈ 42.424 — minimum uniform E budget

PROJECTIVE E=2 / L=16 / score ≈ 42.222 — minimum uniform E budget

$U = -(-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=36 / score ≈ 90.805 — minimum uniform E budget

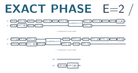


PROJECTIVE E=2 / L=36 / score ≈ 90.405 — minimum uniform E budget

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=36 / score ≈ 90.001 — minimum uniform E budget

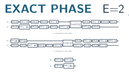


PROJECTIVE E=2 / L=36 / score ≈ 89.601 — minimum uniform E budget

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=2 / L=38 / score ≈ 95.601 — minimum uniform E budget

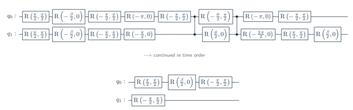


PROJECTIVE E=2 / L=38 / score ≈ 95.201 — minimum uniform E budget

$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=20 / score ≈ 53.880 — minimum uniform E budget



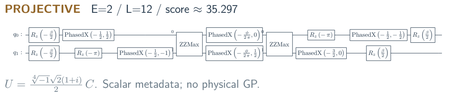
iswap_u

EXACT PHASE E=2 / L=10 / score ≈ 31.260 — minimum uniform E budget

quantinuum_zzmax

EXACT PHASE E=5 / L=31 / score ≈ 89.555 — E upper bound

PROJECTIVE E=2 / L=12 / score ≈ 35.297 — minimum uniform E budget



$U = \frac{\sqrt[8]{-1}\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=2 / L=21 / score ≈ 55.723 — minimum uniform E budget

PROJECTIVE E=2 / L=21 / score ≈ 55.123 — minimum uniform E budget

$U = -i\,C$. Scalar metadata; no physical GP.

rigetti_cz

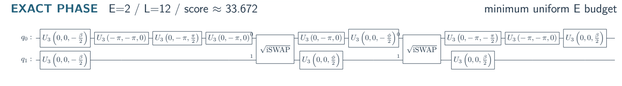
EXACT PHASE E=2 / L=22 / score ≈ 54.022 — minimum uniform E budget

PROJECTIVE E=2 / L=21 / score ≈ 52.022 — minimum uniform E budget

$U = \sqrt[4]{-1}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=2 / L=12 / score ≈ 33.672 — minimum uniform E budget



sqrt_swap_u

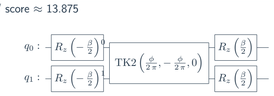
EXACT PHASE E=4 / L=20 / score ≈ 61.720 — E upper bound

PROJECTIVE E=4 / L=20 / score ≈ 61.320 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

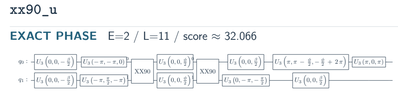
tket_tk2

EXACT PHASE E=1 / L=4 / score ≈ 13.875 — minimum uniform E budget



xx90_u

EXACT PHASE E=2 / L=11 / score ≈ 32.066 — minimum uniform E budget



PROJECTIVE E=2 / L=9 / score ≈ 27.654 — minimum uniform E budget

$U = \eta_{633}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

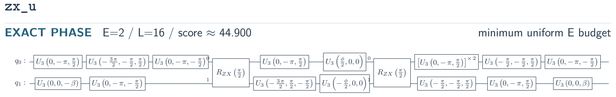
EXACT PHASE E=2 / L=37 / score ≈ 87.078 — minimum uniform E budget

PROJECTIVE E=2 / L=27 / score ≈ 67.258 — minimum uniform E budget

$U = -i\,C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=2 / L=16 / score ≈ 44.900 — minimum uniform E budget



zz_u

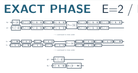
EXACT PHASE E=2 / L=16 / score ≈ 43.700 — minimum uniform E budget

xx_plus_yy (ϕ, β)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=2 / L=37 / score ≈ 90.310 — minimum uniform E budget



PROJECTIVE E=2 / L=21 / score ≈ 54.866 — minimum uniform E budget

$U = -i\,C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=2 / L=14 / score ≈ 40.084 — minimum uniform E budget

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

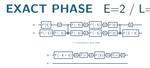
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=24 / score ≈ 58.428 — minimum uniform E budget

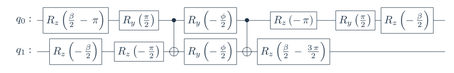


PROJECTIVE E=2 / L=17 / score ≈ 44.222 — minimum uniform E budget

$U = e^{\frac{i\phi}{2}}\,C$. Scalar metadata; no physical GP.

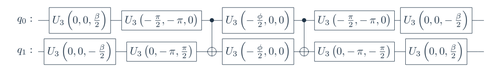
cx_rz_ry

EXACT PHASE E=2 / L=10 / score ≈ 29.820 — minimum uniform E budget



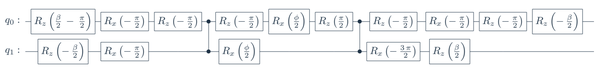
cx_u

EXACT PHASE E=2 / L=10 / score ≈ 30.060 — minimum uniform E budget



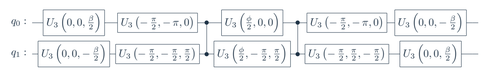
cz_rz_rx

EXACT PHASE E=2 / L=15 / score ≈ 41.030 — minimum uniform E budget



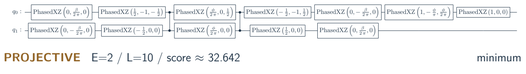
cz_u

EXACT PHASE E=2 / L=10 / score ≈ 31.260 — minimum uniform E budget



google_cz

EXACT PHASE E=2 / L=12 / score ≈ 37.514 — minimum uniform E budget

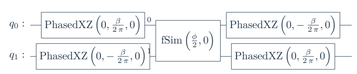


PROJECTIVE E=2 / L=10 / score ≈ 32.642 — minimum uniform E budget

$U = e^{-\frac{i\phi}{2}}\,C$. Scalar metadata; no physical GP.

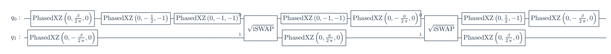
google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 14.349 — minimum uniform E budget



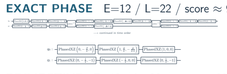
google_sqrt_iswap

EXACT PHASE E=2 / L=10 / score ≈ 31.642 — minimum uniform E budget



google_syc

EXACT PHASE E=12 / L=22 / score ≈ 98.314 — E upper bound

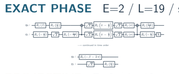


PROJECTIVE E=12 / L=20 / score ≈ 93.441 — E upper bound

$U = e^{\frac{i\phi}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=19 / score ≈ 47.624 — minimum uniform E budget

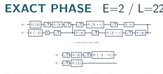


PROJECTIVE E=2 / L=18 / score ≈ 45.624 — minimum uniform E budget

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=2 / L=22 / score ≈ 54.022 — minimum uniform E budget



PROJECTIVE E=2 / L=21 / score ≈ 52.022 — minimum uniform E budget

$U = \sqrt[4]{-\mathbb{I}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=16 / score ≈ 42.424 — minimum uniform E budget

PROJECTIVE E=2 / L=16 / score ≈ 42.222 — minimum uniform E budget

$U = -(-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=36 / score ≈ 90.805 — minimum uniform E budget

PROJECTIVE E=2 / L=36 / score ≈ 90.405 — minimum uniform E budget

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=36 / score ≈ 90.001 — minimum uniform E budget

PROJECTIVE E=2 / L=36 / score ≈ 89.601 — minimum uniform E budget

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

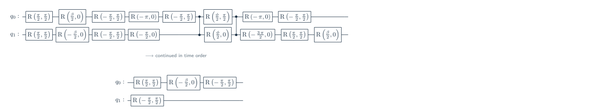
EXACT PHASE E=2 / L=38 / score ≈ 95.601 — minimum uniform E budget

PROJECTIVE E=2 / L=38 / score ≈ 95.201 — minimum uniform E budget

$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=20 / score ≈ 53.880 — minimum uniform E budget



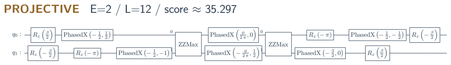
iswap_u

EXACT PHASE E=2 / L=10 / score ≈ 31.260 — minimum uniform E budget

quantinuum_zzmax

EXACT PHASE E=5 / L=31 / score ≈ 89.555 — E upper bound

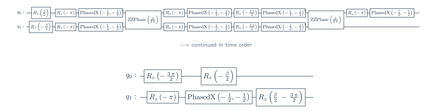
PROJECTIVE E=2 / L=12 / score ≈ 35.297 — minimum uniform E budget



$U = \frac{\sqrt[4]{-1}\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

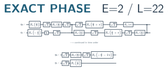
quantinuum_zzphase

EXACT PHASE E=2 / L=21 / score ≈ 55.723 — minimum uniform E budget



rigetti_cz

EXACT PHASE E=2 / L=22 / score ≈ 54.022 — minimum uniform E budget



PROJECTIVE E=2 / L=21 / score ≈ 52.022 — minimum uniform E budget

$U = \sqrt[4]{-\mathbb{I}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=2 / L=10 / score ≈ 29.660 — minimum uniform E budget

sqrt_swap_u

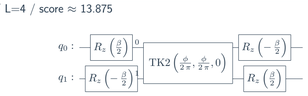
EXACT PHASE E=4 / L=20 / score ≈ 61.720 — E upper bound

PROJECTIVE E=4 / L=20 / score ≈ 61.320 — E upper bound

$U = -i\,C$. Scalar metadata; no physical GP.

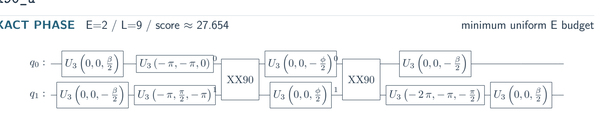
tket_tk2

EXACT PHASE E=1 / L=4 / score ≈ 13.875 — minimum uniform E budget



xx90_u

EXACT PHASE E=2 / L=9 / score ≈ 27.654 — minimum uniform E budget



xx_rz_ry

EXACT PHASE E=2 / L=37 / score ≈ 87.078 — minimum uniform E budget

PROJECTIVE E=2 / L=29 / score ≈ 71.662 — minimum uniform E budget

$U = -i\,C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=2 / L=16 / score ≈ 44.900 — minimum uniform E budget

zz_u

EXACT PHASE E=2 / L=16 / score ≈ 43.700 — minimum uniform E budget

XXPow $(t, global_{shift})$

2 qubits / 32 preserved native bases / alphabetical panel order

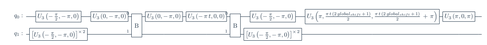
aqt_rxx

PROJECTIVE E=1 / L=0 / score ≈ 4.002 — minimum uniform E budget

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=11 / score ≈ 31.666 — minimum uniform E budget



PROJECTIVE E=2 / L=9 / score ≈ 27.254 — minimum uniform E budget

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

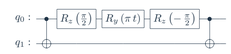
EXACT PHASE E=2 / L=11 / score ≈ 30.010 — minimum uniform E budget

PROJECTIVE E=2 / L=3 / score ≈ 14.002 — minimum uniform E budget

$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

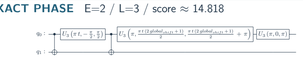
PROJECTIVE E=2 / L=3 / score ≈ 14.406 — minimum uniform E budget



$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=2 / L=3 / score ≈ 14.818 — minimum uniform E budget



PROJECTIVE E=2 / L=1 / score ≈ 10.406 — minimum uniform E budget

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.

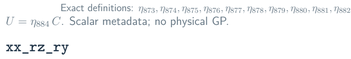
cz_rz_rx

PROJECTIVE E=2 / L=7 / score ≈ 23.214 — minimum uniform E budget

$U = -e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=5 / score ≈ 19.230 — minimum uniform E budget

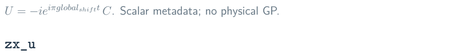


PROJECTIVE E=2 / L=3 / score ≈ 14.818 — minimum uniform E budget

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=2 / L=5 / score ≈ 18.830 — minimum uniform E budget



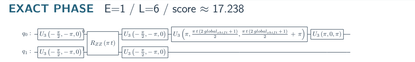
PROJECTIVE E=2 / L=3 / score ≈ 14.818 — minimum uniform E budget

$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=4 / score ≈ 16.632 — E upper bound

PROJECTIVE E=2 / L=2 / score ≈ 12.420 — E upper bound



$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=2 / L=14 / score ≈ 38.875 — minimum uniform E budget

Exact definitions: $\eta_{852}, \eta_{853}, \eta_{854}, \eta_{855}, \eta_{856}, \eta_{857}, \eta_{858}, \eta_{859}, \eta_{860}, \eta_{861}, \eta_{862}, \eta_{863}, \eta_{864}, \eta_{865}, \eta_{866}, \eta_{867}, \eta_{868}, \eta_{869}$ in the expression glossary.

PROJECTIVE E=2 / L=13 / score ≈ 36.869 — minimum uniform E budget

Exact definitions: $\eta_{854}, \eta_{855}, \eta_{856}, \eta_{857}, \eta_{858}, \eta_{859}, \eta_{860}, \eta_{861}, \eta_{862}, \eta_{863}, \eta_{864}, \eta_{865}, \eta_{866}, \eta_{867}, \eta_{868}, \eta_{869}$ in the expression glossary.
$U = \eta_{870}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=15 / score ≈ 81.290 — E upper bound

PROJECTIVE E=12 / L=13 / score ≈ 77.278 — E upper bound

$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

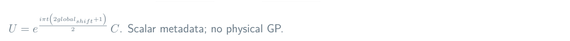
ibm_cx

PROJECTIVE E=2 / L=5 / score ≈ 18.406 — minimum uniform E budget

$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=2 / L=11 / score ≈ 30.810 — minimum uniform E budget



$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

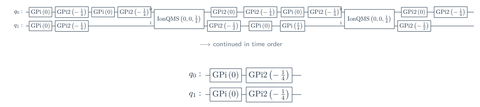
ibm_ecr

PROJECTIVE E=2 / L=4 / score ≈ 16.004 — minimum uniform E budget

$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

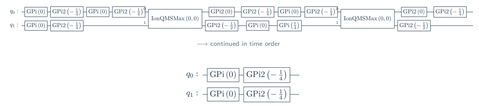
PROJECTIVE E=2 / L=20 / score ≈ 52.852 — E upper bound



$U = ie^{i\pi t\left(global_{shift}+\frac{1}{2}\right)}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=2 / L=20 / score ≈ 52.048 — minimum uniform E budget



$U = ie^{i\pi t\left(global_{shift}+\frac{1}{2}\right)}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=1 / L=8 / score ≈ 21.818 — minimum uniform E budget

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=2 / L=5 / score ≈ 18.820 — minimum uniform E budget

$U = -e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

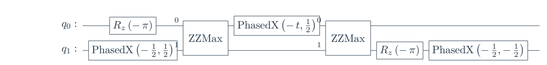
EXACT PHASE E=2 / L=6 / score ≈ 20.836 — minimum uniform E budget

PROJECTIVE E=2 / L=4 / score ≈ 16.424 — minimum uniform E budget

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

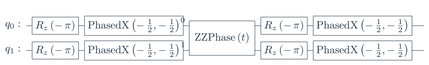
PROJECTIVE E=2 / L=5 / score ≈ 19.016 — minimum uniform E budget



$U = -ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=1 / L=8 / score ≈ 21.626 — minimum uniform E budget



$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=2 / L=11 / score ≈ 30.810 — minimum uniform E budget

$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=2 / L=14 / score ≈ 38.484 — minimum uniform E budget

Exact definitions: $\eta_{871}, \eta_{872}, \eta_{873}, \eta_{874}, \eta_{875}, \eta_{876}, \eta_{877}, \eta_{878}, \eta_{879}, \eta_{880}, \eta_{881}, \eta_{882}, \eta_{883}$ in the expression glossary.

PROJECTIVE E=2 / L=13 / score ≈ 36.478 — minimum uniform E budget

Exact definitions: $\eta_{873}, \eta_{874}, \eta_{875}, \eta_{876}, \eta_{877}, \eta_{878}, \eta_{879}, \eta_{880}, \eta_{881}, \eta_{882}, \eta_{883}$ in the expression glossary.
$U = \eta_{884}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=15 / score ≈ 49.690 — E upper bound

q_0 :
q_1 :
$U_3(\eta_{0,0}, \eta_{0,1}, \eta_{0,2})$
$U_3(\eta_{1,0}, \eta_{1,1}, \eta_{1,2})$
$U_3(\eta_{2,0}, \eta_{2,1}, \eta_{2,2})$
$U_3(\eta_{3,0}, \eta_{3,1}, \eta_{3,2})$
$U_3(\eta_{4,0}, \eta_{4,1}, \eta_{4,2})$
$U_3(\eta_{5,0}, \eta_{5,1}, \eta_{5,2})$
$U_3(\eta_{6,0}, \eta_{6,1}, \eta_{6,2})$
$U_3(\eta_{7,0}, \eta_{7,1}, \eta_{7,2})$
$U_3(\eta_{8,0}, \eta_{8,1}, \eta_{8,2})$
$U_3(\eta_{9,0}, \eta_{9,1}, \eta_{9,2})$
$U_3(\eta_{10,0}, \eta_{10,1}, \eta_{10,2})$
$U_3(\eta_{11,0}, \eta_{11,1}, \eta_{11,2})$
$U_3(\eta_{12,0}, \eta_{12,1}, \eta_{12,2})$
$U_3(\eta_{13,0}, \eta_{13,1}, \eta_{13,2})$
$U_3(\eta_{14,0}, \eta_{14,1}, \eta_{14,2})$
$U_3(\eta_{15,0}, \eta_{15,1}, \eta_{15,2})$
$U_3(\eta_{16,0}, \eta_{16,1}, \eta_{16,2})$
$U_3(\eta_{17,0}, \eta_{17,1}, \eta_{17,2})$
$U_3(\eta_{18,0}, \eta_{18,1}, \eta_{18,2})$
$U_3(\eta_{19,0}, \eta_{19,1}, \eta_{19,2})$
$U_3(\eta_{20,0}, \eta_{20,1}, \eta_{20,2})$
$U_3(\eta_{21,0}, \eta_{21,1}, \eta_{21,2})$
$U_3(\eta_{22,0}, \eta_{22,1}, \eta_{22,2})$
$U_3(\eta_{23,0}, \eta_{23,1}, \eta_{23,2})$
$U_3(\eta_{24,0}, \eta_{24,1}, \eta_{24,2})$
$U_3(\eta_{25,0}, \eta_{25,1}, \eta_{25,2})$
$U_3(\eta_{26,0}, \eta_{26,1}, \eta_{26,2})$
$U_3(\eta_{27,0}, \eta_{27,1}, \eta_{27,2})$
$U_3(\eta_{28,0}, \eta_{28,1}, \eta_{28,2})$
$U_3(\eta_{29,0}, \eta_{29,1}, \eta_{29,2})$
$U_3(\eta_{30,0}, \eta_{30,1}, \eta_{30,2})$
$U_3(\eta_{31,0}, \eta_{31,1}, \eta_{31,2})$
$U_3(\eta_{32,0}, \eta_{32,1}, \eta_{32,2})$
$U_3(\eta_{33,0}, \eta_{33,1}, \eta_{33,2})$
$U_3(\eta_{34,0}, \eta_{34,1}, \eta_{34,2})$
$U_3(\eta_{35,0}, \eta_{35,1}, \eta_{35,2})$
$U_3(\eta_{36,0}, \eta_{36,1}, \eta_{36,2})$
$U_3(\eta_{37,0}, \eta_{37,1}, \eta_{37,2})$
$U_3(\eta_{38,0}, \eta_{38,1}, \eta_{38,2})$
$U_3(\eta_{39,0}, \eta_{39,1}, \eta_{39,2})$
$U_3(\eta_{40,0}, \eta_{40,1}, \eta_{40,2})$
$U_3(\eta_{41,0}, \eta_{41,1}, \eta_{41,2})$
$U_3(\eta_{42,0}, \eta_{42,1}, \eta_{42,2})$
$U_3(\eta_{43,0}, \eta_{43,1}, \eta_{43,2})$
$U_3(\eta_{44,0}, \eta_{44,1}, \eta_{44,2})$
$U_3(\eta_{45,0}, \eta_{45,1}, \eta_{45,2})$
$U_3(\eta_{46,0}, \eta_{46,1}, \eta_{46,2})$
$U_3(\eta_{47,0}, \eta_{47,1}, \eta_{47,2})$
$U_3(\eta_{48,0}, \eta_{48,1}, \eta_{48,2})$
$U_3(\eta_{49,0}, \eta_{49,1}, \eta_{49,2})$
$U_3(\eta_{50,0}, \eta_{50,1}, \eta_{50,2})$
$U_3(\eta_{51,0}, \eta_{51,1}, \eta_{51,2})$
$U_3(\eta_{52,0}, \eta_{52,1}, \eta_{52,2})$
$U_3(\eta_{53,0}, \eta_{53,1}, \eta_{53,2})$
$U_3(\eta_{54,0}, \eta_{54,1}, \eta_{54,2})$
$U_3(\eta_{55,0}, \eta_{55,1}, \eta_{55,2})$
$U_3(\eta_{56,0}, \eta_{56,1}, \eta_{56,2})$
$U_3(\eta_{57,0}, \eta_{57,1}, \eta_{57,2})$
$U_3(\eta_{58,0}, \eta_{58,1}, \eta_{58,2})$
$U_3(\eta_{59,0}, \eta_{59,1}, \eta_{59,2})$
$U_3(\eta_{60,0}, \eta_{60,1}, \eta_{60,2})$
$U_3(\eta_{61,0}, \eta_{61,1}, \eta_{61,2})$
$U_3(\eta_{62,0}, \eta_{62,1}, \eta_{62,2})$
$U_3(\eta_{63,0}, \eta_{63,1}, \eta_{63,2})$
$U_3(\eta_{64,0}, \eta_{64,1}, \eta_{64,2})$
$U_3(\eta_{65,0}, \eta_{65,1}, \eta_{65,2})$
$U_3(\eta_{66,0}, \eta_{66,1}, \eta_{66,2})$
$U_3(\eta_{67,0}, \eta_{67,1}, \eta_{67,2})$
$U_3(\eta_{68,0}, \eta_{68,1}, \eta_{68,2})$
$U_3(\eta_{69,0}, \eta_{69,1}, \eta_{69,2})$
$U_3(\eta_{70,0}, \eta_{70,1}, \eta_{70,2})$
$U_3(\eta_{71,0}, \eta_{71,1}, \eta_{71,2})$
$U_3(\eta_{72,0}, \eta_{72,1}, \eta_{72,2})$
$U_3(\eta_{73,0}, \eta_{73,1}, \eta_{73,2})$
$U_3(\eta_{74,0}, \eta_{74,1}, \eta_{74,2})$
$U_3(\eta_{75,0}, \eta_{75,1}, \eta_{75,2})$
$U_3(\eta_{76,0}, \eta_{76,1}, \eta_{76,2})$
$U_3(\eta_{77,0}, \eta_{77,1}, \eta_{77,2})$
$U_3(\eta_{78,0}, \eta_{78,1}, \eta_{78,2})$
$U_3(\eta_{79,0}, \eta_{79,1}, \eta_{79,2})$
$U_3(\eta_{80,0}, \eta_{80,1}, \eta_{80,2})$
$U_3(\eta_{81,0}, \eta_{81,1}, \eta_{81,2})$
$U_3(\eta_{82,0}, \eta_{82,1}, \eta_{82,2})$
$U_3(\eta_{83,0}, \eta_{83,1}, \eta_{83,2})$
$U_3(\eta_{84,0}, \eta_{84,1}, \eta_{84,2})$
$U_3(\eta_{85,0}, \eta_{85,1}, \eta_{85,2})$
$U_3(\eta_{86,0}, \eta_{86,1}, \eta_{86,2})$
$U_3(\eta_{87,0}, \eta_{87,1}, \eta_{87,2})$
$U_3(\eta_{88,0}, \eta_{88,1}, \eta_{88,2})$
$U_3(\eta_{89,0}, \eta_{89,1}, \eta_{89,2})$
$U_3(\eta_{90,0}, \eta_{90,1}, \eta_{90,2})$
$U_3(\eta_{91,0}, \eta_{91,1}, \eta_{91,2})$
$U_3(\eta_{92,0}, \eta_{92,1}, \eta_{92,2})$
$U_3(\eta_{93,0}, \eta_{93,1}, \eta_{93,2})$
$U_3(\eta_{94,0}, \eta_{94,1}, \eta_{94,2})$
$U_3(\eta_{95,0}, \eta_{95,1}, \eta_{95,2})$
$U_3(\eta_{96,0}, \eta_{96,1}, \eta_{96,2})$
$U_3(\eta_{97,0}, \eta_{97,1}, \eta_{97,2})$
$U_3(\eta_{98,0}, \eta_{98,1}, \eta_{98,2})$
$U_3(\eta_{99,0}, \eta_{99,1}, \eta_{99,2})$
$U_3(\eta_{100,0}, \eta_{100,1}, \eta_{100,2})$
$U_3(\eta_{101,0}, \eta_{101,1}, \eta_{101,2})$
$U_3(\eta_{102,0}, \eta_{102,1}, \eta_{102,2})$
$U_3(\eta_{103,0}, \eta_{103,1}, \eta_{103,2})$
$U_3(\eta_{104,0}, \eta_{104,1}, \eta_{104,2})$
$U_3(\eta_{105,0}, \eta_{105,1}, \eta_{105,2})$
$U_3(\eta_{106,0}, \eta_{106,1}, \eta_{106,2})$
$U_3(\eta_{107,0}, \eta_{107,1}, \eta_{107,2})$
$U_3(\eta_{108,0}, \eta_{108,1}, \eta_{108,2})$
$U_3(\eta_{109,0}, \eta_{109,1}, \eta_{109,2})$
$U_3(\eta_{110,0}, \eta_{110,1}, \eta_{110,2})$
$U_3(\eta_{111,0}, \eta_{111,1}, \eta_{111,2})$
$U_3(\eta_{112,0}, \eta_{112,1}, \eta_{112,2})$
$U_3(\eta_{113,0}, \eta_{113,1}, \eta_{113,2})$
$U_3(\eta_{114,0}, \eta_{114,1}, \eta_{114,2})$
$U_3(\eta_{115,0}, \eta_{115,1}, \eta_{115,2})$
$U_3(\eta_{116,0}, \eta_{116,1}, \eta_{116,2})$
$U_3(\eta_{117,0}, \eta_{117,1}, \eta_{117,2})$
$U_3(\eta_{118,0}, \eta_{118,1}, \eta_{118,2})$
$U_3(\eta_{119,0}, \eta_{119,1}, \eta_{119,2})$
$U_3(\eta_{120,0}, \eta_{120,1}, \eta_{120,2})$
$U_3(\eta_{121,0}, \eta_{121,1}, \eta_{121,2})$
$U_3(\eta_{122,0}, \eta_{122,1}, \eta_{122,2})$
$U_3(\eta_{123,0}, \eta_{123,1}, \eta_{123,2})$
$U_3(\eta_{124,0}, \eta_{124,1}, \eta_{124,2})$
$U_3(\eta_{125,0}, \eta_{125,1}, \eta_{125,2})$
$U_3(\eta_{126,0}, \eta_{126,1}, \eta_{126,2})$
$U_3(\eta_{127,0}, \eta_{127,1}, \eta_{127,2})$
$U_3(\eta_{128,0}, \eta_{128,1}, \eta_{128,2})$
$U_3(\eta_{129,0}, \eta_{129,1}, \eta_{129,2})$
$U_3(\eta_{130,0}, \eta_{130,1}, \eta_{130,2})$
$U_3(\eta_{131,0}, \eta_{131,1}, \eta_{131,2})$
$U_3(\eta_{132,0}, \eta_{132,1}, \eta_{132,2})$
$U_3(\eta_{133,0}, \eta_{133,1}, \eta_{133,2})$
$U_3(\eta_{134,0}, \eta_{134,1}, \eta_{134,2})$
$U_3(\eta_{135,0}, \eta_{135,1}, \eta_{135,2})$
$U_3(\eta_{136,0}, \eta_{136,1}, \eta_{136,2})$
$U_3(\eta_{137,0}, \eta_{137,1}, \eta_{137,2})$
$U_3(\eta_{138,0}, \eta_{138,1}, \eta_{138,2})$
$U_3(\eta_{139,0}, \eta_{139,1}, \eta_{139,2})$
$U_3(\eta_

Y

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_h_phase

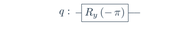
EXACT PHASE E=0 / L=5 / score ≈ 10.406 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.004 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

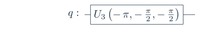
PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum



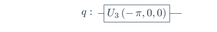
$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum



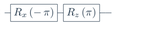
PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum



$U = -i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved E minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_fsim

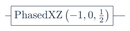
EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

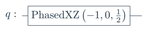


PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved E minimum

ionq_aria

EXACT PHASE E=0 / L=1 / score ≈ 2.402 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=1 / score ≈ 2.402 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=1 / score ≈ 2.402 — proved E minimum

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

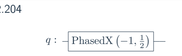
EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

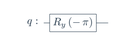
EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum



$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

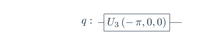
PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum



$U = -i\,C$. Scalar metadata; no physical GP.

Y_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

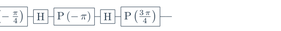
PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.806 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.004 — proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

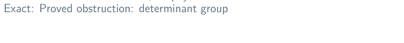
PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

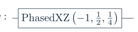
PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

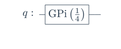
EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum



$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

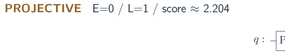
quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Y_q2

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.204 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

clifford_only

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

clifford_t_cz

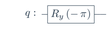
EXACT PHASE E=0 / L=6 / score ≈ 12.000 — proved E minimum

cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.004 — proved E minimum

cx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum



cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.004 — proved E minimum

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

google_cz

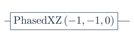
EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum



google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.002 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.002 — proved E minimum

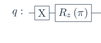
PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.002 — proved E minimum

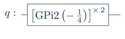
PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum



$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

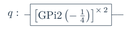


PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.204 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.204 — proved E minimum

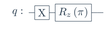
quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.204 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.002 — proved E minimum

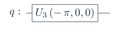
PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum



$U = i\,C$. Scalar metadata; no physical GP.

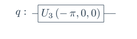
sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum



sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum



tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.204 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

Y_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.806 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.004 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Y_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.406 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.004 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.206 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved E minimum

ionq_aria

EXACT PHASE E=0 / L=1 / score ≈ 2.402 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=1 / score ≈ 2.402 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=1 / score ≈ 2.402 — proved E minimum

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.002 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

Y_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

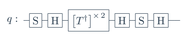
EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum



PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.806 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.004 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

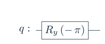
PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

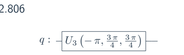
PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum



$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Y_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved E minimum

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

clifford_only

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved E minimum

clifford_t_cz

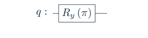
EXACT PHASE E=0 / L=6 / score ≈ 12.000 proved E minimum

cx_h_phase

EXACT PHASE E=0 / L=4 / score ≈ 8.004 proved E minimum

cx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved E minimum



cx_u

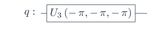
EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.004 proved E minimum

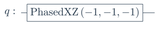
cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum



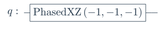
google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum



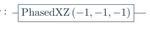
google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum



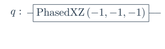
google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum



google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum



ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.002 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.002 proved E minimum

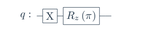
PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.002 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved E minimum



$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

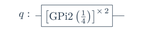
EXACT PHASE E=0 / L=2 / score ≈ 4.804 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 proved E minimum

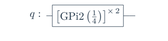


PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.804 proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.402 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved E minimum

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.002 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.204 proved E minimum

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved E minimum

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

Y_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=6 / score ≈ 12.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=5 / score ≈ 10.806 — proved E minimum

PROJECTIVE E=0 / L=4 / score ≈ 8.004 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

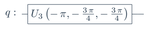
cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

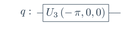
$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=2 / score ≈ 4.004 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

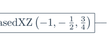
EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

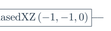
$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum



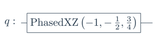
PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

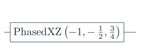


PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

google_syc

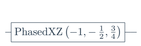
EXACT PHASE E=0 / L=1 / score ≈ 2.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

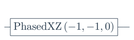
$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved E minimum



PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum



$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=1 / score ≈ 2.402 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

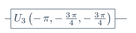
PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum



$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

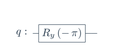
xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\, C$. Scalar metadata; no physical GP.

YPow $(t, global_{shift})$

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved E minimum

$q:$ — $\mathrm{R}\left(\pi\, t, \frac{\pi}{2}\right)$ —

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved E minimum

$q:$ — $U_3(\pi\, t, 0, 0)$ — $U_3\left(\pi, \frac{\pi\,(2\, global_{shift}+1)}{2}, \frac{\pi\,(2\, global_{shift}+1)}{2}+\pi\right)$ — $U_3(\pi, 0, \pi)$ —

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$q:$ — $U_3(\pi\, t, 0, 0)$ —

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=12 / score ≈ 24.412 proved E minimum

$q:$ — $\mathrm{P}\left(-\frac{\pi}{2}\right)$ — H — $\mathrm{P}(\pi\, t)$ — H — $\mathrm{P}\left(\pi\ \left(global_{shift}\, t+\frac{1}{2}\right)\right)$ — H — $\mathrm{P}(\pi)$ — H — $\mathrm{P}(\pi\, global_{shift}\, t)$ — H — $\mathrm{P}(\pi)$ — H —

PROJECTIVE E=0 / L=5 / score ≈ 10.406 proved E minimum

$q:$ — $\mathrm{P}\left(-\frac{\pi}{2}\right)$ — H — $\mathrm{P}(\pi\, t)$ — H — $\mathrm{P}\left(\frac{\pi}{2}\right)$ —

$U = e^{i\pi global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$q:$ — $R_y(\pi\, t)$ —

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved E minimum

$q:$ — $U_3(\pi\, t, 0, 0)$ — $U_3\left(\pi, \frac{\pi\,(2\, global_{shift}+1)}{2}, \frac{\pi\,(2\, global_{shift}+1)}{2}+\pi\right)$ — $U_3(\pi, 0, \pi)$ —

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$q:$ — $U_3(\pi\, t, 0, 0)$ —

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=3 / score ≈ 6.406 proved E minimum

$q:$ — $R_z\left(-\frac{\pi}{2}\right)$ — $R_x(\pi\, t)$ — $R_z\left(\frac{\pi}{2}\right)$ —

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved E minimum

$q:$ — $U_3(\pi\, t, 0, 0)$ — $U_3\left(\pi, \frac{\pi\,(2\, global_{shift}+1)}{2}, \frac{\pi\,(2\, global_{shift}+1)}{2}+\pi\right)$ — $U_3(\pi, 0, \pi)$ —

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$q:$ — $U_3(\pi\, t, 0, 0)$ —

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.218 proved E minimum

$q:$ — $\mathrm{PhasedXZ}\left(t, 0, \frac{1}{2}\right)$ — $\mathrm{PhasedXZ}\left(1, 2\, global_{shift}\, t, -global_{shift}\, t\right)$ — $\mathrm{PhasedXZ}(1, 0, 0)$ —

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved E minimum

$q:$ — $\mathrm{PhasedXZ}\left(t, 0, \frac{1}{2}\right)$ —

$U = e^{i\pi global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=3 / score ≈ 6.218 proved E minimum

$q:$ — $\mathrm{PhasedXZ}\left(t, 0, \frac{1}{2}\right)$ — $\mathrm{PhasedXZ}\left(1, 2\, global_{shift}\, t, -global_{shift}\, t\right)$ — $\mathrm{PhasedXZ}(1, 0, 0)$ —

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved E minimum

$q:$ — $\mathrm{PhasedXZ}\left(t, 0, \frac{1}{2}\right)$ —

$U = e^{i\pi global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=3 / score ≈ 6.218 proved E minimum

$q:$ — $\mathrm{PhasedXZ}\left(t, 0, \frac{1}{2}\right)$ — $\mathrm{PhasedXZ}\left(1, 2\, global_{shift}\, t, -global_{shift}\, t\right)$ — $\mathrm{PhasedXZ}(1, 0, 0)$ —

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved E minimum

$q:$ — $\mathrm{PhasedXZ}\left(t, 0, \frac{1}{2}\right)$ —

$U = e^{i\pi global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=3 / score ≈ 6.218 proved E minimum

$q:$ — $\mathrm{PhasedXZ}\left(t, 0, \frac{1}{2}\right)$ — $\mathrm{PhasedXZ}\left(1, 2\, global_{shift}\, t, -global_{shift}\, t\right)$ — $\mathrm{PhasedXZ}(1, 0, 0)$ —

PROJECTIVE E=0 / L=1 / score ≈ 2.206 proved E minimum

$q:$ — $\mathrm{PhasedXZ}\left(t, 0, \frac{1}{2}\right)$ —

$U = e^{i\pi global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=0 / L=4 / score ≈ 8.004 proved E minimum

$q:$ — $\sqrt{X}$ — $R_z(\pi\, t+\pi)$ — $\sqrt{X}$ — $R_z(\pi)$ —

$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=0 / L=4 / score ≈ 8.004 proved E minimum

$q:$ — $\sqrt{X}$ — $R_z(\pi\, t+\pi)$ — $\sqrt{X}$ — $R_z(\pi)$ —

$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=0 / L=4 / score ≈ 8.004 proved E minimum

$q:$ — $\sqrt{X}$ — $R_z(\pi\, t+\pi)$ — $\sqrt{X}$ — $R_z(\pi)$ —

$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=0 / L=4 / score ≈ 8.608 proved E minimum

$q:$ — $\mathrm{GPi2}(0)$ — $\mathrm{GPi}(0)$ — $\mathrm{GPi}\left(\frac{1}{4}\right)$ — $\mathrm{GPi2}\left(\frac{1}{2}\right)$ —

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=4 / score ≈ 8.608 proved E minimum

$q:$ — $\mathrm{GPi2}(0)$ — $\mathrm{GPi}(0)$ — $\mathrm{GPi}\left(\frac{1}{4}\right)$ — $\mathrm{GPi2}\left(\frac{1}{2}\right)$ —

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=4 / score ≈ 8.608 proved E minimum

$q:$ — $\mathrm{GPi2}(0)$ — $\mathrm{GPi}(0)$ — $\mathrm{GPi}\left(\frac{1}{4}\right)$ — $\mathrm{GPi2}\left(\frac{1}{2}\right)$ —

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved E minimum

$q:$ — $\mathrm{R}\left(\pi\, t, \frac{\pi}{2}\right)$ —

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved E minimum

$q:$ — $U_3(\pi\, t, 0, 0)$ — $U_3\left(\pi, \frac{\pi\,(2\, global_{shift}+1)}{2}, \frac{\pi\,(2\, global_{shift}+1)}{2}+\pi\right)$ — $U_3(\pi, 0, \pi)$ —

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$q:$ — $U_3(\pi\, t, 0, 0)$ —

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved E minimum

$q:$ — $\mathrm{PhasedX}\left(t, \frac{1}{2}\right)$ —

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved E minimum

$q:$ — $\mathrm{PhasedX}\left(t, \frac{1}{2}\right)$ —

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=0 / L=4 / score ≈ 8.004 proved E minimum

$q:$ — $\sqrt{X}$ — $R_z(\pi\, t+\pi)$ — $\sqrt{X}$ — $R_z(\pi)$ —

$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved E minimum

$q:$ — $U_3(\pi\, t, 0, 0)$ — $U_3\left(\pi, \frac{\pi\,(2\, global_{shift}+1)}{2}, \frac{\pi\,(2\, global_{shift}+1)}{2}+\pi\right)$ — $U_3(\pi, 0, \pi)$ —

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$q:$ — $U_3(\pi\, t, 0, 0)$ —

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved E minimum

$q:$ — $U_3(\pi\, t, 0, 0)$ — $U_3\left(\pi, \frac{\pi\,(2\, global_{shift}+1)}{2}, \frac{\pi\,(2\, global_{shift}+1)}{2}+\pi\right)$ — $U_3(\pi, 0, \pi)$ —

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$q:$ — $U_3(\pi\, t, 0, 0)$ —

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.204 proved E minimum

$q:$ — $\mathrm{PhasedX}\left(t, \frac{1}{2}\right)$ —

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved E minimum

$q:$ — $U_3(\pi\, t, 0, 0)$ — $U_3\left(\pi, \frac{\pi\,(2\, global_{shift}+1)}{2}, \frac{\pi\,(2\, global_{shift}+1)}{2}+\pi\right)$ — $U_3(\pi, 0, \pi)$ —

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$q:$ — $U_3(\pi\, t, 0, 0)$ —

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$q:$ — $R_y(\pi\, t)$ —

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved E minimum

$q:$ — $U_3(\pi\, t, 0, 0)$ — $U_3\left(\pi, \frac{\pi\,(2\, global_{shift}+1)}{2}, \frac{\pi\,(2\, global_{shift}+1)}{2}+\pi\right)$ — $U_3(\pi, 0, \pi)$ —

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$q:$ — $U_3(\pi\, t, 0, 0)$ —

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.418 proved E minimum

$q:$ — $U_3(\pi\, t, 0, 0)$ — $U_3\left(\pi, \frac{\pi\,(2\, global_{shift}+1)}{2}, \frac{\pi\,(2\, global_{shift}+1)}{2}+\pi\right)$ — $U_3(\pi, 0, \pi)$ —

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$q:$ — $U_3(\pi\, t, 0, 0)$ —

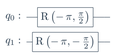
$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.

YY

2 qubits / 32 preserved native bases / alphabetical panel order

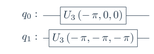
aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.408 — E upper bound



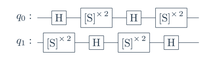
berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved E minimum



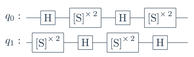
clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — E upper bound



clifford_t_cx

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — E upper bound



clifford_t_cz

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — E upper bound

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.008 — E upper bound

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.004 — E upper bound

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound

cz_rz_rx

EXACT PHASE E=0 / L=4 / score ≈ 8.008 — E upper bound

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved E minimum

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound

ibm_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.004 — E upper bound

ibm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.004 — E upper bound

ibm_ecr

EXACT PHASE E=0 / L=4 / score ≈ 8.004 — E upper bound

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — E upper bound

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — E upper bound

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.804 — E upper bound

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.408 — E upper bound

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound

quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.408 — E upper bound

quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.408 — E upper bound

rigetti_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.004 — E upper bound

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound

tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.408 — E upper bound

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound

xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.004 — E upper bound

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound

zz_u

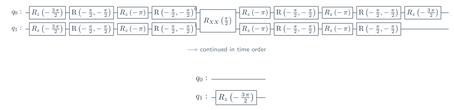
EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound

YY90

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=1 / L=18 / score ≈ 44.254 — proved E minimum



PROJECTIVE E=1 / L=8 / score ≈ 20.000 — proved E minimum

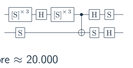
$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=2 / L=4 / score ≈ 16.024 — proved E minimum

clifford_only

EXACT PHASE E=1 / L=12 / score ≈ 28.000 — proved E minimum



PROJECTIVE E=1 / L=8 / score ≈ 20.000 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=1 / L=12 / score ≈ 28.000 — proved E minimum

PROJECTIVE E=1 / L=8 / score ≈ 20.000 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

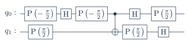
EXACT PHASE E=1 / L=19 / score ≈ 42.000 — proved E minimum

PROJECTIVE E=1 / L=14 / score ≈ 32.000 — proved E minimum

$U = \frac{\sqrt{2}(-1-i)}{2}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=1 / L=8 / score ≈ 21.010 — proved E minimum



PROJECTIVE E=1 / L=8 / score ≈ 20.808 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

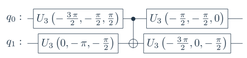
EXACT PHASE E=2 / L=11 / score ≈ 31.822 — E upper bound

PROJECTIVE E=1 / L=8 / score ≈ 21.616 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

cx_u

EXACT PHASE E=1 / L=4 / score ≈ 13.624 — proved E minimum



cz_rz_rx

EXACT PHASE E=2 / L=20 / score ≈ 51.040 — E upper bound

PROJECTIVE E=1 / L=6 / score ≈ 17.212 — proved E minimum

$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=1 / L=5 / score ≈ 16.030 — proved E minimum

PROJECTIVE E=1 / L=4 / score ≈ 14.024 — proved E minimum

$U = \frac{\sqrt{2}i(1+i)}{2}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=1 / L=5 / score ≈ 15.630 — proved E minimum

PROJECTIVE E=1 / L=4 / score ≈ 13.624 — proved E minimum

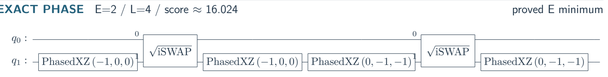
$U = \frac{\sqrt{2}(-1-i)}{2}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=2 / score ≈ 12.820 — E upper bound

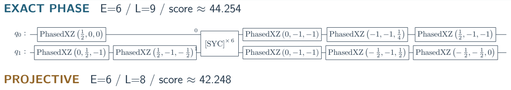
google_sqrt_iswap

EXACT PHASE E=2 / L=4 / score ≈ 16.024 — proved E minimum

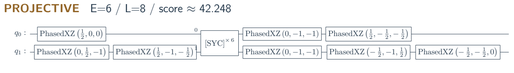


google_syc

EXACT PHASE E=6 / L=9 / score ≈ 44.254 — E upper bound



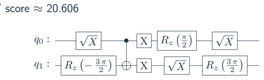
PROJECTIVE E=6 / L=8 / score ≈ 42.248 — E upper bound



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

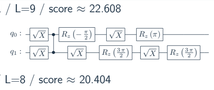
ibm_cx

EXACT PHASE E=1 / L=8 / score ≈ 20.606 — proved E minimum



ibm_cz

EXACT PHASE E=1 / L=9 / score ≈ 22.608 — proved E minimum



PROJECTIVE E=1 / L=8 / score ≈ 20.404 — proved E minimum

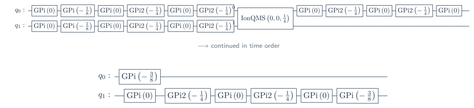
$U = \frac{\sqrt{2}(-1-i)}{2}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=1 / L=6 / score ≈ 16.406 — proved E minimum

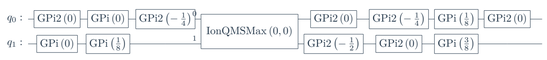
ionq_aria

EXACT PHASE E=1 / L=24 / score ≈ 58.054 — proved E minimum



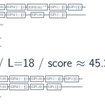
ionq_aria_full

EXACT PHASE E=1 / L=12 / score ≈ 30.828 — proved E minimum



ionq_forte

EXACT PHASE E=1 / L=19 / score ≈ 47.240 — proved E minimum

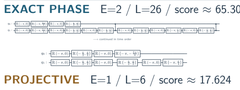


PROJECTIVE E=1 / L=18 / score ≈ 45.238 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=26 / score ≈ 65.304 — E upper bound

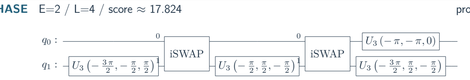


PROJECTIVE E=1 / L=6 / score ≈ 17.624 — proved E minimum

$U = \frac{\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.

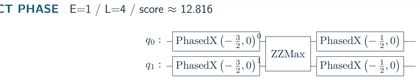
iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 17.824 — proved E minimum



quantinuum_zzmax

EXACT PHASE E=1 / L=4 / score ≈ 12.816 — proved E minimum



quantinuum_zzphase

EXACT PHASE E=1 / L=10 / score ≈ 26.630 — proved E minimum

rigetti_cz

EXACT PHASE E=1 / L=9 / score ≈ 22.608 — proved E minimum

PROJECTIVE E=1 / L=8 / score ≈ 20.404 — proved E minimum

$U = \frac{\sqrt{2}(-1-i)}{2}\,C$. Scalar metadata; no physical GP.

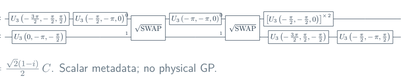
sqrt_iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 16.024 — proved E minimum

sqrt_swap_u

EXACT PHASE E=2 / L=8 / score ≈ 27.648 — proved E minimum

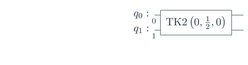
PROJECTIVE E=2 / L=8 / score ≈ 26.848 — proved E minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

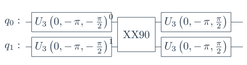
tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.206 — proved E minimum



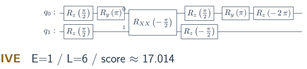
xx90_u

EXACT PHASE E=1 / L=4 / score ≈ 12.824 — proved E minimum



xx_rz_ry

EXACT PHASE E=1 / L=7 / score ≈ 19.016 — proved E minimum



PROJECTIVE E=1 / L=6 / score ≈ 17.014 — proved E minimum

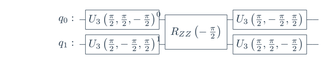
$U = -1\,C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=1 / L=4 / score ≈ 14.226 — proved E minimum

zz_u

EXACT PHASE E=1 / L=4 / score ≈ 14.626 — proved E minimum

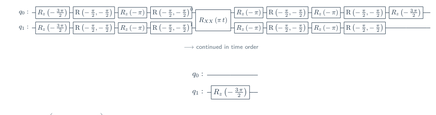


YYPow $(t, global_{shift})$

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=1 / L=18 / score ≈ 44.054 — minimum uniform E budget



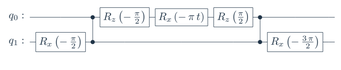
$U = e^{\frac{i\pi t\left(2g\mathrm{lobal}_{\mathrm{shift}}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cz_rz_rx

PROJECTIVE E=2 / L=5 / score ≈ 18.810 — minimum uniform E budget



$U = -e^{\frac{i\pi t\left(2g\mathrm{lobal}_{\mathrm{shift}}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

google_sqrt_iswap

EXACT PHASE E=2 / L=16 / score ≈ 43.287 — minimum uniform E budget

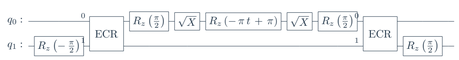
Exact definitions: $\eta_{854}, \eta_{855}, \eta_{859}, \eta_{860}, \eta_{861}, \eta_{862}, \eta_{863}, \eta_{864}, \eta_{865}, \eta_{868}, \eta_{869}, \eta_{885}, \eta_{886}, \eta_{887}, \eta_{888}, \eta_{889}, \eta_{890}, \eta_{891}$ in the expression glossary.

PROJECTIVE E=2 / L=15 / score ≈ 41.281 — minimum uniform E budget

Exact definitions: $\eta_{854}, \eta_{855}, \eta_{859}, \eta_{860}, \eta_{861}, \eta_{862}, \eta_{863}, \eta_{864}, \eta_{865}, \eta_{868}, \eta_{869}, \eta_{887}, \eta_{888}, \eta_{889}, \eta_{890}, \eta_{891}$ in the expression glossary.
$U = \eta_{870}\, C$. Scalar metadata; no physical GP.

ibm_ecr

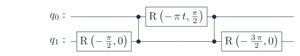
PROJECTIVE E=2 / L=7 / score ≈ 22.810 — minimum uniform E budget



$U = ie^{\frac{i\pi t\left(2g\mathrm{lobal}_{\mathrm{shift}}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

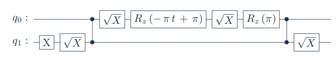
PROJECTIVE E=2 / L=3 / score ≈ 14.612 — minimum uniform E budget



$U = -e^{\frac{i\pi t\left(2g\mathrm{lobal}_{\mathrm{shift}}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

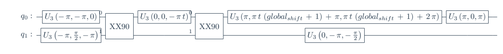
PROJECTIVE E=2 / L=7 / score ≈ 22.004 — minimum uniform E budget



$U = ie^{\frac{i\pi t\left(2g\mathrm{lobal}_{\mathrm{shift}}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=2 / L=6 / score ≈ 20.436 — minimum uniform E budget



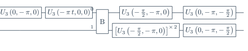
PROJECTIVE E=2 / L=4 / score ≈ 16.424 — minimum uniform E budget

$U = \eta_{897}\, C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=2 / L=15 / score ≈ 40.490 — minimum uniform E budget

PROJECTIVE E=2 / L=13 / score ≈ 36.078 — minimum uniform E budget



$U = e^{\frac{i\pi t\left(2g\mathrm{lobal}_{\mathrm{shift}}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.

cx_h_phase

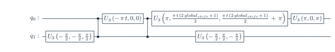
EXACT PHASE E=2 / L=15 / score ≈ 38.818 — minimum uniform E budget

PROJECTIVE E=2 / L=7 / score ≈ 22.810 — minimum uniform E budget

$U = e^{i\pi g\mathrm{lobal}_{\mathrm{shift}}t}\, C$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=2 / L=5 / score ≈ 19.630 — minimum uniform E budget

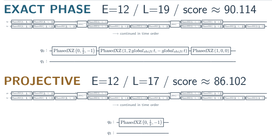


PROJECTIVE E=2 / L=3 / score ≈ 15.218 — minimum uniform E budget

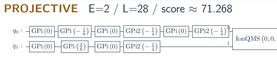
$U = e^{\frac{i\pi t\left(2g\mathrm{lobal}_{\mathrm{shift}}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=19 / score ≈ 90.114 — E upper bound



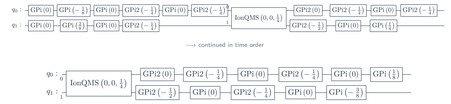
PROJECTIVE E=12 / L=17 / score ≈ 86.102 — E upper bound



$U = e^{i\pi g\mathrm{lobal}_{\mathrm{shift}}t}\, C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=2 / L=28 / score ≈ 71.268 — E upper bound



$U = ie^{i\pi t\left(\mathrm{global}_{\mathrm{shift}}t+\frac{1}{2}\right)}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 22.036 — minimum uniform E budget

PROJECTIVE E=2 / L=4 / score ≈ 17.624 — minimum uniform E budget

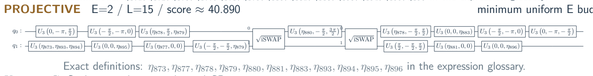
$U = \eta_{892}\, C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=2 / L=16 / score ≈ 42.896 — minimum uniform E budget

Exact definitions: $\eta_{871}, \eta_{872}, \eta_{873}, \eta_{877}, \eta_{878}, \eta_{879}, \eta_{880}, \eta_{881}, \eta_{883}, \eta_{893}, \eta_{894}, \eta_{895}, \eta_{896}$ in the expression glossary.

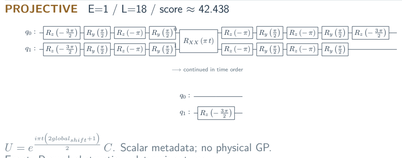
PROJECTIVE E=2 / L=15 / score ≈ 40.890 — minimum uniform E budget



Exact definitions: $\eta_{873}, \eta_{877}, \eta_{878}, \eta_{879}, \eta_{880}, \eta_{881}, \eta_{883}, \eta_{893}, \eta_{894}, \eta_{895}, \eta_{896}$ in the expression glossary.
$U = \eta_{884}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=1 / L=18 / score ≈ 42.438 — minimum uniform E budget



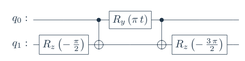
$U = e^{\frac{i\pi t\left(2g\mathrm{lobal}_{\mathrm{shift}}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_rz_ry

PROJECTIVE E=2 / L=3 / score ≈ 14.406 — minimum uniform E budget



$U = -e^{\frac{i\pi t\left(2g\mathrm{lobal}_{\mathrm{shift}}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

google_cz

EXACT PHASE E=2 / L=5 / score ≈ 18.630 — minimum uniform E budget

PROJECTIVE E=2 / L=3 / score ≈ 14.618 — minimum uniform E budget

$U = e^{i\pi t(g\mathrm{lobal}_{\mathrm{shift}}+1)}\, C$. Scalar metadata; no physical GP.

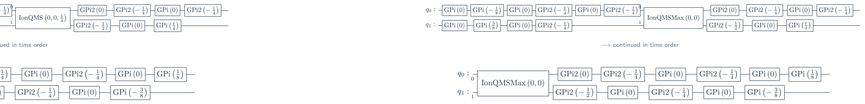
ibm_cx

PROJECTIVE E=2 / L=6 / score ≈ 20.408 — minimum uniform E budget

$U = ie^{\frac{i\pi t\left(2g\mathrm{lobal}_{\mathrm{shift}}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=2 / L=28 / score ≈ 70.464 — minimum uniform E budget



$U = ie^{i\pi t\left(\mathrm{global}_{\mathrm{shift}}t+\frac{1}{2}\right)}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzmax

PROJECTIVE E=2 / L=5 / score ≈ 18.416 — minimum uniform E budget

$U = \eta_{499}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_swap_u

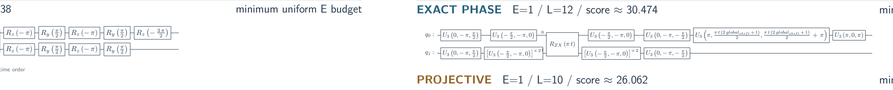
EXACT PHASE E=4 / L=19 / score ≈ 58.514 — E upper bound

PROJECTIVE E=4 / L=17 / score ≈ 54.102 — E upper bound

$U = -ie^{i\pi g\mathrm{lobal}_{\mathrm{shift}}t}\, C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=1 / L=12 / score ≈ 30.474 — minimum uniform E budget



PROJECTIVE E=1 / L=10 / score ≈ 26.062 — minimum uniform E budget

$U = e^{\frac{i\pi t\left(2g\mathrm{lobal}_{\mathrm{shift}}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.

ionq_forte

PROJECTIVE E=1 / L=16 / score ≈ 40.234 — minimum uniform E budget

$U = -e^{\frac{i\pi t\left(2g\mathrm{lobal}_{\mathrm{shift}}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

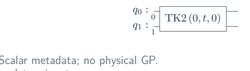
quantinuum_zzphase

PROJECTIVE E=1 / L=10 / score ≈ 26.430 — minimum uniform E budget

$U = e^{\frac{i\pi t\left(2g\mathrm{lobal}_{\mathrm{shift}}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

tket_tk2

PROJECTIVE E=1 / L=0 / score ≈ 4.006 — minimum uniform E budget



$U = e^{\frac{i\pi t\left(2g\mathrm{lobal}_{\mathrm{shift}}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zz_u

EXACT PHASE E=1 / L=10 / score ≈ 26.062 — minimum uniform E budget

PROJECTIVE E=1 / L=8 / score ≈ 21.650 — minimum uniform E budget

$U = e^{\frac{i\pi t\left(2g\mathrm{lobal}_{\mathrm{shift}}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.

Z

1 qubits / 32 preserved native bases / alphabetical panel order

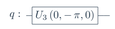
aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum



clifford_only

EXACT PHASE E=0 / L=2 / score ≈ 4.000 proved E minimum

clifford_t_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.000 proved E minimum

clifford_t_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.000 proved E minimum

cx_h_phase

EXACT PHASE E=0 / L=1 / score ≈ 2.002 proved E minimum

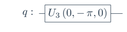
cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum



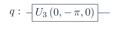
cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

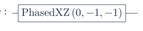
cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum



google_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum



google_fsim

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

google_sqrt_iswap

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

google_syc

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.002 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.002 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.002 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.406 proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.208 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.002 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 proved E minimum

Z_q1

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.208 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Z_q2

1 qubits / 32 preserved native bases / alphabetical panel order

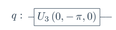
aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum



$U = i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

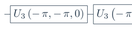
PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum



$U = i\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

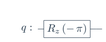
cz_rz_rx

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum



$U = i\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

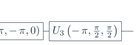
PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum



$U = i\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.208 — proved E minimum

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

Z_q3

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.208 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Z_q4

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.006 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -1\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=0 / L=3 / score ≈ 6.406 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=0 / L=3 / score ≈ 6.406 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=0 / L=3 / score ≈ 6.406 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.208 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.002 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

sqrt_swap_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = i\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

zz_u

EXACT PHASE E=0 / L=1 / score ≈ 2.006 — proved E minimum

Z_q5

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=14 / score ≈ 28.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.806 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

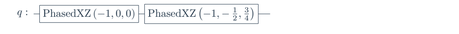
cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

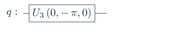
$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

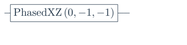
PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

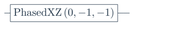


$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

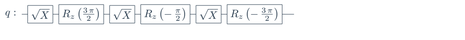
PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum



$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.208 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

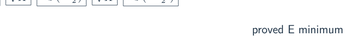
tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

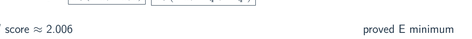
xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

Z_q6

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=12 / score ≈ 24.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=7 / score ≈ 14.406 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

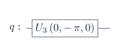
cz_rz_rx

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum



$U = -iC$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.212 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

ibm_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

ibm_ecr

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

ionq_aria

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

ionq_aria_full

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

ionq_forte

EXACT PHASE E=0 / L=2 / score ≈ 4.404 — proved E minimum

iqm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.208 — proved E minimum

iswap_u

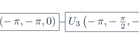
EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum



quantinuum_zzphase

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

rigetti_cz

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

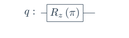
xx_rz_ry

EXACT PHASE E=0 / L=1 / score ≈ 2.002 — proved E minimum

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum



$U = -iC$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.412 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

Z_q7

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=0 / L=10 / score ≈ 20.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=0 / L=8 / score ≈ 16.000 — proved E minimum

PROJECTIVE E=0 / L=2 / score ≈ 4.000 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=0 / L=6 / score ≈ 12.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.612 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=0 / L=2 / score ≈ 4.208 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

EXACT PHASE E=0 / L=7 / score ≈ 14.606 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 — proved E minimum

$U = e^{-\frac{3i\pi}{4}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.812 — proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 — proved E minimum

$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

ZPow $(t, global_{shift})$

1 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

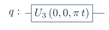
PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum



$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=0 / L=8 / score ≈ 16.008 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=0 / L=2 / score ≈ 4.404 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

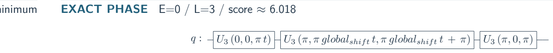
iqm_cz

PROJECTIVE E=0 / L=3 / score ≈ 6.812 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum



PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

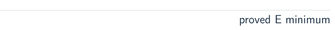
quantinuum_zzphase

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum



$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=0 / L=1 / score ≈ 2.002 proved E minimum

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=0 / L=3 / score ≈ 6.018 proved E minimum

PROJECTIVE E=0 / L=1 / score ≈ 2.006 proved E minimum

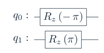
$U = e^{i\pi global_{shift} t}\, C$. Scalar metadata; no physical GP.

ZZ

2 qubits / 32 preserved native bases / alphabetical panel order

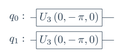
aqt_rxx

EXACT PHASE E=0 / L=2 / score ≈ 4.004 — E upper bound



berkeley_b

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved E minimum



clifford_only

EXACT PHASE E=0 / L=4 / score ≈ 8.000 — E upper bound

clifford_t_cx

EXACT PHASE E=0 / L=4 / score ≈ 8.000 — E upper bound

clifford_t_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.000 — E upper bound

cx_h_phase

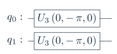
EXACT PHASE E=0 / L=2 / score ≈ 4.004 — E upper bound

cx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.004 — E upper bound

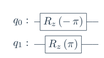
cx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound



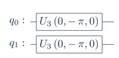
cz_rz_rx

EXACT PHASE E=0 / L=2 / score ≈ 4.004 — E upper bound



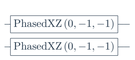
cz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound



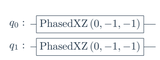
google_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound



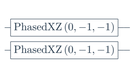
google_fsim

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound



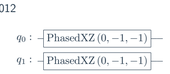
google_sqrt_iswap

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved E minimum



google_syc

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound

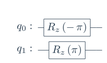


ibm_cx

EXACT PHASE E=0 / L=2 / score ≈ 4.004 — E upper bound

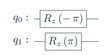
ibm_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.004 — E upper bound



ibm_ecr

EXACT PHASE E=0 / L=2 / score ≈ 4.004 — E upper bound

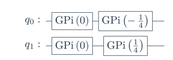


ionq_aria

EXACT PHASE E=0 / L=4 / score ≈ 8.808 — E upper bound

ionq_aria_full

EXACT PHASE E=0 / L=4 / score ≈ 8.808 — E upper bound

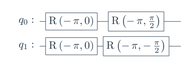


ionq_forte

EXACT PHASE E=0 / L=4 / score ≈ 8.808 — E upper bound

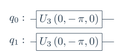
iqm_cz

EXACT PHASE E=0 / L=4 / score ≈ 8.416 — E upper bound



iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound

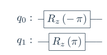


quantinuum_zzmax

EXACT PHASE E=0 / L=2 / score ≈ 4.004 — E upper bound

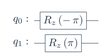
quantinuum_zzphase

EXACT PHASE E=0 / L=2 / score ≈ 4.004 — E upper bound



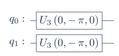
rigetti_cz

EXACT PHASE E=0 / L=2 / score ≈ 4.004 — E upper bound



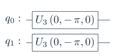
sqrt_iswap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — proved E minimum



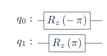
sqrt_swap_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound



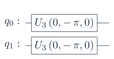
tket_tk2

EXACT PHASE E=0 / L=2 / score ≈ 4.004 — E upper bound



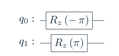
xx90_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound



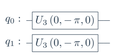
xx_rz_ry

EXACT PHASE E=0 / L=2 / score ≈ 4.004 — E upper bound



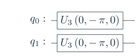
zx_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound



zz_u

EXACT PHASE E=0 / L=2 / score ≈ 4.012 — E upper bound

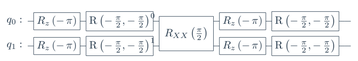


ZZ90

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

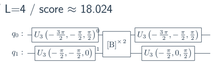
EXACT PHASE E=1 / L=8 / score ≈ 21.826 — proved E minimum



berkeley_b

EXACT PHASE E=2 / L=6 / score ≈ 22.436 — proved E minimum

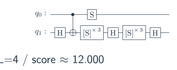
PROJECTIVE E=2 / L=4 / score ≈ 18.024 — proved E minimum



$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=1 / L=10 / score ≈ 24.000 — proved E minimum

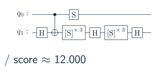


PROJECTIVE E=1 / L=4 / score ≈ 12.000 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=1 / L=10 / score ≈ 24.000 — proved E minimum



PROJECTIVE E=1 / L=4 / score ≈ 12.000 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

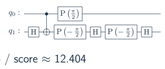
EXACT PHASE E=1 / L=10 / score ≈ 24.000 — proved E minimum

PROJECTIVE E=1 / L=2 / score ≈ 8.000 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=1 / L=6 / score ≈ 16.606 — proved E minimum



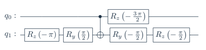
PROJECTIVE E=1 / L=4 / score ≈ 12.404 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=2 / L=1 / score ≈ 10.202 — E upper bound

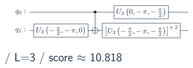
PROJECTIVE E=1 / L=5 / score ≈ 14.810 — proved E minimum



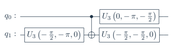
$U = \frac{\sqrt{2}i(1-i)}{2}\,C$. Scalar metadata; no physical GP.

cx_u

EXACT PHASE E=1 / L=4 / score ≈ 13.224 — proved E minimum



PROJECTIVE E=1 / L=3 / score ≈ 10.818 — proved E minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=2 / L=11 / score ≈ 32.222 — E upper bound

PROJECTIVE E=1 / L=2 / score ≈ 8.404 — proved E minimum

$U = \frac{\sqrt{2}i(1-i)}{2}\,C$. Scalar metadata; no physical GP.

cz_u

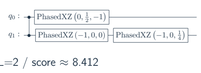
EXACT PHASE E=1 / L=3 / score ≈ 11.018 — proved E minimum

PROJECTIVE E=1 / L=2 / score ≈ 8.412 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=1 / L=3 / score ≈ 10.618 — proved E minimum

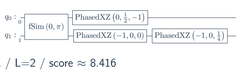


PROJECTIVE E=1 / L=2 / score ≈ 8.412 — proved E minimum

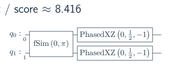
$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=3 / score ≈ 10.622 — proved E minimum



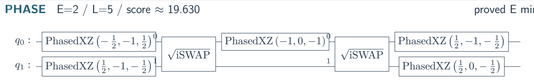
PROJECTIVE E=1 / L=2 / score ≈ 8.416 — proved E minimum



$U = -(-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=2 / L=5 / score ≈ 19.630 — proved E minimum



google_syc

EXACT PHASE E=6 / L=8 / score ≈ 42.048 — E upper bound

PROJECTIVE E=6 / L=7 / score ≈ 40.042 — E upper bound

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=1 / L=7 / score ≈ 18.606 — proved E minimum

ibm_cz

EXACT PHASE E=1 / L=7 / score ≈ 18.606 — proved E minimum

PROJECTIVE E=1 / L=2 / score ≈ 8.404 — proved E minimum

$U = \frac{\sqrt{2}i(-1+i)}{2}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=1 / L=7 / score ≈ 18.404 — proved E minimum

ionq_aria

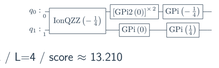
EXACT PHASE E=1 / L=8 / score ≈ 22.022 — proved E minimum

ionq_aria_full

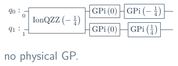
EXACT PHASE E=1 / L=11 / score ≈ 28.626 — proved E minimum

ionq_forte

EXACT PHASE E=1 / L=5 / score ≈ 15.212 — proved E minimum



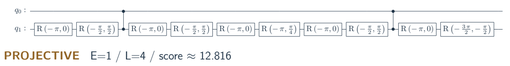
PROJECTIVE E=1 / L=4 / score ≈ 13.210 — proved E minimum



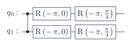
$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=10 / score ≈ 30.040 — E upper bound



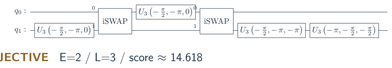
PROJECTIVE E=1 / L=4 / score ≈ 12.816 — proved E minimum



$U = \frac{\sqrt{2}i(1-i)}{2}\,C$. Scalar metadata; no physical GP.

iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 17.024 — proved E minimum



PROJECTIVE E=2 / L=3 / score ≈ 14.618 — proved E minimum

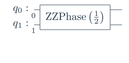
$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=1 / L=0 / score ≈ 4.202 — proved E minimum



rigetti_cz

EXACT PHASE E=1 / L=7 / score ≈ 18.606 — proved E minimum

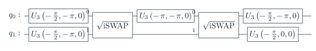
PROJECTIVE E=1 / L=2 / score ≈ 8.404 — proved E minimum

$U = \frac{\sqrt{2}i(-1+i)}{2}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=2 / L=7 / score ≈ 24.442 — proved E minimum

PROJECTIVE E=2 / L=5 / score ≈ 18.830 — proved E minimum



$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=2 / L=7 / score ≈ 24.442 — proved E minimum

PROJECTIVE E=2 / L=7 / score ≈ 24.042 — proved E minimum

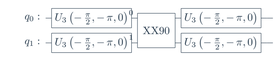
$U = \eta_l\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.206 — proved E minimum

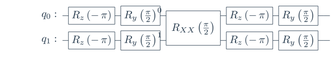
xx90_u

EXACT PHASE E=1 / L=4 / score ≈ 12.824 — proved E minimum



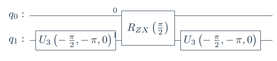
xx_rz_ry

EXACT PHASE E=1 / L=8 / score ≈ 21.018 — proved E minimum



zx_u

EXACT PHASE E=1 / L=2 / score ≈ 8.614 — proved E minimum



zz_u

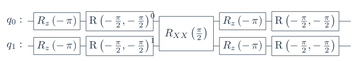
EXACT PHASE E=1 / L=0 / score ≈ 4.202 — proved E minimum

ZZMax

2 qubits / 32 preserved native bases / alphabetical panel order

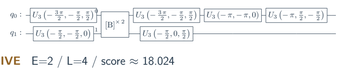
aqt_rxx

EXACT PHASE E=1 / L=8 / score ≈ 21.826 — proved E minimum

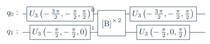


berkeley_b

EXACT PHASE E=2 / L=6 / score ≈ 22.436 — proved E minimum



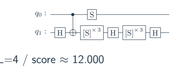
PROJECTIVE E=2 / L=4 / score ≈ 18.024 — proved E minimum



$U = -i\,C$. Scalar metadata; no physical GP.

clifford_only

EXACT PHASE E=1 / L=10 / score ≈ 24.000 — proved E minimum

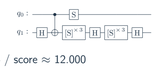


PROJECTIVE E=1 / L=4 / score ≈ 12.000 — proved E minimum

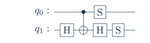
$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

clifford_t_cx

EXACT PHASE E=1 / L=10 / score ≈ 24.000 — proved E minimum



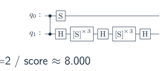
PROJECTIVE E=1 / L=4 / score ≈ 12.000 — proved E minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=1 / L=10 / score ≈ 24.000 — proved E minimum



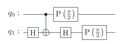
PROJECTIVE E=1 / L=2 / score ≈ 8.000 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=1 / L=6 / score ≈ 16.606 — proved E minimum

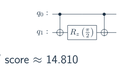
PROJECTIVE E=1 / L=4 / score ≈ 12.404 — proved E minimum



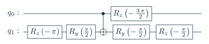
$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=2 / L=1 / score ≈ 10.202 — E upper bound



PROJECTIVE E=1 / L=5 / score ≈ 14.810 — proved E minimum

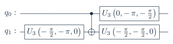


$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

cx_u

EXACT PHASE E=1 / L=4 / score ≈ 13.224 — proved E minimum

PROJECTIVE E=1 / L=3 / score ≈ 10.818 — proved E minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=2 / L=11 / score ≈ 32.222 — E upper bound

PROJECTIVE E=1 / L=2 / score ≈ 8.404 — proved E minimum

$U = \frac{\sqrt{2}i(1-i)}{2}\,C$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=1 / L=3 / score ≈ 11.018 — proved E minimum

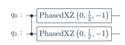
PROJECTIVE E=1 / L=2 / score ≈ 8.412 — proved E minimum

$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=1 / L=3 / score ≈ 10.618 — proved E minimum

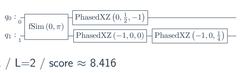
PROJECTIVE E=1 / L=2 / score ≈ 8.412 — proved E minimum



$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=3 / score ≈ 10.622 — proved E minimum

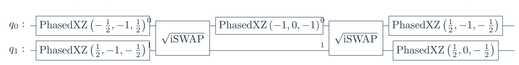


PROJECTIVE E=1 / L=2 / score ≈ 8.416 — proved E minimum

$U = -(-1)^{\frac{3}{4}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

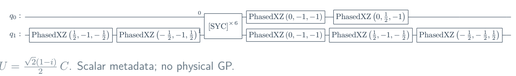
EXACT PHASE E=2 / L=5 / score ≈ 19.630 — proved E minimum



google_syc

EXACT PHASE E=6 / L=8 / score ≈ 42.048 — E upper bound

PROJECTIVE E=6 / L=7 / score ≈ 40.042 — E upper bound



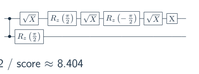
$U = \frac{\sqrt{2}(1-i)}{2}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=1 / L=7 / score ≈ 18.606 — proved E minimum

ibm_cz

EXACT PHASE E=1 / L=7 / score ≈ 18.606 — proved E minimum

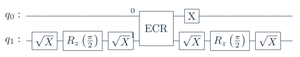


PROJECTIVE E=1 / L=2 / score ≈ 8.404 — proved E minimum

$U = \frac{\sqrt{2}i(-1+i)}{2}\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=1 / L=7 / score ≈ 18.404 — proved E minimum



ionq_aria

EXACT PHASE E=1 / L=8 / score ≈ 22.022 — proved E minimum

ionq_aria_full

EXACT PHASE E=1 / L=11 / score ≈ 28.626 — proved E minimum

ionq_forte

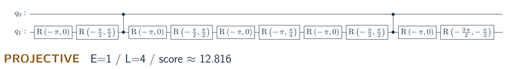
EXACT PHASE E=1 / L=5 / score ≈ 15.212 — proved E minimum

PROJECTIVE E=1 / L=4 / score ≈ 13.210 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

iqm_cz

EXACT PHASE E=2 / L=10 / score ≈ 30.040 — E upper bound



PROJECTIVE E=1 / L=4 / score ≈ 12.816 — proved E minimum

$U = \frac{\sqrt{2}i(1-i)}{2}\,C$. Scalar metadata; no physical GP.

iswap_u

EXACT PHASE E=2 / L=4 / score ≈ 17.024 — proved E minimum

PROJECTIVE E=2 / L=3 / score ≈ 14.618 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

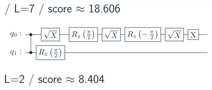
EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

quantinuum_zzphase

EXACT PHASE E=1 / L=0 / score ≈ 4.202 — proved E minimum

rigetti_cz

EXACT PHASE E=1 / L=7 / score ≈ 18.606 — proved E minimum



PROJECTIVE E=1 / L=2 / score ≈ 8.404 — proved E minimum

$U = \frac{\sqrt{2}i(-1+i)}{2}\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

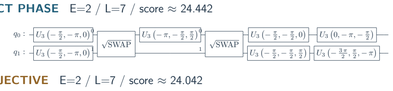
EXACT PHASE E=2 / L=7 / score ≈ 24.442 — proved E minimum

PROJECTIVE E=2 / L=5 / score ≈ 18.830 — proved E minimum

$U = -i\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=2 / L=7 / score ≈ 24.442 — proved E minimum



PROJECTIVE E=2 / L=7 / score ≈ 24.042 — proved E minimum

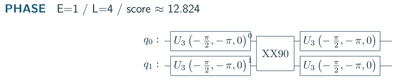
$U = \eta_l\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.206 — proved E minimum

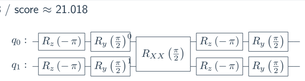
xx90_u

EXACT PHASE E=1 / L=4 / score ≈ 12.824 — proved E minimum



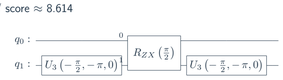
xx_rz_ry

EXACT PHASE E=1 / L=8 / score ≈ 21.018 — proved E minimum



zx_u

EXACT PHASE E=1 / L=2 / score ≈ 8.614 — proved E minimum



zz_u

EXACT PHASE E=1 / L=0 / score ≈ 4.202 — proved E minimum

ZZPhase (α)

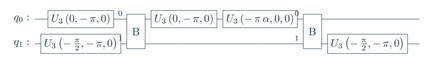
2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=1 / L=8 / score ≈ 21.626 — minimum uniform E budget

berkeley_b

EXACT PHASE E=2 / L=5 / score ≈ 18.430 — minimum uniform E budget



clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

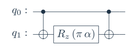
EXACT PHASE E=2 / L=9 / score ≈ 26.410 — minimum uniform E budget

PROJECTIVE E=2 / L=1 / score ≈ 10.002 — minimum uniform E budget

$U = e^{-\frac{i\pi\alpha}{2}}\,C$. Scalar metadata; no physical GP.

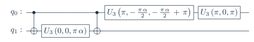
cx_rz_ry

EXACT PHASE E=2 / L=1 / score ≈ 10.002 — minimum uniform E budget



cx_u

EXACT PHASE E=2 / L=3 / score ≈ 14.418 — minimum uniform E budget



PROJECTIVE E=2 / L=1 / score ≈ 10.006 — minimum uniform E budget

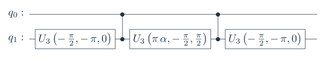
$U = e^{-\frac{i\pi\alpha}{2}}\,C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=2 / L=7 / score ≈ 23.214 — minimum uniform E budget

cz_u

EXACT PHASE E=2 / L=3 / score ≈ 14.818 — minimum uniform E budget



google_cz

EXACT PHASE E=2 / L=5 / score ≈ 19.030 — minimum uniform E budget

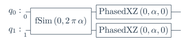
PROJECTIVE E=2 / L=3 / score ≈ 14.818 — minimum uniform E budget

$U = e^{-\frac{i\pi\alpha}{2}}\,C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 12.228 — minimum uniform E budget

PROJECTIVE E=1 / L=2 / score ≈ 8.016 — minimum uniform E budget



$U = e^{-\frac{i\pi\alpha}{2}}\,C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=2 / L=10 / score ≈ 30.581 — minimum uniform E budget

Exact definitions: $\eta_{808}, \eta_{809}, \eta_{900}, \eta_{901}, \eta_{902}, \eta_{903}, \eta_{904}, \eta_{905}, \eta_{906}, \eta_{907}, \eta_{908}, \eta_{909}, \eta_{910}, \eta_{911}$ in the expression glossary.

PROJECTIVE E=2 / L=9 / score ≈ 28.575 — minimum uniform E budget

Exact definitions: $\eta_{903}, \eta_{904}, \eta_{905}, \eta_{906}, \eta_{907}, \eta_{908}, \eta_{909}, \eta_{910}, \eta_{911}, \eta_{912}, \eta_{913}, \eta_{914}$ in the expression glossary.

$U = \eta_{915}\,C$. Scalar metadata; no physical GP.

google_syc

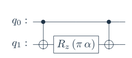
EXACT PHASE E=12 / L=11 / score ≈ 71.866 — E upper bound

PROJECTIVE E=12 / L=9 / score ≈ 67.654 — E upper bound

$U = e^{-\frac{i\pi\alpha}{2}}\,C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=2 / L=1 / score ≈ 10.002 — minimum uniform E budget



ibm_cz

EXACT PHASE E=2 / L=11 / score ≈ 31.214 — minimum uniform E budget

PROJECTIVE E=2 / L=11 / score ≈ 30.810 — minimum uniform E budget

$U = i\,C$. Scalar metadata; no physical GP.

ibm_ecr

EXACT PHASE E=2 / L=8 / score ≈ 24.008 — minimum uniform E budget

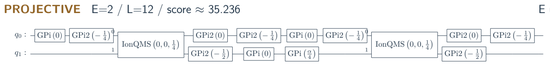
PROJECTIVE E=2 / L=4 / score ≈ 16.004 — minimum uniform E budget

$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=2 / L=12 / score ≈ 35.636 — E upper bound

PROJECTIVE E=2 / L=12 / score ≈ 35.236 — E upper bound

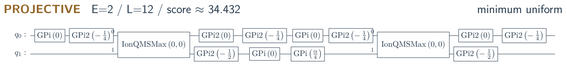


$U = i\,C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=2 / L=12 / score ≈ 34.832 — minimum uniform E budget

PROJECTIVE E=2 / L=12 / score ≈ 34.432 — minimum uniform E budget



$U = i\,C$. Scalar metadata; no physical GP.

ionq_forte

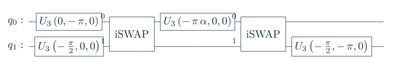
EXACT PHASE E=1 / L=0 / score ≈ 4.202 — minimum uniform E budget

iqm_cz

EXACT PHASE E=2 / L=5 / score ≈ 18.820 — minimum uniform E budget

iswap_u

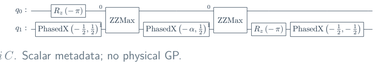
EXACT PHASE E=2 / L=4 / score ≈ 16.424 — minimum uniform E budget



quantinuum_zzmax

EXACT PHASE E=5 / L=24 / score ≈ 73.274 — E upper bound

PROJECTIVE E=2 / L=5 / score ≈ 19.016 — minimum uniform E budget



$U = -i\,C$. Scalar metadata; no physical GP.

quantinuum_zzphase

EXACT PHASE E=1 / L=0 / score ≈ 4.002 — minimum uniform E budget

rigetti_cz

EXACT PHASE E=2 / L=11 / score ≈ 31.214 — minimum uniform E budget

PROJECTIVE E=2 / L=11 / score ≈ 30.810 — minimum uniform E budget

$U = i\,C$. Scalar metadata; no physical GP.

sqrt_iswap_u

EXACT PHASE E=2 / L=10 / score ≈ 30.060 — minimum uniform E budget

Exact definitions: $\eta_{916}, \eta_{917}, \eta_{918}, \eta_{919}, \eta_{920}, \eta_{921}, \eta_{922}, \eta_{923}, \eta_{924}, \eta_{925}$ in the expression glossary.

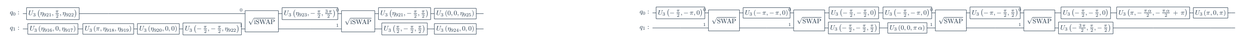
PROJECTIVE E=2 / L=9 / score ≈ 28.054 — minimum uniform E budget

Exact definitions: $\eta_{920}, \eta_{921}, \eta_{922}, \eta_{923}, \eta_{924}, \eta_{925}, \eta_{926}, \eta_{927}$ in the expression glossary.

$U = \eta_{928}\,C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=11 / score ≈ 41.266 — E upper bound



PROJECTIVE E=4 / L=9 / score ≈ 36.454 — E upper bound

$U = -ie^{-\frac{i\pi\alpha}{2}}\,C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.006 — minimum uniform E budget

xx90_u

EXACT PHASE E=2 / L=4 / score ≈ 16.824 — minimum uniform E budget

PROJECTIVE E=2 / L=4 / score ≈ 16.424 — minimum uniform E budget

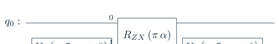
$U = i\,C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=1 / L=8 / score ≈ 20.818 — minimum uniform E budget

zx_u

EXACT PHASE E=1 / L=2 / score ≈ 8.414 — minimum uniform E budget



zz_u

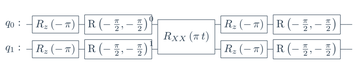
EXACT PHASE E=1 / L=0 / score ≈ 4.002 — minimum uniform E budget

ZZPow $(t, global_{shift})$

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

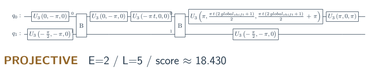
PROJECTIVE E=1 / L=8 / score ≈ 21.626 — minimum uniform E budget



$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=7 / score ≈ 22.842 — minimum uniform E budget



PROJECTIVE E=2 / L=5 / score ≈ 18.430 — minimum uniform E budget

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

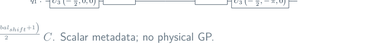
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=2 / L=9 / score ≈ 26.010 — minimum uniform E budget



PROJECTIVE E=2 / L=1 / score ≈ 10.002 — minimum uniform E budget

$U = e^{i\pi global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=2 / L=1 / score ≈ 10.002 — minimum uniform E budget

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=2 / L=3 / score ≈ 14.018 — minimum uniform E budget

PROJECTIVE E=2 / L=1 / score ≈ 10.006 — minimum uniform E budget

$U = e^{i\pi global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

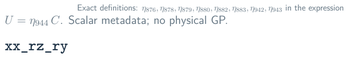
cz_rz_rx

PROJECTIVE E=2 / L=7 / score ≈ 23.214 — minimum uniform E budget

$U = -e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=2 / L=5 / score ≈ 19.230 — minimum uniform E budget

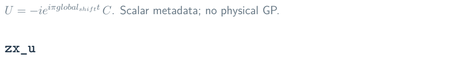


PROJECTIVE E=2 / L=3 / score ≈ 14.818 — minimum uniform E budget

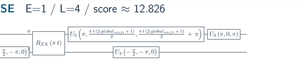
$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=2 / L=5 / score ≈ 18.830 — minimum uniform E budget



PROJECTIVE E=2 / L=3 / score ≈ 14.818 — minimum uniform E budget



$U = e^{i\pi global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 12.028 — minimum uniform E budget

PROJECTIVE E=1 / L=2 / score ≈ 8.016 — minimum uniform E budget

$U = e^{i\pi global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=2 / L=10 / score ≈ 30.581 — minimum uniform E budget

Exact definitions: η_{557}, η_{560}, η_{561}, η_{562}, η_{563}, η_{564}, η_{566}, η_{568}, η_{569}, η_{929}, η_{930}, η_{931}, η_{932}, η_{933} in the expression glossary.

PROJECTIVE E=2 / L=9 / score ≈ 28.575 — minimum uniform E budget

Exact definitions: η_{557}, η_{560}, η_{561}, η_{562}, η_{563}, η_{564}, η_{566}, η_{568}, η_{569}, η_{934}, η_{935}, η_{936} in the expression glossary.
$U = \eta_{937}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=12 / L=11 / score ≈ 71.666 — E upper bound

PROJECTIVE E=12 / L=9 / score ≈ 67.654 — E upper bound

$U = e^{i\pi global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

ibm_cx

PROJECTIVE E=2 / L=1 / score ≈ 10.002 — minimum uniform E budget

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=2 / L=11 / score ≈ 30.810 — minimum uniform E budget

$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=2 / L=4 / score ≈ 16.004 — minimum uniform E budget

$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=2 / L=12 / score ≈ 35.236 — E upper bound

$U = ie^{i\pi t\left(global_{shift}+\frac{1}{2}\right)}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=2 / L=12 / score ≈ 34.432 — minimum uniform E budget

$U = ie^{i\pi t\left(global_{shift}+\frac{1}{2}\right)}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=1 / L=0 / score ≈ 4.202 — minimum uniform E budget

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=2 / L=5 / score ≈ 18.820 — minimum uniform E budget

$U = -e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=2 / L=6 / score ≈ 20.836 — minimum uniform E budget

PROJECTIVE E=2 / L=4 / score ≈ 16.424 — minimum uniform E budget

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=2 / L=5 / score ≈ 19.016 — minimum uniform E budget

$U = -ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=1 / L=0 / score ≈ 4.002 — minimum uniform E budget

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=2 / L=11 / score ≈ 30.810 — minimum uniform E budget

$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=2 / L=10 / score ≈ 30.060 — minimum uniform E budget

Exact definitions: η_{676}, η_{678}, η_{679}, η_{640}, η_{642}, η_{643}, η_{938}, η_{939}, η_{940}, η_{941} in the expression glossary.

PROJECTIVE E=2 / L=9 / score ≈ 28.054 — minimum uniform E budget

Exact definitions: η_{676}, η_{678}, η_{679}, η_{640}, η_{642}, η_{643}, η_{942}, η_{943} in the expression glossary.
$U = \eta_{944}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=4 / L=11 / score ≈ 40.866 — E upper bound

PROJECTIVE E=4 / L=9 / score ≈ 36.454 — E upper bound

$U = -ie^{i\pi global_{shift}\, t}\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=0 / score ≈ 4.006 — minimum uniform E budget

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=2 / L=6 / score ≈ 21.236 — minimum uniform E budget

PROJECTIVE E=2 / L=4 / score ≈ 16.424 — minimum uniform E budget

$U = ie^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=1 / L=8 / score ≈ 20.818 — minimum uniform E budget

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=1 / L=4 / score ≈ 12.826 — minimum uniform E budget

PROJECTIVE E=1 / L=2 / score ≈ 8.414 — minimum uniform E budget

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=1 / L=2 / score ≈ 8.414 — minimum uniform E budget

PROJECTIVE E=1 / L=0 / score ≈ 4.002 — minimum uniform E budget

$U = e^{\frac{i\pi t\left(2 global_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{1} = \frac{(-1)^{\frac{3}{8}}\sqrt{2}(-1+i)}{2}$$

$$\eta_{2} = -\operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{3} = \frac{2\operatorname{atan}\left(\frac{\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right)}{\pi}$$

$$\eta_{4} = -\frac{\operatorname{atan}\left(\frac{\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right)}{\pi}$$

$$\eta_{5} = \frac{2\operatorname{atan}\left(\frac{\frac{\sqrt[4]{2}}{2}}{2\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}\right)}{\pi}$$

$$\eta_{6} = \frac{-\frac{3\pi}{2} - 4\operatorname{atan}\left(\frac{\frac{\sqrt[4]{2}}{2}}{2\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}\right) + 6\operatorname{atan}\left(\frac{\frac{\sqrt[4]{2}}{2}}{\sqrt{2-\sqrt{2}}}\right)}{\pi}$$

$$\eta_{7} = \frac{-3\operatorname{atan}\left(\frac{\frac{\sqrt[4]{2}}{2}}{\sqrt{2-\sqrt{2}}}\right) + 2\operatorname{atan}\left(\frac{\frac{\sqrt[4]{2}}{2}}{2\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}\right)}{\pi} + \frac{1}{2}$$

$$\eta_{8} = -\frac{2\operatorname{atan}\left(\frac{\frac{\sqrt[4]{2}}{2}}{2\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}\right)}{\pi}$$

$$\eta_{9} = \frac{\pi - 2\operatorname{atan}\left(\frac{\frac{\sqrt[4]{2}}{2}}{2\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}\right)}{\pi}$$

$$\eta_{10} = e^{i\left(-\pi + 3\operatorname{atan}\left(\frac{\frac{\sqrt[4]{2}}{2}}{\sqrt{2-\sqrt{2}}}\right)\right)}$$

$$\eta_{11} = \pi - 2\operatorname{atan}\left(\frac{\frac{\sqrt[4]{2}}{2}}{2\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}\right)$$

$$\eta_{12} = -2\operatorname{atan}\left(\frac{\frac{\sqrt[4]{2}}{2}}{2\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}\right)$$

$$\eta_{13} = -\operatorname{atan}\left(\frac{\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{14} = \pi - \operatorname{atan}\left(\frac{\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{15} = -\pi + \operatorname{atan}\left(\frac{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{-\sqrt{\sqrt{2}+2}-\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{16} = \operatorname{atan}\left(\frac{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{-\sqrt{\sqrt{2}+2}-\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{17} = \frac{(-1)^{\frac{3}{8}}\sqrt{2}(-1+i)}{2}$$

$$\eta_{18} = -\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta) + \sin^2(\theta)\cos^2(\phi)} = 0 \\ \operatorname{atan}_2\left(\sin(\phi)\sin(\theta), \sin(\theta)\cos(\phi)\right) & \text{otherwise} \end{cases} - \pi$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{19} = -\frac{\operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},\cos(\theta)\right)}{2}$$

$$\eta_{20} = \pi - \frac{\operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},\cos(\theta)\right)}{2}$$

$$\eta_{21} = -\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)} = 0 \\ \operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},0\right) & \text{otherwise} \end{cases}$$

$$\eta_{22} = \operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},\cos(\theta)\right)$$

$$\eta_{23} = \begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)} = 0 \\ \operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},0\right) & \text{otherwise} \end{cases}$$

$$\eta_{24} = \begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)} = 0 \\ \operatorname{atan_2}(\sin(\phi)\sin(\theta),\sin(\theta)\cos(\phi)) & \text{otherwise} \end{cases}$$

$$\eta_{25} = \alpha - \operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},\cos(\theta)\right)$$

$$\eta_{26} = -\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)} = 0 \\ \operatorname{atan_2}(\sin(\phi)\sin(\theta),\sin(\theta)\cos(\phi)) & \text{otherwise} \end{cases}$$

$$\eta_{27} = e^{-\frac{i\operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},\cos(\theta)\right)}{2}}$$

$$\eta_{28} = \frac{-\pi\left(\frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)} = 0 \\ \operatorname{atan_2}(\sin(\phi)\sin(\theta),\sin(\theta)\cos(\phi)) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}\right) - \frac{\pi}{2}}{\pi}$$

$$\eta_{29} = \frac{\pi\left(\frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)} = 0 \\ \operatorname{atan_2}(\sin(\phi)\sin(\theta),\sin(\theta)\cos(\phi)) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}\right) + \frac{\pi}{2}}{\pi} + \frac{1}{2}$$

$$\eta_{30} = \frac{-\frac{\begin{cases} 2\operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},\cos(\theta)\right) & \text{for } \cos(\theta) \geq 0 \\ 2\operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},\cos(\theta)\right) - 2\pi & \text{otherwise} \end{cases}}{2} - 4\operatorname{atan_2}\left(\sqrt{\frac{1}{2}-\frac{|\cos(\theta)|}{2}}\left(\begin{cases} 1 & \text{for } \cos(\theta) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right)\cdot\sqrt{\frac{|\cos(\theta)|}{2}+\frac{1}{2}+\sqrt{|\cos(\theta)|}}\right) + 2\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{|\cos(\theta)|}{2}}\left(\begin{cases} 1 & \text{for } \cos(\theta) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right)\cdot\sqrt{|\cos(\theta)|}\right) - \pi}{\pi}$$

$$\eta_{31} = \frac{\frac{\begin{cases} 2\operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},\cos(\theta)\right) & \text{for } \cos(\theta) \geq 0 \\ 2\operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},\cos(\theta)\right) - 2\pi & \text{otherwise} \end{cases}}{4} + 2\operatorname{atan_2}\left(\sqrt{\frac{1}{2}-\frac{|\cos(\theta)|}{2}}\left(\begin{cases} 1 & \text{for } \cos(\theta) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right)\cdot\sqrt{\frac{|\cos(\theta)|}{2}+\frac{1}{2}+\sqrt{|\cos(\theta)|}}\right) - \operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{|\cos(\theta)|}{2}}\left(\begin{cases} 1 & \text{for } \cos(\theta) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right)\cdot\sqrt{|\cos(\theta)|}\right)}{\pi} + \frac{1}{2}$$

$$\eta_{32} = -\frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)} = 0 \\ \operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},0\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{33} = \frac{2\operatorname{atan_2}\left(\sqrt{\frac{1}{2}-\frac{|\cos(\theta)|}{2}}\left(\begin{cases} 1 & \text{for } \cos(\theta) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right)\cdot\sqrt{\frac{|\cos(\theta)|}{2}+\frac{1}{2}+\sqrt{|\cos(\theta)|}}\right)}{\pi}$$

$$\eta_{34} = \frac{\begin{cases} 2\operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},\cos(\theta)\right) & \text{for } \cos(\theta) \geq 0 \\ 2\operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},\cos(\theta)\right) - 2\pi & \text{otherwise} \end{cases}}{\frac{2}{\pi}} + \pi$$

$$\eta_{35} = -\frac{\begin{cases} 2\operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},\cos(\theta)\right) & \text{for } \cos(\theta) \geq 0 \\ 2\operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},\cos(\theta)\right) - 2\pi & \text{otherwise} \end{cases}}{\frac{2}{\pi}} - \frac{\pi}{2} + \frac{1}{2}$$

$$\eta_{36} = \frac{\begin{cases} 2\operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},\cos(\theta)\right) & \text{for } \cos(\theta) \geq 0 \\ 2\operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},\cos(\theta)\right) - 2\pi & \text{otherwise} \end{cases}}{2\pi}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{37} = -\frac{2\,\mathrm{atan2}\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{|\cos(\theta)|}{2}}\left(\begin{cases}1 & \text{for } \cos(\theta)\geq 0\\ -1 & \text{otherwise}\end{cases}\right),\sqrt{|\cos(\theta)|}\right)}{\pi}$$

$$\eta_{38} = \frac{\begin{cases}0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)}=0\\ \mathrm{atan2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},0\right) & \text{otherwise}\end{cases}}{\pi}$$

$$\eta_{39} = \frac{\begin{cases}0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)}=0\\ \mathrm{atan2}\left(\sin(\phi)\sin(\theta),\sin(\theta)\cos(\phi)\right) & \text{otherwise}\end{cases}}{\pi}$$

$$\eta_{40} = -\frac{\begin{cases}0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)}=0\\ \mathrm{atan2}\left(\sin(\phi)\sin(\theta),\sin(\theta)\cos(\phi)\right) & \text{otherwise}\end{cases}}{\pi}+\frac{1}{2}$$

$$\eta_{41} = \frac{\alpha-\mathrm{atan2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},\cos(\theta)\right)}{\pi}$$

$$\eta_{42} = \frac{-\alpha+\mathrm{atan2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},\cos(\theta)\right)}{\pi}+\frac{1}{2}$$

$$\eta_{43} = -\frac{\begin{cases}0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)}=0\\ \mathrm{atan2}\left(\sin(\phi)\sin(\theta),\sin(\theta)\cos(\phi)\right) & \text{otherwise}\end{cases}}{\pi}$$

$$\eta_{44} = \frac{\begin{cases}0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)}=0\\ \mathrm{atan2}\left(\sin(\phi)\sin(\theta),\sin(\theta)\cos(\phi)\right) & \text{otherwise}\end{cases}}{\pi}+\frac{1}{2}$$

$$\eta_{45} = e^{i\left(-\frac{\begin{cases}2\,\mathrm{atan2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},\cos(\theta)\right) & \text{for } \cos(\theta)\geq 0\\ 2\,\mathrm{atan2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},\cos(\theta)\right)-2\pi & \text{otherwise}\end{cases}}{4}-2\,\mathrm{atan2}\left(\sqrt{\frac{1}{2}-\frac{|\cos(\theta)|}{2}}\left(\begin{cases}1 & \text{for } \cos(\theta)\geq 0\\ -1 & \text{otherwise}\end{cases}\right)\cdot\sqrt{\frac{|\cos(\theta)|}{2}+\frac{1}{2}}+\sqrt{|\cos(\theta)|}\right)+\mathrm{atan2}\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{|\cos(\theta)|}{2}}\left(\begin{cases}1 & \text{for } \cos(\theta)\geq 0\\ -1 & \text{otherwise}\end{cases}\right),\sqrt{|\cos(\theta)|}\right)\right)}$$

$$\eta_{46} = -\frac{\begin{cases}2\,\mathrm{atan2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},\cos(\theta)\right) & \text{for } \cos(\theta)\geq 0\\ 2\,\mathrm{atan2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},\cos(\theta)\right)-2\pi & \text{otherwise}\end{cases}}{4}$$

$$\eta_{47} = \pi-\frac{\begin{cases}2\,\mathrm{atan2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},\cos(\theta)\right) & \text{for } \cos(\theta)\geq 0\\ 2\,\mathrm{atan2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},\cos(\theta)\right)-2\pi & \text{otherwise}\end{cases}}{4}$$

$$\eta_{48} = 2\,\mathrm{atan2}\left(\sqrt{\frac{1}{2}-\frac{|\cos(\theta)|}{2}}\left(\begin{cases}1 & \text{for } \cos(\theta)\geq 0\\ -1 & \text{otherwise}\end{cases}\right)\cdot\sqrt{\frac{|\cos(\theta)|}{2}+\frac{1}{2}}+\sqrt{|\cos(\theta)|}\right)$$

$$\eta_{49} = \frac{\begin{cases}2\,\mathrm{atan2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},\cos(\theta)\right) & \text{for } \cos(\theta)\geq 0\\ 2\,\mathrm{atan2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},\cos(\theta)\right)-2\pi & \text{otherwise}\end{cases}}{2}+\frac{\pi}{2}$$

$$\eta_{50} = -2\,\mathrm{atan2}\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{|\cos(\theta)|}{2}}\left(\begin{cases}1 & \text{for } \cos(\theta)\geq 0\\ -1 & \text{otherwise}\end{cases}\right),\sqrt{|\cos(\theta)|}\right)$$

$$\eta_{51} = e^{-\frac{i\left(\begin{cases}2\,\mathrm{atan2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},\cos(\theta)\right) & \text{for } \cos(\theta)\geq 0\\ 2\,\mathrm{atan2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)+\sin^2(\theta)\cos^2(\phi)},\cos(\theta)\right)-2\pi & \text{otherwise}\end{cases}\right)}{4}}$$

$$\eta_{52} = -2\pi+\mathrm{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\cos\left(\frac{\pi}{16}\right)}\right)$$

$$\eta_{53} = -\frac{\pi}{32}+\mathrm{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\cos\left(\frac{\pi}{16}\right)+1}\right)$$

$$\eta_{54} = \mathrm{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\cos\left(\frac{\pi}{16}\right)+1}\right)+\frac{31\pi}{32}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{55} = -2i \log\left(\frac{\frac{\sqrt{2}}{2} + \frac{\sqrt{2}i\sqrt{(-1)^{\frac{7}{16}}\left(\sin\left(\frac{\pi}{16}\right) - i\cos\left(\frac{\pi}{16}\right)\right)}}{2}}{\sqrt{\frac{1}{2} + \frac{(-1)^{\frac{7}{16}}\left(\sin\left(\frac{\pi}{16}\right) - i\cos\left(\frac{\pi}{16}\right)\right)}{2}}}\right)$$

$$\eta_{56} = \operatorname{atan}\left(\frac{\sin\left(\frac{7\pi}{16}\right)}{\cos\left(\frac{7\pi}{16}\right)}\right)$$

$$\eta_{57} = -2i \log\left(\frac{\sqrt{\frac{-(-1)^{\frac{1}{16}}\left(1+e^{\frac{i\pi}{16}}\right)^2}{4}} + \frac{\sqrt{(-1)^{\frac{1}{16}}\left(1-\sqrt[4]{-1}\right)\left(1-e^{\frac{i\pi}{16}}\right)}}{2}}{\sqrt{\frac{(-1)^{\frac{1}{16}}\left(1-\sqrt[4]{-1}\right)\left(1-e^{\frac{i\pi}{16}}\right)}{4} - \frac{(-1)^{\frac{1}{16}}\left(1+e^{\frac{i\pi}{16}}\right)^2}{4}}}\right)$$

$$\eta_{58} = -\operatorname{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{1-\cos\left(\frac{\pi}{16}\right)}\right) - \operatorname{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\cos\left(\frac{\pi}{16}\right)+1}\right) + \pi$$

$$\eta_{59} = -2i \log\left(\frac{\frac{\sqrt{2}}{2} + \frac{\sqrt{2}i\sqrt{\frac{i\sqrt[4]{-1}e^{-\frac{i\pi}{16}}}{2}}}{2}}{\sqrt{\frac{1}{2} + \frac{i\sqrt[4]{-1}e^{-\frac{i\pi}{16}}}{2}}}\right)$$

$$\eta_{60} = \operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{-2+\sqrt{\sqrt{2}+2}}\right) + \pi$$

$$\eta_{61} = -2i \log\left(\frac{\sqrt{\frac{\left(1-(-1)^{\frac{1}{16}}\right)\left(1+e^{\frac{i\pi}{16}}\right)}{4}} + \frac{\sqrt{\left(1+(-1)^{\frac{1}{16}}\right)\left(1-e^{\frac{i\pi}{16}}\right)}}{2}}{\sqrt{\frac{\left(1+(-1)^{\frac{1}{16}}\right)\left(1-e^{\frac{i\pi}{16}}\right)}{4} + \frac{\left(1-(-1)^{\frac{1}{16}}\right)\left(1+e^{\frac{i\pi}{16}}\right)}{4}}}\right)$$

$$\eta_{62} = -\pi + \operatorname{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\cos\left(\frac{\pi}{16}\right)+1}\right) + \operatorname{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{1-\cos\left(\frac{\pi}{16}\right)}\right)$$

$$\eta_{63} = -2i \log\left(\frac{\sqrt{\frac{-(-1)^{\frac{1}{16}}\left(1+e^{\frac{i\pi}{16}}\right)^2}{4}} + \frac{\sqrt{\left(1-\sqrt[4]{-1}\right)\left(1-e^{-\frac{i\pi}{16}}\right)}}{2}}{\sqrt{\frac{\left(1-\sqrt[4]{-1}\right)\left(1-e^{-\frac{i\pi}{16}}\right)}{4} - \frac{(-1)^{\frac{1}{16}}\left(1+e^{\frac{i\pi}{16}}\right)^2}{4}}}\right)$$

$$\eta_{64} = -\operatorname{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{1-\cos\left(\frac{\pi}{16}\right)}\right) - \operatorname{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\cos\left(\frac{\pi}{16}\right)+1}\right)$$

$$\eta_{65} = -\pi + \operatorname{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\cos\left(\frac{\pi}{16}\right)}\right)$$

$$\eta_{66} = e^{i\left(-\frac{\pi}{32} + \operatorname{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\cos\left(\frac{\pi}{16}\right)+1}\right)\right)}$$

$$\eta_{67} = -\pi - \operatorname{atan}\left(\frac{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{68} = 2\operatorname{atan}\left(\frac{\sqrt{\frac{1}{2} - \frac{\sqrt{\sqrt{2}+2}}{4}}}{\sqrt{\frac{\sqrt{\sqrt{2}+2}}{4} + \frac{1}{2}}}\right)$$

$$\eta_{69} = \operatorname{atan}\left(\frac{-\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}+2}\right) + \operatorname{atan}\left(\frac{-\sqrt{2-\sqrt{2}}}{2-\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{70} = -\pi + \operatorname{atan}\left(\frac{-\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}+2}\right) + \operatorname{atan}\left(\frac{-\sqrt{2-\sqrt{2}}}{2-\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{71} = -\operatorname{atan}\left(\frac{-\sqrt{2-\sqrt{2}}}{2-\sqrt{\sqrt{2}+2}}\right) - \operatorname{atan}\left(\frac{-\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}+2}\right)$$

$$\eta_{72} = -\operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{2-\sqrt{\sqrt{2}+2}}\right) - \operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}+2}\right) + \pi$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{73} = e^{-2i\,\mathrm{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}+2}\right)}$$

$$\eta_{74} = -\pi + \mathrm{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{75} = \mathrm{atan}\left(\frac{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{76} = -\,\mathrm{atan}\left(\frac{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{77} = \mathrm{atan}\left(\frac{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}\right) + \pi$$

$$\eta_{78} = -\frac{15\pi}{16} + \mathrm{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{2-\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{79} = -\,\mathrm{atan}\left(\frac{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{80} = -\pi + \mathrm{atan}\left(\frac{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{81} = -\,\mathrm{atan}\left(\frac{2-\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right) + \mathrm{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}+2}\right) + \pi$$

$$\eta_{82} = -\,\mathrm{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}+2}\right) + \mathrm{atan}\left(\frac{2-\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{83} = (-1)^{\frac{1}{16}}\, e^{-i\,\mathrm{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}+2}\right)}$$

$$\eta_{84} = -\pi - \mathrm{acos}\left(\frac{\sqrt{3}}{3}\right)$$

$$\eta_{85} = -\pi + \mathrm{acos}\left(\frac{\sqrt{3}}{3}\right)$$

$$\eta_{86} = -\frac{\pi}{2} + \mathrm{acos}\left(\frac{\sqrt{3}}{3}\right)$$

$$\eta_{87} = -\frac{\pi}{2} - \mathrm{acos}\left(\frac{\sqrt{3}}{3}\right)$$

$$\eta_{88} = -\frac{37\pi}{32} + \mathrm{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\cos\left(\frac{\pi}{16}\right)+1}\right)$$

$$\eta_{89} = -\frac{5\pi}{32} + \mathrm{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\cos\left(\frac{\pi}{16}\right)+1}\right)$$

$$\eta_{90} = 2\,\mathrm{atan}\left(\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{91} = -2i\log\left(\frac{\frac{\sqrt{2}}{2}+\frac{\frac{1}{2}\sqrt{2}i\sqrt{\sqrt[3]{-1}(1+i)}}{2}}{\sqrt{\frac{1}{2}+\frac{\sqrt[3]{-1}\sqrt{2}(1+i)}{4}}}\right)$$

$$\eta_{92} = -2i\log\left(\frac{\frac{\sqrt{2}\sqrt{(1-\sqrt[3]{-1})(-\sqrt{2}+2+\sqrt{2}i)}}{4}+\frac{\sqrt{2}i\sqrt{\left(-1-(-1)^{\frac{2}{3}}\right)(\sqrt{2}+\sqrt{2}i+2i)}}{4}}{\sqrt{\frac{(1-\sqrt[3]{-1})(-\sqrt{2}+2+\sqrt{2}i)}{8}+\frac{\left((-1)^{\frac{2}{3}}+1\right)(\sqrt{2}+\sqrt{2}i+2i)}{8}}}\right)$$

$$\eta_{93} = -2i\log\left(\frac{\frac{\sqrt{2}\sqrt{(1-\sqrt[3]{-1})(-\sqrt{2}+2+\sqrt{2}i)}}{4}+\frac{\sqrt{2}i\sqrt{\left(1-(-1)^{\frac{2}{3}}\right)(\sqrt{2}+2+\sqrt{2}i)}}{4}}{\sqrt{\frac{(1-\sqrt[3]{-1})(-\sqrt{2}+2+\sqrt{2}i)}{8}+\frac{\left(1-(-1)^{\frac{2}{3}}\right)(\sqrt{2}+2+\sqrt{2}i)}{8}}}\right)$$

$$\eta_{94} = -2i\log\left(\frac{\frac{\sqrt{-(-1)^{\frac{2}{3}}\left(1+e^{\frac{i\pi}{4}}\right)^2}}{2}+\frac{i\sqrt{(-1)^{\frac{2}{3}}(1-\sqrt[3]{-1})\left(1-e^{\frac{i\pi}{4}}\right)}}{2}}{\sqrt{\frac{(-1)^{\frac{2}{3}}(1-\sqrt[3]{-1})\left(1-e^{\frac{i\pi}{4}}\right)}{4}-\frac{(-1)^{\frac{2}{3}}\left(1+e^{\frac{i\pi}{4}}\right)^2}{4}}}\right)$$

$$\eta_{95} = -2i\log\left(\frac{\frac{\sqrt{2}\sqrt{\left(1-(-1)^{\frac{2}{3}}\right)(\sqrt{2}+2+\sqrt{2}i)}}{4}+\frac{\sqrt{2}i\sqrt{\left(-1-(-1)^{\frac{2}{3}}\right)(-2+\sqrt{2}+\sqrt{2}i)}}{4}}{\sqrt{\frac{\left(-1-(-1)^{\frac{2}{3}}\right)(-2+\sqrt{2}+\sqrt{2}i)}{8}+\frac{\left(1-(-1)^{\frac{2}{3}}\right)(\sqrt{2}+2+\sqrt{2}i)}{8}}}\right)$$

$$\eta_{96} = -2i\log\left(\frac{\frac{\sqrt{2}\sqrt{(1-\sqrt[3]{-1})(-\sqrt{2}+2+\sqrt{2}i)}}{4}+\frac{\sqrt{2}i\sqrt{(1+\sqrt[3]{-1})(\sqrt{2}+2-\sqrt{2}i)}}{4}}{\sqrt{\frac{(1-\sqrt[3]{-1})(-\sqrt{2}+2+\sqrt{2}i)}{8}+\frac{(1+\sqrt[3]{-1})(\sqrt{2}+2-\sqrt{2}i)}{8}}}\right)$$

$$\eta_{97} = e^{i\left(-\frac{37\pi}{32}+\operatorname{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\cos\left(\frac{\pi}{16}\right)+1}\right)\right)}$$

$$\eta_{98} = -\operatorname{atan}\left(\frac{\cos\left(\frac{\pi}{16}\right)}{\sin\left(\frac{\pi}{16}\right)}\right)$$

$$\eta_{99} = \operatorname{atan}\left(\frac{-2+\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{100} = \operatorname{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\cos\left(\frac{\pi}{16}\right)}\right)$$

$$\eta_{101} = \operatorname{atan}\left(\frac{2-\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{102} = -\operatorname{atan}\left(\frac{2-\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{103} = -\pi+\operatorname{atan}\left(\frac{2-\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{104} = \frac{(-1)^{\frac{3}{4}}\sqrt{2}(1+i)^3}{4}$$

$$\eta_{105} = \frac{\pi\left(\left\lfloor\frac{2b}{\pi}+\frac{1}{2}\right\rfloor+\left\lfloor\frac{2c}{\pi}+\frac{1}{2}\right\rfloor\right)}{2}+\begin{cases}0 & \text{for } \sqrt{(-1)^{\left\lfloor\frac{2b}{\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor\frac{2c}{\pi}+\frac{1}{2}\right\rfloor}\cos(2b)\cos(2c)+2\sin^2\left(b-\frac{\pi\left\lfloor\frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos^2\left(c-\frac{\pi\left\lfloor\frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)}=0\\ \operatorname{atan}_2\left(\sqrt{(-1)^{\left\lfloor\frac{2b}{\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor\frac{2c}{\pi}+\frac{1}{2}\right\rfloor}\cos(2b)\cos(2c)},\sqrt{2}\sin\left(b-\frac{\pi\left\lfloor\frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos\left(c-\frac{\pi\left\lfloor\frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise}\end{cases}-\left(\frac{3\pi\left(\left\lfloor\frac{2b}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)}{2}\right)+\pi$$

$$\eta_{106} = \frac{\pi\left(\left\lfloor\frac{2b}{\pi}+\frac{1}{2}\right\rfloor+\left\lfloor\frac{2c}{\pi}+\frac{1}{2}\right\rfloor\right)}{2}+\begin{cases}0 & \text{for } \sqrt{(-1)^{\left\lfloor\frac{2b}{\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor\frac{2c}{\pi}+\frac{1}{2}\right\rfloor}\cos(2b)\cos(2c)+2\sin^2\left(b-\frac{\pi\left\lfloor\frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos^2\left(c-\frac{\pi\left\lfloor\frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)}=0\\ \operatorname{atan}_2\left(\sqrt{(-1)^{\left\lfloor\frac{2b}{\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor\frac{2c}{\pi}+\frac{1}{2}\right\rfloor}\cos(2b)\cos(2c)},\sqrt{2}\sin\left(b-\frac{\pi\left\lfloor\frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos\left(c-\frac{\pi\left\lfloor\frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise}\end{cases}+\frac{\begin{cases}0 & \text{for } \cos\left(b-c-\frac{\pi\left\lfloor\frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}+\frac{\pi\left\lfloor\frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)=0\\ \operatorname{atan}_2\left(-\sqrt{(-1)^{\left\lfloor\frac{2b}{\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor\frac{2c}{\pi}+\frac{1}{2}\right\rfloor}\cos(2b)\cos(2c)},\sin\left(b+c-\frac{\pi\left\lfloor\frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}-\frac{\pi\left\lfloor\frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise}\end{cases}}{2}+\frac{\begin{cases}0 & \text{for } \cos\left(b+c-\frac{\pi\left\lfloor\frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}-\frac{\pi\left\lfloor\frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)=0\\ \operatorname{atan}_2\left(-\sqrt{(-1)^{\left\lfloor\frac{2b}{\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor\frac{2c}{\pi}+\frac{1}{2}\right\rfloor}\cos(2b)\cos(2c)},\sin\left(b-c-\frac{\pi\left\lfloor\frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}+\frac{\pi\left\lfloor\frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise}\end{cases}}{2}-\left(\frac{3\pi\left(\left\lfloor\frac{2b}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)}{2}\right)+3\pi$$

$$\eta_{107} = -\frac{\begin{cases}0 & \text{for } \cos\left(b-c-\frac{\pi\left\lfloor\frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}+\frac{\pi\left\lfloor\frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)=0\\ \operatorname{atan}_2\left(-\sqrt{(-1)^{\left\lfloor\frac{2b}{\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor\frac{2c}{\pi}+\frac{1}{2}\right\rfloor}\cos(2b)\cos(2c)},\sin\left(b+c-\frac{\pi\left\lfloor\frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}-\frac{\pi\left\lfloor\frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise}\end{cases}}{2}+\frac{\begin{cases}0 & \text{for } \cos\left(b+c-\frac{\pi\left\lfloor\frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}-\frac{\pi\left\lfloor\frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)=0\\ \operatorname{atan}_2\left(-\sqrt{(-1)^{\left\lfloor\frac{2b}{\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor\frac{2c}{\pi}+\frac{1}{2}\right\rfloor}\cos(2b)\cos(2c)},\sin\left(b-c-\frac{\pi\left\lfloor\frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}+\frac{\pi\left\lfloor\frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise}\end{cases}}{2}+\pi$$

$$\eta_{108} = -2\operatorname{atan}_2\left(\sqrt{2}\sin\left(c-\frac{\pi\left\lfloor\frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos\left(b-\frac{\pi\left\lfloor\frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}\right),\sqrt{(-1)^{\left\lfloor\frac{2b}{\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor\frac{2c}{\pi}+\frac{1}{2}\right\rfloor}\cos(2b)\cos(2c)+2\sin^2\left(b-\frac{\pi\left\lfloor\frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos^2\left(c-\frac{\pi\left\lfloor\frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)}\right)$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{109} = -\begin{cases} 0 & \text{for } \sqrt{(-1)^{\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}\,(-1)^{\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}\cos(2b)\cos(2c) + 2\sin^2\left(b - \frac{\pi\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}{2}\right)\cos^2\left(c - \frac{\pi\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\sqrt{(-1)^{\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}\,(-1)^{\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}\cos(2b)\cos(2c)}, \sqrt{2}\sin\left(b - \frac{\pi\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}{2}\right)\cos\left(c - \frac{\pi\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{110} = \left(-\frac{\pi\left(1-\left(2\left(\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor \bmod 2\right)\right)\right)}{2}\right) + \left(\pi\left(\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor \bmod 2\right)\right)$$

$$\eta_{111} = \left(\pi\left(\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor \bmod 2\right)\right) + \left(\pi\left(\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor \bmod 2\right)\right)$$

$$\eta_{112} = -\frac{\begin{cases} 0 & \text{for } \cos\left(b - c - \frac{\pi\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}{2} + \frac{\pi\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}\,(-1)^{\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}\cos(2b)\cos(2c)}, \sin\left(b + c - \frac{\pi\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}{2} - \frac{\pi\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\begin{cases} 0 & \text{for } \cos\left(b + c - \frac{\pi\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}{2} - \frac{\pi\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}\,(-1)^{\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}\cos(2b)\cos(2c)}, \sin\left(b - c - \frac{\pi\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}{2} + \frac{\pi\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \left(\pi\left(\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor \bmod 2\right)\right) - 2\pi$$

$$\eta_{113} = \left(\left(1 - \left(2\left(\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor \bmod 2\right)\right)\right)\left(\frac{\begin{cases} 0 & \text{for } \cos\left(b - c - \frac{\pi\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}{2} + \frac{\pi\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}\,(-1)^{\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}\cos(2b)\cos(2c)}, \sin\left(b + c - \frac{\pi\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}{2} - \frac{\pi\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\begin{cases} 0 & \text{for } \cos\left(b + c - \frac{\pi\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}{2} - \frac{\pi\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}\,(-1)^{\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}\cos(2b)\cos(2c)}, \sin\left(b - c - \frac{\pi\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}{2} + \frac{\pi\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \pi\right)\right) + \left(\pi\left(\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor \bmod 2\right)\right)$$

$$\eta_{114} = \left(-\frac{\pi\left(1-\left(2\left(\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor \bmod 2\right)\right)\right)}{2}\right) + \left(\pi\left(\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor \bmod 2\right)\right) + \left(\pi\left(\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor \bmod 2\right)\right)$$

$$\eta_{115} = \left(\pi\left(\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor \bmod 2\right)\right) + \frac{\pi}{2}$$

$$\eta_{116} = -\frac{\begin{cases} 0 & \text{for } \cos\left(b - c - \frac{\pi\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}{2} + \frac{\pi\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}\,(-1)^{\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}\cos(2b)\cos(2c)}, \sin\left(b + c - \frac{\pi\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}{2} - \frac{\pi\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\begin{cases} 0 & \text{for } \cos\left(b + c - \frac{\pi\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}{2} - \frac{\pi\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}\,(-1)^{\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}\cos(2b)\cos(2c)}, \sin\left(b - c - \frac{\pi\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}{2} + \frac{\pi\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \pi$$

$$\eta_{117} = e^{i\left(\frac{\pi\left(\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor + \lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor\right)}{2} + \begin{cases} 0 & \text{for } \sqrt{(-1)^{\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}\,(-1)^{\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}\cos(2b)\cos(2c) + 2\sin^2\left(b - \frac{\pi\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}{2}\right)\cos^2\left(c - \frac{\pi\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\sqrt{(-1)^{\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}\,(-1)^{\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}\cos(2b)\cos(2c)}, \sqrt{2}\sin\left(b - \frac{\pi\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}{2}\right)\cos\left(c - \frac{\pi\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}{2}\right)\right) & \text{otherwise} \end{cases} + \frac{\begin{cases} 0 & \text{for } \cos\left(b - c - \frac{\pi\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}{2} + \frac{\pi\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}\,(-1)^{\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}\cos(2b)\cos(2c)}, \sin\left(b + c - \frac{\pi\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}{2} - \frac{\pi\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\begin{cases} 0 & \text{for } \cos\left(b + c - \frac{\pi\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}{2} - \frac{\pi\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}\,(-1)^{\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}\cos(2b)\cos(2c)}, \sin\left(b - c - \frac{\pi\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor}{2} + \frac{\pi\lfloor \frac{2c}{\pi}+\frac{1}{2} \rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \left(\frac{3\pi\left(\lfloor \frac{2b}{\pi}+\frac{1}{2} \rfloor \bmod 2\right)}{2}\right) + 2\pi\right)}$$

$$\eta_{118} = \frac{\sqrt{2}(-1+i)e^{i(a+b+c)}}{2}$$

$$\eta_{119} = \sin(a+b+c) - i\cos(a+b+c)$$

$$\eta_{120} = \operatorname{atan}\left(\frac{\sqrt{\sqrt{2}+2}}{-2+\sqrt{2}\sqrt{\sqrt{2}+2}}\right) + \pi$$

$$\eta_{121} = -2\pi + \operatorname{atan}\left(\frac{\left(1-\sqrt{2-\sqrt{2}}\right)\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}+\sqrt{2}}\right) + \operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{122} = 2\operatorname{atan}\left(\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}\right)$$

$$\eta_{123} = \operatorname{atan}\left(\frac{\sqrt{\sqrt{2}+2}}{-2+\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{124} = \frac{(1+i)^a e^{i\pi t\left(global_{shift}+\frac{1}{8}\right)}}{8}$$

$$\eta_{125} = -(-1)^{\frac{3}{8}}\, e^{i\pi t\left(global_{shift}+\frac{1}{8}\right)}$$

$$\eta_{126} = \frac{(1+i)^a e^{\frac{i(\gamma+\pi)}{4}}}{8}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{127} = -\frac{(1+i)^2 e^{\frac{i(\gamma+\pi)}{4}}}{2}$$

$$\eta_{128} = \frac{\pi - 2\operatorname{atan_2}\left(\sin\left(\frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{4}\right),\cos\left(\frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{4}\right) + \sqrt{\cos\left(\frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{2}\right)}\right)}{\pi}$$

$$\eta_{129} = \frac{-\pi\left(-\frac{\frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{2}}{\pi} - \frac{\pi}{2} + \frac{1}{2}\right) - \frac{\pi}{2}}{\pi}$$

$$\eta_{130} = \frac{\pi\left(-\frac{\frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{2}}{\pi} - \frac{\pi}{2} + \frac{1}{2}\right) + \frac{\pi}{2}}{\pi} + \frac{1}{2}$$

$$\eta_{131} = \frac{\pi\left(-\frac{\frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{4}}{\pi} - \frac{\pi}{2} + \frac{\frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{4}}{\pi} + \pi + \frac{1}{2}\right) + 2\operatorname{atan_2}\left(\sqrt{2}\sin\left(\frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{4}\right),\sqrt{\cos\left(\frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{2}\right)}\right) - \frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi)) - 3\pi}{2}}{\pi}$$

$$\eta_{132} = \frac{-\operatorname{atan_2}\left(\sqrt{2}\sin\left(\frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{4}\right),\sqrt{\cos\left(\frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{2}\right)}\right) + \frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{4}}{\pi} + \frac{1}{2}$$

$$\eta_{133} = \frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{2\pi}$$

$$\eta_{134} = -\frac{\frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{2}}{\pi} - \frac{\pi}{2} + \frac{1}{2}$$

$$\eta_{135} = -\frac{2\operatorname{atan_2}\left(\sqrt{2}\sin\left(\frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{4}\right),\sqrt{\cos\left(\frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{2}\right)}\right)}{\pi}$$

$$\eta_{136} = \frac{2\operatorname{atan_2}\left(\sin\left(\frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{4}\right),\cos\left(\frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{4}\right) + \sqrt{\cos\left(\frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{2}\right)}\right)}{\pi}$$

$$\eta_{137} = \frac{\frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{4} + \pi}{\pi}$$

$$\eta_{138} = e^{i\left(\operatorname{atan_2}\left(\sqrt{2}\sin\left(\frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{4}\right),\sqrt{\cos\left(\frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{2}\right)}\right) - \frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{4} - 2\operatorname{atan_2}\left(\sin\left(\frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{4}\right),\cos\left(\frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{4}\right) + \sqrt{\cos\left(\frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{2}\right)}\right)\right)}$$

$$\eta_{139} = \pi - 2\operatorname{atan_2}\left(\sin\left(\frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{4}\right),\cos\left(\frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{4}\right) + \sqrt{\cos\left(\frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{2}\right)}\right)$$

$$\eta_{140} = \frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{2} - \frac{\pi}{2}$$

$$\eta_{141} = -\frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{4} + \frac{\pi}{2}$$

$$\eta_{142} = \pi - \frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{4}$$

$$\eta_{143} = \frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{2} + \frac{\pi}{2}$$

$$\eta_{144} = -2\operatorname{atan_2}\left(\sqrt{2}\sin\left(\frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{4}\right),\sqrt{\cos\left(\frac{\operatorname{atan_2}(\sin(\phi),\cos(\phi))}{2}\right)}\right)$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{145} = 2\,\mathrm{atan}_2\left(\sin\left(\frac{\mathrm{atan}_2\left(\sin\left(\phi\right),\cos\left(\phi\right)\right)}{4}\right),\cos\left(\frac{\mathrm{atan}_2\left(\sin\left(\phi\right),\cos\left(\phi\right)\right)}{4}\right)+\sqrt{\cos\left(\frac{\mathrm{atan}_2\left(\sin\left(\phi\right),\cos\left(\phi\right)\right)}{2}\right)}\right)$$

$$\eta_{146} = e^{-\frac{i\,\mathrm{atan}_2\left(\sin\left(\phi\right),\cos\left(\phi\right)\right)}{4}}$$

$$\eta_{147} = -\begin{cases}0 & \text{for } \sqrt{\sin^2\left(\phi\right)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\phi\right)}=0\\ \mathrm{atan}_2\left(\sin\left(\phi\right)\sin\left(\frac{\theta}{2}\right),\sin\left(\frac{\theta}{2}\right)\cos\left(\phi\right)\right) & \text{otherwise}\end{cases}-\pi$$

$$\eta_{148} = -\frac{\mathrm{atan}_2\left(\sqrt{\sin^2\left(\phi\right)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\phi\right)},\cos\left(\frac{\theta}{2}\right)\right)}{2}$$

$$\eta_{149} = \pi-\frac{\mathrm{atan}_2\left(\sqrt{\sin^2\left(\phi\right)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\phi\right)},\cos\left(\frac{\theta}{2}\right)\right)}{2}$$

$$\eta_{150} = -\begin{cases}0 & \text{for } \sqrt{\sin^2\left(\phi\right)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\phi\right)}=0\\ \mathrm{atan}_2\left(\sqrt{\sin^2\left(\phi\right)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\phi\right)},0\right) & \text{otherwise}\end{cases}$$

$$\eta_{151} = \mathrm{atan}_2\left(\sqrt{\sin^2\left(\phi\right)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\phi\right)},\cos\left(\frac{\theta}{2}\right)\right)$$

$$\eta_{152} = \begin{cases}0 & \text{for } \sqrt{\sin^2\left(\phi\right)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\phi\right)}=0\\ \mathrm{atan}_2\left(\sqrt{\sin^2\left(\phi\right)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\phi\right)},0\right) & \text{otherwise}\end{cases}$$

$$\eta_{153} = \begin{cases}0 & \text{for } \sqrt{\sin^2\left(\phi\right)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\phi\right)}=0\\ \mathrm{atan}_2\left(\sin\left(\phi\right)\sin\left(\frac{\theta}{2}\right),\sin\left(\frac{\theta}{2}\right)\cos\left(\phi\right)\right) & \text{otherwise}\end{cases}$$

$$\eta_{154} = -\,\mathrm{atan}_2\left(\sqrt{\sin^2\left(\phi\right)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\phi\right)},\cos\left(\frac{\theta}{2}\right)\right)$$

$$\eta_{155} = -\begin{cases}0 & \text{for } \sqrt{\sin^2\left(\phi\right)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\phi\right)}=0\\ \mathrm{atan}_2\left(\sin\left(\phi\right)\sin\left(\frac{\theta}{2}\right),\sin\left(\frac{\theta}{2}\right)\cos\left(\phi\right)\right) & \text{otherwise}\end{cases}$$

$$\eta_{156} = e^{-\frac{i\,\mathrm{atan}_2\left(\sqrt{\sin^2\left(\phi\right)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\phi\right)},\cos\left(\frac{\theta}{2}\right)\right)}{2}}$$

$$\eta_{157} = \frac{-\pi\left(\frac{\begin{cases}0 & \text{for } \sqrt{\sin^2\left(\phi\right)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\phi\right)}=0\\ \mathrm{atan}_2\left(\sin\left(\phi\right)\sin\left(\frac{\theta}{2}\right),\sin\left(\frac{\theta}{2}\right)\cos\left(\phi\right)\right) & \text{otherwise}\end{cases}}{\pi}+\frac{1}{2}\right)-\frac{\pi}{2}}{\pi}$$

$$\eta_{158} = \frac{\pi\left(\frac{\begin{cases}0 & \text{for } \sqrt{\sin^2\left(\phi\right)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\phi\right)}=0\\ \mathrm{atan}_2\left(\sin\left(\phi\right)\sin\left(\frac{\theta}{2}\right),\sin\left(\frac{\theta}{2}\right)\cos\left(\phi\right)\right) & \text{otherwise}\end{cases}}{\pi}+\frac{1}{2}\right)+\frac{\pi}{2}}{\pi}+\frac{1}{2}$$

$$\eta_{159} = \frac{\begin{cases}2\,\mathrm{atan}_2\left(\sqrt{\sin^2\left(\phi\right)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\phi\right)},\cos\left(\frac{\theta}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right)\geq 0\\ 2\,\mathrm{atan}_2\left(\sqrt{\sin^2\left(\phi\right)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\phi\right)},\cos\left(\frac{\theta}{2}\right)\right)-2\pi & \text{otherwise}\end{cases}}{2}-4\,\mathrm{atan}_2\left(\sqrt{\frac{1}{2}-\frac{\left|\cos\left(\frac{\theta}{2}\right)\right|}{2}}\left(\frac{\begin{cases}1 & \text{for } \cos\left(\frac{\theta}{2}\right)\geq 0\\ -1 & \text{otherwise}\end{cases}}{\pi}\right),\sqrt{\frac{\left|\cos\left(\frac{\theta}{2}\right)\right|}{2}+\frac{1}{2}+\sqrt{\left|\cos\left(\frac{\theta}{2}\right)\right|}}\right)+2\,\mathrm{atan}_2\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\cos\left(\frac{\theta}{2}\right)\right|}{2}}\left(\frac{\begin{cases}1 & \text{for } \cos\left(\frac{\theta}{2}\right)\geq 0\\ -1 & \text{otherwise}\end{cases}}{\pi}\right),\sqrt{\left|\cos\left(\frac{\theta}{2}\right)\right|}\right)-\pi$$

$$\eta_{160} = \frac{\begin{cases}2\,\mathrm{atan}_2\left(\sqrt{\sin^2\left(\phi\right)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\phi\right)},\cos\left(\frac{\theta}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right)\geq 0\\ 2\,\mathrm{atan}_2\left(\sqrt{\sin^2\left(\phi\right)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\phi\right)},\cos\left(\frac{\theta}{2}\right)\right)-2\pi & \text{otherwise}\end{cases}}{4}+2\,\mathrm{atan}_2\left(\sqrt{\frac{1}{2}-\frac{\left|\cos\left(\frac{\theta}{2}\right)\right|}{2}}\left(\frac{\begin{cases}1 & \text{for } \cos\left(\frac{\theta}{2}\right)\geq 0\\ -1 & \text{otherwise}\end{cases}}{\pi}\right),\sqrt{\frac{\left|\cos\left(\frac{\theta}{2}\right)\right|}{2}+\frac{1}{2}+\sqrt{\left|\cos\left(\frac{\theta}{2}\right)\right|}}\right)-\mathrm{atan}_2\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\cos\left(\frac{\theta}{2}\right)\right|}{2}}\left(\frac{\begin{cases}1 & \text{for } \cos\left(\frac{\theta}{2}\right)\geq 0\\ -1 & \text{otherwise}\end{cases}}{\pi}\right),\sqrt{\left|\cos\left(\frac{\theta}{2}\right)\right|}\right)+\frac{1}{2}$$

$$\eta_{161} = -\frac{\begin{cases}0 & \text{for } \sqrt{\sin^2\left(\phi\right)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\phi\right)}=0\\ \mathrm{atan}_2\left(\sqrt{\sin^2\left(\phi\right)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\phi\right)},0\right) & \text{otherwise}\end{cases}}{\pi}$$

$$\eta_{162} = \frac{2\,\mathrm{atan}_2\left(\sqrt{\frac{1}{2}-\frac{\left|\cos\left(\frac{\theta}{2}\right)\right|}{2}}\left(\frac{\begin{cases}1 & \text{for } \cos\left(\frac{\theta}{2}\right)\geq 0\\ -1 & \text{otherwise}\end{cases}}{\pi}\right),\sqrt{\frac{\left|\cos\left(\frac{\theta}{2}\right)\right|}{2}+\frac{1}{2}+\sqrt{\left|\cos\left(\frac{\theta}{2}\right)\right|}}\right)}{\pi}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{163} = \frac{\begin{cases} 2\operatorname{atan}_2\left(\sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)}, \cos\left(\frac{\theta}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right) \geq 0 \\ 2\operatorname{atan}_2\left(\sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)}, \cos\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2} + \pi$$

$$\eta_{164} = \frac{\begin{cases} 2\operatorname{atan}_2\left(\sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)}, \cos\left(\frac{\theta}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right) \geq 0 \\ 2\operatorname{atan}_2\left(\sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)}, \cos\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2} - \frac{\pi}{2} + \frac{1}{2}$$

$$\eta_{165} = \frac{\begin{cases} 2\operatorname{atan}_2\left(\sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)}, \cos\left(\frac{\theta}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right) \geq 0 \\ 2\operatorname{atan}_2\left(\sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)}, \cos\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2\pi}$$

$$\eta_{166} = -\frac{2\operatorname{atan}_2\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{|\cos\left(\frac{\theta}{2}\right)|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\theta}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{|\cos\left(\frac{\theta}{2}\right)|}\right)}{\pi}$$

$$\eta_{167} = \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan}_2\left(\sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)}, 0\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{168} = \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan}_2\left(\sin(\phi)\sin\left(\frac{\theta}{2}\right), \sin\left(\frac{\theta}{2}\right)\cos(\phi)\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{169} = -\frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan}_2\left(\sin(\phi)\sin\left(\frac{\theta}{2}\right), \sin\left(\frac{\theta}{2}\right)\cos(\phi)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}$$

$$\eta_{170} = -\frac{\operatorname{atan}_2\left(\sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)}, \cos\left(\frac{\theta}{2}\right)\right)}{\pi}$$

$$\eta_{171} = \frac{\operatorname{atan}_2\left(\sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)}, \cos\left(\frac{\theta}{2}\right)\right)}{\pi} + \frac{1}{2}$$

$$\eta_{172} = -\frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan}_2\left(\sin(\phi)\sin\left(\frac{\theta}{2}\right), \sin\left(\frac{\theta}{2}\right)\cos(\phi)\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{173} = \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan}_2\left(\sin(\phi)\sin\left(\frac{\theta}{2}\right), \sin\left(\frac{\theta}{2}\right)\cos(\phi)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}$$

$$\eta_{174} = e^{i\left(-\frac{\begin{cases} 2\operatorname{atan}_2\left(\sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)}, \cos\left(\frac{\theta}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right) \geq 0 \\ 2\operatorname{atan}_2\left(\sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)}, \cos\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4} - 2\operatorname{atan}_2\left(\sqrt{\frac{1}{2} - \frac{|\cos\left(\frac{\theta}{2}\right)|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\theta}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right)\cdot\sqrt{\frac{|\cos\left(\frac{\theta}{2}\right)|}{2} + \frac{1}{2}} + \sqrt{|\cos\left(\frac{\theta}{2}\right)|}\right) + \operatorname{atan}_2\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{|\cos\left(\frac{\theta}{2}\right)|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\theta}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{|\cos\left(\frac{\theta}{2}\right)|}\right)\right)}$$

$$\eta_{175} = -\frac{\begin{cases} 2\operatorname{atan}_2\left(\sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)}, \cos\left(\frac{\theta}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right) \geq 0 \\ 2\operatorname{atan}_2\left(\sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)}, \cos\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4}$$

$$\eta_{176} = \pi - \frac{\begin{cases} 2\operatorname{atan}_2\left(\sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)}, \cos\left(\frac{\theta}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right) \geq 0 \\ 2\operatorname{atan}_2\left(\sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)}, \cos\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4}$$

$$\eta_{177} = 2\operatorname{atan}_2\left(\sqrt{\frac{1}{2} - \frac{|\cos\left(\frac{\theta}{2}\right)|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\theta}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{\frac{|\cos\left(\frac{\theta}{2}\right)|}{2} + \frac{1}{2}} + \sqrt{|\cos\left(\frac{\theta}{2}\right)|}\right)$$

$$\eta_{178} = \frac{\begin{cases} 2\operatorname{atan}_2\left(\sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)}, \cos\left(\frac{\theta}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right) \geq 0 \\ 2\operatorname{atan}_2\left(\sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)}, \cos\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2} + \frac{\pi}{2}$$

$$\eta_{179} = -2\operatorname{atan}_2\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{|\cos\left(\frac{\theta}{2}\right)|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\theta}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{|\cos\left(\frac{\theta}{2}\right)|}\right)$$

$$\eta_{180} = e^{-\frac{i\left(\begin{cases} 2\operatorname{atan}_2\left(\sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)}, \cos\left(\frac{\theta}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right) \geq 0 \\ 2\operatorname{atan}_2\left(\sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)}, \cos\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}\right)}{4}}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{181} = \pi - 2\operatorname{atan}_2\left(\sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)}, \left|\cos\left(\frac{\theta}{2}\right)\right|\right)$$

$$\eta_{182} = -\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan}_2\left(\sin\left(\frac{\theta}{2}\right)\cos(\phi), -\sin(\phi)\sin\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} - \begin{cases} 0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan}_2\left(0, \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} - \pi$$

$$\eta_{183} = \begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan}_2\left(\sin\left(\frac{\theta}{2}\right)\cos(\phi), -\sin(\phi)\sin\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} - \frac{\operatorname{atan}_2\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right)+4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, -\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)-\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)+\cos^2\left(\frac{\theta}{2}\right)\right)}{2}$$

$$\eta_{184} = \begin{cases} 0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan}_2\left(0, \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} - \frac{\operatorname{atan}_2\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right)+4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, -\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)-\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)+\cos^2\left(\frac{\theta}{2}\right)\right)}{2} + \pi$$

$$\eta_{185} = -\begin{cases} 0 & \text{for } \sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right)+4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)} = 0 \\ \operatorname{atan}_2\left(2\sin(\phi)\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right), 2\sin\left(\frac{\theta}{2}\right)\cos(\phi)\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{186} = -\begin{cases} 0 & \text{for } \sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right)+4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)} = 0 \\ \operatorname{atan}_2\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right)+4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, 0\right) & \text{otherwise} \end{cases}$$

$$\eta_{187} = \operatorname{atan}_2\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right)+4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, -\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)-\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)+\cos^2\left(\frac{\theta}{2}\right)\right)$$

$$\eta_{188} = \begin{cases} 0 & \text{for } \sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right)+4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)} = 0 \\ \operatorname{atan}_2\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right)+4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, 0\right) & \text{otherwise} \end{cases}$$

$$\eta_{189} = \begin{cases} 0 & \text{for } \sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right)+4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)} = 0 \\ \operatorname{atan}_2\left(2\sin(\phi)\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right), 2\sin\left(\frac{\theta}{2}\right)\cos(\phi)\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{190} = -\operatorname{atan}_2\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right)+4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, -\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)-\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)+\cos^2\left(\frac{\theta}{2}\right)\right)$$

$$\eta_{191} = 2\operatorname{atan}_2\left(\sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)}, \left|\cos\left(\frac{\theta}{2}\right)\right|\right)$$

$$\eta_{192} = \begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan}_2\left(\sin\left(\frac{\theta}{2}\right)\cos(\phi), -\sin(\phi)\sin\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} - \begin{cases} 0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan}_2\left(0, \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{193} = -\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan}_2\left(\sin\left(\frac{\theta}{2}\right)\cos(\phi), -\sin(\phi)\sin\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} - \begin{cases} 0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan}_2\left(0, \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{194} = e^{i\left(\begin{cases} 0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan}_2\left(0, \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} - \frac{\operatorname{atan}_2\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right)+4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, -\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)-\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)+\cos^2\left(\frac{\theta}{2}\right)\right)}{2}\right)}$$

$$\eta_{195} = \frac{\pi - 2\operatorname{atan}_2\left(\sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)}, \left|\cos\left(\frac{\theta}{2}\right)\right|\right)}{\pi}$$

$$\eta_{196} = \frac{-\pi\left(\frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan}_2\left(\sin\left(\frac{\theta}{2}\right)\cos(\phi), -\sin(\phi)\sin\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{\begin{cases} 0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan}_2\left(0, \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}\right) - \frac{\pi}{2}}{\pi}$$

$$\eta_{197} = \frac{\pi\left(\frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan}_2\left(\sin\left(\frac{\theta}{2}\right)\cos(\phi), -\sin(\phi)\sin\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{\begin{cases} 0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan}_2\left(0, \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}\right) + \frac{\pi}{2}}{\pi} + \frac{1}{2}$$

$$\eta_{198} = \frac{\pi\left(\frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan}_2\left(\sin\left(\frac{\theta}{2}\right)\cos(\phi), -\sin(\phi)\sin\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} - \frac{\begin{cases} 0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan}_2\left(0, \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + 1\right) + 2\left(\begin{cases} 0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan}_2\left(0, \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}\right) - \begin{cases} 2\operatorname{atan}_2\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right)+4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, -\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)-\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)+\cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)-\cos^2\left(\frac{\theta}{2}\right) \le 0 \\ 2\operatorname{atan}_2\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right)+4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, -\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)-\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)+\cos^2\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{\pi} - 4\operatorname{atan}_2\left(\sqrt{1 - \frac{\left[\cos^2(x)\cos^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\cos^2(x)-\cos^2\left(\frac{\theta}{2}\right)\right]}{x}}\left(\begin{cases} 1 & \text{for } \sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)-\cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\right)\sqrt{\frac{\left[\cos^2(x)\cos^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\cos^2(x)-\cos^2\left(\frac{\theta}{2}\right)\right]}{x}} + \frac{1}{2} + \sqrt{\left[\sin^2(x)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(x)-\cos^2\left(\frac{\theta}{2}\right)\right]}\right) + 2\operatorname{atan}_2\left(\sqrt{2}\sqrt{1 - \frac{\left[\cos^2(x)\cos^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\cos^2(x)-\cos^2\left(\frac{\theta}{2}\right)\right]}{x}}\left(\begin{cases} 1 & \text{for } \sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)-\cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\right)\sqrt{\left[\sin^2(x)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(x)-\cos^2\left(\frac{\theta}{2}\right)\right]}\right) + 2\operatorname{atan}_2\left(\sqrt{\sin^2(x)\sin^2\left(\frac{\theta}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2(x)}, \left|\cos\left(\frac{\theta}{2}\right)\right|\right) - \frac{3\pi}{2}}{\pi}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{199} = -\frac{\begin{cases} 0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{\begin{cases} 2\operatorname{atan_2}\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right) + 4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, -\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) - \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) + \cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ 2\operatorname{atan_2}\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right) + 4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, -\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) - \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) + \cos^2\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4} + 2\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\frac{\left|\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right)\right|}{2}} + \frac{1}{2} + \sqrt{\left|\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right)\right|}\right)}{\pi} - \operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\left|\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right)\right|}\right) - \operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)}, \left|\cos\left(\frac{\theta}{2}\right)\right|\right)}{\pi} + \frac{1}{2}$$

$$\eta_{200} = -\frac{\begin{cases} 0 & \text{for } \sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right) + 4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)} = 0 \\ \operatorname{atan_2}\left(2\sin(\phi)\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right), 2\sin\left(\frac{\theta}{2}\right)\cos(\phi)\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{201} = \frac{\begin{cases} 0 & \text{for } \sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right) + 4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)} = 0 \\ \operatorname{atan_2}\left(2\sin(\phi)\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right), 2\sin\left(\frac{\theta}{2}\right)\cos(\phi)\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}$$

$$\eta_{202} = -\frac{\begin{cases} 0 & \text{for } \sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right) + 4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right) + 4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, 0\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{203} = \frac{2\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\frac{\left|\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right)\right|}{2}} + \frac{1}{2} + \sqrt{\left|\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right)\right|}\right)}{\pi}$$

$$\eta_{204} = \frac{\begin{cases} 2\operatorname{atan_2}\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right) + 4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, -\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) - \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) + \cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ 2\operatorname{atan_2}\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right) + 4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, -\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) - \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) + \cos^2\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2\pi} + \pi$$

$$\eta_{205} = \frac{\begin{cases} 2\operatorname{atan_2}\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right) + 4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, -\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) - \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) + \cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ 2\operatorname{atan_2}\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right) + 4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, -\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) - \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) + \cos^2\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2\pi} - \frac{\frac{5}{2}}{\pi} + \frac{1}{2}$$

$$\eta_{206} = \frac{\begin{cases} 2\operatorname{atan_2}\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right) + 4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, -\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) - \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) + \cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ 2\operatorname{atan_2}\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right) + 4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, -\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) - \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) + \cos^2\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2\pi}$$

$$\eta_{207} = -\frac{2\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\left|\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right)\right|}\right)}{\pi}$$

$$\eta_{208} = \frac{\begin{cases} 0 & \text{for } \sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right) + 4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right) + 4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, 0\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{209} = \frac{\begin{cases} 0 & \text{for } \sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right) + 4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)} = 0 \\ \operatorname{atan_2}\left(2\sin(\phi)\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right), 2\sin\left(\frac{\theta}{2}\right)\cos(\phi)\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{210} = -\frac{\begin{cases} 0 & \text{for } \sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right) + 4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)} = 0 \\ \operatorname{atan_2}\left(2\sin(\phi)\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right), 2\sin\left(\frac{\theta}{2}\right)\cos(\phi)\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}$$

$$\eta_{211} = -\frac{\operatorname{atan_2}\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right) + 4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, -\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) - \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) + \cos^2\left(\frac{\theta}{2}\right)\right)}{\pi}$$

$$\eta_{212} = \frac{\operatorname{atan_2}\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right) + 4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, -\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) - \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) + \cos^2\left(\frac{\theta}{2}\right)\right)}{\pi} + \frac{1}{2}$$

$$\eta_{213} = \frac{2\operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)}, \left|\cos\left(\frac{\theta}{2}\right)\right|\right)}{\pi}$$

$$\eta_{214} = -\frac{2\left(\begin{cases} 0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}\right)}{\pi}$$

$$\eta_{215} = \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan_2}\left(\sin\left(\frac{\theta}{2}\right)\cos(\phi), -\sin(\phi)\sin\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} + \begin{cases} 0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}$$

$$\eta_{216} = e^{i\left(\begin{cases} 0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} - \frac{\begin{cases} 2\operatorname{atan_2}\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right) + 4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, -\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) - \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) + \cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ 2\operatorname{atan_2}\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right) + 4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, -\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) - \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) + \cos^2\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4} - 2\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\frac{\left|\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right)\right|}{2}} + \frac{1}{2} + \sqrt{\left|\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right)\right|}\right) + \operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\left|\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right)\right|}\right) - \operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)}, \left|\cos\left(\frac{\theta}{2}\right)\right|\right)\right)}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{217} = \begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan_2}\left(\sin\left(\frac{\theta}{2}\right)\cos(\phi), -\sin(\phi)\sin\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} - \frac{\begin{cases} 2\operatorname{atan_2}\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right) + 4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, -\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) - \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) + \cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ 2\operatorname{atan_2}\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right) + 4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, -\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) - \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) + \cos^2\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4}$$

$$\eta_{218} = \begin{cases} 0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} - \frac{\begin{cases} 2\operatorname{atan_2}\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right) + 4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, -\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) - \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) + \cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ 2\operatorname{atan_2}\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right) + 4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, -\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) - \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) + \cos^2\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4} + \pi$$

$$\eta_{219} = 2\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}, \sqrt{\frac{\left|\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right)\right|}{2}} + \frac{1}{2} + \sqrt{\left|\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right)\right|}\right)$$

$$\eta_{220} = \frac{\begin{cases} 2\operatorname{atan_2}\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right) + 4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, -\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) - \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) + \cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ 2\operatorname{atan_2}\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right) + 4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, -\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) - \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) + \cos^2\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2} + \frac{\pi}{2}$$

$$\eta_{221} = -2\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}, \sqrt{\left|\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right)\right|}\right)$$

$$\eta_{222} = e^{i\left(\begin{cases} 0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} - \frac{\begin{cases} 2\operatorname{atan_2}\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right) + 4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, -\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) - \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) + \cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) + \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ 2\operatorname{atan_2}\left(\sqrt{4\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right) + 4\sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi)\cos^2\left(\frac{\theta}{2}\right)}, -\sin^2(\phi)\sin^2\left(\frac{\theta}{2}\right) - \sin^2\left(\frac{\theta}{2}\right)\cos^2(\phi) + \cos^2\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4}\right)}$$

$$\eta_{223} = \frac{-\pi\left(\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \sin\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}\right) - \frac{\pi}{2}}{\pi}$$

$$\eta_{224} = \frac{\pi\left(\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \sin\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}\right) + \frac{\pi}{2}}{\pi} + \frac{1}{2}$$

$$\eta_{225} = \frac{-\frac{\begin{cases} 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right) \ge 0 \\ 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2}}{\pi} - 4\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\theta}{2}\right) \ge 0 \\ -1 & \text{otherwise} \end{cases}, \sqrt{\frac{\left|\cos\left(\frac{\theta}{2}\right)\right|}{2}} + \frac{1}{2} + \sqrt{\left|\cos\left(\frac{\theta}{2}\right)\right|}\right) + 2\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\theta}{2}\right) \ge 0 \\ -1 & \text{otherwise} \end{cases}, \sqrt{\left|\cos\left(\frac{\theta}{2}\right)\right|}\right) - \pi$$

$$\eta_{226} = \frac{\frac{\begin{cases} 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right) \ge 0 \\ 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4}}{\pi} + 2\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\theta}{2}\right) \ge 0 \\ -1 & \text{otherwise} \end{cases}, \sqrt{\frac{\left|\cos\left(\frac{\theta}{2}\right)\right|}{2}} + \frac{1}{2} + \sqrt{\left|\cos\left(\frac{\theta}{2}\right)\right|}\right) - \operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\theta}{2}\right) \ge 0 \\ -1 & \text{otherwise} \end{cases}, \sqrt{\left|\cos\left(\frac{\theta}{2}\right)\right|}\right)\right) + \frac{1}{2}$$

$$\eta_{227} = -\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, 0\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{228} = \frac{\frac{\begin{cases} 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right) \ge 0 \\ 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2}}{\pi} + \pi$$

$$\eta_{229} = \frac{-\frac{\begin{cases} 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right) \ge 0 \\ 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2}}{\pi} - \frac{\pi}{2} + \frac{1}{2}$$

$$\eta_{230} = \frac{\begin{cases} 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right) \ge 0 \\ 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2\pi}$$

$$\eta_{231} = -\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, 0\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{232} = \frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \sin\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{233} = -\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \sin\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}$$

$$\eta_{234} = -\frac{\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right)}{\pi}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{235} = \frac{\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right)}{\pi} + \frac{1}{2}$$

$$\eta_{236} = -\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \sin\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{237} = \frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \sin\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}$$

$$\eta_{238} = e^{i\left(-\frac{\begin{cases} 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right) \geq 0 \\ 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4} - 2\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\theta}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{\frac{\left|\cos\left(\frac{\theta}{2}\right)\right|}{2} + \frac{1}{2}} + \sqrt{\left|\cos\left(\frac{\theta}{2}\right)\right|}\right) + \operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\theta}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{\left|\cos\left(\frac{\theta}{2}\right)\right|}\right)\right)}$$

$$\eta_{239} = -\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \sin\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} - \pi$$

$$\eta_{240} = -\frac{\begin{cases} 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right) \geq 0 \\ 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4}$$

$$\eta_{241} = \pi - \frac{\begin{cases} 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right) \geq 0 \\ 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4}$$

$$\eta_{242} = -\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, 0\right) & \text{otherwise} \end{cases}$$

$$\eta_{243} = \frac{\begin{cases} 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right) \geq 0 \\ 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2} + \frac{\pi}{2}$$

$$\eta_{244} = \begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, 0\right) & \text{otherwise} \end{cases}$$

$$\eta_{245} = \begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \sin\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{246} = -\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right)$$

$$\eta_{247} = -\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \sin\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{248} = e^{-\frac{i\left(\begin{cases} 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right) \geq 0 \\ 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}\right)}{4}}$$

$$\eta_{249} = \frac{-\pi\left(\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(\sin\left(\frac{\theta}{2}\right), 0\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}\right) - \frac{\pi}{2}}{\pi}$$

$$\eta_{250} = \frac{\pi\left(\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(\sin\left(\frac{\theta}{2}\right), 0\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}\right) + \frac{\pi}{2}}{\pi} + \frac{1}{2}$$

$$\eta_{251} = \frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(\sin\left(\frac{\theta}{2}\right), 0\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{252} = -\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(\sin\left(\frac{\theta}{2}\right), 0\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{253} = -\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(\sin\left(\frac{\theta}{2}\right), 0\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{254} = -\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(\sin\left(\frac{\theta}{2}\right), 0\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}$$

$$\eta_{255} = -\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(\sin\left(\frac{\theta}{2}\right), 0\right) & \text{otherwise} \end{cases} - \pi$$

$$\eta_{256} = \begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(\sin\left(\frac{\theta}{2}\right), 0\right) & \text{otherwise} \end{cases}$$

$$\eta_{257} = -\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(\sin\left(\frac{\theta}{2}\right), 0\right) & \text{otherwise} \end{cases}$$

$$\eta_{258} = \begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \sin\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} + \pi$$

$$\eta_{259} = -\frac{\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right)}{2}$$

$$\eta_{260} = \pi - \frac{\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right)}{2}$$

$$\eta_{261} = \operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right)$$

$$\eta_{262} = e^{-\frac{i\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right)}{2}}$$

$$\eta_{263} = \frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \sin\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \pi$$

$$\eta_{264} = \frac{-2\left(\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \sin\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}\right) - \frac{\begin{cases} 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right) \geq 0 \\ 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2} - 4\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\theta}{2}\right)\right|}{2}}, \left(\begin{cases} 1 & \text{for } \cos\left(\frac{\theta}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\frac{\left|\cos\left(\frac{\theta}{2}\right)\right|}{2} + \frac{1}{2}} + \sqrt{\left|\cos\left(\frac{\theta}{2}\right)\right|}\right) + 2\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\theta}{2}\right)\right|}{2}}, \left(\begin{cases} 1 & \text{for } \cos\left(\frac{\theta}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\left|\cos\left(\frac{\theta}{2}\right)\right|}\right) - \pi}{\pi}$$

$$\eta_{265} = \frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \sin\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} + \frac{\begin{cases} 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right) \geq 0 \\ 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \cos\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4} + 2\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\theta}{2}\right)\right|}{2}}, \left(\begin{cases} 1 & \text{for } \cos\left(\frac{\theta}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\frac{\left|\cos\left(\frac{\theta}{2}\right)\right|}{2} + \frac{1}{2}} + \sqrt{\left|\cos\left(\frac{\theta}{2}\right)\right|}\right) - \operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\theta}{2}\right)\right|}{2}}, \left(\begin{cases} 1 & \text{for } \cos\left(\frac{\theta}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\left|\cos\left(\frac{\theta}{2}\right)\right|}\right)}{\pi} + \frac{1}{2}$$

$$\eta_{266} = -\frac{7\pi}{8} + \operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{267} = \pi - \operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{268} = \frac{\pi - 2\operatorname{atan}\left(\frac{\sqrt{\frac{1}{2} - \frac{\sqrt{2}}{4}}}{\frac{\sqrt{2}}{4} + \sqrt{\frac{\sqrt{2}}{4} + \frac{1}{2}}}\right)}{\pi}$$

$$\eta_{269} = \frac{-\frac{3\pi}{4} + 2\operatorname{atan}\left(2^{\frac{3}{4}}\sqrt{\frac{1}{2} - \frac{\sqrt{2}}{4}}\right)}{\pi}$$

$$\eta_{270} = \frac{-\operatorname{atan}\left(2^{\frac{3}{4}}\sqrt{\frac{1}{2} - \frac{\sqrt{2}}{4}}\right) + \frac{\pi}{8}}{\pi} + \frac{1}{2}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{271} = -\frac{2\operatorname{atan}\left(2^{\frac{3}{2}}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}\right)}{\pi}$$

$$\eta_{272} = \frac{2\operatorname{atan}\left(\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{\frac{2^{\frac{3}{2}}}{2}+\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)}{\pi}$$

$$\eta_{273} = e^{i\left(-2\operatorname{atan}\left(\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{\frac{2^{\frac{3}{2}}}{2}+\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)-\frac{\pi}{8}+\operatorname{atan}\left(2^{\frac{3}{2}}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}\right)\right)}$$

$$\eta_{274} = \frac{\sqrt[4]{-1}\left(\sqrt{\sqrt{2}+2}-\sqrt{-2+\sqrt{2}}\right)}{2}$$

$$\eta_{275} = \pi - 2\operatorname{atan}\left(\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{\frac{2^{\frac{3}{2}}}{2}+\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)$$

$$\eta_{276} = -2\operatorname{atan}\left(2^{\frac{3}{2}}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}\right)$$

$$\eta_{277} = 2\operatorname{atan}\left(\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{\frac{2^{\frac{3}{2}}}{2}+\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)$$

$$\eta_{278} = \operatorname{atan}\left(\frac{\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{279} = \operatorname{atan}\left(\frac{\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right)+\pi$$

$$\eta_{280} = 2\operatorname{atan}\left(\frac{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right)+2\pi$$

$$\eta_{281} = \pi - \operatorname{atan}\left(\frac{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{282} = \frac{2\operatorname{atan}\left(\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{\frac{2^{\frac{3}{2}}}{2}+\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)+\pi}{\pi}$$

$$\eta_{283} = \frac{-2\operatorname{atan}\left(2^{\frac{3}{2}}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}\right)-\frac{\pi}{4}}{\pi}$$

$$\eta_{284} = \frac{-\frac{\pi}{8}+\operatorname{atan}\left(2^{\frac{3}{2}}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}\right)}{\pi}+\frac{1}{2}$$

$$\eta_{285} = \frac{2\operatorname{atan}\left(2^{\frac{3}{2}}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}\right)}{\pi}$$

$$\eta_{286} = -\frac{2\operatorname{atan}\left(\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{\frac{2^{\frac{3}{2}}}{2}+\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)}{\pi}$$

$$\eta_{287} = e^{i\left(-\operatorname{atan}\left(2^{\frac{3}{2}}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}\right)+\frac{\pi}{8}+2\operatorname{atan}\left(\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{\frac{2^{\frac{3}{2}}}{2}+\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)\right)}$$

$$\eta_{288} = \frac{(-1)^{\frac{3}{4}}\left(-\sqrt{\sqrt{2}+2}-\sqrt{-2+\sqrt{2}}\right)}{2}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{289} = \operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{290} = \operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}}\right) + \pi$$

$$\eta_{291} = 2\operatorname{atan}\left(\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2\sqrt[4]{\frac{1}{2}}+\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right) + \pi$$

$$\eta_{292} = 2\operatorname{atan}\left(2\sqrt[4]{\frac{1}{2}-\frac{\sqrt{2}}{4}}\right)$$

$$\eta_{293} = -2\operatorname{atan}\left(\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2\sqrt[4]{\frac{1}{2}}+\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)$$

$$\eta_{294} = -\operatorname{atan}\left(\frac{\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right) + 2\pi$$

$$\eta_{295} = -\operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}}\right) + \operatorname{atan}\left(\frac{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right) + 2\pi$$

$$\eta_{296} = \frac{(-1)^{\frac{3}{4}}\sqrt{2}(1-i)}{2}$$

$$\eta_{297} = \operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}\right) + \operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{298} = \frac{-\frac{9\pi}{4} - 4\operatorname{atan}\left(\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2\sqrt[4]{\frac{1}{2}}+\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right) + 2\operatorname{atan}\left(2\sqrt[4]{\frac{1}{2}-\frac{\sqrt{2}}{4}}\right)}{\pi}$$

$$\eta_{299} = \frac{1}{2} - \frac{-\frac{5\pi}{4} - 2\operatorname{atan}\left(\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2\sqrt[4]{\frac{1}{2}}+\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right) + \operatorname{atan}\left(2\sqrt[4]{\frac{1}{2}-\frac{\sqrt{2}}{4}}\right)}{\pi}$$

$$\eta_{300} = \frac{-\frac{5\pi}{4} - 4\operatorname{atan}\left(\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2\sqrt[4]{\frac{1}{2}}+\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right) + 2\operatorname{atan}\left(2\sqrt[4]{\frac{1}{2}-\frac{\sqrt{2}}{4}}\right)}{\pi}$$

$$\eta_{301} = \frac{-\operatorname{atan}\left(2\sqrt[4]{\frac{1}{2}-\frac{\sqrt{2}}{4}}\right) + \frac{\pi}{2} + 2\operatorname{atan}\left(\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2\sqrt[4]{\frac{1}{2}}+\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)}{\pi} + \frac{1}{2}$$

$$\eta_{302} = \frac{2\operatorname{atan}\left(\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)}{\pi}$$

$$\eta_{303} = -ie^{\frac{i\left(-4\operatorname{atan}\left(\frac{2\sqrt[4]{2-\sqrt{2}}}{2}\right) - 4\operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}}\right) + 8\operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}}{2\sqrt[4]{2}+\sqrt{\sqrt{2}+2}}\right) + \pi\right)}{4}}$$

$$\eta_{304} = 2\operatorname{atan}\left(\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right) + \pi$$

$$\eta_{305} = -\pi - 2\operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{306} = \frac{\sqrt[4]{-1}\left(\sqrt{2-\sqrt{2}}-\sqrt{-2-\sqrt{2}}\right)}{2}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{307} = 2\,\mathrm{atan}\left(\frac{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right) + \pi$$

$$\eta_{308} = e^{-i\,\mathrm{atan}\left(\frac{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right)}$$

$$\eta_{309} = \mathrm{atan}\left(\frac{2\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{310} = -\,\mathrm{atan}\left(\frac{2\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{\sqrt{\sqrt{2}+2}}\right) + \frac{\pi}{8}$$

$$\eta_{311} = -\frac{\mathrm{atan}\left(\frac{\sqrt{2}-\sqrt{2}}{\sqrt{\sqrt{2}+2}}\right)}{2}$$

$$\eta_{312} = \pi - \frac{\mathrm{atan}\left(\frac{\sqrt{2}-\sqrt{2}}{\sqrt{\sqrt{2}+2}}\right)}{2}$$

$$\eta_{313} = \frac{\pi - 2\,\mathrm{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\sqrt[4]{\frac{\sqrt{2}}{4}+\frac{1}{2}}+\cos\left(\frac{\pi}{16}\right)}\right)}{\pi}$$

$$\eta_{314} = \frac{-\frac{5\pi}{8} + 2\,\mathrm{atan}\left(\frac{\sqrt{2}\sin\left(\frac{\pi}{16}\right)}{\sqrt[4]{\sqrt{2}+1}}\right)}{\pi}$$

$$\eta_{315} = \frac{-\,\mathrm{atan}\left(\frac{\sqrt{2}\sin\left(\frac{\pi}{16}\right)}{\sqrt[4]{\sqrt{2}+1}}\right) + \frac{\pi}{16}}{\pi} + \frac{1}{2}$$

$$\eta_{316} = -\frac{2\,\mathrm{atan}\left(\frac{\sqrt{2}\sin\left(\frac{\pi}{16}\right)}{\sqrt[4]{\sqrt{2}+1}}\right)}{\pi}$$

$$\eta_{317} = \frac{2\,\mathrm{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\sqrt[4]{\sqrt{2}+2}+\cos\left(\frac{\pi}{16}\right)}\right)}{\pi}$$

$$\eta_{318} = e^{i\left(-2\,\mathrm{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\sqrt[4]{\sqrt{2}+2}+\cos\left(\frac{\pi}{16}\right)}\right) - \frac{\pi}{16} + \mathrm{atan}\left(\frac{\sqrt{2}\sin\left(\frac{\pi}{16}\right)}{\sqrt[4]{\sqrt{2}+1}}\right)\right)}$$

$$\eta_{319} = \pi - 2\,\mathrm{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\sqrt[4]{\sqrt{2}+2}+\cos\left(\frac{\pi}{16}\right)}\right)$$

$$\eta_{320} = -2\,\mathrm{atan}\left(\frac{\sqrt{2}\sin\left(\frac{\pi}{16}\right)}{\sqrt[4]{\sqrt{2}+1}}\right)$$

$$\eta_{321} = 2\,\mathrm{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\sqrt[4]{\sqrt{2}+1}+\cos\left(\frac{\pi}{16}\right)}\right)$$

$$\eta_{322} = -\,\mathrm{atan}\left(\frac{2-\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right) - \frac{\pi}{16}$$

$$\eta_{323} = \frac{2\,\mathrm{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\sqrt[4]{\sqrt{2}+1}+\cos\left(\frac{\pi}{16}\right)}\right) + \pi}{\pi}$$

$$\eta_{324} = \frac{-\frac{3\pi}{8} - 2\,\mathrm{atan}\left(\frac{\sqrt{2}\sin\left(\frac{\pi}{16}\right)}{\sqrt[4]{\sqrt{2}+1}}\right)}{\pi}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{325} = \frac{-\frac{\pi}{16}+\operatorname{atan}\left(\frac{\sqrt{2}\sin\left(\frac{\pi}{16}\right)}{\sqrt[4]{\sqrt{2}^2+\frac{1}{2}}}\right)}{\pi}+\frac{1}{2}$$

$$\eta_{326} = \frac{2\operatorname{atan}\left(\frac{\sqrt{2}\sin\left(\frac{\pi}{16}\right)}{\sqrt[4]{\sqrt{2}^2+\frac{1}{2}}}\right)}{\pi}$$

$$\eta_{327} = -\frac{2\operatorname{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\sqrt[4]{\sqrt{2}^2+\frac{1}{2}}+\cos\left(\frac{\pi}{16}\right)}\right)}{\pi}$$

$$\eta_{328} = e^{i\left(-\operatorname{atan}\left(\frac{\sqrt{2}\sin\left(\frac{\pi}{16}\right)}{\sqrt[4]{\sqrt{2}^2+\frac{1}{2}}}\right)+\frac{\pi}{16}+2\operatorname{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\sqrt[4]{\sqrt{2}^2+\frac{1}{2}}+\cos\left(\frac{\pi}{16}\right)}\right)\right)}$$

$$\eta_{329} = 2\operatorname{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\sqrt[4]{\sqrt{2}^2+\frac{1}{2}}+\cos\left(\frac{\pi}{16}\right)}\right)+\pi$$

$$\eta_{330} = 2\operatorname{atan}\left(\frac{\sqrt{2}\sin\left(\frac{\pi}{16}\right)}{\sqrt[4]{\sqrt{2}^2+\frac{1}{2}}}\right)$$

$$\eta_{331} = -2\operatorname{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\sqrt[4]{\sqrt{2}^2+\frac{1}{2}}+\cos\left(\frac{\pi}{16}\right)}\right)$$

$$\eta_{332} = -\frac{(1+i)^2 e^{\frac{i(2\gamma+ibda+\phi)}{4}}}{2}$$

$$\eta_{333} = -\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{ibda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{ibda}{2}-\frac{\phi}{2}\right)}=0 \\ \operatorname{atan}_2\left(\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{ibda}{2}-\frac{\phi}{2}\right),\sin\left(\frac{\theta}{2}\right)\sin\left(\frac{ibda}{2}-\frac{\phi}{2}\right)\right) & \text{otherwise} \end{cases}-\pi$$

$$\eta_{334} = -\frac{\operatorname{atan}_2\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{ibda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{ibda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{ibda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)},\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{ibda}{2}+\frac{\phi}{2}\right)\right)}{2}$$

$$\eta_{335} = \pi-\frac{\operatorname{atan}_2\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{ibda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{ibda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{ibda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)},\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{ibda}{2}+\frac{\phi}{2}\right)\right)}{2}$$

$$\eta_{336} = -\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{ibda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{ibda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{ibda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)}=0 \\ \operatorname{atan}_2\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{ibda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{ibda}{2}-\frac{\phi}{2}\right)},\sin\left(\frac{ibda}{2}+\frac{\phi}{2}\right)\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{337} = \operatorname{atan}_2\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{ibda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{ibda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{ibda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)},\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{ibda}{2}+\frac{\phi}{2}\right)\right)$$

$$\eta_{338} = \begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{ibda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{ibda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{ibda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)}=0 \\ \operatorname{atan}_2\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{ibda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{ibda}{2}-\frac{\phi}{2}\right)},\sin\left(\frac{ibda}{2}+\frac{\phi}{2}\right)\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{339} = \begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{ibda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{ibda}{2}-\frac{\phi}{2}\right)}=0 \\ \operatorname{atan}_2\left(\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{ibda}{2}-\frac{\phi}{2}\right),\sin\left(\frac{\theta}{2}\right)\sin\left(\frac{ibda}{2}-\frac{\phi}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{340} = \gamma+\frac{ibda}{2}+\frac{\phi}{2}-\operatorname{atan}_2\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{ibda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{ibda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{ibda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)},\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{ibda}{2}+\frac{\phi}{2}\right)\right)$$

$$\eta_{341} = -\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{ibda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{ibda}{2}-\frac{\phi}{2}\right)}=0 \\ \operatorname{atan}_2\left(\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{ibda}{2}-\frac{\phi}{2}\right),\sin\left(\frac{\theta}{2}\right)\sin\left(\frac{ibda}{2}-\frac{\phi}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{342} = e^{-\frac{i\operatorname{atan}_2\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{ibda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{ibda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{ibda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)},\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{ibda}{2}+\frac{\phi}{2}\right)\right)}{2}}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{343} = \frac{-\pi \left(\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}-\frac{\phi}{2}\right), \sin\left(\frac{\theta}{2}\right)\sin\left(\frac{tbda}{2}-\frac{\phi}{2}\right)\right) & \text{otherwise} \end{cases} \right)}{\pi} + \frac{1}{2} - \frac{\pi}{2}$$

$$\eta_{344} = \frac{\pi \left(\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}-\frac{\phi}{2}\right), \sin\left(\frac{\theta}{2}\right)\sin\left(\frac{tbda}{2}-\frac{\phi}{2}\right)\right) & \text{otherwise} \end{cases} \right)}{\pi} + \frac{1}{2} + \frac{\pi}{2}$$

$$\eta_{345} = \frac{\begin{cases} 2\operatorname{atan_2}\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)}, \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right) \geq 0 \\ 2\operatorname{atan_2}\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)}, \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2} - 4\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{\frac{\left|\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right|}{2} + \frac{1}{2} + \sqrt{\left|\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right|}}\right) + 2\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{\left|\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right|}\right) - \pi}{\pi}$$

$$\eta_{346} = \frac{\begin{cases} 2\operatorname{atan_2}\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)}, \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right) \geq 0 \\ 2\operatorname{atan_2}\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)}, \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4} + 2\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{\frac{\left|\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right|}{2} + \frac{1}{2} + \sqrt{\left|\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right|}}\right) - \operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{\left|\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right|}\right)}{\pi} + \frac{1}{2}$$

$$\eta_{347} = -\frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)}, \sin\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{348} = \frac{2\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{\frac{\left|\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right|}{2} + \frac{1}{2} + \sqrt{\left|\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right|}}\right)}{\pi}$$

$$\eta_{349} = \frac{\begin{cases} 2\operatorname{atan_2}\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)}, \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right) \geq 0 \\ 2\operatorname{atan_2}\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)}, \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2\pi} + \frac{\pi}{\pi}$$

$$\eta_{350} = \frac{\begin{cases} 2\operatorname{atan_2}\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)}, \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right) \geq 0 \\ 2\operatorname{atan_2}\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)}, \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2\pi} - \frac{\pi}{2} + \frac{1}{2}$$

$$\eta_{351} = \frac{\begin{cases} 2\operatorname{atan_2}\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)}, \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right) \geq 0 \\ 2\operatorname{atan_2}\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)}, \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2\pi}$$

$$\eta_{352} = -\frac{2\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{\left|\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right|}\right)}{\pi}$$

$$\eta_{353} = \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)}, \sin\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{354} = \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}-\frac{\phi}{2}\right), \sin\left(\frac{\theta}{2}\right)\sin\left(\frac{tbda}{2}-\frac{\phi}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{355} = -\frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}-\frac{\phi}{2}\right), \sin\left(\frac{\theta}{2}\right)\sin\left(\frac{tbda}{2}-\frac{\phi}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}$$

$$\eta_{356} = \frac{\gamma + \frac{tbda}{2} + \frac{\phi}{2} - \operatorname{atan_2}\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)}, \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right)}{\pi}$$

$$\eta_{357} = \frac{-\gamma - \frac{tbda}{2} - \frac{\phi}{2} + \operatorname{atan_2}\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)}, \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right)}{\pi} + \frac{1}{2}$$

$$\eta_{358} = -\frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}-\frac{\phi}{2}\right), \sin\left(\frac{\theta}{2}\right)\sin\left(\frac{tbda}{2}-\frac{\phi}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{359} = \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}-\frac{\phi}{2}\right), \sin\left(\frac{\theta}{2}\right)\sin\left(\frac{tbda}{2}-\frac{\phi}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}$$

$$\eta_{360} = e^{i \left(-\frac{\begin{cases} 2\operatorname{atan_2}\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)}, \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right) \geq 0 \\ 2\operatorname{atan_2}\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{tbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)}, \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4} - 2\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{\frac{\left|\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right|}{2} + \frac{1}{2} + \sqrt{\left|\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right|}}\right) + \operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{\left|\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{tbda}{2}+\frac{\phi}{2}\right)\right|}\right) \right)}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{361} = (-1)^{\frac{3}{4}}\, e^{\frac{i(2\gamma+lbda+\phi)}{4}}$$

$$\eta_{362} = \frac{(1+i)^2 e^{\frac{i(2\gamma+lbda+\phi)}{4}}}{2}$$

$$\eta_{363} = -\frac{\begin{cases} 2\,\mathrm{atan2}\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{lbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{lbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)},\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\geq 0 \\ 2\,\mathrm{atan2}\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{lbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{lbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)},\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\right)-2\pi & \text{otherwise} \end{cases}}{4}$$

$$\eta_{364} = \pi - \frac{\begin{cases} 2\,\mathrm{atan2}\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{lbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{lbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)},\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\geq 0 \\ 2\,\mathrm{atan2}\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{lbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{lbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)},\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\right)-2\pi & \text{otherwise} \end{cases}}{4}$$

$$\eta_{365} = 2\,\mathrm{atan2}\left(\sqrt{\frac{1}{2}-\frac{\left|\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\right|}{2}},\left(\begin{cases}1 & \text{for } \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\geq 0\\ -1 & \text{otherwise}\end{cases},\sqrt{\frac{\left|\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\right|}{2}}+\frac{1}{2}+\sqrt{\left|\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\right|}\right)\right)$$

$$\eta_{366} = \frac{\begin{cases} 2\,\mathrm{atan2}\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{lbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{lbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)},\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\geq 0 \\ 2\,\mathrm{atan2}\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{lbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{lbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)},\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\right)-2\pi & \text{otherwise} \end{cases}}{2} + \frac{\pi}{2}$$

$$\eta_{367} = -2\,\mathrm{atan2}\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\right|}{2}},\left(\begin{cases}1 & \text{for } \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\geq 0\\ -1 & \text{otherwise}\end{cases},\sqrt{\left|\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\right|}\right)\right)$$

$$\eta_{368} = e^{-\frac{i\left(\begin{cases} 2\,\mathrm{atan2}\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{lbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{lbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)},\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\right) & \text{for } \cos\left(\frac{\theta}{2}\right)\cos\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\geq 0 \\ 2\,\mathrm{atan2}\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{lbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{lbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)},\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\right)-2\pi & \text{otherwise} \end{cases}\right)}{4}}$$

$$\eta_{369} = \frac{\pi - 2\,\mathrm{atan2}\left(\sin\left(\frac{\mathrm{atan2}\left(\sin\left(\theta\right),\cos\left(\theta\right)\right)}{4}\right),\cos\left(\frac{\mathrm{atan2}\left(\sin\left(\theta\right),\cos\left(\theta\right)\right)}{4}\right)+\sqrt{\cos\left(\frac{\mathrm{atan2}\left(\sin\left(\theta\right),\cos\left(\theta\right)\right)}{2}\right)}\right)}{\pi}$$

$$\eta_{370} = \frac{-\pi\left(-\frac{\mathrm{atan2}\left(\sin\left(\theta\right),\cos\left(\theta\right)\right)}{\pi}-\frac{\pi}{2}+\frac{1}{2}\right)-\frac{\pi}{2}}{\pi}$$

$$\eta_{371} = \frac{\pi\left(-\frac{\mathrm{atan2}\left(\sin\left(\theta\right),\cos\left(\theta\right)\right)}{\pi}-\frac{\pi}{2}+\frac{1}{2}\right)+\frac{\pi}{2}}{\pi}+\frac{1}{2}$$

$$\eta_{372} = \frac{\pi\left(-\frac{\mathrm{atan2}\left(\sin\left(\theta\right),\cos\left(\theta\right)\right)}{\pi}-\frac{\pi}{2}+\frac{\mathrm{atan2}\left(\sin\left(\theta\right),\cos\left(\theta\right)\right)}{\pi}+\pi+\frac{1}{2}\right)+2\,\mathrm{atan2}\left(\sqrt{2}\sin\left(\frac{\mathrm{atan2}\left(\sin\left(\theta\right),\cos\left(\theta\right)\right)}{4}\right),\sqrt{\cos\left(\frac{\mathrm{atan2}\left(\sin\left(\theta\right),\cos\left(\theta\right)\right)}{2}\right)}\right)-\frac{\mathrm{atan2}\left(\sin\left(\theta\right),\cos\left(\theta\right)\right)}{2}-\frac{3\pi}{2}}{\pi}$$

$$\eta_{373} = \frac{-\,\mathrm{atan2}\left(\sqrt{2}\sin\left(\frac{\mathrm{atan2}\left(\sin\left(\theta\right),\cos\left(\theta\right)\right)}{4}\right),\sqrt{\cos\left(\frac{\mathrm{atan2}\left(\sin\left(\theta\right),\cos\left(\theta\right)\right)}{2}\right)}\right)+\frac{\mathrm{atan2}\left(\sin\left(\theta\right),\cos\left(\theta\right)\right)}{4}}{\pi}+\frac{1}{2}$$

$$\eta_{374} = \frac{\mathrm{atan2}\left(\sin\left(\theta\right),\cos\left(\theta\right)\right)}{2\pi}$$

$$\eta_{375} = \frac{-\frac{\mathrm{atan2}\left(\sin\left(\theta\right),\cos\left(\theta\right)\right)}{2}-\frac{\pi}{2}}{\pi}+\frac{1}{2}$$

$$\eta_{376} = -\frac{2\,\mathrm{atan2}\left(\sqrt{2}\sin\left(\frac{\mathrm{atan2}\left(\sin\left(\theta\right),\cos\left(\theta\right)\right)}{4}\right),\sqrt{\cos\left(\frac{\mathrm{atan2}\left(\sin\left(\theta\right),\cos\left(\theta\right)\right)}{2}\right)}\right)}{\pi}$$

$$\eta_{377} = \frac{2\,\mathrm{atan2}\left(\sin\left(\frac{\mathrm{atan2}\left(\sin\left(\theta\right),\cos\left(\theta\right)\right)}{4}\right),\cos\left(\frac{\mathrm{atan2}\left(\sin\left(\theta\right),\cos\left(\theta\right)\right)}{4}\right)+\sqrt{\cos\left(\frac{\mathrm{atan2}\left(\sin\left(\theta\right),\cos\left(\theta\right)\right)}{2}\right)}\right)}{\pi}$$

$$\eta_{378} = \frac{\frac{\mathrm{atan2}\left(\sin\left(\theta\right),\cos\left(\theta\right)\right)}{4}+\pi}{\pi}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{379} = e^{i\left(\operatorname{atan_2}\left(\sqrt{2}\sin\left(\frac{\operatorname{atan_2}(\sin(\theta),\cos(\theta))}{4}\right),\sqrt{\cos\left(\frac{\operatorname{atan_2}(\sin(\theta),\cos(\theta))}{2}\right)}\right)-\frac{\operatorname{atan_2}(\sin(\theta),\cos(\theta))}{4}-2\operatorname{atan_2}\left(\sin\left(\frac{\operatorname{atan_2}(\sin(\theta),\cos(\theta))}{4}\right),\cos\left(\frac{\operatorname{atan_2}(\sin(\theta),\cos(\theta))}{4}\right)+\sqrt{\cos\left(\frac{\operatorname{atan_2}(\sin(\theta),\cos(\theta))}{2}\right)}\right)\right)}$$

$$\eta_{380} = \pi - 2\operatorname{atan_2}\left(\sin\left(\frac{\operatorname{atan_2}(\sin(\theta),\cos(\theta))}{4}\right),\cos\left(\frac{\operatorname{atan_2}(\sin(\theta),\cos(\theta))}{4}\right)+\sqrt{\cos\left(\frac{\operatorname{atan_2}(\sin(\theta),\cos(\theta))}{2}\right)}\right)$$

$$\eta_{381} = \frac{\operatorname{atan_2}(\sin(\theta),\cos(\theta))}{2}-\frac{\pi}{2}$$

$$\eta_{382} = -\frac{\operatorname{atan_2}(\sin(\theta),\cos(\theta))}{4}+\frac{\pi}{2}$$

$$\eta_{383} = \pi - \frac{\operatorname{atan_2}(\sin(\theta),\cos(\theta))}{4}$$

$$\eta_{384} = \frac{\operatorname{atan_2}(\sin(\theta),\cos(\theta))}{2}+\frac{\pi}{2}$$

$$\eta_{385} = -2\operatorname{atan_2}\left(\sqrt{2}\sin\left(\frac{\operatorname{atan_2}(\sin(\theta),\cos(\theta))}{4}\right),\sqrt{\cos\left(\frac{\operatorname{atan_2}(\sin(\theta),\cos(\theta))}{2}\right)}\right)$$

$$\eta_{386} = 2\operatorname{atan_2}\left(\sin\left(\frac{\operatorname{atan_2}(\sin(\theta),\cos(\theta))}{4}\right),\cos\left(\frac{\operatorname{atan_2}(\sin(\theta),\cos(\theta))}{4}\right)+\sqrt{\cos\left(\frac{\operatorname{atan_2}(\sin(\theta),\cos(\theta))}{2}\right)}\right)$$

$$\eta_{387} = e^{-\frac{i\operatorname{atan_2}(\sin(\theta),\cos(\theta))}{4}}$$

$$\eta_{388} = -\frac{(1+i)^2 e^{\frac{i(lbda+\phi)}{4}}}{4}$$

$$\eta_{389} = \frac{lbda}{2}+\frac{\phi}{2}-\operatorname{atan_2}\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{lbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{lbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)},\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\right)$$

$$\eta_{390} = \frac{\frac{lbda}{2}+\frac{\phi}{2}-\operatorname{atan_2}\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{lbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{lbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)},\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\right)}{\pi}$$

$$\eta_{391} = \frac{-\frac{lbda}{2}-\frac{\phi}{2}+\operatorname{atan_2}\left(\sqrt{\sin^2\left(\frac{\theta}{2}\right)\sin^2\left(\frac{lbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{lbda}{2}-\frac{\phi}{2}\right)+\sin^2\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\cos^2\left(\frac{\theta}{2}\right)},\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{lbda}{2}+\frac{\phi}{2}\right)\right)}{\pi}+\frac{1}{2}$$

$$\eta_{392} = (-1)^{\frac{3}{4}} e^{\frac{i(lbda+\phi)}{4}}$$

$$\eta_{393} = \frac{(1+i)^2 e^{\frac{i(lbda+\phi)}{4}}}{2}$$

$$\eta_{394} = -\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0,\sin\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases} - \pi$$

$$\eta_{395} = \pi global_{shift} t - \frac{\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|,\cos\left(\frac{\pi t}{2}\right)\right)}{2}$$

$$\eta_{396} = \pi global_{shift} t - \frac{\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|,\cos\left(\frac{\pi t}{2}\right)\right)}{2}+\pi$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{397} = -\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, 0\right) & \text{otherwise} \end{cases}$$

$$\eta_{398} = \operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \cos\left(\frac{\pi t}{2}\right)\right)$$

$$\eta_{399} = \begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, 0\right) & \text{otherwise} \end{cases}$$

$$\eta_{400} = \begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \sin\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{401} = -\pi global_{shift} t + \frac{\pi t (2 global_{shift} + 1)}{2} - \operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \cos\left(\frac{\pi t}{2}\right)\right)$$

$$\eta_{402} = -\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \sin\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{403} = e^{i\left(\pi global_{shift} t - \frac{\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \cos\left(\frac{\pi t}{2}\right)\right)}{2}\right)}$$

$$\eta_{404} = \frac{-\pi\left(\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \sin\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}\right) - \frac{\pi}{2}}{\pi}$$

$$\eta_{405} = \frac{\pi\left(\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \sin\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}\right) + \frac{\pi}{2}}{\pi} + \frac{1}{2}$$

$$\eta_{406} = \frac{2\pi global_{shift} t - \frac{\begin{cases} 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \cos\left(\frac{\pi t}{2}\right)\right) & \text{for } \cos\left(\frac{\pi t}{2}\right) \geq 0 \\ 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \cos\left(\frac{\pi t}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2} - 4\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\pi t}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\frac{\left|\cos\left(\frac{\pi t}{2}\right)\right|}{2}} + \frac{1}{2} + \sqrt{\left|\cos\left(\frac{\pi t}{2}\right)\right|}\right) + 2\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\pi t}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\left|\cos\left(\frac{\pi t}{2}\right)\right|}\right) - \pi}{\pi}$$

$$\eta_{407} = \frac{-\pi global_{shift} t + \frac{\begin{cases} 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \cos\left(\frac{\pi t}{2}\right)\right) & \text{for } \cos\left(\frac{\pi t}{2}\right) \geq 0 \\ 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \cos\left(\frac{\pi t}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4} + 2\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\pi t}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\frac{\left|\cos\left(\frac{\pi t}{2}\right)\right|}{2}} + \frac{1}{2} + \sqrt{\left|\cos\left(\frac{\pi t}{2}\right)\right|}\right) - \operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\pi t}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\left|\cos\left(\frac{\pi t}{2}\right)\right|}\right)}{\pi} + \frac{1}{2}$$

$$\eta_{408} = -\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, 0\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{409} = \frac{2\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\pi t}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\frac{\left|\cos\left(\frac{\pi t}{2}\right)\right|}{2}} + \frac{1}{2} + \sqrt{\left|\cos\left(\frac{\pi t}{2}\right)\right|}\right)}{\pi}$$

$$\eta_{410} = \frac{\frac{\begin{cases} 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \cos\left(\frac{\pi t}{2}\right)\right) & \text{for } \cos\left(\frac{\pi t}{2}\right) \geq 0 \\ 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \cos\left(\frac{\pi t}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2}}{\pi} + \pi$$

$$\eta_{411} = \frac{-\frac{\frac{\begin{cases} 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \cos\left(\frac{\pi t}{2}\right)\right) & \text{for } \cos\left(\frac{\pi t}{2}\right) \geq 0 \\ 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \cos\left(\frac{\pi t}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2}}{\pi} - \frac{\pi}{2}}{\pi} + \frac{1}{2}$$

$$\eta_{412} = \frac{\begin{cases} 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \cos\left(\frac{\pi t}{2}\right)\right) & \text{for } \cos\left(\frac{\pi t}{2}\right) \geq 0 \\ 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \cos\left(\frac{\pi t}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2\pi}$$

$$\eta_{413} = -\frac{2\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\pi t}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\left|\cos\left(\frac{\pi t}{2}\right)\right|}\right)}{\pi}$$

$$\eta_{414} = \frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, 0\right) & \text{otherwise} \end{cases}}{\pi}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{415} = -\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \sin\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{416} = -\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \sin\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}$$

$$\eta_{417} = \frac{-\pi global_{shift} t + \frac{\pi t\left(2 global_{shift}+1\right)}{2} - \operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \cos\left(\frac{\pi t}{2}\right)\right)}{\pi}$$

$$\eta_{418} = \frac{\pi global_{shift} t - \frac{\pi t\left(2 global_{shift}+1\right)}{2} + \operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \cos\left(\frac{\pi t}{2}\right)\right)}{\pi} + \frac{1}{2}$$

$$\eta_{419} = -\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \sin\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{420} = \frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \sin\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}$$

$$\eta_{421} = e^{i\left(\pi global_{shift} t - \frac{\begin{cases} 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \cos\left(\frac{\pi t}{2}\right)\right) & \text{for } \cos\left(\frac{\pi t}{2}\right) \geq 0 \\ 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \cos\left(\frac{\pi t}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4} - 2\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\pi t}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{\frac{\left|\cos\left(\frac{\pi t}{2}\right)\right|}{2}} + \frac{1}{2} + \sqrt{\left|\cos\left(\frac{\pi t}{2}\right)\right|}\right) + \operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\pi t}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{\left|\cos\left(\frac{\pi t}{2}\right)\right|}\right)\right)}$$

$$\eta_{422} = \pi global_{shift} t - \frac{\begin{cases} 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \cos\left(\frac{\pi t}{2}\right)\right) & \text{for } \cos\left(\frac{\pi t}{2}\right) \geq 0 \\ 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \cos\left(\frac{\pi t}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4}$$

$$\eta_{423} = \pi global_{shift} t - \frac{\begin{cases} 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \cos\left(\frac{\pi t}{2}\right)\right) & \text{for } \cos\left(\frac{\pi t}{2}\right) \geq 0 \\ 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \cos\left(\frac{\pi t}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4} + \pi$$

$$\eta_{424} = 2\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\pi t}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{\frac{\left|\cos\left(\frac{\pi t}{2}\right)\right|}{2}} + \frac{1}{2} + \sqrt{\left|\cos\left(\frac{\pi t}{2}\right)\right|}\right)$$

$$\eta_{425} = -\frac{\begin{cases} 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \cos\left(\frac{\pi t}{2}\right)\right) & \text{for } \cos\left(\frac{\pi t}{2}\right) \geq 0 \\ 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \cos\left(\frac{\pi t}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2} + \frac{\pi}{2}$$

$$\eta_{426} = -2\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\pi t}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{\left|\cos\left(\frac{\pi t}{2}\right)\right|}\right)$$

$$\eta_{427} = e^{i\left(\pi global_{shift} t - \frac{\begin{cases} 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \cos\left(\frac{\pi t}{2}\right)\right) & \text{for } \cos\left(\frac{\pi t}{2}\right) \geq 0 \\ 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \cos\left(\frac{\pi t}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4}\right)}$$

$$\eta_{428} = \frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \sin\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases} + \pi}{\pi}$$

$$\eta_{429} = \frac{2\pi global_{shift} t - 2\left(\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \sin\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases} - \frac{\begin{cases} 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \cos\left(\frac{\pi t}{2}\right)\right) & \text{for } \cos\left(\frac{\pi t}{2}\right) \geq 0 \\ 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \cos\left(\frac{\pi t}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2} - 4\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\pi t}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{\frac{\left|\cos\left(\frac{\pi t}{2}\right)\right|}{2}} + \frac{1}{2} + \sqrt{\left|\cos\left(\frac{\pi t}{2}\right)\right|}\right) + 2\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\pi t}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{\left|\cos\left(\frac{\pi t}{2}\right)\right|}\right) - \pi}{\pi}$$

$$\eta_{430} = \frac{-\pi global_{shift} t + \begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \sin\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases} + \frac{\begin{cases} 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \cos\left(\frac{\pi t}{2}\right)\right) & \text{for } \cos\left(\frac{\pi t}{2}\right) \geq 0 \\ 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \cos\left(\frac{\pi t}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4} + 2\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\pi t}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{\frac{\left|\cos\left(\frac{\pi t}{2}\right)\right|}{2}} + \frac{1}{2} + \sqrt{\left|\cos\left(\frac{\pi t}{2}\right)\right|}\right) - \operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\cos\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \cos\left(\frac{\pi t}{2}\right) \geq 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{\left|\cos\left(\frac{\pi t}{2}\right)\right|}\right)}{\pi} + \frac{1}{2}$$

$$\eta_{431} = -(-1)^{\frac{3}{4}} e^{\frac{i\pi t\left(4 global_{shift}+1\right)}{4}}$$

$$\eta_{432} = \begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \sin\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases} + \pi$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{433} = \frac{\pi - 2\,\mathrm{atan2}\left(\sin\left(\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{4}\right),\cos\left(\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{4}\right) + \sqrt{\cos\left(\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{2}\right)}\right)}{\pi}$$

$$\eta_{434} = \frac{-\pi\left(-\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{\pi} - \frac{\pi}{2} + \frac{1}{2}\right) - \frac{\pi}{2}}{\pi}$$

$$\eta_{435} = \frac{\pi\left(-\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{\pi} - \frac{\pi}{2} + \frac{1}{2}\right) + \frac{\pi}{2}}{\pi} + \frac{1}{2}$$

$$\eta_{436} = \frac{\pi\left(-\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{\pi} - \frac{\pi}{2} + \frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{\pi} + \pi + \frac{1}{2}\right) + 2\,\mathrm{atan2}\left(\sqrt{2}\sin\left(\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{4}\right),\sqrt{\cos\left(\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{2}\right)}\right) - \mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right) - \frac{3\pi}{2}}{\pi}$$

$$\eta_{437} = \frac{-\,\mathrm{atan2}\left(\sqrt{2}\sin\left(\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{4}\right),\sqrt{\cos\left(\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{2}\right)}\right) + \frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{4}}{\pi} + \frac{1}{2}$$

$$\eta_{438} = \frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{2\pi}$$

$$\eta_{439} = \frac{-\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{2} - \frac{\pi}{2}}{\pi} + \frac{1}{2}$$

$$\eta_{440} = -\frac{2\,\mathrm{atan2}\left(\sqrt{2}\sin\left(\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{4}\right),\sqrt{\cos\left(\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{2}\right)}\right)}{\pi}$$

$$\eta_{441} = \frac{2\,\mathrm{atan2}\left(\sin\left(\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{4}\right),\cos\left(\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{4}\right) + \sqrt{\cos\left(\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{2}\right)}\right)}{\pi}$$

$$\eta_{442} = \frac{\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{4} + \pi}{\pi}$$

$$\eta_{443} = e^{i\left(\mathrm{atan2}\left(\sqrt{2}\sin\left(\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{4}\right),\sqrt{\cos\left(\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{2}\right)}\right) - \frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{4} - 2\,\mathrm{atan2}\left(\sin\left(\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{4}\right),\cos\left(\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{4}\right) + \sqrt{\cos\left(\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{2}\right)}\right)\right)}$$

$$\eta_{444} = \pi - 2\,\mathrm{atan2}\left(\sin\left(\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{4}\right),\cos\left(\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{4}\right) + \sqrt{\cos\left(\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{2}\right)}\right)$$

$$\eta_{445} = \frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{2} - \frac{\pi}{2}$$

$$\eta_{446} = -\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{4} + \frac{\pi}{2}$$

$$\eta_{447} = \pi - \frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{4}$$

$$\eta_{448} = \frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{2} + \frac{\pi}{2}$$

$$\eta_{449} = -2\,\mathrm{atan2}\left(\sqrt{2}\sin\left(\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{4}\right),\sqrt{\cos\left(\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{2}\right)}\right)$$

$$\eta_{450} = 2\,\mathrm{atan2}\left(\sin\left(\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{4}\right),\cos\left(\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{4}\right) + \sqrt{\cos\left(\frac{\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{2}\right)}\right)$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{451} = e^{-\frac{i\,\mathrm{atan2}\left(\sin\left(\pi t\right),\cos\left(\pi t\right)\right)}{4}}$$

$$\eta_{452} = \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\phi\right)\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)+\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)\cos^2\left(\phi\right)}=0 \\ \mathrm{atan2}\left(-\sin\left(\theta\right)\cos\left(\frac{\gamma}{2}\right)\cos\left(\phi\right),-\sin\left(\phi\right)\sin\left(\theta\right)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2+\sin^2\left(\frac{\gamma}{2}\right)}=0 \\ \mathrm{atan2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2},-\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{5\pi}{4}$$

$$\eta_{453} = -\frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\phi\right)\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)+\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)\cos^2\left(\phi\right)}=0 \\ \mathrm{atan2}\left(-\sin\left(\theta\right)\cos\left(\frac{\gamma}{2}\right)\cos\left(\phi\right),-\sin\left(\phi\right)\sin\left(\theta\right)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2+\sin^2\left(\frac{\gamma}{2}\right)}=0 \\ \mathrm{atan2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2},-\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\pi}{4}$$

$$\eta_{454} = -\mathrm{atan2}\left(\sqrt{\sin^2\left(\phi\right)\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)+\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)\cos^2\left(\phi\right)},\sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2+\sin^2\left(\frac{\gamma}{2}\right)}\right)$$

$$\eta_{455} = \frac{\gamma}{2} - \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\phi\right)\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)+\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)\cos^2\left(\phi\right)}=0 \\ \mathrm{atan2}\left(-\sin\left(\theta\right)\cos\left(\frac{\gamma}{2}\right)\cos\left(\phi\right),-\sin\left(\phi\right)\sin\left(\theta\right)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2+\sin^2\left(\frac{\gamma}{2}\right)}=0 \\ \mathrm{atan2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2},-\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\pi}{4}$$

$$\eta_{456} = \frac{\gamma}{2} + \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\phi\right)\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)+\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)\cos^2\left(\phi\right)}=0 \\ \mathrm{atan2}\left(-\sin\left(\theta\right)\cos\left(\frac{\gamma}{2}\right)\cos\left(\phi\right),-\sin\left(\phi\right)\sin\left(\theta\right)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2+\sin^2\left(\frac{\gamma}{2}\right)}=0 \\ \mathrm{atan2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2},-\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\pi}{4}$$

$$\eta_{457} = \frac{\frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\phi\right)\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)+\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)\cos^2\left(\phi\right)}=0 \\ \mathrm{atan2}\left(-\sin\left(\theta\right)\cos\left(\frac{\gamma}{2}\right)\cos\left(\phi\right),-\sin\left(\phi\right)\sin\left(\theta\right)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2}}{\pi} + \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2+\sin^2\left(\frac{\gamma}{2}\right)}=0 \\ \mathrm{atan2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2},-\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{3\pi}{4}$$

$$\eta_{458} = -\frac{\frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\phi\right)\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)+\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)\cos^2\left(\phi\right)}=0 \\ \mathrm{atan2}\left(-\sin\left(\theta\right)\cos\left(\frac{\gamma}{2}\right)\cos\left(\phi\right),-\sin\left(\phi\right)\sin\left(\theta\right)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2}}{\pi} - \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2+\sin^2\left(\frac{\gamma}{2}\right)}=0 \\ \mathrm{atan2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2},-\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{3\pi}{4} + \frac{1}{2}$$

$$\eta_{459} = -\frac{\frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\phi\right)\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)+\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)\cos^2\left(\phi\right)}=0 \\ \mathrm{atan2}\left(-\sin\left(\theta\right)\cos\left(\frac{\gamma}{2}\right)\cos\left(\phi\right),-\sin\left(\phi\right)\sin\left(\theta\right)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2}}{\pi} - \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2+\sin^2\left(\frac{\gamma}{2}\right)}=0 \\ \mathrm{atan2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2},-\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\pi}{4}$$

$$\eta_{460} = \frac{\frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\phi\right)\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)+\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)\cos^2\left(\phi\right)}=0 \\ \mathrm{atan2}\left(-\sin\left(\theta\right)\cos\left(\frac{\gamma}{2}\right)\cos\left(\phi\right),-\sin\left(\phi\right)\sin\left(\theta\right)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2}}{\pi} + \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2+\sin^2\left(\frac{\gamma}{2}\right)}=0 \\ \mathrm{atan2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2},-\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\pi}{4} + \frac{1}{2}$$

$$\eta_{461} = \frac{\pi - \mathrm{atan2}\left(\sqrt{\sin^2\left(\phi\right)\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)+\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)\cos^2\left(\phi\right)},\sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2+\sin^2\left(\frac{\gamma}{2}\right)}\right)}{\pi}$$

$$\eta_{462} = \frac{\mathrm{atan2}\left(\sqrt{\sin^2\left(\phi\right)\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)+\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)\cos^2\left(\phi\right)},\sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2+\sin^2\left(\frac{\gamma}{2}\right)}\right) - \pi}{\pi} + \frac{1}{2}$$

$$\eta_{463} = \frac{\mathrm{atan2}\left(\sqrt{\sin^2\left(\phi\right)\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)+\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)\cos^2\left(\phi\right)},\sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2+\sin^2\left(\frac{\gamma}{2}\right)}\right)}{\pi}$$

$$\eta_{464} = -\frac{\mathrm{atan2}\left(\sqrt{\sin^2\left(\phi\right)\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)+\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)\cos^2\left(\phi\right)},\sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2+\sin^2\left(\frac{\gamma}{2}\right)}\right)}{\pi} + \frac{1}{2}$$

$$\eta_{465} = \frac{\frac{\gamma}{2} - \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\phi\right)\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)+\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)\cos^2\left(\phi\right)}=0 \\ \mathrm{atan2}\left(-\sin\left(\theta\right)\cos\left(\frac{\gamma}{2}\right)\cos\left(\phi\right),-\sin\left(\phi\right)\sin\left(\theta\right)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2}}{\pi} + \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2+\sin^2\left(\frac{\gamma}{2}\right)}=0 \\ \mathrm{atan2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2},-\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{3\pi}{4}$$

$$\eta_{466} = \frac{-\frac{\gamma}{2} + \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\phi\right)\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)+\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)\cos^2\left(\phi\right)}=0 \\ \mathrm{atan2}\left(-\sin\left(\theta\right)\cos\left(\frac{\gamma}{2}\right)\cos\left(\phi\right),-\sin\left(\phi\right)\sin\left(\theta\right)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2}}{\pi} - \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2+\sin^2\left(\frac{\gamma}{2}\right)}=0 \\ \mathrm{atan2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2},-\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{3\pi}{4} + \frac{1}{2}$$

$$\eta_{467} = \frac{\frac{\gamma}{2} + \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\phi\right)\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)+\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)\cos^2\left(\phi\right)}=0 \\ \mathrm{atan2}\left(-\sin\left(\theta\right)\cos\left(\frac{\gamma}{2}\right)\cos\left(\phi\right),-\sin\left(\phi\right)\sin\left(\theta\right)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2}}{\pi} - \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2+\sin^2\left(\frac{\gamma}{2}\right)}=0 \\ \mathrm{atan2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2},-\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{3\pi}{4}$$

$$\eta_{468} = \frac{-\frac{\gamma}{2} - \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\phi\right)\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)+\sin^2\left(\theta\right)\cos^2\left(\frac{\gamma}{2}\right)\cos^2\left(\phi\right)}=0 \\ \mathrm{atan2}\left(-\sin\left(\theta\right)\cos\left(\frac{\gamma}{2}\right)\cos\left(\phi\right),-\sin\left(\phi\right)\sin\left(\theta\right)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2}}{\pi} + \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2+\sin^2\left(\frac{\gamma}{2}\right)}=0 \\ \mathrm{atan2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2},-\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{3\pi}{4} + \frac{1}{2}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{469} = \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right) + \sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan_2}\left(-\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\cos(\phi), -\sin(\phi)\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2 + \sin^2\left(\frac{\gamma}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}, -\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{3\pi}{4}$$

$$\eta_{470} = -\frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right) + \sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan_2}\left(-\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\cos(\phi), -\sin(\phi)\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2 + \sin^2\left(\frac{\gamma}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}, -\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\pi}{4}$$

$$\eta_{471} = \pi - \operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right) + \sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)\cos^2(\phi)}, \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2 + \sin^2\left(\frac{\gamma}{2}\right)}\right)$$

$$\eta_{472} = \operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right) + \sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)\cos^2(\phi)}, \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2 + \sin^2\left(\frac{\gamma}{2}\right)}\right)$$

$$\eta_{473} = \frac{\gamma}{2} - \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right) + \sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan_2}\left(-\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\cos(\phi), -\sin(\phi)\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2 + \sin^2\left(\frac{\gamma}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}, -\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{3\pi}{4}$$

$$\eta_{474} = \frac{\gamma}{2} + \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right) + \sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan_2}\left(-\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\cos(\phi), -\sin(\phi)\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2 + \sin^2\left(\frac{\gamma}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}, -\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{3\pi}{4}$$

$$\eta_{475} = -\frac{2\operatorname{atan}\left(\frac{\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right)}{\pi}$$

$$\eta_{476} = \frac{\operatorname{atan}\left(\frac{\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right)}{\pi}$$

$$\eta_{477} = -\pi + \operatorname{atan}\left(\frac{\cos\left(\frac{3\pi}{16}\right)}{\sin\left(\frac{3\pi}{16}\right)}\right)$$

$$\eta_{478} = \operatorname{atan}\left(\frac{\cos\left(\frac{3\pi}{16}\right)}{\sin\left(\frac{3\pi}{16}\right)}\right)$$

$$\eta_{479} = \operatorname{atan}\left(\frac{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right) + \pi$$

$$\eta_{480} = \operatorname{atan}\left(\frac{-2+\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{481} = \operatorname{atan}\left(\frac{-2+\sqrt{2-\sqrt{2}}}{\sqrt{\sqrt{2}+2}}\right) + \pi$$

$$\eta_{482} = \frac{\sqrt{2}i(1+i)^3 e^{\frac{i(4\pi-\phi)}{4}}}{4}$$

$$\eta_{483} = \frac{\phi}{4} + \frac{\pi\left(\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor + \left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor\right)}{2} + \begin{cases} 0 & \text{for } \sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta) + 2\sin^2\left(\frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos^2\left(\frac{\phi}{4} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)}, -\sqrt{2}\sin\left(\frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos\left(\frac{\phi}{4} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases} - \left(\frac{3\pi\left(\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)}{2}\right) + \pi$$

$$\eta_{484} = \frac{\phi}{4} + \frac{\pi\left(\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor + \left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor\right)}{2} + \begin{cases} 0 & \text{for } \sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta) + 2\sin^2\left(\frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos^2\left(\frac{\phi}{4} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)}, -\sqrt{2}\sin\left(\frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos\left(\frac{\phi}{4} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases} + \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\phi}{4} - \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} - \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)}, -\sin\left(\frac{\phi}{4} + \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\phi}{4} + \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)}, \sin\left(\frac{\phi}{4} - \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} - \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \left(\frac{3\pi\left(\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)}{2}\right) + 3\pi$$

$$\eta_{485} = -\frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\phi}{4} - \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} - \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)}, -\sin\left(\frac{\phi}{4} + \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\phi}{4} + \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)}, \sin\left(\frac{\phi}{4} - \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} - \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \pi$$

$$\eta_{486} = -2\operatorname{atan_2}\left(-\sqrt{2}\sin\left(\frac{\phi}{4} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos\left(\frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right), \sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta) + 2\sin^2\left(\frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos^2\left(\frac{\phi}{4} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2}\right)}\right)$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{487} = -\begin{cases} 0 & \text{for } \sqrt{(-1)^{\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}\,(-1)^{\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)+2\sin^2\left(\frac{\theta}{2}+\frac{\pi\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}{2}\right)\cos^2\left(\frac{\phi}{4}+\frac{\pi\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\sqrt{(-1)^{\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}\,(-1)^{\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)},-\sqrt{2}\sin\left(\frac{\theta}{2}+\frac{\pi\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}{2}\right)\cos\left(\frac{\phi}{4}+\frac{\pi\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{488} = \left(-\frac{\pi\left(1-\left(2\left(\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right)\right)}{2}\right) + \left(\pi\left(\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right)$$

$$\eta_{489} = -\frac{\phi}{2} + \left(\pi\left(\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right) + \left(\pi\left(\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right)$$

$$\eta_{490} = -\frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\phi}{4}-\frac{\theta}{2}+\frac{\pi\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}{2}-\frac{\pi\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}{2}\right)=0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}\,(-1)^{\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)},-\sin\left(\frac{\phi}{4}+\frac{\theta}{2}+\frac{\pi\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}{2}+\frac{\pi\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\phi}{4}+\frac{\theta}{2}+\frac{\pi\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}{2}+\frac{\pi\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}{2}\right)=0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}\,(-1)^{\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)},\sin\left(\frac{\phi}{4}-\frac{\theta}{2}+\frac{\pi\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}{2}-\frac{\pi\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \left(\pi\left(\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right) - 2\pi$$

$$\eta_{491} = \left(\left(1-\left(2\left(\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right)\right)\left(\frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\phi}{4}-\frac{\theta}{2}+\frac{\pi\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}{2}-\frac{\pi\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}{2}\right)=0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}\,(-1)^{\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)},-\sin\left(\frac{\phi}{4}+\frac{\theta}{2}+\frac{\pi\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}{2}+\frac{\pi\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\phi}{4}+\frac{\theta}{2}+\frac{\pi\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}{2}+\frac{\pi\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}{2}\right)=0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}\,(-1)^{\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)},\sin\left(\frac{\phi}{4}-\frac{\theta}{2}+\frac{\pi\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}{2}-\frac{\pi\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \pi\right) + \left(\pi\left(\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right)$$

$$\eta_{492} = -\frac{\phi}{2} - \left(\frac{\pi\left(1-\left(2\left(\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right)\right)}{2}\right) + \left(\pi\left(\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right) + \left(\pi\left(\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right)$$

$$\eta_{493} = \left(\pi\left(\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right) + \frac{\pi}{2}$$

$$\eta_{494} = -\frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\phi}{4}-\frac{\theta}{2}+\frac{\pi\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}{2}-\frac{\pi\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}{2}\right)=0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}\,(-1)^{\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)},-\sin\left(\frac{\phi}{4}+\frac{\theta}{2}+\frac{\pi\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}{2}+\frac{\pi\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\phi}{4}+\frac{\theta}{2}+\frac{\pi\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}{2}+\frac{\pi\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}{2}\right)=0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}\,(-1)^{\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)},\sin\left(\frac{\phi}{4}-\frac{\theta}{2}+\frac{\pi\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}{2}-\frac{\pi\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \pi$$

$$\eta_{495} = e^{i\left(\frac{\phi}{4}+\frac{\pi\left(\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor+\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor\right)}{2}+\begin{cases} 0 & \text{for } \sqrt{(-1)^{\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}\,(-1)^{\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)+2\sin^2\left(\frac{\theta}{2}+\frac{\pi\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}{2}\right)\cos^2\left(\frac{\phi}{4}+\frac{\pi\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\sqrt{(-1)^{\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}\,(-1)^{\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)},-\sqrt{2}\sin\left(\frac{\theta}{2}+\frac{\pi\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}{2}\right)\cos\left(\frac{\phi}{4}+\frac{\pi\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}{2}\right)\right) & \text{otherwise} \end{cases} + \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\phi}{4}-\frac{\theta}{2}+\frac{\pi\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}{2}-\frac{\pi\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}{2}\right)=0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}\,(-1)^{\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)},-\sin\left(\frac{\phi}{4}+\frac{\theta}{2}+\frac{\pi\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}{2}+\frac{\pi\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\phi}{4}+\frac{\theta}{2}+\frac{\pi\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}{2}+\frac{\pi\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}{2}\right)=0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}\,(-1)^{\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)},\sin\left(\frac{\phi}{4}-\frac{\theta}{2}+\frac{\pi\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\rfloor}{2}-\frac{\pi\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \left(\frac{2\pi\left(\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)}{2}\right) + 2\pi\right)}$$

$$\eta_{496} = \frac{\sqrt{2}(1+i)^3 e^{\frac{i(\pi-\phi)}{4}}}{4}$$

$$\eta_{497} = \frac{\sqrt{2}i(1+i)e^{\frac{i(\phi-2\theta)}{4}}}{2}$$

$$\eta_{498} = \frac{\sqrt{2}(1-i)e^{i\pi\left(global_{shift}-\frac{3}{4}\right)}}{2}$$

$$\eta_{499} = \frac{\sqrt{-1}\sqrt{2}(1+i)e^{\frac{i\pi\left(2global_{shift}+1\right)}{2}}}{2}$$

$$\eta_{500} = \frac{\sqrt{2}i(-1+i)e^{i\pi\left(global_{shift}-\frac{1}{4}\right)}}{2}$$

$$\eta_{501} = \frac{-\pi\left(2\phi_0+\frac{1}{2}\right)-\frac{\pi}{2}}{\pi}$$

$$\eta_{502} = \frac{\pi\left(2\phi_0+\frac{1}{2}\right)+\frac{\pi}{2}}{\pi} + \frac{1}{2}$$

$$\eta_{503} = -\frac{2\left(\begin{cases} 0 & \text{for } |\cos(\pi\theta)|=0 \\ \operatorname{atan_2}(0,\cos(\pi\theta)) & \text{otherwise} \end{cases} - \frac{\begin{cases} 2\operatorname{atan_2}\left(\sqrt{4\sin^2(2\pi\phi_1)\sin^2(\pi\theta)\cos^2(\pi\theta)+4\sin^4(\pi\theta)\cos^2(2\pi\phi_1)\cos^2(\pi\theta)},-\sin^2(2\pi\phi_1)\sin^2(\pi\theta)-\sin^2(\pi\theta)\cos^2(2\pi\phi_1)+\cos^2(\pi\theta)\right) & \text{for } \sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)\le 0 \\ 2\operatorname{atan_2}\left(\sqrt{4\sin^2(2\pi\phi_1)\sin^2(\pi\theta)\cos^2(\pi\theta)+4\sin^4(\pi\theta)\cos^2(2\pi\phi_1)\cos^2(\pi\theta)},-\sin^2(2\pi\phi_1)\sin^2(\pi\theta)-\sin^2(\pi\theta)\cos^2(2\pi\phi_1)+\cos^2(\pi\theta)\right)-2\pi & \text{otherwise} \end{cases}}{\pi} - 4\operatorname{atan_2}\left(\sqrt{\frac{1}{2}-\frac{\left|\cos^2(2\pi\phi_1)\cos^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)\le 0 \\ -1 & \text{otherwise} \end{cases}\right)\sqrt{\frac{\left|\cos^2(2\pi\phi_1)\cos^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)\right|}{2}}+\frac{1}{2}+\sqrt{\left|\sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)\right|}\right) + 2\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\cos^2(2\pi\phi_1)\cos^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)\le 0 \\ -1 & \text{otherwise} \end{cases}\right)\sqrt{\left|\sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)\right|}\right) - 2\operatorname{atan_2}\left(\sqrt{\sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)},|\cos(\pi\theta)|\right)}{\pi}$$

$$\eta_{504} = -\frac{\begin{cases} 0 & \text{for } |\cos(\pi\theta)|=0 \\ \operatorname{atan_2}(0,\cos(\pi\theta)) & \text{otherwise} \end{cases} + \frac{\begin{cases} 2\operatorname{atan_2}\left(\sqrt{4\sin^2(2\pi\phi_1)\sin^2(\pi\theta)\cos^2(\pi\theta)+4\sin^4(\pi\theta)\cos^2(2\pi\phi_1)\cos^2(\pi\theta)},-\sin^2(2\pi\phi_1)\sin^2(\pi\theta)-\sin^2(\pi\theta)\cos^2(2\pi\phi_1)+\cos^2(\pi\theta)\right) & \text{for } \sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)\le 0 \\ 2\operatorname{atan_2}\left(\sqrt{4\sin^2(2\pi\phi_1)\sin^2(\pi\theta)\cos^2(\pi\theta)+4\sin^4(\pi\theta)\cos^2(2\pi\phi_1)\cos^2(\pi\theta)},-\sin^2(2\pi\phi_1)\sin^2(\pi\theta)-\sin^2(\pi\theta)\cos^2(2\pi\phi_1)+\cos^2(\pi\theta)\right)-2\pi & \text{otherwise} \end{cases}}{\pi} + 2\operatorname{atan_2}\left(\sqrt{\frac{1}{2}-\frac{\left|\cos^2(2\pi\phi_1)\cos^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)\le 0 \\ -1 & \text{otherwise} \end{cases}\right)\sqrt{\left|\sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)\right|}\right) - \operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\cos^2(2\pi\phi_1)\cos^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)\le 0 \\ -1 & \text{otherwise} \end{cases}\right)\sqrt{\left|\sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)\right|}\right) + 6\operatorname{atan_2}\left(\sqrt{\sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)},|\cos(\pi\theta)|\right) - \frac{5}{2}}{\pi} + \frac{1}{2}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{505} = \frac{2\,\mathrm{atan}_2\left(\sqrt{\sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)},|\cos(\pi\theta)|\right)}{\pi}$$

$$\eta_{506} = -\frac{2\left(\begin{cases}0 & \text{for } |\cos(\pi\theta)|=0\\ \mathrm{atan}_2(0,\cos(\pi\theta)) & \text{otherwise}\end{cases}\right)}{\pi}$$

$$\eta_{507} = \frac{\begin{cases}0 & \text{for } \sqrt{\sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)}=0\\ \mathrm{atan}_2(-\sin(\pi\theta)\cos(2\pi\phi_1),\sin(2\pi\phi_1)\sin(\pi\theta)) & \text{otherwise}\end{cases}}{\pi}+\begin{cases}0 & \text{for } |\cos(\pi\theta)|=0\\ \mathrm{atan}_2(0,\cos(\pi\theta)) & \text{otherwise}\end{cases}+\frac{1}{2}$$

$$\eta_{508} = -\frac{\begin{cases}0 & \text{for } \sqrt{4\sin^2(2\pi\phi_1)\sin^2(\pi\theta)\cos^2(\pi\theta)+4\sin^2(\pi\theta)\cos^2(2\pi\phi_1)\cos^2(\pi\theta)}=0\\ \mathrm{atan}_2(-2\sin(2\pi\phi_1)\sin(\pi\theta)\cos(\pi\theta),-2\sin(\pi\theta)\cos(2\pi\phi_1)\cos(\pi\theta)) & \text{otherwise}\end{cases}}{\pi}$$

$$\eta_{509} = \frac{\begin{cases}0 & \text{for } \sqrt{4\sin^2(2\pi\phi_1)\sin^2(\pi\theta)\cos^2(\pi\theta)+4\sin^2(\pi\theta)\cos^2(2\pi\phi_1)\cos^2(\pi\theta)}=0\\ \mathrm{atan}_2(-2\sin(2\pi\phi_1)\sin(\pi\theta)\cos(\pi\theta),-2\sin(\pi\theta)\cos(2\pi\phi_1)\cos(\pi\theta)) & \text{otherwise}\end{cases}}{\pi}+\frac{1}{2}$$

$$\eta_{510} = -\frac{\begin{cases}0 & \text{for } \sqrt{4\sin^2(2\pi\phi_1)\sin^2(\pi\theta)\cos^2(\pi\theta)+4\sin^2(\pi\theta)\cos^2(2\pi\phi_1)\cos^2(\pi\theta)}=0\\ \mathrm{atan}_2\left(\sqrt{4\sin^2(2\pi\phi_1)\sin^2(\pi\theta)\cos^2(\pi\theta)+4\sin^2(\pi\theta)\cos^2(2\pi\phi_1)\cos^2(\pi\theta)},0\right) & \text{otherwise}\end{cases}}{\pi}$$

$$\eta_{511} = \frac{2\,\mathrm{atan}_2\left(\sqrt{\frac{1}{2}-\frac{|\sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)|}{2}}\left(\begin{cases}1 & \text{for } \sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)\le 0\\ -1 & \text{otherwise}\end{cases}\right)\sqrt{\frac{|\sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)|}{2}}+\frac{1}{2}+\sqrt{|\sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)|}\right)}{\pi}$$

$$\eta_{512} = \frac{\begin{cases}2\,\mathrm{atan}_2\left(\sqrt{4\sin^2(2\pi\phi_1)\sin^2(\pi\theta)\cos^2(\pi\theta)+4\sin^2(\pi\theta)\cos^2(2\pi\phi_1)\cos^2(\pi\theta)},-\sin^2(2\pi\phi_1)\sin^2(\pi\theta)-\sin^2(\pi\theta)\cos^2(2\pi\phi_1)+\cos^2(\pi\theta)\right) & \text{for } \sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)\le 0\\ 2\,\mathrm{atan}_2\left(\sqrt{4\sin^2(2\pi\phi_1)\sin^2(\pi\theta)\cos^2(\pi\theta)+4\sin^2(\pi\theta)\cos^2(2\pi\phi_1)\cos^2(\pi\theta)},-\sin^2(2\pi\phi_1)\sin^2(\pi\theta)-\sin^2(\pi\theta)\cos^2(2\pi\phi_1)+\cos^2(\pi\theta)\right)-2\pi & \text{otherwise}\end{cases}}{\frac{2}{\pi}+\pi}$$

$$\eta_{513} = \frac{\begin{cases}2\,\mathrm{atan}_2\left(\sqrt{4\sin^2(2\pi\phi_1)\sin^2(\pi\theta)\cos^2(\pi\theta)+4\sin^2(\pi\theta)\cos^2(2\pi\phi_1)\cos^2(\pi\theta)},-\sin^2(2\pi\phi_1)\sin^2(\pi\theta)-\sin^2(\pi\theta)\cos^2(2\pi\phi_1)+\cos^2(\pi\theta)\right) & \text{for } \sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)\le 0\\ 2\,\mathrm{atan}_2\left(\sqrt{4\sin^2(2\pi\phi_1)\sin^2(\pi\theta)\cos^2(\pi\theta)+4\sin^2(\pi\theta)\cos^2(2\pi\phi_1)\cos^2(\pi\theta)},-\sin^2(2\pi\phi_1)\sin^2(\pi\theta)-\sin^2(\pi\theta)\cos^2(2\pi\phi_1)+\cos^2(\pi\theta)\right)-2\pi & \text{otherwise}\end{cases}}{\frac{2}{\pi}-\frac{2}{\pi}}+\frac{1}{2}$$

$$\eta_{514} = \frac{\begin{cases}2\,\mathrm{atan}_2\left(\sqrt{4\sin^2(2\pi\phi_1)\sin^2(\pi\theta)\cos^2(\pi\theta)+4\sin^2(\pi\theta)\cos^2(2\pi\phi_1)\cos^2(\pi\theta)},-\sin^2(2\pi\phi_1)\sin^2(\pi\theta)-\sin^2(\pi\theta)\cos^2(2\pi\phi_1)+\cos^2(\pi\theta)\right) & \text{for } \sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)\le 0\\ 2\,\mathrm{atan}_2\left(\sqrt{4\sin^2(2\pi\phi_1)\sin^2(\pi\theta)\cos^2(\pi\theta)+4\sin^2(\pi\theta)\cos^2(2\pi\phi_1)\cos^2(\pi\theta)},-\sin^2(2\pi\phi_1)\sin^2(\pi\theta)-\sin^2(\pi\theta)\cos^2(2\pi\phi_1)+\cos^2(\pi\theta)\right)-2\pi & \text{otherwise}\end{cases}}{2\pi}$$

$$\eta_{515} = -\frac{2\,\mathrm{atan}_2\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{|\sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)|}{2}}\left(\begin{cases}1 & \text{for } \sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)\le 0\\ -1 & \text{otherwise}\end{cases}\right)\sqrt{|\sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)|}\right)}{\pi}$$

$$\eta_{516} = \frac{\begin{cases}0 & \text{for } \sqrt{4\sin^2(2\pi\phi_1)\sin^2(\pi\theta)\cos^2(\pi\theta)+4\sin^2(\pi\theta)\cos^2(2\pi\phi_1)\cos^2(\pi\theta)}=0\\ \mathrm{atan}_2\left(\sqrt{4\sin^2(2\pi\phi_1)\sin^2(\pi\theta)\cos^2(\pi\theta)+4\sin^2(\pi\theta)\cos^2(2\pi\phi_1)\cos^2(\pi\theta)},0\right) & \text{otherwise}\end{cases}}{\pi}$$

$$\eta_{517} = -\frac{\begin{cases}0 & \text{for } \sqrt{4\sin^2(2\pi\phi_1)\sin^2(\pi\theta)\cos^2(\pi\theta)+4\sin^2(\pi\theta)\cos^2(2\pi\phi_1)\cos^2(\pi\theta)}=0\\ \mathrm{atan}_2(-2\sin(2\pi\phi_1)\sin(\pi\theta)\cos(\pi\theta),-2\sin(\pi\theta)\cos(2\pi\phi_1)\cos(\pi\theta)) & \text{otherwise}\end{cases}}{\pi}$$

$$\eta_{518} = -\frac{\begin{cases}0 & \text{for } \sqrt{4\sin^2(2\pi\phi_1)\sin^2(\pi\theta)\cos^2(\pi\theta)+4\sin^2(\pi\theta)\cos^2(2\pi\phi_1)\cos^2(\pi\theta)}=0\\ \mathrm{atan}_2\left(-2\sin(2\pi\phi_1)\sin(\pi\theta)\cos(\pi\theta),-2\sin(\pi\theta)\cos(2\pi\phi_1)\cos(\pi\theta)\right) & \text{otherwise}\end{cases}}{\pi}+\frac{1}{2}$$

$$\eta_{519} = -\frac{\mathrm{atan}_2\left(\sqrt{4\sin^2(2\pi\phi_1)\sin^2(\pi\theta)\cos^2(\pi\theta)+4\sin^2(\pi\theta)\cos^2(2\pi\phi_1)\cos^2(\pi\theta)},-\sin^2(2\pi\phi_1)\sin^2(\pi\theta)-\sin^2(\pi\theta)\cos^2(2\pi\phi_1)+\cos^2(\pi\theta)\right)}{\pi}$$

$$\eta_{520} = \frac{\mathrm{atan}_2\left(\sqrt{4\sin^2(2\pi\phi_1)\sin^2(\pi\theta)\cos^2(\pi\theta)+4\sin^2(\pi\theta)\cos^2(2\pi\phi_1)\cos^2(\pi\theta)},-\sin^2(2\pi\phi_1)\sin^2(\pi\theta)-\sin^2(\pi\theta)\cos^2(2\pi\phi_1)+\cos^2(\pi\theta)\right)}{\pi}+\frac{1}{2}$$

$$\eta_{521} = e^{i\left(\begin{cases}0 & \text{for } |\cos(\pi\theta)|=0\\ \mathrm{atan}_2(0,\cos(\pi\theta)) & \text{otherwise}\end{cases}-\frac{\begin{cases}2\,\mathrm{atan}_2\left(\sqrt{4\sin^2(2\pi\phi_1)\sin^2(\pi\theta)\cos^2(\pi\theta)+4\sin^2(\pi\theta)\cos^2(2\pi\phi_1)\cos^2(\pi\theta)},-\sin^2(2\pi\phi_1)\sin^2(\pi\theta)-\sin^2(\pi\theta)\cos^2(2\pi\phi_1)+\cos^2(\pi\theta)\right) & \text{for } \sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)\le 0\\ 2\,\mathrm{atan}_2\left(\sqrt{4\sin^2(2\pi\phi_1)\sin^2(\pi\theta)\cos^2(\pi\theta)+4\sin^2(\pi\theta)\cos^2(2\pi\phi_1)\cos^2(\pi\theta)},-\sin^2(2\pi\phi_1)\sin^2(\pi\theta)-\sin^2(\pi\theta)\cos^2(2\pi\phi_1)+\cos^2(\pi\theta)\right)-2\pi & \text{otherwise}\end{cases}}{4}-2\,\mathrm{atan}_2\left(\sqrt{\frac{1}{2}-\frac{|\cos^2(2\pi\phi_1)\cos^2(\pi\theta)+\cos^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)|}{2}}\left(\begin{cases}1 & \text{for } \sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)\le 0\\ -1 & \text{otherwise}\end{cases}\right)\sqrt{\frac{|\cos^2(2\pi\phi_1)\cos^2(\pi\theta)+\cos^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)|}{2}}+\frac{1}{2}+\sqrt{|\sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)|}\right)+4\,\mathrm{atan}_2\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{|\cos^2(2\pi\phi_1)\cos^2(\pi\theta)+\cos^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)|}{2}}\left(\begin{cases}1 & \text{for } \sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)\le 0\\ -1 & \text{otherwise}\end{cases}\right)\sqrt{|\sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)|}\right)-4\,\mathrm{atan}_2\left(\sqrt{\sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)},|\cos(\pi\theta)|\right)+\frac{5}{2}\right)}$$

$$\eta_{522} = \begin{cases}0 & \text{for } |\cos(\pi\theta)|=0\\ \mathrm{atan}_2(0,\cos(\pi\theta)) & \text{otherwise}\end{cases}-\frac{\begin{cases}2\,\mathrm{atan}_2\left(\sqrt{4\sin^2(2\pi\phi_1)\sin^2(\pi\theta)\cos^2(\pi\theta)+4\sin^2(\pi\theta)\cos^2(2\pi\phi_1)\cos^2(\pi\theta)},-\sin^2(2\pi\phi_1)\sin^2(\pi\theta)-\sin^2(\pi\theta)\cos^2(2\pi\phi_1)+\cos^2(\pi\theta)\right) & \text{for } \sin^2(2\pi\phi_1)\sin^2(\pi\theta)+\sin^2(\pi\theta)\cos^2(2\pi\phi_1)-\cos^2(\pi\theta)\le 0\\ 2\,\mathrm{atan}_2\left(\sqrt{4\sin^2(2\pi\phi_1)\sin^2(\pi\theta)\cos^2(\pi\theta)+4\sin^2(\pi\theta)\cos^2(2\pi\phi_1)\cos^2(\pi\theta)},-\sin^2(2\pi\phi_1)\sin^2(\pi\theta)-\sin^2(\pi\theta)\cos^2(2\pi\phi_1)+\cos^2(\pi\theta)\right)-2\pi & \text{otherwise}\end{cases}}{4}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{523} = \begin{cases} 0 & \text{for } \left|\cos\left(\pi\theta\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \cos\left(\pi\theta\right)\right) & \text{otherwise} \end{cases} - \frac{\begin{cases} 2\operatorname{atan_2}\left(\sqrt{4\sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right)\cos^2\left(\pi\theta\right) + 4\sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right)\cos^2\left(\pi\theta\right)}, -\sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right) - \sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right) + \cos^2\left(\pi\theta\right)\right) & \text{for } \sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right) + \sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right) - \cos^2\left(\pi\theta\right) \leq 0 \\ 2\operatorname{atan_2}\left(\sqrt{4\sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right)\cos^2\left(\pi\theta\right) + 4\sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right)\cos^2\left(\pi\theta\right)}, -\sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right) - \sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right) + \cos^2\left(\pi\theta\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4} + \pi$$

$$\eta_{524} = 2\operatorname{atan_2}\left(\sqrt{\sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right) + \sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right)}, \left|\cos\left(\pi\theta\right)\right|\right)$$

$$\eta_{525} = \begin{cases} 0 & \text{for } \sqrt{\sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right) + \sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\pi\theta\right)\cos\left(2\pi\phi_1\right), \sin\left(2\pi\phi_1\right)\sin\left(\pi\theta\right)\right) & \text{otherwise} \end{cases} - \begin{cases} 0 & \text{for } \left|\cos\left(\pi\theta\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \cos\left(\pi\theta\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{526} = -\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right) + \sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\pi\theta\right)\cos\left(2\pi\phi_1\right), \sin\left(2\pi\phi_1\right)\sin\left(\pi\theta\right)\right) & \text{otherwise} \end{cases} - \begin{cases} 0 & \text{for } \left|\cos\left(\pi\theta\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \cos\left(\pi\theta\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{527} = -\begin{cases} 0 & \text{for } \sqrt{4\sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right)\cos^2\left(\pi\theta\right) + 4\sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right)\cos^2\left(\pi\theta\right)} = 0 \\ \operatorname{atan_2}\left(-2\sin\left(2\pi\phi_1\right)\sin\left(\pi\theta\right)\cos\left(\pi\theta\right), -2\sin\left(\pi\theta\right)\cos\left(2\pi\phi_1\right)\cos\left(\pi\theta\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{528} = -\begin{cases} 0 & \text{for } \sqrt{4\sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right)\cos^2\left(\pi\theta\right) + 4\sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right)\cos^2\left(\pi\theta\right)} = 0 \\ \operatorname{atan_2}\left(\sqrt{4\sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right)\cos^2\left(\pi\theta\right) + 4\sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right)\cos^2\left(\pi\theta\right)}, 0\right) & \text{otherwise} \end{cases}$$

$$\eta_{529} = 2\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right) + \sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right) - \cos^2\left(\pi\theta\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right) + \sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right) - \cos^2\left(\pi\theta\right) \leq 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\frac{\left|\sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right) + \sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right) - \cos^2\left(\pi\theta\right)\right|}{2} + \frac{1}{2}} + \sqrt{\left|\sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right) + \sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right) - \cos^2\left(\pi\theta\right)\right|}\right)$$

$$\eta_{530} = \frac{\begin{cases} 2\operatorname{atan_2}\left(\sqrt{4\sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right)\cos^2\left(\pi\theta\right) + 4\sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right)\cos^2\left(\pi\theta\right)}, -\sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right) - \sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right) + \cos^2\left(\pi\theta\right)\right) & \text{for } \sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right) + \sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right) - \cos^2\left(\pi\theta\right) \leq 0 \\ 2\operatorname{atan_2}\left(\sqrt{4\sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right)\cos^2\left(\pi\theta\right) + 4\sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right)\cos^2\left(\pi\theta\right)}, -\sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right) - \sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right) + \cos^2\left(\pi\theta\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2} + \frac{\pi}{2}$$

$$\eta_{531} = -2\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right) + \sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right) - \cos^2\left(\pi\theta\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right) + \sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right) - \cos^2\left(\pi\theta\right) \leq 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\left|\sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right) + \sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right) - \cos^2\left(\pi\theta\right)\right|}\right)$$

$$\eta_{532} = \begin{cases} 0 & \text{for } \sqrt{4\sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right)\cos^2\left(\pi\theta\right) + 4\sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right)\cos^2\left(\pi\theta\right)} = 0 \\ \operatorname{atan_2}\left(\sqrt{4\sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right)\cos^2\left(\pi\theta\right) + 4\sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right)\cos^2\left(\pi\theta\right)}, 0\right) & \text{otherwise} \end{cases}$$

$$\eta_{533} = \begin{cases} 0 & \text{for } \sqrt{4\sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right)\cos^2\left(\pi\theta\right) + 4\sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right)\cos^2\left(\pi\theta\right)} = 0 \\ \operatorname{atan_2}\left(-2\sin\left(2\pi\phi_1\right)\sin\left(\pi\theta\right)\cos\left(\pi\theta\right), -2\sin\left(\pi\theta\right)\cos\left(2\pi\phi_1\right)\cos\left(\pi\theta\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{534} = -\operatorname{atan_2}\left(\sqrt{4\sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right)\cos^2\left(\pi\theta\right) + 4\sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right)\cos^2\left(\pi\theta\right)}, -\sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right) - \sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right) + \cos^2\left(\pi\theta\right)\right)$$

$$\eta_{535} = e^{i\left(\begin{cases} 0 & \text{for } \left|\cos\left(\pi\theta\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \cos\left(\pi\theta\right)\right) & \text{otherwise} \end{cases} - \frac{\begin{cases} 2\operatorname{atan_2}\left(\sqrt{4\sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right)\cos^2\left(\pi\theta\right) + 4\sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right)\cos^2\left(\pi\theta\right)}, -\sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right) - \sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right) + \cos^2\left(\pi\theta\right)\right) & \text{for } \sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right) + \sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right) - \cos^2\left(\pi\theta\right) \leq 0 \\ 2\operatorname{atan_2}\left(\sqrt{4\sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right)\cos^2\left(\pi\theta\right) + 4\sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right)\cos^2\left(\pi\theta\right)}, -\sin^2\left(2\pi\phi_1\right)\sin^2\left(\pi\theta\right) - \sin^2\left(\pi\theta\right)\cos^2\left(2\pi\phi_1\right) + \cos^2\left(\pi\theta\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4}\right)}$$

$$\eta_{536} = \frac{\pi - 2\operatorname{atan_2}\left(0, \sqrt{\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)}\right)}{\pi}$$

$$\eta_{537} = \frac{-\pi\left(\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\pi\theta\right), \cos\left(\pi\theta\right)\right) & \text{otherwise} \end{cases}{\pi} + \frac{1}{2}\right) - \frac{\pi}{2}}{\pi}$$

$$\eta_{538} = \frac{\pi\left(\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\pi\theta\right), \cos\left(\pi\theta\right)\right) & \text{otherwise} \end{cases}{\pi} + \frac{1}{2}\right) + \frac{\pi}{2}}{\pi} + \frac{1}{2}$$

$$\eta_{539} = \frac{\pi\left(-\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\pi\theta\right), \cos\left(\pi\theta\right)\right) & \text{otherwise} \end{cases}{\pi} + \frac{1}{2}\right) + 2\left(\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\pi\theta\right), \cos\left(\pi\theta\right)\right) & \text{otherwise} \end{cases} - \frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\pi\theta\right)\right|\left|\cos\left(\pi\theta\right)\right|, -\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)\right) & \text{for } \sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right) \leq 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\pi\theta\right)\right|\left|\cos\left(\pi\theta\right)\right|, -\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2} + 2\operatorname{atan_2}\left(0, \sqrt{\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)}\right) - 4\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right) \leq 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\frac{\left|\sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right)\right|}{2} + \frac{1}{2}} + \sqrt{\left|\sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right)\right|}\right) + 2\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right) \leq 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\left|\sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right)\right|}\right) - \frac{3\pi}{2}}{\pi}$$

$$\eta_{540} = \frac{-\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\pi\theta\right), \cos\left(\pi\theta\right)\right) & \text{otherwise} \end{cases} + \frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\pi\theta\right)\right|\left|\cos\left(\pi\theta\right)\right|, -\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)\right) & \text{for } \sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right) \leq 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\pi\theta\right)\right|\left|\cos\left(\pi\theta\right)\right|, -\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4} - \operatorname{atan_2}\left(0, \sqrt{\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)}\right) + 2\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right) \leq 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\frac{\left|\sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right)\right|}{2} + \frac{1}{2}} + \sqrt{\left|\sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right)\right|}\right) - \operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right) \leq 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\left|\sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right)\right|}\right)}{\pi} + \frac{1}{2}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{541} = -\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\pi\theta\right)\right|\left|\cos\left(\pi\theta\right)\right| = 0 \\ \operatorname{atan_2}\left(0, -2\sin\left(\pi\theta\right)\cos\left(\pi\theta\right)\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{542} = \frac{2\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right) \leq 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\frac{\left|\sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right)\right|}{2}} + \frac{1}{2} + \sqrt{\left|\sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right)\right|}\right)}{\pi}$$

$$\eta_{543} = \frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\pi\theta\right)\right|\left|\cos\left(\pi\theta\right)\right|, -\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)\right) & \text{for } \sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right) \leq 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\pi\theta\right)\right|\left|\cos\left(\pi\theta\right)\right|, -\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)\right) - 2\pi & \text{otherwise} \end{cases}}{\frac{2}{\pi}} + \pi$$

$$\eta_{544} = -\frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\pi\theta\right)\right|\left|\cos\left(\pi\theta\right)\right|, -\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)\right) & \text{for } \sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right) \leq 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\pi\theta\right)\right|\left|\cos\left(\pi\theta\right)\right|, -\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)\right) - 2\pi & \text{otherwise} \end{cases}}{\frac{2}{\pi}} - \frac{\pi}{2} + \frac{1}{2}$$

$$\eta_{545} = -\frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\pi\theta\right)\right|\left|\cos\left(\pi\theta\right)\right|, -\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)\right) & \text{for } \sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right) \leq 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\pi\theta\right)\right|\left|\cos\left(\pi\theta\right)\right|, -\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2\pi}$$

$$\eta_{546} = -\frac{2\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right) \leq 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\left|\sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right)\right|}\right)}{\pi}$$

$$\eta_{547} = \frac{\begin{cases} 0 & \text{for } \left|\sin\left(\pi\theta\right)\right|\left|\cos\left(\pi\theta\right)\right| = 0 \\ \operatorname{atan_2}\left(0, -2\sin\left(\pi\theta\right)\cos\left(\pi\theta\right)\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{548} = -\frac{\operatorname{atan_2}\left(2\left|\sin\left(\pi\theta\right)\right|\left|\cos\left(\pi\theta\right)\right|, -\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)\right)}{\pi}$$

$$\eta_{549} = \frac{\operatorname{atan_2}\left(2\left|\sin\left(\pi\theta\right)\right|\left|\cos\left(\pi\theta\right)\right|, -\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)\right)}{\pi} + \frac{1}{2}$$

$$\eta_{550} = \frac{2\operatorname{atan_2}\left(0, \sqrt{\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)}\right)}{\pi}$$

$$\eta_{551} = -\frac{2\left(\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\pi\theta\right), \cos\left(\pi\theta\right)\right) & \text{otherwise} \end{cases}\right)}{\pi}$$

$$\eta_{552} = -\frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\pi\theta\right), \cos\left(\pi\theta\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}$$

$$\eta_{553} = e^{i\left(\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\pi\theta\right), \cos\left(\pi\theta\right)\right) & \text{otherwise} \end{cases} - \frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\pi\theta\right)\right|\left|\cos\left(\pi\theta\right)\right|, -\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)\right) & \text{for } \sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right) \leq 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\pi\theta\right)\right|\left|\cos\left(\pi\theta\right)\right|, -\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4} - \operatorname{atan_2}\left(0, \sqrt{\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)}\right) - 2\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right) \leq 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\frac{\left|\sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right)\right|}{2}} + \frac{1}{2} + \sqrt{\left|\sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right)\right|}\right) + \operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right) \leq 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\left|\sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right)\right|}\right)}$$

$$\eta_{554} = \pi - 2\operatorname{atan_2}\left(0, \sqrt{\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)}\right)$$

$$\eta_{555} = -\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\pi\theta\right), \cos\left(\pi\theta\right)\right) & \text{otherwise} \end{cases} - \pi$$

$$\eta_{556} = -\frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\pi\theta\right)\right|\left|\cos\left(\pi\theta\right)\right|, -\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)\right) & \text{for } \sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right) \leq 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\pi\theta\right)\right|\left|\cos\left(\pi\theta\right)\right|, -\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4}$$

$$\eta_{557} = \begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\pi\theta\right), \cos\left(\pi\theta\right)\right) & \text{otherwise} \end{cases} - \frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\pi\theta\right)\right|\left|\cos\left(\pi\theta\right)\right|, -\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)\right) & \text{for } \sin^2\left(\pi\theta\right) - \cos^2\left(\pi\theta\right) \leq 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\pi\theta\right)\right|\left|\cos\left(\pi\theta\right)\right|, -\sin^2\left(\pi\theta\right) + \cos^2\left(\pi\theta\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4} + \pi$$

$$\eta_{558} = -\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\pi\theta\right)\right|\left|\cos\left(\pi\theta\right)\right| = 0 \\ \operatorname{atan_2}\left(0, -2\sin\left(\pi\theta\right)\cos\left(\pi\theta\right)\right) & \text{otherwise} \end{cases}}{\pi}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{559} = 2\,\mathrm{atan}_2\left(\sqrt{\tfrac{1}{2} - \tfrac{|\sin^2(\pi\theta)-\cos^2(\pi\theta)|}{2}}\left(\begin{cases}1 & \text{for } \sin^2(\pi\theta)-\cos^2(\pi\theta)\le 0\\ -1 & \text{otherwise}\end{cases}\right), \sqrt{\tfrac{|\sin^2(\pi\theta)-\cos^2(\pi\theta)|}{2}} + \tfrac{1}{2} + \sqrt{|\sin^2(\pi\theta)-\cos^2(\pi\theta)|}\right)$$

$$\eta_{560} = \frac{\begin{cases}2\,\mathrm{atan}_2\left(2\,|\sin(\pi\theta)|\,|\cos(\pi\theta)|, -\sin^2(\pi\theta)+\cos^2(\pi\theta)\right) & \text{for } \sin^2(\pi\theta)-\cos^2(\pi\theta)\le 0\\ 2\,\mathrm{atan}_2\left(2\,|\sin(\pi\theta)|\,|\cos(\pi\theta)|, -\sin^2(\pi\theta)+\cos^2(\pi\theta)\right)-2\pi & \text{otherwise}\end{cases}}{2} + \tfrac{\pi}{2}$$

$$\eta_{561} = -2\,\mathrm{atan}_2\left(\sqrt{2}\sqrt{\tfrac{1}{2} - \tfrac{|\sin^2(\pi\theta)-\cos^2(\pi\theta)|}{2}}\left(\begin{cases}1 & \text{for } \sin^2(\pi\theta)-\cos^2(\pi\theta)\le 0\\ -1 & \text{otherwise}\end{cases}\right), \sqrt{|\sin^2(\pi\theta)-\cos^2(\pi\theta)|}\right)$$

$$\eta_{562} = \begin{cases}0 & \text{for } |\sin(\pi\theta)|\,|\cos(\pi\theta)| = 0\\ \mathrm{atan}_2\left(0, -2\sin(\pi\theta)\cos(\pi\theta)\right) & \text{otherwise}\end{cases}$$

$$\eta_{563} = -\,\mathrm{atan}_2\left(2\,|\sin(\pi\theta)|\,|\cos(\pi\theta)|, -\sin^2(\pi\theta)+\cos^2(\pi\theta)\right)$$

$$\eta_{564} = 2\,\mathrm{atan}_2\left(0, \sqrt{\sin^2(\pi\theta)+\cos^2(\pi\theta)}\right)$$

$$\eta_{565} = -\begin{cases}0 & \text{for } \sqrt{\sin^2(\pi\theta)+\cos^2(\pi\theta)} = 0\\ \mathrm{atan}_2\left(-\sin(\pi\theta), \cos(\pi\theta)\right) & \text{otherwise}\end{cases}$$

$$\eta_{566} = e^{i\left(\begin{cases}0 & \text{for } \sqrt{\sin^2(\pi\theta)+\cos^2(\pi\theta)} = 0\\ \mathrm{atan}_2\left(-\sin(\pi\theta), \cos(\pi\theta)\right) & \text{otherwise}\end{cases} - \frac{\begin{cases}2\,\mathrm{atan}_2\left(2\,|\sin(\pi\theta)|\,|\cos(\pi\theta)|, -\sin^2(\pi\theta)+\cos^2(\pi\theta)\right) & \text{for } \sin^2(\pi\theta)-\cos^2(\pi\theta)\le 0\\ 2\,\mathrm{atan}_2\left(2\,|\sin(\pi\theta)|\,|\cos(\pi\theta)|, -\sin^2(\pi\theta)+\cos^2(\pi\theta)\right)-2\pi & \text{otherwise}\end{cases}}{4}\right)}$$

$$\eta_{567} = \frac{\sqrt[4]{-1}\sqrt{2}(1+i)e^{i\,\mathrm{global_shift}}}{2}$$

$$\eta_{568} = 2\,\mathrm{atan}\left(\frac{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right) + \tfrac{3\pi}{2}$$

$$\eta_{569} = \mathrm{atan}\left(\frac{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}\right) + \mathrm{atan}\left(\frac{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right) + \tfrac{3\pi}{2}$$

$$\eta_{570} = 2\,\mathrm{atan}\left(\frac{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}\right) + \tfrac{3\pi}{2}$$

$$\eta_{571} = e^{\frac{3i\left(2\,\mathrm{atan}\left(\frac{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}\right)+2\,\mathrm{atan}\left(\frac{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right)+\pi\right)}{4}}$$

$$\eta_{572} = -\pi - 2\,\mathrm{atan}\left(\frac{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{573} = \mathrm{atan}\left(\frac{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}\right) - \mathrm{atan}\left(\frac{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{574} = -\,\mathrm{atan}\left(\frac{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right) - \mathrm{atan}\left(\frac{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{575} = -\pi + \mathrm{atan}\left(\frac{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right) - \mathrm{atan}\left(\frac{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{576} = \mathrm{atan}\left(\frac{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}\right) + \mathrm{atan}\left(\frac{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right) + \pi$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{577} = -\operatorname{atan}\left(\frac{-(2-\sqrt{2})^{\frac{3}{2}}+\sqrt{2}\sqrt{\sqrt{2}+2}}{\sqrt{2}\sqrt{2-\sqrt{2}}+(2-\sqrt{2})\sqrt{\sqrt{2}+2}}\right) - \operatorname{atan}\left(\frac{-(\sqrt{2}+2)^{\frac{3}{2}}+\sqrt{2}\sqrt{2-\sqrt{2}}}{\sqrt{2}\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}(\sqrt{2}+2)}\right)$$

$$\eta_{578} = -\pi - \operatorname{atan}\left(\frac{-(\sqrt{2}+2)^{\frac{3}{2}}+\sqrt{2}\sqrt{2-\sqrt{2}}}{\sqrt{2}\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}(\sqrt{2}+2)}\right) + \operatorname{atan}\left(\frac{\sqrt{2}\sqrt{2-\sqrt{2}}+(2-\sqrt{2})\sqrt{\sqrt{2}+2}}{-(2-\sqrt{2})^{\frac{3}{2}}+\sqrt{2}\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{579} = \operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}\right) + \operatorname{atan}\left(\frac{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{580} = -2i\log\left(\frac{\frac{\sqrt{(-1)^{\frac{3}{4}}\left(\sqrt{2-\sqrt{2}}-i\sqrt{\sqrt{2}+2}\right)}}{2}+\frac{i\sqrt{\sqrt[4]{-1}\left(\sqrt{\sqrt{2}+2}-i\sqrt{2-\sqrt{2}}\right)}}{2}}{\sqrt{\frac{\sqrt[4]{-1}\left(\sqrt{\sqrt{2}+2}-i\sqrt{2-\sqrt{2}}\right)}{4}}+\frac{(-1)^{\frac{3}{4}}\left(\sqrt{2-\sqrt{2}}-i\sqrt{\sqrt{2}+2}\right)}{4}}\right)$$

$$\eta_{581} = \operatorname{atan}\left(\frac{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right) - \operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{582} = -\pi - \operatorname{atan}\left(\frac{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right) - \operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{583} = -2i\log\left(\frac{\frac{\sqrt{\sqrt[4]{-1}\left(\sqrt{\sqrt{2}+2}-i\sqrt{2-\sqrt{2}}\right)}}{2}+\frac{i\sqrt{(-1)^{\frac{3}{4}}\left(\sqrt{2-\sqrt{2}}-i\sqrt{\sqrt{2}+2}\right)}}{2}}{\sqrt{\frac{\sqrt[4]{-1}\left(\sqrt{\sqrt{2}+2}-i\sqrt{2-\sqrt{2}}\right)}{4}}+\frac{(-1)^{\frac{3}{4}}\left(\sqrt{2-\sqrt{2}}-i\sqrt{\sqrt{2}+2}\right)}{4}}\right)$$

$$\eta_{584} = \operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}\right) - \operatorname{atan}\left(\frac{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right) + \pi$$

$$\eta_{585} = -2i\log\left(\frac{\frac{\sqrt{\sqrt[4]{-1}\left(\sqrt{\sqrt{2}+2}-i\sqrt{2-\sqrt{2}}\right)}}{2}+\frac{i\sqrt{(-1)^{\frac{3}{4}}\left(\sqrt{2-\sqrt{2}}-\sqrt{-2-\sqrt{2}}\right)}}{2}}{\sqrt{\frac{\sqrt[4]{-1}\left(\sqrt{\sqrt{2}+2}-i\sqrt{2-\sqrt{2}}\right)}{4}}+\frac{(-1)^{\frac{3}{4}}\left(\sqrt{2-\sqrt{2}}-\sqrt{-2-\sqrt{2}}\right)}{4}}\right)$$

$$\eta_{586} = -\pi + 2\operatorname{atan}\left(\frac{\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{587} = -\pi - \operatorname{atan}\left(\frac{\sqrt{\sqrt{2}+2}+(1+\sqrt{2})\sqrt{2-\sqrt{2}}}{-\sqrt{2-\sqrt{2}}+(1+\sqrt{2})\sqrt{\sqrt{2}+2}}\right) - \operatorname{atan}\left(\frac{-\sqrt{2-\sqrt{2}}+(1-\sqrt{2})\sqrt{\sqrt{2}+2}}{(1-\sqrt{2})\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{588} = -2i\log\left(\frac{\frac{\sqrt{2}\sqrt{(-1)^{\frac{3}{4}}\left(\sqrt{2-\sqrt{2}}-\sqrt{-2-\sqrt{2}}\right)}\left((-1)^{\frac{3}{4}}(1+i)+1-i\right)\left(-\sqrt{2}(1-i)(1+i)+2+2i\right)}}{8}+\frac{\sqrt{2}i\sqrt{-(-1)^{\frac{3}{4}}\left(\sqrt{2-\sqrt{2}}-i\sqrt{\sqrt{2}+2}\right)(2+\sqrt{2}(1-i)(1+i)+2i)\left((-1)^{\frac{3}{4}}(1+i)-1+i\right)}}{8}}{\sqrt{\frac{(-1)^{\frac{3}{4}}\left(\sqrt{2-\sqrt{2}}-\sqrt{-2-\sqrt{2}}\right)\left((-1)^{\frac{3}{4}}(1+i)+1-i\right)\left(-\sqrt{2}(1-i)(1+i)+2+2i\right)}{32}}-\frac{(-1)^{\frac{3}{4}}\left(\sqrt{2-\sqrt{2}}-i\sqrt{\sqrt{2}+2}\right)(2+\sqrt{2}(1-i)(1+i)+2i)\left((-1)^{\frac{3}{4}}(1+i)-1+i\right)}{32}}\right)$$

$$\eta_{589} = -\pi - \operatorname{atan}\left(\frac{-\sqrt{2-\sqrt{2}}+(1-\sqrt{2})\sqrt{\sqrt{2}+2}}{(1-\sqrt{2})\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right) + \operatorname{atan}\left(\frac{-\sqrt{2-\sqrt{2}}+(1+\sqrt{2})\sqrt{\sqrt{2}+2}}{\sqrt{\sqrt{2}+2}+(1+\sqrt{2})\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{590} = e^{i\left(-\frac{\operatorname{atan}\left(\frac{\sqrt{\sqrt{2}+2}+(1+\sqrt{2})\sqrt{2-\sqrt{2}}}{-\sqrt{2-\sqrt{2}}+(1+\sqrt{2})\sqrt{\sqrt{2}+2}}\right)}{2}+\frac{\operatorname{atan}\left(\frac{-\sqrt{2-\sqrt{2}}+(1+\sqrt{2})\sqrt{\sqrt{2}+2}}{\sqrt{\sqrt{2}+2}+(1+\sqrt{2})\sqrt{2-\sqrt{2}}}\right)}{2}+\frac{7\pi}{8}\right)}$$

$$\eta_{591} = \frac{\sqrt{2}i(1+i)^{\frac{3}{4}}e^{\frac{i(-4\gamma-\phi+\pi)}{4}}}{4}$$

$$\eta_{592} = -\tfrac{\lambda}{2} - \gamma - \tfrac{\phi}{2} + \tfrac{\zeta}{2} + \left(\pi\left(\left\lfloor -\tfrac{\phi}{2\pi}+\tfrac{1}{2}\right\rfloor \bmod 2\right)\right) + \left(\pi\left(\left\lfloor -\tfrac{\theta}{\pi}+\tfrac{1}{2}\right\rfloor \bmod 2\right)\right)$$

$$\eta_{593} = \tfrac{\lambda}{2} - \gamma - \tfrac{\phi}{2} - \tfrac{\zeta}{2} - \left(\frac{\pi\left(1-\left(2\left(\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right)\right)}{2}\right) + \left(\pi\left(\left\lfloor -\tfrac{\phi}{2\pi}+\tfrac{1}{2}\right\rfloor \bmod 2\right)\right) + \left(\pi\left(\left\lfloor -\tfrac{\theta}{\pi}+\tfrac{1}{2}\right\rfloor \bmod 2\right)\right)$$

$$\eta_{594} = -(-1)^{\frac{3}{4}}e^{-\frac{i(4\gamma+\phi)}{4}}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{595} = \frac{\chi}{2} - \gamma - \frac{\phi}{2} - \frac{\zeta}{2} - \frac{\pi}{2}$$

$$\eta_{596} = \frac{\frac{\chi}{2} - \gamma - \frac{\phi}{2} - \frac{\zeta}{2}}{\pi}$$

$$\eta_{597} = \frac{-\frac{\chi}{2} - \gamma - \frac{\phi}{2} + \frac{\zeta}{2}}{\pi}$$

$$\eta_{598} = \frac{\chi}{2} - \gamma - \frac{\phi}{2} - \frac{\zeta}{2} - \frac{3\pi}{2}$$

$$\eta_{599} = \frac{\sqrt{2}(1+i)^3 e^{\frac{i(-4\gamma-\phi+\pi)}{4}}}{4}$$

$$\eta_{600} = \frac{\pi(-2phase_{exponent}+t(2global_{shift}-1))}{2}$$

$$\eta_{601} = \pi\left(axis_{phaseexponent} + z_{exponent} - \tfrac{1}{2}\right)$$

$$\eta_{602} = \pi\left(axis_{phaseexponent} + z_{exponent} + \tfrac{1}{2}\right)$$

$$\eta_{603} = ie^{\frac{i\pi(e_{exponent}+z_{exponent})}{2}}$$

$$\eta_{604} = \frac{(-1)^{\frac{3}{16}}\sqrt{2}(-1-i)}{2}$$

$$\eta_{605} = \frac{(-1)^{\frac{3}{16}}\sqrt{2}(1+i)}{2}$$

$$\eta_{606} = \frac{(-1)^{\frac{3}{16}}\sqrt{2}(-1+i)}{2}$$

$$\eta_{607} = \frac{(-1)^{\frac{3}{16}}\sqrt{2}(1-i)}{2}$$

$$\eta_{608} = -\frac{\pi}{2} - 2\operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{609} = -2\operatorname{atan}\left(\frac{\sqrt{2}}{\sqrt{2}+2}\right) + 2\pi$$

$$\eta_{610} = -\pi + \operatorname{atan}\left(\frac{\sqrt{2}}{2}\right)$$

$$\eta_{611} = -\begin{cases} 0 & \text{for } \sqrt{\frac{vx^2\operatorname{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \frac{vy^2\operatorname{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4}} = 0 \\ \operatorname{atan}_2\left(-\frac{vx\operatorname{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}, \frac{vy\operatorname{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}\right) & \text{otherwise} \end{cases} - \begin{cases} 0 & \text{for } \sqrt{\frac{vz^2\operatorname{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \cos^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)} = 0 \\ \operatorname{atan}_2\left(-\frac{vz\operatorname{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}, \cos\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{612} = 2\operatorname{atan}_2\left(\sqrt{\frac{vx^2\operatorname{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \frac{vy^2\operatorname{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4}}, \sqrt{\frac{vz^2\operatorname{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \cos^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}\right)$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{613} = \begin{cases} 0 & \text{for } \sqrt{\frac{vx^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \frac{vy^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4}} = 0 \\ \mathrm{atan}_2\left(-\frac{vx\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}, \frac{vy\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}\right) & \text{otherwise} \end{cases} - \begin{cases} 0 & \text{for } \sqrt{\frac{vz^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \cos^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)} = 0 \\ \mathrm{atan}_2\left(-\frac{vz\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}, \cos\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{614} = \pi - 2\,\mathrm{atan}_2\left(\sqrt{\frac{vx^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \frac{vy^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4}}, \sqrt{\frac{vz^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \cos^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}\right)$$

$$\eta_{615} = -\begin{cases} 0 & \text{for } \sqrt{\frac{vx^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \frac{vy^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4}} = 0 \\ \mathrm{atan}_2\left(-\frac{vx\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}, \frac{vy\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}\right) & \text{otherwise} \end{cases} - \begin{cases} 0 & \text{for } \sqrt{\frac{vz^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \cos^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)} = 0 \\ \mathrm{atan}_2\left(-\frac{vz\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}, \cos\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)\right) & \text{otherwise} \end{cases} - \pi$$

$$\eta_{616} = \begin{cases} 0 & \text{for } \sqrt{\frac{vx^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \frac{vy^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4}} = 0 \\ \mathrm{atan}_2\left(-\frac{vx\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}, \frac{vy\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}\right) & \text{otherwise} \end{cases}$$

$$\eta_{617} = \begin{cases} 0 & \text{for } \sqrt{\frac{vz^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \cos^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)} = 0 \\ \mathrm{atan}_2\left(-\frac{vz\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}, \cos\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)\right) & \text{otherwise} \end{cases} + \pi$$

$$\eta_{618} = \begin{cases} 1 & \text{for } \sqrt{vz^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right) + 2\cos\left(\sqrt{vx^2+vy^2+vz^2}\right) + 2} = 0 \\ e^{\,i\,\mathrm{atan}_2\left(-\frac{vz\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}, \cos\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)\right)} & \text{otherwise} \end{cases}$$

$$\eta_{619} = -\begin{cases} 0 & \text{for } \sqrt{\frac{vx^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \frac{vy^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4}} = 0 \\ \mathrm{atan}_2\left(-\frac{vx\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}, \frac{vy\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}\right) & \text{otherwise} \end{cases} - \begin{cases} 0 & \text{for } \sqrt{\frac{vz^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \cos^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)} = 0 \\ \mathrm{atan}_2\left(-\frac{vz\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}, \cos\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)\right) & \text{otherwise} \end{cases} - \frac{\pi}{2}$$

$$\eta_{620} = \begin{cases} 0 & \text{for } \sqrt{\frac{vx^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \frac{vy^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4}} = 0 \\ \mathrm{atan}_2\left(-\frac{vx\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}, \frac{vy\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}\right) & \text{otherwise} \end{cases} - \begin{cases} 0 & \text{for } \sqrt{\frac{vz^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \cos^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)} = 0 \\ \mathrm{atan}_2\left(-\frac{vz\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}, \cos\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)\right) & \text{otherwise} \end{cases} + \frac{\pi}{2}$$

$$\eta_{621} = \begin{cases} 0 & \text{for } \sqrt{\frac{vz^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \cos^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)} = 0 \\ \mathrm{atan}_2\left(-\frac{vz\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}, \cos\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)\right) & \text{otherwise} \end{cases} - \mathrm{atan}_2\left(\sqrt{\frac{vx^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \frac{vy^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4}}, \sqrt{\frac{vz^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \cos^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}\right)$$

$$\eta_{622} = \begin{cases} -i\,\mathrm{atan}_2\left(\frac{\sqrt{vx^2+vy^2}\left|\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)\right|}{2}, \frac{\sqrt{vz^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)+2\cos\left(\sqrt{vx^2+vy^2+vz^2}\right)+2}}{2}\right) \\ e \\ i\left(\mathrm{atan}_2\left(-\frac{vz\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}, \cos\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)\right) - \mathrm{atan}_2\left(\frac{\sqrt{vx^2+vy^2}\left|\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)\right|}{2}, \frac{\sqrt{vz^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)+2\cos\left(\sqrt{vx^2+vy^2+vz^2}\right)+2}}{2}\right)\right) \\ e \end{cases} \begin{array}{l} \text{for } \sqrt{vz^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right) + 2\cos\left(\sqrt{vx^2+vy^2+vz^2}\right) + 2} = 0 \\ \text{otherwise} \end{array}$$

$$\eta_{623} = \frac{-\begin{cases} 0 & \text{for } \sqrt{\frac{vx^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \frac{vy^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4}} = 0 \\ \mathrm{atan}_2\left(-\frac{vx\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}, \frac{vy\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}\right) & \text{otherwise} \end{cases} - \begin{cases} 0 & \text{for } \sqrt{\frac{vz^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \cos^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)} = 0 \\ \mathrm{atan}_2\left(-\frac{vz\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}, \cos\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{624} = \frac{2\,\mathrm{atan}_2\left(\sqrt{\frac{vx^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \frac{vy^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4}}, \sqrt{\frac{vz^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \cos^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}\right)}{\pi}$$

$$\eta_{625} = \frac{\begin{cases} 0 & \text{for } \sqrt{\frac{vx^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \frac{vy^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4}} = 0 \\ \mathrm{atan}_2\left(-\frac{vx\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}, \frac{vy\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}\right) & \text{otherwise} \end{cases} - \begin{cases} 0 & \text{for } \sqrt{\frac{vz^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \cos^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)} = 0 \\ \mathrm{atan}_2\left(-\frac{vz\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}, \cos\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{626} = \frac{2\left(\begin{cases} 0 & \text{for } \sqrt{\frac{vz^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \cos^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)} = 0 \\ \mathrm{atan}_2\left(-\frac{vz\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}, \cos\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)\right) & \text{otherwise} \end{cases}\right) - 2\,\mathrm{atan}_2\left(\sqrt{\frac{vx^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \frac{vy^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4}}, \sqrt{\frac{vz^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \cos^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}\right)}{\pi}$$

$$\eta_{627} = \frac{-\begin{cases} 0 & \text{for } \sqrt{\frac{vz^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \cos^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)} = 0 \\ \mathrm{atan}_2\left(-\frac{vz\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}, \cos\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)\right) & \text{otherwise} \end{cases} + \mathrm{atan}_2\left(\sqrt{\frac{vx^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \frac{vy^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4}}, \sqrt{\frac{vy^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \cos^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}\right)}{\pi}$$

$$\eta_{628} = 2\,\mathrm{atan}_2\left(\sqrt{\frac{vx^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \frac{vy^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4}}, \sqrt{\frac{vz^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \cos^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}\right) + \pi$$

$$\eta_{629} = \begin{cases} 0 & \text{for } \sqrt{\frac{vx^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \frac{vy^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4}} = 0 \\ \mathrm{atan}_2\left(-\frac{vx\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}, \frac{vy\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}\right) & \text{otherwise} \end{cases} - \begin{cases} 0 & \text{for } \sqrt{\frac{vz^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \cos^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)} = 0 \\ \mathrm{atan}_2\left(-\frac{vz\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}, \cos\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)\right) & \text{otherwise} \end{cases} + \pi$$

$$\eta_{630} = \frac{-\begin{cases} 0 & \text{for } \sqrt{\frac{vx^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \frac{vy^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4}} = 0 \\ \mathrm{atan}_2\left(-\frac{vx\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}, \frac{vy\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}\right) & \text{otherwise} \end{cases} - \begin{cases} 0 & \text{for } \sqrt{\frac{vz^2\,\mathrm{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4} + \cos^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)} = 0 \\ \mathrm{atan}_2\left(-\frac{vz\,\mathrm{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}, \cos\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)\right) & \text{otherwise} \end{cases}}{4\pi}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{631} = \frac{\operatorname{atan_2}\left(\sqrt{\frac{vx^2\operatorname{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4}+\frac{vy^2\operatorname{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4}},\sqrt{\frac{vz^2\operatorname{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4}+\cos^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}\right)}{2\pi}$$

$$\eta_{632} = \frac{\begin{cases}0 & \text{for } \sqrt{\frac{vx^2\operatorname{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4}+\frac{vy^2\operatorname{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4}} = 0\\ \operatorname{atan_2}\left(-\frac{vx\operatorname{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2},\frac{vy\operatorname{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2}\right) & \text{otherwise}\end{cases}}{4\pi} - \frac{\begin{cases}0 & \text{for } \sqrt{\frac{vz^2\operatorname{sinc}^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{4}+\cos^2\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)} = 0\\ \operatorname{atan_2}\left(-\frac{vz\operatorname{sinc}\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)}{2},\cos\left(\frac{\sqrt{vx^2+vy^2+vz^2}}{2}\right)\right) & \text{otherwise}\end{cases}}{4\pi}$$

$$\eta_{633} = \frac{\sqrt{2}i(-1+i)e^{-\frac{i(2\phi+\pi)}{4}}}{2}$$

$$\eta_{634} = \frac{2\left(\begin{cases}0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0\\ \operatorname{atan_2}\left(0,\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise}\end{cases}\right) - \frac{\begin{cases}2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\le 0\\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right)-2\pi & \text{otherwise}\end{cases}}{2} - 4\operatorname{atan_2}\left(\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}},\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right)\cdot\sqrt{\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}+\frac{1}{2}+\sqrt{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}}\right) + 2\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}},\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right)\cdot\sqrt{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}\right) - 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|,\left|\cos\left(\frac{\theta}{2}\right)\right|\right) - 2\pi}{\pi}$$

$$\eta_{635} = \frac{-\begin{cases}0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0\\ \operatorname{atan_2}\left(0,\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise}\end{cases} + \frac{\begin{cases}2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\le 0\\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right)-2\pi & \text{otherwise}\end{cases}}{4} + 2\operatorname{atan_2}\left(\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}},\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right)\cdot\sqrt{\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}+\frac{1}{2}+\sqrt{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}}\right) - \operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}},\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right)\cdot\sqrt{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}\right) + \operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|,\left|\cos\left(\frac{\theta}{2}\right)\right|\right)}{\pi} + \frac{1}{2}$$

$$\eta_{636} = \frac{2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|,\left|\cos\left(\frac{\theta}{2}\right)\right|\right)}{\pi}$$

$$\eta_{637} = \frac{\begin{cases}0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0\\ \operatorname{atan_2}\left(-\sin\left(\frac{\theta}{2}\right),0\right) & \text{otherwise}\end{cases} + \begin{cases}0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0\\ \operatorname{atan_2}\left(0,\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise}\end{cases}}{\pi} + \frac{1}{2}$$

$$\eta_{638} = -\frac{\begin{cases}0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right| = 0\\ \operatorname{atan_2}\left(0,-2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise}\end{cases}}{\pi}$$

$$\eta_{639} = \frac{\begin{cases}0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right| = 0\\ \operatorname{atan_2}\left(0,-2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise}\end{cases}}{\pi} + \frac{1}{2}$$

$$\eta_{640} = -\frac{\begin{cases}0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right| = 0\\ \operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,0\right) & \text{otherwise}\end{cases}}{\pi}$$

$$\eta_{641} = \frac{2\operatorname{atan_2}\left(\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}},\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right)\cdot\sqrt{\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}+\frac{1}{2}+\sqrt{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}}\right)}{\pi}$$

$$\eta_{642} = \frac{\begin{cases}2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\le 0\\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right)-2\pi & \text{otherwise}\end{cases}}{2} + \pi}{\pi}$$

$$\eta_{643} = \frac{-\frac{\begin{cases}2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\le 0\\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right)-2\pi & \text{otherwise}\end{cases}}{2} - \pi}{\pi} + \frac{1}{2}$$

$$\eta_{644} = \frac{\begin{cases}2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\le 0\\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right)-2\pi & \text{otherwise}\end{cases}}{2\pi}$$

$$\eta_{645} = -\frac{2\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}},\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right)\cdot\sqrt{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}\right)}{\pi}$$

$$\eta_{646} = \frac{\begin{cases}0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right| = 0\\ \operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,0\right) & \text{otherwise}\end{cases}}{\pi}$$

$$\eta_{647} = \frac{\begin{cases}0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right| = 0\\ \operatorname{atan_2}\left(0,-2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise}\end{cases}}{\pi}$$

$$\eta_{648} = -\frac{\begin{cases}0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right| = 0\\ \operatorname{atan_2}\left(0,-2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise}\end{cases}}{\pi} + \frac{1}{2}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{649} = -\frac{\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right)}{\pi}$$

$$\eta_{650} = \frac{\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right)}{\pi}+\frac{1}{2}$$

$$\eta_{651} = e^{i\left(\begin{cases}0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right|=0\\ \operatorname{atan_2}\left(0,\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise}\end{cases}-\frac{\begin{cases}2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\le 0\\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right)-2\pi & \text{otherwise}\end{cases}}{4}-2\operatorname{atan_2}\left(\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right),\sqrt{\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}+\frac{1}{2}}+\sqrt{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}\right)+\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right),\sqrt{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}\right)-\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|,\left|\cos\left(\frac{\theta}{2}\right)\right|\right)+\frac{\pi}{2}\right)}$$

$$\eta_{652} = \begin{cases}0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right|=0\\ \operatorname{atan_2}\left(0,\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise}\end{cases}-\frac{\begin{cases}2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\le 0\\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right)-2\pi & \text{otherwise}\end{cases}}{4}-\pi$$

$$\eta_{653} = \begin{cases}0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right|=0\\ \operatorname{atan_2}\left(0,\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise}\end{cases}-\frac{\begin{cases}2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\le 0\\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right)-2\pi & \text{otherwise}\end{cases}}{4}+\pi$$

$$\eta_{654} = 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|,\left|\cos\left(\frac{\theta}{2}\right)\right|\right)$$

$$\eta_{655} = \begin{cases}0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right|=0\\ \operatorname{atan_2}\left(-\sin\left(\frac{\theta}{2}\right),0\right) & \text{otherwise}\end{cases}-\begin{cases}0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right|=0\\ \operatorname{atan_2}\left(0,\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise}\end{cases}$$

$$\eta_{656} = -\begin{cases}0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right|=0\\ \operatorname{atan_2}\left(-\sin\left(\frac{\theta}{2}\right),0\right) & \text{otherwise}\end{cases}-\begin{cases}0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right|=0\\ \operatorname{atan_2}\left(0,\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise}\end{cases}$$

$$\eta_{657} = -\begin{cases}0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|=0\\ \operatorname{atan_2}\left(0,-2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise}\end{cases}$$

$$\eta_{658} = -\begin{cases}0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|=0\\ \operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,0\right) & \text{otherwise}\end{cases}$$

$$\eta_{659} = 2\operatorname{atan_2}\left(\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right),\sqrt{\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}+\frac{1}{2}}+\sqrt{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}\right)$$

$$\eta_{660} = \frac{\begin{cases}2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\le 0\\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right)-2\pi & \text{otherwise}\end{cases}}{2}+\frac{\pi}{2}$$

$$\eta_{661} = -2\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right),\sqrt{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}\right)$$

$$\eta_{662} = \begin{cases}0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|=0\\ \operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,0\right) & \text{otherwise}\end{cases}$$

$$\eta_{663} = \begin{cases}0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|=0\\ \operatorname{atan_2}\left(0,-2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise}\end{cases}$$

$$\eta_{664} = -\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right)$$

$$\eta_{665} = e^{i\left(\begin{cases}0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right|=0\\ \operatorname{atan_2}\left(0,\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise}\end{cases}-\frac{\begin{cases}2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\le 0\\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right)-2\pi & \text{otherwise}\end{cases}}{4}\right)}$$

$$\eta_{666} = \frac{2\left(\begin{cases}0 & \text{for } \sqrt{\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)}=0\\ \operatorname{atan_2}\left(-\sin\left(\frac{\theta}{2}\right),\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise}\end{cases}-\frac{\begin{cases}2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\le 0\\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right)-2\pi & \text{otherwise}\end{cases}}{2}-2\operatorname{atan_2}\left(0,\sqrt{\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)}\right)-4\operatorname{atan_2}\left(\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right),\sqrt{\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}+\frac{1}{2}}+\sqrt{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}\right)+2\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right),\sqrt{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}\right)-2\pi}{\pi}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{667} = \frac{-\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\theta}{2}\right),\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} = 0 + \frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right) \le 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right)-2\pi & \text{otherwise} \end{cases}}{4} + \operatorname{atan_2}\left(0,\sqrt{\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)}\right) + 2\operatorname{atan_2}\left(\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\cdot\sqrt{\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}}+\frac{1}{2}+\sqrt{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}\right) - \operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\cdot\sqrt{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}\right)}{\pi} + \frac{1}{2}$$

$$\eta_{668} = \frac{2\operatorname{atan_2}\left(0,\sqrt{\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)}\right)}{\pi}$$

$$\eta_{669} = -\frac{2\left(\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\theta}{2}\right),\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}\right)}{\pi}$$

$$\eta_{670} = \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\theta}{2}\right),\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}$$

$$\eta_{671} = e^{i\left(\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\theta}{2}\right),\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} - \frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right) \le 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right)-2\pi & \text{otherwise} \end{cases}}{4} - \operatorname{atan_2}\left(0,\sqrt{\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)}\right) - 2\operatorname{atan_2}\left(\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\cdot\sqrt{\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}}+\frac{1}{2}+\sqrt{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}\right) + \operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\cdot\sqrt{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}\right) + \frac{\pi}{2}\right)}$$

$$\eta_{672} = \begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\theta}{2}\right),\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} - \frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right) \le 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right)-2\pi & \text{otherwise} \end{cases}}{4} - \pi$$

$$\eta_{673} = \begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\theta}{2}\right),\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} - \frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right) \le 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right)-2\pi & \text{otherwise} \end{cases}}{4} + \pi$$

$$\eta_{674} = 2\operatorname{atan_2}\left(0,\sqrt{\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)}\right)$$

$$\eta_{675} = -\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\theta}{2}\right),\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{676} = e^{i\left(\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\theta}{2}\right),\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} - \frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right) \le 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right)-2\pi & \text{otherwise} \end{cases}}{4}\right)}$$

$$\eta_{677} = \frac{2\left(\begin{cases} 0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0,\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}\right) - \frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right) \le 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right)-2\pi & \text{otherwise} \end{cases}}{2} - 4\operatorname{atan_2}\left(\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\cdot\sqrt{\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}}+\frac{1}{2}+\sqrt{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}\right) + 2\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\cdot\sqrt{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}\right) - 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|,\left|\cos\left(\frac{\theta}{2}\right)\right|\right) - \pi}{\pi}$$

$$\eta_{678} = \frac{-\begin{cases} 0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0,\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} + \frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right) \le 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|,-\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right)-2\pi & \text{otherwise} \end{cases}}{4} + 2\operatorname{atan_2}\left(\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\cdot\sqrt{\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}}+\frac{1}{2}+\sqrt{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}\right) - \operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\cdot\sqrt{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}\right) + \operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|,\left|\cos\left(\frac{\theta}{2}\right)\right|\right) - \frac{\pi}{2}}{\pi} + \frac{1}{2}$$

$$\eta_{679} = \frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0,\sin\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} + \begin{cases} 0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0,\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}$$

$$\eta_{680} = -\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(-2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right),0\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{681} = \frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(-2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right),0\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}$$

$$\eta_{682} = -\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(-2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right),0\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{683} = -\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(-2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right),0\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}$$

$$\eta_{684} = \begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0,\sin\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} - \begin{cases} 0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0,\cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{685} = -\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \sin\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} - \begin{cases} 0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{686} = -\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(-2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right), 0\right) & \text{otherwise} \end{cases}$$

$$\eta_{687} = \begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(-2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right), 0\right) & \text{otherwise} \end{cases}$$

$$\eta_{688} = \frac{\sqrt{2}i(-1+i)e^{\frac{i(2\theta-\pi)}{4}}}{2}$$

$$\eta_{689} = \frac{\pi - 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \left|\cos\left(\frac{\theta}{2}\right)\right|\right)}{\pi}$$

$$\eta_{690} = \frac{-\pi\left(\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\theta}{2}\right), 0\right) & \text{otherwise} \end{cases}}{\pi} + \frac{\begin{cases} 0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}\right) - \frac{\pi}{2}}{\pi}$$

$$\eta_{691} = \frac{\pi\left(\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\theta}{2}\right), 0\right) & \text{otherwise} \end{cases}}{\pi} + \frac{\begin{cases} 0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}\right) + \frac{\pi}{2}}{\pi} + \frac{1}{2}$$

$$\eta_{692} = \frac{\pi\left(\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\theta}{2}\right), 0\right) & \text{otherwise} \end{cases}}{\pi} + \frac{\begin{cases} 0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} - \frac{2\left(\begin{cases} 0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}\right)}{\pi} + \frac{1}{2}\right) + 2\left(\begin{cases} 0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}\right) - \frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right| \left|\cos\left(\frac{\theta}{2}\right)\right|, -\sin^2\left(\frac{\theta}{2}\right) + \cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right| \left|\cos\left(\frac{\theta}{2}\right)\right|, -\sin^2\left(\frac{\theta}{2}\right) + \cos^2\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2} - 4\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{\frac{\left|\sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right)\right|}{2}} + \frac{1}{2} + \sqrt{\left|\sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right)\right|}\right) + 2\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{\left|\sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right)\right|}\right) + 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \left|\cos\left(\frac{\theta}{2}\right)\right|\right) - \frac{3\pi}{2}}{\pi}$$

$$\eta_{693} = -\frac{\begin{cases} 0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} + \frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right| \left|\cos\left(\frac{\theta}{2}\right)\right|, -\sin^2\left(\frac{\theta}{2}\right) + \cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right| \left|\cos\left(\frac{\theta}{2}\right)\right|, -\sin^2\left(\frac{\theta}{2}\right) + \cos^2\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4} + 2\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{\frac{\left|\sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right)\right|}{2}} + \frac{1}{2} + \sqrt{\left|\sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right)\right|}\right) - \operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{\left|\sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right)\right|}\right) - \operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \left|\cos\left(\frac{\theta}{2}\right)\right|\right)}{\pi} + \frac{1}{2}$$

$$\eta_{694} = e^{i\left(\begin{cases} 0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} - \frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right| \left|\cos\left(\frac{\theta}{2}\right)\right|, -\sin^2\left(\frac{\theta}{2}\right) + \cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right| \left|\cos\left(\frac{\theta}{2}\right)\right|, -\sin^2\left(\frac{\theta}{2}\right) + \cos^2\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4} - 2\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{\frac{\left|\sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right)\right|}{2}} + \frac{1}{2} + \sqrt{\left|\sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right)\right|}\right) + \operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{\left|\sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right)\right|}\right) - \operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \left|\cos\left(\frac{\theta}{2}\right)\right|\right)\right)}$$

$$\eta_{695} = \pi - 2\operatorname{atan_2}\left(\left|\sin\left(\frac{\theta}{2}\right)\right|, \left|\cos\left(\frac{\theta}{2}\right)\right|\right)$$

$$\eta_{696} = -\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\theta}{2}\right), 0\right) & \text{otherwise} \end{cases} - \begin{cases} 0 & \text{for } \left|\cos\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} - \pi$$

$$\eta_{697} = \begin{cases} 0 & \text{for } \left|\sin\left(\frac{\theta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\theta}{2}\right), 0\right) & \text{otherwise} \end{cases} - \frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right| \left|\cos\left(\frac{\theta}{2}\right)\right|, -\sin^2\left(\frac{\theta}{2}\right) + \cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right| \left|\cos\left(\frac{\theta}{2}\right)\right|, -\sin^2\left(\frac{\theta}{2}\right) + \cos^2\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4}$$

$$\eta_{698} = \frac{\pi - 2\operatorname{atan_2}\left(0, \sqrt{\sin^2\left(\frac{\theta}{2}\right) + \cos^2\left(\frac{\theta}{2}\right)}\right)}{\pi}$$

$$\eta_{699} = \frac{-\pi\left(\frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\theta}{2}\right) + \cos^2\left(\frac{\theta}{2}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\theta}{2}\right), \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}\right) - \frac{\pi}{2}}{\pi}$$

$$\eta_{700} = \frac{\pi\left(\frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\theta}{2}\right) + \cos^2\left(\frac{\theta}{2}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\theta}{2}\right), \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}\right) + \frac{\pi}{2}}{\pi} + \frac{1}{2}$$

$$\eta_{701} = \frac{\pi\left(-\frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\theta}{2}\right) + \cos^2\left(\frac{\theta}{2}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\theta}{2}\right), \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}\right) + 2\left(\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\theta}{2}\right) + \cos^2\left(\frac{\theta}{2}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\theta}{2}\right), \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases}\right) - \frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right| \left|\cos\left(\frac{\theta}{2}\right)\right|, -\sin^2\left(\frac{\theta}{2}\right) + \cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right| \left|\cos\left(\frac{\theta}{2}\right)\right|, -\sin^2\left(\frac{\theta}{2}\right) + \cos^2\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2} + 2\operatorname{atan_2}\left(0, \sqrt{\sin^2\left(\frac{\theta}{2}\right) + \cos^2\left(\frac{\theta}{2}\right)}\right) - 4\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{\frac{\left|\sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right)\right|}{2}} + \frac{1}{2} + \sqrt{\left|\sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right)\right|}\right) + 2\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{\left|\sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right)\right|}\right) - \frac{3\pi}{2}}{\pi}$$

$$\eta_{702} = -\frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\theta}{2}\right) + \cos^2\left(\frac{\theta}{2}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\theta}{2}\right), \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} + \frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right| \left|\cos\left(\frac{\theta}{2}\right)\right|, -\sin^2\left(\frac{\theta}{2}\right) + \cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right| \left|\cos\left(\frac{\theta}{2}\right)\right|, -\sin^2\left(\frac{\theta}{2}\right) + \cos^2\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4} - \operatorname{atan_2}\left(0, \sqrt{\sin^2\left(\frac{\theta}{2}\right) + \cos^2\left(\frac{\theta}{2}\right)}\right) + 2\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{\frac{\left|\sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right)\right|}{2}} + \frac{1}{2} + \sqrt{\left|\sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right)\right|}\right) - \operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\right), \sqrt{\left|\sin^2\left(\frac{\theta}{2}\right) - \cos^2\left(\frac{\theta}{2}\right)\right|}\right)}{\pi} + \frac{1}{2}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{703} = e^{i\left(\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\theta}{2}\right), \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} - \frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|, -\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right) \le 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|, -\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4} - \operatorname{atan_2}\left(0, \sqrt{\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)}\right) - 2\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}}, \begin{cases} 1 & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases} \cdot \sqrt{\frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2} + \frac{1}{2} + \sqrt{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}}\right) + \operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}{2}}, \begin{cases} 1 & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases} \cdot \sqrt{\left|\sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right)\right|}\right)}$$

$$\eta_{704} = \pi - 2\operatorname{atan_2}\left(0, \sqrt{\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)}\right)$$

$$\eta_{705} = -\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\theta}{2}\right), \cos\left(\frac{\theta}{2}\right)\right) & \text{otherwise} \end{cases} - \pi$$

$$\eta_{706} = -\frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|, -\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right) & \text{for } \sin^2\left(\frac{\theta}{2}\right)-\cos^2\left(\frac{\theta}{2}\right) \le 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\theta}{2}\right)\right|\left|\cos\left(\frac{\theta}{2}\right)\right|, -\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4}$$

$$\eta_{707} = -\pi + \operatorname{atan}\left(\frac{\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{708} = -\frac{\pi}{4} - \operatorname{atan}\left(3-2\sqrt{2}\right)$$

$$\eta_{709} = -\operatorname{atan}\left(3-2\sqrt{2}\right) + \frac{3\pi}{4}$$

$$\eta_{710} = -\frac{3\pi}{2} + \operatorname{atan}\left(\frac{\sqrt{2}}{2}\right)$$

$$\eta_{711} = \operatorname{atan}\left(\frac{\sqrt{2}}{2}\right) + \frac{\pi}{2}$$

$$\eta_{712} = \operatorname{atan}\left(3-2\sqrt{2}\right) + \frac{\pi}{4}$$

$$\eta_{713} = -\frac{3\pi}{4} + \operatorname{atan}\left(3-2\sqrt{2}\right)$$

$$\eta_{714} = -\pi - \operatorname{atan}\left(\frac{\sqrt{2}}{2}\right)$$

$$\eta_{715} = \operatorname{atan}\left(3-2\sqrt{2}\right) + \frac{5\pi}{4}$$

$$\eta_{716} = -\operatorname{atan}\left(\frac{\sqrt{2}}{2}\right) + \frac{3\pi}{2}$$

$$\eta_{717} = \frac{(-1)^{\frac{3}{4}}\sqrt{2}(1-i)}{2}$$

$$\eta_{718} = -\operatorname{acos}\left(\frac{\sqrt{3}}{3}\right) + \frac{3\pi}{2}$$

$$\eta_{719} = e^{-i\left(-\frac{\pi}{2} + \operatorname{acos}\left(\frac{\sqrt{3}}{3}\right)\right)}$$

$$\eta_{720} = \frac{-\frac{9\pi}{4} + 6\operatorname{atan}\left(\frac{\sqrt{2}-1}{\sqrt{2-\sqrt{2}}}\right)}{\pi}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{721} = \frac{-3\,\mathrm{atan}\left(\frac{\frac{1}{2}\sqrt{2}}{\sqrt{2-\sqrt{2}}}\right)+\frac{7\pi}{8}}{\pi}+\frac{1}{2}$$

$$\eta_{722} = \frac{-2\,\mathrm{atan}\left(\frac{\frac{1}{2}\sqrt{2}}{2\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}\right)+\frac{\pi}{2}}{\pi}$$

$$\eta_{723} = e^{i\left(-\frac{3\pi}{8}+3\,\mathrm{atan}\left(\frac{\frac{1}{2}\sqrt{2}}{\sqrt{2-\sqrt{2}}}\right)\right)}$$

$$\eta_{724} = \frac{(-1)^{\frac{3}{8}}\sqrt{2}(1+i)}{2}$$

$$\eta_{725} = -2\,\mathrm{atan}\left(\frac{\frac{1}{2}\sqrt{2}}{2\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}\right)+\frac{\pi}{2}$$

$$\eta_{726} = \mathrm{acos}\left(\frac{\sqrt{3}}{3}\right)+\frac{\pi}{2}$$

$$\eta_{727} = e^{i\left(-\frac{\pi}{2}+\mathrm{acos}\left(\frac{\sqrt{3}}{3}\right)\right)}$$

$$\eta_{728} = \frac{-6\,\mathrm{atan}\left(\frac{\frac{1}{2}\sqrt{2}}{\sqrt{2-\sqrt{2}}}\right)+\frac{9\pi}{8}}{\pi}$$

$$\eta_{729} = \frac{-\frac{11\pi}{8}+3\,\mathrm{atan}\left(\frac{\frac{1}{2}\sqrt{2}}{\sqrt{2-\sqrt{2}}}\right)}{\pi}+\frac{1}{2}$$

$$\eta_{730} = \frac{-\frac{\pi}{2}+2\,\mathrm{atan}\left(\frac{\frac{1}{2}\sqrt{2}}{2\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}\right)}{\pi}$$

$$\eta_{731} = \frac{-\pi+2\,\mathrm{atan}\left(\frac{\frac{1}{2}\sqrt{2}}{2\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}\right)}{\pi}$$

$$\eta_{732} = e^{i\left(-3\,\mathrm{atan}\left(\frac{\frac{1}{2}\sqrt{2}}{\sqrt{2-\sqrt{2}}}\right)+\frac{3\pi}{8}\right)}$$

$$\eta_{733} = -\frac{\pi}{2}+2\,\mathrm{atan}\left(\frac{\frac{1}{2}\sqrt{2}}{2\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}\right)$$

$$\eta_{734} = -\pi+2\,\mathrm{atan}\left(\frac{\frac{1}{2}\sqrt{2}}{2\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}\right)$$

$$\eta_{735} = \frac{\sqrt{2}i(1+i)^3 e^{\frac{i\pi\left(i\left(4g\mathrm{lobal}_{\mathrm{shift}}+1\right)+1\right)}{4}}}{4}$$

$$\eta_{736} = \frac{\pi t\left(4g\mathrm{lobal}_{\mathrm{shift}}+1\right)}{4}+\begin{cases}0 & \text{for } \sqrt{2\sin^2\left(\frac{\pi t}{4}+\frac{\pi\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}{2}\right)\cos^2\left(\frac{\pi t}{4}+\frac{\pi\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)}=0\\ \mathrm{atan}_2\left(\left|\cos\left(\frac{\pi t}{2}\right)\right|,-\sqrt{2}\sin\left(\frac{\pi t}{4}+\frac{\pi\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}{2}\right)\cos\left(\frac{\pi t}{4}+\frac{\pi\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}{2}\right)\right) & \text{otherwise}\end{cases}-\left(\frac{3\pi\left(\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor \bmod 2\right)}{2}\right)+\pi\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor+\pi$$

$$\eta_{737} = \frac{\pi t\left(4g\mathrm{lobal}_{\mathrm{shift}}+1\right)}{4}+\frac{\begin{cases}0 & \text{for } (-1)^{\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}\cos\left(\frac{\pi t}{2}\right)=0\\ \mathrm{atan}_2\left(-\left|\cos\left(\frac{\pi t}{2}\right)\right|,0\right) & \text{otherwise}\end{cases}}{2}+\begin{cases}0 & \text{for } \sqrt{2\sin^2\left(\frac{\pi t}{4}+\frac{\pi\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}{2}\right)\cos^2\left(\frac{\pi t}{4}+\frac{\pi\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)}=0\\ \mathrm{atan}_2\left(\left|\cos\left(\frac{\pi t}{2}\right)\right|,-\sqrt{2}\sin\left(\frac{\pi t}{4}+\frac{\pi\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}{2}\right)\cos\left(\frac{\pi t}{4}+\frac{\pi\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}{2}\right)\right) & \text{otherwise}\end{cases}-\left(\frac{3\pi\left(\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor \bmod 2\right)}{2}\right)+\frac{\mathrm{atan}_2\left(-\left|\cos\left(\frac{\pi t}{2}\right)\right|,-(-1)^{\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}\sin\left(\frac{\pi t}{2}\right)\right)}{2}+\pi\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor+3\pi$$

$$\eta_{738} = \frac{\begin{cases}0 & \text{for } (-1)^{\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}\cos\left(\frac{\pi t}{2}\right)=0\\ \mathrm{atan}_2\left(-\left|\cos\left(\frac{\pi t}{2}\right)\right|,0\right) & \text{otherwise}\end{cases}}{2}-\frac{\mathrm{atan}_2\left(-\left|\cos\left(\frac{\pi t}{2}\right)\right|,-(-1)^{\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}\sin\left(\frac{\pi t}{2}\right)\right)}{2}+\pi$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{739} = -2\,\mathrm{atan}_2\left(-\sqrt{2}\sin\left(\tfrac{\pi t}{4}+\tfrac{\pi\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}{2}\right)\cos\left(\tfrac{\pi t}{4}+\tfrac{\pi\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}{2}\right),\sqrt{2\sin^2\left(\tfrac{\pi t}{4}+\tfrac{\pi\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}{2}\right)\cos^2\left(\tfrac{\pi t}{4}+\tfrac{\pi\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}{2}\right)+\cos^2\left(\tfrac{\pi t}{2}\right)}\right)$$

$$\eta_{740} = -\begin{cases} 0 & \text{for } \sqrt{2\sin^2\left(\tfrac{\pi t}{4}+\tfrac{\pi\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}{2}\right)\cos^2\left(\tfrac{\pi t}{4}+\tfrac{\pi\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}{2}\right)+\cos^2\left(\tfrac{\pi t}{2}\right)} = 0 \\ \mathrm{atan}_2\left(\left|\cos\left(\tfrac{\pi t}{2}\right)\right|,-\sqrt{2}\sin\left(\tfrac{\pi t}{4}+\tfrac{\pi\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}{2}\right)\cos\left(\tfrac{\pi t}{4}+\tfrac{\pi\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{741} = \left(-\tfrac{\pi\left(1-\left(2\left(\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor \bmod 2\right)\right)\right)}{2}\right) + \left(\pi\left(\left\lfloor\tfrac{1}{2}-\tfrac{t}{2}\right\rfloor \bmod 2\right)\right)$$

$$\eta_{742} = 2\pi\left(\left\lfloor\tfrac{1}{2}-\tfrac{t}{2}\right\rfloor \bmod 2\right)$$

$$\eta_{743} = -\frac{\begin{cases} 0 & \text{for } (-1)^{\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}\cos\left(\tfrac{\pi t}{2}\right) = 0 \\ \mathrm{atan}_2\left(-\left|\cos\left(\tfrac{\pi t}{2}\right)\right|,0\right) & \text{otherwise} \end{cases}}{2} + \left(\pi\left(\left\lfloor\tfrac{1}{2}-\tfrac{t}{2}\right\rfloor \bmod 2\right)\right) - \frac{\mathrm{atan}_2\left(-\left|\cos\left(\frac{\pi t}{2}\right)\right|,-(-1)^{\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}\sin\left(\frac{\pi t}{2}\right)\right)}{2} - 2\pi$$

$$\eta_{744} = \left(\left(1-\left(2\left(\left\lfloor\tfrac{1}{2}-\tfrac{t}{2}\right\rfloor \bmod 2\right)\right)\right)\left(-\frac{\begin{cases} 0 & \text{for } (-1)^{\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}\cos\left(\tfrac{\pi t}{2}\right) = 0 \\ \mathrm{atan}_2\left(-\left|\cos\left(\tfrac{\pi t}{2}\right)\right|,0\right) & \text{otherwise} \end{cases}}{2} + \frac{\mathrm{atan}_2\left(-\left|\cos\left(\frac{\pi t}{2}\right)\right|,-(-1)^{\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}\sin\left(\frac{\pi t}{2}\right)\right)}{2} + \pi\right)\right) + \left(\pi\left(\left\lfloor\tfrac{1}{2}-\tfrac{t}{2}\right\rfloor \bmod 2\right)\right)$$

$$\eta_{745} = \left(-\tfrac{\pi\left(1-\left(2\left(\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor \bmod 2\right)\right)\right)}{2}\right) + \left(2\pi\left(\left\lfloor\tfrac{1}{2}-\tfrac{t}{2}\right\rfloor \bmod 2\right)\right)$$

$$\eta_{746} = \left(\pi\left(\left\lfloor\tfrac{1}{2}-\tfrac{t}{2}\right\rfloor \bmod 2\right)\right) + \tfrac{\pi}{2}$$

$$\eta_{747} = -\frac{\begin{cases} 0 & \text{for } (-1)^{\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}\cos\left(\tfrac{\pi t}{2}\right) = 0 \\ \mathrm{atan}_2\left(-\left|\cos\left(\tfrac{\pi t}{2}\right)\right|,0\right) & \text{otherwise} \end{cases}}{2} - \frac{\mathrm{atan}_2\left(-\left|\cos\left(\frac{\pi t}{2}\right)\right|,-(-1)^{\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}\sin\left(\frac{\pi t}{2}\right)\right)}{2} - \pi$$

$$\eta_{748} = e^{i\left(\frac{\pi t\left(4\mathrm{global}_{\mathrm{shift}}+1\right)}{4} + \frac{\begin{cases} 0 & \text{for } (-1)^{\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}\cos\left(\tfrac{\pi t}{2}\right) = 0 \\ \mathrm{atan}_2\left(-\left|\cos\left(\tfrac{\pi t}{2}\right)\right|,0\right) & \text{otherwise} \end{cases}}{2} + \begin{cases} 0 & \text{for } \sqrt{2\sin^2\left(\tfrac{\pi t}{4}+\tfrac{\pi\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}{2}\right)\cos^2\left(\tfrac{\pi t}{4}+\tfrac{\pi\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}{2}\right)+\cos^2\left(\tfrac{\pi t}{2}\right)} = 0 \\ \mathrm{atan}_2\left(\left|\cos\left(\tfrac{\pi t}{2}\right)\right|,-\sqrt{2}\sin\left(\tfrac{\pi t}{4}+\tfrac{\pi\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}{2}\right)\cos\left(\tfrac{\pi t}{4}+\tfrac{\pi\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases} - \left(\frac{2\pi\left(\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor \bmod 2\right)}{2}\right) + \frac{\mathrm{atan}_2\left(-\left|\cos\left(\frac{\pi t}{2}\right)\right|,-(-1)^{\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor}\sin\left(\frac{\pi t}{2}\right)\right)}{2} + \pi\left\lfloor\tfrac{1}{2}-\tfrac{t}{2}\right\rfloor + 2\pi\right)}$$

$$\eta_{749} = -e^{\frac{i\pi\left(t\left(4\mathrm{global}_{\mathrm{shift}}+1\right)+1\right)}{4}}$$

$$\eta_{750} = -\frac{t\left(2\mathrm{global}_{\mathrm{shift}}-1\right)}{2} - \frac{3}{4}$$

$$\eta_{751} = \frac{\sqrt{2}(-1+i)e^{i\pi t\left(\mathrm{global}_{\mathrm{shift}}-\frac{1}{2}\right)}}{2}$$

$$\eta_{752} = \frac{-\frac{\pi t\left(2\mathrm{global}_{\mathrm{shift}}-1\right)}{2}-\pi}{\pi}$$

$$\eta_{753} = e^{\frac{i\pi\left(t\left(4\mathrm{global}_{\mathrm{shift}}+1\right)-3t-2\right)}{4}}$$

$$\eta_{754} = -ie^{\frac{i\pi\left(t\left(4\mathrm{global}_{\mathrm{shift}}+1\right)+1\right)}{4}}$$

$$\eta_{755} = \frac{\sqrt{2}(1+i)^2 e^{\frac{i\pi\left(t\left(4\mathrm{global}_{\mathrm{shift}}+1\right)+1\right)}{4}}}{4}$$

$$\eta_{756} = \frac{\sqrt{2}i(1+i)e^{i\pi\mathrm{global}_{\mathrm{shift}}t}}{2}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{757} = \frac{\sqrt[8]{-1}\sqrt{2}(-1+i)e^{ir_{global_shift}t}}{2}$$

$$\eta_{758} = \frac{\sqrt[8]{-1}\sqrt{2}(1-i)e^{ir_{global_shift}t}}{2}$$

$$\eta_{759} = \frac{(-1)^{\frac{3}{8}}\sqrt{2}(1-i)}{2}$$

$$\eta_{760} = -\pi + \operatorname{atan}\left(\frac{-\sqrt{3\sqrt{2}+6}+\sqrt{2-\sqrt{2}}+\sqrt{6-3\sqrt{2}}+\sqrt{\sqrt{2}+2}}{-\sqrt{2-\sqrt{2}}+\sqrt{3}\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}+\sqrt{3}\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{761} = \operatorname{atan}\left(\frac{-\sqrt{3\sqrt{2}+6}+\sqrt{2-\sqrt{2}}+\sqrt{6-3\sqrt{2}}+\sqrt{\sqrt{2}+2}}{-\sqrt{2-\sqrt{2}}+\sqrt{3}\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}+\sqrt{3}\sqrt{\sqrt{2}+2}}\right) + \pi$$

$$\eta_{762} = \pi - 2\operatorname{atan}\left(\frac{-\sqrt{3}\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}+\sqrt{3}\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}{-\sqrt{2-\sqrt{2}}+\sqrt{3}\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}+\sqrt{3}\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{763} = \frac{\operatorname{atan}\left(\frac{2\sqrt{\sqrt{2}+\sqrt{6}}}{-\sqrt{\sqrt{2}+2}+\sqrt{3}\sqrt{2-\sqrt{2}}}\right)+\operatorname{atan}\left(\frac{2\sqrt{-\sqrt{2}+\sqrt{6}}}{-\sqrt{2-\sqrt{2}}+\sqrt{3}\sqrt{\sqrt{2}+2}}\right)+\frac{7\pi}{12}-2\operatorname{atan}\left(\frac{2\sqrt[8]{3}\sqrt{1+\sqrt{3}}}{-\sqrt{\sqrt{2}+2}+\sqrt{3}\sqrt{2-\sqrt{2}}}\right)}{\pi}$$

$$\eta_{764} = \frac{-\frac{13\pi}{24}+\operatorname{atan}\left(\frac{2\sqrt[8]{3}\sqrt{1+\sqrt{3}}}{-\sqrt{\sqrt{2}+2}+\sqrt{3}\sqrt{2-\sqrt{2}}}\right)-\frac{\operatorname{atan}\left(\frac{2\sqrt{-\sqrt{2}+\sqrt{6}}}{-\sqrt{2-\sqrt{2}}+\sqrt{3}\sqrt{\sqrt{2}+2}}\right)}{2}-\frac{\operatorname{atan}\left(\frac{3\sqrt{\sqrt{2}+\sqrt{6}}}{-\sqrt{\sqrt{2}+2}+\sqrt{3}\sqrt{2-\sqrt{2}}}\right)}{2}}{\pi}+\frac{1}{2}$$

$$\eta_{765} = \frac{-\frac{\pi}{2}-\frac{\operatorname{atan}\left(\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\frac{\sqrt{3}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{2}}\right)}{2}-\frac{\operatorname{atan}\left(\frac{\sqrt{-\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{\sqrt{3}\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{2}}\right)}{2}}{\pi}$$

$$\eta_{766} = \frac{1}{2}-\frac{-\frac{\pi}{2}-\frac{\operatorname{atan}\left(\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\frac{\sqrt{3}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{2}}\right)}{2}-\frac{\operatorname{atan}\left(\frac{\sqrt{-\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{\sqrt{3}\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{2}}\right)}{2}}{\pi}$$

$$\eta_{767} = \frac{-\frac{\pi}{2}-\frac{\operatorname{atan}\left(\frac{\sqrt{-\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{\sqrt{3}\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{2}}\right)}{2}+\frac{\operatorname{atan}\left(\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\frac{\sqrt{3}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{2}}\right)}{2}}{\pi}$$

$$\eta_{768} = \frac{-\frac{\operatorname{atan}\left(\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\frac{\sqrt{3}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{2}}\right)}{2}+\frac{\operatorname{atan}\left(\frac{\sqrt{-\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{\sqrt{3}\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{2}}\right)}{2}+\frac{\pi}{2}}{\pi}+\frac{1}{2}$$

$$\eta_{769} = \frac{-\operatorname{atan}\left(\frac{\sqrt{2}\sqrt{\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\sqrt{3}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}+\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)-\operatorname{atan}\left(\frac{\sqrt{3}\sqrt{-\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}+\sqrt{3}\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)}{\pi}$$

$$\eta_{770} = \frac{-\operatorname{atan}\left(\frac{\sqrt{2}\sqrt{\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\sqrt{3}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}+\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)+\operatorname{atan}\left(\frac{\sqrt{3}\sqrt{-\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}+\sqrt{3}\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)}{\pi}$$

$$\eta_{771} = \frac{\frac{\operatorname{atan}\left(\frac{\sqrt{-\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{\sqrt{3}\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{2}}\right)}{2}+\frac{\operatorname{atan}\left(\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\frac{\sqrt{3}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{2}}\right)}{2}}{\pi}$$

$$\eta_{772} = \frac{-\frac{\operatorname{atan}\left(\frac{\sqrt{-\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{\sqrt{3}\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{2}}\right)}{2}+\frac{\operatorname{atan}\left(\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\frac{\sqrt{3}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{2}}\right)}{2}}{\pi}$$

$$\eta_{773} = e^{i\left(\frac{\operatorname{atan}\left(\frac{2\sqrt{\sqrt{2}+\sqrt{6}}}{-\sqrt{\sqrt{2}+2}+\sqrt{3}\sqrt{2-\sqrt{2}}}\right)}{2}+\frac{\pi}{24}+\frac{\operatorname{atan}\left(\frac{2\sqrt{-\sqrt{2}+\sqrt{6}}}{-\sqrt{2-\sqrt{2}}+\sqrt{3}\sqrt{\sqrt{2}+2}}\right)}{2}-\operatorname{atan}\left(\frac{2\sqrt[8]{3}\sqrt{1+\sqrt{3}}}{-\sqrt{\sqrt{2}+2}+\sqrt{3}\sqrt{2-\sqrt{2}}}\right)\right)}$$

$$\eta_{774} = \frac{(-1)^{\frac{5}{8}}\sqrt{2}(1+i)}{2}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{775} = \frac{\operatorname{atan}\left(\frac{2\sqrt{-\sqrt{2}+\sqrt{6}}}{-\sqrt{2}-\sqrt{2}+\sqrt{3}\sqrt{\sqrt{2}+2}}\right)}{2} + \frac{13\pi}{24}$$

$$\eta_{776} = \frac{\operatorname{atan}\left(\frac{2\sqrt{-\sqrt{2}+\sqrt{6}}}{-\sqrt{2}-\sqrt{2}+\sqrt{3}\sqrt{\sqrt{2}+2}}\right)}{2} + \frac{25\pi}{24}$$

$$\eta_{777} = -\frac{\pi}{2} - \frac{\operatorname{atan}\left(\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\frac{\sqrt{3}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{2}}\right)}{2} - \frac{\operatorname{atan}\left(\frac{\sqrt{-\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{\sqrt{3}\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{2}}\right)}{2}$$

$$\eta_{778} = -\frac{\pi}{2} - \frac{\operatorname{atan}\left(\frac{\sqrt{-\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{\sqrt{3}\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{2}}\right)}{2} + \frac{\operatorname{atan}\left(\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\frac{\sqrt{3}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{2}}\right)}{2}$$

$$\eta_{779} = -\operatorname{atan}\left(\frac{\sqrt{2}\sqrt{\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\sqrt{3}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}+\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right) - \operatorname{atan}\left(\frac{\sqrt{2}\sqrt{-\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}+\sqrt{3}\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)$$

$$\eta_{780} = -\operatorname{atan}\left(\frac{\sqrt{2}\sqrt{\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\sqrt{3}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}+\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right) + \operatorname{atan}\left(\frac{\sqrt{2}\sqrt{-\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}+\sqrt{3}\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)$$

$$\eta_{781} = \frac{\operatorname{atan}\left(\frac{\sqrt{-\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{\sqrt{3}\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{2}}\right)}{2} + \frac{\operatorname{atan}\left(\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\frac{\sqrt{3}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{2}}\right)}{2}$$

$$\eta_{782} = -\frac{\operatorname{atan}\left(\frac{\sqrt{-\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{\sqrt{3}\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{2}}\right)}{2} + \frac{\operatorname{atan}\left(\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\frac{\sqrt{3}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{2}}\right)}{2}$$

$$\eta_{783} = e^{i\left(\frac{\pi}{24}+\frac{\operatorname{atan}\left(\frac{2\sqrt{-\sqrt{2}+\sqrt{6}}}{-\sqrt{2}-\sqrt{2}+\sqrt{3}\sqrt{\sqrt{2}+2}}\right)}{2}\right)}$$

$$\eta_{784} = \operatorname{atan}\left(\frac{-\sqrt{3}\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{3}\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{785} = \operatorname{atan}\left(\frac{-\sqrt{3}\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{3}\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right) + \pi$$

$$\eta_{786} = \frac{\pi\left(\left\lfloor\frac{1}{2}-b\right\rfloor+\left\lfloor\frac{1}{2}-c\right\rfloor\right)}{2} + \begin{cases} 0 & \text{for } \sqrt{(-1)^{\left\lfloor\frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor\frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c) + 2\sin^2\left(\frac{\pi b}{2}+\frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2}\right)\cos^2\left(\frac{\pi c}{2}+\frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)} = 0 \\ \operatorname{atan}_2\left(\sqrt{(-1)^{\left\lfloor\frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor\frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)}, -\sqrt{2}\sin\left(\frac{\pi b}{2}+\frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2}\right)\cos\left(\frac{\pi c}{2}+\frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases} - \left(\frac{3\pi\left(\left\lfloor\frac{1}{2}-b\right\rfloor \bmod 2\right)}{2}\right) + \pi$$

$$\eta_{787} = \frac{\pi\left(\left\lfloor\frac{1}{2}-b\right\rfloor+\left\lfloor\frac{1}{2}-c\right\rfloor\right)}{2} + \begin{cases} 0 & \text{for } \sqrt{(-1)^{\left\lfloor\frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor\frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c) + 2\sin^2\left(\frac{\pi b}{2}+\frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2}\right)\cos^2\left(\frac{\pi c}{2}+\frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)} = 0 \\ \operatorname{atan}_2\left(\sqrt{(-1)^{\left\lfloor\frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor\frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)}, -\sqrt{2}\sin\left(\frac{\pi b}{2}+\frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2}\right)\cos\left(\frac{\pi c}{2}+\frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases} + \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\pi b}{2}-\frac{\pi c}{2}+\frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2}-\frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right) = 0 \\ \operatorname{atan}_2\left(-\sqrt{(-1)^{\left\lfloor\frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor\frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)}, -\sin\left(\frac{\pi b}{2}+\frac{\pi c}{2}+\frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2}+\frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\pi b}{2}+\frac{\pi c}{2}+\frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2}+\frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right) = 0 \\ \operatorname{atan}_2\left(-\sqrt{(-1)^{\left\lfloor\frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor\frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)}, -\sin\left(\frac{\pi b}{2}-\frac{\pi c}{2}+\frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2}-\frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \left(\frac{3\pi\left(\left\lfloor\frac{1}{2}-b\right\rfloor \bmod 2\right)}{2}\right) + 3\pi$$

$$\eta_{788} = -\frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\pi b}{2}-\frac{\pi c}{2}+\frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2}-\frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right) = 0 \\ \operatorname{atan}_2\left(-\sqrt{(-1)^{\left\lfloor\frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor\frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)}, -\sin\left(\frac{\pi b}{2}+\frac{\pi c}{2}+\frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2}+\frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\pi b}{2}+\frac{\pi c}{2}+\frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2}+\frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right) = 0 \\ \operatorname{atan}_2\left(-\sqrt{(-1)^{\left\lfloor\frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor\frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)}, -\sin\left(\frac{\pi b}{2}-\frac{\pi c}{2}+\frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2}-\frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \pi$$

$$\eta_{789} = -2\operatorname{atan}_2\left(-\sqrt{2}\sin\left(\frac{\pi c}{2}+\frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)\cos\left(\frac{\pi b}{2}+\frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2}\right), \sqrt{(-1)^{\left\lfloor\frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor\frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c) + 2\sin^2\left(\frac{\pi b}{2}+\frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2}\right)\cos^2\left(\frac{\pi c}{2}+\frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)}\right)$$

$$\eta_{790} = -\begin{cases} 0 & \text{for } \sqrt{(-1)^{\left\lfloor\frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor\frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c) + 2\sin^2\left(\frac{\pi b}{2}+\frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2}\right)\cos^2\left(\frac{\pi c}{2}+\frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)} = 0 \\ \operatorname{atan}_2\left(\sqrt{(-1)^{\left\lfloor\frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor\frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)}, -\sqrt{2}\sin\left(\frac{\pi b}{2}+\frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2}\right)\cos\left(\frac{\pi c}{2}+\frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{791} = \left(-\frac{\pi\left(1-\left(2\left(\left\lfloor\frac{1}{2}-b\right\rfloor \bmod 2\right)\right)\right)}{2}\right) + \left(\pi\left(\left\lfloor\frac{1}{2}-b\right\rfloor \bmod 2\right)\right)$$

$$\eta_{792} = \left(\pi\left(\left\lfloor\frac{1}{2}-b\right\rfloor \bmod 2\right)\right) + \left(\pi\left(\left\lfloor\frac{1}{2}-c\right\rfloor \bmod 2\right)\right)$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{793} = -\frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\pi b}{2} - \frac{\pi c}{2} + \frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2} - \frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor\frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor\frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)}, -\sin\left(\frac{\pi b}{2} + \frac{\pi c}{2} + \frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2} + \frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\pi b}{2} + \frac{\pi c}{2} + \frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2} + \frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor\frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor\frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)}, -\sin\left(\frac{\pi b}{2} - \frac{\pi c}{2} + \frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2} - \frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \left(\pi\left(\left\lfloor\frac{1}{2}-b\right\rfloor \bmod 2\right)\right) - 2\pi$$

$$\eta_{794} = \left(\left(1 - \left(2\left(\left\lfloor\frac{1}{2}-b\right\rfloor \bmod 2\right)\right)\right)\left(\frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\pi b}{2} - \frac{\pi c}{2} + \frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2} - \frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor\frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor\frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)}, -\sin\left(\frac{\pi b}{2} + \frac{\pi c}{2} + \frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2} + \frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\pi b}{2} + \frac{\pi c}{2} + \frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2} + \frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor\frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor\frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)}, -\sin\left(\frac{\pi b}{2} - \frac{\pi c}{2} + \frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2} - \frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \pi\right)\right) + \left(\pi\left(\left\lfloor\frac{1}{2}-b\right\rfloor \bmod 2\right)\right)$$

$$\eta_{795} = \left(-\frac{\pi\left(1-\left(2\left(\left\lfloor\frac{1}{2}-b\right\rfloor \bmod 2\right)\right)\right)}{2}\right) + \left(\pi\left(\left\lfloor\frac{1}{2}-b\right\rfloor \bmod 2\right)\right) + \left(\pi\left(\left\lfloor\frac{1}{2}-c\right\rfloor \bmod 2\right)\right)$$

$$\eta_{796} = \left(\pi\left(\left\lfloor\frac{1}{2}-b\right\rfloor \bmod 2\right)\right) + \frac{\pi}{2}$$

$$\eta_{797} = -\frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\pi b}{2} - \frac{\pi c}{2} + \frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2} - \frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor\frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor\frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)}, -\sin\left(\frac{\pi b}{2} + \frac{\pi c}{2} + \frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2} + \frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\pi b}{2} + \frac{\pi c}{2} + \frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2} + \frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor\frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor\frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)}, -\sin\left(\frac{\pi b}{2} - \frac{\pi c}{2} + \frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2} - \frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \pi$$

$$\eta_{798} = e^{i\left(\frac{\pi\left(\left\lfloor\frac{1}{2}-b\right\rfloor+\left\lfloor\frac{1}{2}-c\right\rfloor\right)}{2} + \begin{cases} 0 & \text{for } \sqrt{(-1)^{\left\lfloor\frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor\frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c) + 2\sin^2\left(\frac{\pi b}{2} + \frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2}\right)\cos^2\left(\frac{\pi c}{2} + \frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\sqrt{(-1)^{\left\lfloor\frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor\frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)}, -\sqrt{2}\sin\left(\frac{\pi b}{2} + \frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2}\right)\cos\left(\frac{\pi c}{2} + \frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases} + \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\pi b}{2} - \frac{\pi c}{2} + \frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2} - \frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor\frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor\frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)}, -\sin\left(\frac{\pi b}{2} + \frac{\pi c}{2} + \frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2} + \frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\pi b}{2} + \frac{\pi c}{2} + \frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2} + \frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor\frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor\frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)}, -\sin\left(\frac{\pi b}{2} + \frac{\pi c}{2} + \frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2} + \frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \left(\frac{3\pi\left(\left\lfloor\frac{1}{2}-b\right\rfloor \bmod 2\right)}{2}\right) + 2\pi\right)}$$

$$\eta_{799} = \frac{\sqrt{2}(1-i)^2 e^{-\frac{i\pi(a+b+c+1)}{2}}}{4}$$

$$\eta_{800} = \frac{\sqrt{2}i(1+i)e^{-\frac{i\pi b}{2}}}{2}$$

$$\eta_{801} = \frac{\sqrt[4]{-1}\sqrt{2}(-\frac{1}{2}+i)e^{-\frac{i\pi c}{2}}}{2}$$

$$\eta_{802} = \frac{\sqrt[4]{-1}\sqrt{2}(1-i)e^{-\frac{i\pi c}{2}}}{2}$$

$$\eta_{803} = \frac{-4\operatorname{atan}\left(\frac{1}{\sqrt{2}+\sqrt{3}}\right) - \frac{\pi}{3}}{\pi}$$

$$\eta_{804} = \frac{-\frac{\pi}{3} + 2\operatorname{atan}\left(\frac{1}{\sqrt{2}+\sqrt{3}}\right)}{\pi} + \frac{1}{2}$$

$$\eta_{805} = \frac{-\pi + \operatorname{atan}\left(\sqrt{2}\right)}{\pi}$$

$$\eta_{806} = \frac{2\operatorname{atan}\left(\frac{1}{\sqrt{2}+\sqrt{3}}\right)}{\pi}$$

$$\eta_{807} = \frac{\pi - \operatorname{atan}\left(\sqrt{2}\right)}{\pi}$$

$$\eta_{808} = e^{i\left(-2\operatorname{atan}\left(\frac{1}{\sqrt{2}+\sqrt{3}}\right) - \frac{\pi}{6}\right)}$$

$$\eta_{809} = 2\operatorname{atan}\left(\frac{1}{\sqrt{2}+\sqrt{3}}\right)$$

$$\eta_{810} = -\frac{\alpha}{4} - \frac{\beta}{4} + \begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) + \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\alpha}{4} - \frac{\beta}{4}\right), \cos\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right) & \text{otherwise} \end{cases} - \frac{\operatorname{atan_2}\left(2\left|\sin\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right|\left|\cos\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right|, -\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) + \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right)}{2} - \pi$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{811} = -\frac{\alpha}{4} - \frac{\beta}{4} + \begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) + \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)} = 0 \\ \operatorname{atan}_2\left(-\sin\left(\frac{\alpha}{4} - \frac{\beta}{4}\right), \cos\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{812} = 2\operatorname{atan}_2\left(0, \sqrt{\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) + \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)}\right)$$

$$\eta_{813} = -\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) + \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)} = 0 \\ \operatorname{atan}_2\left(-\sin\left(\frac{\alpha}{4} - \frac{\beta}{4}\right), \cos\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{814} = -\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right| \left|\cos\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right| = 0 \\ \operatorname{atan}_2\left(0, -2\sin\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\cos\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{815} = \operatorname{atan}_2\left(2\left|\sin\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right| \left|\cos\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right|, -\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) + \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right)$$

$$\eta_{816} = \begin{cases} 0 & \text{for } \left|\sin\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right| \left|\cos\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right| = 0 \\ \operatorname{atan}_2\left(0, -2\sin\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\cos\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{817} = \frac{\alpha}{2} + \frac{\beta}{2} - \operatorname{atan}_2\left(2\left|\sin\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right| \left|\cos\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right|, -\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) + \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right)$$

$$\eta_{818} = e^{i\left(-\frac{\alpha}{2} - \frac{\beta}{4} + \begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) + \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)} = 0 \\ \operatorname{atan}_2\left(-\sin\left(\frac{\alpha}{4} - \frac{\beta}{4}\right), \cos\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right) & \text{otherwise} \end{cases}\right)}$$

$$\eta_{819} = \frac{-\frac{\alpha}{2} - \frac{\beta}{4} + 2\left(\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) + \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)} = 0 \\ \operatorname{atan}_2\left(-\sin\left(\frac{\alpha}{4} - \frac{\beta}{4}\right), \cos\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right) & \text{otherwise} \end{cases}\right) - \begin{cases} 2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right| \left|\cos\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right|, -\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) + \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right) & \text{for } \sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) - \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) \le 0 \\ 2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right| \left|\cos\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right|, -\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) + \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right) - 2\pi & \text{otherwise} \end{cases} - 2\operatorname{atan}_2\left(0, \sqrt{\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) + \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)}\right) - 4\operatorname{atan}_2\left(\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) - \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right|}{2}}, \left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) - \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\frac{\left|\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) - \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right|}{2} + \frac{1}{2}} + \sqrt{\left|\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) - \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right|}\right) + 2\operatorname{atan}_2\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) - \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) - \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\left|\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) - \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right|}\right) - 2\pi}{\pi}$$

$$\eta_{820} = \frac{\frac{\alpha}{4} + \frac{\beta}{4} - \begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) + \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)} = 0 \\ \operatorname{atan}_2\left(-\sin\left(\frac{\alpha}{4} - \frac{\beta}{4}\right), \cos\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right) & \text{otherwise} \end{cases} + \frac{\begin{cases} 2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right| \left|\cos\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right|, -\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) + \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right) & \text{for } \sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) - \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) \le 0 \\ 2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right| \left|\cos\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right|, -\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) + \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4} + \operatorname{atan}_2\left(0, \sqrt{\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) + \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)}\right) + 2\operatorname{atan}_2\left(\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) - \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) - \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\frac{\left|\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) - \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right|}{2} + \frac{1}{2}} + \sqrt{\left|\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) - \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right|}\right) - \operatorname{atan}_2\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) - \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) - \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\left|\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) - \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right|}\right)}{\pi} + \frac{1}{2}$$

$$\eta_{821} = \frac{2\operatorname{atan}_2\left(0, \sqrt{\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) + \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)}\right)}{\pi}$$

$$\eta_{822} = -\frac{2\left(\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) + \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)} = 0 \\ \operatorname{atan}_2\left(-\sin\left(\frac{\alpha}{4} - \frac{\beta}{4}\right), \cos\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right) & \text{otherwise} \end{cases}\right)}{\pi}$$

$$\eta_{823} = \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) + \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)} = 0 \\ \operatorname{atan}_2\left(-\sin\left(\frac{\alpha}{4} - \frac{\beta}{4}\right), \cos\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}$$

$$\eta_{824} = -\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right| \left|\cos\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right| = 0 \\ \operatorname{atan}_2\left(0, -2\sin\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\cos\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{825} = \frac{2\operatorname{atan}_2\left(\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) - \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) - \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\right) \cdot \sqrt{\frac{\left|\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) - \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right|}{2} + \frac{1}{2}} + \sqrt{\left|\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) - \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right|}\right)}{\pi}$$

$$\eta_{826} = \frac{\begin{cases} 2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right| \left|\cos\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right|, -\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) + \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right) & \text{for } \sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) - \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) \le 0 \\ 2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right| \left|\cos\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right|, -\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) + \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{\frac{2}{\pi}} + \pi$$

$$\eta_{827} = \frac{-\frac{\begin{cases} 2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right| \left|\cos\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right|, -\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) + \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right) & \text{for } \sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) - \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) \le 0 \\ 2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right| \left|\cos\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right|, -\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) + \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{\frac{2}{\pi}} - \frac{\pi}{2}}{2} + \frac{1}{2}$$

$$\eta_{828} = \frac{\begin{cases} 2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right| \left|\cos\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right|, -\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) + \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right) & \text{for } \sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) - \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) \le 0 \\ 2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right| \left|\cos\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right|, -\sin^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right) + \cos^2\left(\frac{\alpha}{4} - \frac{\beta}{4}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2\pi}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{829} = -\frac{2\,\mathrm{atan}_2\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}{2}}\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right),\sqrt{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}\right)}{\pi}$$

$$\eta_{830} = \frac{\begin{cases}0 & \text{for } \left|\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|\left|\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|=0\\ \mathrm{atan}_2\left(0,-2\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right) & \text{otherwise}\end{cases}}{\pi}$$

$$\eta_{831} = \frac{\frac{\alpha}{2}+\frac{\beta}{2}-\mathrm{atan}_2\left(2\left|\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|\left|\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|,-\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right)}{\pi}$$

$$\eta_{832} = \frac{-\frac{\alpha}{2}-\frac{\beta}{2}+\mathrm{atan}_2\left(2\left|\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|\left|\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|,-\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right)}{\pi}+\frac{1}{2}$$

$$\eta_{833} = e^{i\left(-\frac{\alpha}{2}-\frac{\beta}{2}+\begin{cases}0 & \text{for } \sqrt{\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)}=0\\ \mathrm{atan}_2\left(-\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right),\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right) & \text{otherwise}\end{cases}-\frac{\begin{cases}2\,\mathrm{atan}_2\left(2\left|\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|\left|\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|,-\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right) & \text{for } \sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\le 0\\ 2\,\mathrm{atan}_2\left(2\left|\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|\left|\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|,-\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right)-2\pi & \text{otherwise}\end{cases}}{4}\right)}$$

$$\eta_{834} = -\frac{\alpha}{4}-\frac{\beta}{4}+\begin{cases}0 & \text{for } \sqrt{\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)}=0\\ \mathrm{atan}_2\left(-\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right),\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right) & \text{otherwise}\end{cases}-\frac{\begin{cases}2\,\mathrm{atan}_2\left(2\left|\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|\left|\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|,-\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right) & \text{for } \sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\le 0\\ 2\,\mathrm{atan}_2\left(2\left|\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|\left|\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|,-\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right)-2\pi & \text{otherwise}\end{cases}}{4}-\pi$$

$$\eta_{835} = -\frac{\alpha}{4}-\frac{\beta}{4}+\begin{cases}0 & \text{for } \sqrt{\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)}=0\\ \mathrm{atan}_2\left(-\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right),\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right) & \text{otherwise}\end{cases}-\frac{\begin{cases}2\,\mathrm{atan}_2\left(2\left|\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|\left|\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|,-\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right) & \text{for } \sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\le 0\\ 2\,\mathrm{atan}_2\left(2\left|\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|\left|\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|,-\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right)-2\pi & \text{otherwise}\end{cases}}{4}+\pi$$

$$\eta_{836} = 2\,\mathrm{atan}_2\left(\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}{2}}\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right),\sqrt{\frac{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}{2}}+\frac{1}{2}+\sqrt{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}\right)$$

$$\eta_{837} = \frac{\begin{cases}2\,\mathrm{atan}_2\left(2\left|\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|\left|\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|,-\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right) & \text{for } \sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\le 0\\ 2\,\mathrm{atan}_2\left(2\left|\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|\left|\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|,-\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right)-2\pi & \text{otherwise}\end{cases}}{2}+\frac{\pi}{2}$$

$$\eta_{838} = -2\,\mathrm{atan}_2\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}{2}}\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right),\sqrt{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}\right)$$

$$\eta_{839} = e^{i\left(-\frac{\alpha}{2}-\frac{\beta}{2}+\begin{cases}0 & \text{for } \sqrt{\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)}=0\\ \mathrm{atan}_2\left(-\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right),\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right) & \text{otherwise}\end{cases}-\frac{\begin{cases}2\,\mathrm{atan}_2\left(2\left|\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|\left|\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|,-\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right) & \text{for } \sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\le 0\\ 2\,\mathrm{atan}_2\left(2\left|\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|\left|\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|,-\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right)-2\pi & \text{otherwise}\end{cases}}{4}\right)}$$

$$\eta_{840} = \frac{-\frac{\alpha}{2}-\frac{\beta}{2}+2\left(\begin{cases}0 & \text{for } \sqrt{\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)}=0\\ \mathrm{atan}_2\left(-\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right),\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right) & \text{otherwise}\end{cases}-\frac{\begin{cases}2\,\mathrm{atan}_2\left(2\left|\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|\left|\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|,-\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right) & \text{for } \sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\le 0\\ 2\,\mathrm{atan}_2\left(2\left|\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|\left|\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|,-\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right)-2\pi & \text{otherwise}\end{cases}}{2}-2\,\mathrm{atan}_2\left(0,\sqrt{\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)}\right)-4\,\mathrm{atan}_2\left(\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}{2}}\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right),\sqrt{\frac{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}{2}}+\frac{1}{2}+\sqrt{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}\right)+2\,\mathrm{atan}_2\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}{2}}\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right),\sqrt{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}\right)-\pi}{\pi}$$

$$\eta_{841} = \frac{\frac{\alpha}{4}+\frac{\beta}{4}-\begin{cases}0 & \text{for } \sqrt{\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)}=0\\ \mathrm{atan}_2\left(-\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right),\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right) & \text{otherwise}\end{cases}+\frac{\begin{cases}2\,\mathrm{atan}_2\left(2\left|\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|\left|\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|,-\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right) & \text{for } \sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\le 0\\ 2\,\mathrm{atan}_2\left(2\left|\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|\left|\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|,-\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right)-2\pi & \text{otherwise}\end{cases}}{4}+\mathrm{atan}_2\left(0,\sqrt{\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)}\right)+2\,\mathrm{atan}_2\left(\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}{2}}\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right),\sqrt{\frac{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}{2}}+\frac{1}{2}+\sqrt{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}\right)-\mathrm{atan}_2\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}{2}}\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right),\sqrt{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}\right)-\frac{\pi}{2}}{\pi}+\frac{1}{2}$$

$$\eta_{842} = \pi-2\,\mathrm{atan}_2\left(0,\sqrt{\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)}\right)$$

$$\eta_{843} = -\begin{cases}0 & \text{for } \sqrt{\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)}=0\\ \mathrm{atan}_2\left(-\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right),\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right) & \text{otherwise}\end{cases}-\pi$$

$$\eta_{844} = -\frac{\alpha}{4}-\frac{\beta}{4}-\frac{\mathrm{atan}_2\left(2\left|\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|\left|\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|,-\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right)}{2}$$

$$\eta_{845} = \frac{\pi-2\,\mathrm{atan}_2\left(0,\sqrt{\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)}\right)}{\pi}$$

$$\eta_{846} = \frac{-\pi\left(\begin{cases}0 & \text{for } \sqrt{\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)}=0\\ \frac{\mathrm{atan}_2\left(-\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right),\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right)}{\pi}+\frac{1}{2} & \text{otherwise}\end{cases}\right)-\frac{\pi}{2}}{\pi}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{847} = \frac{\pi\left(\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right),\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right) & \text{otherwise} \end{cases}+\frac{1}{2}\right)+\frac{\pi}{2}}{\pi}+\frac{1}{2}$$

$$\eta_{848} = \frac{-\frac{\alpha}{2}-\frac{\beta}{2}+\pi\left(\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right),\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right) & \text{otherwise} \end{cases}+\frac{1}{2}\right)+2\left(\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right),\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right) & \text{otherwise} \end{cases} = 0\right)\left(\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|\left|\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|,-\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right) & \text{for } \sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\le 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|\left|\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|,-\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right)-2\pi & \text{otherwise} \end{cases}\right)+2\operatorname{atan_2}\left(0,\sqrt{\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)}\right)-4\operatorname{atan_2}\left(\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\le 0 \\ -1 & \text{otherwise} \end{cases}\right),\sqrt{\frac{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}{2}}+\frac{1}{2}+\sqrt{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}\right)+2\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\le 0 \\ -1 & \text{otherwise} \end{cases}\right),\sqrt{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}\right)-\frac{3\pi}{2}}{\pi}$$

$$\eta_{849} = \frac{\frac{\alpha}{4}+\frac{\beta}{4}-\left(\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right),\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right) & \text{otherwise} \end{cases} = 0\right)+\frac{\left(\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|\left|\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|,-\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right) & \text{for } \sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\le 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|\left|\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|,-\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right)-2\pi & \text{otherwise} \end{cases}\right)}{4}-\operatorname{atan_2}\left(0,\sqrt{\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)}\right)+2\operatorname{atan_2}\left(\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\le 0 \\ -1 & \text{otherwise} \end{cases}\right),\sqrt{\frac{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}{2}}+\frac{1}{2}+\sqrt{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}\right)-\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\le 0 \\ -1 & \text{otherwise} \end{cases}\right),\sqrt{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}\right)}{\pi}+\frac{1}{2}$$

$$\eta_{850} = e^{i\left(-\frac{\alpha}{2}-\frac{\beta}{2}+\left(\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right),\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right) & \text{otherwise} \end{cases} = 0\right)-\frac{\left(\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|\left|\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|,-\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right) & \text{for } \sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\le 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|\left|\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|,-\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right)-2\pi & \text{otherwise} \end{cases}\right)}{4}-\operatorname{atan_2}\left(0,\sqrt{\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)}\right)-2\operatorname{atan_2}\left(\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\le 0 \\ -1 & \text{otherwise} \end{cases}\right),\sqrt{\frac{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}{2}}+\frac{1}{2}+\sqrt{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}\right)+\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\le 0 \\ -1 & \text{otherwise} \end{cases}\right),\sqrt{\left|\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|}\right)\right)}$$

$$\eta_{851} = -\frac{\alpha}{4}-\frac{\beta}{4}-\frac{\left(\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|\left|\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|,-\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right) & \text{for } \sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)-\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\le 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|\left|\cos\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right|,-\sin^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)+\cos^2\left(\frac{\alpha}{4}-\frac{\beta}{4}\right)\right)-2\pi & \text{otherwise} \end{cases}\right)}{4}$$

$$\eta_{852} = \frac{\pi t(2global_{shift}+1)+2\left(\begin{cases} 0 & \text{for } \left|\cos\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0,\cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases} = 0\right)-\frac{\left(\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|,-\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)\right) & \text{for } \sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\le 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|,-\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)\right)-2\pi & \text{otherwise} \end{cases}\right)}{2}-4\operatorname{atan_2}\left(\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\le 0 \\ -1 & \text{otherwise} \end{cases}\right),\sqrt{\frac{\left|\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\right|}{2}}+\frac{1}{2}+\sqrt{\left|\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\right|}\right)+2\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\le 0 \\ -1 & \text{otherwise} \end{cases}\right),\sqrt{\left|\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\right|}\right)-2\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|,\left|\cos\left(\frac{\pi t}{2}\right)\right|\right)-2\pi}{\pi}$$

$$\eta_{853} = \frac{-\frac{\pi t\left(2global_{shift}+1\right)}{2}-\left(\begin{cases} 0 & \text{for } \left|\cos\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0,\cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases}+\frac{\left(\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|,-\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)\right) & \text{for } \sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\le 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|,-\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)\right)-2\pi & \text{otherwise} \end{cases}\right)}{4}+2\operatorname{atan_2}\left(\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\le 0 \\ -1 & \text{otherwise} \end{cases}\right),\sqrt{\frac{\left|\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\right|}{2}}+\frac{1}{2}+\sqrt{\left|\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\right|}\right)-\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\le 0 \\ -1 & \text{otherwise} \end{cases}\right),\sqrt{\left|\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\right|}\right)+\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|,\left|\cos\left(\frac{\pi t}{2}\right)\right|\right)}{\pi}+\frac{1}{2}$$

$$\eta_{854} = \frac{2\operatorname{atan_2}\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|,\left|\cos\left(\frac{\pi t}{2}\right)\right|\right)}{\pi}$$

$$\eta_{855} = -\frac{2\left(\begin{cases} 0 & \text{for } \left|\cos\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0,\cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases}\right)}{\pi}$$

$$\eta_{856} = \frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\pi t}{2}\right),0\right) & \text{otherwise} \end{cases}+\begin{cases} 0 & \text{for } \left|\cos\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0,\cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi}+\frac{1}{2}$$

$$\eta_{857} = -\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0,-2\sin\left(\frac{\pi t}{2}\right)\cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{858} = \frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0,-2\sin\left(\frac{\pi t}{2}\right)\cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi}+\frac{1}{2}$$

$$\eta_{859} = -\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|,0\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{860} = \frac{2\operatorname{atan_2}\left(\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\le 0 \\ -1 & \text{otherwise} \end{cases}\right),\sqrt{\frac{\left|\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\right|}{2}}+\frac{1}{2}+\sqrt{\left|\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\right|}\right)}{\pi}$$

$$\eta_{861} = \frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|,-\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)\right) & \text{for } \sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\le 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|,-\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)\right)-2\pi & \text{otherwise} \end{cases}}{\frac{\pi}{2}}+\pi}{\pi}$$

$$\eta_{862} = \frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|,-\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)\right) & \text{for } \sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\le 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|,-\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)\right)-2\pi & \text{otherwise} \end{cases}}{\frac{\pi}{2}}-\frac{\pi}{2}}{\pi}+\frac{1}{2}$$

$$\eta_{863} = \frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|,-\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)\right) & \text{for } \sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\le 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|,-\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)\right)-2\pi & \text{otherwise} \end{cases}}{2\pi}$$

$$\eta_{864} = -\frac{2\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\le 0 \\ -1 & \text{otherwise} \end{cases}\right),\sqrt{\left|\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\right|}\right)}{\pi}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{865} = -\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right| \left|\cos\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right| \left|\cos\left(\frac{\pi t}{2}\right)\right|, 0\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{866} = -\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right| \left|\cos\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan}_2\left(0, -2\sin\left(\frac{\pi t}{2}\right)\cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi}$$

$$\eta_{867} = -\frac{\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right| \left|\cos\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan}_2\left(0, -2\sin\left(\frac{\pi t}{2}\right)\cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}$$

$$\eta_{868} = -\frac{\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|, -\sin^2\left(\frac{\pi t}{2}\right) + \cos^2\left(\frac{\pi t}{2}\right)\right)}{\pi}$$

$$\eta_{869} = \frac{\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|, -\sin^2\left(\frac{\pi t}{2}\right) + \cos^2\left(\frac{\pi t}{2}\right)\right)}{\pi} + \frac{1}{2}$$

$$\eta_{870} = e^{i\left(\frac{\pi t\left(2 global_{shift}+1\right)}{2} + \begin{cases} 0 & \text{for } \left|\cos\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan}_2\left(0, \cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases} - \frac{\begin{cases} 2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right| \left|\cos\left(\frac{\pi t}{2}\right)\right|, -\sin^2\left(\frac{\pi t}{2}\right) + \cos^2\left(\frac{\pi t}{2}\right)\right) & \text{for } \sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right) \le 0 \\ 2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right| \left|\cos\left(\frac{\pi t}{2}\right)\right|, -\sin^2\left(\frac{\pi t}{2}\right) + \cos^2\left(\frac{\pi t}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4} - 2\operatorname{atan}_2\left(\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right)\right|}{2}}, \begin{cases} 1 & \text{for } \sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}\sqrt{\frac{\left|\sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right)\right|}{2} + \frac{1}{2}} + \sqrt{\left|\sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right)\right|}\right) + \operatorname{atan}_2\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}, \sqrt{\left|\sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right)\right|}\right) - \operatorname{atan}_2\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \left|\cos\left(\frac{\pi t}{2}\right)\right|\right) + \frac{\pi}{2}\right)\right)}$$

$$\eta_{871} = \frac{\pi t\left(2 global_{shift}+1\right)}{2} + \begin{cases} 0 & \text{for } \left|\cos\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan}_2\left(0, \cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases} - \frac{\begin{cases} 2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right| \left|\cos\left(\frac{\pi t}{2}\right)\right|, -\sin^2\left(\frac{\pi t}{2}\right) + \cos^2\left(\frac{\pi t}{2}\right)\right) & \text{for } \sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right) \le 0 \\ 2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right| \left|\cos\left(\frac{\pi t}{2}\right)\right|, -\sin^2\left(\frac{\pi t}{2}\right) + \cos^2\left(\frac{\pi t}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4} - \pi$$

$$\eta_{872} = \frac{\pi t\left(2 global_{shift}+1\right)}{2} + \begin{cases} 0 & \text{for } \left|\cos\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan}_2\left(0, \cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases} - \frac{\begin{cases} 2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right| \left|\cos\left(\frac{\pi t}{2}\right)\right|, -\sin^2\left(\frac{\pi t}{2}\right) + \cos^2\left(\frac{\pi t}{2}\right)\right) & \text{for } \sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right) \le 0 \\ 2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right| \left|\cos\left(\frac{\pi t}{2}\right)\right|, -\sin^2\left(\frac{\pi t}{2}\right) + \cos^2\left(\frac{\pi t}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4} + \pi$$

$$\eta_{873} = 2\operatorname{atan}_2\left(\left|\sin\left(\frac{\pi t}{2}\right)\right|, \left|\cos\left(\frac{\pi t}{2}\right)\right|\right)$$

$$\eta_{874} = \begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan}_2\left(-\sin\left(\frac{\pi t}{2}\right), 0\right) & \text{otherwise} \end{cases} - \begin{cases} 0 & \text{for } \left|\cos\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan}_2\left(0, \cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{875} = -\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan}_2\left(-\sin\left(\frac{\pi t}{2}\right), 0\right) & \text{otherwise} \end{cases} - \begin{cases} 0 & \text{for } \left|\cos\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan}_2\left(0, \cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{876} = -\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right| \left|\cos\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan}_2\left(0, -2\sin\left(\frac{\pi t}{2}\right)\cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{877} = -\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right| \left|\cos\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right| \left|\cos\left(\frac{\pi t}{2}\right)\right|, 0\right) & \text{otherwise} \end{cases}$$

$$\eta_{878} = 2\operatorname{atan}_2\left(\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}, \sqrt{\frac{\left|\sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right)\right|}{2} + \frac{1}{2}} + \sqrt{\left|\sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right)\right|}\right)\right)$$

$$\eta_{879} = \frac{\begin{cases} 2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right| \left|\cos\left(\frac{\pi t}{2}\right)\right|, -\sin^2\left(\frac{\pi t}{2}\right) + \cos^2\left(\frac{\pi t}{2}\right)\right) & \text{for } \sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right) \le 0 \\ 2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right| \left|\cos\left(\frac{\pi t}{2}\right)\right|, -\sin^2\left(\frac{\pi t}{2}\right) + \cos^2\left(\frac{\pi t}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2} + \frac{\pi}{2}$$

$$\eta_{880} = -2\operatorname{atan}_2\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}, \sqrt{\left|\sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right)\right|}\right)\right)$$

$$\eta_{881} = \begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right| \left|\cos\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right| \left|\cos\left(\frac{\pi t}{2}\right)\right|, 0\right) & \text{otherwise} \end{cases}$$

$$\eta_{882} = \begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right| \left|\cos\left(\frac{\pi t}{2}\right)\right| = 0 \\ \operatorname{atan}_2\left(0, -2\sin\left(\frac{\pi t}{2}\right)\cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{883} = -\operatorname{atan}_2\left(2\left|\sin\left(\tfrac{\pi t}{2}\right)\right|\left|\cos\left(\tfrac{\pi t}{2}\right)\right|, -\sin^2\left(\tfrac{\pi t}{2}\right)+\cos^2\left(\tfrac{\pi t}{2}\right)\right)$$

$$\eta_{884} = e^{i\left(\frac{\pi t\left(2\mathit{global}_{\mathit{shift}}+1\right)}{2}+\begin{cases}0 & \text{for } \left|\cos\left(\frac{\pi t}{2}\right)\right|=0\\ \operatorname{atan}_2\left(0,\cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise}\end{cases}-\frac{\begin{cases}2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|, -\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)\right) & \text{for } \sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\le 0\\ 2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|, -\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)\right)-2\pi & \text{otherwise}\end{cases}}{4}\right)}$$

$$\eta_{885} = \frac{\pi t\left(2\mathit{global}_{\mathit{shift}}+1\right)+2\left(\begin{cases}0 & \text{for } \left|\cos\left(\frac{\pi t}{2}\right)\right|=0\\ \operatorname{atan}_2\left(0,\cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise}\end{cases}-\frac{\begin{cases}2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|, -\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)\right) & \text{for } \sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\le 0\\ 2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|, -\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)\right)-2\pi & \text{otherwise}\end{cases}}{2}\right)-4\operatorname{atan}_2\left(\sqrt{\tfrac{1}{2}-\frac{\left|\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right)\cdot\sqrt{\frac{\left|\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\right|}{2}}+\tfrac{1}{2}+\sqrt{\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)}\right)+2\operatorname{atan}_2\left(\sqrt{2}\sqrt{\tfrac{1}{2}-\frac{\left|\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right)\cdot\sqrt{\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)}\right)-2\operatorname{atan}_2\left(\left|\sin\left(\tfrac{\pi t}{2}\right)\right|,\left|\cos\left(\tfrac{\pi t}{2}\right)\right|\right)-\pi}{\pi}$$

$$\eta_{886} = \frac{-\frac{\pi t\left(2\mathit{global}_{\mathit{shift}}+1\right)}{2}-\begin{cases}0 & \text{for } \left|\cos\left(\frac{\pi t}{2}\right)\right|=0\\ \operatorname{atan}_2\left(0,\cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise}\end{cases}+\frac{\begin{cases}2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|, -\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)\right) & \text{for } \sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\le 0\\ 2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|, -\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)\right)-2\pi & \text{otherwise}\end{cases}}{4}+2\operatorname{atan}_2\left(\sqrt{\tfrac{1}{2}-\frac{\left|\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right)\cdot\sqrt{\frac{\left|\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\right|}{2}}+\tfrac{1}{2}+\sqrt{\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)}\right)-\operatorname{atan}_2\left(\sqrt{2}\sqrt{\tfrac{1}{2}-\frac{\left|\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right)\cdot\sqrt{\left|\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\right|}\right)+\operatorname{atan}_2\left(\left|\sin\left(\tfrac{\pi t}{2}\right)\right|,\left|\cos\left(\tfrac{\pi t}{2}\right)\right|\right)-\frac{\pi}{2}}{\pi}+\tfrac{1}{2}$$

$$\eta_{887} = -\frac{\begin{cases}0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right|=0\\ \operatorname{atan}_2\left(0,\sin\left(\frac{\pi t}{2}\right)\right) & \text{otherwise}\end{cases}}{\pi}+\frac{\begin{cases}0 & \text{for } \left|\cos\left(\frac{\pi t}{2}\right)\right|=0\\ \operatorname{atan}_2\left(0,\cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise}\end{cases}}{\pi}+\tfrac{1}{2}$$

$$\eta_{888} = -\frac{\begin{cases}0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|=0\\ \operatorname{atan}_2\left(-2\sin\left(\frac{\pi t}{2}\right)\cos\left(\frac{\pi t}{2}\right),0\right) & \text{otherwise}\end{cases}}{\pi}$$

$$\eta_{889} = \frac{\begin{cases}0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|=0\\ \operatorname{atan}_2\left(-2\sin\left(\frac{\pi t}{2}\right)\cos\left(\frac{\pi t}{2}\right),0\right) & \text{otherwise}\end{cases}}{\pi}+\tfrac{1}{2}$$

$$\eta_{890} = \frac{\begin{cases}0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|=0\\ \operatorname{atan}_2\left(-2\sin\left(\frac{\pi t}{2}\right)\cos\left(\frac{\pi t}{2}\right),0\right) & \text{otherwise}\end{cases}}{\pi}$$

$$\eta_{891} = -\frac{\begin{cases}0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|=0\\ \operatorname{atan}_2\left(-2\sin\left(\frac{\pi t}{2}\right)\cos\left(\frac{\pi t}{2}\right),0\right) & \text{otherwise}\end{cases}}{\pi}+\tfrac{1}{2}$$

$$\eta_{892} = \frac{\sqrt{2}(1-i)e^{\frac{i\pi\left(2t\left(2\mathit{global}_{\mathit{shift}}+1\right)-3\right)}{4}}}{2}$$

$$\eta_{893} = \begin{cases}0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right|=0\\ \operatorname{atan}_2\left(0,\sin\left(\frac{\pi t}{2}\right)\right) & \text{otherwise}\end{cases}-\begin{cases}0 & \text{for } \left|\cos\left(\frac{\pi t}{2}\right)\right|=0\\ \operatorname{atan}_2\left(0,\cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise}\end{cases}$$

$$\eta_{894} = -\begin{cases}0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right|=0\\ \operatorname{atan}_2\left(0,\sin\left(\frac{\pi t}{2}\right)\right) & \text{otherwise}\end{cases}-\begin{cases}0 & \text{for } \left|\cos\left(\frac{\pi t}{2}\right)\right|=0\\ \operatorname{atan}_2\left(0,\cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise}\end{cases}$$

$$\eta_{895} = -\begin{cases}0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|=0\\ \operatorname{atan}_2\left(-2\sin\left(\frac{\pi t}{2}\right)\cos\left(\frac{\pi t}{2}\right),0\right) & \text{otherwise}\end{cases}$$

$$\eta_{896} = \begin{cases}0 & \text{for } \left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|=0\\ \operatorname{atan}_2\left(-2\sin\left(\frac{\pi t}{2}\right)\cos\left(\frac{\pi t}{2}\right),0\right) & \text{otherwise}\end{cases}$$

$$\eta_{897} = \frac{\sqrt{2}i(-1+i)e^{\frac{i\pi\left(2t\left(2\mathit{global}_{\mathit{shift}}+1\right)+2t-1\right)}{4}}}{2}$$

$$\eta_{898} = \frac{\pi-2\operatorname{atan}_2\left(0,\sqrt{\sin^2\left(\frac{\pi\alpha}{2}\right)+\cos^2\left(\frac{\pi\alpha}{2}\right)}\right)}{\pi}$$

$$\eta_{899} = \frac{-\pi\left(\frac{\begin{cases}0 & \text{for } \sqrt{\sin^2\left(\frac{\pi\alpha}{2}\right)+\cos^2\left(\frac{\pi\alpha}{2}\right)}=0\\ \operatorname{atan}_2\left(-\sin\left(\frac{\pi\alpha}{2}\right),\cos\left(\frac{\pi\alpha}{2}\right)\right) & \text{otherwise}\end{cases}}{\pi}+\tfrac{1}{2}\right)-\frac{\pi}{2}}{\pi}$$

$$\eta_{900} = \frac{\pi\left(\frac{\begin{cases}0 & \text{for } \sqrt{\sin^2\left(\frac{\pi\alpha}{2}\right)+\cos^2\left(\frac{\pi\alpha}{2}\right)}=0\\ \operatorname{atan}_2\left(-\sin\left(\frac{\pi\alpha}{2}\right),\cos\left(\frac{\pi\alpha}{2}\right)\right) & \text{otherwise}\end{cases}}{\pi}+\tfrac{1}{2}\right)+\frac{\pi}{2}}{\pi}+\tfrac{1}{2}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{901} = \frac{\pi\left(-\frac{\begin{cases}0 & \text{for } \sqrt{\sin^2\left(\frac{\pi\alpha}{2}\right)+\cos^2\left(\frac{\pi\alpha}{2}\right)}=0\\ \operatorname{atan}_2\left(-\sin\left(\frac{\pi\alpha}{2}\right),\cos\left(\frac{\pi\alpha}{2}\right)\right) & \text{otherwise}\end{cases}}{\pi}+\frac{1}{2}\right)+2\left(\begin{cases}0 & \text{for } \sqrt{\sin^2\left(\frac{\pi\alpha}{2}\right)+\cos^2\left(\frac{\pi\alpha}{2}\right)}=0\\ \operatorname{atan}_2\left(-\sin\left(\frac{\pi\alpha}{2}\right),\cos\left(\frac{\pi\alpha}{2}\right)\right) & \text{otherwise}\end{cases}\right)-\frac{\begin{cases}2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi\alpha}{2}\right)\right|\left|\cos\left(\frac{\pi\alpha}{2}\right)\right|,-\sin^2\left(\frac{\pi\alpha}{2}\right)+\cos^2\left(\frac{\pi\alpha}{2}\right)\right) & \text{for } \sin^2\left(\frac{\pi\alpha}{2}\right)-\cos^2\left(\frac{\pi\alpha}{2}\right)\le 0\\ 2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi\alpha}{2}\right)\right|\left|\cos\left(\frac{\pi\alpha}{2}\right)\right|,-\sin^2\left(\frac{\pi\alpha}{2}\right)+\cos^2\left(\frac{\pi\alpha}{2}\right)\right)-2\pi & \text{otherwise}\end{cases}}{2}+2\operatorname{atan}_2\left(0,\sqrt{\sin^2\left(\frac{\pi\beta}{2}\right)+\cos^2\left(\frac{\pi\beta}{2}\right)}\right)-4\operatorname{atan}_2\left(\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\pi\beta}{2}\right)-\cos^2\left(\frac{\pi\beta}{2}\right)\right|}{2}}\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\pi\alpha}{2}\right)-\cos^2\left(\frac{\pi\alpha}{2}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right)\cdot\sqrt{\frac{\left|\sin^2\left(\frac{\pi\beta}{2}\right)-\cos^2\left(\frac{\pi\beta}{2}\right)\right|}{2}}+\frac{1}{2}+\sqrt{\left|\sin^2\left(\frac{\pi\beta}{2}\right)-\cos^2\left(\frac{\pi\beta}{2}\right)\right|}\right)+2\operatorname{atan}_2\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\pi\beta}{2}\right)-\cos^2\left(\frac{\pi\beta}{2}\right)\right|}{2}}\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\pi\alpha}{2}\right)-\cos^2\left(\frac{\pi\alpha}{2}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right)\cdot\sqrt{\left|\sin^2\left(\frac{\pi\beta}{2}\right)-\cos^2\left(\frac{\pi\beta}{2}\right)\right|}\right)-\frac{3\pi}{2}}{\pi}$$

$$\eta_{902} = \frac{-\frac{\begin{cases}0 & \text{for } \sqrt{\sin^2\left(\frac{\pi\alpha}{2}\right)+\cos^2\left(\frac{\pi\alpha}{2}\right)}=0\\ \operatorname{atan}_2\left(-\sin\left(\frac{\pi\alpha}{2}\right),\cos\left(\frac{\pi\alpha}{2}\right)\right) & \text{otherwise}\end{cases}}{2}+\frac{\begin{cases}2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi\alpha}{2}\right)\right|\left|\cos\left(\frac{\pi\alpha}{2}\right)\right|,-\sin^2\left(\frac{\pi\alpha}{2}\right)+\cos^2\left(\frac{\pi\alpha}{2}\right)\right) & \text{for } \sin^2\left(\frac{\pi\alpha}{2}\right)-\cos^2\left(\frac{\pi\alpha}{2}\right)\le 0\\ 2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi\alpha}{2}\right)\right|\left|\cos\left(\frac{\pi\alpha}{2}\right)\right|,-\sin^2\left(\frac{\pi\alpha}{2}\right)+\cos^2\left(\frac{\pi\alpha}{2}\right)\right)-2\pi & \text{otherwise}\end{cases}}{4}-\operatorname{atan}_2\left(0,\sqrt{\sin^2\left(\frac{\pi\beta}{2}\right)+\cos^2\left(\frac{\pi\beta}{2}\right)}\right)+2\operatorname{atan}_2\left(\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\pi\beta}{2}\right)-\cos^2\left(\frac{\pi\beta}{2}\right)\right|}{2}}\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\pi\alpha}{2}\right)-\cos^2\left(\frac{\pi\alpha}{2}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right)\cdot\sqrt{\frac{\left|\sin^2\left(\frac{\pi\beta}{2}\right)-\cos^2\left(\frac{\pi\beta}{2}\right)\right|}{2}}+\frac{1}{2}+\sqrt{\left|\sin^2\left(\frac{\pi\beta}{2}\right)-\cos^2\left(\frac{\pi\beta}{2}\right)\right|}\right)-\operatorname{atan}_2\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\pi\beta}{2}\right)-\cos^2\left(\frac{\pi\beta}{2}\right)\right|}{2}}\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\pi\alpha}{2}\right)-\cos^2\left(\frac{\pi\alpha}{2}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right)\cdot\sqrt{\left|\sin^2\left(\frac{\pi\beta}{2}\right)-\cos^2\left(\frac{\pi\beta}{2}\right)\right|}\right)}{\pi}+\frac{1}{2}$$

$$\eta_{903} = -\frac{\begin{cases}0 & \text{for } \left|\sin\left(\frac{\pi\alpha}{2}\right)\right|\left|\cos\left(\frac{\pi\alpha}{2}\right)\right|=0\\ \operatorname{atan}_2\left(0,-2\sin\left(\frac{\pi\alpha}{2}\right)\cos\left(\frac{\pi\alpha}{2}\right)\right) & \text{otherwise}\end{cases}}{\pi}$$

$$\eta_{904} = \frac{2\operatorname{atan}_2\left(\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\pi\beta}{2}\right)-\cos^2\left(\frac{\pi\beta}{2}\right)\right|}{2}}\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\pi\alpha}{2}\right)-\cos^2\left(\frac{\pi\alpha}{2}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right)\cdot\sqrt{\frac{\left|\sin^2\left(\frac{\pi\beta}{2}\right)-\cos^2\left(\frac{\pi\beta}{2}\right)\right|}{2}}+\frac{1}{2}+\sqrt{\left|\sin^2\left(\frac{\pi\beta}{2}\right)-\cos^2\left(\frac{\pi\beta}{2}\right)\right|}\right)}{\pi}$$

$$\eta_{905} = \frac{\frac{\begin{cases}2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi\alpha}{2}\right)\right|\left|\cos\left(\frac{\pi\alpha}{2}\right)\right|,-\sin^2\left(\frac{\pi\alpha}{2}\right)+\cos^2\left(\frac{\pi\alpha}{2}\right)\right) & \text{for } \sin^2\left(\frac{\pi\alpha}{2}\right)-\cos^2\left(\frac{\pi\alpha}{2}\right)\le 0\\ 2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi\alpha}{2}\right)\right|\left|\cos\left(\frac{\pi\alpha}{2}\right)\right|,-\sin^2\left(\frac{\pi\alpha}{2}\right)+\cos^2\left(\frac{\pi\alpha}{2}\right)\right)-2\pi & \text{otherwise}\end{cases}}{2}}{\pi}+\pi$$

$$\eta_{906} = \frac{-\frac{\begin{cases}2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi\alpha}{2}\right)\right|\left|\cos\left(\frac{\pi\alpha}{2}\right)\right|,-\sin^2\left(\frac{\pi\alpha}{2}\right)+\cos^2\left(\frac{\pi\alpha}{2}\right)\right) & \text{for } \sin^2\left(\frac{\pi\alpha}{2}\right)-\cos^2\left(\frac{\pi\alpha}{2}\right)\le 0\\ 2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi\alpha}{2}\right)\right|\left|\cos\left(\frac{\pi\alpha}{2}\right)\right|,-\sin^2\left(\frac{\pi\alpha}{2}\right)+\cos^2\left(\frac{\pi\alpha}{2}\right)\right)-2\pi & \text{otherwise}\end{cases}}{2}}{\pi}-\frac{\pi}{2}}{\pi}+\frac{1}{2}$$

$$\eta_{907} = \frac{\begin{cases}2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi\alpha}{2}\right)\right|\left|\cos\left(\frac{\pi\alpha}{2}\right)\right|,-\sin^2\left(\frac{\pi\alpha}{2}\right)+\cos^2\left(\frac{\pi\alpha}{2}\right)\right) & \text{for } \sin^2\left(\frac{\pi\alpha}{2}\right)-\cos^2\left(\frac{\pi\alpha}{2}\right)\le 0\\ 2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi\alpha}{2}\right)\right|\left|\cos\left(\frac{\pi\alpha}{2}\right)\right|,-\sin^2\left(\frac{\pi\alpha}{2}\right)+\cos^2\left(\frac{\pi\alpha}{2}\right)\right)-2\pi & \text{otherwise}\end{cases}}{2\pi}$$

$$\eta_{908} = -\frac{2\operatorname{atan}_2\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\pi\beta}{2}\right)-\cos^2\left(\frac{\pi\beta}{2}\right)\right|}{2}}\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\pi\alpha}{2}\right)-\cos^2\left(\frac{\pi\alpha}{2}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right)\cdot\sqrt{\left|\sin^2\left(\frac{\pi\beta}{2}\right)-\cos^2\left(\frac{\pi\beta}{2}\right)\right|}\right)}{\pi}$$

$$\eta_{909} = -\frac{\begin{cases}0 & \text{for } \left|\sin\left(\frac{\pi\alpha}{2}\right)\right|\left|\cos\left(\frac{\pi\alpha}{2}\right)\right|=0\\ \operatorname{atan}_2\left(0,-2\sin\left(\frac{\pi\alpha}{2}\right)\cos\left(\frac{\pi\alpha}{2}\right)\right) & \text{otherwise}\end{cases}}{\pi}$$

$$\eta_{910} = -\frac{\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi\beta}{2}\right)\right|\left|\cos\left(\frac{\pi\beta}{2}\right)\right|,-\sin^2\left(\frac{\pi\beta}{2}\right)+\cos^2\left(\frac{\pi\beta}{2}\right)\right)}{\pi}$$

$$\eta_{911} = \frac{\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi\beta}{2}\right)\right|\left|\cos\left(\frac{\pi\beta}{2}\right)\right|,-\sin^2\left(\frac{\pi\beta}{2}\right)+\cos^2\left(\frac{\pi\beta}{2}\right)\right)}{\pi}+\frac{1}{2}$$

$$\eta_{912} = \frac{2\operatorname{atan}_2\left(0,\sqrt{\sin^2\left(\frac{\pi\beta}{2}\right)+\cos^2\left(\frac{\pi\beta}{2}\right)}\right)}{\pi}$$

$$\eta_{913} = -\frac{2\left(\begin{cases}0 & \text{for } \sqrt{\sin^2\left(\frac{\pi\alpha}{2}\right)+\cos^2\left(\frac{\pi\alpha}{2}\right)}=0\\ \operatorname{atan}_2\left(-\sin\left(\frac{\pi\alpha}{2}\right),\cos\left(\frac{\pi\alpha}{2}\right)\right) & \text{otherwise}\end{cases}\right)}{\pi}$$

$$\eta_{914} = \frac{\begin{cases}0 & \text{for } \sqrt{\sin^2\left(\frac{\pi\alpha}{2}\right)+\cos^2\left(\frac{\pi\alpha}{2}\right)}=0\\ \operatorname{atan}_2\left(-\sin\left(\frac{\pi\alpha}{2}\right),\cos\left(\frac{\pi\alpha}{2}\right)\right) & \text{otherwise}\end{cases}}{\pi}+\frac{1}{2}$$

$$\eta_{915} = e^{i\left(\begin{cases}0 & \text{for } \sqrt{\sin^2\left(\frac{\pi\alpha}{2}\right)+\cos^2\left(\frac{\pi\alpha}{2}\right)}=0\\ \operatorname{atan}_2\left(-\sin\left(\frac{\pi\alpha}{2}\right),\cos\left(\frac{\pi\alpha}{2}\right)\right) & \text{otherwise}\end{cases}-\frac{\begin{cases}2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi\alpha}{2}\right)\right|\left|\cos\left(\frac{\pi\alpha}{2}\right)\right|,-\sin^2\left(\frac{\pi\alpha}{2}\right)+\cos^2\left(\frac{\pi\alpha}{2}\right)\right) & \text{for } \sin^2\left(\frac{\pi\alpha}{2}\right)-\cos^2\left(\frac{\pi\alpha}{2}\right)\le 0\\ 2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi\alpha}{2}\right)\right|\left|\cos\left(\frac{\pi\alpha}{2}\right)\right|,-\sin^2\left(\frac{\pi\alpha}{2}\right)+\cos^2\left(\frac{\pi\alpha}{2}\right)\right)-2\pi & \text{otherwise}\end{cases}}{4}-\operatorname{atan}_2\left(0,\sqrt{\sin^2\left(\frac{\pi\beta}{2}\right)+\cos^2\left(\frac{\pi\beta}{2}\right)}\right)-2\operatorname{atan}_2\left(\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\pi\beta}{2}\right)-\cos^2\left(\frac{\pi\beta}{2}\right)\right|}{2}}\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\pi\alpha}{2}\right)-\cos^2\left(\frac{\pi\alpha}{2}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right)\cdot\sqrt{\frac{\left|\sin^2\left(\frac{\pi\beta}{2}\right)-\cos^2\left(\frac{\pi\beta}{2}\right)\right|}{2}}+\frac{1}{2}+\sqrt{\left|\sin^2\left(\frac{\pi\beta}{2}\right)-\cos^2\left(\frac{\pi\beta}{2}\right)\right|}\right)+\operatorname{atan}_2\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\pi\beta}{2}\right)-\cos^2\left(\frac{\pi\beta}{2}\right)\right|}{2}}\left(\begin{cases}1 & \text{for } \sin^2\left(\frac{\pi\alpha}{2}\right)-\cos^2\left(\frac{\pi\alpha}{2}\right)\le 0\\ -1 & \text{otherwise}\end{cases}\right)\cdot\sqrt{\left|\sin^2\left(\frac{\pi\beta}{2}\right)-\cos^2\left(\frac{\pi\beta}{2}\right)\right|}\right)}{4}\right)}$$

$$\eta_{916} = \pi - 2\operatorname{atan}_2\left(0,\sqrt{\sin^2\left(\frac{\pi\alpha}{2}\right)+\cos^2\left(\frac{\pi\alpha}{2}\right)}\right)$$

$$\eta_{917} = -\begin{cases}0 & \text{for } \sqrt{\sin^2\left(\frac{\pi\alpha}{2}\right)+\cos^2\left(\frac{\pi\alpha}{2}\right)}=0\\ \operatorname{atan}_2\left(-\sin\left(\frac{\pi\alpha}{2}\right),\cos\left(\frac{\pi\alpha}{2}\right)\right) & \text{otherwise}\end{cases}-\pi$$

$$\eta_{918} = -\frac{\begin{cases}2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi\alpha}{2}\right)\right|\left|\cos\left(\frac{\pi\alpha}{2}\right)\right|,-\sin^2\left(\frac{\pi\alpha}{2}\right)+\cos^2\left(\frac{\pi\alpha}{2}\right)\right) & \text{for } \sin^2\left(\frac{\pi\alpha}{2}\right)-\cos^2\left(\frac{\pi\alpha}{2}\right)\le 0\\ 2\operatorname{atan}_2\left(2\left|\sin\left(\frac{\pi\alpha}{2}\right)\right|\left|\cos\left(\frac{\pi\alpha}{2}\right)\right|,-\sin^2\left(\frac{\pi\alpha}{2}\right)+\cos^2\left(\frac{\pi\alpha}{2}\right)\right)-2\pi & \text{otherwise}\end{cases}}{4}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{919} = \begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\pi\alpha}{2}\right) + \cos^2\left(\frac{\pi\alpha}{2}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\pi\alpha}{2}\right), \cos\left(\frac{\pi\alpha}{2}\right)\right) & \text{otherwise} \end{cases} - \frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi\alpha}{2}\right)\right|\left|\cos\left(\frac{\pi\alpha}{2}\right)\right|, -\sin^2\left(\frac{\pi\alpha}{2}\right) + \cos^2\left(\frac{\pi\alpha}{2}\right)\right) & \text{for } \sin^2\left(\frac{\pi\alpha}{2}\right) - \cos^2\left(\frac{\pi\alpha}{2}\right) \le 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi\alpha}{2}\right)\right|\left|\cos\left(\frac{\pi\alpha}{2}\right)\right|, -\sin^2\left(\frac{\pi\alpha}{2}\right) + \cos^2\left(\frac{\pi\alpha}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4} + \pi$$

$$\eta_{920} = -\begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi\alpha}{2}\right)\right|\left|\cos\left(\frac{\pi\alpha}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, -2\sin\left(\frac{\pi\alpha}{2}\right)\cos\left(\frac{\pi\alpha}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{921} = 2\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\frac{\pi\beta}{2}\right) - \cos^2\left(\frac{\pi\beta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\pi\alpha}{2}\right) - \cos^2\left(\frac{\pi\alpha}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}, \sqrt{\frac{\left|\sin^2\left(\frac{\pi\beta}{2}\right) - \cos^2\left(\frac{\pi\beta}{2}\right)\right|}{2}} + \frac{1}{2} + \sqrt{\left|\sin^2\left(\frac{\pi\alpha}{2}\right) - \cos^2\left(\frac{\pi\alpha}{2}\right)\right|}\right)\right)$$

$$\eta_{922} = \frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi\alpha}{2}\right)\right|\left|\cos\left(\frac{\pi\alpha}{2}\right)\right|, -\sin^2\left(\frac{\pi\alpha}{2}\right) + \cos^2\left(\frac{\pi\alpha}{2}\right)\right) & \text{for } \sin^2\left(\frac{\pi\alpha}{2}\right) - \cos^2\left(\frac{\pi\alpha}{2}\right) \le 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi\alpha}{2}\right)\right|\left|\cos\left(\frac{\pi\alpha}{2}\right)\right|, -\sin^2\left(\frac{\pi\alpha}{2}\right) + \cos^2\left(\frac{\pi\alpha}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2} + \frac{\pi}{2}$$

$$\eta_{923} = -2\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\frac{\pi\beta}{2}\right) - \cos^2\left(\frac{\pi\beta}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\pi\alpha}{2}\right) - \cos^2\left(\frac{\pi\alpha}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}, \sqrt{\left|\sin^2\left(\frac{\pi\alpha}{2}\right) - \cos^2\left(\frac{\pi\alpha}{2}\right)\right|}\right)\right)$$

$$\eta_{924} = \begin{cases} 0 & \text{for } \left|\sin\left(\frac{\pi\beta}{2}\right)\right|\left|\cos\left(\frac{\pi\beta}{2}\right)\right| = 0 \\ \operatorname{atan_2}\left(0, -2\sin\left(\frac{\pi\beta}{2}\right)\cos\left(\frac{\pi\beta}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{925} = -\operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi\alpha}{2}\right)\right|\left|\cos\left(\frac{\pi\alpha}{2}\right)\right|, -\sin^2\left(\frac{\pi\alpha}{2}\right) + \cos^2\left(\frac{\pi\alpha}{2}\right)\right)$$

$$\eta_{926} = 2\operatorname{atan_2}\left(0, \sqrt{\sin^2\left(\frac{\pi\beta}{2}\right) + \cos^2\left(\frac{\pi\beta}{2}\right)}\right)$$

$$\eta_{927} = -\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\pi\alpha}{2}\right) + \cos^2\left(\frac{\pi\alpha}{2}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\pi\alpha}{2}\right), \cos\left(\frac{\pi\alpha}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{928} = e^{i\left(\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\pi\alpha}{2}\right) + \cos^2\left(\frac{\pi\alpha}{2}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\pi\alpha}{2}\right), \cos\left(\frac{\pi\alpha}{2}\right)\right) & \text{otherwise} \end{cases} - \frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi\alpha}{2}\right)\right|\left|\cos\left(\frac{\pi\alpha}{2}\right)\right|, -\sin^2\left(\frac{\pi\alpha}{2}\right) + \cos^2\left(\frac{\pi\alpha}{2}\right)\right) & \text{for } \sin^2\left(\frac{\pi\alpha}{2}\right) - \cos^2\left(\frac{\pi\alpha}{2}\right) \le 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi\alpha}{2}\right)\right|\left|\cos\left(\frac{\pi\alpha}{2}\right)\right|, -\sin^2\left(\frac{\pi\alpha}{2}\right) + \cos^2\left(\frac{\pi\alpha}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4}\right)}$$

$$\eta_{929} = \frac{\pi - 2\operatorname{atan_2}\left(0, \sqrt{\sin^2\left(\frac{\pi\beta}{2}\right) + \cos^2\left(\frac{\pi\beta}{2}\right)}\right)}{\pi}$$

$$\eta_{930} = \frac{-\pi\left(\frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\pi\beta}{2}\right) + \cos^2\left(\frac{\pi\beta}{2}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\pi\beta}{2}\right), \cos\left(\frac{\pi\beta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}\right)^{-\frac{\pi}{2}}}{\pi}$$

$$\eta_{931} = \frac{\pi\left(\frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\pi\beta}{2}\right) + \cos^2\left(\frac{\pi\beta}{2}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\pi\beta}{2}\right), \cos\left(\frac{\pi\beta}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}\right)^{+\frac{\pi}{2}}}{\pi} + \frac{1}{2}$$

$$\eta_{932} = \frac{\pi t\left(2global_{shift}+1\right)+\pi\left(\frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\pi t}{2}\right) + \cos^2\left(\frac{\pi t}{2}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\pi t}{2}\right), \cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}\right)^{+2\left(\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\pi t}{2}\right) + \cos^2\left(\frac{\pi t}{2}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\pi t}{2}\right), \cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases} - \frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|, -\sin^2\left(\frac{\pi t}{2}\right) + \cos^2\left(\frac{\pi t}{2}\right)\right) & \text{for } \sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right) \le 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|, -\sin^2\left(\frac{\pi t}{2}\right) + \cos^2\left(\frac{\pi t}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{2} + 2\operatorname{atan_2}\left(0, \sqrt{\sin^2\left(\frac{\pi t}{2}\right) + \cos^2\left(\frac{\pi t}{2}\right)}\right) - 4\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}, \sqrt{\frac{\left|\sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right)\right|}{2}} + \frac{1}{2} + \sqrt{\left|\sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right)\right|}\right) + 2\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}, \sqrt{\left|\sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right)\right|}\right) - \frac{3\pi}{2}}{\pi}$$

$$\eta_{933} = \frac{-\frac{\pi t\left(2global_{shift}+1\right)}{2} - \begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\pi t}{2}\right) + \cos^2\left(\frac{\pi t}{2}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\pi t}{2}\right), \cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases} + \frac{\begin{cases} 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|, -\sin^2\left(\frac{\pi t}{2}\right) + \cos^2\left(\frac{\pi t}{2}\right)\right) & \text{for } \sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right) \le 0 \\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|, -\sin^2\left(\frac{\pi t}{2}\right) + \cos^2\left(\frac{\pi t}{2}\right)\right) - 2\pi & \text{otherwise} \end{cases}}{4} - \operatorname{atan_2}\left(0, \sqrt{\sin^2\left(\frac{\pi t}{2}\right) + \cos^2\left(\frac{\pi t}{2}\right)}\right) + 2\operatorname{atan_2}\left(\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}, \sqrt{\frac{\left|\sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right)\right|}{2}} + \frac{1}{2} + \sqrt{\left|\sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right)\right|}\right) - \operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2} - \frac{\left|\sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right)\right|}{2}}\left(\begin{cases} 1 & \text{for } \sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right) \le 0 \\ -1 & \text{otherwise} \end{cases}, \sqrt{\left|\sin^2\left(\frac{\pi t}{2}\right) - \cos^2\left(\frac{\pi t}{2}\right)\right|}\right)}{\pi} + \frac{1}{2}$$

$$\eta_{934} = \frac{2\operatorname{atan_2}\left(0, \sqrt{\sin^2\left(\frac{\pi t}{2}\right) + \cos^2\left(\frac{\pi t}{2}\right)}\right)}{\pi}$$

$$\eta_{935} = -\frac{2\left(\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\pi t}{2}\right) + \cos^2\left(\frac{\pi t}{2}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\pi t}{2}\right), \cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases}\right)}{\pi}$$

$$\eta_{936} = \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2\left(\frac{\pi t}{2}\right) + \cos^2\left(\frac{\pi t}{2}\right)} = 0 \\ \operatorname{atan_2}\left(-\sin\left(\frac{\pi t}{2}\right), \cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise} \end{cases}}{\pi} + \frac{1}{2}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{937} = e^{i\left(\frac{\pi(2 global_shift+1)}{2}+\begin{cases}0 & \text{for } \sqrt{\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)}=0\\ \operatorname{atan_2}\left(-\sin\left(\frac{\pi t}{2}\right),\cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise}\end{cases}-\frac{\begin{cases}2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|,-\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)\right) & \text{for } \sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\le 0\\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|,-\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)\right)-2\pi & \text{otherwise}\end{cases}}{4}-\operatorname{atan_2}\left(0,\sqrt{\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)}\right)-2\operatorname{atan_2}\left(\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\right|}{2}}\begin{cases}1 & \text{for } \sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\le 0\\ -1 & \text{otherwise}\end{cases},\sqrt{\frac{\left|\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\right|}{2}+\frac{1}{2}}+\sqrt{\left|\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\right|}\right)+\operatorname{atan_2}\left(\sqrt{2}\sqrt{\frac{1}{2}-\frac{\left|\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\right|}{2}}\begin{cases}1 & \text{for } \sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\le 0\\ -1 & \text{otherwise}\end{cases},\sqrt{\left|\sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\right|}\right)\right)}$$

$$\eta_{938} = \pi - 2\operatorname{atan_2}\left(0,\sqrt{\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)}\right)$$

$$\eta_{939} = -\begin{cases}0 & \text{for } \sqrt{\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)}=0\\ \operatorname{atan_2}\left(-\sin\left(\frac{\pi t}{2}\right),\cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise}\end{cases}-\pi$$

$$\eta_{940} = \frac{\pi(2 global_shift+1)}{2}-\frac{\begin{cases}2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|,-\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)\right) & \text{for } \sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\le 0\\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|,-\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)\right)-2\pi & \text{otherwise}\end{cases}}{4}$$

$$\eta_{941} = \frac{\pi(2 global_shift+1)}{2}+\begin{cases}0 & \text{for } \sqrt{\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)}=0\\ \operatorname{atan_2}\left(-\sin\left(\frac{\pi t}{2}\right),\cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise}\end{cases}-\frac{\begin{cases}2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|,-\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)\right) & \text{for } \sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\le 0\\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|,-\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)\right)-2\pi & \text{otherwise}\end{cases}}{4}+\pi$$

$$\eta_{942} = 2\operatorname{atan_2}\left(0,\sqrt{\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)}\right)$$

$$\eta_{943} = -\begin{cases}0 & \text{for } \sqrt{\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)}=0\\ \operatorname{atan_2}\left(-\sin\left(\frac{\pi t}{2}\right),\cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise}\end{cases}$$

$$\eta_{944} = e^{i\left(\frac{\pi(2 global_shift+1)}{2}+\begin{cases}0 & \text{for } \sqrt{\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)}=0\\ \operatorname{atan_2}\left(-\sin\left(\frac{\pi t}{2}\right),\cos\left(\frac{\pi t}{2}\right)\right) & \text{otherwise}\end{cases}-\frac{\begin{cases}2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|,-\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)\right) & \text{for } \sin^2\left(\frac{\pi t}{2}\right)-\cos^2\left(\frac{\pi t}{2}\right)\le 0\\ 2\operatorname{atan_2}\left(2\left|\sin\left(\frac{\pi t}{2}\right)\right|\left|\cos\left(\frac{\pi t}{2}\right)\right|,-\sin^2\left(\frac{\pi t}{2}\right)+\cos^2\left(\frac{\pi t}{2}\right)\right)-2\pi & \text{otherwise}\end{cases}}{4}\right)}$$